



WILSON
ENGINEERING

Eastsound Sewer and Water District

Eastsound, Washington

Contract Documents

for the Construction of

EASTSOUND WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

**PART 1 – BIDDING REQUIREMENT
PART 2 – CONTRACTING REQUIREMENTS
PART 3 – TECHNICAL SPECIFICATIONS
PART 4 – REFERENCE DOCUMENTS
PART 5 – DRAWINGS**

Wilson Project # 2020-070

This project is funded in part by the Washington State
Department of Ecology
SRF Loan Agreement No. WQC-2021-EaSoWd-00208

Prepared by:
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Dec. 30, 2022

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ENGINEER'S STATEMENT

THE CONTRACT DOCUMENTS HAVE BEEN PREPARED UNDER THE DIRECTION OF THE PROFESSIONAL ENGINEERS,
REGISTERED IN THE STATE OF WASHINGTON, WHOSE SEALS AND SIGNATURES APPEAR BELOW.



12-30-2022

JEFF CHRISTNER, PE
PROJECT PRINCIPAL
WILSON ENGINEERING, LLC

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END OF SECTION

PART 1 - BIDDING REQUIREMENTS

SECTION 00 11 16 – INVITATION TO BID

ADVERTISEMENT FOR BIDS

Project Name: Eastsound Sewer and Water District
Wastewater Treatment Plant Upgrade – Phase I
Bid Date: February 8, 2023 – 10:00 AM
Pre-Bid Meeting: January 18, 2023 – 9:30 AM
Engineer: Wilson Engineering, LLC

NOTICE TO BIDDERS: Sealed bids will be received from contractors by the Eastsound Sewer and Water District, 143 Cessna Rd, Eastsound, WA 98245 until 10:00 AM, Wednesday, February 8, 2023 for the ESWD WWTP UPGRADE - PHASE I. All bids shall be received in sealed envelopes with “ESWD WWTP UPGRADE - PHASE I” marked plainly thereon. The Project involves the proposed wastewater treatment plant upgrades per contract documents. Said bids will then and there be opened and read aloud. Bidders and other properly interested parties are invited to be present at the bid opening. Bids received after the time fixed for opening cannot be considered.

Please contact Jeff Christner or Sam Rodriguez, Wilson Engineering, (360) 733-6100, for project information. Only bids from bidders who have obtained the Contract Documents and have requested to be listed on the Planholders’ List, will be accepted. Copies of plans and specifications are on file for review at the Eastsound Sewer and Water District, 143 Cessna Rd, Eastsound, WA 98245. Hard copies are available for purchase (\$200 non-refundable fee), or may be downloaded from the project website at <https://wilsonengineering.com/bidding-documents/>

A deposit in the form of a postal money order, cashier’s check, or bond in the amount of 5% of the greatest amount bid must be submitted with each bid proposal. Should the successful bidder fail to enter into a contract or furnish a satisfactory contract bond within the time stated in the specifications, the deposit shall be forfeited to the District.

There will be non-mandatory, pre-bid meeting for the Project held at 9:30 AM, Wednesday, January 18, 2023, at the WWTP, 143 Cessna Rd, Eastsound, WA 98245. A site visit of the Eastsound WWTP will follow the meeting.

The District shall reject any bid not accompanied by bid security. The District reserves the right to reject any or all bids if such action is in the best interest of the District. The Eastsound Sewer and Water District is an equal opportunity and affirmative action employer. Small, Minority and Women-owned businesses are encouraged to submit bids.

It is anticipated that this project will be funded in part by the Washington State Department of Ecology. Neither the State of Washington nor any of its departments or employees are, or shall be, a party to this contract or any subcontract resulting from this solicitation for bids. All bidders must be licensed contractors registered in the State of Washington. All work performed on this project will be subject to prevailing state wage rates.

In addition, this project is partially funded through the Washington State Community Development Block Grant program with federal funds from the U.S. Department of Housing and Urban Development.

The Successful bidder will be required to conform to the wage requirements prescribed by the federal Davis-Bacon and Related Acts which requires that all laborers and mechanics employed by contractors and subcontractors performing on contracts funded in whole or in part by SRF appropriations in excess of \$2000 pay their laborers and mechanics not less than the prevailing wage rates and fringe benefits, and determined by the Secretary of Labor, for corresponding classes of laborers and mechanics employed on similar projects in the area.

All work performed on this project will be subject to the higher of the prevailing state or federal wage rates.

*****END OF SECTION*****

SECTION 00 21 13 – INSTRUCTIONS TO BIDDERS

INSTRUCTIONS TO BIDDERS

It is anticipated that this project will be funded in part by the Washington State Department of Ecology. Neither the State of Washington nor any of its departments or employees are, or shall be, a party to this contract or any subcontract.

1. Bidder Qualifications

- A. Prospective Bidders shall be registered by the Washington State Department of Labor and Industries in accordance with state law.
- B. Corporations shall be registered with the State of Washington, Office of the Secretary of State.
- C. Bidders shall be regularly employed in the type of work contemplated herein.

2. Bidder's Representations

Submittal of a bid shall be deemed conclusive evidence that the bidder has:

- A. Carefully examined the proposed work site, become familiar with conditions impacting the work, and incorporated such observations into the bid.
- B. Read and understands the bidding and contract documents.
- C. produced a bid that is without exception based on the materials, equipment and systems required by the bidding documents.
- D. produced a bid that is made based on a complete set of Bidding Documents. The Owner is not responsible for any bidding errors resulting from the use of incomplete documents.

3. Document Interpretation

- A. The bidder shall carefully study and review the Bid Documents and promptly report any errors or omissions to the Engineer.
- B. Bidders or sub-bidders shall make any requests for clarification to the Engineer. If so directed, the Engineer may require the Bidder to submit requests in writing.
- C. Interpretations, corrections and changes to the Bidding Documents shall be made by Addendum. Interpretations, corrections and changes to the Bidding Documents made in any other manner will not be binding, and Bidders shall not rely on them.
- D. Substitutions shall not be considered prior to the receipt of bids. The Owner is not responsible for any bidding errors resulting from the use of substitutions.

4. Addenda

- A. Addenda will be mailed, emailed, delivered or faxed to all who are known by the Engineer to have received a complete set of Bidding Documents. Copies will also be provided to the locations where plans are available for review.
- B. The Bidder shall acknowledge receipt of addenda in their bid.

5. Bidding Procedures

- A. To be considered responsive, bids shall be submitted on the enclosed form and shall be filled in by typewriter or manually in ink.
- B. The Bid form shall include the Bidder's legal name exactly as it appears on his/her registration. Form shall be signed by the individual authorized to represent the Bidder.
- C. A list of subcontractors individually accounting for more than 10-percent of the Contract Sum and the work said subcontractor will perform shall be submitted with the bid or within one hour of the published bid time.

6. Pre-Bid Meeting

- A. There will be a pre-bid meeting at date and time shown on the Invitation to Bid.
- B. Prior to attending the pre-bid meeting, bidders shall have carefully studied and compared all drawings, specifications and other instructions to identify any inconsistency or omission. Also any discrepancies between the contract documents and the physical condition of the locality shall be identified. The intent is to identify any questions or concerns regarding the proposed improvements that the bidders may have.

7. Bid Security

- A. Each Bid shall be accompanied by a Bid Security in the form of a cashier's check, certified check or surety bond equal to 5-percent of the total Bid amount. Security shall pledge that the Bidder shall enter into a contract with the Owner in accordance with the terms of the Bid Documents including furnishing payment and performance bonds.
- B. In the event a Bidder refuses to enter into such contract or fail to furnish such bonds as required, the bid security shall be forfeited to the Owner as liquidated damages.
- C. The Owner may retain bid securities submitted with the bid until such time as; (1) the contract has been executed and bonds received, (2) 30-days have elapsed, (3) all Bids have been rejected.

8. Submission of Bids

- A. Bids shall be submitted in a sealed envelope. Envelopes shall clearly show (1) the project's name and owner as it appears on the Bid Solicitation, (2) the Contractor's name and registration number, and (3) the time and date of the bid opening.
- B. Bids received after the published bid time and date will be returned unopened.
- C. Bids submitted by mail shall conform with the above requirements and be sent to Eastsound Sewer and Water District – ESWD WWTP 143 Cessna Rd, EASTSOUND, WA 98245, All bids shall be received in sealed envelopes with "ESWD WWTP UPGRADE - PHASE I" marked plainly thereon. Bidder shall assume full responsibility for timely delivery of bid documents and the Owner is not responsible for bids received late.
- D. Oral, facsimile or telegraphic bids, modifications, or adjustments are not valid and will not receive consideration.

9. Modification or Withdrawal of Bid

- A. After the bid opening, bids shall not be withdrawn, modified or canceled by the Bidder during the stipulated time period.
- B. Bids submitted by mail prior to the bid opening may be modified or withdrawn by notice to the Owner. Such notice shall be in writing and signed by the same authorized individual signing the bid form. If such modifications or withdrawals are transmitted electronically, the original document shall be mailed and postmarked on or before the date and time of the bid opening.
- C. Withdrawn bids may be resubmitted up until the date and time of the bid opening and in accordance with these Instructions to Bidders.
- D. Bid security shall be in an amount sufficient for the bid as modified or resubmitted.

10. Opening of Bids

- A. Bids received on time will be opened and read aloud at the time and place stipulated in the Bid Solicitation. An abstract or tabulation will be made available to Bidders.
- B. Should a Bidder discover an error in his/her bid after submittal, the Bidder may request withdrawal of the bid with the following conditions:
- C. The Bidder must document the error(s) for the Owner. The Owner will review documentation and determine if the bid withdrawal and release of the Bid Security will be allowed.
- D. The Owner must receive the Bidder's intent to withdraw his/her bid submittal in writing no more than 30-hours after the bid opening (faxed notice is acceptable).
- E. The Owner alone will approve or disapprove the request for withdrawal. If approved, the Bidder will no longer be considered for Contract award and the Bid Security will be returned.
- F. If the Bidder fails to notify the Owner in accordance of an error as set forth above, and the Owner awards the Bidder the Contract, the Bidder shall either execute the Contract for the bid amount or withdraw the bid and forfeit the Bid Security.

11. Rejection of Bids

- A. The Owner reserves the right to reject any or all bids, reject a bid not accompanied by a proper bid security or other material required by the Bidding Documents, or reject a bid which is in anyway irregular or incomplete.

12. Acceptance of Bids

- A. The Owner intends to award the Contract to the lowest responsible responsive bidder whose bid submittal does not exceed available funds and conforms with the requirements described herein. The Owner shall have the right to waive informalities or irregularities in a bid submittal and to accept the bid that, in the opinion of the Owner, is in the Owner's best interest.
- B. After determination of the successful bidder based on the lowest responsible responsive bidder and other factors set forth in these instructions, the award may be made to said successful bidder on its base Bid and any combination of its additive bid items for which Owner determines funds will be available at the time of award.

- C. In evaluating whether a bidder is responsible, Owner will consider the qualifications of the bidder and many consider the qualifications and experience of subcontractors and suppliers purposed for those portions of the work for which the identity of subcontractors and suppliers must be submitted as provided in the bidding documents.

13. Contract Bond

- A. Bidders shall provide a contract bond as attached. Contract bond shall be signed by an approved surety or sureties, be in the full contract amount, and cover the faithful performance of the work described in the Contract Documents. The Contract Bond shall be in full effect until one year after Substantial Completion.

14. Contract Agreement and Award

- A. Owner's execution of the contract is contingent on the timely receipt of the Contract Bond and other submittals required by the Contract Documents.
- B. The award of the Contract, if it be awarded, shall be made within 45-days of the bid opening to the Bidder deemed by the Owner to be the lowest responsible responsive bidder.
- C. The 45-day period may be extended by mutual consent of the bidder and the Owner. If, after the 45-day period and no agreement to time extension has been made, the Contractor may withdraw his bid.
- D. The Owner reserves the right to award the bid schedules and bid alternates in any combination.

15. Execution of Contract

- A. The Bidder to whom the contract has been awarded shall sign the contract and return it and other submittals within 10 working days of the award.
- B. The Owner shall have the right to reject a contract submitted by a bidder if it is qualified by reservations or conditions stipulated by the bidder or its surety.
- C. No bid is binding on the Owner until executed by the Eastsound Sewer and Water District. No work shall be performed within the project site prior to the Notice to Proceed. Material or equipment orders or work undertaken away from the project site prior to contract execution shall be at the sole risk of the bidder.

16. Failure to Execute Contract

- A. If the bidder to whom award has been made fails to sign the contract and furnish satisfactory bonds within 10 calendar days of the award, or declares in writing its intent not to execute the contract, the bid security will be forfeited to the Owner and the second lowest responsible bidder will be notified of its receipt of award.
- B. If the second lowest responsible responsive bidder fails to execute the contract and furnish bonds within 20 calendar days after such notification, forfeiture of its bid security shall also be made and the third lowest responsible responsive bidder will be notified of its receipt of award, and in like manner until either (1) the contract and bond are executed by a responsible responsive bidder, (2) or further bid submittals are rejected, or (3) the number of bids submitted is exhausted.
- C. If the contract is not executed by the Contractor and Owner within the stipulated time, and it is evident that circumstances warrant an extension of time, the Owner may extend the time for executing the contract and/or bond for a period not to exceed 10 additional calendar days.

17. Return of Bid Security

- A. When bid submittals have been examined, bid securities and deposits accompanying submittals ineligible from further consideration will be returned.
- B. All other bid securities and deposits will be held until the contract has been properly executed, after which bid securities and deposits except those subject to forfeiture will be returned.

*****END OF SECTION*****

SECTION 00 24 13 – SCOPES OF BIDS

BID SCHEDULE –

EASTSOUND SEWER AND WATER DISTRICT – ESWD WWTP UPGRADE - PHASE I

SCOPES OF BID

This section outlines the individual bid items listed on the Bid Schedule in Section 00 41 00 - Bid Form. The descriptions are not all-inclusive, but generally indicate where costs should be allocated within the bid proposal. Descriptions represent work that shall be complete, in-place, tested, and in full operation prior to Owner's acceptance.

Each item is to be paid on a lump sum or unit price basis and shall include furnishing all necessary planning, labor, equipment, materials, and supplies required to furnish, install and test the improvements covered under the item. Each item shall include, as applicable, work shown on the plans including all excavations, back-fill, back-fill materials, compaction, pavement removal, disposal of waste material at contractor's site, locating and protecting existing utilities and services, base and top course, paving, trenching, imported backfill, pipe bedding, cleaning, testing, surface restoration and landscaping. The scope of each bid item is outlined below. It is not intended to include all of the appurtenances of an item in the description. See appropriate Specification or WSDOT Standard Specifications and as shown on the Drawings for a more complete representation of the work. It is the responsibility of the Bidders to include all costs for the completed project in the bid items listed.

BASE BID ITEM

1. WWTP Upgrade, Lump Sum (LS)

- A. Measurement for payment for individual items of the WWTP Upgrade will be based upon the unit values listed in the Schedule of Values submitted by the CONTRACTOR and reviewed and approved by the Engineer.
- B. Payment for the WWTP Upgrade will be made at the Lump Sum Price shown on PROPOSAL for Bid Item 1, which will constitute full compensation for all WORK as described in the Contract Documents.

UNIT QUANTITY BID ITEMS

2. Trench Safety System, Lump Sum (LS)

This work consists of installing trench safety systems in accordance with WSDOT, OSHA, and other applicable rules and regulations.

3. Bollards, price based on Each (EA)

Construction of bollards (at site locations chosen by the Owner) shall be considered unit quantity work which the Contractor shall be compensated for as described below.

- A. Measurement for payment for bollards will be based on the actual quantity, each, of finished bollards.
- B. Payment for bollards, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

4. Preparation of SWPPP, Lump Sum (LS)

This includes all costs associated with determining, developing, preparing, and submitting a Stormwater Pollution Prevention Plan (SWPPP). SWPPP shall be prepared in accordance with Section 31 32 11 – Soil Surface Erosion Control, TESC Plans (see civil sheets), and the NPDES permit. Measurement and partial payments will be based on approximate percentage of completion of SWPPP. This work also includes transferring the NPDES Construction Stormwater General Permit to Contractor's name, and other requirements identified in Section 31 32 11 – Soil Surface Erosion Control.

5. Maintenance Work for SWPPP, Lump Sum (LS)

This Includes all costs associated with implementing, adjusting, inspections, reporting, responding to precipitation events, and maintaining effective erosion and sediment control measures for the WWTP Upgrade throughout the life of the project in accordance with the Stormwater Pollution Prevention Plan and the NPDES permit, including quarry spall construction entrance, track clean plates siltation ponds, silt fencing, straw, and other sediment trapping devices; slope stabilization measures; low-impact construction practices; and project sequencing. Work also includes monitoring stormwater discharges, reporting, and all cost associated with adaptive management of the one site erosion control, dewatering, and water management measures. Work also includes updating the SWPPP and keeping a current version available on-site at all times for reference. Measurement and partial payments will be based on approximate percentage of completion of the WWTP Upgrade.

6. Electrical, Lump Sum (LS)

This includes all work shown on the E sheets and in specifications Division 26. Detailed breakdowns are provided in Sections 26 05 00 and 26 80 00. Measurement and partial payments will be based on approximate percentage of completion of the time period needed for the Electrical work.

7. Minor Changes, Estimate (EST)

- A. This item shall consist of any minor changes or deviations from the Plans & Specs for work to remedy unforeseen conditions, utility conflicts, or special surface restoration. All work under Minor Changes shall be negotiated and pre-approved in writing by the Engineer prior to commencing work.
- B. Payment or credits for changes amounting to \$100,000 or less may be made under the Bid Item MINOR CHANGES. At the discretion of the Owner, this procedure for Minor Changes may be used in lieu of the more formal procedure as outlined in the General Conditions. All Minor Change work will be within the scope of the Contract Work and will not change Contract Time. For the purpose of providing a common Proposal for all Bidders, the Owner has entered an amount for MINOR CHANGES in the Proposal to become part of the total Bid by the Contractor.

EXTRA WORK ITEMS

8. HMA Pavement, price based on Tons (TN)

- A. Measurement for payment for extra Hot Mixed Asphalt Pavement (HMA Class ½" PG 58H-22 per Section 32 12 16) will be based on the actual quantity, tons, of finished extra HMA Pavement.
- B. Payment for extra HMA Pavement, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

9. Structural Concrete, price based on Cubic Yards (CY)

- C. Measurement for payment for extra structural concrete (poured in place structural concrete with reinforcing similar to the digester basin walls as directed by the Engineer) will be based on the actual quantity, cubic yards, of finished extra structural concrete.
- D. Payment for extra structural concrete, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

10. Over Excavation, price based on Cubic Yards (CY)

- A. Measurement for payment for over excavation (over excavation due to poor subsurface soil conditions), as directed by the Engineer, will be based on the actual quantity, cubic yards, of finished over excavation. Over excavated material is to be hauled off-site and disposed of at a certified disposal site per specifications.
- B. Payment for over excavation, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications

11. Crushed Surfacing Top Course, price based on Tons (TN)

- A. Measurement for payment for extra crushed surfacing top course, as directed by the Engineer, will be based on the actual quantity, tons, of finished material. Includes all costs associated with providing, spreading, and compacting Crushed Surfacing Top Course per WSDOT Section 9-03.9(3) Crushed Surfacing Top Course, minimum density 95%.
- B. Payment for extra crushed surfacing top course, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications

12. Crushed Surfacing Base Course, price based on Tons (TN)

- A. Measurement for payment for extra crushed surfacing base course, as directed by the Engineer, will be based on the actual quantity, tons, of finished material. Includes all costs associated with providing, spreading, and compacting Crushed Surfacing Base Course per WSDOT Section 9-03.9(3) Crushed Surfacing Base Course, minimum density 95%.
- B. Payment for extra crushed surfacing base course, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

13. Gravel Base, price based on Tons (TN)

- A. Measurement for payment for extra gravel base, as directed by the Engineer, will be based on the actual quantity, tons, of finished material. Includes all costs associated with providing, spreading, and compacting Gravel Base per WSDOT Section 9-03.10 Gravel Base, minimum density 95%.
- B. Payment for extra gravel base, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

14. Gravel Backfill for Pipe Zone Bedding, price based on Tons (TN)

- A. Measurement for payment for extra gravel backfill for pipe zone bedding, as directed by the Engineer, will be based on the actual quantity, tons, of finished material. Includes all costs associated with providing, spreading, and compacting Gravel Backfill for Pipe Zone Bedding per WSDOT Section 9-03.12(3) Gravel Backfill for Pipe Zone Bedding, minimum density 90%.
- B. Payment for extra gravel backfill for pipe zone bedding, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

15. Gravel Borrow, price based on Tons (TN)

- A. Measurement for payment for extra Gravel Borrow, as directed by the Engineer, will be based on the actual quantity, tons, of finished material. Includes all costs associated with providing, spreading, and compacting Gravel Borrow per WSDOT Section 9-03.14(1) Gravel Borrow, minimum density 95%.
- B. Payment for extra gravel borrow, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

16. Bank Run Gravel for Trench Backfill, price based on Tons (TN)

- A. Measurement for payment for extra Bank Run Gravel for Trench Backfill, as directed by the Engineer, will be based on the actual quantity, tons, of finished material. Includes all costs associated with providing, spreading, and compacting Bank Run Gravel for Trench Backfill per WSDOT Section 9-03.19 Bank Run Gravel for Trench Backfill, minimum density 95%.
- B. Payment for extra bank run gravel for trench backfill, if any, will be made at the Unit Price shown on Proposal, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications.

17. Extra SCADA/PLC Programmer Services, Force Account (FA)

- A. Measurement for payment for Extra SCADA/PLC Programmer Services by Contractor's System Integrator will be based on the actual force account total of finished Extra SCADA/PLC Programmer Services (including controls improvements) as directed by District Staff. All Extra SCADA/PLC Programmer Services must be agreed on, prior to scheduling the work. This bid item is only for additional work, above and beyond the work included in

the contract documents. Force account work to be per WSDOT Standard Specifications, Section 1-09.6.

- B. Payment for Extra Programmer Services will be made at the total sum of the actual force account, said payment will constitute full compensation for all WORK which shall be in accordance with the applicable specifications, including 2022 WSDOT requirements.

BID ALTERNATE 1

18. Train 3 – Stainless Steel Fabrication Above Water Surface, Lump Sum (LS)

- A. This price shall include all additional costs for work and material, as needed, to provide Type 316 stainless steel for all in-basin steel components of the Train 3 equipment that will be located above the water surface (and to a depth 1-ft below water surface), excluding FRP components (such as the effluent weirs and covers). Submerged steel components are to be painted steel. See Specification Section 46 53 00 – Biological Treatment System. This Bid Alternate is only to include the additional cost associated with this improvement, if selected by the Owner.

BID ALTERNATE 2

19. Train 3 – Stainless Steel Fabrication Throughout, Lump Sum (LS)

- B. This price shall include all additional costs for work and material, as needed, to provide Type 316 stainless steel for all in-basin steel components of the Train 3 equipment, excluding FRP components (such as the effluent weirs and covers). This includes exposed and submerged steel components; however, it is noted that the structural supports for these components may be provided as painted steel. See Specification Section 46 53 00 – Biological Treatment System. This Bid Alternate is only to include the additional cost associated with this improvement, if selected by the Owner.

*****END OF SECTION*****

SECTION 00 31 13 – PRELIMINARY PROJECT PHASES

The following Preliminary Project Phasing Plan is provided for planning purposes. This phasing plan is not meant to dictate means and methods to perspective Bidders or take the place any required planning on the part of the Bidder to provide a responsive Bid. This phasing plan is simply an outline of the work to be performed that takes into the account the lead-time and critical path nature of the submittals, ordering, and delivery of the project equipment.

Permit Schedule

- A. County Building Permit Application Schedule
 - 1. Submit Applications to San Juan County: November 2021
 - 2. Permit Approvals from County: June 2022

Bid Phase

- A. Preliminary Bid Phase Schedule
 - 1. Advertise For Bids: January 4, 2023
 - 2. Pre-Bid Meeting (on-site): January 18, 2023
 - 3. Open Bids: February 8, 2023
 - 4. Evaluate Bids, Issue Award: February 15, 2023
 - 5. Preconstruction Meeting: March 15, 2023

Construction Phasing Plan

- A. Work Summary
 - 1. Construction Submittals: Beginning after Award (Feb 2023). Prioritize all long lead items.
 - 2. Mobilize on-site: March 2023
 - 3. Prepare SWPPP and Establish Erosion Control Measures: March 2023
 - 4. Prepare Dewatering Plan for Excavations: March 2023
 - 5. Set-Up Temporary Bypass Pipe (per Interim Plan of Operation) from the existing chlorine contact tank outlet to the existing 8" plant outlet pipe, as shown on the Plan of Interim Operation: April 2023
 - 6. Begin construction on Blower Building, Digester, UV Disinfection Chamber, Sludge Pumps, Yard Piping, Associated Vaults, and Electrical: April 2023
 - 7. Begin Construction on new 2W Air Gap Water System: May 2023
 - 8. Begin Construction on new Influent Pump Station: June 2023
 - 9. Complete New Blower Building Structure: July 2023
 - 10. Begin construction on the third Train: August 2023
 - 11. Rent Temporary 480V Generator: Sept 25, 2023
 - 12. Complete New 480V Service, MCC, and PLC Panel Installations: Oct 4, 2023
 - 13. Complete New UV Disinfection System: Oct 7, 2023

14. Place New UV Disinfection System On-Line: Oct 11, 2023 (Contractor to schedule the switch over from temporary 8" effluent piping to new 8" effluent piping, 8" effluent flow meter, and UV Disinfection Channels #1 and #2.)
15. Complete New 2W Water System: Oct 18, 2023
16. Complete New Influent Pump Station: Oct 25, 2023
17. Place New Influent Pump Station On-Line: Oct 25, 2023 (Contractor to schedule the switch over from existing piping to new influent pump station, influent flow splitter, and influent sampler. All flow to be routed to Train #1 until influent piping is reconnected to Train #2.)
18. Major Piping Work Completion Deadline: Nov 1, 2023
19. Complete New Aerobic Digester: Nov 1, 2023
20. Complete New Dewatering System: Feb 9, 2024
21. Permanent 480V Generator Installed/Tested/On-Line (Remove Rental): April 2024
22. Complete Construction of Train #3: July 8, 2024
23. Seed new Train #3 Treatment Facility with Activated Sludge. Pump 10,000 gallons of activated sludge from either Train #1 or #2: July 8-12, 2024
24. Substantial Completion Deadline: July 26, 2024
25. Punch List Work: July-October 2024
26. Final Surfacing/Pavement Work (entire site): August 2024
27. Final Completion Deadline: October 16, 2024

*****END OF SECTION*****

SECTION 00 41 00 – BID PROPOSAL

Name of Bidder: _____

To: Eastsound Sewer and Water District
Project: ESWD WWTP UPGRADE - PHASE I
143 Cessna Rd
Eastsound, Washington 98245

BASE BID – WWTP Expansion

ITEM		APPROX. QTY	UNIT	UNIT PRICE	AMOUNT
1	WWTP Upgrade	1	LS		
	BASE BID ITEM 1				

UNIT QUANTITY BID ITEMS

ITEM		APPROX. QTY	UNIT	UNIT PRICE	AMOUNT
2	Trench Safety Excavation Provisions	1	LS		
3	Bollards	12	EA		
4	SWPPP Preparation	1	LS		
5	Maintenance Work for SWPPP	1	LS		
6	Electrical	1	LS		
7	Minor Changes	1	EST.	\$100,000.00	\$100,000.00
	SUBTOTAL UNIT QUANTITY BID ITEMS 2-7				

EXTRA WORK ITEMS

ITEM		APPROX. QTY	UNIT	UNIT PRICE	AMOUNT
8	HMA Pavement	10	TN		
9	Structural Concrete	10	CY		
10	Over Excavation	10	CY		
11	Crushed Surfacing Top Course	10	TN		
12	Crushed Surfacing Base Course	10	TN		
13	Gravel Base	10	TN		
14	Gravel Backfill for Pipe Zone Bedding	10	TN		
15	Gravel Borrow	10	TN		
16	Bank Run Gravel for Trench Backfill	10	TN		
17	Extra SCADA/PLC Programmer Services	1	FA	\$100,000.00	\$100,000.00
	SUBTOTAL EXTRA WORK ITEMS 8-17				

BID ALTERNATE 1:

ITEM		APPROX. QTY	UNIT	UNIT PRICE	AMOUNT
18	Train 3 – Stainless Steel Fabrication Above Water Surface	1	LS		
	SUBTOTAL BID ALTERNATE 1, ITEM 18				

BID ALTERNATE 2:

ITEM		APPROX. QTY	UNIT	UNIT PRICE	AMOUNT
19	Train 3 – Stainless Steel Fabrication Throughout	1	LS		
	SUBTOTAL BID ALTERNATE 2, ITEM 19				

	SUBTOTAL ITEMS 1-19		\$
	8.3% SALES TAX (Eastsound Sewer and Water District, San Juan County)		\$
	TOTAL BID		\$

LIST OF MANUFACTURERS

The named manufacturer for some equipment items are listed below. Contractor is to circle his selected manufacturer, when a choice is available. Contractor's Base Bid Item #1 - WWTP Upgrade is to be based on the following:

<u>Equipment</u>	<u>Base Bid Manufacturer</u>
A. Submersible Pump Equipment	<u>Flygt</u>
B. Biological Treatment Sys. Equipment	<u>Smith and Loveless</u>
C. Rotary Fan Press Dewatering Equipment	<u>Fournier, Prime Solutions</u>
D. UV Disinfection Equipment	<u>Trojan</u>
E. Generator	<u>Cummins, Kohler, Caterpillar, Generac</u>

EQUIPMENT ALTERNATES:

Contractors may submit an approval package for equipment of alternate manufacturers to the base bid manufacturer for review with their bid and included as an Equipment Alternate on this Bid Proposal. A pre-approval review prior to bid will not be done. Contractors proposing alternate manufacturers will be responsible for all costs associated with system evaluation and redesign including all electrical, mechanical, and civil aspects of the installation.

- A. Furnish Submersible Pump Equipment other than specified.

Lump Sum Deduction \$ - Deduction _____

Amount in Words: _____

Manufacturer & Model No. _____

- B. Furnish Biological Treatment System Equipment other than specified.

Lump Sum Deduction \$ - Deduction _____

Amount in Words: _____

Manufacturer & Model No. _____

- C. Furnish Rotary Fan Press Dewatering Equipment other than specified.

Lump Sum Deduction \$ - Deduction _____

Amount in Words: _____

Manufacturer & Model No. _____

- D. Furnish UV Disinfection Equipment other than specified.

Lump Sum Deduction \$ - Deduction _____

Amount in Words: _____

Manufacturer & Model No. _____

E. Furnish Generator Equipment other than specified.

Lump Sum Deduction \$ - Deduction _____

Amount in Words: _____

Manufacturer & Model No. _____

Note 1: Mobilization items and partial payments for Mobilization shall be in accordance with WSDOT Standard Specifications Section 1-09.7.

Note 2: See Section 00 24 13 –Scopes of Bids for more description of each bid item.

Note 3: Payments will be made for actual quantities of bid items installed on the project. Estimated contract quantities provide contingencies with respect to rock and miscellaneous alignment adjustments; and to establish a not-to-exceed level of construction cost for Owner's planning purposes. If required by actual field conditions, bidders should be prepared to perform all expected and contingency work reflected in the estimated contract quantities and in the quantity ranges identified for additive and deductive unit prices. Bidders should not, however, expect to be paid the total contract price if the contingency quantities are not required to be installed by actual field conditions.

The Owner reserves the right to accept or reject any or all bid prices within sixty (60) days of the bid date.

Time for Completion

See Supplementary Condition 15 - Completion Date for completion time requirements.

Liquidated Damages

The undersigned agrees to pay the Owner as liquidated damages the sum as specified in the General Conditions for each consecutive calendar day that is in default after the Contract Time. Liquidated damages shall be deducted from the contract by change order or from the Contractor's application for payment as determined by Owner in its sole discretion.

Contractor is required to pay Washington State Prevailing Wages and Federal Prevailing Wages. All work performed on this project will be subject to the higher of the prevailing state or federal wage rates.

Receipt of Addenda

Receipt of the following addenda is acknowledged:

Addendum No. _____

Addendum No. _____

Addendum No. _____

Addendum No. _____

Addendum No. _____

Addendum No. _____

Name of Firm _____

NOTE: If bidder is a corporation, write State of Incorporation; if a partnership, give full names and addresses of all parties below.

Non-Collusion Declaration: By signing below, I hereby declare that I, firm, association or corporation has (have) not, either directly or indirectly, entered into any agreement, participated in any collusion, or otherwise taken any action restraining free competitive bidding for this project.

Signed by _____, Official Capacity _____

Print Name _____

Address _____

City _____ State _____ Zip Code _____

Date _____ Telephone _____ FAX _____

State of Washington Contractor's License No. _____

Federal Tax ID # _____ e-mail address: _____

Employment Security Department No. _____

***** END OF SECTION *****

SECTION 00 43 13 – BID BOND FORM

Deposit Statement

Herewith find a deposit in the form of certified check, or cashier's check, in the amount of
Five percent (5%) of maximum amount bid (Total for all Bid Items + sales tax) in the attached Proposal.

Bid Bond

KNOW ALL PEOPLE BY THESE PRESENTS, that _____ the

CONTRACTOR, hereinafter known as PRINCIPAL, and _____ hereinafter
known as SURETY, are held and firmly bound to the Eastsound Sewer and Water District hereinafter known as

OWNER, in the penal sum of _____

_____ dollars (not less than 5% of Base Bid plus Additive Alternates including Washington State Sales Tax) for the
payment of which sum well and truly to be made, we do jointly and severally bind ourselves, our heirs, executors,
administrators, successors and assigns firmly by these presents.

WHEREAS, the PRINCIPAL has submitted a bid for

(Project Title): Eastsound Sewer and water District WWTP Upgrades _____

NOW, THEREFORE, the condition of this obligation is such that if the OWNER accepts the bid of the
PRINCIPAL, and

- a. the PRINCIPAL executes such contract documents required by the terms of the bid and provides required
Bonds for the performance of the contract and for the prompt payment of labor and material furnished for
the project as may be specified in the bid then this obligation is satisfied, or
- b. in the event of the failure of the PRINCIPAL to execute such contract documents and provide such Bonds
required by the terms of the bid, the PRINCIPAL shall pay and forfeit to the OWNER the full penal sum
hereof, then this obligation shall be null and void; otherwise this obligation remains in full force and effect
and the SURETY shall forthwith pay and forfeit to the OWNER, as a penalty and liquidated damages, the
amount of this bond.

SIGNED, SEALED AND DATED THIS _____ day of _____, 20____.

PRINCIPAL

SURETY

By

By

Title

Title

Address of PRINCIPAL

Address of SURETY

Note: If PRINCIPAL is Partnership, all Partners should execute bond. Surety companies executing bonds must
appear on the Treasury Department's most current list (Circular 570 as amended) and be authorized to
transact business in the State of Washington. A power of attorney must be provided which appoints the
SURETY's true and lawful attorney-in-fact to make, execute, seal and deliver this bond.

*** END OF SECTION ***

SECTION 00 43 93 – BID SUBMITTAL CHECKLIST

The bidder is advised to use the following list to assemble all forms required to be submitted with their bids. In accordance with RCW 39.30.060, bidders shall submit the required documentation listed below.

Bid must be received prior to **2:00 P.M. PST, Wednesday, February 8, 2023.**

Bid Submittal Checklist

Part 1 – (to be submitted with the bid)

- _____ Bid Proposal (Section 00 41 00 BID PROPOSAL).
- _____ Bid Guarantee (Section 00 43 13 BID BOND or other type of Bid Guarantee).
- _____ Contractors Qualifications (Section 00 45 13).
- _____ Non-Collusion Affidavit (Section 00 45 19).
- _____ Certification of Compliance with Wage Payment Statutes (Section 00 45 29).
- _____ Attachment 3 Certification of Non-Segregated Facilities (See Section 00 53 00, page 25 of 26).
- _____ Attachment 4 Notice to Labor Unions or Other Organization of Workers: Non-Discrimination in Employment (See Section 00 53 00, page 26 of 26).

Part 2 – (to be submitted either with the bid or within 1-hour of the bid)

- _____ List of Subcontractors (Section 00 45 33).
- _____ Complete Bidders List (as described in Section 00 53 00, page 6 of 26).
- _____ DBE Subcontractor Performance Form (WA Ecology Form 6100-3, Included in Part 4 Appendix G)
- _____ DBE Subcontractor Utilization Form (WA Ecology Form 6100-4, Included in Part 4 Appendix G)

Statement of Qualifications (within 24 hours of the bid opening)

Within 24 hours of the bid opening, the apparent low bidder shall furnish the Engineer with the Statement of Qualifications for the Contractor (Section 00 22 13 SUPPLEMENTAL BIDDER RESPONSIBILITY CRITERIA) and listed Subcontractors (Section 00 45 43 SUBCONTRACTOR QUALIFICATIONS). The second and third low bidders shall furnish this information only when requested by the Engineer.

***** END OF SECTION *****

SECTION 00 45 13 - CONTRACTORS QUALIFICATIONS

CONTRACTORS QUALIFICATIONS

The below listed reference information shall be submitted with the Bid.

Bidder to list three previous wastewater facility projects with similar value (\$2,000,000+) completed by Bidder as prime contractor. Bidder shall have successfully completed with their own equipment and personnel a minimum of three similar projects in the last ten years to be considered qualified.

1. Project: _____
(Name and Location)
Contract Amount: _____

Reference: _____
(Company Name, Contact & Telephone)
2. Project: _____
(Name and Location)
Contract Amount: _____

Reference: _____
(Company Name, Contact & Telephone)
3. Project: _____
(Name and Location)
Contract Amount: _____

Reference: _____
(Company Name, Contact & Telephone)

Bidder shall provide the following information.

1. Resume of superintendent proposed for project.
2. List and provide references (Owner and Engineer) for any project within the last three years which have involved disputes for which the Contractor filed a claim resulting in formal dispute resolution, third-party mediation or arbitration, or a lawsuit.
3. List and provide references (Owner and Engineer) for all public works contracts in which the Contractor was sued by the Owner.

***** END OF SECTION *****

SECTION 00 45 19 – NON-COLLUSION AFFIDAVIT

NON-COLLUSION AFFIDAVIT

STATE OF WASHINGTON)
) ss.
COUNTY OF WHATCOM)

The undersigned, being duly sworn, deposes and says that the person, firm, association, co-partnership or corporation herein named, has not, either directly or indirectly, entered into any agreement, participated in any collusion, or otherwise taken any action in the restraining of free competitive bidding in the preparation and submission of a proposal to the Eastsound Sewer and Water District for consideration in the award of a contract on the improvement named above.

Contractor

Subscribed and sworn to before me this ____ day of _____, 20__.

Notary Public in and for the
State of Washington, residing at

*****END OF SECTION*****

SECTION 00 45 29 – CERTIFICATION OF COMPLIANCE WITH WAGE PAYMENT STATUTES

The bidder hereby certifies that, within the three-year period immediately preceding the bid solicitation date (February 8, 2023), the bidder is not a “willful” violator, as defined in RCW 49.48.082, of any provision of chapters 49.46, 49.48, or 49.52 RCW, as determined by a final and binding citation and notice of assessment issued by the Department of Labor and Industries or through a civil judgment entered by a court of limited or general jurisdiction.

I declare under penalty of perjury under the laws of the State of Washington that the foregoing is true and correct.

Bidder’s Business Name

Signature of Authorized Official*

Printed Name

Title

Date

City

State or country

Check One:

Sole Proprietorship ☐ Partnership ☐ Joint Venture ☐ Corporation ☐

State of Incorporation, or if not a corporation, State where business entity was formed:

If a co-partnership, give firm name under which business is transacted:

** If a corporation, proposal must be executed in the corporate name by the president or vice-president (or any other corporate officer accompanied by evidence of authority to sign). If a co-partnership, proposal must be executed by a partner.*

*****END OF SECTION*****

SECTION 00 45 33 – LIST OF SUBCONTRACTORS - BIDS ON PUBLIC WORKS - IDENTIFICATION, SUBSTITUTION OF SUBCONTRACTORS

The prime contractor shall submit as part of the bid, or within one hour after the published bid submittal time, the names of the subcontractors with whom the bidder, if awarded the contract, will subcontract for performance of the work of: HVAC; Plumbing; and Electrical, or to name itself for the work. The prime contract bidder shall not list more than one subcontractor for each category of work identified, unless subcontractors vary with bid alternates, in which case the prime contract bidder must indicate which subcontractor will be used for which alternate. Failure of the prime contract bidder to submit as part of the bid the names of such subcontractors or to name itself to perform such work or the naming of two or more subcontractors to perform the same work shall render the prime contract bidder's bid nonresponsive and, therefore, void.

HVAC Subcontractor: _____

Address: _____ **Phone:** _____

Plumbing Subcontractor: _____

Address: _____ **Phone:** _____

Electrical Subcontractor: _____

Address: _____ **Phone:** _____

The prime contractor shall submit as part of the bid, or within 48 hours after the published bid submittal time, the names of the subcontractors with whom the bidder, if awarded the contract, will subcontract for performance of the work of: Structural Steel Installation and Rebar Installation, or to name itself for the work. The prime contract bidder shall not list more than one subcontractor for each category of work identified, unless subcontractors vary with bid alternates, in which case the prime contract bidder must indicate which subcontractor will be used for which alternate. Failure of the prime contract bidder to submit as part of the bid the names of such subcontractors or to name itself to perform such work or the naming of two or more subcontractors to perform the same work shall render the prime contract bidder's bid nonresponsive and, therefore, void.

Structural Steel Installation Subcontractor: _____

Address: _____ **Phone:** _____

Rebar Installation Subcontractor: _____

Address: _____ **Phone:** _____

Contractor shall also provide a list of all subcontractors whose work exceeds ten (10) percent of the bid and the name of the Control System Integrator. Additional sheets may be used if necessary. This combined subcontractor list must be submitted with the bid OR within the time frame allotted, as described in Section 00 43 93 – BID SUBMITTAL CHECKLIST.

Control System Integrator:

Name: _____

Address: _____

Telephone Number: _____

Portion of Work: _____

Subcontractors performing more than 10% of the bid price:

Name: _____

Address: _____

Telephone Number: _____

Portion of Work: _____

Name: _____

Address: _____

Telephone Number: _____

Portion of Work: _____

Name: _____

Address: _____

Telephone Number: _____

Portion of Work: _____

Name: _____

Address: _____

Telephone Number: _____

Portion of Work: _____

***** END OF SECTION *****

SECTION 00 45 43 – SUBCONTRACTOR QUALIFICATIONS

The below listed reference information will be required 24 hours after the bid opening for all listed subcontractors of the apparent low bidder. The information may also be asked of the subcontractors of the next two low bidders at that time.

Bidder to list the following information for **three** projects for **each** of the subcontractors listed in Section 00 45 33 LIST OF SUBCONTRACTORS. The selected projects must be of equivalent size and scope to the portion of work the subcontractor will complete on the **ESWD WWTP Upgrade - Phase I Project**, and the subcontractor must have completed the work using his/her own personnel and equipment.

(This sheet shall be duplicated for each Subcontractor)

Name of Subcontractor: _____

1. Project: _____
(Name and Location)

Contract Amount: _____

Reference: _____
(Company Name, Contact & Telephone)

2. Project: _____
(Name and Location)

Contract Amount: _____

Reference: _____
(Company Name, Contact & Telephone)

3. Project: _____
(Name and Location)

Contract Amount: _____

Reference: _____
(Company Name, Contact & Telephone)

***** END OF SECTION *****

PART 2

CONTRACTING REQUIREMENTS

SECTION 00 51 00 – NOTICE OF AWARD

NOTICE OF AWARD

To: _____.
_____.
_____.
_____.

For: Eastsound Sewer and Water District
ESWD WWTP Upgrade - Phase I Project

The Owner has considered the BID submitted by you for the above described WORK in response to its Advertisement for Bids and Information for Bidders.

You are hereby notified that your BID has been ACCEPTED in accordance with your proposal for the amount of \$_____.

You are required by the Information for Bidders to execute the Contract and furnish the required Bond(s) and certificates of insurance within ten (10) calendar days from the date of this Notice of Award.

If you fail to execute said Contract and furnish said Bond(s) within ten (10) working days from the date of this Notice, the District will be entitled to consider all your rights arising out of the District's acceptance of your BID as abandoned and as a forfeiture of your BID BOND. The Owner will be entitled to such other rights as may be granted by law.

You are required to return an acknowledged copy of this NOTICE OF AWARD to the District within 3 days of its receipt.

Dated this _____ day of _____, 20__

Eastsound Sewer and Water District
Owner

By _____
Title _____

=====

ACCEPTANCE OF NOTICE:

Receipt of this NOTICE OF AWARD is hereby

acknowledged: By _____.

Dated this _____ *day of* _____, 20__

By _____ *Printed Name*

*****END OF SECTION*****

SECTION 00 52 00 – AGREEMENT FORM

THIS AGREEMENT is made and entered into at Eastsound, Washington, this _____ day of _____, 2023, by and between Eastsound Sewer and Water District, hereinafter designated as the OWNER, and _____, hereinafter designated as the CONTRACTOR.

It is made with reference to the following facts:

1. OWNER has heretofore caused to be prepared certain Contract Documents including Bidding Requirements, Contracting Requirements, Technical Specifications, Miscellaneous Documents and Plans for the construction of the **ESWD WWTP Upgrade - Phase I Project**.
2. CONTRACTOR filed with the OWNER on _____, 20____, a proposal to complete said work.
3. Contractor agreed to accept as payment therefor the sum fully stated and set forth in the Proposal.
4. The Contract Documents fully and accurately describe the terms and conditions upon which the CONTRACTOR proposed to furnish said equipment, labor, material and appurtenances and perform said work, together with the manner and time of furnishing same.
5. Third-Party Beneficiary: The State of Washington shall be, and is hereby, named as an express third-party beneficiary of this contract, with full rights as such.

IT IS THEREFORE AGREED, first, that a copy of said Contract Documents as aforesaid, does in all particulars become a part of the Agreement by and between the parties hereto in all matters and things therein set forth and described; and further, that the OWNER and CONTRACTOR hereby accept and agree to the terms and conditions of said Contract Documents as filed completely as if said terms and conditions and plans were herein set out in full.

This Agreement shall be binding upon all parties hereto and their respective heirs, executors, administrators and assigns.

IN WITNESS WHEREOF, the parties hereto have execute, or caused to be executed by their duly authorized officials, this Agreement in triplicate each of which shall be deemed an original on the day first above written.

OWNER: EASTSOUND SEWER AND WATER DISTRICT

By: _____

Name: _____

Title: _____

(SEAL)

ATTEST:

Name: _____

TITLE: _____

APPROVED AS TO FORM:

By: _____

Name: _____

Dated: _____

CONTRACTOR _____

By: _____

Name: _____

Title: _____

(SEAL)

ATTEST:

Name: _____

Title: _____

***** END OF SECTION *****

SECTION 00 53 00 – WPCRF INSERTS

WASHINGTON STATE DEPARTMENT OF ECOLOGY WATER POLLUTION CONTROL REVOLVING FUND SPECIFICATIONS INSERT

Revised 1/22/21

The following clauses will be incorporated into construction contracts receiving financial assistance from the Washington State Department of Ecology Water Pollution Control Revolving Fund. In the event of conflict within the contract these clauses shall take precedence

Required Bid Submittals

The following submittals are required to be submitted with the bid proposal:

- Certification Of Nonsegregated Facilities (attachment 3)
- DBE Subcontractor Utilization Form (EPA Form 6100-4)
- One copy of DBE Subcontractor Performance Form (EPA Form 6100-3) for each DBE subcontractor.
- Complete Bidders List.

Compliance with State and Local Laws

The Contractor shall assure compliance with all applicable federal, state, and local laws, requirements, and ordinances as they pertain to the design, implementation, and administration of the approved project.

State Interest Exclusion

It is anticipated that this project will be funded in part by the Washington State Department of Ecology . Neither the State Of Washington nor any of its departments or employees are, or shall be, a party to this contract or any subcontract.

Third Party Beneficiary

Partial funding of this project is being provided through the Washington State Department of Ecology Water Pollution Control Revolving Fund. All parties agree that the State of Washington shall be, and is hereby, named as an express third-party beneficiary of this contract, with full rights as such.

Access to the construction site and to records

The contractor shall provide for the safe access to the construction site and to the contractor's records by Washington State Department of Ecology and Environmental Protection Agency (EPA) personnel.

The Contractor shall maintain accurate records and accounts to facilitate the Owner's audit requirements and shall ensure that all subcontractors maintain auditable records.

These Project records shall be separate and distinct from the Contractor's other records and accounts.

All such records shall be available to the Owner and to Washington State Department of Ecology and EPA personnel for examination. All records pertinent to this project shall be retained by the Contractor for a period of three (3) years after the final audit.

Protection of the Environment

No construction related activity shall contribute to the degradation of the environment, allow material to enter surface or ground waters, or allow particulate emissions to the atmosphere, which exceed state or federal standards. Any actions that potentially allow a discharge to state waters must have prior approval of the Washington State Department of Ecology.

Funding Recognition

All site-specific projects must have a sign of sufficient size to be seen from nearby roadways acknowledging department financial assistance and left in place throughout the life of the project. Department logos must be on all signs and documents. Logos will be provided as needed.

Inadvertent Discovery Of Archeological Resources

The contractor shall obtain a copy of the Inadvertent Discovery Plan from the Project Owner. The contractor shall keep a copy of the inadvertent discovery plan for the project on the work site at all times. The contractor shall immediately stop all work if human remains, cultural, or archeological resources are discovered in the course of construction. The contractor shall follow the inadvertent discovery plan in dealing with the human remains, cultural, or archeological resources.

Use Of American Iron And Steel

This provision applies to projects for the construction, alteration, maintenance, or repair of a "treatment works" as defined in the Federal Water Pollution Control Act (33 USC 1381 et seq.). This provision does not apply if the engineering plans and specifications for the project were approved by the Ecology prior to January 17, 2014.

The Contractor acknowledges to and for the benefit of the Project Owner and the State of Washington that it understands the goods and services under this Agreement are being funded with monies made available by the Water Pollution Control Revolving Fund which contains provisions commonly known as "American Iron and Steel;" that requires all of the iron and steel products used in the project be produced in the United States ("American Iron and Steel Requirements") including iron and steel products provided by the Contractor pursuant to this Agreement. "Iron and Steel products" means the following products made primarily of iron or steel: lined or unlined pipes and fittings, manhole covers and other municipal castings, hydrants, tanks, flanges, pipe clamps and restraints, valves, structural steel, reinforced precast concrete, and construction materials.

The Contractor hereby represents and warrants to and for the benefit of the Project Owner and the State that:

- (a) the Contractor has reviewed and understands the American Iron and Steel Requirements,
- (b) all of the iron and steel products used in the project will be and/or have been produced in the United States in a manner that complies with the American Iron and Steel Requirements, unless a waiver of the requirements is approved, and

(c) the Contractor will provide any further verified information, certification or assurance of compliance with this paragraph, or information necessary to support a waiver of the American Iron and Steel Requirements, as may be requested by the Project Owner or the State.

Notwithstanding any other provision of this Agreement, any failure to comply with this paragraph by the Contractor shall permit the Project Owner or State to recover as damages against the Contractor any loss, expense or cost (including without limitation attorney's fees) incurred by the Project Owner or State resulting from any such failure (including without limitation any impairment or loss of funding, whether in whole or in part, from the State or any damages owed to the State by the Project Owner). While the Contractor has no direct contractual privity with the State, as a lender to the Project Owner for the funding of its project, the Project Owner and the Contractor agree that the State is a third-party beneficiary and neither this paragraph (nor any other provision of the Agreement necessary to give this paragraph force or effect shall be amended or waived without the prior written consent of the State.

Prevailing Wage

The work performed under this contract is subject to the wage requirements of the Davis-Bacon Act. The Contractor shall conform to the wage requirements prescribed by the federal Davis-Bacon and Related Acts which requires that all laborers and mechanics employed by contractors and subcontractors performing on contracts funded in whole or in part by SRF appropriations in excess of \$2000 pay their laborers and mechanics not less than the prevailing wage rates and fringe benefits, and determined by the Secretary of Labor, for corresponding classes of laborers and mechanics employed on similar projects in the area. Attachment 1 to this specification insert and an up to date wage determination shall be included in full into this contract and in any subcontract in excess of \$2,000. Wage determinations can be found at <http://www.wdol.gov>.

The Contractor agrees that the Contractor is legally and financially responsible for compliance with the Davis-Bacon Act wage rules. All laborers and mechanics employed by contractors and subcontractors employed as part of this contract shall be paid wages at rates not less than those prevailing on projects of a character similar in the locality as determined by the Secretary of Labor in accordance with subchapter IV of chapter 31 of title 40, United States Code.

Certification Regarding Suspension, Debarment, Ineligibility Or Voluntary Exclusion

1. The CONTRACTOR, by signing this agreement, certifies that it is not suspended, debarred, proposed for debarment, declared ineligible or otherwise excluded from contracting with the federal government, or from receiving contracts paid for with federal funds. If the CONTRACTOR is unable to certify to the statements contained in the certification, they must provide an explanation as to why they cannot.
2. The CONTRACTOR shall provide immediate written notice to the Department if at any time the CONTRACTOR learns that its certification was erroneous when submitted or had become erroneous by reason of changed circumstances.
3. The terms covered transaction, debarred, suspended, ineligible, lower tier covered

transaction, participant, person, primary covered transaction, principal, proposal, and voluntarily excluded, as used in this clause, have the meaning set out in the Definitions and Coverage sections of rules implementing Executive Order 12549. You may contact the department for assistance in obtaining a copy of those regulations..

4. The CONTRACTOR agrees it shall not knowingly enter into any lower tier covered transaction with a person who is proposed for debarment under the applicable Code of Federal Regulations, debarred, suspended, declared ineligible, or voluntarily excluded from participation in this covered transaction.
5. The CONTRACTOR further agrees by signing this agreement, that it will include this clause titled "Certification Regarding Suspension, Debarment, Ineligibility Or Voluntary Exclusion" without modification in all lower tier covered transactions and in all solicitations for lower tier covered transactions.
6. Pursuant to 2CFR180.330, the CONTRACTOR is responsible for ensuring that any lower tier covered transaction complies with certification of suspension and debarment requirements.
7. CONTRACTOR acknowledges that failing to disclose the information required in the Code of Federal Regulations may result in the delay or negation of this funding agreement, or pursuance of legal remedies, including suspension and debarment.
8. CONTRACTOR agrees to keep proof in its agreement file, that it, and all lower tier recipients or contractors, are not suspended or debarred, and will make this proof available to the Department upon request. RECIPIENT/CONTRACTOR must run a search in <http://www.sam.gov/> and print a copy of completed searches to document proof of compliance.

This term and condition supersedes EPA Form 5700-49, "Certification Regarding Debarment, Suspension, and Other Responsibility Matters."

Disadvantaged Business Enterprises

General Compliance (40 CFR Part 33).

The contractor shall comply with the requirements of the Environmental Protection Agency's Program for Participation By Disadvantaged Business Enterprises (DBE) 40 CFR Part 33.

Non-discrimination Provision (40CFR Appendix A to Part 33).

The contractor shall not discriminate on the basis of race, color, national origin or sex in the performance of this contract. The contractor shall carry out applicable requirements of 40 CFR part 33 in the award and administration of contracts awarded under EPA financial assistance agreements. Failure by the contractor to carry out these requirements is a material breach of this contract which may result in the termination of this contract or other legally available remedies.

The contractor shall comply with all federal and state nondiscrimination laws, including, but not limited to Title VI and VII of the Civil Rights Act of 1964, Section 504 of the Rehabilitation Act of 1973, Title IX of the Education Amendments of 1972, the Age Discrimination Act of 1975, and

Chapter 49.60 RCW, Washington's Law Against Discrimination, and 42 U.S.C. 12101 et seq, the Americans with Disabilities Act (ADA).

Six Good Faith Efforts (40 CFR Part 33 Subpart C).

The contractor agrees to make the following good faith efforts whenever procuring subcontracts, equipment, services and supplies. The contractor shall retain records documenting compliance with the following six good faith efforts.

1. Ensuring Disadvantaged Business Enterprises are made aware of contracting opportunities to the fullest extent practicable through outreach and recruitment activities. For Indian Tribal, State and Local and Government recipients, this will include placing Disadvantaged Business Enterprises on solicitation lists and soliciting them whenever they are potential sources. Qualified Women and Minority business enterprises may be found on the Internet at www.omwbe.wa.gov or by contacting the Washington State Office of Minority and Women's Enterprises at (866) 208-1064.
2. Making information on forthcoming opportunities available to Disadvantaged Business Enterprises and arrange time frames for contracts and establish delivery schedules, where the requirements permit, in a way that encourages and facilitates participation by Disadvantaged Business Enterprises in the competitive process. This includes, whenever possible, posting solicitations for bids or proposals for a minimum of thirty (30) calendar days before the bid or proposal closing date.
3. Considering in the contracting process whether firms competing for large contracts could subcontract with Disadvantaged Business Enterprises. For Indian Tribal, State and local Government recipients, this will include dividing total requirements when economically feasible into smaller tasks or quantities to permit maximum participation by Disadvantaged Business Enterprises in the competitive process.
4. Encourage contracting with a consortium of Disadvantaged Business Enterprises when a contract is too large for one of these firms to handle individually.
5. Using services and assistance of the Small Business Administration and the Minority Business Development Agency of the Department of Commerce.
6. If the prime contractor awards subcontracts, requiring the subcontractors to take the six good faith efforts in paragraphs 1 through 5 above.

MBE/WBE Reporting (40 CFR Part 33 Parts 33.302, 33.502 and 33.503).

1. The contractor shall complete the DBE Subcontractor Utilization Form (EPA Form 6100-4).
2. The contractor shall require all DBE subcontractors to complete the DBE Subcontractor Performance Form (EPA Form 6100-3). The DBE Subcontractor Performance Form is only required to be completed by certified DBE subcontractors.
3. The contractor shall submit DBE Subcontractor Utilization Form (EPA Form 6100-4) and all completed DBE Subcontractor Performance Form(s) (EPA Form 6100-3) as part of the bid, or within one hour after the published bid submittal time (consistent with RCW 39.30.060)
4. The contractor shall provide DBE Subcontractor Participation Form (EPA Form 6100-2) to all DBE subcontractors. These subcontractors may submit Subcontractor Participation Form (EPA Form 6100-2) to the EPA Region 10 DBE coordinator in order to document issues or concerns with their usage or payment for a subcontract.

The 6100 forms can be found at:

<https://ecology.wa.gov/About-us/How-we-operate/Grants-loans/Find-a-grant-or-loan/Water-Quality-grants-and-loans/Facility-project-resources>

Bidders List (40 CFR Part 33 part 33.501)

All bidders shall submit the following information for all firms that bid or quote on subcontracts (including both DBE and non-DBE firms) as part of the bid, or within one hour after the published bid submittal time (consistent with RCW 39.30.060).

1. Firm's name with point of contact;
2. Firm's mailing address, telephone number, and e-mail address;
3. The work on which the firm bid or quoted, and when the firm bid or quoted; and
4. Firm's status as an MBE/WBE or non-MBE/WBE.

Contract Administration Provisions (40 CFR part 33.302).

The contractor shall comply with the contract administration provisions of 40 CFR, Part 33.302.

1. The contractor shall pay its subcontractor for satisfactory performance no more than 30 days from the contractor's receipt of payment.
2. The contractor shall notify the owner in writing prior to any termination of a DBE subcontractor.
3. If a DBE subcontractor fails to complete work under the subcontract for any reason, the contractor shall employ the six good faith efforts when soliciting a replacement subcontractor.
4. The contractor shall employ the six good faith efforts even if the contractor has achieved its fair share objectives.

Equal Opportunity (EEO)

If this Contract exceeds \$10,000, the Contractor shall comply with Executive Order 11246, "Equal Employment Opportunity," as amended by Executive Order 11375, "Amending Executive Order 11246 Relating to Equal Employment Opportunity," and as supplemented by regulations at 41 CFR part 60.

Contractor's compliance with Executive Order 11246 shall be based on implementation of the Equal Opportunity Clause, and specific affirmative active obligations required by the Standard Federal Equal Employment Opportunity Construction Contract Specifications, as set forth in 41 CFR Part 60-4.

Equal Opportunity Clause (41 CFR part 60-1.4(b))

During the performance of this contract, the contractor agrees as follows:

1. The contractor will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin. The contractor will take affirmative action to ensure that applicants are employed, and that employees are treated during employment without regard to their race, color, religion, sex, or national origin. Such action shall include, but not be limited to the following: Employment, upgrading, demotion, or transfer; recruitment or recruitment advertising; layoff or termination; rates of pay or other

forms of compensation; and selection for training, including apprenticeship. The contractor agrees to post in conspicuous places, available to employees and applicants for employment, notices to be provided setting forth the provisions of this nondiscrimination clause.

2. The contractor will, in all solicitations or advertisements for employees placed by or on behalf of the contractor, state that all qualified applicants will receive considerations for employment without regard to race, color, religion, sex, or national origin.
3. The contractor will send to each labor union or representative of workers with which he has a collective bargaining agreement or other contract or understanding, a notice to be provided advising the said labor union or workers' representatives of the contractor's commitments under this section, and shall post copies of the notice in conspicuous places available to employees and applicants for employment.
4. The contractor will comply with all provisions of Executive Order 11246 of September 24, 1965, and of the rules, regulations, and relevant orders of the Secretary of Labor.
5. The contractor will furnish all information and reports required by Executive Order 11246 of September 24, 1965, and by rules, regulations, and orders of the Secretary of Labor, or pursuant thereto, and will permit access to his books, records, and accounts by the administering agency and the Secretary of Labor for purposes of investigation to ascertain compliance with such rules, regulations, and orders.
6. In the event of the contractor's noncompliance with the nondiscrimination clauses of this contract or with any of the said rules, regulations, or orders, this contract may be canceled, terminated, or suspended in whole or in part and the contractor may be declared ineligible for further Government contracts or federally assisted construction contracts in accordance with procedures authorized in Executive Order 11246 of September 24, 1965, and such other sanctions may be imposed and remedies invoked as provided in Executive Order 11246 of September 24, 1965, or by rule, regulation, or order of the Secretary of Labor, or as otherwise provided by law.
7. The contractor will include the portion of the sentence immediately preceding paragraph (1) and the provisions of paragraphs (1) through (7) in every subcontract or purchase order unless exempted by rules, regulations, or orders of the Secretary of Labor issued pursuant to section 204 of Executive Order 11246 of September 24, 1965, so that such provisions will be binding upon each subcontractor or vendor. The contractor will take such action with respect to any subcontract or purchase order as the administering agency may direct as a means of enforcing such provisions, including sanctions for noncompliance: Provided, however, That in the event a contractor becomes involved in, or is threatened with, litigation with a subcontractor or vendor as a result of such direction by the administering agency the contractor may request the United States to enter into such litigation to protect the interests of the United States.

Federal Equal Employment Opportunity Construction Contract Specifications
(Executive Order 11246 and 41 CFR part 60-4.3)

1. As used in these specifications:
 - a. "Covered area" means the geographical area described in the solicitation from which this contract resulted;
 - b. "Director" means Director, Office of Federal Contract Compliance Programs, United States Department of Labor, or any person to whom the Director delegates authority;
 - c. "Employer identification number" means the Federal Social Security number used on the Employer's Quarterly Federal Tax Return, U.S. Treasury Department Form 941.

- d. "Minority" includes:
- i. Black (all persons having origins in any of the Black African racial groups not of Hispanic origin);
 - ii. Hispanic (all persons of Mexican, Puerto Rican, Cuban, Central or South American or other Spanish Culture or origin, regardless of race);
 - iii. Asian and Pacific Islander (all persons having origins in any of the original peoples of the Far East, Southeast Asia, the Indian Subcontinent, or the Pacific Islands); and
 - iv. American Indian or Alaskan Native (all persons having origins in any of the original peoples of North America and maintaining identifiable tribal affiliations through membership and participation or community identification).
2. Whenever the Contractor, or any Subcontractor at any tier, subcontracts a portion of the work involving any construction trade, it shall physically include in each subcontract in excess of \$10,000 the provisions of these specifications and the Notice which contains the applicable goals for minority and female participation and which is set forth in the solicitations from which this contract resulted.
 3. If the Contractor is participating (pursuant to 41 CFR 60-4.5) in a Hometown Plan approved by the U.S. Department of Labor in the covered area either individually or through an association, its affirmative action obligations on all work in the Plan area (including goals and timetables) shall be in accordance with that Plan for those trades which have unions participating in the Plan. Contractors must be able to demonstrate their participation in and compliance with the provisions of any such Hometown Plan. Each Contractor or Subcontractor participating in an approved Plan is individually required to comply with its obligations under the EEO clause, and to make a good faith effort to achieve each goal under the Plan in each trade in which it has employees. The overall good faith performance by other Contractors or Subcontractors toward a goal in an approved Plan does not excuse any covered Contractor's or Subcontractor's failure to take good faith efforts to achieve the Plan goals and timetables.
 4. The Contractor shall implement the specific affirmative action standards provided in paragraphs 7 a through p of these specifications. The goals set forth in the solicitation from which this contract resulted are expressed as percentages of the total hours of employment and training of minority and female utilization the Contractor should reasonably be able to achieve in each construction trade in which it has employees in the covered area. Covered Construction contractors performing construction work in geographical areas where they do not have a Federal or federally assisted construction contract shall apply the minority and female goals established for the geographical area where the work is being performed. Goals are published periodically in the Federal Register in notice form, and such notices may be obtained from any Office of Federal Contract Compliance Programs office or from Federal procurement contracting officers. The Contractor is expected to make substantially uniform progress in meeting its goals in each craft during the period specified.
 5. Neither the provisions of any collective bargaining agreement, nor the failure by a union with whom the Contractor has a collective bargaining agreement, to refer either minorities or women shall excuse the Contractor's obligations under these specifications, Executive Order 11246, or the regulations promulgated pursuant thereto.
 6. In order for the nonworking training hours of apprentices and trainees to be counted in meeting the goals, such apprentices and trainees must be employed by the Contractor during the training period, and the Contractor must have made a commitment to employ the apprentices and trainees at the completion of their training, subject to the availability of

employment opportunities. Trainees must be trained pursuant to training programs approved by the U.S. Department of Labor.

7. The Contractor shall take specific affirmative actions to ensure equal employment opportunity. The evaluation of the Contractor's compliance with these specifications shall be based upon its effort to achieve maximum results from its actions. The Contractor shall document these efforts fully, and shall implement affirmative action steps at least as extensive as the following:
 - a. Ensure and maintain a working environment free of harassment, intimidation, and coercion at all sites, and in all facilities at which the Contractor's employees are assigned to work. The Contractor, where possible, will assign two or more women to each construction project. The Contractor shall specifically ensure that all foremen, superintendents, and other on-site supervisory personnel are aware of and carry out the Contractor's obligation to maintain such a working environment, with specific attention to minority or female individuals working at such sites or in such facilities.
 - b. Establish and maintain a current list of minority and female recruitment sources, provide written notification to minority and female recruitment sources and to community organizations when the Contractor or its unions have employment opportunities available, and maintain a record of the organizations' responses.
 - c. Maintain a current file of the names, addresses and telephone numbers of each minority and female off-the-street applicant and minority or female referral from a union, a recruitment source or community organization and of what action was taken with respect to each such individual. If such individual was sent to the union hiring hall for referral and was not referred back to the Contractor by the union or, if referred, not employed by the Contractor, this shall be documented in the file with the reason therefor, along with whatever additional actions the Contractor may have taken.
 - d. Provide immediate written notification to the Director when the union or unions with which the Contractor has a collective bargaining agreement has not referred to the Contractor a minority person or woman sent by the Contractor, or when the Contractor has other information that the union referral process has impeded the Contractor's efforts to meet its obligations.
 - e. Develop on-the-job training opportunities and/or participate in training programs for the area which expressly include minorities and women, including upgrading programs and apprenticeship and trainee programs relevant to the Contractor's employment needs, especially those programs funded or approved by the Department of Labor. The Contractor shall provide notice of these programs to the sources compiled under 7b above.
 - f. Disseminate the Contractor's EEO policy by providing notice of the policy to unions and training programs and requesting their cooperation in assisting the Contractor in meeting its EEO obligations; by including it in any policy manual and collective bargaining agreement; by publicizing it in the company newspaper, annual report, etc.; by specific review of the policy with all management personnel and with all minority and female employees at least once a year; and by posting the company EEO policy on bulletin boards accessible to all employees at each location where construction work is performed.
 - g. Review, at least annually, the company's EEO policy and affirmative action obligations under these specifications with all employees having any responsibility for hiring, assignment, layoff, termination or other employment decisions including specific review of these items with onsite supervisory personnel such as Superintendents, General

- Foremen, etc., prior to the initiation of construction work at any job site. A written record shall be made and maintained identifying the time and place of these meetings, persons attending, subject matter discussed, and disposition of the subject matter.
- h. Disseminate the Contractor's EEO policy externally by including it in any advertising in the news media, specifically including minority and female news media, and providing written notification to and discussing the Contractor's EEO policy with other Contractors and Subcontractors with whom the Contractor does or anticipates doing business.
 - i. Direct its recruitment efforts, both oral and written, to minority, female and community organizations, to schools with minority and female students and to minority and female recruitment and training organizations serving the Contractor's recruitment area and employment needs. Not later than one month prior to the date for the acceptance of applications for apprenticeship or other training by any recruitment source, the Contractor shall send written notification to organizations such as the above, describing the openings, screening procedures, and tests to be used in the selection process.
 - j. Encourage present minority and female employees to recruit other minority persons and women and, where reasonable, provide after school, summer and vacation employment to minority and female youth both on the site and in other areas of a Contractor's work force.
 - k. Validate all tests and other selection requirements where there is an obligation to do so under 41 CFR part 60-3.
 - l. Conduct, at least annually, an inventory and evaluation at least of all minority and female personnel for promotional opportunities and encourage these employees to seek or to prepare for, through appropriate training, etc., such opportunities.
 - m. Ensure that seniority practices, job classifications, work assignments and other personnel practices, do not have a discriminatory effect by continually monitoring all personnel and employment related activities to ensure that the EEO policy and the Contractor's obligations under these specifications are being carried out.
 - n. Ensure that all facilities and company activities are nonsegregated except that separate or single-user toilet and necessary changing facilities shall be provided to assure privacy between the sexes.
 - o. Document and maintain a record of all solicitations of offers for subcontracts from minority and female construction contractors and suppliers, including circulation of solicitations to minority and female contractor associations and other business associations.
 - p. Conduct a review, at least annually, of all supervisors' adherence to and performance under the Contractor's EEO policies and affirmative action obligations.
8. Contractors are encouraged to participate in voluntary associations which assist in fulfilling one or more of their affirmative action obligations (7a through p). The efforts of a contractor association, joint contractor-union, contractor-community, or other similar group of which the contractor is a member and participant, may be asserted as fulfilling any one or more of its obligations under 7a through p of these Specifications provided that the contractor actively participates in the group, makes every effort to assure that the group has a positive impact on the employment of minorities and women in the industry, ensures that the concrete benefits of the program are reflected in the Contractor's minority and female workforce participation, makes a good faith effort to meet its individual goals and timetables, and can provide access to documentation which demonstrates the effectiveness of actions taken on behalf of the Contractor. The obligation to comply, however, is the Contractor's and failure of such a group to fulfill an obligation shall not be a defense for the

Contractor's noncompliance.

9. A single goal for minorities and a separate single goal for women have been established. The Contractor, however, is required to provide equal employment opportunity and to take affirmative action for all minority groups, both male and female, and all women, both minority and non-minority. Consequently, the Contractor may be in violation of the Executive Order if a particular group is employed in a substantially disparate manner (for example, even though the Contractor has achieved its goals for women generally, the Contractor may be in violation of the Executive Order if a specific minority group of women is underutilized).
10. The Contractor shall not use the goals and timetables or affirmative action standards to discriminate against any person because of race, color, religion, sex, or national origin.
11. The Contractor shall not enter into any Subcontract with any person or firm debarred from Government contracts pursuant to Executive Order 11246.
12. The Contractor shall carry out such sanctions and penalties for violation of these specifications and of the Equal Opportunity Clause, including suspension, termination and cancellation of existing subcontracts as may be imposed or ordered pursuant to Executive Order 11246, as amended, and its implementing regulations, by the Office of Federal Contract Compliance Programs. Any Contractor who fails to carry out such sanctions and penalties shall be in violation of these specifications and Executive Order 11246, as amended.
13. The Contractor, in fulfilling its obligations under these specifications, shall implement specific affirmative action steps, at least as extensive as those standards prescribed in paragraph 7 of these specifications, so as to achieve maximum results from its efforts to ensure equal employment opportunity. If the Contractor fails to comply with the requirements of the Executive Order, the implementing regulations, or these specifications, the Director shall proceed in accordance with 41 CFR 60-4.8.
14. The Contractor shall designate a responsible official to monitor all employment related activity to ensure that the company EEO policy is being carried out, to submit reports relating to the provisions hereof as may be required by the Government and to keep records. Records shall at least include for each employee the name, address, telephone numbers, construction trade, union affiliation if any, employee identification number when assigned, social security number, race, sex, status (e.g., mechanic, apprentice trainee, helper, or laborer), dates of changes in status, hours worked per week in the indicated trade, rate of pay, and locations at which the work was performed. Records shall be maintained in an easily understandable and retrievable form; however, to the degree that existing records satisfy this requirement, contractors shall not be required to maintain separate records.
15. Nothing herein provided shall be construed as a limitation upon the application of other laws which establish different standards of compliance or upon the application of requirements for the hiring of local or other area residents (e.g., those under the Public Works Employment Act of 1977 and the Community Development Block Grant Program).

Reporting Requirements (EEO-1)

On or before September 30 of each year, a contractor that is subject to Title VII of the Civil Rights Act of 1964, as amended, and that has 100 or more employees, shall file with the EEOC or its delegate an "Employer Information Report EEO-1". Instructions on how to file are available on the EEOC's website at <http://www.eeoc.gov/employers/eo1survey/howtofile.cfm>. The contractor shall retain a copy of the most recent report filed.

Segregated Facilities (41 CFR part 60-1.8)

The contractor shall ensure that facilities provided for employees are provided in such a manner that segregation on the basis of race, color, religion, sex or national origin cannot result. The contractor may neither require such segregated use by written or oral policies nor tolerate such use by employee custom. The contractor's obligation extends further to ensuring that its employees are not assigned to perform their services at any location, under the contractor's control, where the facilities are segregated. This obligation extends to all contracts containing the equal opportunity clause regardless of the amount of the contract. The term "facilities," as used in this section, means waiting rooms, work areas, restaurants and other eating areas, time clocks, restrooms, wash rooms, locker rooms, and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing provided for employees; Provided, That separate or single-user restrooms and necessary dressing or sleeping areas shall be provided to assure privacy between the sexes.

Attachments:

1. Wage Rate Requirements For Subrecipients
2. Current Wage Rate Determination (to be provided by project owner)
3. Certification Of Nonsegregated Facilities
4. Notice To Labor Unions Or Other Organization Of Workers: Non-Discrimination In Employment

EPA Form 6100-4, EPA Form 6100.3, EPA Form 6100-2

ATTACHMENT 1 - WAGE RATE REQUIREMENTS FOR SUBRECIPIENTS. **(To be included in full in any contract in excess of \$2,000)**

The following terms and conditions specify how recipients will assist EPA in meeting its Davis-Bacon (DB) responsibilities when DB applies to EPA awards of financial assistance under the FY 2013 Continuing Resolution with respect to State recipients and subrecipients that are governmental entities. If a subrecipient has questions regarding when DB applies, obtaining the correct DB wage determinations, DB provisions, or compliance monitoring, it may contact the State recipient. If a State recipient needs guidance, the recipient may contact Lorraine Fleury at fleury.lorraine@epa.gov or at 215-814-2341 of EPA, Region III Grants and Audit Management Branch for guidance. The recipient or subrecipient may also obtain additional guidance from DOL's web site at <http://www.dol.gov/whd/>

1. Applicability of the Davis- Bacon (DB) prevailing wage requirements.

Under the FY 2013 Appropriations Act, DB prevailing wage requirements apply to the construction, alteration, and repair of treatment works carried out in whole or in part with assistance made available by a State water pollution control revolving fund and to any construction project carried out in whole or in part by assistance made available by a drinking water treatment revolving loan fund. If a subrecipient encounters a unique situation at a site that presents uncertainties regarding DB applicability, the subrecipient must discuss the situation with the recipient State before authorizing work on that site.

2. Obtaining Wage Determinations.

(a) Subrecipients shall obtain the wage determination for the locality in which a covered activity subject to DB will take place prior to issuing requests for bids, proposals, quotes or other methods for soliciting contracts (solicitation) for activities subject to DB. These wage determinations shall be incorporated into solicitations and any subsequent contracts. Prime contracts must contain a provision requiring that subcontractors follow the wage determination incorporated into the prime contract.

(i) While the solicitation remains open, the subrecipient shall monitor www.wdol.gov weekly to ensure that the wage determination contained in the solicitation remains current. The subrecipients shall amend the solicitation if DOL issues a modification more than 10 days prior to the closing date (i.e. bid opening) for the solicitation. If DOL modifies or supersedes the applicable wage determination less than 10 days prior to the closing date, the subrecipients may request a finding from the State recipient that there is not a reasonable time to notify interested contractors of the modification of the wage determination. The State recipient will provide a report of its findings to the subrecipient.

(ii) If the subrecipient does not award the contract within 90 days of the closure of the solicitation, any modifications or supersedes DOL makes to the wage determination contained in the solicitation shall be effective unless the State recipient, at the request of the subrecipient, obtains an extension of the 90 day period from DOL pursuant to 29 CFR 1.6(c)(3)(iv). The subrecipient shall monitor www.wdol.gov on a weekly basis if it does not award the contract within 90 days of closure of the solicitation to ensure that wage determinations contained in the solicitation remain current.

(b) If the subrecipient carries out activity subject to DB by issuing a task order, work assignment or similar instrument to an existing contractor (ordering instrument) rather than by publishing a solicitation, the subrecipient shall insert the appropriate DOL wage determination from www.wdol.gov into the ordering instrument.

(c) Subrecipients shall review all subcontracts subject to DB entered into by prime contractors to verify that the prime contractor has required its subcontractors to include the applicable wage determinations.

(d) As provided in 29 CFR 1.6(f), DOL may issue a revised wage determination applicable to a subrecipient's contract after the award of a contract or the issuance of an ordering instrument if DOL determines that the subrecipient has failed to incorporate a wage determination or has used a wage determination that clearly does not apply to the contract or ordering instrument. If this occurs, the subrecipient shall either terminate the contract or ordering instrument and issue a revised solicitation or ordering instrument or incorporate DOL's wage determination retroactive to the beginning of the contract or ordering instrument by change order. The subrecipient's contractor must be compensated for any increases in wages resulting from the use of DOL's revised wage determination.

3. Contract and Subcontract provisions.

(a) The Recipient shall insure that the subrecipient(s) shall insert in full in any contract in excess of \$2,000 which is entered into for the actual construction, alteration and/or repair, including painting and decorating, of a treatment work under the CWSRF or a construction project under the DWSRF financed in whole or in part from Federal funds or in accordance with guarantees of a Federal agency or financed from funds obtained by pledge of any contract of a Federal agency to make a loan, grant or annual contribution (except where a different meaning is expressly indicated), and which is subject to the labor standards provisions of any of the acts listed in § 5.1 or the FY 2012 Appropriations Act , the following clauses:

(1) Minimum wages.

(i) All laborers and mechanics employed or working upon the site of the work will be paid unconditionally and not less often than once a week, and without subsequent deduction or rebate on any account (except such payroll deductions as are permitted by regulations issued by the Secretary of Labor under the Copeland Act (29 CFR part 3)), the full amount of wages and bona fide fringe benefits (or cash equivalents thereof) due at time of payment computed at rates not less than those contained in the wage determination of the Secretary of Labor which is attached hereto and made a part hereof, regardless of any contractual relationship which may be alleged to exist between the contractor and such laborers and mechanics.

Contributions made or costs reasonably anticipated for bona fide fringe benefits under section 1(b)(2) of the Davis-Bacon Act on behalf of laborers or mechanics are considered wages paid to such laborers or mechanics, subject to the provisions of paragraph (a)(1)(iv) of this section; also, regular contributions made or costs incurred for more than a weekly period (but not less often than quarterly) under plans, funds, or programs which cover the particular weekly period, are deemed to be constructively made or incurred during such weekly period. Such laborers and mechanics shall be paid the appropriate wage rate and fringe benefits on the wage determination for the classification of work actually performed, without regard to skill, except as provided in § 5.5(a)(4). Laborers or mechanics performing work in more than one classification may be compensated at the rate specified for each classification for the time actually worked therein: Provided that the employer's payroll records accurately set forth the time spent in each classification in which work is performed. The wage determination (including any additional classification and wage rates conformed under paragraph (a)(1)(ii) of this section) and the Davis-Bacon poster (WH-1321) shall be posted at all times by the contractor and its subcontractors at the site of the work in a prominent and accessible place where it can be easily seen by the workers.

Subrecipients may obtain wage determinations from the U.S. Department of Labor's web site, www.wdol.gov.

(ii)(A) The subrecipient(s), on behalf of EPA, shall require that any class of laborers or mechanics, including helpers, which is not listed in the wage determination and which is to be employed under the contract shall be classified in conformance with the wage determination. The State award official

shall approve a request for an additional classification and wage rate and fringe benefits therefore only when the following criteria have been met:

- (1) The work to be performed by the classification requested is not performed by a classification in the wage determination; and
- (2) The classification is utilized in the area by the construction industry; and
- (3) The proposed wage rate, including any bona fide fringe benefits, bears a reasonable relationship to the wage rates contained in the wage determination.

(B) If the contractor and the laborers and mechanics to be employed in the classification (if known), or their representatives, and the subrecipient(s) agree on the classification and wage rate (including the amount designated for fringe benefits where appropriate), documentation of the action taken and the request, including the local wage determination shall be sent by the subrecipient (s) to the State award official. The State award official will transmit the request, to the Administrator of the Wage and Hour Division, Employment Standards Administration, U.S. Department of Labor, Washington, DC 20210 and to the EPA DB Regional Coordinator concurrently. The Administrator, or an authorized representative, will approve, modify, or disapprove every additional classification request within 30 days of receipt and so advise the State award official or will notify the State award official within the 30-day period that additional time is necessary.

(C) In the event the contractor, the laborers or mechanics to be employed in the classification or their representatives, and the subrecipient(s) do not agree on the proposed classification and wage rate (including the amount designated for fringe benefits, where appropriate), the award official shall refer the request and the local wage determination, including the views of all interested parties and the recommendation of the State award official, to the Administrator for determination. The request shall be sent to the EPA DB Regional Coordinator concurrently. The Administrator, or an authorized representative, will issue a determination within 30 days of receipt of the request and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(D) The wage rate (including fringe benefits where appropriate) determined pursuant to paragraphs (a)(1)(ii)(B) or (C) of this section, shall be paid to all workers performing work in the classification under this contract from the first day on which work is performed in the classification.

(iii) Whenever the minimum wage rate prescribed in the contract for a class of laborers or mechanics includes a fringe benefit which is not expressed as an hourly rate, the contractor shall either pay the benefit as stated in the wage determination or shall pay another bona fide fringe benefit or an hourly cash equivalent thereof.

(iv) If the contractor does not make payments to a trustee or other third person, the contractor may consider as part of the wages of any laborer or mechanic the amount of any costs reasonably anticipated in providing bona fide fringe benefits under a plan or program, Provided, That the Secretary of Labor has found, upon the written request of the contractor, that the applicable standards of the Davis-Bacon Act have been met. The Secretary of Labor may require the contractor to set aside in a separate account assets for the meeting of obligations under the plan or program.

(2) Withholding. The subrecipient(s), shall upon written request of the EPA Award Official or an authorized representative of the Department of Labor, withhold or cause to be withheld from the

contractor under this contract or any other Federal contract with the same prime contractor, or any other federally-assisted contract subject to Davis-Bacon prevailing wage requirements, which is held by the same prime contractor, so much of the accrued payments or advances as may be considered necessary to pay laborers and mechanics, including apprentices, trainees, and helpers, employed by the contractor or any subcontractor the full amount of wages required by the contract. In the event of failure to pay any laborer or mechanic, including any apprentice, trainee, or helper, employed or working on the site of the work, all or part of the wages required by the contract, the (Agency) may, after written notice to the contractor, sponsor, applicant, or owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds until such violations have ceased.

(3) Payrolls and basic records.

(i) Payrolls and basic records relating thereto shall be maintained by the contractor during the course of the work and preserved for a period of three years thereafter for all laborers and mechanics working at the site of the work. Such records shall contain the name, address, and social security number of each such worker, his or her correct classification, hourly rates of wages paid (including rates of contributions or costs anticipated for bona fide fringe benefits or cash equivalents thereof of the types described in section 1(b)(2)(B) of the Davis-Bacon Act), daily and weekly number of hours worked, deductions made and actual wages paid. Whenever the Secretary of Labor has found under 29 CFR 5.5(a)(1)(iv) that the wages of any laborer or mechanic include the amount of any costs reasonably anticipated in providing benefits under a plan or program described in section 1(b)(2)(B) of the Davis-Bacon Act, the contractor shall maintain records which show that the commitment to provide such benefits is enforceable, that the plan or program is financially responsible, and that the plan or program has been communicated in writing to the laborers or mechanics affected, and records which show the costs anticipated or the actual cost incurred in providing such benefits. Contractors employing apprentices or trainees under approved programs shall maintain written evidence of the registration of apprenticeship programs and certification of trainee programs, the registration of the apprentices and trainees, and the ratios and wage rates prescribed in the applicable programs.

(ii)(A) The contractor shall submit weekly, for each week in which any contract work is performed, a copy of all payrolls to the subrecipient, that is, the entity that receives the sub-grant or loan from the State capitalization grant recipient. Such documentation shall be available on request of the State recipient or EPA. As to each payroll copy received, the subrecipient shall provide written confirmation in a form satisfactory to the State indicating whether or not the project is in compliance with the requirements of 29 CFR 5.5(a)(1) based on the most recent payroll copies for the specified week. The payrolls shall set out accurately and completely all of the information required to be maintained under 29 CFR 5.5(a)(3)(i), except that full social security numbers and home addresses shall not be included on the weekly payrolls. Instead the payrolls shall only need to include an individually identifying number for each employee (e.g., the last four digits of the employee's social security number). The required weekly payroll information may be submitted in any form desired. Optional Form WH-347 is available for this purpose from the Wage and Hour Division Web site at <http://www.dol.gov/whd/programs/dbra/wh347.htm> or its successor site. The prime contractor is responsible for the submission of copies of payrolls by all subcontractors. Contractors and subcontractors shall maintain the full social security number and current address of each covered worker, and shall provide them upon request to the subrecipient(s) for transmission to the State or EPA if requested by EPA, the State, the contractor, or the Wage and Hour Division of the Department of Labor for purposes of an investigation or audit of compliance with prevailing wage requirements. It is not a violation of this section for a prime contractor to require a subcontractor to provide addresses and social security numbers to the prime contractor for its own records, without weekly submission to the subrecipient(s).

(B) Each payroll submitted shall be accompanied by a "Statement of Compliance," signed by the contractor or subcontractor or his or her agent who pays or supervises the payment of the persons employed under the contract and shall certify the following:

- (1) That the payroll for the payroll period contains the information required to be provided under § 5.5 (a)(3)(ii) of Regulations, 29 CFR part 5, the appropriate information is being maintained under § 5.5 (a)(3)(i) of Regulations, 29 CFR part 5, and that such information is correct and complete;
- (2) That each laborer or mechanic (including each helper, apprentice, and trainee) employed on the contract during the payroll period has been paid the full weekly wages earned, without rebate, either directly or indirectly, and that no deductions have been made either directly or indirectly from the full wages earned, other than permissible deductions as set forth in Regulations, 29 CFR part 3;
- (3) That each laborer or mechanic has been paid not less than the applicable wage rates and fringe benefits or cash equivalents for the classification of work performed, as specified in the applicable wage determination incorporated into the contract.

(C) The weekly submission of a properly executed certification set forth on the reverse side of Optional Form WH-347 shall satisfy the requirement for submission of the "Statement of Compliance" required by paragraph (a)(3)(ii)(B) of this section.

(D) The falsification of any of the above certifications may subject the contractor or subcontractor to civil or criminal prosecution under section 1001 of title 18 and section 231 of title 31 of the United States Code.

(iii) The contractor or subcontractor shall make the records required under paragraph (a)(3)(i) of this section available for inspection, copying, or transcription by authorized representatives of the State, EPA or the Department of Labor, and shall permit such representatives to interview employees during working hours on the job. If the contractor or subcontractor fails to submit the required records or to make them available, the Federal agency or State may, after written notice to the contractor, sponsor, applicant, or owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds. Furthermore, failure to submit the required records upon request or to make such records available may be grounds for debarment action pursuant to 29 CFR 5.12.

(4) Apprentices and trainees--

(i) Apprentices. Apprentices will be permitted to work at less than the predetermined rate for the work they performed when they are employed pursuant to and individually registered in a bona fide apprenticeship program registered with the U.S. Department of Labor, Employment and Training Administration, Office of Apprenticeship Training, Employer and Labor Services, or with a State Apprenticeship Agency recognized by the Office, or if a person is employed in his or her first 90 days of probationary employment as an apprentice in such an apprenticeship program, who is not individually registered in the program, but who has been certified by the Office of Apprenticeship Training, Employer and Labor Services or a State Apprenticeship Agency (where appropriate) to be eligible for probationary employment as an apprentice. The allowable ratio of apprentices to journeymen on the job site in any craft classification shall not be greater than the ratio permitted to the contractor as to the entire work force under the registered program. Any worker listed on a payroll at an apprentice wage rate, who is not registered or otherwise employed as stated above, shall

be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any apprentice performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. Where a contractor is performing construction on a project in a locality other than that in which its program is registered, the ratios and wage rates (expressed in percentages of the journeyman's hourly rate) specified in the contractor's or subcontractor's registered program shall be observed. Every apprentice must be paid at not less than the rate specified in the registered program for the apprentice's level of progress, expressed as a percentage of the journeymen hourly rate specified in the applicable wage determination. Apprentices shall be paid fringe benefits in accordance with the provisions of the apprenticeship program. If the apprenticeship program does not specify fringe benefits, apprentices must be paid the full amount of fringe benefits listed on the wage determination for the applicable classification. If the Administrator determines that a different practice prevails for the applicable apprentice classification, fringes shall be paid in accordance with that determination. In the event the Office of Apprenticeship Training, Employer and Labor Services, or a State Apprenticeship Agency recognized by the Office, withdraws approval of an apprenticeship program, the contractor will no longer be permitted to utilize apprentices at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(ii) Trainees. Except as provided in 29 CFR 5.16, trainees will not be permitted to work at less than the predetermined rate for the work performed unless they are employed pursuant to and individually registered in a program which has received prior approval, evidenced by formal certification by the U.S. Department of Labor, Employment and Training Administration. The ratio of trainees to journeymen on the job site shall not be greater than permitted under the plan approved by the Employment and Training Administration. Every trainee must be paid at not less than the rate specified in the approved program for the trainee's level of progress, expressed as a percentage of the journeyman hourly rate specified in the applicable wage determination. Trainees shall be paid fringe benefits in accordance with the provisions of the trainee program. If the trainee program does not mention fringe benefits, trainees shall be paid the full amount of fringe benefits listed on the wage determination unless the Administrator of the Wage and Hour Division determines that there is an apprenticeship program associated with the corresponding journeyman wage rate on the wage determination which provides for less than full fringe benefits for apprentices. Any employee listed on the payroll at a trainee rate who is not registered and participating in a training plan approved by the Employment and Training Administration shall be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any trainee performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. In the event the Employment and Training Administration withdraws approval of a training program, the contractor will no longer be permitted to utilize trainees at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(iii) Equal employment opportunity. The utilization of apprentices, trainees and journeymen under this part shall be in conformity with the equal employment opportunity requirements of Executive Order 11246, as amended and 29 CFR part 30.

(5) Compliance with Copeland Act requirements. The contractor shall comply with the requirements of 29 CFR part 3, which are incorporated by reference in this contract.

(6) Subcontracts. The contractor or subcontractor shall insert in any subcontracts the clauses contained in 29 CFR 5.5(a)(1) through (10) and such other clauses as the EPA determines may be appropriate, and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The

prime contractor shall be responsible for the compliance by any subcontractor or lower tier subcontractor with all the contract clauses in 29 CFR 5.5.

(7) Contract termination; debarment. A breach of the contract clauses in 29 CFR 5.5 may be grounds for termination of the contract, and for debarment as a contractor and a subcontractor as provided in 29 CFR 5.12.

(8) Compliance with Davis-Bacon and Related Act requirements. All rulings and interpretations of the Davis-Bacon and Related Acts contained in 29 CFR parts 1, 3, and 5 are herein incorporated by reference in this contract.

(9) Disputes concerning labor standards. Disputes arising out of the labor standards provisions of this contract shall not be subject to the general disputes clause of this contract. Such disputes shall be resolved in accordance with the procedures of the Department of Labor set forth in 29 CFR parts 5, 6, and 7. Disputes within the meaning of this clause include disputes between the contractor (or any of its subcontractors) and Subrecipient(s), State, EPA, the U.S. Department of Labor, or the employees or their representatives.

(10) Certification of eligibility.

(i) By entering into this contract, the contractor certifies that neither it (nor he or she) nor any person or firm who has an interest in the contractor's firm is a person or firm ineligible to be awarded Government contracts by virtue of section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1).

(ii) No part of this contract shall be subcontracted to any person or firm ineligible for award of a Government contract by virtue of section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1).

(iii) The penalty for making false statements is prescribed in the U.S. Criminal Code, 18 U.S.C. 1001.

4. Contract Provision for Contracts in Excess of \$100,000.

(a) Contract Work Hours and Safety Standards Act. The subrecipient shall insert the following clauses set forth in paragraphs (a)(1), (2), (3), and (4) of this section in full in any contract in an amount in excess of \$100,000 and subject to the overtime provisions of the Contract Work Hours and Safety Standards Act. These clauses shall be inserted in addition to the clauses required by Item 3, above or 29 CFR 4.6. As used in this paragraph, the terms laborers and mechanics include watchmen and guards.

(1) Overtime requirements. No contractor or subcontractor contracting for any part of the contract work which may require or involve the employment of laborers or mechanics shall require or permit any such laborer or mechanic in any workweek in which he or she is employed on such work to work in excess of forty hours in such workweek unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of forty hours in such workweek.

(2) Violation; liability for unpaid wages; liquidated damages. In the event of any violation of the clause set forth in paragraph (a)(1) of this section the contractor and any subcontractor responsible therefore shall be liable for the unpaid wages. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such District or to such territory), for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or mechanic, including watchmen and guards, employed in violation of the clause set forth in paragraph (a)(1) of this section, in the sum of \$10 for each calendar day on which

such individual was required or permitted to work in excess of the standard workweek of forty hours without payment of the overtime wages required by the clause set forth in paragraph (a)(1) of this section.

(3) Withholding for unpaid wages and liquidated damages. The subrecipient, upon written request of the EPA Award Official or an authorized representative of the Department of Labor, shall withhold or cause to be withheld, from any moneys payable on account of work performed by the contractor or subcontractor under any such contract or any other Federal contract with the same prime contractor, or any other federally-assisted contract subject to the Contract Work Hours and Safety Standards Act, which is held by the same prime contractor, such sums as may be determined to be necessary to satisfy any liabilities of such contractor or subcontractor for unpaid wages and liquidated damages as provided in the clause set forth in paragraph (b)(2) of this section.

(4) Subcontracts. The contractor or subcontractor shall insert in any subcontracts the clauses set forth in paragraph (a)(1) through (4) of this section and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in paragraphs (a)(1) through (4) of this section.

(b) In addition to the clauses contained in Item 3, above, in any contract subject only to the Contract Work Hours and Safety Standards Act and not to any of the other statutes cited in 29 CFR 5.1, the Subrecipient shall insert a clause requiring that the contractor or subcontractor shall maintain payrolls and basic payroll records during the course of the work and shall preserve them for a period of three years from the completion of the contract for all laborers and mechanics, including guards and watchmen, working on the contract. Such records shall contain the name and address of each such employee, social security number, correct classifications, hourly rates of wages paid, daily and weekly number of hours worked, deductions made, and actual wages paid. Further, the Subrecipient shall insert in any such contract a clause providing that the records to be maintained under this paragraph shall be made available by the contractor or subcontractor for inspection, copying, or transcription by authorized representatives of the (write the name of agency) and the Department of Labor, and the contractor or subcontractor will permit such representatives to interview employees during working hours on the job.

5. Compliance Verification

(a) The subrecipient shall periodically interview a sufficient number of employees entitled to DB prevailing wages (covered employees) to verify that contractors or subcontractors are paying the appropriate wage rates. As provided in 29 CFR 5.6(a)(6), all interviews must be conducted in confidence. The subrecipient must use Standard Form 1445 (SF 1445) or equivalent documentation to memorialize the interviews. Copies of the SF 1445 are available from EPA on request.

(b) The subrecipient shall establish and follow an interview schedule based on its assessment of the risks of noncompliance with DB posed by contractors or subcontractors and the duration of the contract or subcontract. Subrecipients must conduct more frequent interviews if the initial interviews or other information indicates that there is a risk that the contractor or subcontractor is not complying with DB. Subrecipients shall immediately conduct necessary interviews in response to an alleged violation of the prevailing wage requirements. All interviews shall be conducted in confidence.

(c) The subrecipient shall periodically conduct spot checks of a representative sample of weekly payroll data to verify that contractors or subcontractors are paying the appropriate wage rates. The subrecipient shall establish and follow a spot check schedule based on its assessment of the risks of noncompliance with DB posed by contractors or subcontractors and the duration of the contract or subcontract. At a minimum, if practicable, the subrecipient should spot check payroll data within two weeks of each contractor or

subcontractor's submission of its initial payroll data and two weeks prior to the completion date the contract or subcontract . Subrecipients must conduct more frequent spot checks if the initial spot check or other information indicates that there is a risk that the contractor or subcontractor is not complying with DB. In addition, during the examinations the subrecipient shall verify evidence of fringe benefit plans and payments thereunder by contractors and subcontractors who claim credit for fringe benefit contributions.

(d) The subrecipient shall periodically review contractors and subcontractors use of apprentices and trainees to verify registration and certification with respect to apprenticeship and training programs approved by either the U.S Department of Labor or a state, as appropriate, and that contractors and subcontractors are not using disproportionate numbers of, laborers, trainees and apprentices. These reviews shall be conducted in accordance with the schedules for spot checks and interviews described in Item 5(b) and (c) above.

(e) Subrecipients must immediately report potential violations of the DB prevailing wage requirements to the EPA DB contact listed above and to the appropriate DOL Wage and Hour District Office listed at <http://www.dol.gov/whd/america2.htm>.

ATTACHMENT 2

DAVIS-BACON WAGE RATE DETERMINATION

[SRF Assistance Recipient to insert applicable wage determinations here]

How to obtain a Wage Determination:

1. www.wdol.gov
2. Click “Selecting DBA WDs”
3. Select the State and county where the work will be performed
4. Select the “Construction Type”: Heavy, Building, Highway, or Residential
5. Click on one of the wage determinations. Verify that the wage determination displayed is the correct wage determination, and not for “Heavy Dredging”.
6. Select the text box displaying the Wage Determination and copy the text of the Wage Determination.
7. Click “Sign Up for Alert Service” to receive notification if the Wage Determination is updated.

When to update the wage determination:

1. If DOL updates the Wage Determination, you must update the Wage Determination through an addendum to the bid specifications.
2. If the update occurs less than 10 days prior to the date of bid opening, you are not required to update the Wage Determination.

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ATTACHMENT 3

CERTIFICATION OF NONSEGREGATED FACILITIES

(Applicable to federally assisted construction contracts and related subcontracts exceeding \$10,000 which are not exempt from the Equal Opportunity clause.)

The federally assisted construction contractor certifies that he does not maintain or provide for his employees any segregated facilities at any of his establishments, and that he does not permit his employees to perform their services at any location, under his control, where segregated facilities are maintained. The federally assisted construction contractor certifies, further that he will not maintain or provide for his employees any segregated facilities at any of his establishments, and that he will not permit his employees to perform their services at any location, under his control, where segregated facilities are maintained. The federally assisted construction contractor agrees that a breach of this certification is a violation of the Equal Opportunity clause in this contract.

As used in this certification, the term "segregated facilities" means any waiting rooms, work area, rest rooms and wash rooms, restaurants and other eating areas, time clocks, locker rooms and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing facilities provided for employees which are segregated by explicit directive or area, in fact, segregated on the basis of race, creed, color, or national origin, because of habit, local custom, or otherwise. The federally assisted construction contractor agrees that (except where he has obtained identical certifications from proposed contractors for specific time periods) he will obtain identical certifications from proposed subcontractors prior to the award of subcontracts exceeding \$10,000 which are not exempt from the provisions of the Equal Opportunity clause, and that he will retain such, certification in this file.

Signature

Date

Name and title of signer (please type)

[THIS FORM SHALL BE COMPLETED IN FULL AND SUBMITTED WITH THE BID PROPOSAL]

ATTACHMENT 4

NOTICE TO LABOR UNIONS OR OTHER ORGANIZATION OF WORKERS: NON-DISCRIMINATION IN EMPLOYMENT

TO: _____
(name of union or organization of worker)

The undersigned currently holds contract(s) with _____
(name of applicant)
_____ involving funds or credit of the U.S. Government or (a) subcontract(s)
with a prime contractor holding such contract(s).

You are advised that under the provisions of the above contract(s) or subcontract(s) and in accordance with Section 202 of Executive Order 11246 dated September 24, 1965, the undersigned is obliged not to discriminate against any employee or applicant for employment because of race, color, creed, or national origin. This obligation not to discriminate in employment includes, but is not limited to, the following:

EMPLOYMENT, UPGRADING, TRANSFER OR DEMOTION

RECRUITMENT AND ADVERTISING

RATES OF PAY OR OTHER FORMS OF COMPENSATION

SELECTION FOR TRAINING INCLUDING APPRENTICESHIP, LAYOFF OR
TERMINATION

This notice is furnished you pursuant to the provisions of the above contract(s) or subcontract(s) and Executive Order 11246.

Copies of this notice will be posted by the undersigned in conspicuous places available to employees or applicants for employment.

(contractor or subcontractor(s))

(Date)

SECTION 00 54 00 – CDBG CONDITIONS

Washington State Department of Commerce
Community Development Block Grant Program

CDBG CONDITIONS [Attachment 7-A(1)]

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Access to Records

The Contractor agrees to keep such records as the Owner may require. All such records shall be available to the Owner and duly authorized officials of the state for examination. All records pertinent to this project shall be retained by the Contractor for a period of three (3) years after the final audit.

Contract Security

For contracts in excess of \$100,000, the Contractor shall furnish a performance bond in an amount at least equal to one hundred percent (100%) of the contract prices as security for the faithful performance of this contract and also a payment bond in an amount not less than one hundred percent (100%) of the contract price or in a penal sum not less than that prescribed by state, territorial or local law, as security for the payment of all persons performing labor on the project under this contract and furnishing materials in connection with this contract. The performance bond and the payment bond may be in one or in separate instruments in accordance with local law.

Lands and Rights-of-Way

Prior to the start of construction, the Owner shall obtain all lands and rights-of-way necessary for the carrying out and completion of work to be performed under this contract. Any property acquisition shall comply with Title II and III of the Uniform Relocation Assistance and Real Property Acquisition Policies of 1970 (42 U.S.C. 4630).

Other Prohibited Interests

No official of the Owner who is authorized in such capacity and on behalf of the Owner to negotiate, make, accept or approve, or to take part in negotiating, making, accepting, or approving any architectural, engineering, inspection, construction or material supply become directly or indirectly interested personally in this contract or in any part hereof. No officer, employee, architect, attorney, engineer or inspector of or for the Owner who is authorized in such capacity and on behalf of the Owner to exercise any legislative, executive, supervisory or other similar functions in connection with the construction of the project, shall become directly or indirectly interested personally in this contract or in any part thereof, any material supply contract, subcontract, insurance contract, or any other contract pertaining to the project.

Ineligible Subcontractors

The Contractor shall not subcontract any part of the work covered by this Contract or permit subcontracted work to be further subcontracted without the Local Public Agency's or Public Body's prior written approval of the subcontractor. The Local Public Agency or Public Body will not approve any subcontractor for work covered by this contract who is at the time ineligible under the provisions of any applicable regulations issued by the Secretary of Labor, U. S. Department of Labor or the Secretary of Housing and Urban Development, to receive such subcontract. Verifications of Subcontractors need to be obtained by the Contractor from www.sam.gov.

Federal Labor Standards Provisions

U.S. Department of Housing and Urban Development Office of Davis-Bacon and Labor Standard

Form HUD-4010 (07/2021) – Ref. Handbook 1344.1 – Previous editions are obsolete

A. APPLICABILITY

The Project or Program to which the construction work covered by this Contract pertains is being assisted by the United States of America, and the following Federal Labor Standards Provisions are included in this Contract pursuant to the provisions applicable to such Federal assistance.

(1) MINIMUM WAGES

(i) All laborers and mechanics employed or working upon the site of the work will be paid unconditionally and not less often than once a week, and without subsequent deduction or rebate on any account (except such payroll deductions as are permitted by regulations issued by the Secretary of Labor under the Copeland Act (29 CFR Part 3)), the full amount of wages and bona fide fringe benefits (or cash equivalents thereof) due at time of payment, computed at rates not less than those contained in the wage determination of the Secretary of Labor (which is attached hereto and made a part hereof), regardless of any contractual relationship which may be alleged to exist between the contractor and such laborers and mechanics. Contributions made or costs reasonably anticipated for bona fide fringe benefits under Section 1(b)(2) of the Davis-Bacon Act on behalf of laborers or mechanics are considered wages paid to such laborers or mechanics, subject to the provisions of 29 CFR 5.5(a)(1)(iv); also, regular contributions made or costs incurred for more than a weekly period (but not less often than quarterly) under plans, funds, or programs, which cover the particular weekly period, are deemed to be constructively made or incurred during such weekly period.

Such laborers and mechanics shall be paid the appropriate wage rate and fringe benefits on the wage determination for the classification of work actually performed, without regard to skill, except as provided in 29 CFR 5.5(a)(4). Laborers or mechanics performing work in more than one classification may be compensated at the rate specified for each classification for the time actually worked therein: Provided, that the employer's payroll records accurately set forth the time spent in each classification in which work is performed. The wage determination (including any additional classification and wage rates conformed under 29 CFR 5.5(a)(1)(ii) and the Davis-Bacon poster (WH1321)) shall be posted at all times by the contractor and its subcontractors at the site of the work in a prominent and accessible place, where it can be easily seen by the workers.

(ii) Additional Classifications.

(A) Any class of laborers or mechanics which is not listed in the wage determination and which is to be employed under the contract shall be classified in conformance with the wage determination. HUD shall approve an additional classification and wage rate and fringe benefits therefor only when the following criteria have been met:

- (1) The work to be performed by the classification requested is not performed by a classification in the wage determination;
- (2) The classification is utilized in the area by the construction industry; and
- (3) The proposed wage rate, including any bona fide fringe benefits, bears a reasonable relationship to the wage rates contained in the wage determination.

(B) If the contractor, the laborers and mechanics to be employed in the classification (if known), or their representatives, and HUD or its designee agree on the proposed classification and wage rate (including the amount designated for fringe benefits, where appropriate), a report of the action taken shall be sent by HUD or its designee to the Administrator of the Wage and Hour Division ("Administrator"), Employment Standards Administration, U.S. Department of Labor, Washington, D.C. 20210. The Administrator, or an authorized representative, will approve, modify, or disapprove every additional classification action within 30 days of receipt and so advise HUD or its designee or will notify HUD or its designee within the 30-day period that additional time is necessary. (Approved by the Office of Management and Budget ("OMB") under OMB control number 1235-0023.)

- (C) In the event the contractor, the laborers or mechanics to be employed in the classification or their representatives, or HUD or its designee do not agree on the proposed classification and wage rate (including the amount designated for fringe benefits, where appropriate), HUD or its designee shall refer the questions, including the views of all interested parties and the recommendation of HUD or its designee, to the Administrator for determination. The Administrator, or an authorized representative, will issue a determination within 30 days of receipt or its designee or will notify HUD or its designee within the 30-day period that additional time is necessary. (Approved by the Office of Management and Budget under OMB Control Number 1235-0023.)
- (D) The wage rate (including fringe benefits, where appropriate) determined pursuant to subparagraphs (1)(ii)(B) or (C) of this paragraph, shall be paid to all workers performing work in the classification under this Contract from the first day on which work is performed in the classification.

(iii) Whenever the minimum wage rate prescribed in the contract for a class of laborers or mechanics includes a fringe benefit which is not expressed as an hourly rate, the contractor shall either pay the benefit as stated in the wage determination or shall pay another bona fide fringe benefit or an hourly cash equivalent thereof.

(iv) If the contractor does not make payments to a trustee or other third person, the contractor may consider as part of the wages of any laborer or mechanic the amount of any costs reasonably anticipated in providing bona fide fringe benefits under a plan or program, Provided, that the Secretary of Labor has found, upon the written request of the contractor, that the applicable standards of the Davis-Bacon Act have been met. The Secretary of Labor may require the contractor to set aside in a separate account assets for the meeting of obligations under the plan or program. (Approved by the Office of Management and Budget under OMB Control Number 1235-0023.)

(2) Withholding. HUD or its designee shall, upon its own action or upon written request of an authorized representative of the U.S. Department of Labor, withhold or cause to be withheld from the contractor under this contract or any other Federal contract with the same prime contractor, or any other Federally-assisted contract subject to Davis-Bacon prevailing wage requirements which is held by the same prime contractor, so much of the accrued payments or advances as may be considered necessary to pay laborers and mechanics, including apprentices, trainees and helpers, employed by the contractor or any subcontractor the full amount of wages required by the contract. In the event of failure to pay any laborer or mechanic, including any apprentice, trainee or helper, employed or working on the site of the work, all or part of the wages required by the contract, HUD or its designee may, after written notice to the contractor, sponsor, applicant, or owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds until such violations have ceased. HUD or its designee may, after written notice to the contractor, disburse such amounts withheld for and on account of the contractor or subcontractor to the respective employees to whom they are due. The Department of Labor shall make such disbursements in the case of direct Davis-Bacon Act contracts.

(3) Payrolls and basic records.

(i) Maintaining Payroll Records. Payrolls and basic records relating thereto shall be maintained by the contractor during the course of the work and preserved for a period of three years thereafter for all laborers and mechanics working at the site of the work. Such records shall contain the name, address, and social security number of each such worker, his or her correct classification(s), hourly rates of wages paid (including rates of contributions or costs anticipated for bona fide fringe benefits or cash equivalents thereof of the types described in Section 1(b)(2)(B) of the Davis-Bacon Act), daily and weekly number of hours worked, deductions made, and actual wages paid.

Whenever the Secretary of Labor has found, under 29 CFR 5.5(a)(1)(iv), that the wages of any laborer or mechanic include the amount of any costs reasonably anticipated in providing benefits under a plan or program described in Section 1(b)(2)(B) of the Davis-Bacon Act, the contractor shall maintain records which show that the commitment to provide such benefits is enforceable, that the plan or program is financially responsible, and that the plan or program has been communicated in writing to the laborers or mechanics affected, and records which show the costs anticipated or the actual cost incurred in providing such benefits.

Contractors employing apprentices or trainees under approved programs shall maintain written evidence of the registration of apprenticeship programs and certification of trainee programs,

the registration of the apprentices and trainees, and the ratios and wage rates prescribed in the applicable programs. (Approved by the Office of Management and Budget under OMB Control Numbers 1235-0023 and 1215-0018.)

(ii) Certified Payroll Reports.

(A) The contractor shall submit weekly, for each week in which any contract work is performed, a copy of all payrolls to HUD or its designee if the agency is a party to the contract, but if the agency is not such a party, the contractor will submit the payrolls to the applicant sponsor, or owner, as the case may be, for transmission to HUD or its designee. The payrolls submitted shall set out accurately and completely all of the information required to be maintained under 29 CFR 5.5(a)(3)(i), except that full social security numbers and home addresses shall not be included on weekly transmittals. Instead, the payrolls only need to include an individually identifying number for each employee (e.g., the last four digits of the employee's social security number). The required weekly payroll information may be submitted in any form desired. Optional Form WH-347 is available for this purpose from the Wage and Hour Division Web site at <https://www.dol.gov/agencies/whd/forms> or its successor site. The prime contractor is responsible for the submission of copies of payrolls by all subcontractors.

Contractors and subcontractors shall maintain the full social security number and current address of each covered worker, and shall provide them upon request to HUD or its designee if the agency is a party to the contract, but if the agency is not such a party, the contractor will submit the payrolls to the applicant sponsor, or owner, as the case may be, for transmission to HUD or its designee, the contractor, or the Wage and Hour Division of the U.S. Department of Labor for purposes of an investigation or audit of compliance with prevailing wage requirements. It is not a violation of this subparagraph for a prime contractor to require a subcontractor to provide addresses and social security numbers to the prime contractor for its own records, without weekly submission to HUD or its designee. (Approved by the Office of Management and Budget under OMB Control Number 1235-0008.)

(B) Each payroll submitted shall be accompanied by a "Statement of Compliance," signed by the contractor or subcontractor or his or her agent who pays or supervises the payment of the persons employed under the contract and shall certify the following:

(1) That the payroll for the payroll period contains the information required to be provided under 29 CFR 5.5(a)(3)(ii), the appropriate information is being maintained under 29 CFR 5.5(a)(3)(i), and that such information is correct and complete;

(2) That each laborer or mechanic (including each helper, apprentice, and trainee) employed on the contract during the payroll period has been paid the full weekly wages earned, without rebate, either directly or indirectly, and that no deductions have been made either directly or indirectly from the full wages earned, other than permissible deductions as set forth in 29 CFR Part 3;

(3) That each laborer or mechanic has been paid not less than the applicable wage rates and fringe benefits or cash equivalents for the classification of work performed, as specified in the applicable wage determination incorporated into the contract; and

(C) The weekly submission of a properly executed certification set forth on the reverse side of Optional Form WH-347 shall satisfy the requirement for submission of the "Statement of Compliance" required by subparagraph (a)(3)(ii)(b).

(D) The falsification of any of the above certifications may subject the contractor or subcontractor to civil or criminal prosecution under Section 1001 of Title 18 and Section 3729 of Title 31 of the United States Code.

(iii) The contractor or subcontractor shall make the records required under subparagraph (a)(3)(i) available for inspection, copying, or transcription by authorized representatives of HUD or its designee or the U.S. Department of Labor, and shall permit such representatives to interview employees during working hours on the job. If the contractor or subcontractor fails to submit the required records or to make them available, HUD or its designee may, after written notice to the contractor, sponsor, applicant, or owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds. Furthermore, failure to

submit the required records upon request or to make such records available may be grounds for debarment action pursuant to 29 CFR 5.12.

(4) Apprentices and Trainees.

- (i) Apprentices.** Apprentices will be permitted to work at less than the predetermined rate for the work they performed when they are employed pursuant to and individually registered in a bona fide apprenticeship program registered with the U.S. Department of Labor, Employment and Training Administration, Office of Apprenticeship Training, Employer and Labor Services, or with a State Apprenticeship Agency recognized by the Office, or if a person is employed in his or her first 90 days of probationary employment as an apprentice in such an apprenticeship program, who is not individually registered in the program, but who has been certified by the Office of Apprenticeship Training, Employer and Labor Services, or a State Apprenticeship Agency (where appropriate), to be eligible for probationary employment as an apprentice.

The allowable ratio of apprentices to journeymen on the job site in any craft classification shall not be greater than the ratio permitted to the contractor as to the entire work force under the registered program. Any worker listed on a payroll at an apprentice wage rate, who is not registered or otherwise employed as stated above, shall be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any apprentice performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. Where a contractor is performing construction on a project in a locality other than that in which its program is registered, the ratios and wage rates (expressed in percentages of the journeyman's hourly rate) specified in the contractor's or subcontractor's registered program shall be observed.

Every apprentice must be paid at not less than the rate specified in the registered program for the apprentice's level of progress, expressed as a percentage of the journeymen hourly rate specified in the applicable wage determination. Apprentices shall be paid fringe benefits in accordance with the provisions of the apprenticeship program.

If the apprenticeship program does not specify fringe benefits, apprentices must be paid the full amount of fringe benefits listed on the wage determination for the applicable classification. If the Administrator determines that a different practice prevails for the applicable apprentice classification, fringe benefits shall be paid in accordance with that determination. In the event the Office of Apprenticeship Training, Employer and Labor Services, or a State Apprenticeship Agency recognized by the Office, withdraws approval of an apprenticeship program, the contractor will no longer be permitted to utilize apprentices at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

- (ii) Trainees.** Except as provided in 29 CFR 5.16, trainees will not be permitted to work at less than the predetermined rate for the work performed, unless they are employed pursuant to and individually registered in a program which has received prior approval, evidenced by formal certification by the U.S. Department of Labor, Employment and Training Administration. The ratio of trainees to journeymen on the job site shall not be greater than permitted under the plan approved by the Employment and Training Administration. Every trainee must be paid at not less than the rate specified in the approved program for the trainee's level of progress, expressed as a percentage of the journeyman hourly rate specified in the applicable wage determination. Trainees shall be paid fringe benefits in accordance with the provisions of the trainee program. If the trainee program does not mention fringe benefits, trainees shall be paid the full amount of fringe benefits listed on the wage determination unless the Administrator of the Wage and Hour Division determines that there is an apprenticeship program associated with the corresponding journeyman wage rate on the wage determination which provides for less than full fringe benefits for apprentices. Any employee listed on the payroll at a trainee rate who is not registered and participating in a training plan approved by the Employment and Training Administration shall be paid not less than the applicable wage rate on the wage determination for the work actually performed.

In addition, any trainee performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. In the event the Employment and Training Administration withdraws approval of a training program, the contractor will no longer be permitted to utilize trainees at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(iii) Equal employment opportunity. The utilization of apprentices, trainees, and journeymen under 29 CFR Part 5 shall be in conformity with the equal employment opportunity requirements of Executive Order 11246, as amended, and 29 CFR Part 30.

(5) Compliance with Copeland Act requirements. The contractor shall comply with the requirements of 29 CFR Part 3, which are incorporated by reference in this Contract.

(6) Subcontracts. The contractor or subcontractor will insert in any subcontracts the clauses contained in subparagraphs (1) through (11) in this paragraph (a) and such other clauses as HUD or its designee may, by appropriate instructions, require, and a copy of the applicable prevailing wage decision, and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for the compliance by any subcontractor or lower tier subcontractor with all the contract clauses in this paragraph.

(7) Contract termination; debarment. A breach of the contract clauses in 29 CFR 5.5 may be grounds for termination of the contract and for debarment as a contractor and a subcontractor as provided in 29 CFR 5.12.

(8) Compliance with Davis-Bacon and Related Act Requirements. All rulings and interpretations of the Davis-Bacon and Related Acts contained in 29 CFR Parts 1, 3, and 5 are herein incorporated by reference in this Contract.

(9) Disputes concerning labor standards. Disputes arising out of the labor standards provisions of this Contract shall not be subject to the general disputes clause of this Contract. Such disputes shall be resolved in accordance with the procedures of the U.S. Department of Labor set forth in 29 CFR Parts 5, 6, and 7. Disputes within the meaning of this clause include disputes between the contractor (or any of its subcontractors) and HUD or its designee, the U.S. Department of Labor, or the employees or their representatives.

(10) Certification of Eligibility.

(i) By entering into this Contract, the contractor certifies that neither it (nor he or she) nor any person or firm who has an interest in the contractor's firm is a person or firm ineligible to be awarded Government contracts by virtue of Section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1) or to be awarded HUD contracts or participate in HUD programs pursuant to 24 CFR Part 24.

(ii) No part of this Contract shall be subcontracted to any person or firm ineligible for award of a Government contract by virtue of Section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1) or to be awarded HUD contracts or participate in HUD programs pursuant to 24 CFR Part 24.

(iii) Anyone who knowingly makes, presents, or submits a false, fictitious, or fraudulent statement, representation or certification is subject to criminal, civil and/or administrative sanctions, including fines, penalties, and imprisonment (e.g., 18 U.S.C. §§ 287, 1001, 1010, 1012; 31 U.S.C. §§ 3729, 3802).

(11) Complaints, Proceedings, or Testimony by Employees. No laborer or mechanic, to whom the wage, salary, or other labor standards provisions of this Contract are applicable, shall be discharged or in any other manner discriminated against by the contractor or any subcontractor because such employee has filed any complaint or instituted or caused to be instituted any proceeding or has testified or is about to testify in any proceeding under or relating to the labor standards applicable under this Contract to his employer.

B. CONTRACT WORK HOURS AND SAFETY STANDARDS ACT

The provisions of this paragraph (b) are applicable where the amount of the prime contract exceeds **\$100,000**. As used in this paragraph, the terms "laborers" and "mechanics" include watchmen and guards.

(1) Overtime requirements. No contractor or subcontractor contracting for any part of the contract work, which may require or involve the employment of laborers or mechanics, shall require or permit any such laborer or mechanic in any workweek in which the individual is employed on such work to work in excess of 40 hours in such workweek, unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of 40 hours in such workweek.

(2) Violation; liability for unpaid wages; liquidated damages. In the event of any violation of the clause set forth in subparagraph B(1) of this paragraph, the contractor, and any subcontractor responsible therefor, shall be liable for the unpaid wages. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such

District or to such territory) for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or mechanic, including watchmen and guards, employed in violation of the clause set forth in subparagraph B(1) of this paragraph, in the sum of **\$27** for each calendar day on which such individual was required or permitted to work in excess of the standard workweek of 40 hours without payment of the overtime wages required by the clause set forth in subparagraph B(1) of this paragraph. In accordance with the Federal Civil Penalties Inflation Adjustment Act of 1990 (28 U.S.C. § 2461 Note), the Department of Labor adjusts this civil monetary penalty for inflation no later than January 15 each year.

(3) Withholding for unpaid wages and liquidated damages. HUD or its designee shall, upon its own action or upon written request of an authorized representative of the U.S. Department of Labor, withhold or cause to be withheld from any moneys payable on account of work performed by the contractor or subcontractor under any such contract, or any other Federal contract with the same prime contract, or any other Federally-assisted contract subject to the Contract Work Hours and Safety Standards Act which is held by the same prime contractor, such sums as may be determined to be necessary to satisfy any liabilities of such contractor or subcontractor for unpaid wages and liquidated damages, as provided in the clause set forth in subparagraph B(2) of this paragraph.

(4) Subcontracts. The contractor or subcontractor shall insert in any subcontracts the clauses set forth in subparagraph B(1) through (4) of this paragraph and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in subparagraphs B(1) through (4) of this paragraph.

C. HEALTH AND SAFETY

The provisions of this paragraph (c) are applicable where the amount of the prime contract exceeds **\$100,000**.

- (1) No laborer or mechanic shall be required to work in surroundings or under working conditions which are unsanitary, hazardous, or dangerous to his or her health and safety, as determined under construction safety and health standards promulgated by the Secretary of Labor by regulation.
- (2) The contractor shall comply with all regulations issued by the Secretary of Labor pursuant to 29 CFR Part 1926 and failure to comply may result in imposition of sanctions pursuant to the Contract Work Hours and Safety Standards Act, (Public Law 91-54, 83 Stat 96), 40 U.S.C. § 3701 et seq.
- (3) The contractor shall include the provisions of this paragraph in every subcontract, so that such provisions will be binding on each subcontractor. The contractor shall take such action with respect to any subcontractor as the Secretary of Housing and Urban Development or the Secretary of Labor shall direct as a means of enforcing such provisions.

End of Federal Labor Standards Provisions

Compliance with State Prevailing Wage Requirements (RCW 39.12)

Before any payment is made of any sums due under this Contract, the Local Government Body must receive from the Contractor and each subcontractor a copy of the "Statement of Intent to Pay Prevailing Wages" approved by the Washington State Department of Labor and Industries. Also following the acceptance of the project, the Local Government Body must receive from the Contractor and each subcontractor a copy of the "Affidavit of Wages Paid" approved by the State Department of Labor and Industries. Forms may be obtained from the Department of Labor and Industries. The Contractor and each subcontractor shall pay all fees associated with and make all applications directly to the Department of Labor and Industries. These affidavits will be required before any funds retained, according to the provisions of RCW 60.28.010, are released to the Contractor. Payment by the Contractor or subcontractor of any fees shall be considered incidental to the construction and all costs shall be included in other pay items of the project.

Title VI of the Civil Rights Act of 1964

No person shall, on the grounds of race, color or national origin, be excluded from participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving Federal financial assistance. (Public Law 88-352, Title VI of the Civil Rights Act of 1964, 42 U.S.C. 2000d et. seq.)

Section 109 of the Housing and Community Development Act of 1974, as Amended

No person in the United States shall on the grounds of race, color, national origin, sex or religion be excluded from participation in, be denied the benefits of, or be subjected to discrimination under any program or activity funded in whole or in part with funds made available under this title.

Age Discrimination Act of 1975, as Amended

No person shall be excluded from participation, denied program benefits, or subjected to discrimination on the basis of age under any program or activity receiving federal funding assistance. (42 U.S.C. 610 et. seq.)

Section 504 of the Rehabilitation Act of 1973, as Amended

No otherwise qualified individual shall, solely by reason of his or her handicap, be excluded from participation (including employment), denied program benefits, or subjected to discrimination under any program or activity receiving Federal funds. (29 U.S.C. 794)

Section 3 Clause of the Housing and Community Development Act of 1968

a. This is a Section 3 covered project. Section 3 projects means housing rehabilitation, housing construction, and other public construction projects assisted under HUD programs that provide housing and community development financial assistance when the total amount of assistance to the project exceeds a threshold of \$200,000. The project is the site or sites together with any building(s) and improvements located on the site(s) that are under common ownership, management, and financing.

b. The work to be performed under this contract is subject to the requirements of Section 3 of the Housing and Urban Development Act of 1968, as amended, 12 U.S.C. 1701u (Section 3). The purpose of Section 3 is to ensure that employment and other economic opportunities generated by the U.S. Department of Housing and Urban Development (HUD) assistance or HUD assisted

projects covered by Section 3, shall, to the greatest extent feasible, be directed to low- and very low-income persons.

c. The parties to this contract will comply with HUD's regulations as set forth in 24 CFR Part 75, which implement Section 3. As evidenced by their execution of this contract, the parties to this contract certify that they are under no contractual or other impediment that would prevent them from complying with the Part 75 regulations.

d. The Section 3 requirements apply to all contractors and subcontractors performing work in connection with a Section 3 covered project. Contractor means any entity entering into a contract with (a) a recipient to perform work in connection with work in connection with a Section 3 project; or (b) a subrecipient for work in connection with a Section 3 project. Subcontractor means any entity that has a contract with a Contractor to undertake a portion of the contractor's obligation to perform work in connection with a Section 3 project.

e. The contractor agrees to include this Section 3 Clause in every subcontract subject to compliance with regulations in 24 CFR Part 75, and agrees to take appropriate action upon a finding that the subcontractor is in violation of the regulations in 24 CFR Part 75. The contractor will not subcontract with any subcontractor where the contractor has notice or knowledge that the subcontractor has been found in violation of regulations under 24 CFR Part 75.

f. The contractor will certify that any vacant employment positions, including training positions, that are filled (1) after the contractor is selected, but before the contract is executed, and (2) with persons other than those to whom the regulations of 24 CFR Part 75 require employment opportunities to be directed, where not filled to circumvent the contractor's obligations under 24 CFR Part 75.

g. Noncompliance with HUD's regulations in 24 CFR Part 75 may result in sanctions, termination of this contract for default, and debarment or suspension from future HUD assisted contracts.

Executive Order 11246 - Equal Opportunity Clause

During the performance of this Contract, the Contractor agrees as follows:

- (a) The Contractor will not discriminate against any employee or applicant for employment because of race, color, religion, sex or national origin. The Contractor will take affirmative action to ensure that applicants are employed and that employees are treated during employment, without regard to their race, color, religion, sex or national origin. Such action shall include, but not be limited to the following: employment, upgrading, demotion or transfer; recruitment or recruitment advertising; layoff or termination, rates of pay or other forms of compensation; and selection for training, including apprenticeship. The Contractor agrees to post in conspicuous places, available to employees and applicants for employment, notices to be provided setting forth the provision of this nondiscrimination clause;
- (b) The Contractor will, in all solicitations or advertisement for employees placed by or on behalf of the Contractor, state that all qualified applicants will receive consideration without regard to race, color, religion, sex or national origin;
- (c) The Contractor will send to each labor union or representative of workers with which he has a collective bargaining agreement or other contract or understanding, a notice to be provided by the Contract Compliance Officer advising the said labor union or workers' representatives of the Contractor's commitment under this Section, and shall post copies of the notice in conspicuous places available to employees and applicants for employment;

- (d) The Contractor will comply with all provisions of Executive Order 11246 of September 24, 1965, and of the rules, regulations and relevant orders of the Secretary of Labor;
- (e) The Contractor will furnish all information and reports required by Executive Order 11246 of September 24, 1965, and by rules, regulations and orders of the Secretary of Labor, or pursuant thereto, and will permit access to his books, records and accounts by the Department and the Secretary of Labor for purposes of investigation to ascertain compliance with such rules, regulations and orders;
- (f) In the event of the Contractor's noncompliance with the nondiscrimination clauses of this Contract or with any of the said rules, regulations or orders, this Contract may be canceled, terminated or suspended in whole or in part and the Contractor may be declared ineligible for further government contracts in accordance with procedures authorized in Executive Order 11246 of September 24, 1965 or by rule, regulation or order of the Secretary of Labor, or as otherwise provided by law; and
- (g) In the event of the Contractor's noncompliance with the nondiscrimination clauses of this Contract or with any of the said rules, regulations or orders, this Contract may be canceled, terminated or suspended in whole or in part and the Contractor may be declared ineligible for further government contracts in accordance with procedures authorized in Executive Order 11246 of September 24, 1965, so that such provisions will be binding upon each subcontractor or vendor. The Contractor will take such action with respect to any subcontract or purchase order as the Department may direct as a means of enforcing such provisions, including sanctions for noncompliance. Provided, however, that in the event a Contractor becomes involved in, or is threatened with, litigation with a subcontractor or vendor as a result of such direction by the Department, the Contractor may request the United States to enter into such litigation to protect the interest of the United States (Section 202 Equal Opportunity Clause).

Executive Order 11246 - Notice Requirement for Affirmative Action

- (a) The Bidder's attention is called to the "Equal Opportunity Clause" and the "Standard Federal Equal Employment Opportunity Construction Contract Specifications" set forth herein.
- (b) The Goals and timetables for minority and female participation, expressed in percentage terms for the Contractor's aggregate work force in each trade on all construction work in the covered area, are as follows:
 - The female participation goal in each trade is 6.9 percent for every county in Washington State.
 - Refer to <https://www.dol.gov/sites/dolgov/files/ofccp/ParticipationGoals.pdf> for minority participation percentages.

These goals are applicable to all the Contractor's construction work (whether or not it is federal or federally assisted) performed in the covered area.

The Contractor's compliance with the Executive Order and the regulations in 41 CFR Part 60-4 shall be based on its implementation of the Equal Opportunity Clause, specific affirmative action obligations required by the specifications set forth in 41 CFR Part 60-

4.3(a), and its efforts to meet the goals established for the geographical area where the contract resulting from this solicitation is to be performed. The hours of minority and female employment and training must be substantially uniform throughout the length of the contract and in each trade, and the Contractor shall make a good faith effort to employ minorities and women evenly on each of its projects. The transfer of minority or female employees or trainees from Contractor to Contractor or from project to project for the sole purpose of meeting the Contractor's goals shall be a violation of contract, the Executive Order and the regulations in 41 CFR Part 60-4. Compliance with the goals will be measured against the total work hours performed.

- (c) As used in this Notice, and in the contract resulting from this solicitation, the "covered area" is (insert description of the geographical areas where the contract is to be performed giving the state, county and city, if any).

Executive Order 11246 - Construction Contract Specifications

- (a) As used in these specifications:

- (1) "Covered area" means the geographical area described in the solicitation from which this Contract resulted;
- (2) "Director" means Director, Office of Federal Contract Compliance Programs, United States Department of Labor, or any person to whom the Director delegates authority;
- (3) "Employer identification number" means the Federal Social Security number used on the Employer's Quarterly Federal Tax Return, U. S. Treasury Department Form 941; and
- (4) "Minority" includes:
 - Black (all persons having origins in any of the Black African racial groups not of Hispanic origin);
 - Hispanic (all persons of Mexican, Puerto Rican, Cuban, Central or South American or other Spanish Culture or origin, regardless of race);
 - Asian and Pacific Islander (all persons having origins in any of the original peoples of the Far East, Southeast Asia and Indian Subcontinent, or the Pacific Islands); and
 - American Indian or Alaskan Native (all persons having origins in any of the original peoples of North America and maintaining identifiable tribal affiliations through membership and participation or community identification).

- (b) Whenever the Contractor, or any subcontractor at any tier, subcontracts a portion of the work involving any construction trade, it shall physically include in each subcontract in excess of \$10,000 the provisions of these specifications and the notice which contains the applicable goals for minority and female participation and which is set forth in the solicitation from which this Contract resulted.

- (c) If the Contractor is participating (pursuant to 41 CFR Part 60-4.5) in a Hometown Plan approved by the U. S. Department of Labor in the covered area either individually or through an association, its affirmative action obligations on all work in the plan area (including goals and timetables) shall be in accordance with that plan for those trades which have unions participating in the plan. Contractors must be able to demonstrate their participation in and compliance with the provisions of any such Hometown Plan. Each Contractor or subcontractor participating in an approved plan is individually required to comply with its obligations under the EEO clause, and to make a good faith effort to achieve each goal under the plan in each trade in which it has employees. The overall good faith performance by other contractors or subcontractors toward a goal in an approved plan does not excuse any covered Contractor's or subcontractor's failure to take good faith efforts to achieve the plan goals and timetables.
- (d) The Contractor shall implement the specific affirmative action standards provided in paragraphs 82 (g) (1) through (16) of these specifications. The goals set forth in the solicitation from which this contract resulted are expressed as percentages of the total hours of employment and training of minority and female utilization the Contractor should reasonably be able to achieve in each construction trade in which it has employees in the covered area. The Contractor is expected to make substantially uniform progress toward its goals in each craft during the period specified. Covered construction contractors performing contracts in geographical areas where they do not have a federal or federally-assisted construction contract shall apply the minority and female goals established for the geographic area where the Contract is being performed. Goals are published periodically in the Federal Register in notice form and such notices may be obtained from any Federal Contract Compliance Programs office or from Federal procurement contracting officers. The Contractor is expected to make substantially uniform progress in meeting its goals in each craft during the period specified.
- (e) Neither the provisions of any collective bargaining agreement, nor the failure by a union with whom the Contractor has a collective bargaining agreement, to refer either minorities or women shall excuse the Contractor's obligations under these specifications, Executive Order 11246, or the regulations promulgated pursuant thereto.
- (f) In order for the nonworking training hours of apprentices and trainees to be counted in meeting the goals, such apprentices and trainees must be employed by the Contractor during the training period, and the Contractor must have made a commitment to employ the apprentices and trainees at the completion of their training, subject to the availability of employment opportunities. Trainees must be trained pursuant to training programs approved by the U. S. Department of Labor.
- (g) The Contractor shall take specific affirmative actions to ensure equal employment opportunity. The evaluation of the Contractor's compliance with these specifications shall be based upon its effort to achieve maximum results from its actions. The Contractor shall document these efforts fully, and shall implement affirmative action steps at least as extensive as the following:

- (1) Ensure and maintain a working environment free of harassment, intimidation and coercion at all sites, and in all facilities at which the Contractor's employees are assigned to work. The Contractor, where possible, will assign two or more women to each construction project. The Contractor shall specifically ensure that all foremen, superintendents and other on-site supervisory personnel are aware of and carry out the Contractor's obligation to maintain such a working environment, with specific attention to minority or female individuals working at such site or in such facilities;
- (2) Establish and maintain a current list of minority and female recruitment sources, provide written notification to minority and female recruitment sources, and to community organizations when the Contractor or its unions have employment opportunities available, and maintain a record of the organizations' responses;
- (3) Maintain a current file of the names, addresses and telephone numbers of each minority or female referral from a union, a recruitment source or community organization and of what action was taken with respect to each such individual. If such individual was sent to the union hiring hall for referral and was not referred back to the Contractor, this shall be documented in the file with the reason therefore, along with whatever additional actions the Contractor may have taken;
- (4) Provide immediate written notification to the Director when the union or unions with which the Contractor has a collective bargaining agreement has not referred to the Contractor a minority person or woman sent by the Contractor, or when the Contractor has other information that the union referral process has impeded the Contractor's efforts to meet its obligations;
- (5) Develop on-the-job training opportunities and/or participate in training programs for the area which expressly include minorities and women, including upgrading programs and apprenticeship and trainee programs relevant to the Contractor's employment needs, especially those programs funded or approved by the Department of Labor. The Contractor shall provide notice of these programs to the sources complied under 7b above;
- (6) Disseminate the Contractor's EEO Policy by providing notice of the policy to unions and training programs and requesting their cooperation in assisting the Contractor in meeting its EEO obligations; by including it in any policy manual and collective bargaining agreement; by publicizing it in the company newspaper, annual report, etc.; by specific review of the policy with all management personnel and with all minority and female employees at least once a year; and by posting the company EEO policy on bulletin boards accessible to all employees at each location where construction work is performed;
- (7) Review, at least annually, the company's EEO policy and affirmative action obligations under these specifications with all employees having any responsibility for hiring, assignment, layoff, termination or other employment decisions, including specific review of these items with on-site supervisory personnel such as Superintendents, General Foremen, etc., prior to the initiation of construction work at any job site. A written record shall be made and maintained identifying the time

and place of these meetings, persons attending, subject matter discussed and disposition of the subject matter;

- (8) Disseminate the Contractor's EEO policy externally by including it in any advertising in the news media, specifically including minority and female news media, and providing written notification to and discussing the Contractor's EEO policy with other contractors and subcontractors with whom the Contractor does or anticipates doing business;
 - (9) Direct its recruitment efforts, both oral and written, to minority, female and community organizations, to schools with minority and female students, and to minority and female recruitment and training organizations serving the Contractor's recruitment area and employment needs. Not later than one month prior to the date for the acceptance of applications for apprenticeship or other training by any recruitment source, the Contractor shall send written notification to organizations such as the above, describing the openings, screening procedures and tests to be used in the selection process;
 - (10) Encourage present minority and female employees to recruit other minority persons and women and, where reasonable, provide after school, summer and vacation employment to minority and female youth both on the site and in other areas of a Contractor's work force;
 - (11) Validate all tests and other selection requirements where there is an obligation to do so under 41 CFR Part 60-3;
 - (12) Conduct, at least annually, an inventory and evaluation at least of all minority and female personnel for promotional opportunities and encourage these employees to seek or to prepare for, through appropriate training, etc., such opportunities;
 - (13) Ensure that seniority practices, job classifications, work assignments and other personnel practices, do not have a discriminatory effect by continually monitoring all personnel and employment related activities to ensure that the EEO policy and the Contractor's obligations under these specifications are being carried out;
 - (14) Ensure that all facilities and company activities are non-segregated except that separate or single-user toilet and necessary changing facilities shall be provided to assure privacy between the sexes;
 - (15) Document and maintain a record of all solicitations of offers for subcontracts from minority and female construction contractors and suppliers, including circulation of solicitations to minority and female contractor associations and other business obligations; and
 - (16) Conduct a review, at least annually, of all supervisors' adherence to and performance under the Contractor's EEO policies and affirmative action obligations.
- (h) Contractors are encouraged to participate in voluntary associations, which assist in fulfilling one or more of their affirmative action obligations (g1 through g16). The efforts of a contractor association, joint contractor-union, contractor-community, or other similar groups of which the Contractor is a member and participant, may be asserted as fulfilling any one or more of its obligations under (g1 through g16) of these

specifications provided that the Contractor actively participates in the group, makes every effort to assure that the group has a positive impact on the employment of minorities and women in the industry, ensures that the concrete benefits of the program are reflected in the Contractor's minority and female work force participation, makes a good faith effort to meet its individual goals and timetables, and can provide access to documentation which demonstrates the effectiveness of actions taken on behalf of the Contractor. The obligation shall not be a defense for the Contractor's noncompliance.

- (i) A single goal for minorities and a separate single goal for women have been established. The Contractor, however, is required to provide equal employment opportunities and to take affirmative action for all minority groups, both male and female, and all women, both minority and non-minority. Consequently, the Contractor may be in violation of the Executive Order if a particular group is employed in a substantially disparate manner (for example, even though the Contractor has achieved its goals for women generally, the Contractor may be in violation of the Executive Order if a specific minority of women is underutilized).
- (j) The Contractor shall not use the goals and timetables or affirmative action standards to discriminate against any person because of race, color, religion, sex or national origin.
- (k) The Contractor shall not enter into any subcontract with any person or firm debarred from government contracts pursuant to Executive Order 11246.
- (l) The Contractor shall carry out such sanctions and penalties for violation of these specifications and of the Equal Opportunity Clause, including suspension, termination and cancellation of existing subcontracts as may be imposed or ordered pursuant to Executive Order 11246, as amended, and its implementing regulations, by the Office of Federal Contract Compliance Programs. Any Contractor who fails to carry out such sanctions and penalties shall be in violation of these specifications and Executive Order 11246, as amended.
- (m) The Contractor, in fulfilling its obligations under these specifications, shall implement specific affirmative action steps, at least as extensive as those standards prescribed in paragraph 7 of these specifications, so as to achieve maximum results from its efforts to ensure equal employment opportunity. If the Contractor fails to comply with the requirements of the Executive Order, the implementing regulations or these specifications, the Director shall proceed in accordance with 41 CFR Part 60-4.8.
- (n) The Contractor shall designate a responsible official to monitor all employment related activity to ensure that the company's EEO policy is being carried out, to submit reports relating to the provisions hereof as may be required by the government and to keep records. Records shall at least include for each employee the name, address, telephone numbers, construction trade, union affiliation, if any, employee identification number when assigned, social security number, race, sex, status (e.g., mechanic, apprentice trainee, helper or laborer), dates of changes in status, hours worked per week in the indicated trade, rate of pay, and locations at which the work was performed. Records shall be maintained in an easily understandable and retrievable form; however, to the

degree that existing records satisfy this requirement, Contractors shall not be required to maintain separate records.

- (o) Nothing herein provided shall be construed as a limitation upon the application of other laws, which establish different standards of compliance or upon the application of requirements for the hiring of local or other area residents (e.g., those under the Public Works Employment Act of 1977 and the Community Development Block Grant Program).

Americans with Disabilities Act of 1990

Subject to the provisions of this title, no qualified individual with a disability shall, by reason of such disability, be excluded from participation in or be denied the benefits of the services, programs or activities of a public entity, or be subjected to discrimination by any such entity.

SECTION 00 55 00 – NOTICE TO PROCEED

NOTICE TO PROCEED

DATE _____
CONTACT _____
CONTRACTOR _____
ADDRESS _____
ADDRESS _____

RE: Notice to Proceed
ESWD WWTP Upgrade - Phase I Project

Dear CONTACT:

The Eastsound Sewer and Water District has reviewed and approved the contract bond and evidence of insurance for the aforementioned Project. Therefore, the contract has been executed.

This notice shall constitute the Notice to Proceed on the above referenced project. Contract time will begin on _____, 2023. The contracted work shall be completed by no later than _____, 2024 per contract requirements.

If you have comments, questions, or require further information, please do not hesitate to contact me at (360) -376-2720 DISTRICT PHONE NUMBER.

Sincerely,

EASTSOUND SEWER AND WATER DISTRICT

NAME
District Manager

NAME
Project Manager

CC. file
Jeff Christner, PE – Wilson Engineering
Scott Wilson, PE – Wilson Engineering
Ken Ziebart, PE – Department of Ecology

*****END OF SECTION*****

SECTION 00 61 13 – PERFORMANCE AND PAYMENT BOND FORMS

PERFORMANCE BOND to the Eastsound Sewer and Water District

KNOW ALL PEOPLE BY THESE PRESENTS, That we _____
_____ the Contractor named in the Contract hereinafter referred
to as PRINCIPAL, and _____ as SURETY, are jointly and severally held
and firmly bound to the Eastsound Sewer and Water District, hereinafter referred to as OWNER named in said
Contract WWTP UPGRADE, EASTSOUND, Washington, for the penal sum of,

DOLLARS (\$ _____),
lawful money of the United States, for the payment of which sum well and truly to be made, we bind ourselves, our
heirs, assigns, administrators and successors jointly and severally, firmly by these presents.

THE CONDITION OF THIS OBLIGATION IS SUCH, that Whereas, the Principal entered into a contract with
the Owner, dated the ____ day of _____, 20____, for such construction work with the Eastsound Sewer and
Water District, Washington.

NOW, THEREFORE, if the Principal shall well, truly and faithfully perform all of the provisions and fulfill all of
the undertakings, covenants, terms, conditions and agreements of said contract during the period of the original
contract and any extensions thereof that may be granted by the Owner, with or without notices to the surety; and
during the life of any guaranty required under the contract; and shall also well and truly perform and fulfill all of the
undertakings, covenants, terms, conditions and agreements of any and all duly authorized modifications of said
contract that may hereafter be made; notice of which modifications to the surety being hereby waived, shall
indemnify and save harmless owner from all cost and damage by reason of the principal's default of failure to do so,
and shall pay the State of Washington sales and use taxes, and amounts due said state pursuant to Titles 50 and 51 of
the Revised Code of Washington then this obligation to be void, otherwise to remain in full force and effect.

IN WITNESS WHEREOF, the above bonded parties have executed this instrument under their separate seals this
____ day of _____, 20____, the name and corporate seal of each corporate party hereto affixed, and these
presents duly signed by its undersigned representatives pursuant to authority of its governing body.

Corporate Seal:

PRINCIPAL

ATTEST: (If Corporation)

By: _____

Title: _____

Corporate Seal:

SURETY

By: _____

Title: _____

PAYMENT BOND
to the Eastsound Sewer and Water District

KNOW ALL PEOPLE BY THESE PRESENTS: that

(Name of Contractor)

(Address of Contractor)

a _____, hereinafter called Principal,
(Corporation, Partnership or Individual)

and _____
(Name of Surety)

(Address of surety)

hereinafter called **SURETY**, are held and firmly bound unto _____

(Name of Owner)

(Address of Owner)

hereinafter called **OWNER**, in the penal sum of _____ Dollars, \$(_____)

in lawful money of the United States, for the payment of which sum well and truly to be made, we bind ourselves, successors, and assigns, jointly and severally, firmly by these presents.

THE CONDITION OF THIS OBLIGATION is such that whereas, the Principal entered into a certain contract with the **OWNER**, dated the _____ day of _____ 20____, a copy of which is hereto attached and made a part hereof for the construction of:

NOW, THEREFORE, if the Principal shall promptly make payment to all persons, firms, **SUBCONTRACTORS**, and corporations furnishing materials for or performing labor in the prosecution of the **WORK** provided for in such contract, and any authorized extension or modification thereof including all amounts due for materials, lubricants, oil, gasoline, coal, and coke, repairs on machinery, equipment and tools, consumed or used in connection with the construction of such **WORK**, and all Insurance premiums on said **WORK**, and for all labor, performed in such **WORK** whether by **SUBCONTRACTOR** or otherwise, then this obligation shall be void; otherwise to remain in full force and effect.

PAYMENT BOND (cont.)

PROVIDED, FURTHER, that the said **SURETY** for value received hereby stipulates and agrees that no change, extension of time, alteration or addition to the terms of the contract or to the **WORK** to be performed thereunder or the **SPECIFICATIONS** accompanying the same shall in any wise affect its obligation on this **BOND**, and it does hereby waive notice of any such change, extension of time, alteration or addition to the terms of the contract or to the **WORK** or to the **SPECIFICATIONS**.

PROVIDED, FURTHER, that no final settlement between the **OWNER** and the **CONTRACTOR** shall abridge the right of any beneficiary hereunder, whose claim may be unsatisfied.

IN WITNESS WHEREOF, this instrument is executed in _____ counterparts, each one of which
(number)

shall be deemed an original, this the _____ day of _____

ATTEST:

Principal

(Principal) Secretary

(SEAL) By _____(s)

(Address)

Witness as to Principal

(Address)

(Surety)

ATTEST: By _____
(Attorney –in-Fact)

Witness as to Surety
(Address)

(Address)

NOTE: Date of **BOND** must not be prior to date of Contract.
If **CONTRACTOR** is Partnership, all partners should execute **BOND**.

IMPORTANT: Surety companies executing **BONDS** must appear on the Treasury Department's most current list (Circular 570 as amended) and be authorized to transact business in the State where the **PROJECT** is locate

***** END OF SECTION *****

SECTION 00 61 23 – RETAINAGE BOND FORM

EASTSOUND SEWER AND WATER DISTRICT RETAINAGE INVESTMENT OPTION

CONTRACTOR: _____

PROJECT NAME: _____

DATE: _____

Pursuant to Chapter 60.28 RCW, you may choose how your retainage under this contract will be held and invested. Please complete and sign this form indicating your preference. If you fail to do so, the Eastsound Sewer and Water District (the District) will hold your retainage as described in "Current Expense", option 1 below.

- _____ 1. Current Expense: The District will retain your money in its Current Expense Fund Account until thirty days following final acceptance of the improvement or work as completed. You will not receive interest earned on this money.
- _____ 2. Interest Bearing Account: The District will deposit retainage checks in an interest-bearing account in a bank, mutual savings bank, or savings and loan association, not subject to withdrawal until after the final acceptance of the improvement or work as completed or until agreed to by both parties. Interest on the account will be paid to you.

BONDS AND SECURITIES ACCEPTABLE BY THE EASTSOUND SEWER AND WATER DISTRICT :

1. Bills, certificates, notes or bonds of the United States.
2. Other obligations of the United States or its agencies.
3. Indebtedness of the Federal national Mortgage Association.
4. Time Deposits in commercial banks.

Designate below the type of investment selected:

- _____ 3. Bond-in-Lieu: With the consent of the District, the contractor may submit a bond for all or any portion of the amount of funds retained by the District in a form acceptable to the District and from a bonding company meeting standards established by the District, if any. Unless otherwise indicated, the contractor elects to submit a bond for the entire 5% retainage amount. Such bond and any proceeds there from shall be made subject to all claims and liens and in the same manner and priority as set forth for retained percentages in Chapter 60.28 RCW. Whenever the District accepts a bond-in-lieu of retained funds from a contractor, the contractor shall accept like bonds from any subcontractors or suppliers from which the contractor has retained funds. The contractor shall then release the funds retained from the subcontractor or supplier, to the subcontractor or supplier, within thirty days of the contractor's receipt of the retained funds from the District.

Retainage is normally released 30 - 45 days after final acceptance of work by the District, or following receipt Employment Security / Department of Revenue clearance, whichever takes longer.

(Contractor's Signature) Date

Title: _____

*** END OF SECTION ***

SECTION 00 73 00 – SUPPLEMENTARY CONDITIONS

SUPPLEMENTAL CONDITIONS

The following supplementary conditions modify WSDOT Standard Specifications. If there are any conflicts between these Supplemental Conditions and the Standard Specifications, these Supplemental Conditions shall take precedence.

1. DOCUMENTS INCORPORATED BY REFERENCE

The following documents are incorporated by reference, to include, but not be limited to:

- Specifications
- Proposal
- Drawings
- Contract
- WSDOT Standard Specifications Plans for Road, Bridge and Municipal Construction, 2022 Edition
- Washington State Department of Ecology Water Pollution Control Revolving Fund Specifications Insert revised 1/22/2021 (included in Section 00 53 00)

2. CONFLICT AND PRECEDENCE

In the event of any conflicting provisions or requirements between the component parts of the Contract Documents, the component parts shall take precedence in the following order:

1. Change Orders
2. Contract Form
3. Addenda
4. Permits and requirements from governmental agencies
5. Drawings
6. Supplemental Conditions
7. Technical Specifications
8. EASTSOUND District Standards
9. WSDOT Standard Drawings & Details
10. WSDOT Standard Specifications

3. CONTRACT PLANS AND SPECIFICATIONS

Five (5) sets of Contract Documents, three (3) sets of 11"x 17" plans, Two (2) sets of 24"x 36" plans, and a CD with Contract Documents and plans in PDF will be furnished to the Contractor free of charge. Additional sets may be purchased at the advertised price per set.

4. EXAMINATION OF PLANS, SPECIFICATIONS AND SITE OF WORK

The bidder shall carefully examine the proposed work site (including material sites), and the contract documents. Submittal of a bid shall be conclusive evidence that the bidder has made these examinations and understands all requirements for the performance of the completed work.

The Contractor shall make deductions and conclusions as to the nature of the materials to be excavated, the difficulties which may arise from subsurface conditions, and of doing any other work affected by the subsurface conditions and shall accept full responsibility. The accuracy of information furnished by the Owner and/or Engineer and/or the plans and specifications as to underground structures, foundation conditions, character of soil, position and quantity of surface and ground water, etc., is not guaranteed. Bidders must satisfy themselves by personal examination and by such other means as they desire with respect to actual conditions in regard to existing groundwater or surface structures. Unforeseen conditions shall not constitute a claim for additional payment under the terms of the contract or constitute a basis for cancellation thereof.

The Specifications do not necessarily discuss complete details of construction, work or materials, performance or installation, and do not necessarily cover construction details or other items of work or fixtures of equipment may affect any installation. These details must be ascertained by the Contractor and correlated to bring the parts together to a completed whole.

Where alternate methods have not been brought to the Owner's attention, it is assumed that the Contractor has figured the costlier method or methods.

5. WORK AND MATERIALS

In addition to the requirements stated in this contract document, the following shall apply:

All work and materials under this contract shall conform to the 2022 Edition of *Standard Specifications for Road, Bridge and Municipal Construction* as prepared by Washington State Department of Transportation (WSDOT) and Washington State Chapter of American Public Works Association (APWA), and according to the instructions and recommendations of the manufacturer of the material concerned. In case of a conflict between any of the above referenced Standards, the more stringent shall apply.

References throughout the above mentioned Standard Specifications to "State" or "Owner" shall refer to the Eastsound Sewer and Water District.

6. OMISSIONS AND DISCREPANCIES

Upon receipt of Award of Contract, the Contractor shall carefully study and compare all drawings, specifications and other instructions and shall, prior to ordering material or performing work, report in writing to the Owner any error, inconsistency or omission not discovered at the pre-bid meeting. If during the accomplishment of the work, a discrepancy is found between the drawings and the physical condition of the locality, it shall be the Contractor's duty to inform the Owner in writing, and the Owner shall promptly verify the same. Any work done after such discovery, until authorized, will be done at the Contractor's risk.

Minor items of work or material omitted from the original plans or specifications, but clearly inferable from the information presented and which are called for by accepted good practice, shall be provided and/or performed by the Contractor as part of the original bid.

7. SURVEYS, PERMITS, REGULATIONS

The Contractor is responsible for providing construction staking for the project. The Engineer has established horizontal references and vertical grade datum for the Contractor's use. The Contractor shall be responsible for protection and preservation of the established reference points.

The bidder shall be familiar with all Federal, State, and local requirements that affect the completion of work in any way (such as laws, ordinances, or rules affecting employees, subcontractors, materials, equipment or procedures). In addition, the Contractor must comply with the following Washington State Laws, including without limitation: Chapter 60.28 RCW (retainage); 39.08 RCW (bond requirements); 18.27 RCW (contractor registration); 35.22.650 RCW (equal opportunity); and 70.92 RCW (handicapped). The Owner will not consider any plea of misunderstanding or ignorance of such requirements.

The Owner will assist with coordinating District permit applications, if needed. The Contractor is to pick up the Land Disturbance permit from the District and fill-out remaining information required, prior to mobilization. However, the Contractor will be responsible for providing submittal information, as needed (including shop drawing, mechanical, and plumbing information) to the Engineer/District (if requested). Temporary permits, easements, and other Non-District permits shall be acquired by the Contractor (if needed).

Contractor is required to pick up the Building Permit from San Juan County and fill-out remaining information required, prior to construction. The County is currently processing a building permit and a land use permit for this project. The new Building Permit is #21-0385.

8. EXISTING UTILITIES

The location of all existing utilities shown on the plans is per the best available information, and is therefore approximate only. The Owner/Engineer does not guarantee the accuracy of this information. The contractor shall take whatever measures deemed necessary to verify the accuracy of this information and the cost of such shall be incidental to the bid.

Forty-eight (48) hours prior to starting construction, the Contractor shall contact the Eastsound Sewer and Water District and Underground Utility Locate (if needed). All costs incurred by the Contractor in complying with the requirements of this Section shall be incidental to the entire project and shall be included in the contract price.

9. CONNECTIONS TO EXISTING MAINS (WATER MAINS AND SEWER FORCE MAINS)

Connection to existing mains is the full responsibility of the Contractor. Temporary routing of existing pipelines or services, shoring, temporary thrust blocks, extra fittings required to route the pipe over or under existing or new pipe or other utilities and all other work and materials required for making complete, permanent and workable connections are incidental to other items of work.

The Contractor shall be responsible for determining which residents will be affected by shutoffs, and will notify them in writing (with a copy provided to the District) 24 hours in advance. In addition, the Contractor shall notify private property owners or tenants, by having a representative of the Contractor personally contact the private property owner or tenant. If the property owner or tenant is not available, the Contractor shall leave a door hanger notice indicating the commencement date of work, duration of work, the type of work being done, and the Contractor's and Engineer's phone number and address for questions and concerns. The Engineer shall be provided adequate time to review, comment, and approve the door hanger notice prior to the Contractor placing any notices.

The Contractor shall locate and verify the type of pipe, size and depth prior to making the connection. Detailed sketches and plans of the connection proposed by the Contractor shall be given to the Engineer not less than one week prior to the expected construction. The Eastsound Sewer and Water District shall be notified not less than two (2) working days prior to connection to existing mains.

10. SUBSURFACE CONDITIONS

The CONTRACTOR shall make deductions and conclusions as to the nature of the materials to be excavated, the difficulties which may arise from subsurface conditions, and of doing any other work affected by the subsurface conditions and shall accept full responsibility. The accuracy of information furnished by the OWNER and/or ENGINEER and/or the plans and specifications as to underground structures, foundation conditions, character of soil, position and quantity of surface and ground water, etc., is not guaranteed. Bidders must satisfy themselves by personal examination and by such other means as they desire with respect to actual conditions in regard to existing groundwater or subsurface structures. Unforeseen conditions shall not constitute a claim for additional payment under the terms of the contract or constitute a basis for cancellation thereof.

PLAN AND PROCEDURES FOR THE UNANTICIPATED DISCOVERY OF CULTURAL RESOURCES AND HUMAN SKELETAL REMAINS - The **Inadvertent Discovery Plan (IDP)** is included in Part 4 – Reference Documents, Appendix G.

11. TRAFFIC CONTROL

Traffic shall be maintained in accordance with WSDOT Section 1-07.23 of the WSDOT Standard Specifications and Manual of Uniform Traffic Control Devices. The Contractor shall not close any roadway without first obtaining authorization from the district. The cost for all necessary traffic control by the Contractor shall be incidental to the entire project and shall be included in the contract price.

12. SUBCONTRACTING

The Contractor shall perform work amounting to a minimum of 50% percent of the Awarded Contract Price using his own personnel and equipment. All subcontracting shall be in conformance with WSDOT Section 1-08.1 of the WSDOT Standard Specifications.

13. PRE-CONSTRUCTION CONFERENCE

A Pre-Construction conference shall be held at a time and place fixed by the Owner which will be within two weeks from the date of notification of award of contract. At a minimum, the Contractor's project manager and field superintendent are required to attend. Sub-contractors, suppliers and others interested are encouraged to attend.

14. HOURS OF WORK

The Contractor shall schedule operations so that the work will be performed during the hours of 7AM to 5PM Monday through Friday, excluding holidays. A normal 40-hour Monday through Friday work week (4 – 10hr or 5 – 8hr days) is intended. The Contractor shall compensate the District \$150 per hour for each hour over 40 hours per week worked to pay for additional inspection time. The Contractor shall obtain prior approval from the District for overtime hours and schedules.

15. COMPLETION DATE

The contracted work is to be completed by no later than Oct. 16, 2024. The Contractor will be limited to 360 working days (72 weeks) on-site work. The Contractor shall plan accordingly to meet this completion requirement.

16. SCHEDULE OF CONSTRUCTION & VALUES

Within 10 working days of receiving the notice to proceed, the contractor shall furnish to the District a Schedule of Values. In addition, the Contractor shall furnish a Schedule of Construction at the Pre-Construction Meeting. The Schedule shall identify the project start and finish dates with a detailed breakdown of the proposed order of work and completion dates for major phases of the work. The schedule shall be developed by a critical path method. Time required for testing, backfiring, inspections, ordering, punch lists, etc. shall be incorporated into the schedule (although they do not necessarily need to be specifically identified).

17. RETAINAGE

The owner will deduct from the partial pay estimate a retainage of five percent (5%). Upon completion of all work, specified training, final inspection, and acceptance by Owner, the amount retained under the Contract will be paid within thirty (30) days following final acceptance by Owner and receipt by the Owner of the following:

- State Department of Labor and Industries Release
- Washington State Department of Revenue Release
- Washington State Employment Security Department Release
- Contractor and Subcontractors Affidavit of Wages Paid

The retainage will not be released if any claim has been filed on the project.

18. LIQUIDATED DAMAGES

Liquidated damages will be assessed in accordance with WSDOT 1-08.9 for each working day beyond the Contracted completion date. Liquidated damages for failure to complete the physical Work of a Contract on time shall be as follows:

$$LD = \frac{0.15C}{T}$$

Where:

LD = liquidated damages per working day (rounded to the nearest dollar)

C = original Contract amount

T = original time for Physical Completion

19. PHYSICAL COMPLETION FOR WWTP UPGRADE

Completion of the WWTP UPGRADE shall be defined as follows, with no exceptions:

Substantial completion of the WWTP UPGRADE shall be defined as follows, with no exceptions:

The new wastewater facility shall be able to be put to beneficial use. This shall include construction of the influent pump station, flow splitter, train 3, digester, sludge dewatering system, blower building, UV disinfection system, yard piping, electrical, instruments, controls, backup generator, SCADA/PLC programming, and misc. wastewater facilities. The new wastewater facility will be on-line and functioning per project requirements.

Final completion of the WWTP UPGRADE shall be defined as follows, with no exceptions:

The entire project shall be complete and put to beneficial use. This shall include removal of temporary pump/process equipment, remaining site work, asphalt pavement, performance testing, and all cleanup complete. The entire wastewater facility will be operable.

20. PAYMENT TO CONTRACTOR

At least five (5) working days before the end of the month, the Contractor shall submit to the Engineer an itemized application for payment, supported by receipt or other vouchers, showing payments for materials and labor, payments to sub-contractors, and such other evidence of the Contractor's right to payment as the Engineer may direct. The Owner's progress payment shall be made approximately 30 days after the date of submittal.

The owner will deduct from the partial pay estimate a retainage as defined above. Upon completion of all work, final inspection, and acceptance by Owner, the amount retained under the Contract will be paid at the expiration of the thirty (30) day period following final acceptance by owner provided the following conditions are met:

- A. Releases have been obtained from the State Department of Labor and Industries, the State of Washington Employment Security Department, the Washington State Department of Revenue, and all other departments and agencies having jurisdiction over the activities of the Contractor.
- B. No claims, as provided by law, have been filed against the retained percentage.
- C. Affidavit of Wages Paid is on file with the Owner for the Contractor and all Subcontractors.
- D. All contract work is complete in every respect, including operations and maintenance manuals, as-built drawings, etc.

21. INDEMNIFICATION

The Contractor agrees to protect, indemnify, and hold harmless the Owner, Engineer and their employees, agents, and staff, from all claims, liabilities, damages, expenses, or rights of action, directly or indirectly attributable to the Contractor's activities in connection with this contract, except for the sole negligence of the Owner or Engineer as outlined in Section WSDOT 1-07.14.

22. RECORD DRAWINGS

Before receiving payment for more than 90% of the work or declaring physical completion of the work, the Contractor will provide the Owner with accurate record information of all construction activity for the entire project (red line drawing on a full-size print). This red line drawing shall include, but not be limited to, any changes to the project and the exact location of all constructed utilities and any other existing utilities discovered during construction that are not identified on existing record information. The red line drawing shall be based on accurate field measurements tied to project benchmarks. The Owner will use this information to prepare Record Drawings. The cost for furnishing this record information shall be considered incidental to the entire project and shall be included in the contract price.

23. BARRIER REQUIREMENTS

During construction, the Contractor shall always maintain satisfactory and substantial temporary fencing, railing, barricades or steel plates at all openings, obstructions or other hazards. All such barriers shall have warning signs or lights as necessary for safety. Safe access to and protection of the construction site and the Contractor's records shall be maintained always.

24. CONTROL OF WORK

The presence or absence of an Inspector at the job site will be at the sole discretion of the Owner and such presence, or absence, of an Inspector will not relieve the Contractor of his responsibility to obtain the construction results specified in the Contract Documents. The Owner, inspector and engineer do not purport to be Safety Engineers and are not engaged in that capacity and shall have neither authority nor responsibility to enforce construction safety laws, rules, regulations, procedures or the safety of persons on and about the construction site. Any personal assistance which an Inspector may give the Contractor will not be construed as the basis of any assumption of responsibility in any manner, financial or otherwise, by the Owner, Inspector, or the Engineer. The Inspector is on site to insure the project is completed in accordance with all plans and specification, to insure the Owner is getting what is required. He is not there to do the Contractor's scheduling or contact his subs or deliver messages.

25. BLASTING

Blasting is not anticipated and will not be permitted without expressed written consent of the Owner. If blasting is permitted, contractor is responsible for obtaining all necessary permits and insurance.

26. INSURANCE

The Contractor shall take out and maintain during the life of this contract Public Liability Insurance for bodily injury and property damage liability including without limitation, coverage for explosion, blasting, collapse and destruction of underground utilities (X.C.U.) and contingent liability, including products and completed operations and blanket contractual liability, as shall protect the Contractor, the Owner and the Engineer. The Contractor shall have the Owner and the Engineer specifically added as additional named insured in said policies (on Form B), all at no cost to the Owner or the Engineer. The above insurance shall cover the Owner, the Engineer, Contractor and Subcontractors for claims or damages for bodily injury, including wrongful death, as well as other claims for property damage which may arise from operations under this contract whether such operations be by themselves or by any subcontractor or anyone directly or indirectly employed by either of them. The Contractor agrees, in addition, to indemnify and save harmless the Owner and

Engineer, either or both, from all suits, claims, demands, judgements, and attorney's fees, expenses or losses occasioned by the performance of this Contract by the Contractor or Subcontractor or persons working directly or indirectly for the Contractor or Subcontractor, or on account of or in consequence of any act or omission of any such person, including but not limited to neglect in safeguarding the work, or failure to conform with the safety standards for construction work adopted by the Safety Division of the Department of Labor and Industry of the State of Washington.

The amount of such insurance shall be as follows:

Bodily injury liability insurance in an amount not less than \$1,000,000.00 for injuries, including wrongful death, to any one person and subject to the same limit for each person, in an amount not less than \$2,000,000.00 on account of any one occurrence, and property damage liability insurance in an amount not less than \$1,000,000.00 for each occurrence. Builders Risk (All Risk Insurance) coverage equal to project bid amount.

The Contractor shall not cause any policy to be canceled or permit it to lapse, and all policies shall include a clause to the effect that the policy or certificate shall not be subject to cancellation or to a reduction in the required limits of liability or amounts of insurance or any other material change until notice has been mailed to the Engineer and Owner stating when, not less than thirty (30) days thereafter, such cancellation or reduction or change shall be effective. In the event notice of cancellation is received by the Owner, the Contractor shall immediately obtain other comparable insurance acceptable to the Owner and provide proof thereof to the Owner. In the event the Contractor is unable to obtain and provide such insurance, the Contractor shall immediately cease all work on the project, save and except that which is necessary to secure the site and prevent injury.

All certificates of insurance, authenticated by the proper officer of the insurer, shall state in particular those insured, the extent of the insurance, the location and operations to which the insurance applies, the expiration date, and the above-mentioned notice of cancellation clause.

Provided, however, the Owner may accept insurance covering a Subcontractor in character and amounts less than the standard requirements set forth under this subsection where such standard requirements appear excessive because of the character or extent of the work to be performed by such subcontractor.

A Certificate of Insurance evidencing coverage and a copy of the endorsement naming the Owner and Engineer as additional insured must be submitted to the Engineer prior to the commencement of the Contract in accordance with WSDOT Section 1-03.3.

The following endorsement for additional insured shall be included in all applicable policies and on the Certificate of Insurance:

The Owner and Engineer are additional named insured for all coverages provided by the policy of insurance and shall be fully and completely protected from all claims and risks by this policy and for any and every injury, death, damage, and/or loss of any sort whatsoever, including consequential damages, sustained by any person, organization or corporation in connection with any activity performed by the Contractor or any subcontractors or by anyone directly or indirectly by virtue of the provisions of that contract between the (Owner name), as Owner and (Contractor's name), entitled (Project Title), dated (date).

The coverages provided by this policy to the Owner or any other named insured shall not be terminated, reduced, or otherwise modified in any respect without providing at least 30 days prior written notice by certified mail to the Owner and other additional named insured. The coverages provided by this policy are primary to any insurance maintained by the Owner.

Third-Party Beneficiary: All parties agree that the State of Washington shall be, and is hereby, named as an express third-party beneficiary of this contract, with full rights as such.

27. CHANGES

The Owner reserves the right to make changes in the work within the general scope of the Contract Documents at any time during the progress of the work. The Contractor shall perform all work in accordance with the changes specified by the Owner.

Changes required by the Owner may include but are not limited to:

- (a) Deletion of any portion of the work.
- (b) Increases or decreases in quantities.
- (c) Changes in specifications and/or designs.
- (d) Method or manner of performance of the work.
- (e) Addition of any new work.
- (f) Acceleration or delay in the performance of the work.

The Owner shall have the option of paying for such changes by one or more of the following methods:

- (1) by the lump sum or unit contract prices set forth in the Proposal;
- (2) by equitable adjustment mutually agreed upon by the Contractor and the Owner; or
- (3) by Force Account in accordance with WSDOT Section 1-09.6

In the case that the Contractor and the Owner are unable to agree on the amount of equitable adjustment, the Owner will unilaterally determine the amount to be paid for the change in accordance with WSDOT Section 1-09.4. The Owner's decision concerning such amount to be paid shall be final as provided in WSDOT Section 1-05.1.

All administrative costs associated with change orders shall be considered to be part of the Contractor's overhead for the work as bid and not a direct cost of the change. Such administrative costs shall include, but not be limited to, costs of defining changed work, determining estimated cost of changed work, preparing proposals for change orders and negotiation of the method and amount of compensation for changed work.

The compensation for each change shall include all direct and indirect costs including, but not limited to, costs of impacts on related and indirect operations and of delay or acceleration of other work resulting from the change. Failure of the Contractor to identify all direct and indirect costs at the time of negotiation of compensation for each changed shall preclude subsequent claim, after formal execution of a change order, by the Contractor for any additional costs associated with the change.

No payment for extra work or any other change in the contract will be made unless the extra work or change has been authorized by the Owner prior to start of the extra work by the Contractor.

For (a) Deletion of any portion of the work, above, the following requirements shall apply:

No payment will be made for items which are deleted from the contract and not performed. No payment will be made for any anticipated profits which would have been earned on work deleted. Payment for costs incurred by the Contractor prior to the deletion of the work shall include and be limited to actual documented costs of field labor, equipment and materials and shall not cover and include overhead as defined in WSDOT Section 1-09.6.

Acceptable materials ordered by the Contractor or delivered on the work prior to the date of cancellation of the work will be either purchased from the Contractor by the Owner at the actual cost and shall become property of the Owner or the Owner will reimburse the Contractor for his actual costs connected with returning these materials to the suppliers.

For (b) Increases or decreases in quantities, above, the following requirements apply:

Payment for all bid items shall be at the unit prices bid, regardless of the actual final quantities of the bid items incorporated into the work and regardless of any increase or decrease from the quantities designated in the Schedule of Contract Prices.

No extra or additional payment will be made for any increase in quantity of any bid item. No extra or additional payment will be made for any decrease in quantity of any bid item. No payment will be made for any anticipated profits which would have been earned on deleted quantities.

For (c) Changes in specifications and/or designs; (d) Addition of any new work; and (e) Acceleration or delay in the performance of the work above, the following requirements shall apply:

If the Engineer determines that the above changes cause an increase or decrease in the Contractor's cost of performance of that portion of the work associated with the change and/or an increase or decrease in the contract time required for performance of the work, the increase or decrease in compensation and/or contract time will be determined by agreement of the parties.

28. INCREASED OR DECREASED QUANTITIES

The Contractor shall not purchase or place orders for full quantities of materials until the work has advanced to a state permitting the determination of the exact quantities required. The original bid item quantities designated on the Proposal and other estimates of quantities of materials furnished by the Engineer shall be considered as approximate and not indicative of the actual quantities required. The Owner will not be responsible for any materials purchased in excess of actual requirements and will not be responsible for any increased costs or extra expense that the Contractor may have on account of materials or work not being ordered at some earlier date.

29. SALES TAX

The work is within the Eastsound Sewer and Water District. The Contractor shall correctly reference on payments of sales tax to the Washington Department of Revenue EASTSOUND's tax code.

30. GUARANTEES

Except where special longer warranties are required, the Contractor shall guarantee all materials and workmanship for a period of one year from the date of Substantial Completion of the project.

Neither final acceptance by the Owner nor partial and final payment nor any provision in the Contract Documents shall relieve the Contractor of responsibility for faulty materials or workmanship.

If, prior to the expiration of one year after the date of the District's acceptance of all work or such longer period of time as may be prescribed by law or by the terms of any applicable special guarantee required by the Contract Documents, any work is found to be defective or not in compliance with the Contract Documents, the Contractor shall promptly, without cost to Owner, either correct such work, or, if it has been rejected by Owner, remove and replace it with acceptable work. If the Contractor does not promptly comply with the notification issued by the Owner for correction of defective and/or non-complying work and have the defect completely repaired within 30 calendar days, the Owner may have the work corrected or removed and replaced and all direct and indirect costs of such removal and replacement, including costs of all professional services, shall be paid by Contractor.

The guarantee shall apply to all elements and parts of the work, regardless of knowledge by the

Owner, engineer and inspector(s) of defects or deficiencies and regardless of failure of the Owner, Engineer and/or inspector(s) to inform the Contractor of known or suspected defects or deficiencies prior to Substantial Completion of the work by the Owner.

All subcontractor's, manufacturers', and suppliers' warranties and guarantees, express or implied, for any part of the work, materials and equipment shall be deemed obtained and shall be enforced by the Contractor for the benefit of the Owner without the necessity of formal transfer or assignment thereof. Warranties and guarantees by subcontractors, manufacturers, and suppliers shall begin on and extend for one year after the date of Substantial Completion of all work.

All work (including materials and equipment) repaired or replaced in accordance with this Section shall be guaranteed for a period of one year after the date of District's acceptance of the repair/replacement work.

31. ECOLOGY CLAUSE

It is anticipated that this project will be funded in part by the Washington State Department of Ecology. Neither the State of Washington nor any of its departments or employees are, or shall be, a party to this contract or any subcontract.

32. THIRD-PARTY BENEFICIARY CLAUSE

All parties agree that the State of Washington shall be, and is hereby, named as an express third-party beneficiary of this contract, with full rights as such.

33. ENVIRONMENTAL PROTECTION CLAUSE

Protection of the Environment: No construction related activity shall contribute to the degradation of the environment, allow material to enter surface or ground waters, or allow particulate emissions to the atmosphere, which exceeds State or Federal standards. Any actions that potentially allow a discharge to State waters must have prior approval of the Washington State Department of Ecology.

34. DAVIS BACON CLAUSE

The Successful bidder will be required to conform to the wage requirements prescribed by the federal Davis-Bacon and Related Acts which requires that all laborers and mechanics employed by contractors and subcontractors performing on contracts funded in whole or in part by SRF appropriations in excess of \$2,000 pay their laborers and mechanics not less than the prevailing wage rates and fringe benefits, as determined by the Secretary of Labor, for corresponding classes of laborers and mechanics employed on similar projects in the area.

35. PREVAILING WAGE RATE CLARIFICATION

All work performed on this project will be subject to the higher of the prevailing state or federal wage rates.



WASHINGTON STATE DEPARTMENT OF ECOLOGY

WATER POLLUTION CONTROL REVOLVING FUND

SPECIFICATIONS INSERT

Revised 1/22/21

The following clauses will be incorporated into construction contracts receiving financial assistance from the Washington State Department of Ecology Water Pollution Control Revolving Fund. In the event of conflict within the contract these clauses shall take precedence

REQUIRED BID SUBMITTALS

The following submittals are required to be submitted with the bid proposal:

- Certification Of Nonsegregated Facilities (attachment 3)
- DBE Subcontractor Utilization Form (EPA Form 6100-4)
- One copy of DBE Subcontractor Performance Form (EPA Form 6100-3) for each DBE subcontractor.
- Complete Bidders List.

COMPLIANCE WITH STATE AND LOCAL LAWS

The Contractor shall assure compliance with all applicable federal, state, and local laws, requirements, and ordinances as they pertain to the design, implementation, and administration of the approved project.

STATE INTEREST EXCLUSION

It is anticipated that this project will be funded in part by the Washington State Department of Ecology . Neither the State Of Washington nor any of its departments or employees are, or shall be, a party to this contract or any subcontract.

THIRD PARTY BENEFICIARY

Partial funding of this project is being provided through the Washington State Department of Ecology Water Pollution Control Revolving Fund. All parties agree that the State of Washington shall be, and is hereby, named as an express third-party beneficiary of this contract, with full rights as such.

ACCESS TO THE CONSTRUCTION SITE AND TO RECORDS

The contractor shall provide for the safe access to the construction site and to the contractor's records by Washington State Department of Ecology and Environmental Protection Agency (EPA) personnel.

The Contractor shall maintain accurate records and accounts to facilitate the Owner's audit requirements and shall ensure that all subcontractors maintain auditable records.

These Project records shall be separate and distinct from the Contractor's other records and accounts.

All such records shall be available to the Owner and to Washington State Department of Ecology and EPA personnel for examination. All records pertinent to this project shall be retained by the Contractor for a period of three (3) years after the final audit.

PROTECTION OF THE ENVIRONMENT

No construction related activity shall contribute to the degradation of the environment, allow material to enter surface or ground waters, or allow particulate emissions to the atmosphere, which exceed state or federal standards. Any actions that potentially allow a discharge to state waters must have prior approval of the Washington State Department of Ecology.

FUNDING RECOGNITION

All site-specific projects must have a sign of sufficient size to be seen from nearby roadways acknowledging department financial assistance and left in place throughout the life of the project. Department logos must be on all signs and documents. Logos will be provided as needed.

INADVERTENT DISCOVERY OF ARCHEOLOGICAL RESOURCES

The contractor shall obtain a copy of the Inadvertent Discovery Plan from the Project Owner. The contractor shall keep a copy of the inadvertent discovery plan for the project on the work site at all times. The contractor shall immediately stop all work if human remains, cultural, or archeological resources are discovered in the course of construction. The contractor shall follow the inadvertent discovery plan in dealing with the human remains, cultural, or archeological resources.

Use Of American Iron And Steel

This provision applies to projects for the construction, alteration, maintenance, or repair of a "treatment works" as defined in the Federal Water Pollution Control Act (33 USC 1381 et seq.). This provision does not apply if the engineering plans and specifications for the project were approved by the Ecology prior to January 17, 2014.

The Contractor acknowledges to and for the benefit of the Project Owner and the State of Washington that it understands the goods and services under this Agreement are being funded with monies made available by the Water Pollution Control Revolving Fund which contains provisions commonly known as "American Iron and Steel;" that requires all of the iron and steel products used in the project be produced in the United States ("American Iron and Steel Requirements") including iron and steel products provided by the Contractor pursuant to this Agreement. "Iron and Steel products" means the following products made primarily of iron or steel: lined or unlined pipes and fittings, manhole covers and other municipal castings, hydrants, tanks, flanges, pipe clamps and restraints, valves, structural steel, reinforced precast concrete, and construction materials.

The Contractor hereby represents and warrants to and for the benefit of the Project Owner and the State that:

- (a) the Contractor has reviewed and understands the American Iron and Steel Requirements,
- (b) all of the iron and steel products used in the project will be and/or have been produced in the United States in a manner that complies with the American Iron and Steel Requirements, unless a waiver of the requirements is approved, and
- (c) the Contractor will provide any further verified information, certification or assurance of compliance with this paragraph, or information necessary to support a waiver of the American Iron and Steel Requirements, as may be requested by the Project Owner or the State.

Notwithstanding any other provision of this Agreement, any failure to comply with this paragraph by the Contractor shall permit the Project Owner or State to recover as damages against the Contractor any loss, expense or cost (including without limitation attorney's fees) incurred by the Project Owner or State resulting from any such failure (including without limitation any impairment or loss of funding, whether in whole or in part, from the State or any damages owed to the State by the Project Owner). While the Contractor has no direct contractual privity with the State, as a lender to the Project Owner for the funding of its project, the Project Owner and the Contractor agree that the State is a third-party beneficiary and neither this paragraph (nor any other provision of the Agreement necessary to give this paragraph force or effect shall be amended or waived without the prior written consent of the State.

PREVAILING WAGE

The work performed under this contract is subject to the wage requirements of the Davis-Bacon Act. The Contractor shall conform to the wage requirements prescribed by the federal Davis-Bacon and Related Acts which requires that all laborers and mechanics employed by contractors and subcontractors performing on contracts funded in whole or in part by SRF appropriations in excess of \$2000 pay their laborers and mechanics not less than the prevailing wage rates and fringe benefits, and determined by the Secretary of Labor, for corresponding classes of laborers and mechanics employed on similar projects in the area. Attachment 1 to this specification insert and an up to date wage determination shall be included in full into this contract and in any subcontract in excess of \$2,000. Wage determinations can be found at <http://www.wdol.gov>.

The Contractor agrees that the Contractor is legally and financially responsible for compliance with the Davis-Bacon Act wage rules. All laborers and mechanics employed by contractors and subcontractors employed as part of this contract shall be paid wages at rates not less than those prevailing on projects of a character similar in the locality as determined by the Secretary of Labor in accordance with subchapter IV of chapter 31 of title 40, United States Code.

CERTIFICATION REGARDING SUSPENSION, DEBARMENT, INELIGIBILITY OR VOLUNTARY EXCLUSION

1. The CONTRACTOR, by signing this agreement, certifies that it is not suspended, debarred, proposed for debarment, declared ineligible or otherwise excluded from contracting with the federal government, or from receiving contracts paid for with federal funds. If the CONTRACTOR is unable to certify to the statements contained in the certification, they must provide an explanation as to why they cannot.
2. The CONTRACTOR shall provide immediate written notice to the Department if at any time the CONTRACTOR learns that its certification was erroneous when submitted or had become

erroneous by reason of changed circumstances.

3. The terms covered transaction, debarred, suspended, ineligible, lower tier covered transaction, participant, person, primary covered transaction, principal, proposal, and voluntarily excluded, as used in this clause, have the meaning set out in the Definitions and Coverage sections of rules implementing Executive Order 12549. You may contact the department for assistance in obtaining a copy of those regulations..
4. The CONTRACTOR agrees it shall not knowingly enter into any lower tier covered transaction with a person who is proposed for debarment under the applicable Code of Federal Regulations, debarred, suspended, declared ineligible, or voluntarily excluded from participation in this covered transaction.
5. The CONTRACTOR further agrees by signing this agreement, that it will include this clause titled "Certification Regarding Suspension, Debarment, Ineligibility Or Voluntary Exclusion" without modification in all lower tier covered transactions and in all solicitations for lower tier covered transactions.
6. Pursuant to 2CFR180.330, the CONTRACTOR is responsible for ensuring that any lower tier covered transaction complies with certification of suspension and debarment requirements.
7. CONTRACTOR acknowledges that failing to disclose the information required in the Code of Federal Regulations may result in the delay or negation of this funding agreement, or pursuance of legal remedies, including suspension and debarment.
8. CONTRACTOR agrees to keep proof in its agreement file, that it, and all lower tier recipients or contractors, are not suspended or debarred, and will make this proof available to the Department upon request. RECIPIENT/CONTRACTOR must run a search in <http://www.sam.gov/> and print a copy of completed searches to document proof of compliance.

This term and condition supersedes EPA Form 5700-49, "Certification Regarding Debarment, Suspension, and Other Responsibility Matters."

DISADVANTAGED BUSINESS ENTERPRISES

General Compliance (40 CFR Part 33).

The contractor shall comply with the requirements of the Environmental Protection Agency's Program for Participation By Disadvantaged Business Enterprises (DBE) 40 CFR Part 33.

Non-discrimination Provision (40CFR Appendix A to Part 33).

The contractor shall not discriminate on the basis of race, color, national origin or sex in the performance of this contract. The contractor shall carry out applicable requirements of 40 CFR part 33 in the award and administration of contracts awarded under EPA financial assistance agreements. Failure by the contractor to carry out these requirements is a material breach of this contract which may result in the termination of this contract or other legally available remedies.

The contractor shall comply with all federal and state nondiscrimination laws, including, but not limited to Title VI and VII of the Civil Rights Act of 1964, Section 504 of the Rehabilitation Act of 1973, Title IX of the Education Amendments of 1972, the Age Discrimination Act of 1975, and Chapter 49.60 RCW, Washington's Law Against Discrimination, and 42 U.S.C. 12101 et seq, the Americans with Disabilities Act (ADA).

Six Good Faith Efforts (40 CFR Part 33 Subpart C).

The contractor agrees to make the following good faith efforts whenever procuring subcontracts, equipment, services and supplies. The contractor shall retain records documenting compliance with the following six good faith efforts.

1. Ensuring Disadvantaged Business Enterprises are made aware of contracting opportunities to the fullest extent practicable through outreach and recruitment activities. For Indian Tribal, State and Local and Government recipients, this will include placing Disadvantaged Business Enterprises on solicitation lists and soliciting them whenever they are potential sources. Qualified Women and Minority business enterprises may be found on the Internet at www.omwbe.wa.gov or by contacting the Washington State Office of Minority and Women's Enterprises at (866) 208-1064.
2. Making information on forthcoming opportunities available to Disadvantaged Business Enterprises and arrange time frames for contracts and establish delivery schedules, where the requirements permit, in a way that encourages and facilitates participation by Disadvantaged Business Enterprises in the competitive process. This includes, whenever possible, posting solicitations for bids or proposals for a minimum of thirty (30) calendar days before the bid or proposal closing date.
3. Considering in the contracting process whether firms competing for large contracts could subcontract with Disadvantaged Business Enterprises. For Indian Tribal, State and local Government recipients, this will include dividing total requirements when economically feasible into smaller tasks or quantities to permit maximum participation by Disadvantaged Business Enterprises in the competitive process.
4. Encourage contracting with a consortium of Disadvantaged Business Enterprises when a contract is too large for one of these firms to handle individually.
5. Using services and assistance of the Small Business Administration and the Minority Business Development Agency of the Department of Commerce.
6. If the prime contractor awards subcontracts, requiring the subcontractors to take the six good faith efforts in paragraphs 1 through 5 above.

MBE/WBE Reporting (40 CFR Part 33 Parts 33.302, 33.502 and 33.503).

1. The contractor shall complete the DBE Subcontractor Utilization Form (EPA Form 6100-4).
2. The contractor shall require all DBE subcontractors to complete the DBE Subcontractor Performance Form (EPA Form 6100-3). The DBE Subcontractor Performance Form is only required to be completed by certified DBE subcontractors.
3. The contractor shall submit DBE Subcontractor Utilization Form (EPA Form 6100-4) and all completed DBE Subcontractor Performance Form(s) (EPA Form 6100-3) as part of the bid, or within one hour after the published bid submittal time (consistent with RCW 39.30.060)
4. The contractor shall provide DBE Subcontractor Participation Form (EPA Form 6100-2) to all DBE subcontractors. These subcontractors may submit Subcontractor Participation Form (EPA Form 6100-2) to the EPA Region 10 DBE coordinator in order to document issues or concerns with their usage or payment for a subcontract.

The 6100 forms can be found at:

<https://ecology.wa.gov/About-us/How-we-operate/Grants-loans/Find-a-grant-or-loan/Water-Quality-grants-and-loans/Facility-project-resources>

Bidders List (40 CFR Part 33 part 33.501)

All bidders shall submit the following information for all firms that bid or quote on subcontracts (including both DBE and non-DBE firms) as part of the bid, or within one hour after the published bid submittal time (consistent with RCW 39.30.060).

1. Firm's name with point of contact;

2. Firm's mailing address, telephone number, and e-mail address;
3. The work on which the firm bid or quoted, and when the firm bid or quoted; and
4. Firm's status as an MBE/WBE or non-MBE/WBE.

Contract Administration Provisions (40 CFR part 33.302).

The contractor shall comply with the contract administration provisions of 40 CFR, Part 33.302.

1. The contractor shall pay its subcontractor for satisfactory performance no more than 30 days from the contractor's receipt of payment.
2. The contractor shall notify the owner in writing prior to any termination of a DBE subcontractor.
3. If a DBE subcontractor fails to complete work under the subcontract for any reason, the contractor shall employ the six good faith efforts when soliciting a replacement subcontractor.
4. The contractor shall employ the six good faith efforts even if the contractor has achieved its fair share objectives.

EQUAL OPPORTUNITY (EEO)

If this Contract exceeds \$10,000, the Contractor shall comply with Executive Order 11246, "Equal Employment Opportunity," as amended by Executive Order 11375, "Amending Executive Order 11246 Relating to Equal Employment Opportunity," and as supplemented by regulations at 41 CFR part 60.

Contractor's compliance with Executive Order 11246 shall be based on implementation of the Equal Opportunity Clause, and specific affirmative active obligations required by the Standard Federal Equal Employment Opportunity Construction Contract Specifications, as set forth in 41 CFR Part 60-4.

Equal Opportunity Clause (41 CFR part 60-1.4(b))

During the performance of this contract, the contractor agrees as follows:

1. The contractor will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin. The contractor will take affirmative action to ensure that applicants are employed, and that employees are treated during employment without regard to their race, color, religion, sex, or national origin. Such action shall include, but not be limited to the following: Employment, upgrading, demotion, or transfer; recruitment or recruitment advertising; layoff or termination; rates of pay or other forms of compensation; and selection for training, including apprenticeship. The contractor agrees to post in conspicuous places, available to employees and applicants for employment, notices to be provided setting forth the provisions of this nondiscrimination clause.
2. The contractor will, in all solicitations or advertisements for employees placed by or on behalf of the contractor, state that all qualified applicants will receive considerations for employment without regard to race, color, religion, sex, or national origin.
3. The contractor will send to each labor union or representative of workers with which he has a collective bargaining agreement or other contract or understanding, a notice to be provided advising the said labor union or workers' representatives of the contractor's commitments under this section, and shall post copies of the notice in conspicuous places available to employees and applicants for employment.
4. The contractor will comply with all provisions of Executive Order 11246 of September 24, 1965, and of the rules, regulations, and relevant orders of the Secretary of Labor.
5. The contractor will furnish all information and reports required by Executive Order 11246 of September 24, 1965, and by rules, regulations, and orders of the Secretary of Labor, or pursuant thereto, and will permit access to his books, records, and accounts by the administering agency and the Secretary of Labor for purposes of investigation to ascertain compliance with such rules, regulations, and orders.
6. In the event of the contractor's noncompliance with the nondiscrimination clauses of this contract

or with any of the said rules, regulations, or orders, this contract may be canceled, terminated, or suspended in whole or in part and the contractor may be declared ineligible for further Government contracts or federally assisted construction contracts in accordance with procedures authorized in Executive Order 11246 of September 24, 1965, and such other sanctions may be imposed and remedies invoked as provided in Executive Order 11246 of September 24, 1965, or by rule, regulation, or order of the Secretary of Labor, or as otherwise provided by law.

7. The contractor will include the portion of the sentence immediately preceding paragraph (1) and the provisions of paragraphs (1) through (7) in every subcontract or purchase order unless exempted by rules, regulations, or orders of the Secretary of Labor issued pursuant to section 204 of Executive Order 11246 of September 24, 1965, so that such provisions will be binding upon each subcontractor or vendor. The contractor will take such action with respect to any subcontract or purchase order as the administering agency may direct as a means of enforcing such provisions, including sanctions for noncompliance: Provided, however, That in the event a contractor becomes involved in, or is threatened with, litigation with a subcontractor or vendor as a result of such direction by the administering agency the contractor may request the United States to enter into such litigation to protect the interests of the United States.

Federal Equal Employment Opportunity Construction Contract Specifications

(Executive Order 11246 and 41 CFR part 60-4.3)

1. As used in these specifications:
 - a. "Covered area" means the geographical area described in the solicitation from which this contract resulted;
 - b. "Director" means Director, Office of Federal Contract Compliance Programs, United States Department of Labor, or any person to whom the Director delegates authority;
 - c. "Employer identification number" means the Federal Social Security number used on the Employer's Quarterly Federal Tax Return, U.S. Treasury Department Form 941.
 - d. "Minority" includes:
 - i. Black (all persons having origins in any of the Black African racial groups not of Hispanic origin);
 - ii. Hispanic (all persons of Mexican, Puerto Rican, Cuban, Central or South American or other Spanish Culture or origin, regardless of race);
 - iii. Asian and Pacific Islander (all persons having origins in any of the original peoples of the Far East, Southeast Asia, the Indian Subcontinent, or the Pacific Islands); and
 - iv. American Indian or Alaskan Native (all persons having origins in any of the original peoples of North America and maintaining identifiable tribal affiliations through membership and participation or community identification).
2. Whenever the Contractor, or any Subcontractor at any tier, subcontracts a portion of the work involving any construction trade, it shall physically include in each subcontract in excess of \$10,000 the provisions of these specifications and the Notice which contains the applicable goals for minority and female participation and which is set forth in the solicitations from which this contract resulted.
3. If the Contractor is participating (pursuant to 41 CFR 60-4.5) in a Hometown Plan approved by the U.S. Department of Labor in the covered area either individually or through an association, its affirmative action obligations on all work in the Plan area (including goals and timetables) shall be in accordance with that Plan for those trades which have unions participating in the Plan. Contractors must be able to demonstrate their participation in and compliance with the provisions of any such Hometown Plan. Each Contractor or Subcontractor participating in an approved Plan is individually required to comply with its obligations under the EEO clause, and to make a good faith effort to achieve each goal under the Plan in each trade in which it has employees. The overall good faith performance by other Contractors or Subcontractors toward a goal in an approved Plan does not excuse any covered Contractor's or Subcontractor's failure to take good faith efforts to achieve the Plan goals and timetables.
4. The Contractor shall implement the specific affirmative action standards provided in paragraphs 7 a through p of these specifications. The goals set forth in the solicitation from which this contract resulted are expressed as percentages of the total hours of employment and training of minority and female utilization the Contractor should reasonably be able to achieve in each construction trade in which it has employees in the covered area. Covered Construction contractors

performing construction work in geographical areas where they do not have a Federal or federally assisted construction contract shall apply the minority and female goals established for the geographical area where the work is being performed. Goals are published periodically in the Federal Register in notice form, and such notices may be obtained from any Office of Federal Contract Compliance Programs office or from Federal procurement contracting officers. The Contractor is expected to make substantially uniform progress in meeting its goals in each craft during the period specified.

5. Neither the provisions of any collective bargaining agreement, nor the failure by a union with whom the Contractor has a collective bargaining agreement, to refer either minorities or women shall excuse the Contractor's obligations under these specifications, Executive Order 11246, or the regulations promulgated pursuant thereto.
6. In order for the nonworking training hours of apprentices and trainees to be counted in meeting the goals, such apprentices and trainees must be employed by the Contractor during the training period, and the Contractor must have made a commitment to employ the apprentices and trainees at the completion of their training, subject to the availability of employment opportunities. Trainees must be trained pursuant to training programs approved by the U.S. Department of Labor.
7. The Contractor shall take specific affirmative actions to ensure equal employment opportunity. The evaluation of the Contractor's compliance with these specifications shall be based upon its effort to achieve maximum results from its actions. The Contractor shall document these efforts fully, and shall implement affirmative action steps at least as extensive as the following:
 - a. Ensure and maintain a working environment free of harassment, intimidation, and coercion at all sites, and in all facilities at which the Contractor's employees are assigned to work. The Contractor, where possible, will assign two or more women to each construction project. The Contractor shall specifically ensure that all foremen, superintendents, and other on-site supervisory personnel are aware of and carry out the Contractor's obligation to maintain such a working environment, with specific attention to minority or female individuals working at such sites or in such facilities.
 - b. Establish and maintain a current list of minority and female recruitment sources, provide written notification to minority and female recruitment sources and to community organizations when the Contractor or its unions have employment opportunities available, and maintain a record of the organizations' responses.
 - c. Maintain a current file of the names, addresses and telephone numbers of each minority and female off-the-street applicant and minority or female referral from a union, a recruitment source or community organization and of what action was taken with respect to each such individual. If such individual was sent to the union hiring hall for referral and was not referred back to the Contractor by the union or, if referred, not employed by the Contractor, this shall be documented in the file with the reason therefor, along with whatever additional actions the Contractor may have taken.
 - d. Provide immediate written notification to the Director when the union or unions with which the Contractor has a collective bargaining agreement has not referred to the Contractor a minority person or woman sent by the Contractor, or when the Contractor has other information that the union referral process has impeded the Contractor's efforts to meet its obligations.
 - e. Develop on-the-job training opportunities and/or participate in training programs for the area which expressly include minorities and women, including upgrading programs and apprenticeship and trainee programs relevant to the Contractor's employment needs, especially those programs funded or approved by the Department of Labor. The Contractor shall provide notice of these programs to the sources compiled under 7b above.
 - f. Disseminate the Contractor's EEO policy by providing notice of the policy to unions and training programs and requesting their cooperation in assisting the Contractor in meeting its EEO obligations; by including it in any policy manual and collective bargaining agreement; by publicizing it in the company newspaper, annual report, etc.; by specific review of the policy with all management personnel and with all minority and female employees at least once a year; and by posting the company EEO policy on bulletin boards accessible to all employees at each location where construction work is performed.
 - g. Review, at least annually, the company's EEO policy and affirmative action obligations under these specifications with all employees having any responsibility for hiring, assignment, layoff, termination or other employment decisions including specific review of these items with on-site supervisory personnel such as Superintendents, General Foremen, etc., prior to the

- initiation of construction work at any job site. A written record shall be made and maintained identifying the time and place of these meetings, persons attending, subject matter discussed, and disposition of the subject matter.
- h. Disseminate the Contractor's EEO policy externally by including it in any advertising in the news media, specifically including minority and female news media, and providing written notification to and discussing the Contractor's EEO policy with other Contractors and Subcontractors with whom the Contractor does or anticipates doing business.
 - i. Direct its recruitment efforts, both oral and written, to minority, female and community organizations, to schools with minority and female students and to minority and female recruitment and training organizations serving the Contractor's recruitment area and employment needs. Not later than one month prior to the date for the acceptance of applications for apprenticeship or other training by any recruitment source, the Contractor shall send written notification to organizations such as the above, describing the openings, screening procedures, and tests to be used in the selection process.
 - j. Encourage present minority and female employees to recruit other minority persons and women and, where reasonable, provide after school, summer and vacation employment to minority and female youth both on the site and in other areas of a Contractor's work force.
 - k. Validate all tests and other selection requirements where there is an obligation to do so under 41 CFR part 60-3.
 - l. Conduct, at least annually, an inventory and evaluation at least of all minority and female personnel for promotional opportunities and encourage these employees to seek or to prepare for, through appropriate training, etc., such opportunities.
 - m. Ensure that seniority practices, job classifications, work assignments and other personnel practices, do not have a discriminatory effect by continually monitoring all personnel and employment related activities to ensure that the EEO policy and the Contractor's obligations under these specifications are being carried out.
 - n. Ensure that all facilities and company activities are nonsegregated except that separate or single-user toilet and necessary changing facilities shall be provided to assure privacy between the sexes.
 - o. Document and maintain a record of all solicitations of offers for subcontracts from minority and female construction contractors and suppliers, including circulation of solicitations to minority and female contractor associations and other business associations.
 - p. Conduct a review, at least annually, of all supervisors' adherence to and performance under the Contractor's EEO policies and affirmative action obligations.
8. Contractors are encouraged to participate in voluntary associations which assist in fulfilling one or more of their affirmative action obligations (7a through p). The efforts of a contractor association, joint contractor-union, contractor-community, or other similar group of which the contractor is a member and participant, may be asserted as fulfilling any one or more of its obligations under 7a through p of these Specifications provided that the contractor actively participates in the group, makes every effort to assure that the group has a positive impact on the employment of minorities and women in the industry, ensures that the concrete benefits of the program are reflected in the Contractor's minority and female workforce participation, makes a good faith effort to meet its individual goals and timetables, and can provide access to documentation which demonstrates the effectiveness of actions taken on behalf of the Contractor. The obligation to comply, however, is the Contractor's and failure of such a group to fulfill an obligation shall not be a defense for the Contractor's noncompliance.
9. A single goal for minorities and a separate single goal for women have been established. The Contractor, however, is required to provide equal employment opportunity and to take affirmative action for all minority groups, both male and female, and all women, both minority and non-minority. Consequently, the Contractor may be in violation of the Executive Order if a particular group is employed in a substantially disparate manner (for example, even though the Contractor has achieved its goals for women generally, the Contractor may be in violation of the Executive Order if a specific minority group of women is underutilized).
10. The Contractor shall not use the goals and timetables or affirmative action standards to discriminate against any person because of race, color, religion, sex, or national origin.
11. The Contractor shall not enter into any Subcontract with any person or firm debarred from Government contracts pursuant to Executive Order 11246.
12. The Contractor shall carry out such sanctions and penalties for violation of these specifications and of the Equal Opportunity Clause, including suspension, termination and cancellation of existing subcontracts as may be imposed or ordered pursuant to Executive Order 11246, as

amended, and its implementing regulations, by the Office of Federal Contract Compliance Programs. Any Contractor who fails to carry out such sanctions and penalties shall be in violation of these specifications and Executive Order 11246, as amended.

13. The Contractor, in fulfilling its obligations under these specifications, shall implement specific affirmative action steps, at least as extensive as those standards prescribed in paragraph 7 of these specifications, so as to achieve maximum results from its efforts to ensure equal employment opportunity. If the Contractor fails to comply with the requirements of the Executive Order, the implementing regulations, or these specifications, the Director shall proceed in accordance with 41 CFR 60-4.8.
14. The Contractor shall designate a responsible official to monitor all employment related activity to ensure that the company EEO policy is being carried out, to submit reports relating to the provisions hereof as may be required by the Government and to keep records. Records shall at least include for each employee the name, address, telephone numbers, construction trade, union affiliation if any, employee identification number when assigned, social security number, race, sex, status (e.g., mechanic, apprentice trainee, helper, or laborer), dates of changes in status, hours worked per week in the indicated trade, rate of pay, and locations at which the work was performed. Records shall be maintained in an easily understandable and retrievable form; however, to the degree that existing records satisfy this requirement, contractors shall not be required to maintain separate records.
15. Nothing herein provided shall be construed as a limitation upon the application of other laws which establish different standards of compliance or upon the application of requirements for the hiring of local or other area residents (e.g., those under the Public Works Employment Act of 1977 and the Community Development Block Grant Program).

Reporting Requirements (EEO-1)

On or before September 30 of each year, a contractor that is subject to Title VII of the Civil Rights Act of 1964, as amended, and that has 100 or more employees, shall file with the EEOC or its delegate an "Employer Information Report EEO-1". Instructions on how to file are available on the EEOC's website at <http://www.eeoc.gov/employers/eeo1survey/howtofile.cfm>. The contractor shall retain a copy of the most recent report filed.

Segregated Facilities (41 CFR part 60-1.8)

The contractor shall ensure that facilities provided for employees are provided in such a manner that segregation on the basis of race, color, religion, sex or national origin cannot result. The contractor may neither require such segregated use by written or oral policies nor tolerate such use by employee custom. The contractor's obligation extends further to ensuring that its employees are not assigned to perform their services at any location, under the contractor's control, where the facilities are segregated. This obligation extends to all contracts containing the equal opportunity clause regardless of the amount of the contract. The term "facilities," as used in this section, means waiting rooms, work areas, restaurants and other eating areas, time clocks, restrooms, wash rooms, locker rooms, and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing provided for employees; Provided, That separate or single-user restrooms and necessary dressing or sleeping areas shall be provided to assure privacy between the sexes.

ATTACHMENTS:

1. Wage Rate Requirements For Subrecipients
2. Current Wage Rate Determination (to be provided by project owner)
3. Certification Of Nonsegregated Facilities
4. Notice To Labor Unions Or Other Organization Of Workers: Non-Discrimination In Employment

EPA Form 6100-4, EPA Form 6100.3, EPA Form 6100-2

ATTACHMENT 1 - WAGE RATE REQUIREMENTS FOR SUBRECIPIENTS. (TO BE INCLUDED IN FULL IN ANY CONTRACT IN EXCESS OF \$2,000)

The following terms and conditions specify how recipients will assist EPA in meeting its Davis-Bacon (DB) responsibilities when DB applies to EPA awards of financial assistance under the FY 2013 Continuing Resolution with respect to State recipients and subrecipients that are governmental entities. If a subrecipient has questions regarding when DB applies, obtaining the correct DB wage determinations, DB provisions, or compliance monitoring, it may contact the State recipient. If a State recipient needs guidance, the recipient may contact Lorraine Fleury at fleury.lorraine@epa.gov or at 215-814-2341 of EPA, Region III Grants and Audit Management Branch for guidance. The recipient or subrecipient may also obtain additional guidance from DOL's web site at <http://www.dol.gov/whd/>

1. Applicability of the Davis- Bacon (DB) prevailing wage requirements.

Under the FY 2013 Appropriations Act, DB prevailing wage requirements apply to the construction, alteration, and repair of treatment works carried out in whole or in part with assistance made available by a State water pollution control revolving fund and to any construction project carried out in whole or in part by assistance made available by a drinking water treatment revolving loan fund. If a subrecipient encounters a unique situation at a site that presents uncertainties regarding DB applicability, the subrecipient must discuss the situation with the recipient State before authorizing work on that site.

2. Obtaining Wage Determinations.

(a) Subrecipients shall obtain the wage determination for the locality in which a covered activity subject to DB will take place prior to issuing requests for bids, proposals, quotes or other methods for soliciting contracts (solicitation) for activities subject to DB. These wage determinations shall be incorporated into solicitations and any subsequent contracts. Prime contracts must contain a provision requiring that subcontractors follow the wage determination incorporated into the prime contract.

(i) While the solicitation remains open, the subrecipient shall monitor www.wdol.gov weekly to ensure that the wage determination contained in the solicitation remains current. The subrecipients shall amend the solicitation if DOL issues a modification more than 10 days prior to the closing date (i.e. bid opening) for the solicitation. If DOL modifies or supersedes the applicable wage determination less than 10 days prior to the closing date, the subrecipients may request a finding from the State recipient that there is not a reasonable time to notify interested contractors of the modification of the wage determination. The State recipient will provide a report of its findings to the subrecipient.

(ii) If the subrecipient does not award the contract within 90 days of the closure of the solicitation, any modifications or supersedes DOL makes to the wage determination contained in the solicitation shall be effective unless the State recipient, at the request of the subrecipient, obtains an extension of the 90 day period from DOL pursuant to 29 CFR 1.6(c)(3)(iv). The subrecipient shall monitor www.wdol.gov on a weekly basis if it does not award the contract within 90 days of closure of the solicitation to ensure that wage determinations contained in the solicitation remain current.

(b) If the subrecipient carries out activity subject to DB by issuing a task order, work assignment or similar instrument to an existing contractor (ordering instrument) rather than by publishing a solicitation, the subrecipient shall insert the appropriate DOL wage determination from www.wdol.gov into the ordering instrument.

(c) Subrecipients shall review all subcontracts subject to DB entered into by prime contractors to verify that the prime contractor has required its subcontractors to include the applicable wage determinations.

(d) As provided in 29 CFR 1.6(f), DOL may issue a revised wage determination applicable to a subrecipient's contract after the award of a contract or the issuance of an ordering instrument if DOL determines that the subrecipient has failed to incorporate a wage determination or has used a wage determination that clearly does not apply to the contract or ordering instrument. If this occurs, the subrecipient shall either terminate the contract or ordering instrument and issue a revised solicitation or ordering instrument or incorporate DOL's wage determination retroactive to the beginning of the contract or ordering instrument by change order. The subrecipient's contractor must be compensated for any increases in wages resulting from the use of DOL's revised wage determination.

3. Contract and Subcontract provisions.

(a) The Recipient shall insure that the subrecipient(s) shall insert in full in any contract in excess of \$2,000 which is entered into for the actual construction, alteration and/or repair, including painting and decorating, of a treatment work under the CWSRF or a construction project under the DWSRF financed in whole or in part from Federal funds or in accordance with guarantees of a Federal agency or financed from funds obtained by pledge of any contract of a Federal agency to make a loan, grant or annual contribution (except where a different meaning is expressly indicated), and which is subject to the labor standards provisions of any of the acts listed in § 5.1 or the FY 2012 Appropriations Act , the following clauses:

(1) Minimum wages.

(i) All laborers and mechanics employed or working upon the site of the work will be paid unconditionally and not less often than once a week, and without subsequent deduction or rebate on any account (except such payroll deductions as are permitted by regulations issued by the Secretary of Labor under the Copeland Act (29 CFR part 3)), the full amount of wages and bona fide fringe benefits (or cash equivalents thereof) due at time of payment computed at rates not less than those contained in the wage determination of the Secretary of Labor which is attached hereto and made a part hereof, regardless of any contractual relationship which may be alleged to exist between the contractor and such laborers and mechanics.

Contributions made or costs reasonably anticipated for bona fide fringe benefits under section 1(b)(2) of the Davis-Bacon Act on behalf of laborers or mechanics are considered wages paid to such laborers or mechanics, subject to the provisions of paragraph (a)(1)(iv) of this section; also, regular contributions made or costs incurred for more than a weekly period (but not less often than quarterly) under plans, funds, or

programs which cover the particular weekly period, are deemed to be constructively made or incurred during such weekly period. Such laborers and mechanics shall be paid the appropriate wage rate and fringe benefits on the wage determination for the classification of work actually performed, without regard to skill, except as provided in § 5.5(a)(4). Laborers or mechanics performing work in more than one classification may be compensated at the rate specified for each classification for the time actually worked therein: Provided that the employer's payroll records accurately set forth the time spent in each classification in which work is performed. The wage determination (including any additional classification and wage rates conformed under paragraph (a)(1)(ii) of this section) and the Davis-Bacon poster (WH-1321) shall be posted at all times by the contractor and its subcontractors at the site of the work in a prominent and accessible place where it can be easily seen by the workers.

Subrecipients may obtain wage determinations from the U.S. Department of Labor's web site, www.wdol.gov.

(ii)(A) The subrecipient(s), on behalf of EPA, shall require that any class of laborers or mechanics, including helpers, which is not listed in the wage determination and which is to be employed under the contract shall be classified in conformance with the wage determination. The State award official shall approve a request for an additional classification and wage rate and fringe benefits therefore only when the following criteria have been met:

- (1) The work to be performed by the classification requested is not performed by a classification in the wage determination; and
- (2) The classification is utilized in the area by the construction industry; and
- (3) The proposed wage rate, including any bona fide fringe benefits, bears a reasonable relationship to the wage rates contained in the wage determination.

(B) If the contractor and the laborers and mechanics to be employed in the classification (if known), or their representatives, and the subrecipient(s) agree on the classification and wage rate (including the amount designated for fringe benefits where appropriate), documentation of the action taken and the request, including the local wage determination shall be sent by the subrecipient (s) to the State award official. The State award official will transmit the request, to the Administrator of the Wage and Hour Division, Employment Standards Administration, U.S. Department of Labor, Washington, DC 20210 and to the EPA DB Regional Coordinator concurrently. The Administrator, or an authorized representative, will approve, modify, or disapprove every additional classification request within 30 days of receipt and so advise the State award official or will notify the State award official within the 30-day period that additional time is necessary.

(C) In the event the contractor, the laborers or mechanics to be employed in the classification or their representatives, and the subrecipient(s) do not agree on the proposed classification and wage rate (including the amount designated for fringe benefits, where appropriate), the award official shall refer the request and the local

wage determination, including the views of all interested parties and the recommendation of the State award official, to the Administrator for determination. The request shall be sent to the EPA DB Regional Coordinator concurrently. The Administrator, or an authorized representative, will issue a determination within 30 days of receipt of the request and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(D) The wage rate (including fringe benefits where appropriate) determined pursuant to paragraphs (a)(1)(ii)(B) or (C) of this section, shall be paid to all workers performing work in the classification under this contract from the first day on which work is performed in the classification.

(iii) Whenever the minimum wage rate prescribed in the contract for a class of laborers or mechanics includes a fringe benefit which is not expressed as an hourly rate, the contractor shall either pay the benefit as stated in the wage determination or shall pay another bona fide fringe benefit or an hourly cash equivalent thereof.

(iv) If the contractor does not make payments to a trustee or other third person, the contractor may consider as part of the wages of any laborer or mechanic the amount of any costs reasonably anticipated in providing bona fide fringe benefits under a plan or program, Provided, That the Secretary of Labor has found, upon the written request of the contractor, that the applicable standards of the Davis-Bacon Act have been met. The Secretary of Labor may require the contractor to set aside in a separate account assets for the meeting of obligations under the plan or program.

(2) Withholding. The subrecipient(s), shall upon written request of the EPA Award Official or an authorized representative of the Department of Labor, withhold or cause to be withheld from the contractor under this contract or any other Federal contract with the same prime contractor, or any other federally-assisted contract subject to Davis-Bacon prevailing wage requirements, which is held by the same prime contractor, so much of the accrued payments or advances as may be considered necessary to pay laborers and mechanics, including apprentices, trainees, and helpers, employed by the contractor or any subcontractor the full amount of wages required by the contract. In the event of failure to pay any laborer or mechanic, including any apprentice, trainee, or helper, employed or working on the site of the work, all or part of the wages required by the contract, the (Agency) may, after written notice to the contractor, sponsor, applicant, or owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds until such violations have ceased.

(3) Payrolls and basic records.

(i) Payrolls and basic records relating thereto shall be maintained by the contractor during the course of the work and preserved for a period of three years thereafter for all laborers and mechanics working at the site of the work. Such records shall contain the name, address, and social security number of each such worker, his or her correct classification, hourly rates of wages paid (including rates of contributions or costs anticipated for bona fide fringe benefits or cash equivalents thereof of the types described in section 1(b)(2)(B) of the Davis-Bacon Act), daily and weekly number of hours worked, deductions made and actual wages paid. Whenever the Secretary of

Labor has found under 29 CFR 5.5(a)(1)(iv) that the wages of any laborer or mechanic include the amount of any costs reasonably anticipated in providing benefits under a plan or program described in section 1(b)(2)(B) of the Davis-Bacon Act, the contractor shall maintain records which show that the commitment to provide such benefits is enforceable, that the plan or program is financially responsible, and that the plan or program has been communicated in writing to the laborers or mechanics affected, and records which show the costs anticipated or the actual cost incurred in providing such benefits. Contractors employing apprentices or trainees under approved programs shall maintain written evidence of the registration of apprenticeship programs and certification of trainee programs, the registration of the apprentices and trainees, and the ratios and wage rates prescribed in the applicable programs.

(ii)(A) The contractor shall submit weekly, for each week in which any contract work is performed, a copy of all payrolls to the subrecipient, that is, the entity that receives the sub-grant or loan from the State capitalization grant recipient. Such documentation shall be available on request of the State recipient or EPA. As to each payroll copy received, the subrecipient shall provide written confirmation in a form satisfactory to the State indicating whether or not the project is in compliance with the requirements of 29 CFR 5.5(a)(1) based on the most recent payroll copies for the specified week. The payrolls shall set out accurately and completely all of the information required to be maintained under 29 CFR 5.5(a)(3)(i), except that full social security numbers and home addresses shall not be included on the weekly payrolls. Instead the payrolls shall only need to include an individually identifying number for each employee (e.g., the last four digits of the employee's social security number). The required weekly payroll information may be submitted in any form desired. Optional Form WH-347 is available for this purpose from the Wage and Hour Division Web site at <http://www.dol.gov/whd/programs/dbra/wh347.htm> or its successor site. The prime contractor is responsible for the submission of copies of payrolls by all subcontractors. Contractors and subcontractors shall maintain the full social security number and current address of each covered worker, and shall provide them upon request to the subrecipient(s) for transmission to the State or EPA if requested by EPA, the State, the contractor, or the Wage and Hour Division of the Department of Labor for purposes of an investigation or audit of compliance with prevailing wage requirements. It is not a violation of this section for a prime contractor to require a subcontractor to provide addresses and social security numbers to the prime contractor for its own records, without weekly submission to the subrecipient(s).

(B) Each payroll submitted shall be accompanied by a "Statement of Compliance," signed by the contractor or subcontractor or his or her agent who pays or supervises the payment of the persons employed under the contract and shall certify the following:

(1) That the payroll for the payroll period contains the information required to be provided under § 5.5 (a)(3)(ii) of Regulations, 29 CFR part 5, the appropriate information is being maintained under § 5.5 (a)(3)(i) of Regulations, 29 CFR part 5, and that such information is correct and complete;

(2) That each laborer or mechanic (including each helper, apprentice, and trainee) employed on the contract during the payroll period has been paid the full weekly wages earned, without rebate, either directly or indirectly, and that no

deductions have been made either directly or indirectly from the full wages earned, other than permissible deductions as set forth in Regulations, 29 CFR part 3;

(3) That each laborer or mechanic has been paid not less than the applicable wage rates and fringe benefits or cash equivalents for the classification of work performed, as specified in the applicable wage determination incorporated into the contract.

(C) The weekly submission of a properly executed certification set forth on the reverse side of Optional Form WH-347 shall satisfy the requirement for submission of the "Statement of Compliance" required by paragraph (a)(3)(ii)(B) of this section.

(D) The falsification of any of the above certifications may subject the contractor or subcontractor to civil or criminal prosecution under section 1001 of title 18 and section 231 of title 31 of the United States Code.

(iii) The contractor or subcontractor shall make the records required under paragraph (a)(3)(i) of this section available for inspection, copying, or transcription by authorized representatives of the State, EPA or the Department of Labor, and shall permit such representatives to interview employees during working hours on the job. If the contractor or subcontractor fails to submit the required records or to make them available, the Federal agency or State may, after written notice to the contractor, sponsor, applicant, or owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds. Furthermore, failure to submit the required records upon request or to make such records available may be grounds for debarment action pursuant to 29 CFR 5.12.

(4) Apprentices and trainees--

(i) Apprentices. Apprentices will be permitted to work at less than the predetermined rate for the work they performed when they are employed pursuant to and individually registered in a bona fide apprenticeship program registered with the U.S. Department of Labor, Employment and Training Administration, Office of Apprenticeship Training, Employer and Labor Services, or with a State Apprenticeship Agency recognized by the Office, or if a person is employed in his or her first 90 days of probationary employment as an apprentice in such an apprenticeship program, who is not individually registered in the program, but who has been certified by the Office of Apprenticeship Training, Employer and Labor Services or a State Apprenticeship Agency (where appropriate) to be eligible for probationary employment as an apprentice. The allowable ratio of apprentices to journeymen on the job site in any craft classification shall not be greater than the ratio permitted to the contractor as to the entire work force under the registered program. Any worker listed on a payroll at an apprentice wage rate, who is not registered or otherwise employed as stated above, shall be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any apprentice performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. Where a contractor is performing construction on a project in a locality other than that in which its program is registered,

the ratios and wage rates (expressed in percentages of the journeyman's hourly rate) specified in the contractor's or subcontractor's registered program shall be observed. Every apprentice must be paid at not less than the rate specified in the registered program for the apprentice's level of progress, expressed as a percentage of the journeymen hourly rate specified in the applicable wage determination. Apprentices shall be paid fringe benefits in accordance with the provisions of the apprenticeship program. If the apprenticeship program does not specify fringe benefits, apprentices must be paid the full amount of fringe benefits listed on the wage determination for the applicable classification. If the Administrator determines that a different practice prevails for the applicable apprentice classification, fringes shall be paid in accordance with that determination. In the event the Office of Apprenticeship Training, Employer and Labor Services, or a State Apprenticeship Agency recognized by the Office, withdraws approval of an apprenticeship program, the contractor will no longer be permitted to utilize apprentices at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(ii) Trainees. Except as provided in 29 CFR 5.16, trainees will not be permitted to work at less than the predetermined rate for the work performed unless they are employed pursuant to and individually registered in a program which has received prior approval, evidenced by formal certification by the U.S. Department of Labor, Employment and Training Administration. The ratio of trainees to journeymen on the job site shall not be greater than permitted under the plan approved by the Employment and Training Administration. Every trainee must be paid at not less than the rate specified in the approved program for the trainee's level of progress, expressed as a percentage of the journeyman hourly rate specified in the applicable wage determination. Trainees shall be paid fringe benefits in accordance with the provisions of the trainee program. If the trainee program does not mention fringe benefits, trainees shall be paid the full amount of fringe benefits listed on the wage determination unless the Administrator of the Wage and Hour Division determines that there is an apprenticeship program associated with the corresponding journeyman wage rate on the wage determination which provides for less than full fringe benefits for apprentices. Any employee listed on the payroll at a trainee rate who is not registered and participating in a training plan approved by the Employment and Training Administration shall be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any trainee performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. In the event the Employment and Training Administration withdraws approval of a training program, the contractor will no longer be permitted to utilize trainees at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(iii) Equal employment opportunity. The utilization of apprentices, trainees and journeymen under this part shall be in conformity with the equal employment opportunity requirements of Executive Order 11246, as amended and 29 CFR part 30.

(5) Compliance with Copeland Act requirements. The contractor shall comply with the requirements of 29 CFR part 3, which are incorporated by reference in this contract.

(6) Subcontracts. The contractor or subcontractor shall insert in any subcontracts the clauses contained in 29 CFR 5.5(a)(1) through (10) and such other clauses as the EPA determines may be appropriate, and also a clause requiring the subcontractors to include

these clauses in any lower tier subcontracts. The prime contractor shall be responsible for the compliance by any subcontractor or lower tier subcontractor with all the contract clauses in 29 CFR 5.5.

(7) Contract termination; debarment. A breach of the contract clauses in 29 CFR 5.5 may be grounds for termination of the contract, and for debarment as a contractor and a subcontractor as provided in 29 CFR 5.12.

(8) Compliance with Davis-Bacon and Related Act requirements. All rulings and interpretations of the Davis-Bacon and Related Acts contained in 29 CFR parts 1, 3, and 5 are herein incorporated by reference in this contract.

(9) Disputes concerning labor standards. Disputes arising out of the labor standards provisions of this contract shall not be subject to the general disputes clause of this contract. Such disputes shall be resolved in accordance with the procedures of the Department of Labor set forth in 29 CFR parts 5, 6, and 7. Disputes within the meaning of this clause include disputes between the contractor (or any of its subcontractors) and Subrecipient(s), State, EPA, the U.S. Department of Labor, or the employees or their representatives.

(10) Certification of eligibility.

(i) By entering into this contract, the contractor certifies that neither it (nor he or she) nor any person or firm who has an interest in the contractor's firm is a person or firm ineligible to be awarded Government contracts by virtue of section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1).

(ii) No part of this contract shall be subcontracted to any person or firm ineligible for award of a Government contract by virtue of section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1).

(iii) The penalty for making false statements is prescribed in the U.S. Criminal Code, 18 U.S.C. 1001.

4. Contract Provision for Contracts in Excess of \$100,000.

(a) Contract Work Hours and Safety Standards Act. The subrecipient shall insert the following clauses set forth in paragraphs (a)(1), (2), (3), and (4) of this section in full in any contract in an amount in excess of \$100,000 and subject to the overtime provisions of the Contract Work Hours and Safety Standards Act. These clauses shall be inserted in addition to the clauses required by Item 3, above or 29 CFR 4.6. As used in this paragraph, the terms laborers and mechanics include watchmen and guards.

(1) Overtime requirements. No contractor or subcontractor contracting for any part of the contract work which may require or involve the employment of laborers or mechanics shall require or permit any such laborer or mechanic in any workweek in which he or she is

employed on such work to work in excess of forty hours in such workweek unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of forty hours in such workweek.

(2) Violation; liability for unpaid wages; liquidated damages. In the event of any violation of the clause set forth in paragraph (a)(1) of this section the contractor and any subcontractor responsible therefore shall be liable for the unpaid wages. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such District or to such territory), for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or mechanic, including watchmen and guards, employed in violation of the clause set forth in paragraph (a)(1) of this section, in the sum of \$10 for each calendar day on which such individual was required or permitted to work in excess of the standard workweek of forty hours without payment of the overtime wages required by the clause set forth in paragraph (a)(1) of this section.

(3) Withholding for unpaid wages and liquidated damages. The subrecipient, upon written request of the EPA Award Official or an authorized representative of the Department of Labor, shall withhold or cause to be withheld, from any moneys payable on account of work performed by the contractor or subcontractor under any such contract or any other Federal contract with the same prime contractor, or any other federally-assisted contract subject to the Contract Work Hours and Safety Standards Act, which is held by the same prime contractor, such sums as may be determined to be necessary to satisfy any liabilities of such contractor or subcontractor for unpaid wages and liquidated damages as provided in the clause set forth in paragraph (b)(2) of this section.

(4) Subcontracts. The contractor or subcontractor shall insert in any subcontracts the clauses set forth in paragraph (a)(1) through (4) of this section and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in paragraphs (a)(1) through (4) of this section.

(b) In addition to the clauses contained in Item 3, above, in any contract subject only to the Contract Work Hours and Safety Standards Act and not to any of the other statutes cited in 29 CFR 5.1, the Subrecipient shall insert a clause requiring that the contractor or subcontractor shall maintain payrolls and basic payroll records during the course of the work and shall preserve them for a period of three years from the completion of the contract for all laborers and mechanics, including guards and watchmen, working on the contract. Such records shall contain the name and address of each such employee, social security number, correct classifications, hourly rates of wages paid, daily and weekly number of hours worked, deductions made, and actual wages paid. Further, the Subrecipient shall insert in any such contract a clause providing that the records to be maintained under this paragraph shall be made available by the contractor or subcontractor for inspection, copying, or transcription by authorized representatives of the (write the name of agency) and the Department of Labor, and the contractor or subcontractor will permit such representatives to interview employees during working hours on the job.

5. Compliance Verification

(a) The subrecipient shall periodically interview a sufficient number of employees entitled to DB prevailing wages (covered employees) to verify that contractors or subcontractors are paying the appropriate wage rates. As provided in 29 CFR 5.6(a)(6), all interviews must be conducted in confidence. The subrecipient must use Standard Form 1445 (SF 1445) or equivalent documentation to memorialize the interviews. Copies of the SF 1445 are available from EPA on request.

(b) The subrecipient shall establish and follow an interview schedule based on its assessment of the risks of noncompliance with DB posed by contractors or subcontractors and the duration of the contract or subcontract. Subrecipients must conduct more frequent interviews if the initial interviews or other information indicates that there is a risk that the contractor or subcontractor is not complying with DB. Subrecipients shall immediately conduct necessary interviews in response to an alleged violation of the prevailing wage requirements. All interviews shall be conducted in confidence.

(c) The subrecipient shall periodically conduct spot checks of a representative sample of weekly payroll data to verify that contractors or subcontractors are paying the appropriate wage rates. The subrecipient shall establish and follow a spot check schedule based on its assessment of the risks of noncompliance with DB posed by contractors or subcontractors and the duration of the contract or subcontract. At a minimum, if practicable, the subrecipient should spot check payroll data within two weeks of each contractor or subcontractor's submission of its initial payroll data and two weeks prior to the completion date the contract or subcontract. Subrecipients must conduct more frequent spot checks if the initial spot check or other information indicates that there is a risk that the contractor or subcontractor is not complying with DB. In addition, during the examinations the subrecipient shall verify evidence of fringe benefit plans and payments thereunder by contractors and subcontractors who claim credit for fringe benefit contributions.

(d) The subrecipient shall periodically review contractors and subcontractors use of apprentices and trainees to verify registration and certification with respect to apprenticeship and training programs approved by either the U.S Department of Labor or a state, as appropriate, and that contractors and subcontractors are not using disproportionate numbers of, laborers, trainees and apprentices. These reviews shall be conducted in accordance with the schedules for spot checks and interviews described in Item 5(b) and (c) above.

(e) Subrecipients must immediately report potential violations of the DB prevailing wage requirements to the EPA DB contact listed above and to the appropriate DOL Wage and Hour District Office listed at <http://www.dol.gov/whd/america2.htm>.

ATTACHMENT 2

DAVIS-BACON WAGE RATE DETERMINATION

[SRF Assistance Recipient to insert applicable wage determinations here]

How to obtain a Wage Determination:

1. www.wdol.gov
2. Click "Selecting DBA WDs"
3. Select the State and county where the work will be performed
4. Select the "Construction Type": Heavy, Building, Highway, or Residential
5. Click on one of the wage determinations. Verify that the wage determination displayed is the correct wage determination, and not for "Heavy Dredging".
6. Select the text box displaying the Wage Determination and copy the text of the Wage Determination.
7. Click "Sign Up for Alert Service" to receive notification if the Wage Determination is updated.

When to update the wage determination:

1. If DOL updates the Wage Determination, you must update the Wage Determination through an addendum to the bid specifications.
2. If the update occurs less than 10 days prior to the date of bid opening, you are not required to update the Wage Determination.

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ATTACHMENT 3

CERTIFICATION OF NONSEGREGATED FACILITIES

(Applicable to federally assisted construction contracts and related subcontracts exceeding \$10,000 which are not exempt from the Equal Opportunity clause.)

The federally assisted construction contractor certifies that he does not maintain or provide for his employees any segregated facilities at any of his establishments, and that he does not permit his employees to perform their services at any location, under his control, where segregated facilities are maintained. The federally assisted construction contractor certified, further that he will not maintain or provide for his employees any segregated facilities at any of his establishments, and that he will not permit his employees to perform their services at any location, under his control, where segregated facilities are maintained. The federally assisted construction contractor agrees that a breach of this certification is a violation of the Equal Opportunity clause in this contract.

As used in this certification, the term "segregated facilities" means any waiting rooms, work area, rest rooms and wash rooms, restaurants and other eating areas, time clocks, locker rooms and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing facilities provided for employees which are segregated by explicit directive or area, in fact, segregated on the basis of race, creed, color, or national origin, because of habit, local custom, or otherwise. The federally assisted construction contractor agrees that (except where he has obtained identical certifications from proposed contractors for specific time periods) he will obtain identical certifications from proposed subcontractors prior to the award of subcontracts exceeding \$10,000 which are not exempt from the provisions of the Equal Opportunity clause, and that he will retain such, certification in this file.

Signature

Date

Name and title of signer (please type)

[THIS FORM SHALL BE COMPLETED IN FULL AND SUBMITTED WITH THE BID PROPOSAL]

ATTACHMENT 4

NOTICE TO LABOR UNIONS OR OTHER ORGANIZATION OF WORKERS: NON-DISCRIMINATION IN EMPLOYMENT

TO: _____
(name of union or organization of worker)

The undersigned currently holds contract(s) with _____
(name of applicant)

_____ involving funds or credit of the U.S. Government or (a) subcontract(s) with a prime contractor holding such contract(s).

You are advised that under the provisions of the above contract(s) or subcontract(s) and in accordance with Section 202 of Executive Order 11246 dated September 24, 1965, the undersigned is obliged not to discriminate against any employee or applicant for employment because of race, color, creed, or national origin. This obligation not to discriminate in employment includes, but is not limited to, the following:

EMPLOYMENT, UPGRADING, TRANSFER OR DEMOTION

RECRUITMENT AND ADVERTISING

RATES OF PAY OR OTHER FORMS OF COMPENSATION

SELECTION FOR TRAINING INCLUDING APPRENTICESHIP, LAYOFF OR TERMINATION

This notice is furnished you pursuant to the provisions of the above contract(s) or subcontract(s) and Executive Order 11246.

Copies of this notice will be posted by the undersigned in conspicuous places available to employees or applicants for employment.

(contractor or subcontractor(s))

(Date)

*** END OF SECTION ***

PART 3 - TECHNICAL SPECIFICATIONS

SECTION 01 11 10 – SUMMARY OF WORK

PART 1. GENERAL

1.01 WORK COVERED BY CONTRACT DOCUMENTS

- A. The work covered by the Contract Documents consists of furnishing all labor, equipment and materials necessary for the construction of all of the WWTP improvements and miscellaneous site improvements as shown on the plans and specified herein.
- B. Contractor shall furnish all labor, tools, equipment and materials not pre-purchased or supplied by the Owner. In addition, the Contractor shall provide demolition, shoring, bracing, sheeting, cribbing, falsework, pumping, dewatering, drainage, forms, and all material as required or necessary to demolish, excavate, backfill, grade, construct, lay, erect, install, test, and clean-up site. The work shall consist of, in general, the expansion of the existing wastewater treatment plant.

A more detailed summary of the work is included in Section 00 24 13 SCOPES OF BID.

1.02 WORK AND RESPONSIBILITIES

- A. Unless otherwise indicated, work and responsibilities include, but are not limited to the following:
 - 1. Providing and paying for labor, materials, equipment, tools, machines, facilities, and services necessary for execution and completion of work.
 - 2. Paying required taxes.
 - 3. Giving required notices.
 - 4. Enforcing strict discipline and good order among employees.
 - 5. Using new materials, except as noted.
 - 6. Maintaining required egress and other requirements in accordance with governing Codes and Ordinances throughout the work.
 - 7. Obtaining and paying for required permits, fees and notices, see General Conditions.

1.03 SEQUENCE/PHASING

- A. These documents are not to be interpreted implicitly or explicitly as definition of procedure and sequence of operations. Order as to procedure and sequence of operations are Contractor options, consistent with contract documents and as approved by Owner. A preliminary construction phasing plan is included in Section 00 31 13 PRELIMINARY PROJECT PHASING.
- B. Site Work: Proposed stockpiling areas must be approved by the Owner.

1.04 COOPERATION AND COORDINATION

- A. Contractor is responsible for coordinating and scheduling work of subcontractors to expedite progress of the Project.
- B. Subcontractor Instructions: Subcontractors to become familiar with Conditions of the Contract and the work of other Sections related to their own work.
- C. Project Coordination and Scheduling Control: Responsibility for coordination and close adherence to time schedules rests solely with the General Contractor who shall maintain coordination and scheduling control at all times.

- D. Each separate contractor and each subcontractor responsible to the General Contractor shall cooperate diligently with the General Contractor in the execution of their work so as to cause no delay in the completion of the Project. This responsibility includes the completion of all work in a timely manner and all items of equipment connected and fully operating at the time of Substantial Completion. Each separate contractor and each subcontractor shall diligently comply with the following requirements:
1. Inform other trades of requirements at proper time to prevent delay or revisions.
 2. Be informed on the requirements of other trades and check own work for conflicts with the work of other trades.
 3. Insure delivery of materials and performance of work on coordinated schedule with other trades.
 4. Contractor is to ensure the subcontractors and equipment suppliers are responsible for compatibility and completeness of the installation and operation of the equipment in their respective Specification Sections including conformance with code requirements. If power, piping, conduit, or other work required for complete installation is not provided by others to equipment location or is not adequate for complete installation, the subcontractor or equipment supplier shall be responsible for providing the necessary connections.
- E. Notification and Correction of Defective Work: Before starting a section of work, each contractor and subcontractor shall carefully examine all preparatory work that has been executed to receive his work. Check carefully, by whatever means required, to ensure that the work and adjacent, related work will finish to proper contours, planes, and levels. Promptly notify the Contractor of any defects or imperfections in preparatory work which will in any way affect satisfactory completion of the work. Under no condition shall a section of work proceed prior to preparatory work having been completed, cured, dried, or otherwise made satisfactory to receive such related work. Correction of defective work shall be the responsibility of the contractor or subcontractor providing the defective work. Correction of work due to underlying defects shall be the responsibility of the contractor or subcontractor providing work.
- F. Intent of Drawings: The work of each contractor and subcontractor shall conform to the intent of the contract drawings. Drawings showing work of other trades are partly diagrammatic and do not intend to show in details all features of work. Each contractor shall carefully review and compare related drawings and shall thoroughly understand the building conditions affecting their work. All changes required in the work caused by failure to do so shall be at no expense to the Owner. The design is based upon dimensions and requirements for the equipment of the "first-named" manufacturer. All changes required in the work caused by the use of an approved "substitute" to the first-named manufacturer shall be at no expense to the Owner.
- G. Interference's and Right-Of-Way: Make proper provisions to minimize interference's. Where conflicts occur, gravity drainage improvements have right-of-way over mechanical and electrical work; electrical work has right-of-way over landscaping work. Submit conflicts which cannot be resolved by right-of-way to Engineer for instructions.
- H. Cooperate and coordinate with any other separate Contractors under Contract with the Owner.

1.05 CONSTRUCTION STAGING AREA

- A. Coordinate staging areas with the District.

1.06 EXISTING UTILITIES

A. Administrative Requirements

1. The Contractor is advised that underground excavation is regulated under RCW Chapter 19.122. Included therein are the following requirements:
 - a. 48-hours before beginning any excavation work, the Contractor shall inform local utilities through the utility one-call locator service at (800) 424-5555 or 811;
 - b. Protect existing utilities in the vicinity of excavation work;
 - c. In the event of any damages, notify the utility purveyor and the utility one-call locator service immediately;
 - d. Immediately repair any damaged utilities deemed to be an emergency;
 - e. Coordinate non-emergency repairs with the utility purveyor;
 - f. Provisions for assigning the financial liability of any repair work.
 - g. Further, the Contractor is required to contact the Engineer, Wilson Engineering, LLC (Jeff Christner, P.E.), 48-hours before starting construction at (360) 733-6100.

B. Field Protection Requirements

1. Utilities of record are shown on the Drawings insofar as possible to do so. These, however are shown for convenience only and the Owner and his representatives assume no responsibility for improper locations or failure to show utility locations on the Drawings. At Contractor's expense, immediately repair utilities damaged during construction.

1.07 MISCELLANEOUS

A. Additional work items include, but are not limited to:

1. Maintaining a pedestrian and vehicular access to and around existing projects.
2. Not unreasonably encumbering site with materials or equipment.
3. Assuming full responsibility for protection and safekeeping of products stored on premises.
4. Moving any stored products interfering with any other Contractors.
5. Obtaining and paying for use of additional storage or work areas needed for operations.
6. Restoration of any damage to existing improvements adjacent to work site.
7. Moving and replacing items incidental to completion of the work including mailboxes, fences, small shrubs and trees, street signs, yard decorations, etc.

*****END OF SECTION*****

SECTION 01 31 00 – PROJECT COORDINATION

PART 1. GENERAL

1.01 DESCRIPTION OF WORK

- A. Project meetings will be held to accomplish the following:
 - 1. Coordinate the work of the project and resolve any conflicts or construction problems.
 - 2. Establish a sound working relationship between the Contractor, Owner, and Engineer.
 - 3. Establish sound working procedures.
 - 4. Review job progress and quality of work.
 - 5. Expedite the work to completion within the scheduled time limit.
- B. Representatives of Contractors, subcontractors, and suppliers attending the meetings shall be qualified and authorized to act on behalf of the entity each represents.

1.02 RELATED SECTIONS:

- A. Related work specified elsewhere:
 - 1. Section 00 25 13 - Pre-Bid Conference
 - 2. Section 00 31 13 - Preliminary Project Phases
 - 3. Section 01 11 00 - Summary of Work
 - 4. Section 01 33 00 - Submittals:

1.03 PRECONSTRUCTION MEETING

- A. The pre-construction meeting will be scheduled within the time frame identified in the General Conditions after the Notice to Proceed has been issued. The Owner will notify the Contractor as to the time and place of the meeting.
- B. Present at the meeting shall be a representative of the Owner, the Engineer, the Contractor, Project Superintendent, and major subcontractors.
- C. The Contractor must be prepared for a thorough discussion and review, as well as revisions which may be deemed necessary in the opinion of the Owner, of the following:
 - 1. General project information
 - 2. Responsibilities of all involved parties
 - 3. Content of the contract
 - 4. Contractor's schedule
 - 5. Project Phasing and Schedule of construction
 - 6. Penalties and Liquidated Damages
 - 7. Subcontracts
 - 8. Status of Owner furnished materials
 - 9. Change order procedures
 - 10. Staking of work

11. Project inspection
12. Acceptance of work
13. Labor standards requirements
14. Rights-of-way and easements
15. Placement of project signs and posters
16. Handling of disputes
17. Additional issues as required.

1.04 PROGRESS MEETINGS

- A. Unless otherwise required, progress meetings will be held by the Owner on a weekly basis at a location near the site. Present at these meetings shall be the Contractor, subcontractors and suppliers as required, the Owner and other interested parties, i.e., material suppliers, public utility, etc.
- B. The Contractor must be prepared for a thorough discussion and review, as well as revisions which may be deemed necessary in the opinion of the Owner, of the following:
 1. Review work since previous meeting.
 2. Make field observations and address any conflicts or problems.
 3. Review material delivery schedules
 4. Review work progress including any issues that may impact project schedule.
 5. Review submittal schedule.
 6. Maintenance, testing and quality standards.
 7. Review any proposed changes.
 8. Review pay requests and procedures.
- C. The Owner shall preside over progress meetings. The Engineer shall be responsible for taking minutes, recording all significant proceedings and decisions. Copies of minutes shall be distributed within one week after the meeting.

1.05 SCHEDULE

- A. The Contractor shall develop and submit a preliminary construction progress schedule for the contracted work. This schedule shall be submitted to the Owner within 10 days of Contract Award.
- B. Schedule shall be a critical path diagram depicting the first day of each week and sized to be legible and permit notations and future revisions.
- C. Schedule shall be arranged chronologically by the start date of each item, and consider the following:
 1. The estimated construction progress schedule shall:
 - a. Show complete sequence of construction by activity.
 - b. Show start and stop dates of each major construction element.
 - c. Show projected percent completion for each major construction element at the first of each month.
 2. Through construction, the Contractor shall record progress of each major construction element.

3. Revisions shall show changes relative to previously submitted schedules and updated projections of progress and completion.
 4. The schedule shall be updated on a monthly basis and submitted with the pay estimate.
- D. The schedule and all subsequent revisions shall be kept at the Contractor's field office with copies available for the Engineer and Owner.

*****END OF SECTION*****

SECTION 01 33 00 – SUBMITTAL PROCEDURES

PART 1. GENERAL

1.01 DESCRIPTION OF WORK

- A. Summarize, but not necessarily a complete listing, submittals required of the Conditions of the Contract and the General Requirements.
- B. General procedures for specification submittals. Specific requirements for submittals are included in the individual sections.

1.02 RELATED SECTIONS

- A. Related work specified elsewhere:
 - 1. Section 00 73 00 – Supplementary Conditions
 - 2. Section 01 70 00 - Contract Closeout

1.03 SUBMITTAL SCHEDULE

This listing of submittals is a checklist for the Contractor's convenience and is not an exhaustive listing of provisions of any law or the requirements of these Contract Documents. The Owner reserves the right to amend this list.

- A. With his bid, the Contractor shall furnish the following:
 - 1. Bid Proposal (Section 00 41 00 BID PROPOSAL)
 - 2. Bid Guarantee (Section 00 43 13 BID BOND or other type of Bid Guarantee),
 - 3. Certification of Nonsegregated Facilities (Attachment 3 of Section 00 73 00 SUPPLEMENTAL CONDITIONS),
 - 4. EPA Form 6100-3 DBE Program Subcontractor Performance Form (Attachment 6 in Section 00 73 00 SUPPLEMENTAL CONDITIONS) for all DBE subcontractors.
 - 5. EPA form 6100-4 DBE Program Subcontractor Utilization Form (Attachment 7 in Section 00 73 00 SUPPLEMENTAL CONDITIONS).
- B. Within 1-hour of the bid, the Contractor shall furnish the following:
 - 1. List of Subcontractors (Section 00 43 36 PROPOSED SUBCONTRACTORS FORM).
- C. Within 24-hours of bid opening, Bidders so directed shall furnish the following:
 - 1. Contractors Qualifications, (as described in Section 00 22 13 SUPPLEMENTAL BIDDER RESPONSIBILITY CRITERIA).
 - 2. Subcontractors Qualifications, (as described in Section 00 45 14 SUBCONTRACTOR QUALIFICATIONS).
- D. Prior to executing the Contract Agreement, (Section 00 52 00 AGREEMENT FORM), the Contractor shall furnish the following:
 - 1. Payment and Performance Bonds, (Section 00 61 13 PERFORMANCE AND PAYMENT BONDS FORMS).
 - 2. Insurance Certificates.
 - 3. Prevailing wage rate requirements.

- E. 10 days after execution of the Agreement, the Contractor shall furnish the following:
 - 1. Construction schedule.
 - 2. Requests for material substitutions.
 - 3. Schedule of Values for the work.
- F. After starting construction, each month the Contractor shall furnish the following:
 - 1. Application for Payment on Owner approved form with breakdown of work performed organized in accordance with the Schedule of Values.
 - 2. Updated construction schedule (submitted with each monthly pay request)
- G. 14-days prior to beginning any work at the WWTP which will necessitate a shutdown, the Contractor shall furnish a bypass pumping and emergency response plan for the site.
- H. Certified Payroll in accordance with current federal wage requirements of the Davis-Bacon Act, (per requirements of Section 00 73 00 SUPPLEMENTAL CONDITIONS).
- I. With the final application for payment, the Contractor shall furnish the following:
 - 1. Contractor's affidavit stating payment of subcontractors
 - 2. Subcontractors' statements of being paid
 - 3. Final location, by each property, of all items on private property for which payment is requested.
- J. Before releasing retained funds, the Contractor shall furnish the following:
 - 1. Record drawings and related contract closeout documents
 - 2. Affidavits of Payment (wages, subcontractors, taxes, etc.)

1.04 GENERAL SUBMITTAL REQUIREMENTS

- A. Identification of Submittals
 - 1. Identify each submittal with Project title and number; clearly define location of submittal in the project and/or its location in the Contract Documents.
 - 2. It is the responsibility of the Contractor to coordinate the work of the various trades involved with the work under this agreement. Contractor shall check all submittals by his subcontractors and mark them with his approval prior to submittal.

1.05 SUBMITTAL OF SHOP DRAWINGS & SAMPLES

- A. General
 - 1. Provide submittals in pdf format.
 - 2. Submittal of shop drawings and samples shall be accompanied by a transmittal letter containing project name, Contractor's name, number of drawings and samples, titles and other pertinent data.
 - 3. Shop drawings shall be at a convenient size. A space shall be provided in the lower right-hand corner for the review stamp.
 - 4. The Contractor is responsible for obtaining and distributing required prints of shop drawings to his subcontractors and suppliers.
 - 5. Contractor shall maintain a complete material list and file of approved submittals at the project site for use as reference by interested parties.

B. Samples

1. Form of Submittal: When samples are specified to be submitted, furnish two samples, except as noted herein, of sufficient size to indicate general visual effect or as otherwise specified in the specifications, and in as nearly the form in which the material will appear on the project as practicable; i.e., submit paint on samples of actual material for which they are specified as a finish; one set of reviewed and selected samples will be retained by the owner.
2. Review:
 - a. The Owner will check submitted samples against file samples and project requirements, will make final selection of colors and finishes from samples, and will approve sample for application on the project in conformance with the Specifications.
 - b. Should a submitted sample not be in conformance with the specifications, resubmit sample which conforms with the requirements of Contract Documents.

C. Catalog Cuts, Data & Brochures

1. Where indicated in the Specifications, catalog cuts and similar data will be accepted in lieu of shop drawings, provided they contain required information and are clearly printed. Submit manufacturer's descriptive data including catalog sheets for materials, equipment and fixtures, showing dimension, performance characteristics and capacities, wiring diagrams and controls, schedules, and other pertinent information as required.

D. Submittal of Product Certificates

1. Where manufacturer certificates are specified to be furnished attesting to conformance with specification requirements, submit certificates in triplicate prior to acceptance of the Work.

E. Test Reports

1. Submittal is classified either as "shop drawing" or "product data", depending upon whether the report is uniquely prepared for the project or a standard publication of regular product or workmanship control testing at the point of production (respectively).
2. Refer to individual sections of the Specifications for specific requirements; furnish 3 copies when required.

F. Warranties

1. Provide warranties, guarantees and/or maintenance agreements where the Specifications require a period longer than the Contractor warranty period.

G. Operation & Maintenance Data

Furnish instructions and data on materials and equipment installed in the work in accordance with requirements of the technical provisions of the specifications and assemble as specified below. These manuals shall be submitted prior to application for payment exceeding 90% of the total contract amount.

1. Provide four (4) hard copy sets and two (2) electronic copy sets of Operation and Maintenance Data. Each hard copy set shall be bound in separate commercial quality three-ring binders with durable and cleanable plastic covers. The words "Operation and Maintenance Manual (or Instruction)" along with the type of

equipment covered shall be typed or neatly printed on the cover. The electronic copy sets shall be in PDF format and stored on either CD or flash drive units.

2. Each set shall be complete with an index and, as a minimum, cover the following items:
 - a. Name, location and telephone number of manufacturer and product's model number.
 - b. Name, location and telephone number of nearest supplier and spare parts warehouse.
 - c. Start-up procedures and normal operating characteristics and instruction.
 - d. Regulation, control, shut-down and emergency instructions.
 - e. Recommended preventative maintenance procedures including a lubrication schedule with recommended lubricants.
 - f. Trouble-shooting guide.
 - g. Complete nomenclature and commercial number of all parts including exploded views of each assembly.
 - h. List of recommended spare parts.
 - i. Complete as-built elementary wiring and outline diagrams.
 - j. Statements of warranty or guarantee.
3. Operation and Maintenance Manuals shall be submitted in at least draft form for Engineer's review with Shop Drawings, Catalog Cuts and other material submittal data. Final drafts, incorporating Engineer's comments, shall be submitted prior to Contractor's application of payment for 75 percent or more of the work.
4. Contractor shall maintain a complete file of all Engineer reviewed Operation and Maintenance Manuals at the project site for use as a reference by interested parties.

*****END OF SECTION*****

SECTION 01 41 00 – REGULATORY REQUIREMENTS

PART 1. GENERAL

1.01 SECTION INCLUDES

As required by General Conditions: "Contractor shall comply with and give notices required by all federal, state, and local laws, ordinances, rules, regulations and lawful orders of public authorities applicable to performance of the Work." Except where otherwise expressly required by applicable Laws and regulations, neither OWNER nor ENGINEER will be responsible for monitoring CONTRACTOR'S compliance with any Laws and Regulations. Contractor is responsible for keeping the District, Labor & Industries and other authorities completely informed of any changes in the work in a timely manner, and is responsible for informing them of any changes in the work which may affect codes and laws. This includes contract modifications, amendments, additions, shop drawings, and the like, current as of Project Manual date.

- A. Make any and all adjustments and modifications as required to conform to ordinances, and regulations.
- B. Referenced codes establish minimum requirement levels. Where provisions of various codes or standards conflict, the more stringent provisions govern. Promptly submit to Engineer written notice of observed contract document variations from legal requirements.
- C. Compliance requirements include, but are not limited to following:
 - 1. International Building Code and Related Standards, most recent edition, published by the International Conference of Building Officials.
 - 2. State Rules and Regulations for Barrier Free Design/WAC 51-10.
 - 3. The Americans with Disabilities Act (ADA) "Accessibility Guidelines for Buildings and Facilities."
 - 4. Washington State Department of Labor and Industries Regulations.
 - 5. Electrical Work:
 - a. Underwriters' Laboratories (UL).
 - b. National Electrical Manufacturers' Association (NEMA).
 - c. NFPA, National Electric Code (NEC), National Electric Safety Code, and above electrical listings, as applicable.
 - d. State Electrical Construction Code.
 - 6. Environmental Requirements: All work to be performed in compliance with relevant statutes and regulations dealing with prevention of environmental pollution and preservation of public natural resources.
 - 7. Standard Specifications for Road and Bridge Construction, Washington State Department of Transportation, (WSDOT) latest edition.
 - 8. Standard Specifications for Municipal Public Works Construction, Washington State Chapter, American Public Works Association, latest edition.
 - 9. San Juan County Standards, latest editions.
 - 10. Eastsound Sewer and Water District Standards, latest edition.

1.02 MISCELLANEOUS EXPLANATIONS/INTENT

- A. Number of Specified Items Required: Wherever in these Specifications an article, device, or piece of equipment is referred to in the singular number, the reference applies to as many such articles as are shown on the Drawings or required to complete the installation.
- B. Drawings/Diagrammatic:
 - 1. Drawings are in part diagrammatic and do not necessarily show complete details of construction, work or materials, performance or installation. And they do not necessarily show how construction details, other items or work, fixtures, and equipment may affect any particular installation. Contractor is required to ascertain and correlate the work to bring the parts together into a satisfactory and completed whole.
 - 2. Furnish and install work not covered under any heading, Section, branch, class or trade of the project manual, but shown on or reasonably inferable from the Drawings. This includes all work necessary to produce the intended results. Install similarly for items more positively indicated.
- C. Wording of these Specifications: These Specifications are of the abbreviated or streamlined type and may include incomplete sentences.
 - 1. Words such as "shall", "the Contractor shall", "shall be", and similar mandatory phrases, are required to be supplied by inference in the same manner as they are in a note on the Drawings.
 - 2. Provide all items, articles, materials, and operations listed, including all labor, materials, equipment and incidentals, required for their completion.
- D. Tense, Gender, Singular, Plural: Present tense words include future tense. Words in masculine gender include feminine and neuter genders. Words in the singular include plural. Plural words include singular.
- E. All, Entire, and the Like: For brevity throughout the documents, these words may be omitted. Read their implications into all work.
- F. Specifications by Reference: Any material specified by reference or number, symbol or title of a specified standard, such as commercial standard, ANSI and ASTM documents, Federal Specifications, trade association standard, or the like, shall comply with the following:
 - 1. The latest revision requirements thereof, and any amendment or supplement thereto, in effect on Bid date or date of Owner-Contractor Agreement when there are no bids.
- G. Dimensions and Measurements on Drawings: Dimensions govern. Do not scale. Contractor is to check all dimensions in the field and verify them with respect to adjacent or incorporated work. Large scale drawings take precedence over plans, elevations, and cross sections.
- H. First Class Workmanship: First Class Workmanship is expected.
 - 1. Prior to installing any item or material, verify that receiving surfaces are plumb, level, true to line, and straight to the degree necessary to achieve tolerances specified or required. Perform without extra cost all shimmering, blocking,

grinding, or patching required to make such surfaces plumb, level, true to line, and straight.

2. Take care in attention to details and fitting at intersections and junctures of materials. All joints are to be tight, straight, even, and smooth.

I. Presence of Engineer/Owner: Do not misconstrue presence of this person or any of his representatives at the site as assuring compliance with Contract Documents.

PART 2. MATERIALS (NOT USED)

PART 3. EXECUTION (NOT USED)

*****END OF SECTION*****

SECTION 01 45 00 – QUALITY CONTROL

PART 1. GENERAL

1.01 DESCRIPTION OF WORK

- A. Inspection and testing laboratory qualifications, duties and responsibilities.
- B. Contractor's quality control requirements.

1.02 RELATED SECTIONS

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 33 00 – Submittal Procedures
 - 2. Section 01 66 00 – Product Storage and Handling Requirements
 - 3. Section 01 70 00 – Execution and Closeout Requirements
 - 4. Technical Specifications include quality control requirements for certain portions of the work.

1.03 APPLICABLE PUBLICATIONS AND REGULATORY REQUIREMENTS

- A. ASTM E329: Inspection and Testing Agencies for Concrete, Steel, and Bituminous Materials as used in Construction.
- B. WSDOT and ASHTO: Applicable sections that pertain to compaction testing for subgrade, base and top course, and asphalt testing.
- C. Washington State Building Code and International Building Code Standards.
- D. Nothing in the Drawings or Specifications shall be construed to permit Work not conforming to applicable laws, ordinances, rules or regulations.
- E. When Drawings or Specifications exceed requirements of applicable laws, ordinances, rules, or regulations, comply with documents establishing the more stringent requirements.

1.04 DEFINITIONS

- A. Factory Tests: Tests made on various products and component parts prior to shipment to the job site, including but not limited to such items as pumps, valves, miscellaneous piping equipment, electrical equipment, and precast concrete.
- B. Field Tests: Tests or analyses made at, or in the vicinity of the job site in connection with the actual construction.
- C. Product: The term "product" includes the plural thereof, and means a type or a category of manufactured goods, constructions, installations and natural and processed materials or those associated services whose characterizations, classification or functional performances determination is specified by standards.
- D. Person: The term "person" means associations, companies, corporations, educational institutions, firms, government agencies, at the Federal, State and Local level, partnerships, and societies, as well as divisions thereof, and individuals.
- E. Testing Laboratory: The term "testing laboratory" means and "person", as defined above, whose functions include testing, analyzing, or inspecting "products" as defined above, and/or evaluating the designs or specifications of such "products" according to the requirements of applicable standards.

- F. **Certified Test Reports:** Certified test reports are reports of tests signed by a qualified professional attesting that tests were performed in accordance with the test method specified, that the test results reported are accurate, and that items tested either meet or fail to meet the stated minimum requirements. These test reports include those performed by Factory Mutual, Underwriter's Laboratories, Inc., and others.
- G. **Certified Inspection Reports:** Certified inspection reports are those signed by approved inspectors attesting that the items inspected meet the specification requirements other than any exceptions included in the report.
- H. **Manufacturer's Certificate of Conformance or Compliance:** A certificate signed by an authorized manufacturer's official attesting that the material or equipment delivered meets the specification requirements.

1.05 QUALITY CONTROL REQUIREMENTS

- A. All work under the contract shall be inspected and tested as specified herein. The Contractor shall maintain records of all inspections and tests. Approvals shall be obtained before delivery of materials to the project site.
- B. Responsibility for field testing is defined below. The contractor is responsible for notifying Owner/Engineer and coordinating test requests as discussed elsewhere in these documents. The contractor is responsible for the costs of any repeat tests required where failed tests were obtained, including those tests that were originally the responsibility of the Owner/Engineer.
 - 1. **Contractor Responsibility** – Services to be coordinated by the Contractor, performed by the Contractor's Testing Agency, and Paid by the Contractor:
 - a. Pipe Pressure Testing – Per Specs
 - b. Potable Water Pipe Disinfection & Testing – Per Specs
 - c. Equipment Testing – Per Equipment Specs
 - d. Clean Water Leakage Testing
 - e. Clean Water Mechanical Testing (pumps, diffusers, mixers, air lift assemblies, etc.)
 - 2. **Owner/Engineer Responsibility** – Services to be coordinated by the Contractor, performed by the Owner's Testing Agency, and Paid by the Owner / Engineer:
 - a. Special Inspections for Buildings
 - (i) Concrete Work
 - (ii) Structural Steel and Bolting, where needed.
 - b. Earthwork
 - (i) Inspection / Testing soils below footings for adequate subgrade bearing conditions
 - (ii) Perform sieve / proctor tests for each source of fill and native soils
 - (iii) Test in-place density of fills/backfills for buildings, pavement bases and utilities.
 - c. Concrete
 - (i) Review concrete batch tickets and verify compliance with approved mix design.

- (ii) Periodic inspection of placement of reinforcing steel.
 - (iii) Continuous inspection during placement, including sampling, slump, air, temperature, and taking concrete specimens.
 - (iv) Continuous inspection during grouting of bolts, rebar dowels, and anchors.
 - (v) Test concrete for compressive strength.
 - d. Asphalt Pavement
 - (i) Rice density tests. Compaction tests. Lab: Asphalt extraction/gradation
 - e. Painting / Coating Systems
 - (i) Periodic inspection of field work
 - f. Structural Steel
 - (i) Periodic inspection of field welding and bolted connections.
- C. If required, contractor responsibility for quality control testing shall be as follows:
 - 1. Factory Tests: Unless otherwise specified, the Contractor will arrange and pay for factory tests when required by the contract documents.
 - 2. Factory Inspection: Unless otherwise specified, the Contractor will arrange and pay for factory inspection when required by the contract documents.
 - 3. Field Inspection and Tests by the Contractor: Unless otherwise specified, the Contractor shall furnish all equipment, instruments, qualified personnel, and facilities necessary to inspect all work and perform all tests when required by the contract documents. All inspections and tests performed and test results shall be promptly submitted to the Owner.
 - 4. Approval of Testing Laboratories: All laboratory work under this contract shall be performed by a laboratory approved by the Owner.
- D. Laboratory Reports: Reports shall cite the contract requirements, the test or analysis procedures used, the actual test results, and include a statement that the item tested or analyzed conforms or fails to conform to the specifications requirements. All test reports shall be signed by a representative of the testing laboratory authorized to sign certified test reports. The Contractor shall arrange for immediate and direct delivery of the signed original of all reports, certifications, and other documentation to the Owner.
- E. Repeated Tests and Inspections: The Contractor shall repeat tests and inspections after each failed test until passing test results are obtained. The retesting and reinspection shall be performed at no additional cost to the Owner and the Contractor shall reimburse the Owner for their, or their representative's, time and expenses due to the failed test results, including those tests that were originally the responsibility of the Owner/Engineer.

1.06 CONTRACTOR'S RESPONSIBILITY

- A. Access. Furnish free access to various parts of the work and assist testing inspection personnel in performance of their duties at no additional cost to the Owner.

- B. Concealed Work. When directed by the Owner, the Contractor shall open for inspection any part of the work which has been concealed. Should the Contractor refuse or neglect such a request, the Owner may employ any other person to open up the same or do so himself. If any part of the work has been concealed in violation of the Owner's instruction or, if on being opened, it is found not to be in accordance with the terms of the Contract Documents the expense of opening and recovering, whether done by the Contractor or not, shall be charged to the Contractor. If the work has been concealed but not in violation of the Owner's instructions and is found to be in accordance with the terms of the Contract Documents the actual necessary expense of opening and recovering is done by the Contractor it shall be considered as extra work and paid for accordingly.
- C. Notices. The Contractor shall notify the Owner not less than 48 hours, unless otherwise noted, before work requiring inspection is started. The Contractor shall schedule portions of the work requiring inspection and testing, so that the agency's time on the project is continuous and as brief as possible.

1.07 CONSTRUCTION SURVEILLANCE BY OWNER

- A. Appointment. The Owner may appoint an on-site representative for surveillance of any and all portions of the work. Such surveillance may extend to any or all parts of the work, and to the preparation or manufacture of materials to be used.
- B. Authority of On-Site Representative.
 - 1. On-site representative is not authorized to revoke, alter, enlarge or relax the provisions of the Contract Documents, and is placed on the work site to keep the Owner informed as to the progress of the work and the manner in which it is being done.
 - 2. On-site representative may also call the attention of the Contractor to any deviations from the plans or specifications. Failure of the Owner or his representative to call the attention of the Contractor to faulty work or deviation from the Contract Documents shall not constitute acceptance of said work.
 - 3. The representative is not authorized to approve or accept any portions of the work or to issue instructions contrary to the Contract Documents.
 - 4. The representative will exercise only such additional authority as may be specially delegated to him by the Owner, notice of which will be given in writing to the Contractor.

1.08 DEFECTIVE WORK

- A. Remove and replace any work found defective or not complying with requirements of Contract Documents, at no additional cost to Owner. Work will be checked as it progresses, but failure to detect any defective work or materials shall not in any way prevent later rejection when such defect is discovered, nor shall it obligate the Owner for final acceptance.

1.09 CLEAN WATER LEAKAGE TESTING

- A. Prior to startup, the Contractor is responsible for obtaining water and piping water to each newly constructed structures as necessary for a clean water leakage test. Contractor shall completely fill to the rim each structure, and the structure shall remain filled for 2 hours with no visible leakage. Structures requiring clean water leakage test include:
 - 1. Influent Pump Station
 - 2. Flow Splitter

3. Aeration Basins
4. Anoxic Basins
5. Clarifier
6. UV Disinfection Chamber
7. Digester
8. 2W Water System Pump Station

PART 2. MATERIAL (NOT USED)

PART 3. EXECUTION (NOT USED)

(Summary of Quality Control Testing Services Follows)

QUALITY CONTROL TESTING SERVICES

Services to be coordinated by the Contractor, performed by Owner's Testing Lab, and Paid by Owner/Engineer.

Special Inspections for Buildings

- Concrete Work.
- Structural Steel and Bolting, where needed.
- Air Barrier Testing per WA Energy Code Section 402.

Earthwork

- Inspect/test soils below footings for adequate subgrade bearing conditions.
- Perform sieve/proctor tests for each source of fill and native soils.
- Test in-place density of fills/backfills for buildings, pavement bases and utilities.

Concrete

- Review concrete batch tickets and verify compliance with approved mix design.
- Periodic inspection of placement of reinforcing steel.
- Continuous inspection during placement, including sampling, slump, air, temperature and taking concrete specimens.
- Continuous inspection during grouting of bolts, rebar dowels and anchors.
- Test concrete for compressive strength.

Asphalt Pavement

- Rice Density Tests. Compaction Tests. Lab: asphalt extraction/gradation.

Coating Systems

- Periodic inspection of field work.

Structural Steel

- Periodic inspection of field welding and bolted connections.

Services to be provided by the Contractor

Pipe Pressure Testing

- See Specifications

Potable Water Pipe Disinfection Testing

- See Specifications

Equipment Testing

- See Division 46 - Equipment Specifications

*****END OF SECTION*****

SECTION 01 50 00 – TEMPORARY FACILITIES

PART 1. GENERAL

1.01 DESCRIPTION OF SECTION

- A. The Contractor shall provide all arrangements, material and labor needed for obtaining temporary utility services.
- B. The Contractor is encouraged but not required to maintain a field office, but water and sanitation facilities must be provided for the Contractor's employees and subcontractors.
- C. Make all connections to the utility purveyor's requirements and in accordance with code requirements; remove from site upon completion of all work or when directed.
- D. Providing Temporary Facilities:
 - 1. Provide temporary construction, devices, equipment, power and convenience utilities for use, convenience and safety of personnel engaged in the work of the contract.
 - 2. Provide wastewater bypass systems for both effluent and influent wastewater.
 - 3. Provide temporary utilities and access during construction to existing home owners at all times.

1.02 RELATED SECTIONS

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 11 00 – Summary of Work
 - 2. Section 01 70 00 - Execution and Closeout Requirements
 - 3. Section 31 32 11 – Soil Surface Erosion Control

1.03 REGULATIONS

- A. Health and Safety: Conform with "Safety Standards for Construction Work, Chapter 296-155 WAC" by State of Washington Department of Labor and Industries.
- B. Construction Codes: Comply with regulatory construction codes as applicable.
- C. Washington State Department of Health: Comply with all applicable codes for temporary sewer and water service.

1.04 TEMPORARY FACILITIES

- A. Temporary Electrical Light & Power:
 - 1. Provide all temporary lighting and power, including pole or poles, transformer if required, for construction purposes.
 - 2. Provide temporary connections to closest utility source.
 - 3. Provide all required extension cords, lighting outlets and power outlets (grounding type), lamps, and other required equipment and accessories necessary only for adequate temporary lighting and power for construction purposes.
 - 4. Remove temporary lighting and power equipment and their connections at completion of the work or sooner if approved or directed.

B. Water for Construction Purposes:

1. The Contractor is responsible for obtaining and providing water as required for the work and for testing.
2. If agreed, Contractor to make temporary connections with metered connection with backflow preventers to utility piping as required for the work and provide meter, piping, hoses, nozzles and other accessories required.
3. At completion, or before as directed, disconnect temporary connections and piping and remove from site.
4. Provide secure system to prevent unauthorized use during Contractor's absence.

C. Clean Water Testing

1. Prior to startup, the Contractor is responsible for obtaining water and piping water to each newly constructed structure as necessary for a clean water leakage test. Contractor shall completely fill to the rim each structure, and the structure shall remain filled for 2 hours with no visible leakage. Structures requiring clean water leakage test include:
 - a. Influent Pump Station
 - b. Flow Splitter
 - c. Train #3 (including Aeration Basins, Anoxic Basins, and Clarifier)
 - d. UV Disinfection Channel #1 and #2
 - e. Digester
 - f. 2W Water system Pump Station Wet Well

D. Sanitary Facilities:

1. The Contractor shall provide temporary restroom services at the field office location, or other centrally located site. Service may be provided by contract service. Facilities shall be regularly serviced and maintained, and kept reasonably clean. Facilities shall be promptly removed at the conclusion of the work.

E. Drinking water:

1. Provide from proven safe source, for all those connected with the work in accordance with WISHA and Health Department requirements.
2. Pipe and transport in such manner as to keep it clean and fresh; serve in single containers or provide sanitary drinking fountains.

F. Residential and Commercial Access:

1. Provide access to residential homes and commercial facilities (wastewater treatment plant and water treatment plant facilities) at all times.
2. Provide access to the area at all times for emergency and service vehicles.
3. Provide through access to emergency service vehicles to the helicopter pad.

G. Equipment Storage

1. Contractor is fully responsible for safe storage of all materials and equipment.
2. Provide all fences, gates, locks, covers, weather protection, surveillance, etc. to assure safe storage.
3. Protect all materials and equipment from the weather.

H. Temporary Wastewater Systems

Contractor shall maintain continuous oversight of bypass pumping operations, and other temporary process equipment, at all times using either onsite personnel or the implementation of an alarming system with an automatic cellular notification system, which informs the Contractor when an alarm is in progress.

The Contractor shall have personnel available for call out during problems or emergencies. Call out personnel shall be familiar and experienced with the bypass system, and sufficiently qualified to make necessary repairs or adjustments to ensure the operation performs suitably when backups or overflows occur. Call out personnel shall be available 24-hours per day, 7 days per week and able to respond on-site within 30 minutes of receiving an alarm.

Additionally, backup alarms and 'Normal' or 'Running' equipment status shall be wired to the District's existing control system at the WWTP. Programming and other work related to the PLC and SCADA for these temporary status and alarm monitoring will be provided by the PLC/SCADA Programmer and will be considered incidental to work required for this project.

I. First Aid

1. In accordance with requirements of 296-24 WAC, furnish personnel trained in first aid and certified as approved by Washington Department of Labor and Industries.

1.05 MISCELLANEOUS PROVISIONS

A. Cleaning Up:

1. General: The Contractor and each subcontractor at all times shall keep the premises free from accumulation of waste materials or rubbish caused by his operations. Clean up work areas as required at the end of each day's work.
2. Trash removal: Remove all trash and debris from site and dispose of at Contractor's expense. Allow no debris, broken or open cartons, or other refuse to collect in the project or around it; allow no inflammable or hazardous materials to be stored on the site without approved protection precautions and procedures.
3. Street and parking area cleaning: Immediately clean all spilled material which results from the work of this contract and waste hauling operations; use motorized equipment and hand labor as required. Remove from streets, driveways or parking areas in time to prevent such materials from affecting traffic or clogging street drainage system; clean any drains contaminated.

- B. Noise Control: During the period of construction, provide satisfactory means, as approved by the Owner, of controlling noise originating from construction work and equipment.

- C. Dust Control: During the period of construction, provide satisfactory means of controlling dust and dirt, including application of water to control dust but not cause erosion.
- D. Temporary Erosion and Sedimentation Control: The Contractor shall provide sedimentation and erosion control in accordance with the Contract Plans and Section 31 32 11 SOIL SURFACE EROSION CONTROL in the Contract Specifications

1.06 DEBRIS CONTROL

- A. Cleaning during construction: Maintain all areas free of extraneous debris.
- B. Prevent accumulation of debris at construction site, storage and parking areas, and along access roads and haul routes.
- C. Keep storm sewers free of debris or extraneous materials.
- D. Offsite Cleanup: Prevent any leaking of materials from the vehicle used to haul offsite and clean haul routes daily.

1.07 POLLUTION CONTROL

- A. Provide all method, means and facilities required to prevent any contamination of adjacent areas, drainages, waterways, and/or tidal areas adjacent to the project site. Contractor will be expected to respond immediately to any spills and to take whatever measures are necessary to prevent further contamination and clean up accidental contamination. Contractor will be solely responsible for any and all costs of clean up in the event of discharge (of any kind). In the event that the Contractor is slow in responding, the Owner may elect to pay for clean-up costs directly, and all costs incurred from this, including labor, overhead, materials, management, etc., will be deducted from the next pay request.
- B. Provide methods, means, and facilities required to prevent contamination of soil, water, or atmosphere. Allow no discharge of noxious substances from construction operations.
- C. Provide systems for control of atmospheric pollutants in accordance with Federal/State/Local published rules and regulations.

1.08 BARRIER REQUIREMENTS

- A. During construction, the Contractor shall at all times maintain satisfactory and substantial temporary fencing, railing, barricades or steel plates at all excavations, obstructions or other hazards. All such barriers shall have warning signs or lights as necessary for safety.

PART 2. PRODUCTS (NOT USED)

PART 3. EXECUTION (NOT USED)

*****END OF SECTION*****

SECTION 01 60 00 – PRODUCT REQUIREMENTS

PART 1. GENERAL

1.01 DESCRIPTION OF SECTION

- A. General requirements for providing transportation, handling, storage, and protection of materials and equipment.
- B. Contractor's options in selection of products and manufacturers, and procedures for consideration of proposed substitutions.
- C. All material and equipment incorporated into the work:
 - 1. Shall be new, free from defects and of equal or superior quality as specified herein and on the drawings.
 - 2. Shall be the products of established manufacturers regularly engaged in the fabrication of such equipment.
 - 3. Shall comply with the size, type and quality specified and shall be designed for use in the particular application.
 - 4. Shall be designed, fabricated and assembled in accordance with standard engineering and shop practice.
 - 5. Shall be complete with accessories, trim, finish, safety guards, and other devices and details needed for a complete installation and for intended use and effect.

1.02 RELATED SECTIONS

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 30 00 - Submittals
 - 2. Section 01 45 00 - Quality Control

1.03 MANUFACTURER'S INSTRUCTIONS

- A. Installation of all materials and equipment shall comply with manufacturer's printed instructions. The Contractor shall have the responsibility to distribute copies of such instructions to all parties involved in the installation, including the Owner. One complete set of instructions shall be maintained on the job site during installation and until completion.
- B. All materials and equipment shall be handled, installed, connected, cleaned, conditioned and adjusted in strict accordance with such instructions and in conformance with the specified requirements. The Owner should be immediately notified should job conditions or specified requirements conflict with the manufacturer's instructions.

1.04 TRANSPORTATION AND HANDLING

- A. All materials and equipment shall be transported and handled in such a manner as to prevent any damage.
- B. Deliveries of products shall be in accordance with construction schedules as to cause no delay in the work or to conflict with work and conditions at the site.

- C. Products shall be delivered in the manufacturer's original containers with identifying labels intact and legible. Where materials are specified to conform to ASTM, Federal or other reference specifications, the materials shall be delivered to the site bearing the manufacturer's label stating that the materials meet the requirement of such referenced specifications.
- D. Products shall be inspected immediately upon delivery to assure compliance with specified requirements and approved submittals and that products are properly protected and undamaged.
- E. The Contractor shall provide personnel and equipment to receive and unload products delivered to the site. No products shall be delivered to the site unless such forces are available.

1.05 STORAGE AND PROTECTION

- A. Contractor is fully responsible for safe storage of all materials and equipment.
- B. All products shall be stored in strict accordance with the manufacturer's instructions, with seals and labels intact and legible.
- C. All products shall be arranged in a neat order and protected from damage from the weather, traffic and construction operations. Easy access for periodic inspection shall be provided.

1.06 PRODUCTS AND SUBSTITUTIONS

- A. Products:
 - 1. Where available, provide standard products of types which have been produced and used previously and successfully on other projects and in similar application.
 - 2. Where additional amounts of a product, by nature of its application, are likely to be needed by Owner at a later date for maintenance and repair or replacement work, provide a standard, domestically produced product which is likely to be available to Owner at such later date.
 - 3. For Products specified only by a reference standard, the Contractor may select any product meeting that standard.
 - 4. Where the make or name of a material is specified in the written documents or on the drawings, it is to establish a quality standard in that particular field of manufacture. Requests for substitutions of materials of other makes or names must be submitted to the Owner and must receive favorable written response from the Owner prior to ordering, furnishing or installing the proposed substitution item.
- B. Requests for Substitutions:
 - 1. For a period of thirty (30) days after the Contract Date, the Owner will consider written requests from the Contractor for substitution of Products.
 - 2. Requests for each Product substitution shall be submitted separately. Requests for substitutions will be received and considered when revisions to contract documents are not required, and the product or material is in keeping with the general intent of the Contract Documents.

3. A request for substitution by the Contractor constitutes a representation that the Contractor:
 - a. Will provide the same warranties or bonds for the substituted item as for the Product specified.
 - b. Will coordinate the installation of an accepted substitution into the work and make all other changes as required to make the work complete in all respects.
 4. Submit six (6) copies of requests for substitutions, fully identified for Product or method being replaced by substitution, including related specification section and drawing number(s), and fully documented to show compliance with requirements for substitutions.
 5. Include product data/drawings, description of methods, samples where applicable, Contractor's detailed comparison of significant qualities between specified item and proposed substitution, statement of effect on construction time and coordination with other affected work, cost information or proposal, and Contractor's statement to the effect that proposed substitution will result in overall work equal-to-or-better-than work originally indicated.
 6. The contractor agrees to pay all Engineering costs accruing as a result of checking and/or redesign due to substitutions. These costs will be charged to the Contractor and will be considered incidental to the contract price.
- C. Owner's Review
1. Within two weeks of receipt of request, or within one week of receipt of requested additional information or documentation (whichever is later), the Owner will notify the Contractor of either his acceptance or his rejection of the proposed substitution. Rejection will include statement of the reasons for rejection (non-compliance with the requirements for requested substitutions, or other reasons as detailed.)

PART 2. MATERIALS (NOT USED)

PART 3. EXECUTION (NOT USED)

*****END OF SECTION*****

SECTION 01 66 00 – PRODUCT STORAGE AND HANDLING REQUIREMENTS

PART 1. GENERAL

1.01 DESCRIPTION OF SECTION

- A. General requirements for providing transportation, handling, storage, and protection of materials and equipment.
- B. Contractor's options in selection of products and manufacturers, and procedures for consideration of proposed substitutions.
- C. All material and equipment incorporated into the work:
 - 1. Shall be new, free from defects and of equal or superior quality as specified herein and on the drawings.
 - 2. Shall be the products of established manufacturers regularly engaged in the fabrication of such equipment.
 - 3. Shall comply with the size, type and quality specified and shall be designed for use in the particular application.
 - 4. Shall be designed, fabricated and assembled in accordance with standard engineering and shop practice.
 - 5. Shall be complete with accessories, trim, finish, safety guards, and other devices and details needed for a complete installation and for intended use and effect.

1.02 RELATED SECTIONS

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 30 00 - Submittals
 - 2. Section 01 45 00 - Quality Control

1.03 MANUFACTURER'S INSTRUCTIONS

- A. Installation of all materials and equipment shall comply with manufacturer's printed instructions. The Contractor shall have the responsibility to distribute copies of such instructions to all parties involved in the installation, including the Owner. One complete set of instructions shall be maintained on the job site during installation and until completion.
- B. All materials and equipment shall be handled, installed, connected, cleaned, conditioned and adjusted in strict accordance with such instructions and in conformance with the specified requirements. The Owner should be immediately notified should job conditions or specified requirements conflict with the manufacturer's instructions.

1.04 TRANSPORTATION AND HANDLING

- A. All materials and equipment shall be transported and handled in such a manner as to prevent any damage.
- B. Deliveries of products shall be in accordance with construction schedules as to cause no delay in the work or to conflict with work and conditions at the site.
- C. Products shall be delivered in the manufacturer's original containers with identifying labels intact and legible. Where materials are specified to conform to ASTM, Federal or other reference specifications, the materials shall be delivered to the site bearing the

manufacturer's label stating that the materials meet the requirement of such referenced specifications.

- D. Products shall be inspected immediately upon delivery to assure compliance with specified requirements and approved submittals and that products are properly protected and undamaged.
- E. The Contractor shall provide personnel and equipment to receive and unload products delivered to the site. No products shall be delivered to the site unless such forces are available.

1.05 STORAGE AND PROTECTION

- A. Contractor is fully responsible for safe storage of all materials and equipment.
- B. All products shall be stored in strict accordance with the manufacturer's instructions, with seals and labels intact and legible.
- C. All products shall be arranged in a neat order and protected from damage from the weather, traffic and construction operations. Easy access for periodic inspection shall be provided.

1.06 PRODUCTS AND SUBSTITUTIONS

- A. Products:
 - 1. Where available, provide standard products of types which have been produced and used previously and successfully on other projects and in similar application.
 - 2. Where additional amounts of a product, by nature of its application, are likely to be needed by Owner at a later date for maintenance and repair or replacement work, provide a standard, domestically produced product which is likely to be available to Owner at such later date.
 - 3. For Products specified only by a reference standard, the Contractor may select any product meeting that standard.
 - 4. Where the make or name of a material is specified in the written documents or on the drawings, it is to establish a quality standard in that particular field of manufacture. Requests for substitutions of materials of other makes or names must be submitted to the Owner and must receive favorable written response from the Owner prior to ordering, furnishing or installing the proposed substitution item.
- B. Requests for Substitutions:
 - 1. For a period of thirty (30) days after the Contract Date, the Owner will consider written requests from the Contractor for substitution of Products.
 - 2. Requests for each Product substitution shall be submitted separately. Requests for substitutions will be received and considered when revisions to contract documents are not required, and the product or material is in keeping with the general intent of the Contract Documents.
 - 3. A request for substitution by the Contractor constitutes a representation that the Contractor:
 - a. Will provide the same warranties or bonds for the substituted item as for the Product specified.

- b. Will coordinate the installation of an accepted substitution into the work and make all other changes as required to make the work complete in all respects.
 - 4. Submit six (6) copies of requests for substitutions, fully identified for Product or method being replaced by substitution, including related specification section and drawing number(s), and fully documented to show compliance with requirements for substitutions.
 - 5. Include product data/drawings, description of methods, samples where applicable, Contractor's detailed comparison of significant qualities between specified item and proposed substitution, statement of effect on construction time and coordination with other affected work, cost information or proposal, and Contractor's statement to the effect that proposed substitution will result in overall work equal-to-or-better-than work originally indicated.
 - 6. The contractor agrees to pay all Engineering costs accruing as a result of checking and/or redesign due to substitutions. These costs will be charged to the Contractor and will be considered incidental to the contract price.
- C. Owner's Review
- 1. Within two weeks of receipt of request, or within one week of receipt of requested additional information or documentation (whichever is later), the Owner will notify the Contractor of either his acceptance or his rejection of the proposed substitution. Rejection will include statement of the reasons for rejection (non-compliance with the requirements for requested substitutions, or other reasons as detailed.)

PART 2. MATERIALS (NOT USED)

PART 3. EXECUTION (NOT USED)

*****END OF SECTION*****

SECTION 01 70 00 – EXECUTION AND CLOSEOUT REQUIREMENTS

PART 1. GENERAL

1.01 DESCRIPTION OF SECTION

- A. Specific administrative procedures, and closeout submittals at substantial completion and at final acceptance of the work.
- B. Requirements for record documents and start-up procedures.
- C. The listing of procedures and submittals is given generally as a checklist for the Contractor's convenience. The Owner reserves the right to add to this list. This list is not an exhaustive listing of either all applicable laws or of the provisions of any law.

The Contractor shall comply with all contract requirements prior to contract closeout. Specific administrative procedures, and closeout submittals at substantial completion and at final acceptance of the work.

1.02 RELATED SECTIONS

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 33 00 – Submittal Procedures

1.03 SUBSTANTIAL COMPLETION

- A. Prior to submitting for substantial completion, the Contractor shall have:
 - 1. Delivered tools, spare parts, extra stocks of materials, and similar physical items to Owner.
 - 2. Made final changeover of locks and transmit keys to Owner.
 - 3. Completed start-up testing of systems, and performed instructions for Owner's operating/maintenance personnel. Discontinued (or change over) and removed from project site temporary facilities and services.
 - 4. Provided record information to the owner of the as-constructed facilities.
 - 5. Completed final cleaning up requirements, including but not limited to, touch-up of marred surfaces, grading, installation of handrails, etc.
 - 6. Provided O&M Manuals for all equipment, controls, telemetry and SCADA.
 - 7. Provide Compliance Documentation per 2015 Washington State Energy Code (WSEC) close out requirements per section C103.6.3 as follows:

C103.6.3 Compliance Documentation. All energy code compliance forms and calculations shall be delivered in one document to the building owner as part of the project record drawings, manuals, or as a standalone document. This document shall include the specific energy code year utilized for compliance determination for each system, NFRC certificates for the installed windows, list of total area for each NFRC certificate, the interior lighting power compliance path (building area, space by space) used to calculate the lighting power allowance. For projects complying with Section C401.2 item 1, the documentation shall include:

 - (i) The envelope insulation compliance path (prescriptive or component performance).

- (ii) The completed code compliance forms, and all compliance calculations including, but not limited to those required by section C402.1.5, C403.2.12.1, C405.4, and C405.5.

- B. When the Contractor considers the work to be substantially complete, he shall submit to the Owner:
 - 1. Written notice that the work, or designated portion thereof, is substantially complete. (The term "substantially complete" shall be defined as in accordance with the WSDOT General Specifications and Sections 00 72 00 and 00 73 00a of these documents).
 - 2. List of items to be completed or corrected and reasons for being incomplete. The failure to include any items on such list does not alter the responsibility of the Contractor to complete all work in accordance with the Contract Documents.
 - 3. Progress payment request coincident with or first following date claimed, show either 100% completion for portion of work claimed as "substantially complete", or list incomplete items and the value of the incomplete work.
 - 4. Submit statement showing accounting of changes to the Contract Sum.
 - 5. Specific warranties, workmanship/maintenance bonds, maintenance agreements, final certification and similar documents.
 - 6. Obtain and submit releases enabling Owner's full and unrestricted use of the work and access to services and utilities, including (where required) certificate of occupancy permits, operating certificates, and similar releases.
 - 7. Record (as-built) drawings, project manual, manual of materials, operation and maintenance manuals, and similar final record information.
- C. Upon receipt of Contractor's request, the Owner will either proceed with inspection or advise Contractor of prerequisites not fulfilled. Following initial inspection, Owner will either prepare certificate of substantial completion, or advise Contractor of work which must be performed prior to issuance of certificate; and repeat inspection when requested and assured that work has been substantially completed. Results of completed inspection will form the initial "punch list" for final acceptance.
- D. When the Engineer, on the basis of an inspection, concurs that the work is substantially complete, he will:
 - 1. Prepare and deliver to the Contractor a certificate of Substantial Completion accompanied by the Contractor's list of items to be completed or corrected. The Certificate of Substantial Completion shall state the responsibilities of the Contractor for security, maintenance, heat, damages to the work and insurance and shall fix the time within which the Contractor shall complete the items listed therein. Warranties and guarantees required by the Contract Documents shall commence on the Date of Substantial Completion.
 - 2. The Certificate of Substantial Completion is submitted to the Contractor for their written acceptance of their responsibilities as stated therein.

1.04 FINAL INSPECTION

- A. When the Contractor considers the work to be complete, he shall submit written notice to the Owner that the work has been completed and inspected in compliance with the Contract Documents including punchlist items, and equipment and systems have been tested and are operational; and requesting a contract completion inspection.
- B. When the Engineer, on the basis of an inspection, concurs that the work is acceptable under the Contract Documents, he will notify the Contractor in writing and request the Contractor to provide remaining submittals.
- C. Should the Engineer determine that the work is not acceptable under the Contract Documents:
 - 1. The Engineer will promptly notify the Contractor in writing giving the reasons therefor.
 - 2. The Contractor shall remedy the deficiencies in the work and submit a new written notice for final inspection to the Owner.

1.05 FINAL PAYMENT

- A. When the Contractor has satisfied all requirements of this section and all other conditions of the Contract Documents, the Contractor may submit a final Application for Payment. Should the Owner determine the Work acceptable under the Contract Documents and the Agreement fully performed, he will promptly issue a final Certificate for Payment stating that to the best of his knowledge, the Work has been completed in accordance with the terms and conditions of the Contract Documents and that the entire balance due the Contractor, and as noted in the final certificate, is due and payable.
- B. The accumulated retainage shall not be paid until the Contractor submits to the Owner:
 - 1. Affidavit that all payrolls, bills for materials and equipment, and other indebtedness connected with the work for which the Owner might in any way be responsible, have been paid or otherwise settled.
 - 2. Release of Lien. One will be required from each lien holder who has duly filed a notice of claim with the Owner. If any liens remain unsatisfied after the expiration of the statutory lien period, the Contractor shall refund the Owner all amounts that the Owner may be compelled to pay in discharging such lien including all costs and reasonable attorney's fees.
 - 3. State Department of Revenue form that all taxes have been paid.
 - 4. State Department of Labor and Industry affidavit of wages paid.
 - 5. State Department of Employment Security Contractor release.
- C. The making of final payment shall constitute a waiver of all claims by the Owner except those arising from:
 - 1. Unsettled liens or disputes.
 - 2. Faulty or defective work appearing after Substantial Completion under the project guarantee and equipment warranty period.
 - 3. Failure of the work to comply with the requirements of the Contract Documents.

4. Terms of any special warranties required by the Contract Documents.

The acceptance of final payment shall constitute a waiver of all claims by the Contractor except those previously made in writing and identified by the Contractor as unsettled at the time of the final Application for Payment.

1.06 FIELD TESTS AND ADJUSTMENTS

- A. All mechanical and electrical equipment, as required under the separate section headings, shall be tested by the Contractor to the satisfaction of the Engineer before any facility is put into operation. Tests shall be as specified herein and shall be made to determine whether the equipment has been properly assembled, aligned and connected. Any changes, adjustments or replacements required to make the equipment operate as specified shall be performed by the Contractor as part of the Work.
- B. At least 14 days before the time allowed in the construction schedule for commencing testing and start-up procedures, the Contractor shall submit to the Engineer details of the procedure proposed for testing and start-up of all mechanical and electrical equipment, except when such procedures have been covered in the specifications.
- C. The Contractor's testing and start-up procedures shall include detailed descriptions of all preoperational electrical, mechanical and instrumentation testing work. Each control device, item of mechanical, electrical and instrumentation equipment, and all control circuits shall be considered in the testing procedures, which shall be designed, in a stepwise, logical sequence to ensure that all equipment has been properly serviced, aligned, connected, calibrated and adjusted prior to operation. The Contractor is advised that failure to observe these precautions may place the acceptability of the subject equipment in question, and he may either be required to demonstrate that the equipment has not been damaged, or replace it as determined by the Engineer. Testing procedures shall be designed to duplicate as nearly as possible all conditions of operations, and shall be carefully selected to ensure that the equipment is not damaged. Once the testing procedures have been accepted by the Engineer, the Contractor shall produce checkout, alignment and adjustment, and calibration sign-off forms for each item of equipment, which shall be used in the field by the Contractor and the Engineer jointly, to ensure that each item has been properly installed and tested. All testing must be performed in the presence of the Engineer.
- D. During the testing of the mechanical, instrumentation and electrical equipment, the Contractor shall make available, as necessary, representatives of the manufacturers of all the various pieces of equipment, or other qualified persons, who shall instruct the Owner's personnel in the operation and care thereof. Instructions shall include written step-by-step operation and trouble-shooting procedures with a complete description of all necessary test equipment and all protective device settings. Upon completion of testing, the manufacturer's representative shall provide the Engineer with a letter stating that the specific piece of equipment has been properly installed and tested and will satisfy the requirements of the Contract Documents.

1.07 RECORD (AS-BUILT) DRAWING INFORMATION

- A. During the construction period, the Contractor shall maintain a complete set of prints for the sole purpose of maintaining a day-by-day record of installed information. This information shall include, but not limited to: the size and location of all concealed or underground piping, conduit, and ductwork; all approved deviations from the specifications and drawings; the location of any visible objects relocated due to interference's or requested relocations submitted and approved on shop drawings. Such relocations shall be dimensioned.

- B. Addenda, bulletins, field orders, and change orders shall be posted and referenced in the record set of prints.

1.08 RECORD PROJECT MANUALS

- A. Maintain one copy of the Contract Documents, including addenda, change orders and similar modifications issued in printed form during construction, and mark-up variations (of substance) in actual work in comparison with text of the Project Manual and modifications as issued.
- B. Give particular attention to substitutions, selection of options, and similar information on work where it is concealed or cannot otherwise be readily discerned at a later date by direct observation. Not related record drawing information and product data, where applicable.

1.09 MISCELLANEOUS RECORD SUBMITTALS

- A. Refer to other sections of these specifications for requirements of miscellaneous record-keeping and submittals in connection with actual performance of the work. Immediately prior to date(s) of substantial completion, complete miscellaneous records and place in good order, properly identified and bound or filed, ready for continued use and reference. Submit to Engineer for Owner's records.

1.010 FINAL CLEAN-UP

- A. At the completion of the work, the Contractor shall leave the premises in a neat and unobstructed condition, ready for Owner occupancy. The buildings shall be left in a dust free condition and all equipment and materials in perfect repair and adjustment.
- B. After all trades have completed their work and just before final acceptance and occupancy by owner, thoroughly clean all surfaces of project. Clean lighting fixtures and electrical equipment, including washing and polishing lenses inside and out. Wash and polish all exposed metal surfaces. Broom clean exterior paved areas and rake clear other surfaces of the grounds. All waste building materials, pipe, etc. shall be removed from the site and disposed of.

*****END OF SECTION*****

SECTION 01 91 00 – COMMISSIONING

PART 1. GENERAL

1.01 GENERAL

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 33 00 – Submittals Procedure
 - 2. Section 01 70 00 – Execution and Closeout Requirements
- B. Description of Section:
 - 1. Specific commissioning procedures and start-up requirements.
 - 2. Requirements for equipment settings and field verification.
 - 3. The listing of procedures and field tests is given generally as a checklist for the Contractor's convenience. The Owner reserves the right to add to this list. This list is not an exhaustive listing of all applicable settings and requirements to achieve specified results.
 - 4. The Contractor shall comply with all contract requirements prior to contract closeout. Specific administrative procedures, and closeout submittals at substantial completion and at final acceptance of the work.

1.02 COMMISSIONING INFLUENT PUMP STATION EQUIPMENT:

- A. The Submersible Pump equipment supplier is to provide a minimum of five (5) days of start-up supervision and operator training. Supervision and training duties to include the following:
 - 1. Inspect installation for concurrence to design
 - 2. Check the amperage draw from the pumps
 - 3. Make sure equipment is installed to allow easy access.
 - 4. Make sure all operations of the controls are working properly
 - 5. Review Operation and Maintenance Manual with personnel
 - 6. Demonstration and Training on controls

1.03 COMMISSIONING BIOLOGICAL TREATMENT (TRAIN 3) PROCESS EQUIPMENT:

- A. The Biological Treatment process equipment supplier is to provide a minimum of two (2) weeks of start-up supervision and operator training. Supervision and training duties to include the following:
 - 1. Inspect installation for concurrence to design
 - 2. Check the amperage draw from all motors (clarifier drive, mixers, etc.)
 - 3. Make sure equipment is installed to allow easy access.
 - 4. Make sure all operations of the controls, including valves, dissolved oxygen sensors, pH sensors, and air lift pumps are working properly

5. Verify weirs and baffles are installed level and configured for smooth and even flow throughout the clarifier perimeter.
6. Review Operation and Maintenance Manual with personnel
7. Demonstration and Training on controls

1.04 COMMISSIONING BLOWER EQUIPMENT:

- A. The blower equipment supplier is to provide a minimum of two (2) days of start-up supervision and operator training. Supervision and training duties to include the following:
 1. Inspect installation for concurrence to design
 2. Check the amperage draw from the blower drives
 3. Make sure equipment is installed to allow easy access.
 4. Make sure all operations of the controls are working properly
 5. Review Operation and Maintenance Manual with personnel
 6. Demonstration and Training on controls

1.05 COMMISSIONING UV DISINFECTION SYSTEM

- A. The UV disinfection system equipment supplier is to provide a minimum of two (2) days of start-up supervision and operator training. Supervision and training duties to include the following:
 1. Inspect installation for concurrence to design
 2. Check the amperage draw from the UV modules
 3. Make sure equipment is installed to allow easy access.
 4. Make sure all operations of the controls are working properly including intensity monitoring.
 5. Review Operation and Maintenance Manual with personnel
 6. Demonstration and Training on controls

1.06 COMMISSIONING SLUDGE PUMP EQUIPMENT (WAS AND DIGESTED SLUDGE):

- A. The Sludge Pump (peristaltic pump) equipment supplier is to provide a minimum of two (2) days of start-up supervision and operator training. Supervision and training duties to include the following:
 1. Inspect installation for concurrence to design
 2. Check the amperage draw from the pumps
 3. Make sure equipment is installed to allow easy access.
 4. Make sure all operations of the controls are working properly
 5. Review Operation and Maintenance Manual with personnel
 6. Demonstration and Training on controls

1.07 COMMISSIONING DEWATERING EQUIPMENT

- A. The dewatering Equipment suppliers are to each provide a minimum of one (2) days of start-up supervision and operator training. Supervision and training duties to include the following:
1. Inspect installation for concurrence to design
 2. Check the amperage draw from all motors (conveyor, press, pumps, etc.)
 3. Make sure equipment is installed to allow easy access.
 4. Make sure all operations of the controls are working properly
 5. Review Operation and Maintenance Manual with personnel
 6. Demonstration and Training on controls

1.10 COMMISSIONING 2W WATER SYSTEM EQUIPMENT:

- A. The 2W Water System supplier is to provide a minimum of one (1) day of start-up supervision and operator training. Supervision and training duties to include the following:
1. Inspect installation for concurrence to design
 2. Check the amperage draw from the pumps.
 3. Make sure equipment is installed to allow easy access.
 4. Verify control is set to maintain constant pressure (60 to 90 psi), and pumps will turn on/off and ramp speed up/down, as needed, to maintain constant pressure setpoint as flow varies from 0 to 50 gpm.
 5. Verify automatic valve (at air gap location) is operating correctly
 6. Make sure all operations of the controls, including floats, are working properly
 7. Review Operation and Maintenance Manual with personnel
 8. Demonstration and Training on controls

1.11 COMMISSIONING MAGNETIC FLOW METER EQUIPMENT:

- B. The Flow Meter equipment supplier is to provide a minimum of two (2) days of start-up supervision and operator training. Supervision and training duties to include the following:
1. Inspect installation for concurrence to design
 2. Check the amperage draw from the flow meters.
 3. Make sure equipment is installed to allow easy access.
 4. Verify all units are configured for specified environments. Some units may be submerged in water, and it is critical that units are sealed and rated for environments.

5. Make sure all operations of the controls are working properly
6. Review Operation and Maintenance Manual with personnel
7. Demonstration and Training on controls

*****END OF SECTION*****

SECTION 02 41 00 - DEMOLITION

PART 1. GENERAL

1.01 DESCRIPTION

- A. Related Work Specified Elsewhere
 - 1. Scopes of Bids: Section 00 24 13
 - 2. Summary of Work: Section 01 11 00
- B. Description of System: The work covered by this section includes the furnishing of all labor, equipment, and materials necessary for the demolition, removal, rehabilitation and equipment salvage of all construction as specified herein and as shown on the drawings.

1.02 JOB CONDITIONS

- A. All removed equipment, materials, and debris, unless otherwise noted or requested by the Owner, shall become the property of the Contractor. The Contractor shall deliver all items to be salvaged (as directed by the Owner), to the storage area designated by the Owner. **The items that the Owner has selected for retention include the following:** any miscellaneous valves, piping and electronic components as directed by Owner.
- B. Protection: Ensure the safe passage of persons around the area of demolition. Conduct operations to prevent injury to adjacent buildings, structures, other facilities, and people and livestock.

PART 2. PRODUCTS

Not Used.

PART 3. EXECUTION

3.01 DEMOLITION

- A. Pollution Controls:
 - 1. Use water sprinkling, temporary enclosures, and other suitable methods to limit the amount of dust and dirt rising and scattering in the air to the lowest practical level.
 - 2. Comply with governing regulations pertaining to environmental protection.
- B. Removal Requirements:
 - 1. Provide complete removal and disposal of all structures identified for demolition. All pipes connected to abandoned structures are to be plugged/grouted in an approved manner, preventing any potential water and/or sewer leaks. Salvage items as directed by the Owner.
 - 2. Proposed wastewater equipment is to be purchased and on-hand, prior to removal of specified structures. Contractor is to coordinate removal/demolition with Owner's staff (a minimum of 2 week) prior to all removal/demolition work.

- C. Structures to be Removed (demolished):
1. Existing septic tank.
 2. Existing pump sump pump station.
 3. Existing miscellaneous valves, piping, concrete, and accessories as detailed on the Contract Plans. (salvage items as directed by Owner's staff).
 4. See electrical drawings and specifications for additional demolition items/requirements.

3.02 DISPOSAL OF DEMOLISHED MATERIALS

- A. General. Remove from the site debris, rubbish, and other materials resulting from demolition operations. Burning of removed materials from demolished structures will not be permitted on the site. Comply with all federal, state and local regulations regarding hauling and disposal.
- B. Removal. Transport materials removed from demolished structures and dispose of at a legal disposal site.

*****END OF SECTION*****

SECTION 02 83 00 – CHAIN LINK FENCES AND GATES

PART 1 - GENERAL

1.01 DESCRIPTION OF WORK

- A. Work consists of furnishing all labor, materials, and incidentals necessary to erect all 6-foot high galvanized chain-link fence and gates with 3 strand barbed wire on top (7-foot total height) at the location shown on the drawings. Barbed wire is to be aligned in vertical position on gates and aligned in an outward angled position on rest of fencing. Construction is to provide a rigid, taut fence closely conforming to the surface of the ground. It is noted that bottom rails are to be included for preventing access under fence fabric.
- B. Work included
 - 1. Fabric, line posts, end, corner and pull posts, gate posts, gate frames, top rails, bottom rails, and post braces and accessories.

1.02 RELATED SECTIONS

- A. Section 05500 - Metal Fabrications

1.03 REFERENCE STANDARDS

- A. American Society for Testing and Materials (ASTM)
 - 1. A 53 Standard Specification for Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses.
 - 2. A 153 Standard Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware.
 - 3. A 392 Standard Specification for Zinc-Coated Steel Chain-Link Fence Fabric.

1.04 SUBMITTALS

- A. Three samples, approximately 6 inches long, or 6 inches square, of fabric material (standard galvanized), post section and typical accessories.
- B. Submit shop drawings showing fence height, type of fabric, and location and size of posts and gates, including details of post tops, rails, braces, foundations, footings, gate posts, hinges, frames, latches, ties and other accessories.

1.05 QUALIFICATION OF INSTALLER

- A. Installer must be experienced in fence installations and must examine conditions under which fence and gates are to be installed. The Contractor shall notify the Engineer in writing of improper conditions of work, and shall not proceed with work until unsatisfactory conditions have been corrected.

1.06 QUALITY ASSURANCE

- A. Certifications

PART 2 - PRODUCTS

2.01 FENCES, POSTS, RAILS AND BRACES

- A. All steel tubular members shall comply with provisions of ASTM A 53, Schedule 40, for weight and coating.

2.02 FABRIC

- A. Chain link fabric to conform to ASTM A 392, No. 9 gage wire, 2-inch mesh, Class II galvanizing.
- B. Fabric galvanized after weaving.
- C. Fabric knuckled at bottom selvage and twisted and barbed at top.

2.03 LINE POSTS

- A. Posts of galvanized steel.
- B. Posts round in section, with 2.375-inch outside diameter and weighing 3.65 lb/ft.

2.04 END, CORNER PULL POSTS

- A. Posts of galvanized steel.
- B. Posts round in section, with 2.875-inch outside diameter and weighing 5.79 lb/ft.

2.05 GATE POSTS

- A. Posts of galvanized steel.
- B. Gate leaves over 6 feet - 0 inch and up to and including 13 feet - 0 inch wide: 4 inches O.D. Schedule 40 pipe and weighing 9.1 lb/ft.

- C. Gate leaves over 13 feet 0 inch and up to and including 18 feet 0 inch: 6-5/8-inch O.D. Schedule 40 pipe and weighing 18.97 lb/ft.

2.06 GATE FRAMES FOR CHAIN LINK FENCING

- A. Frames of galvanized steel.
- B. Frames round in section, with 1.9-inch outside diameter, and weighing 2.72 lb/ft.
- C. Frames shall have intermediate members and/or diagonal truss rods for gate leaves more than 8 feet wide.
- D. Gate frame joints shall be made by welding or by means of heavy fittings making rigid and watertight connections.

2.07 TOP RAILS, BOTTOM RAILS, AND POST BRACES

- A. Top rails, bottom rails, and post braces of galvanized steel.
- B. Top rails, bottom rails, and post braces round in section, with 1.66-inch outside diameter, and weighing 2.27 -lb/ft.

2.08 BARBED WIRE

- A. Barbed wire shall be double strand twisted 12-1/2 gauge galvanized with 14 gauge, 4 point barbs spaced on approximately 5-inch centers. Extension arms to accommodate barbed wire shall withstand a 250-pound pulldown load from end of arm.

2.09 ACCESSORIES AND ATTACHMENTS

- A. Stretcher bars: Galvanized steel 3/16 by 3/4-inch in cross section, or equivalent cross section with length equal to full height of fabric.
- B. Truss rods: Galvanized steel, 3/8-inch-diameter, or equivalent cross section, and shall have suitable adjustment.
- C. Post tops: Caps of pressed galvanized steel. Provide with a hole suitable for through-passage of the top rail. Fit snugly to the post, have means for attaching securely to the post and exclude moisture from tabular posts.
- D. Gates swing: Swing type, complete with latches, stops, keepers, hinges, locks and fabric. Fabric to match fence. Hinges of adequate strength to support gate and not twist or turn under action of gate. Latches of plunger bar type and full gate height located in a manner that will engage the gate stop. Forked latches used for single gates less than 10 feet wide. Latches shall provide for locking. Stops shall consist of a flush plate with anchor placed in concrete to engage the plunger bar

of the latch. Other approved types of stops may be used for single gates less than 10 feet wide. Keepers shall be substantial devices for securing and supporting the free end of the gate in open position.

- E. Rolling Gates: Non motorized rolling type, complete with latches, stops, keepers, locks and fabric. Fabric to match fence.
- F. Top rail and bottom rail couplings: Outside sleeve type at least 6 inches long. At least 20% of the couplings shall have an internal heavy spring to take up expansion and contraction.
- G. Brace wire, tie wire, and tension wire:
 - 1. Galvanized wire meeting requirements of ASTM A 12 1, Class 3 coating.
 - 2. Unless otherwise designated, size of wire shall not be smaller than the following:

Tension wire	No. 7
Brace wire	No. 9
Tie wires or clips for fastening field fence to steel posts	No. 12
 - 3. Tie wires for chain-link fence of size and type recommended by manufacturer, but not smaller than No. 9 for post ties or No. 12 for rail and brace ties. Equivalent galvanized steel clips or aluminum wire or clips may be used as accepted by the Engineer.
- H. Galvanizing: All pipe sections galvanized after fabrication shall be in accordance with ASTM A 53. All other items incidental to erection of fence except fabric and wire fabric ties galvanized after fabrication in accordance with ASTM A 153. Wire fabric ties will have not less than 0.8 ounce of zinc per square foot.

2.10 CONCRETE FOOTINGS

- A. Concrete shall be mixed and placed in strict accord with Section 03300.

PART 3 - EXECUTION

3.01 CLEARING AND GRADING

- A. Contractor shall perform such clearing and grading as necessary to construct fence to required alignment and provide a reasonably smooth ground profile at the fence line.

3.02 POST ASSEMBLIES

- A. End, corner, gate, and pull or intermediate anchor posts placed at designated locations.

- B. Posts securely braced and holes filled with concrete. Form not required for post encasement

3.03 HORIZONTAL DEFLECTION

- A. At points of deflection where fence changes alignment by more than 5 degrees provide a post brace and truss rod in each fence panel to the post located at the angle point.
- B. Footings for all posts located at points where the change in alignment exceeds 5 degrees shall be constructed as specified for end posts.

3.04 LINE POSTS

- A. Line posts spaced at not more than 10-foot centers.

3.05 POST BRACES

- A. A brace and truss assembly shall support each gate, comer, pull, or end post for chain link fencing. Brace shall extend to each adjacent line post at mid-height of fabric. Truss shall extend from line post back to gate, corner, pull, or end post.

3.06 FABRIC

- A. Fabric shall not be erected until 5 days after the time of setting the posts in concrete. Fabric shall be fastened to line posts with clips or bands spaced approximately 12 inches apart and to top rail with bands or tie wires at approximately 24-inch intervals. Pull fabric taut and tie to posts, rails and tension wires. Install fabric on security side of fence and anchor to framework so that fabric remains in tension after pulling force is released.

3.07 TENSION WIRES

- A. Tension wires installed at bottom of fabric before stretching fabric and tied to each post with wire ties or clips.

3.08 ELECTRICAL GROUNDS

- A. Chain-link fence which crosses beneath any primary electrical power transmission line, other than a secondary feeder line for individual customer service, shall be properly grounded. Grounding shall consist of placing one ground rod at point of crossing and one 25 to 50 feet in each direction from the crossing.
 - 1. Chain-link fence erected adjacent to and within 50 feet of a primary power line shall be grounded by placing ground rods at not more than 500-foot intervals.

2. Each applicable straight section of fence shall have at least one ground. Engineer may require installation of an additional ground at terminus of a section of fence or at other locations near areas of pedestrian traffic.
3. Ground rod shall be connected to fence.

END OF SECTION

SECTION 02 92 00 - LANDSCAPING

PART 1 - GENERAL

1.1 DESCRIPTION OF WORK

- A. Work in this section shall include all labor, equipment and materials necessary for re-establishing grass vegetation in areas disturbed during construction. Work shall include stripping, excavation, hauling, stockpiling, placing topsoil, placing compost, and hydro-seeding.

1.2 RELATED SPECIFICATIONS

- A. Section 31 20 00 – *Earthwork*
- B. Section 31 32 11 – *Soil surface Erosion Control*

1.3 SEEDING GUARANTEE

- A. During the one (1) year guarantee period should any seeded area show signs of failure such as dead or dying areas of grass, bare spots, dead or dying plants, etc., the Contractor shall repair or replace all deficient items to the satisfaction of the Engineer.
- B. All graded areas not seeded or paved by October 15 shall be covered with two (2) inch depth straw to prevent erosion. Straw to be provided and installed by the Contractor. Do not perform planting or seeding when ground is frozen, snow covered, muddy or in an otherwise unsatisfactory condition. When unforeseen conditions detrimental to plant growth are encountered, such as adverse drainage conditions, obstructions, compaction, or toxified soils, notify the Engineer before proceeding.

1.03 SUBMITTALS

- A. The Contractor shall submit product specifications and installation recommendations for all materials to be provided under this section.
- B. Submit seed vendor's blue tag certification for required grass seed mixture, indicating percentage by weight, and percentages of purity, germination, and weed seed for each grass species.
- C. Upon seeded areas acceptance, submit written maintenance instructions recommending procedures for maintenance of seeded areas.

PART 2 - PRODUCTS

2.1 TOPSOIL MATERIAL

- A. Topsoil shall conform to Section 9.14.1(2) of the Standard Specifications.

2.2 GRASS SEED

- A. Seed shall conform to Table 1 and Table 2 below.
- B. Seed mix for R.I. Galleries and Site Restoration Areas (including constructed slopes, disturbed areas, and field behind recreation center) shall be:

Seed of the following composition, proportion, and quality shall be applied at a rate of 8 pounds per 1000 square feet:

Table 1	
Common Name	% by Weight
Nobility Perennial Ryegrass	30
Amazing GS Perennial Ryegrass	30
Longfellow II Chewings Fescue	20
Gibilbralter Creeping Red Fescue	20

This mixture is provided by Sunmark Seeds under the name DOT Multipurpose Mixture. Seeds shall be certified "Weed Free" indicating there are no noxious or nuisance weeds in the seed.

2.3 FERTILIZER

- A. Fertilizer shall be a granular, non-burning product composed of not less than 50% organic, slow acting, guaranteed analysis professional fertilizer. Seeded area starter fertilizer containing 20% nitrogen, 26% phosphoric acid, and 6% potash by weight, or similar approved composition applied at a rate of 6.5 lbs/1000 SF.

2.4 WOOD FIBER MULCH FOR HYDRO-SEEDING

- A. Hydro-seeding to be applied to all areas disturbed and/or regraded (which will not be protected with quarry spalls, gravel, and/or pavement) during construction. Commercially prepared wood fiber mulch specifically manufactured for hydro-seeding application shall be used.
- B. Dispersing agents may be added at Contractor's option provided that the additive is not harmful to the mixture.

2.5 WATER

- A. The Contractor shall furnish water as required for planting and establishing vegetation in seeded areas. Provide all necessary hoses, equipment, attachments, and accessories for adequate watering of seeded areas.

PART 3 - EXECUTION

3.1 SEEDING

- A. Inspect all subgrades for debris and adverse drainage conditions. Remove all debris including rocks 1-inch in diameter and larger, sticks, roots, sod and other deleterious material. Notify the Owner of any grades or conditions which might create adverse or undesirable drainage patterns.
- B. Smoothly blend and feather topsoil into existing surrounding grades. Rake or lightly harrow topsoil until the soil is friable and of uniform texture and satisfactory for seed placement.
- C. After seeding, topsoil shall be rolled for compaction and shall be minus ½-inch below all adjacent paved or graveled surfaces. Irrigate immediately until soil is damp to about 6".
- D. The hydro-seeding operation shall include the installation of seed, fertilizer, mulch, and tackifier with a tracer to verify uniform application.
- E. Hydro-seeding shall be done in accordance with WSDOT Spec. 8-01.3(2)B.
- F. Seed immediately after preparation of seed bed. Seeding may occur August 15 to October 15.
- G. Seed shall be applied at a rate listed above.
- H. Mulch shall be applied at a rate of 2,000 pounds per acre.

3.2 MAINTENANCE AND WATERING

- A. Patch, repair and re-seed any and all damaged or barren areas observed prior to final project acceptance at no additional cost to the Owner.
- B. The Contractor shall protect and care for all seeded areas until fully established and hearty. Care shall include equipment and labor necessary to provide sufficient and continuous watering of all seeded areas until final acceptance.

END OF SECTION

SECTION 03 30 00 – CAST-IN-PLACE CONCRETE

PART 1. GENERAL

1.01 DESCRIPTION

- A. Work includes but is not limited to the following:
 - 1. All material, labor, and equipment to prepare subgrade, build forms, install rebar, and place concrete for all cast-in-place concrete shown on the Plans. Compensation for all costs associated with cast-in-place concrete shall be incidental to the lump sum bid items in which the work is identified.

1.02 RELATED SECTIONS

- A. Coordinate related work specified in other parts of the Project Manual, including but not limited to the following:
 - 1. Section 31 20 00 Earthwork
 - 2. Section 03 40 00 Precast Concrete

1.03 EXTRA WORK ITEMS (written authorization required)

- A. Overexcavation. See Section 31 20 00 Earth Moving.
- B. Structural Fill. See Section 31 20 00 Earth Moving.

1.04 REGULATORY REQUIREMENTS

- A. Comply with the requirements of Section 01 41 00 REGULATORY REQUIREMENTS and the following. Note that where conflicts may arise, comply with the most stringent requirement:
 - 1. WSDOT *Standard Specifications for Road, Bridge and Municipal Construction*, M41-10, 2022 Edition.
 - 2. Washington State Department of Ecology, *Stormwater Management Manual for Western Washington*, 2019.
 - 3. Codes and Standards listed in Section 1.05 below are incorporated as regulatory requirements by reference.

1.05 CODES AND STANDARDS

- A. Comply with the Drawing Structural Notes. They are part of the contract documents and take precedence over less specific requirements in these specifications.
- B. American Concrete Institute (ACI)
 - 1. ACI-301-16 – *Specifications for Structural Concrete*
 - 2. ACI-318-14 – *Building Code Requirements for Structural Concrete*
 - 3. ACI-117-10 – *Specification for Tolerances for Concrete Construction*
 - 4. ACI-347-04 – *Recommended Practice for Concrete Formwork*
 - 5. ACI-350R-06 – *Requirements for Environmental Engineering Concrete Structures*
 - 6. ACI-SP-066-04 – *ACI Detailing Manual*
 - 7. ACI-305.1-14 – *Specification for Hot Weather Concreting*
 - 8. ACI-306.1-90 – *Standard Specification for Cold Weather Concreting (Reapproved 2002)*

- C. Concrete Reinforcing Steel Institute (CRSI), *Manual of Standard Practice*, 27th Edition.
- D. International Code Council, *International Building Code*, 2018.

1.06 SUBMITTALS

- A. Submit shop drawings that conform to ACI-SP-066-04 detailing standards. Failure to conform to ACI detailing standards may constitute grounds for rejection without review, with resubmittal required.
- B. Submit mix design, to meet criteria per the Structural Notes.
- C. Material Certificates: Signed by manufacturer and Contractor; Submit in lieu of laboratory test reports as acceptable to Engineer, demonstrate compliance with requirements.
 - 1. Certification of admixtures compatibility by Contractor's approved technician.
- D. Mix Design: Written proposal for each concrete mix to meet criteria per the Structural Notes. Submit each design a minimum 15 working days prior to start of Work that includes the mix. Each mix design shall include following:
 - 1. All materials and admixtures and their proportions.
 - 2. Water-cement ratio, slump, aggregate grading, and maximum aggregate size.
 - 3. Evidence that mix design meets the strength requirements and other requirements. Compression and other test data (field experience method) or results of testing (trial batch method) used to establish mix proportions.
 - 4. State (WSDOT) materials sources for principal constituents, including cement and aggregate.
 - 5. Certificates signed by manufacturer and Contractor stating compatibility of admixtures
- E. The contractor shall submit a detailed plan for dealing with hot and cold weather work, prepared by a registered Professional Engineer. The plan shall be subject to review by the Engineer of Record. ACI 305 and 306 shall be the basis of the plan. At a minimum, the plan shall include
 - 1. Protect concrete with curing blankets or by other means for a minimum of 72 hours when temperatures are expected to be below 38 degrees F for more than 3 hours. The plan shall include specific measures for securing the protection against wind, rain, snow, or other inclement weather.
 - 2. When temperatures are expected to be above 85 degrees F for more than 12 hours, use curing compound or continuous wetting techniques to reduce cracking. The plan shall include specific measures dealing with hot, sunny, and windy weather.
- F. The contractor shall submit plans and calculations for formwork necessary at the Clarifier, digester and chlorine contact basin structures, sealed by a Washington State Structural Engineer. These shall comply to ACI 347R-14, Guide to Formwork for Concrete. The submittal of these documents is intended to show good faith in planning and execution of the work. It does not relieve the contractor of its responsibility for diligent and safe prosecution of the work at all times.
- G. The contractor shall submit plans indicating locations and types of all proposed construction, control and expansion joints. Spacing shall not exceed the maximum allowable spacing indicated on the plans. If control joint spacing limits are not indicated on the plans, control joint spacing shall not exceed 30 times the slab thickness unless approved in writing by the Engineer of Record.

1.07 QUALITY ASSURANCE

- A. Regulatory Requirements: See referenced Codes, ordinances and the like as well as Section 01 45 00 QUALITY CONTROL.
- B. Testing and Inspection. Owner will provide independent testing and inspection outside of the contract. Contractor shall coordinate and cooperate with Owner's independent testing and inspection agency. The Contractor shall give adequate notice of upcoming work to the testing agency and/or the on-site agent of the Engineer of Record. Owner's independent testing and inspection agency will perform the following tasks:
 - 1. Compaction and Moisture Control Tests on foundation subgrade prior to placement reinforcing.
 - 2. Reinforcing placement, lap length, cover, and bar sizes, concrete placement taking of test cylinders, and cylinder testing.

1.08 PROJECT SITE CONDITIONS

- A. Coordination: Notify the Engineer upon completion of preparatory work and of intended schedule for placement.
- B. Establish and maintain required lines and elevations.

PART 2. PRODUCTS

2.01 GENERAL

- A. Comply with "Quality Assurance" provisions, "References," and Specifications. Where these may be in conflict, the more specific requirements govern.
- B. All products to be installed in accordance with "Quality Assurance" provisions, "References," Specifications, and Manufacturer's directions. Where these may be in conflict, the more specific requirements govern.

2.02 FORM-FACING MATERIALS

- A. Smooth-Formed Finished Concrete: Form-facing panels that provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints. Applies to slabs and interior exposed concrete wall conditions and raised concrete stem walls.
- B. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit. Applies to concrete surface conditions hidden below grade.
- C. Form Liner Finish Concrete: Coordinate with drawings for locations, and Section 3.02 for installation requirements.

2.03 STEEL REINFORCEMENT

- A. See the Structural Notes and the Structural Drawings for reinforcing materials.
- B. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded-wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice."

2.04 CAST-IN-PLACE CONCRETE MATERIALS

- A. Aggregates: All aggregates shall be from Washington Department of Transportation approved sources and shall be free from marine life shells and residues of salt.
- B. Concrete: Mix design shall be per the Structural Notes (per Section 1.06, Submittals).
- C. Reinforcing steel shall be as per the Structural Notes.
- D. Joint Sealant. Joint Sealant shall be GREY IN COLOR approximating matching color of concrete and meet WSDOT Standard Specification 9-04.2.
- E. Premolded Joint Filler for Expansion Joints. Pro-Flex manufactured by Oscoda or approved; 100 percent recycled vinyl.
- F. Waterstop Hydrophilic bentonite or modified chloroprene rubber; Cetco Waterstop – RX101T Greenstreak Hydrotite CJ0725, or equal. Maintain 3” minimum cover from face of waterstop to face of concrete. Remove all dirt, coatings, and debris from concrete surface. Install waterstop per manufacturer’s instructions.
- G. Adhesive Doweling Accessories Accepted products shall be as stated in the Structural Notes in the Contract Plans.

2.05 MOISTURE RETARDERS FOR INTERIOR SLABS-ON-GRADE

- A. Sheet Vapor Retarder: ASTM E 1745, Class A. Include manufacturer's recommended adhesive or pressure-sensitive tape.
 - 1. Stegowrap Vapor Barrier, 15 mil sheeting and associated tape and accessories for complete installation.

2.06 CURING MATERIALS FOR INTERIOR SLABS-ON-GRADE

- A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.
- B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.
- C. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.
- D. Water: Potable.
- E. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, non-dissipating, certified by curing compound manufacturer to not interfere with bonding of floor covering.

2.07 RELATED MATERIALS

- A. Expansion- and Isolation-Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber.

PART 3. EXECUTION

3.01 EXAMINATION

- A. Verify installation conditions as satisfactory to receive work of this Section. Do not install until unsatisfactory conditions are corrected. Beginning work constitutes acceptance of conditions as satisfactory.
- B. Follow the plan for hot and cold weather work per Section 1.06.
 - 1. The Owner shall have authority to require such protections based on National Weather Service hourly forecasts.

3.02 FORMWORK INSTALLATION

- A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.
- B. Construct formwork of well-fitted, strong, rigid, and leak-proof materials so bulges, fins, or voids are not created in the finished work, and concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117.
- C. Chamfer exterior corners and edges of permanently exposed concrete.

3.03 EMBEDDED ITEM INSTALLATION

- A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
- B. Anchor bolts for steel columns, foundation bolts, and anchor rods for machinery (including but not limited to the skimmer on the clarifiers) shall be set to the tolerances of ANSI/AISC 303-16, Code of Standard Practice for Steel Buildings, Section 7.5 by a Registered Land Surveyor. The Owner's designated representative for construction shall conduct a survey of the as-built locations of the other embedded items and shall verify that all items covered in Section 7.5 meet the corresponding tolerances.

3.04 VAPOR-RETARDER INSTALLATION

- A. Sheet Vapor Retarders: Place, protect, and repair sheet vapor retarder according to ASTM E 1643 and manufacturer's written instructions.
 - 1. Lap joints 6 inches and seal with manufacturer's recommended tape.
 - 2. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.

3.05 STEEL REINFORCEMENT INSTALLATION

- A. Bar detailing not shown otherwise on drawings shall comply with the CRSI Manual of Standard Practice.
- B. Lap all bars 40 diameters unless shown otherwise on the drawings. Bars parallel to the line of the wall shall be continuous, i.e. lapped or with corner "L" bar laps, or otherwise terminated at the ends of the wall with 90-degree x 12 bar diameter hooks.
- C. Support reinforcing bars on chairs or other purpose-made devices so that they are securely held in place and maintain tolerances during placement and consolidation of concrete. Pulling up reinforcing to "float" in slabs as concrete is placed is unacceptable, and any such work is subject to rejection, demolition, and rework.

3.06 JOINTS IN BUILDING SLABS

- A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.
- B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by Architect or Engineer of Record.
- C. Contraction Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of concrete thickness as follows:
 - 1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint to a radius of 1/8 inch. Repeat grooving of contraction joints after applying surface finishes. Eliminate groover tool marks on concrete surfaces.
 - 2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inch-wide joints into concrete when cutting action does not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks.
- D. Isolation Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.

3.07 CONCRETE PLACEMENT

- A. All work associated with the manufacture, transport, and placement of concrete shall comply with References listed in parts 1.04 and 1.05 of this specification section.
 - 1. Contractor to plan for and accommodate any embedded items such as wall pipe, railings, ladders, hatches etc. which are not or may not be suitable for subsequent installation into cured concrete.
 - 2. This plant is designed for a long service life with a minimum of maintenance. Penetrations of concrete walls or slabs require prior approval to core drill through. It is the responsibility of the contractor to coordinate work and provide sleeves as required. Failure in coordination of piping with concrete work may result in rejection of work to be demolished and redone at no cost to the Owner.
- B. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections are completed.
- C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete is placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.
 - 1. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.

3.08 FINISHING FORMED SURFACES

- A. General: Vibrate to compact, screed, level, and tamp with a grid tamper to raise a thin mortar bed to the surface. Trowel after concrete has hardened sufficiently to prevent drawing moisture to the surface. Do not dust with dry materials.
- B. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie-holes, defects, and depressions 1/16-inch or larger in depth or width repaired and filled with mortar. The mortar shall consist of 1 part cement and 1-1/2 parts fine (passing No. 100 screen) mixed with enough water and an emulsified bonding agent to have the

consistency of a thick cream. The surfaces shall be brush sandblasted prior to filling holes to expose all holes near the surface. Thoroughly wet surfaces and rub mortar on with burlap, sponge rubber floats, or trowels while surface is damp. Wipe surface clean and moist cure. Remove fins and other projections that exceed specified limits on formed-surface irregularities.

1. Apply to concrete surfaces not exposed to public view.
- C. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Tie-holes, defects, and depressions 1/16-inch or larger in depth or width repaired and filled with mortar. The mortar shall consist of 1 part cement and 1-1/2 parts fine (passing No. 100 screen) mixed with enough water and an emulsified bonding agent to have the consistency of a thick cream. The surfaces shall be brush sandblasted prior to filling holes to expose all holes near the surface. Thoroughly wet surfaces and rub mortar on with burlap, sponge rubber floats, or trowels while surface is damp. Wipe surface clean and moist cure. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
1. Apply to concrete surfaces exposed to public view.
- D. Form Liner Finish: As cast texture imparted by reusable Form Liners with a minimum of joints.
- E. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces unless otherwise indicated.
- F. Sandblast: Provide light sand blasted finish at non-form liner wall finishes at exterior ramp walls.

3.09 FINISHING FLOORS AND SLABS

- A. General: Comply with ACI 302.1R recommendations for screeding, restraighening, and finishing operations for concrete surfaces.
1. Interior Floor Slabs: Steel trowel and install joints straight and true. Do not apply curing compounds. Damp cure only. Slope concrete slabs with Portland Cement Grout 1/8" per foot to sumps or as indicated on plans. Apply slip-resistant rake finish to wetwell bottoms.
 2. Sidewalks, Exterior Slabs on Grade and Curbs: Steel trowel and medium broom finish.
- B. Trowel Finish: After applying float finish, apply first troweling and consolidate concrete by hand or power-driven trowel. Continue troweling passes and restraighen until surface is free of trowel marks and uniform in texture and appearance. Grind smooth any surface defects that would telegraph through applied coatings or floor coverings.
1. Apply a trowel finish to surfaces exposed to view or to be covered with resilient flooring, carpet, ceramic or quarry tile set over a cleavage membrane, paint, or another thin-film-finish coating system.
 2. Finish and measure surface, so gap at any point between concrete surface and an unlevelled, freestanding, 10-ft.-long straightedge resting on two high spots and placed anywhere on the surface does not exceed 1/4 inch.
- C. Broom Finish: Apply a broom finish to exterior concrete platforms, steps, ramps, and elsewhere as indicated, including entire interior slab at UV/maintenance shop building, and at mudroom and at Lab Building.

1. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route. Coordinate required final finish with Architect before application.

3.010 CONCRETE PROTECTING AND CURING

- A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 305 for hot-weather protection during curing.
- B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.
- C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for remainder of curing period.
- D. Cure concrete according to ACI 308.1, by one or a combination of the following methods:
 1. All structural concrete shall be moist cured for a minimum of 90 days, unless otherwise instructed by Engineer.
 2. Moisture Curing
 - a. Building Slabs: Keep surfaces continuously moist for a minimum of seven (7) days.
 3. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period, using cover material and waterproof tape.
 - a. Curing for Building Slabs: Cure for a minimum of seven (7) days.
 4. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.
 - a. Removal where floor coverings will be installed: After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound does not interfere with bonding of floor covering used on Project.
 5. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hours later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

3.011 CONCRETE SURFACE REPAIRS

- A. Defective Concrete: Repair and patch defective areas when directed by Engineer of Record (EOR). Remove and replace concrete that cannot be repaired and patched to EOR's approval.

3.012 FIELD QUALITY CONTROL

- A. Special Inspections: Owner will engage a qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.

*****END OF SECTION*****

SECTION 03 40 00 – PRECAST CONCRETE

PART 1 - GENERAL

1.01 SECTION INCLUDES

- A. Work includes but is not limited to the following:
 - 1. All material, labor, and equipment to design, prepare, furnish, deliver, and install all precast concrete structures shown on the drawings, including but not limited to concrete vaults, top slabs, and the effluent wet well.
 - 2. Includes seals, coatings, or other related accessories per the drawings.

1.02 RELATED SECTIONS

- A. Coordinate related work specified in other parts of the Project Manual, including but not limited to the following:
 - 1. Section 31 20 00 Earthwork
 - 2. Section 03 30 00 Cast-in-Place Concrete

1.03 EXTRA WORK ITEMS (written authorization required)

- A. Structural Fill. See Section 31 20 00 Earthwork.

1.04 REFERENCES

- A. Comply with the requirements of Section 01 41 00 REGULATORY REQUIREMENTS, and as listed herein.
- B. Comply with the following references (C – D). Where conflicts may arise, comply with the more stringent requirements.
- C. WSDOT Standard Specifications for Road, Bridge and Municipal Construction, M 41-10, 2022 Edition.
- D. Washington State Department of Ecology 2019 Stormwater Management Manual for Western Washington.

1.05 CODES AND STANDARDS

- A. American Concrete Institute (ACI) ACI-318-14, Building Code Requirements for Structural Concrete.
- B. ACI 117-10 Specification for Tolerances for Concrete Construction. Note that where there may be conflicts regarding tolerances between this standard and WSDOT Standard Specifications for Road, Bridge and Municipal Construction, M 41-10, the more stringent will govern.
- C. ACI SP-066 2004, ACI Detailing Manual.
- D. Precast/Prestressed Concrete Institute (PCI) MNL 116 Manual for Quality Control for Plants and Production of Structural Precast Concrete Products, 4th Edition.
- D. PCI Design Handbook 7th Edition.

1.06 SUBMITTALS

- A. Shop drawings and calculations sealed by an engineer registered in Washington are required for all precast concrete structures. The calculations shall consider dead, live, soil, hydraulic, and other loadings, as well as the stresses of transportation and handling. Minimum force requirements given in the contract documents are unfactored loads.
- B. Shop drawings shall include lifting loops, hangers, and other accessories are subject to the requirements of paragraph A, above. In addition, the locations of embeds and anchor bolts shall be coordinated to eliminate conflicts.
- C. Shop drawings shall conform to ACI SP-066 detailing standards.
- D. Submit mix design, showing adequate strength to meet the loadings per the calculations (paragraph A, above). The mix shall meet all criteria per the Structural Notes.
- E. Material Certificates: Signed by manufacturer and Contractor; Submit in lieu of laboratory test reports as acceptable to Engineer, demonstrate compliance with requirements.
 - 1. Certification of admixtures compatibility by Contractor's approved technician.
- F. Mix Design: Written proposal for each concrete mix and strength required submit 15 days prior to start of Work. The mix design shall list the following
 - 1. All materials and admixtures and their proportions.
 - 2. Water-cement ratio, slump, and aggregate grading.
 - 3. Evidence that mix design meets the strength requirements: Compression test data (field experience method) or results of testing (trial batch method) used to establish mix proportions.

1.07 QUALITY ASSURANCE

- A. The producer shall submit certificates showing compliance with the National Precast Concrete Association (NPCA) quality control program, for the last five years.
- B. The Owner and the Engineer reserve the right to inspect or observe fabrication at the precast plant.

PART 2 - PRECAST STRUCTURES

2.01 GENERAL

- A. Comply with "Quality Assurance" provisions, "References," and Specifications. Where these may be in conflict, the more stringent requirements govern.

2.02 PRECAST STRUCTURES

- A. Sizes and configurations are noted on drawings. Do not use alternate sizes without approval of the Civil Engineer.
- B. Design all components of the system for the loads given on the structural notes. See Section 1.06 of this specification section regarding submittals.

- C. Details of precast framing and connections on the contract documents depict the intent. Mild reinforcing shown on the drawings is the minimum required; determine additional reinforcing per the above requirements.
- D. Top Slab Schedule:
1. Influent Pump Station Top Slab – Top slab to be constructed per the details included on the Contract Plans and fitted with a new, 3'-0" x 5'-0", aluminum access hatch with safety grates, Flygt "FLED-13" light duty aluminum hatch or approved equal.
- E. Precast Wet Wells and Vaults
1. Precast manhole and vault components shall have 28-day compressive strength of 5,000 psi. Rebar shall be ASTM A615 Grade 60. Mesh reinforcing, if any, shall be ASTM A-185 Grade 65. Design shall be per ACI 318-14 and ASTM C-857 "Minimum Structural Design Loading for Underground Precast Concrete Utility Structures."
 2. Precast vaults shall be as described in the following table:

Vault Name	Reference Drawing	Min. Interior Dimensions	Precast Vault Manufacturer and Model	Hatch Specifications	Hatch Manufacturer and Model
2W Plant Water Pump Station	C4.2	8' Inner Diameter, depth as needed	Oldcastle Infrastructure 96" Diameter Wet Well	(48" x 30"), H-20 rated aluminum hatch w/ safety grate system	LW Products Model HD-3C
Influent Pump Station	C4.1	8' Inner Diameter, depth as needed	Oldcastle Infrastructure 96" Diameter Wet Well	48" x 30", H-20 rated aluminum hatch w/ safety grate system	LW Products Model HD-3C
Influent Flow Meter Valve Vault	C7.4	4'L x 4'W, depth as needed	Oldcastle Infrastructure 4686-LA	36" x 36", H-20 rated Opening, Single Leaf Hatch	Halliday Products H1R3636
Effluent Flow Meter Valve Vault	C7.4	4'L x 4'W, depth as needed	Oldcastle Infrastructure 4686-LA	36" x 36", H-20 rated Opening, Single Leaf Hatch	Halliday Products H1R3636
Plants 2 & 3 Flow Meter Vault	C7.4	4'L x 4'W, depth as needed	Oldcastle Infrastructure 4686-LA	36" x 36", H-20 rated Opening, Single Leaf Hatch	Halliday Products H1R3636
Influent Pump Station Meter Vaults	C4.1	4'L x 4'W, depth as needed	Oldcastle Infrastructure 4686-LA	36" x 36", H-20 rated Opening, Single Leaf Hatch	Halliday Products H1R3636

F. Cement Grouts:

1. Anchor Bolt and Plate Grout for Non-Mechanical Equipment:
2. Cement grout to fill anchor bolt pockets, handrail pockets, and under non-mechanical equipment base plates shall be non-shrinking, non-staining, non-metallic grout with a minimum 6,000 psi compressive strength in seven days. Allow for ¾" grout where not shown otherwise. Clean all contact surfaces and place grout completely under plate, without voids.
3. Portland Cement Grout shall be used to provide slopes between wall and slabs shall be one part Portland Cement to two parts sand with an expansive agent to limit shrinkage. Sufficient water shall be added for placement while maintaining a minimum 4,000 psi 28-day compressive strength.

G. Base Plate and Mechanical Pad Grout:

1. Epoxy grout to be used for mechanical equipment base pads such as pump bases. Epoxy shall be a moisture-tolerant three-component mix: two-part epoxy resin and aggregate. Epoxy grout shall be specifically labeled for long-term support and precision alignment of machinery and conform to ASTM C 881, TYPE IV, Grade 1, such as Master Builders Technologies MASTERFLOW 648 CP PLUS or Sika Corporation SIKADUR 42 GROUT PAK.

H. Sealant Gaskets:

1. The sealant gaskets shall be pre-formed, continuous rope form plastic material, protected by a removable two-piece wrapper.
2. Sealing compound shall be reinforced hydrocarbon resins blended with plasticizing compounds and reinforced with inert mineral filler. The sealing compound shall have no solvents, irritating fumes or obnoxious odors.
3. The adhesive and cohesive strength of the sealant gaskets shall not be dependent upon oxidizing, evaporating or chemical action.
4. Sealant Gasket shall conform to Federal Specification SS-S-210.
5. Sealant gaskets shall be RAM-NEK as manufactured by K.T. Snyder Company, Inc. of Houston, TX; QUIKSEAL as supplied by Associated Concrete Products, Santa Ana, CA; or approved equal.

I. External Precast Vault Joint Sleeves:

1. Joint sleeves shall be heat-shrinkable sleeves constructed of irradiated and cross-linked polyethylene impermeable backing coated with protective heat-activated adhesive.
2. Joint sleeves shall be capable of bonding to primed concrete, metal, and fiberglass surfaces.
3. Joint sleeve material shall be compatible with concrete, steel, iron, and fiberglass.
4. Joint sleeves shall be supplied with a separate closure seal to secure sleeve in place during installation and seal overlap area.
5. Joint sleeve primer of the same manufacturer as the joint sleeve material shall be provided to prime concrete and steel surfaces prior to joint sleeve installation.
6. Joint sleeves shall be WrapidSeal Manhole Encapsulation System as manufactured by Canusa, Division of Shaw Resource Services, Inc. of The Woodlands, TX.; or approved equal.

PART 3. EXECUTION

3.01 EXAMINATION

- A. Verify installation conditions as satisfactory to receive work of this Section. Do not install until unsatisfactory conditions are corrected. Beginning work constitutes acceptance of conditions as satisfactory.

3.02 PRECAST CONCRETE - INSTALLATION

- A. Install in accordance with "Quality Assurance" provisions, "References," Specifications, and Manufacturer's directions. Where these may be in conflict, the more stringent requirements govern.
- B. Where indicated on the drawings, set precast units on cast-in-place bases level and true.
- C. Apply primer to joint surfaces in accordance with manufacturer's instructions. Make all joints watertight with sealant gaskets.
- D. Apply primer and heat-shrinkable joint sleeves in accordance with manufacturer's instructions. Ensure surfaces are clean, dry, and free of frost, surface rust, foreign objects, sharp edges, and projections that could damage manhole encapsulation systems. The Contractor shall provide a minimum of 3-days' notice prior to the beginning application of the joint sleeves. Joint sleeve installer shall be experienced with the required installation techniques and have attended a minimum of 1 day of training at the manufacturer's facility or on-site with manufacturer's representative.
- E. Backfill with approved gravel backfill material. Compact the backfill material to 95% of relative density from the pipe bedding and base slab up to final finish grade, over an area defined as being within a distance of 4 feet from the exterior walls of the vault.
- F. Accurately locate vault frames to within 1/4-inch vertical elevation. Coordinate the activities of all trades so that this tolerance is achieved.
- G. Install the vault hatches and frames. Machine as necessary to obtain a solid fit, without rattling under load.

*****END OF SECTION*****

SECTION 05 50 00 – METAL FABRICATIONS

1.01 DESCRIPTION

- A. Work consists of furnishing labor, materials and equipment for the fabrication and erection of metal fabrications shown which are not part of structural steel or other metal systems specified in this document.
- B. Section includes embedded and nonembedded metal work, aluminum shapes, clip angles, rungs, tubing, rods guides, inserts, brackets, anchor bolts, beams, bracing, steel framing and supports for mechanical and electrical equipment, shelf angles, metal ladders/ship ladders, metal pipe and tube railings, and similar items.
- C. Products furnished, but not installed, under this Section:
 - 1. Anchor bolts, steel pipe sleeves, slotted-channel inserts, and wedge-type inserts indicated to be cast into concrete or built into unit masonry.
 - 2. Steel weld plates and angles for casting into concrete for applications where they are not specified in other Sections.

1.02 RELATED SECTIONS

- A. Coordinate related work specified in other parts of the Project Manual, including but not limited to the following:
 - 1. Section 03 30 00 Cast-in-Place Concrete
 - 2. Section 13 12 20 Metal Building Systems
 - 3. Section 09 90 00 Painting

1.03 REFERENCES

- A. American Society for Testing and Materials (ASTM):
 - 1. A 36 – *Specification for Structural Steel*
 - 2. A 153 – *Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware*
 - 3. A 307 – *Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile*
 - 4. A 325 – *Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength*
 - 5. A 490 – *Specification for High-Strength Steel Bolts, Classes 10.9 and 10.9.3, for Structural Steel Joints*
 - 6. B 6 – *Specification for Zinc (Slab Zinc)*
- B. American Welding Society (AWS).
 - 1. D1.1 – *Structural Welding Code*
- C. The Society for Protective Coatings (SSPC).
 - 1. *Painting Manual*

1.04 SUBMITTALS

- A. Submittals shall be in accordance with Section 01 33 00. In addition, the following specific information shall be provided:
 - 1. Shop Drawings showing fabrication and installation details for all fabricated items including all connections, field joints, and finishes.
 - a. Include plans, elevations, sections, and details of metal fabrications and their connections. Show anchorage and accessory items.
 - 2. Product Data for the following:
 - a. Nonslip aggregates and nonslip-aggregate surface finishes.
 - b. Premanufactured metal assemblies.
 - c. Paint products.
 - d. Grout.
 - 3. Samples for Verification for each type and finish of extruded nosing and tread.
 - 4. Qualification Data for qualified professional engineer.
 - 5. Mill Certificates signed by manufacturers of stainless-steel certifying that products furnished comply with requirements.
 - 6. Paint Compatibility Certificates from manufacturers of topcoats applied over shop primers certifying that shop primers are compatible with topcoats.
 - 7. Welder's certificates.

1.05 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. Metal Fabrications: Meet applicable requirements of IBC Chapter 27.
- B. Welding Qualifications: Qualify procedures and personnel according to the following:
 - 1. AWS D1.1/D1.1M, "Structural Welding Code – Steel"
 - 2. AWS D1.6, "Structural Welding Code – Stainless Steel"

1.06 PROJECT CONDITIONS

- A. Field Measurements: Verify actual locations of walls and other construction contiguous with metal fabrications by field measurements before fabrication.

1.07 COORDINATION

- A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers' written recommendations to ensure that shop primers and topcoats are compatible with one another.
- B. Coordinate installation of anchorages and steel weld plates and angles for casting into concrete. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

2.01 MATERIALS FOR BUILDINGS

A. Metals for Buildings, General

1. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For metal fabrications exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes. Edges shall be free of burns.

B. Ferrous Metals for Buildings

1. Recycled Content of Steel Products: Provide products with average recycled content of steel products so postconsumer recycled content plus one-half of preconsumer recycled content is not less than 25 percent.
2. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.
3. Stainless-Steel Sheet, Strip, and Plate: ASTM A 240/A 240M or ASTM A 666, Type 304 and Type 316L.
4. Stainless-Steel Bars and Shapes: ASTM A 276, Type 304 and Type 316L.
5. Rolled-Steel Floor Plate: ASTM A 786/A 786M, rolled from plate complying with ASTM A 36/A 36M or ASTM A 283/A 283M, Grade C or D.
6. Rolled-Stainless-Steel Floor Plate: ASTM A 793.
7. Abrasive-Surface Floor Plate: Steel plate with abrasive granules rolled into surface or with abrasive material metallurgically bonded to steel.
 - a. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - (i) a. IKG Industries, a division of Harsco Corporation; Mebac.
 - (ii) b. SlipNOT Metal Safety Flooring, a W. S. Molnar company; SlipNOT.
8. Steel Tubing: ASTM A 500, cold-formed steel tubing.
9. Steel Pipe: ASTM A 53/A 53M, standard weight (Schedule 40) unless otherwise indicated.
10. Slotted Channel Framing: Cold-formed metal box channels (struts) complying with MFMA-4.
 - a. Size of Channels: 1-5/8 by 1-5/8 inches (41 by 41 mm) as indicated on plans.
 - b. Material: Galvanized steel, ASTM A 653/A 653M, commercial steel, Type B or structural steel, Grade 33 (Grade 230), with G90 (Z275) coating; 0.064-inch (1.6-mm) nominal thickness.
 - c. Material: Cold-rolled steel, ASTM A 1008/A 1008M, commercial steel, Type B, or structural steel, Grade 33 (Grade 230); minimum thickness; coated with rust-inhibitive, baked-on, acrylic enamel or hot-dip galvanized after fabrication.
11. Cast Iron: Either gray iron, ASTM A 48/A 48M, or malleable iron, ASTM A 47/A 47M, unless otherwise indicated.

C. Nonferrous Metals for Buildings

1. Aluminum Plate and Sheet: ASTM B 209 (ASTM B 209M), Alloy 6061-T6.
2. Aluminium Extrusions: ASTM B 221 (ASTM B 221M), Alloy 6063-T6.
3. Aluminum-Alloy Rolled Tread Plate: ASTM B 632/B 632M, Alloy 6061-T6.
4. Aluminum Castings: ASTM B 26/B 26M, Alloy 443.0-F.
5. Bronze Plate, Sheet, Strip, and Bars: ASTM B 36/B 36M, Alloy UNS No. C28000 (muntz metal, 60 percent copper).

D. Fasteners for Buildings

1. General: Unless otherwise indicated, provide Type 304 or Type 316 stainless-steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, at exterior walls. Select fasteners for type, grade, and class required.
 - a. Provide stainless-steel fasteners for fastening aluminum.
 - b. Provide stainless-steel fasteners for fastening stainless steel.
 - c. Provide stainless-steel fasteners for fastening nickel silver.
 - d. Provide bronze fasteners for fastening bronze.
2. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A 325, Type 3 (ASTM A 325M, Type 3); with hex nuts, ASTM A 563, Grade C3 (ASTM A 563M, Class 8S3); and, where indicated, flat washers.
3. Stainless-Steel Bolts and Nuts: Regular hexagon-head annealed stainless-steel bolts, ASTM F 593 (ASTM F 738M); with hex nuts, ASTM F 594 (ASTM F 836M); and, where indicated, flat washers; Alloy Group 1 (A1).
4. Anchor Bolts: ASTM F 1554, Grade 36, of dimensions indicated; with nuts, ASTM A 563; and, where indicated, flat washers.
 - a. Hot-dip galvanize or provide mechanically deposited, zinc coating where item being fastened is indicated to be galvanized.
5. Eyebolts: ASTM A 489.
6. Machine Screws: ASME B18.6.3 (ASME B18.6.7M).
7. Lag Screws: ASME B18.2.1 (ASME B18.2.3.8M).
8. Wood Screws: Flat head, ASME B18.6.1.
9. Plain Washers: Round, ASME B18.22.1 (ASME B18.22M).
10. Lock Washers: Helical, spring type, ASME B18.21.1 (ASME B18.21.2M).
11. Anchors, General: Anchors capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing according to ASTM E 488, conducted by a qualified independent testing agency.

12. Cast-in-Place Anchors in Concrete: Either threaded type or wedge type unless otherwise indicated; galvanized ferrous castings, either ASTM A 47/A 47M malleable iron or ASTM A 27/A 27M cast steel. Provide bolts, washers, and shims as needed, all hot-dip galvanized per ASTM F 2329.
13. Post-Installed Anchors: Torque-controlled expansion anchors and chemical anchors.
 - a. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B 633 or ASTM F 1941 (ASTM F 1941M), Class Fe/Zn 5, unless otherwise indicated.
 - b. Material for Exterior Locations and Where Stainless Steel is Indicated: Alloy Group 1 (A1) stainless-steel bolts, ASTM F 593 (ASTM F 738M), and nuts, ASTM F 594 (ASTM F 836M).
14. Slotted-Channel Inserts: Cold-formed, hot-dip galvanized-steel box channels (struts) complying with MFMA-4, 1-5/8 by 7/8 inches (41 by 22 mm) by length indicated with anchor straps or studs not less than 3 inches (75 mm) long at not more than 8 inches (200 mm) o.c. Provide with temporary filler and tee-head bolts, complete with washers and nuts, all zinc-plated to comply with ASTM B 633, Class Fe/Zn 5, as needed for fastening to inserts.

E. Miscellaneous Materials for Buildings

1. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.
2. Shop Primers: Provide primers that comply with Division 09 painting Sections.
3. Universal Shop Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with MPI#79 and compatible with topcoat.
 - a. Use primer containing pigments that make it easily distinguishable from zinc-rich primer.
4. Epoxy Zinc-Rich Primer: Complying with MPI#20 and compatible with topcoat.
 - a. Products: Subject to compliance with requirements, provide the following available products that may be incorporated into the Work include, but are not limited to, the following:
5. Galvanizing Repair Paint: High-zinc-dust-content paint complying with SSPC-Paint 20 and compatible with paints specified to be used over it.
6. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187.
7. Nonshrink, Metallic Grout: Factory-packaged, ferrous-aggregate grout complying with ASTM C 1107, specifically recommended by manufacturer for heavy-duty loading applications.
8. Nonshrink, Nonmetallic Grout: Factory-packaged, non-staining, noncorrosive, nongaseous grout complying with ASTM C 1107. Provide grout specifically recommended by manufacturer for interior and exterior applications.
9. Concrete: Comply with requirements in Division 03 Section "Cast-in-Place Concrete" for normal-weight, air-entrained, concrete with a minimum 28-day compressive strength of 3000 psi (20 MPa).

2.02 MATERIALS FOR OTHER APPLICATIONS

A. Steel and Iron

1. Wide Flange Shapes: ASTM A 992.
2. Carbon Steel Shapes (excl. wide flange shapes), Plates, and Bars: ASTM A 36.
3. Steel Pipe: ASTM A 53 Grade B, Schedule 40 unless indicated otherwise on drawings.
4. Steel Tubing, square or rectangular: ASTM A 500 Grade B, welded and seamless.
5. Steel Sheet: ASTM A 366 if not galvanized, ASTM A 526 if galvanized.
6. "Diamond Plate": ASTM A 786 with A 36 mechanical properties.

B. Aluminum

1. Pipe Railings, Sleeves and Balusters: ASTM B 221, Schedule 40 and 80 Pipe, Alloy 6066-T6, finish AAM32.
2. Pipe Railing Sleeve Flange: ASTM B 209, Alloy 6061-T6 finish AAM32.
3. Extruded Structural Shapes and Tread Supports: ASTM B 221 Alloy 6066-T6, null finish.
4. Fabrication Weld balusters to railings.

C. Stainless Steel: ASTM A 240 Type 304 and Type 316.

D. Galvanizing

1. Where items are called to be galvanized, galvanize by the hot-dip process in conformance with ASTM A 123, A 153, and A 525. Two ounces/square foot minimum.

2.03 FASTENERS FOR OTHER APPLICATIONS

A. Metal to metal, unless noted otherwise on drawings:

1. Steel to steel: ASTM A 307, Grade A, hex head, galvanized unless neither steel item is galvanized.
2. Aluminum to aluminum: ASTM F 593, Type 304 stainless steel.
3. Other: Stainless steel; ASTM F 593, Type 304 stainless steel.

B. Metal to concrete:

1. Mechanical Anchors (For use on non-water bearing concrete structures only, such as building slab and concrete walkway locations. Locations to be approved by Engineer):
 - a. Anchor bolts, for ungalvanized steel: ASTM A 307.
 - b. Anchor bolts, all other applications: Stainless steel, ASTM A 193, Grade B8M, Class 2
 - c. Expansion bolts: Stainless steel per Structural Notes.

2. Chemical Anchors (For use on all water bearing concrete structures. In addition, to be used at all locations where available spacing and/or edge offset is less than 3.5”):
 - a. Adhesive anchors: See Structural Notes. Adhere strictly to MFR’s recommended instructions and procedures.
- C. Metal to wood:
 1. Bolts: ASTM A 307, Grade A, hex head, hot-dip galvanized, except stainless steel where noted on Drawings.
 2. Lag screws: Square head type, Federal specification FFB-561, zinc coated for steel, stainless steel for all other.
 3. Metal to hollow construction: Toggle bolts, Federal Specification FF-B-588, zinc-coated (galvanized).
- D. Powder-actuated fasteners: May be used for all types of fastenings where pullout and shear values will not exceed 40% of manufacturer’s specified values; galvanized for steel; stainless steel for all other.
- E. Washers:
 1. Provide washers of the same material and finish as the bolt or lag bolt in the following locations:
 - a. Under all nuts.
 - b. Under bolt heads where bolt material has a yield stress more than 110% of the yield stress of the material being fastened.
 2. Under all high-strength bolts, A325 or A490 or stainless steel, use ASTM F844 washers under head of the bolt and the nut.
- F. Lock washers: Provide spring steel helical lock washers of the same finish as the bolt under nuts and bolt heads of connections subject to vibration.
- G. Nuts: Nuts shall conform to the recommendation of the applicable bolt specification and shall be of the same material as the bolt.

2.04 WELDING

- A. Perform welding in accordance with pertinent recommendations of the American Welding Society. Use low hydrogen electrodes and methods recommended by manufacturer of material being welded. Type, size and spacing of welds in accordance with reviewed Shop Drawings.
 1. Welding shall be done by operators who are State-licensed. The quality of welding shall conform to AWS “Code for Arc Welding in Building Construction”, Section 4 Workmanship.

- B. Welds behind finished surfaces: Use methods to minimize distortion and discoloration of finished surface.
- C. Remove flux and slag from both sides of welds.
- D. Grind accessible welds smooth.
- E. Buff or polish welded surfaces which will be exposed to view in the finished work to match and blend with adjacent parent material.
- F. Complete welding before galvanizing, anodizing or painting.

2.05 GALVANIZING REPAIR PAINT

- A. High zinc dust content paint, Military Specification MIL-P-21035.

2.06 SHOP PRIMED SURFACES AND ASSEMBLIES

- A. Steel Surfaces
 - 1. Shop prime steel with primer for paint system as specified in Section 09 90 00, painting schedule.
 - 2. If metal surfaces are shop primed with a coating other than the one specified in Section 09 90 00, the Contractor shall submit in writing to the Engineer a statement that the primer is compatible with the required top coatings specified in Section 09 90 00. Should the entire coating system fail down to bare metal as a result of the shop primer, the Contractor shall be responsible for removal of existing coatings, and re-priming and re-painting at no additional cost to the Owner.

2.07 SURFACE TREATMENT - FIELD APPLIED

- A. Galvanized surfaces: Items that must be drilled or cut in the field as approved by Engineer shall be coated with galvanizing repair paint before installation.
- B. Painted surfaces: Touch-up damaged shop primed surfaces. Provide and apply field priming and finish painting as specified in Section 09 90 00, painting schedule.
- C. Field preparation and painting provided under Section 09 90 00.

2.08 PIPE HANGER RODS

- A. Rods shall be cut and threaded to fit and galvanized after fabrication. The use of "all thread-rod" will not be accepted unless it is stainless steel in accordance with ASTM A 193 and used throughout the whole project.

2.09 STAIR NOSINGS

- A. Concrete stairs shall have 4-inch wide aluminum abrasive nosings 8 inches less than the width of the tread.
- B. Stair nosings shall be Amcolun, Type DSA4 as manufactured by the American Abrasive Metals Company; Super Grit, Type 242 as manufactured by Wooster Products, Inc., or equal. Color as selected by Engineer.

2.010 CORROSION PROTECTION

- A. Ferrous metals that are not entirely embedded in concrete shall be hot-dip galvanized after fabrication. Other miscellaneous steel Items not specifically described elsewhere shall be hot-dip galvanized. Galvanizing shall be in conformance with ASTM A 123 for shapes and weldments, and A 153 for high strength bolts and associated hardware.

2.011 FABRICATION, GENERAL

- A. Verify dimensions on site prior to shop fabrication.
- B. Fabricate Items with joints neatly fitted and properly secured.
- C. Fit and shop assemble in largest practical sections, for delivery to site. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.
- D. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch (1 mm) unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.
- E. Exposed mechanical fastenings: Flush countersunk screws or bolts shall be unobtrusively located consistent with design of structure, except where specifically noted otherwise.
- F. Bring connected pieces into contact before bolting or otherwise mechanically fastened.
- G. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.
- H. Form exposed work with accurate angles and surfaces and straight edges.
- I. Weld corners and seams continuously to comply with the following:
 - 1. Use materials and methods that minimize distortion and develop strength and not impair corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
- J. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners or welds where possible. Where exposed fasteners are required, use Phillips flat-head (countersunk) fasteners unless otherwise indicated. Locate joints where least conspicuous.
- K. Fabricate seams and other connections that will be exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.
- L. Cut, reinforce, drill, and tap metal fabrications as indicated to receive finish hardware, screws, and similar items.
- M. Supply components required for proper anchorage of metal fabrications. Fabricate anchorage and related components of same material and finish as metal fabrication. Coordinate with supporting structure. Space anchoring devices to secure metal fabrications rigidly in place and to support indicated loads.
 - 1. Where units are indicated to be cast into concrete or built into masonry, equip with integrally welded steel strap anchors, 1/8 by 1-1/2 inches (3.2 by 38 mm), with a minimum 6-inch (150-mm) embedment and 2-inch (50-mm) hook, not less than 8 inches (200 mm) from ends and corners of units and 24 inches (600 mm) o.c., unless otherwise indicated.
- N. Clean surfaces of rust, scale, grease and foreign matter prior to prime painting, galvanizing, anodizing or buffing.
- O. Galvanize or prime paint steel Items as scheduled. Do not shop prime surfaces to be embedded in concrete. Primer is part of paint system specified in Section 09 90 00. Conform with ASTM A 123 and A 153 as applicable. Provide minimum 2.00 ounce/square foot galvanized coating except as otherwise specified therein.

- P. Shop primers that do not contain rust-inhibitive agents are not acceptable.

2.012 MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Provide steel framing and supports not specified in other Sections as needed to complete the Work.
- B. Fabricate units from steel shapes, plates, and bars of welded construction unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction.
 - 1. Fabricate units from slotted channel framing where indicated.
 - 2. Furnish inserts for units installed after concrete is placed.
- C. Fabricate supports for operable partitions from continuous steel beams of sizes recommended by partition manufacturer with attached bearing plates, anchors, and braces as recommended by partition manufacturer. Drill or punch bottom flanges of beams to receive partition track hanger rods; locate holes where indicated on operable partition Shop Drawings.
- D. Fabricate steel girders for wood frame construction from continuous steel shapes of sizes indicated.
 - 1. Provide bearing plates welded to beams where indicated.
 - 2. Drill or punch girders and plates for field-bolted connections where indicated.
 - 3. Where wood nailers are attached to girders with bolts or lag screws, drill or punch holes at 24 inches (600 mm) o.c.
- E. Fabricate steel pipe columns for supporting wood frame construction from steel pipe with steel baseplates and top plates as indicated. Drill or punch baseplates and top plates for anchor and connection bolts and weld to pipe with fillet welds all around. Make welds the same size as pipe wall thickness unless otherwise indicated.
 - 1. Unless otherwise indicated, fabricate from Schedule 40 steel pipe.
 - 2. Unless otherwise indicated, provide 1/2-inch (12.7-mm) baseplates with four 5/8-inch (16-mm) anchor bolts and 1/4-inch (6.4-mm) top plates.
- F. Galvanize miscellaneous framing and supports where indicated.
- G. Prime miscellaneous framing and supports with zinc-rich primer where indicated.

2.013 MISCELLANEOUS STEEL TRIM

- A. Unless otherwise indicated, fabricate units from steel shapes, plates, and bars of profiles shown with continuously welded joints and smooth exposed edges. Miter corners and use concealed field splices where possible.
- B. Provide cutouts, fittings, and anchorages as needed to coordinate assembly and installation with other work.

- C. Provide with integrally welded steel strap anchors for embedding in concrete or masonry construction.
- D. Galvanize exterior miscellaneous steel trim.
- E. Prime interior miscellaneous steel trim with zine-rich primer.

2.014 PIPE RAILING AND TUBING

- A. Handrails and Guardrails: 1-1/4" diameter steel pipe to meet ADA standards, ASTM A53, Grade B, standard weight including wall flanges, bends and elbows.

2.015 FINISHES, GENERAL

- A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.
- B. Finish metal fabrications after assembly.
- C. Finish exposed surfaces to remove tool and die marks and stretch lines, and to blend into surrounding surface.

2.016 STEEL AND IRON FINISHES

- A. Galvanizing: Hot-dip galvanize items as indicated to comply with ASTM A 153/A 153M for steel and iron hardware and with ASTM A 123/A 123M for other steel and iron products.
- B. Do not quench or apply post galvanizing treatments that might interfere with paint adhesion.
- C. Shop prime iron and steel items not indicated to be galvanized unless they are to be embedded in concrete, sprayed-on fireproofing, or masonry, or unless otherwise indicated.
- D. Preparation for Shop Priming: Prepare surfaces to comply SSPC-SP 3, "Power Tool Cleaning." Requirements indicated below:
 - 1. Exterior Items: SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
 - 2. Items Indicated to Receive Zinc-Rich Primer: SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
 - 3. Items Indicated to Receive Primers Specified in Division 09 Section "High-Performance Coatings": SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
 - 4. Other Items: SSPC-SP 3, "Power Tool Cleaning."
- E. Shop Priming: Apply shop primer to comply with SSPC-PA 1, "Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel," for shop painting.
 - 1. Stripe paint corners, crevices, bolts, welds, and sharp edges.

2.017 ALUMINUM FINISHES

- A. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.
- B. As-Fabricated Finish: Prefinished as scheduled.

3.01 INSPECTION

- A. Installer must examine the areas and conditions under which miscellaneous metal items are to be installed and notify the Contractor in writing of conditions detrimental to the timely and proper completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected in a manner satisfactory to the installer.

3.02 PREPARATION

- A. Furnish setting drawings, diagrams, templates, instructions, and directions for installation of anchorages, such as concrete inserts, anchor bolts and miscellaneous items having integral anchors which are to be embedded in concrete or masonry. Coordinate delivery of such items to project site.

3.03 ERECTION

- A. Obtain Engineer's written permission prior to site cutting, welding, or making adjustments which are not part of scheduled work.
- B. Install items square and level, accurately fitted and free from distortion or defects.
- C. Make provision for erection stresses by temporary bracing. Keep work in alignment.
- D. Replace items damaged in course of installation.
- E. Grouting (see Section 04 60 00):
 - 1. Use non-shrink grout under bearing plates, per Structural Notes.
- F. After installation, touch up scratched and damaged prime painted and galvanized surfaces.
 - 1. Use same primer as used for shop priming of painted surfaces.
 - 2. Use galvanizing repair paint for galvanized surfaces.

3.04 PROTECTION OF ALUMINUM

- A. Aluminum in contact with masonry materials:
 - 1. Paint aluminum surfaces in contact with lime mortar, concrete or other masonry material with a heavy coat of bituminous paint.
- B. Aluminum in contact with dissimilar metals:
 - 1. Where aluminum surfaces are in contact with metals other than stainless steel, zinc, or small areas of white bronze, keep aluminum surfaces from direct contact with incompatible metals by any of the following methods:
 - a. Paint the dissimilar metal with bituminous paint
 - b. Apply a good quality caulking material between the aluminum and the dissimilar metal, only where accepted by the Engineer.
 - c. Use a non-absorptive tape or gasket only where accepted by the Engineer.

2. Dissimilar metals shall be coated when used in locations where drainage from them passes over aluminum.
3. Aluminum modifications/fabrication shall not be performed on-site unless Contractor obtains prior approval from Engineer.

3.05 PROTECTION OF STAINLESS STEEL

- A. Stainless steel modifications/fabrication shall not be performed on-site unless Contractor obtains prior approval from Engineer.

3.06 STAIR NOSING INSTALLATION

- A. Install stair nosing with 1/8-inch wide gap located between concrete tread material and nosing end, to provide for thermal expansion, fill gap with sealant.

3.07 INSTALLATION, GENERAL

- A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.
- B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.
- C. Field Welding: Comply with the following requirements:
 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 2. Obtain fusion without undercut or overlap.
 3. Remove welding flux immediately.
 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- D. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag screws, wood screws, and other connectors.
- E. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.
- F. Corrosion Protection: Coat concealed surfaces of aluminum that will come into contact with grout, concrete, masonry, wood, or dissimilar metals with the following:
 1. Cast Aluminum: Heavy coat of bituminous paint.
 2. Extruded Aluminum: Two coats of clear lacquer.

3.08 INSTALLING MISCELLANEOUS FRAMING AND SUPPORTS

- A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings.
- B. Anchor supports for operable partitions securely to and rigidly brace from building structure.

- C. Support steel girders on solid grouted masonry, concrete, or steel pipe columns. Secure girders with anchor bolts embedded in grouted masonry or concrete or with bolts through top plates of pipe columns.
 - 1. Where grout space under bearing plates is indicated for girders supported on concrete or masonry, provide installation details for review by Architect.
- D. Install pipe columns on concrete footings with grouted baseplates. Position and grout column baseplates, provide installation details for review by Architect.
 - 1. Grout baseplates of columns supporting steel girders after girders are installed and leveled.

3.09 INSTALLING BEARING AND LEVELING PLATES

- A. Clean concrete and masonry bearing surfaces of bond-reducing materials, and roughen to improve bond to surfaces. Clean bottom surface of plates.
- B. Set bearing and leveling plates on wedges, shims, or leveling nuts. After bearing members have been positioned and plumbed, tighten anchor bolts. Do not remove wedges or shims but, if protruding, cut off flush with edge of bearing plate before packing with grout.
 - 1. Use nonshrink grout, either metallic or nonmetallic, in concealed locations where not exposed to moisture; use nonshrink, nonmetallic grout in exposed locations unless otherwise indicated.
 - 2. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.010 ADJUSTING AND CLEANING

- A. Touchup Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.
 - 1. Apply by brush or spray to provide a minimum 2.0-mil (0.05-mm) dry film thickness.
- B. Touchup Painting: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint are specified in Division 09 painting Sections.
- C. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

*****END OF SECTION*****

SECTION 05 51 10 – STEEL STAIRS, LADDERS, & GRATING PLATFORMS

PART 1. GENERAL

1.01 SECTION INCLUDES

- A. Work covered by this section includes:
 - 1. Design, detail, fabricate, and install galvanized steel stairs, ladders, and platform structural framing where shown on the plans. Stair and ladder treads shall be non-skid grating, and the platform shall be aluminum grating supported on the framing. The stairs, ladders, and platforms will, when taken together, form the access and platforms needed for a complete and proper installation shown on the drawings.

1.02 RELATED SECTIONS

- A. Coordinate this work with related work shown on the drawings and specifications in the Project Manual, including but not limited to:
 - 1. Section 05 52 00 - Aluminum Handrails and Guardrails
 - 2. Section 05 53 00 – Grating

1.03 SUBMITTALS

- A. Submit calculations and shop drawings, including consideration of the following
 - 1. Drawings sealed by a Registered Professional Engineer showing overall views (“E drawings”) as well as parts and pieces for the framing of stairs and platforms, including member sizes, dimensions and cuts, welds, end plates, bolts, etc. These drawings should describe the size, placement, and orientation of all the stairs and platforms covered by this section. Sets of typical details without reference to specific location on the site are not acceptable, and will be rejected without review.
 - 2. The shop drawings referenced in the preceding paragraph shall be coordinated with the metal grating drawings, or the grating drawings shall be included as part of this shop drawing package. Grating type and span direction shall be clearly indicated. Manufacturer’s data on span rating is acceptable in lieu of calculations regarding grating.
 - 3. Handrails shall be aluminum, as shown in the civil drawings. The stair, ladder, and platform drawings shall be coordinated with the handrail drawings, or the handrail shall be included as part of this shop drawing package. Grating type and span direction shall be clearly indicated. See Specifications Section 05 52 00 and 05 53 00 for further requirements, including calculations and Professional Engineering Seals.

1.04 CODES AND STANDARDS

- A. Comply with the Drawing Structural Notes, they are part of the contract documents and take precedence over less specific requirements in these specifications.
- B. American Institute for Steel Construction (AISC), “Specifications for Steel Buildings, “ANSI/AISC 360.
- C. American Welding Society (AWS) “Structural Welding Code - Steel,” AWS D1.1 2015.
- D. International Code Council, “International Building Code 2018.”

PROJECT SITE CONDITIONS

- E. Coordination: Notify the Engineer upon completion of preparatory work and of intended schedule for placement.
- F. Establish and maintain required lines and elevations.

1.05 DELIVERY, STORAGE AND HANDLING

- A. Deliver all materials to job site properly marked to identify the structure for which they are intended and at such intervals to insure uninterrupted progress of the work. Marking shall correspond to markings indicated on the shop drawings.
- B. Store all members off the ground using pallets, platforms, or other supports.
- C. Do not store materials on the structure in a manner that might cause distortion or damage to the members of the supporting structure.

PART 2 - FABRICATION

2.01 GENERAL REQUIREMENTS

- A. Fit and shop assemble stair and ladder in the largest practical sections for delivery to the job site.
- B. Miter the stringers at changes in direction with joints tightly fitted and secured by continuous welds.
- C. Close and fit the ends of stringers at landings or platforms. Coordinate with the grating requirements to maintain required elevations.
- D. Provide standard prefabricated grating treads with non-slip ribs.
 - 1. Mechanically fasten or weld treads to stringers.
 - 2. Treads with grating bearing bars shall be banded.

2.02 MATERIALS

- A. See the Structural Notes for steel materials, and the specifications for grating.

2.03 GALVANIZING

- A. Hot dip galvanize all steel framing after fabrication, in the largest sections practical given limitations of dipping tanks, with sections to be bolted together at the time of installation.

2.04 DIMENSIONS

- A. Unless shown otherwise on drawings, all stairs, ladders and platforms shall have the following dimensions:
 - 1. 7" Riser, 11" Tread, 1" Nosing
 - 2. Clear width for all stairways shall be 4'-0" Min.
 - 3. Clear width for all access ladders shall be 3'-0" Min.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. Install steel fabricated stairs and platforms in accordance with the shop drawings and in conformance with the project documents.
- B. Install components plumb and level, accurately fitted, free from distortion or defects.
- C. Securely anchor to the structure in accordance with the shop drawings.
- D. Field weld only where specifically approved (in writing) at time of shop drawing review.
- E. Obtain approval prior to site cutting or creating adjustments.

****END OF SECTION****

SECTION 05 52 00 – ALUMINIUM HANDRAIL AND GUARDRAIL

PART 1. GENERAL

1.01 SUMMARY

A. Scope

Aluminum handrailing and guardrailing shall be provided and installed at the locations shown on the drawings. Guardrails, sized to prevent a 6" sphere from passing through, are to be used where shown on drawings. Handrails (with two horizontal bars) is to be used where shown on drawings. General arrangement details, including height, picket spacing, toe plates, and the like are shown on Civil Drawings.

B. Reference Standards

Handrailing and guardrailing shall conform to the standards of the Occupational Safety and Health Administration and to the 2018 International Building Code.

1.02 SUBMITTALS

A. The Contractor shall submit shop drawings in accordance with Section 01 33 00, Submittals indicating compliance with material specifications and showing a dimensional layout of all handrailing and guardrailing. The Contractor shall submit for review the manufacturer's assembly and installation instructions. The submittal shall include shop drawings and calculations by a Washington State Registered Structural Engineer demonstrating that the rail system, including connections, meets OSHA and IBC requirements.

B. The Contractor shall coordinate submittals of the railing systems with slabs, platforms, basins, or other structures shown on the drawings. Partial submittals that do not indicate specifically where parts and pieces are to be used shall be rejected without review. Likewise, submittals not sealed by a Washington State registered Structural Engineer shall be rejected.

PART 2. PRODUCTS

A. Handrails and guardrails shall be aluminum. Prefabricated components shall have a clear satin anodized architectural Class I finish of minimum 0.7 mil thickness. Rails, posts, and stanchions shall be fabricated from 1-1/2" IPS diameter, Schedule 40 cylindrical sections. Prefabricated sections or fittings shall be manufactured by

1. Golden Railing Company, Golden, CO
2. CRL Architectural Products and Services, Los Angeles, CA
3. Ametco Railings, Willoughby, OH
4. Or approved equal

B. Toeboards shall be provided on all guardrails. Toeboards shall be aluminum, minimum thickness of 3/16". Toeboards shall be designed to allow for thermal contraction and expansion.

PART 3. EXECUTION

3.01 FABRICATION

- A. Pipe cuts shall be clean, straight, square and accurate for minimum joint gap. Work shall be done in conformance with the railing manufacturer's instructions.
- B. Handrailing and guardrailing shall be connected by screws or bolts. Holes shall be 1/16" larger than the nominal size of the bolts. Handrail and guardrail components with mismatched holes shall be replaced.
- C. Where a bolted connection to the supporting structure is indicated, bolts, nuts and washers shall be Type 316 stainless steel. Weep holes shall be drilled at the lowest possible elevation in each post and rail.
- D. Where a removable section of railing is shown it shall be independently mounted and not connected to adjacent railing.

3.02 MEASUREMENT AND ASSEMBLY

- A. Dissimilar metals shall be protected from galvanic corrosion by means of pressure tapes, coatings or isolators. Aluminum in contact with concrete or grout shall be protected with a heavy coat of bituminous paint.
- B. Metal to be embedded in concrete shall be placed accurately and held in correct position while the concrete is placed. Recesses or blockouts shall be formed in the concrete, and the metal work shall be grouted in place after the concrete has attained its design strength.
- C. Assemble and install in accordance with printed instructions of the manufacturer.

***** END OF SECTION *****

SECTION 05 53 00 – GRATING

PART 1. GENERAL

1.01 SCOPE

- A. Aluminum grating set in pultruded vinyl ester fiberglass reinforced embedment angles with an integral anchor shall be provided and installed at the locations shown on the drawings.

1.02 QUALITY ASSURANCE

- A. All grating shall conform to Metal Bar Grating Manual, latest edition, of the National Association of Architectural Metal Manufacturers.

1.03 SUBMITTALS

- A. The Contractor shall submit shop drawings in accordance with Section 01 33 00, Submittals, indicating compliance with material specifications and showing a dimensional layout of all handrailing required. The Contractor shall submit for review the manufacturer's assembly and installation instructions.

PART 2. PRODUCTS

2.01 SOLID PLANK GRATING (UV CHANNELS ONLY)

- A. The grating shall be 2" thick all-aluminum solid plank with an aluminum solid slip-resistant surface. Solid grating shall be installed to block all light from leaving the UV channel.
- B. Grating shall be SlipNOT Slip Resistant Solid Aluminum Plank, or approved equal.
- C. Concrete embedment angles for grating shall be as per the grating manufacturer's recommendations.
- D. Where changes in channel direction, openings for gates, ends of grating runs, prohibit adequate support for gratings, additional aluminum supports shall be furnished to provide a full seating surface for the two ends of grating supporting the primary grating span.

2.02 ALUMINUM BAR GRATING

- A. The grating shall support a uniform load of 100 lbs/ft² with a maximum deflection of 1/360 of the clear span of the grating. Minimum depth and weight shall be:

Grating Span	Grating depth (in)	Weight (lbs/ft ²)
2'-6" to 6'-7"	2.0	4.0
- B. Grating shall be I-Bar, with the maximum spacing between bars 1/8" or less. Grating ends and cutouts shall be banded. Where changes in channel direction, openings for gates, ends of grating runs, prohibit adequate support for gratings, additional aluminum support angles shall be furnished to provide a full seating surface for the two primary supporting ends along the primary grating span.
- C. Concrete embedment angles for grating shall be 2.0" x 2.0" x 0.25" pultruded vinyl ester fiberglass reinforced embedment angles with an integral anchor, furnished gray in color as manufactured by IKG Industries (800) 835-8356 or equal. The top of embedment angles shall be flush with the top of grating.
- D. Where changes in channel direction, openings for gates, ends of grating runs, prohibit adequate support for gratings, additional aluminum supports shall be furnished to provide

a full seating surface for the two ends of grating supporting the primary grating span.

PART 3. EXECUTION

3.01 INSTALLATION

- A. Dissimilar metals shall be protected from galvanic corrosion by means of pressure tapes, coatings or isolators. Aluminum in contact with concrete or grout shall be protected with a heavy coat of bituminous paint.
- B. All grating shall be fastened to supports by suitable removable fasteners. Fastener clips shall be aluminum and necessary bolts, nut and washers shall be Type 316 stainless steel.
- C. Assemble and install in accordance with printed instructions of the manufacturer.

****END OF SECTION****

SECTION 06 20 00 - FINISH CARPENTRY

PART 1 - GENERAL

1.1 REFERENCES

- A. American Plywood Association (APA).
- B. Architectural Woodwork Institute (AWI).
- C. United States Product Standard (PS).
 - 1. PS-1 - "Construction and Industrial Plywood."

1.2 SUBMITTALS

- A. Shop Drawings: Show materials, methods of fabrication, and details of installation.
- B. Samples: Furnish required samples with finishes specified.

1.3 QUALITY ASSURANCE

- A. Qualifications: Provide finish carpentry Work in accordance with AWI "Quality Standards," in the grades specified.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Trim:
 - 1. Exterior: Solid stock red cedar, S4S, for painted finish or 5/4" cement fiber trim.
 - 2. Exterior Plywood with P-Lam at exterior window sills.
- B. Telephone Terminal Board: Douglas fir BC Grade interior plywood, 3/4 inch thick, size as shown. Edge band with 1/2 inch hardwood on exposed edges.
- C. Fasteners:
 - 1. As shown, specified, and as required to securely install materials.
 - 2. Fasteners for Exterior Use: Aluminum or Stainless Steel.
 - 3. Size of fasteners for siding and paneling shall be as recommended by manufacturer.
- D. Joint Sealant: As specified in Section 07 92 00.

2.2 FABRICATION

- A. Conform with AWI "Quality Standards," Section 300, Custom Grade requirements as applicable. Standard wood moldings shall conform with Western Wood Product Association WP Series, where applicable.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install all millwork in accordance with reviewed Shop Drawings and AWI "Quality Standards."
- B. Cope internal corners and miter external corners at all standing and running trim.
- C. Provide running trim in as long lengths as practical. Make splices with 45 degree butt joints.
- D. Install materials straight and true. Leave 1/8 inch space between ends of exterior trim, seal joint. Tightly butt ends of interior trim.
- E. In exterior Work drive nail heads flush with surface of siding and trim. Maintain nailing pattern in straight horizontal lines.
- F. In interior Work countersink nails and fill nail holes.
- G. Machine sand trim and finish with hand sanding. Leave free from machine or tool marks that will show through finishes specified. Ease all edges of trim.
- H. Install all finish hardware, accurately fit, securely apply, and carefully adjust to provide smooth and proper operation of all hardware.
- I. Miscellaneous Items: Install all items shown and specified, which are not called for to be installed under other Sections, to plumb, true, and level lines and positions. Install in accordance with details, manufacturer's printed instructions and additional requirements specified. Provide connections and miscellaneous items required to make Work of this Section complete. Securely fasten wall and ceiling mounted items to solid backing or blocking.

3.2 CLEANING

- A. Remove dirt and other foreign matter from installed materials.
- B. Upon completion of installation, leave materials clean and ready for finishing.

****END OF SECTION****

SECTION 07 20 00 - INSULATION

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Thermal insulation and vapor barrier in exterior roof and wall construction.

1.2 QUALITY ASSURANCE

- A. Installation: company specializing in installing metal building insulation, with three years experience and previous experience installing specified vapor barrier and suspension system.
- B. Provide the materials in original manufacturer's packages together with detailed instructions and shop drawings typical of the installation. Materials shall be inspected for damage or shortage upon delivery and stored in a dry, secure manner. Installation shall proceed with care to assure proper sealing of the liner fabric. Insulation shall be placed on or behind the liner fabric in the full specified thickness without voids or compression.
- C. All exposed parts of the system shall have flame spread of 25 or less and smoke developed value of 50 or less based on ASTM E-84 standards. Vapor barrier fabric shall be woven reinforced polyethylene with extrusion-welded seams fabricated to fit not less than the full bay length by the width of the building. Buildings more than 100 feet wide may have field seams on the bottom of a ridge purlin or no less than 50 feet apart. All field seams must be sealed with vapor barrier lap sealant. Wall bays shall have not less than one entire wall bay or end wall column space from ceiling to floor. Edges of the vapor barrier fabric shall be trimmed and sealed to the adjoining steel fabric.

PART 2 - PRODUCTS

2.1 SUSPENSION SYSTEM

- A. Simple Saver System TM interior finish and insulation system manufactured by Thermal Design Inc., or approved equal.
- B. Steel straps; vapor barrier liner fabric; fasteners; sealant; adhesives and tape used to seal suspension fabric or construction seams: Simple Saver System TM, by Thermal Design Inc., or approved equal.

2.2 INSULATION MATERIALS

- A. Metal building insulation: unfaced, batt or blanket type, fiberglass insulation which shall meet or exceed the following:
 - 1. Provide "in-place" thermal resistance of R- (as noted on Drawings) as determined by ASTM C518.
 - 2. Meets Federal Specification HH-1-558B, Form B, Type 1
- B. Thermal block: minimum 1/8" thick by 3" wide extruded polystyrene foam, 25 psi minimum compressive strength.

PART 3 - EXECUTION

3.1 INSULATION INSTALLATION

- A. Install per manufacture's recommendations and installation instructions.
- B. Face fiberglass batts suspended between structural frames, and metal covering sheets are to be fastened directly to frame.
- C. Install insulation materials to provide continuity of thermal barrier at building enclosure elements, without gaps or voids. Place vapor barrier on interior side of insulation. Vapor barrier to be installed continuous without gaps or voids, tape seal all joints and penetrations.

*****END OF SECTION*****

SECTION 07 92 00 - JOINT SEALANTS

PART 1 - GENERAL

1.1 REFERENCES

- A. ASTM International (ASTM):
 - 1. C 920 - "Specification for Elastomeric Joint Sealants."

1.2 SUBMITTALS

- A. Manufacturer's data on materials, and application equipment and procedures.

1.3 QUALITY ASSURANCE

- A. Qualifications: Applicator shall have a minimum of 3 years experience in the satisfactory installation of joint sealants of the types specified.

1.4 WARRANTY

- A. Furnish 2 year special warranty.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Exterior Sealant: Dow Corning 795 silicone; no substitutions, single component, gun-grade, non-sag, silicone sealant, with movement capability of plus or minus 12-1/2 percent. Color: clear.
- B. Interior Sealant: Tremco, Pecora Corporation, Sonneborn Building Products, single component, gun-grade, paintable, acrylic-latex, water-base sealant. Color as selected. At exterior wall conditions: Proscor, R-Guard Air Dam or approved equal.
- C. Sanitary Sealant: Dow-Corning, General Electric, single component, gun-grade, silicone rubber sealant, with movement capability plus or minus 25 percent.
- D. Acoustical Sealer: Tremco, Pecora Corporation, permanently plastic, non-skinning, non-bleeding, non-staining sealant, for use in concealed spaces.
- E. Joint Backing: Closed cell neoprene or polyethylene, compatible with sealant materials, of sizes and shapes as recommended by the joint sealant manufacturer.
- F. Primers, Solvents, and Cleaning Materials: Non-staining and non-injurious to exposed surfaces, of types as recommended by the joint sealant manufacturer.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Surface Preparation:
 - 1. Thoroughly clean and dry surfaces prior to installation.
 - 2. Clean metals with manufacturer's recommended solvent, wipe clean.
 - 3. Remove any dust, dirt, oil, grease, and any other foreign substances from surface areas to receive sealant.
 - 4. Prime surfaces as recommended by sealant manufacturer. Prime all concrete substrates, regardless of manufacturer's requirements.
 - 5. Mask off any adjacent surfaces which are not to receive sealant.

3.2 APPLICATION

- A. General: Comply with sealant manufacturer's installation instructions. Provide bond breaker tape as required to prevent sealant adhesion to backing. Joint depth 1/2 joint width, but not less than 1/4 inch nor more than 1/2 inch, unless otherwise specified by sealant manufacturer.
- B. Joint Backing: Install as recommended by sealant manufacturer to prevent sealant from adhering to rigid, inflexible materials or joint surfaces where such adhesion would result in sealant failure. Sealant shall bond two opposing joint surfaces.
- C. Sealant Application:
 - 1. Do not install sealants during rainfall or very windy conditions when windborne contaminants can become embedded in uncured sealant.
 - 2. Apply materials with hand gun, powered gun, or trowel to completely fill voids and joints, free of wrinkles and skips.
 - 3. Observe temperature control in accordance with sealant manufacturer's written recommendations.
 - 4. Do not allow any air entrapment in sealant.
 - 5. Extrude sealant fully into joint to be sealed, tool sealant to press into joint, assuring full adhesion to sides of joint surfaces, resulting in a uniformly smooth concave profile.
 - 6. Tool sealant using only materials recommended by sealant manufacturer.
 - 7. Remove masking tape immediately after sealant application to produce clean, sharp line.
 - 8. Do not seal weeps or drainage provisions in sill channels.
 - 9. Allow sealants to cure adequately prior to covering with other Work.
 - 10. Coordinate sealant installation with concrete, masonry, and flashing applications.
 - 11. Apply sand to wet sealant surfaces to match finish of adjacent concrete and masonry conditions.

3.3 CLEANING

- A. Remove all empty containers, materials, and debris from the site. Dispose off site in accordance with applicable regulations.
- B. Remove any sealant spills, masking materials, and similar items from all surfaces not intended for their application.
- C. Clean and repair surfaces soiled or damaged by sealant Work.

****END OF SECTION****

SECTION 08 10 00 - METAL DOORS AND FRAMES

PART 1 - GENERAL

1.01 SUMMARY

- A. Products Installed but not Furnished Under this Section: Finish Hardware specified in Section 08 70 00 Finish Hardware.

1.02 REFERENCES

- A. American Society for Testing and Materials (ASTM).
 - 1. A 366 - "Specification for Steel, Carbon, Cold- Rolled Sheet, Commercial Quality."
 - 2. A 526 - "Specification for Steel Sheet, Zinc- Coated (Galvanized) by the Hot-Dip Process, Commercial Quality."
- B. Steel Door Institute (SDI).
 - 1. 100 - "Recommended Specifications, Standard Steel Doors and Frames."
 - 2. 105 - "Recommended Erection instructions for Steel Frames."
 - 3. 107 - "Hardware on Steel Doors (Reinforcement- Application)."

1.03 QUALITY ASSURANCE

- A. Qualifications: Comply with SDI Publication, 100.

PART 2 - PRODUCTS

2.01 MATERIALS

- A. Doors: Commercial quality cold-rolled sheet steel in conformance with ASTM A 366. Stretcher level steel for door faces. Hot-dip galvanize steel for exterior doors in conformance with ASTM A 526, G90 coating designation, factory baked on primer.
- B. Hardware: Comply with SDI 107. Furnished as specified in Section 08 70 00.

2.02 FABRICATION

- A. Exterior Insulated Hollow Metal (INS HM) Doors:
 - 1. Type: Standard-duty, full flush panel, in accordance with SDI 100 for Grade II, Model 1 doors, polyurethane or polyisocyanurate core.
 - 2. Steel Grades: 18-gauge face sheets minimum

- B. Frames: One-piece, welded, 16 gage minimum, galvanized, with integral stops, jambs, and trim in accordance with SDI 100 for Grade II, Model 2 doors.
- C. Tempered Glass Windows: Provide as specified on door schedule.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. Install in accordance with SDI 105, manufacturer's recommendations, and requirements of labeling authority.
- B. Install hardware, adjust, and lubricate for proper operation.

*****END OF SECTION*****

SECTION 08 70 00 - FINISH HARDWARE

PART 1 - GENERAL

1.01 SUMMARY

- A. Description: Work under this section includes the complete finish hardware requirements for the project. Quantities listed are for the contractor's convenience only and are not guaranteed. Items not specifically mentioned but necessary to complete the work shall be furnished, matching the items specified in quality and finish.
- B. The Contractor shall be responsible for proper operation and fitting of hardware in locations specified. Exposed surfaces of hardware shall be covered and well protected during installation so as to avoid damage to finishes.
- C. Work not Included: Hardware not included in this division but specified elsewhere:
 - 1. Metal Fabrications - Section 05 50 00.
 - 2. Metal Doors and Frames - Section 08 10 00.

1.02 QUALITY CONTROL

- A. Supplier: Finish hardware shall be supplied by recognized builders' hardware supplier who has been furnishing hardware in the same area as the project for a period of not less than two years.
- B. Codes:
 - 1. All finish hardware shall comply with applicable local and state current building codes.
 - 2. Provide hardware which meets or exceeds handicapped accessibility per local and state building code requirements.

1.03 SUBMITTALS

- A. Manufacturer's Data: Submit copies of manufacturer's data for each item of finish hardware with each hardware schedule submitted per Section 01 30 00 - Submittals.
- B. Hardware Schedule: Submit final hardware schedule in manner indicated below. Coordinate hardware with doors, frames and related work to ensure proper size, thickness, hand, function and finish of hardware. Based on finish hardware indicated, organize hardware schedule into "hardware sets" indicating complete designations of every item required for each door or opening. Include the following information:

1. Type, style, function, size and finish of each hardware item.
 2. Name of manufacturer of each item.
 3. Fastenings and other pertinent information.
 4. Location of hardware set cross referenced to indications on drawings both on floor plans and in door and frame schedules.
 5. Explanation of all abbreviations, symbols, codes, etc. contained in schedule.
 6. Recommended mounting locations for hardware.
- C. Submittal Sequence: Submit copies of schedule at earliest possible date in accordance with requirements in Division -1, particularly where acceptance of hardware schedule must precede fabrication of other work (e.g., hollow metal frames) which is critical in the project construction schedule. Include with schedule the product data, samples, shop drawings of other work affected by finish hardware, and other information essential to the coordinated review of hardware schedule.
- D. Keying Schedule: Submit separate detailed schedule indicating how the owner's final instructions on keying of locks has been fulfilled.
- E. Templates: Furnish hardware templates to each fabricator of doors, frames and other work to be factory or shop prepared for the installation of hardware. Upon request, check shop drawings of other work, to confirm that adequate provisions are made for proper location and installation of hardware.

1.04 PRODUCT HANDLING

- A. Tag each item or package separately, with identification related to final hardware schedule, and include basic installation instructions with each item or package.
- B. Packaging of hardware is responsibility of supplier. As material is received by hardware supplier from various manufacturers, sort and repackage in containers clearly marked with appropriate hardware set number to match set number of approved hardware schedule. Two or more identical sets may be packed in the same container.
- C. Inventory hardware jointly with representatives of hardware supplier and hardware installer until each is satisfied that count is correct.
- D. Delivery individually packaged hardware items at the proper times to the proper locations (shop or project site) for installation.

1.05 GUARANTEE

- A. Finish hardware shall be guaranteed against defects in workmanship and operation for a period of one year, backed by a factory guarantee of the hardware manufacturer, except that door closers shall be so guaranteed for five years. No

liability shall be assumed by the hardware supplier where faulty operation is due to improper installation or failure to exercise normal maintenance.

PART 2 - MATERIALS

2.01 MANUFACTURERS: See PART FOUR HARDWARE SCHEDULE.

2.02 MATERIALS AND FABRICATION GENERAL:

- A. Hand of door: Drawings show direction of slide, swing or hand of each door leaf. Furnish each item of hardware for proper installation and operation of door movement as shown.
- B. Manufacturer's Name Plate: Do not use manufacturer's products which have manufacturer's name or trade name displayed in a visible location (omit removable name plates), except in conjunction with required UL or FM labels and as otherwise acceptable to the Architect. Manufacturer's identification will be permitted on rim of lock cylinders and latch faceplates only.
- C. Base Metals: Produce hardware units of basic metal and forming method indicated, using manufacturer's standard metal alloy, composition, temper and hardness, but in no case of lesser (commercially recognized) quality than specified for applicable hardware units by applicable ANSI A156 series standard for each type hardware item and with ANSI A156.18 for finish designations indicated. Do not furnish "optional" materials or forming methods for those indicated, except as otherwise specified.
- D. Fasteners: Provide hardware manufactured to conform to published templates, generally prepared for machine screw installation. Do not provide hardware which has been prepared for self tapping sheet metal screws, except as specifically indicated.
- E. Furnish screws for installation, with each hardware item. Provide Phillips flat head screws except as otherwise indicated. Finish exposed (exposed under any condition) screws to match hardware finish or, if exposed in surfaces of other work, to match finishes of such other work as closely as possible, including "prepared for paint" in surfaces to receive painted finish.
- F. Provide concealed fasteners for hardware units which are exposed when door is closed, except to extent no standard units of type specified are available with concealed fasteners. Do not use thru-bolts for installation where bolt head or not on opposite face is exposed in other work, except where it is not feasible to adequately reinforce the work. In such cases, provides leaves for each thru-bolt or use sex screw fasteners.

- G. Tools and Maintenance Instructions for Maintenance: Furnish a complete set of specialized tools and maintenance instructions as needed for Owner's continued adjustment, maintenance, and removal and replacement of finish hardware.

2.03 LOCKSETS AND LATCHSETS

- A. Key-in-knob type, steel cylindrical case with interior parts plated to resist corrosion, with access to cylinder without removal from the door, 2-3/4" backset. Design as specified.
 - 1. Strikes: Standard strikes sized to extend no more than 3/16" beyond door frame or adjacent trim.
 - 2. Keying: Keyed and registered at the factory, 6 pin cylinders, master keyed, keyed alike conforming to Upper Skagit Indian Tribe standard. All keys shall be tagged with room or door identification and delivered by the manufacturer direct to the Owner.

2.04 CLOSERS

- A. Full rack and pinion construction with tamperproof valves to control closing, latching, and backcheck. Furnish proper mounting hardware to suite installation conditions, including all required screws or bolts. Provide thru bolts at wood doors with carriage heads exposed. Except as otherwise specifically indicated, comply with the manufacturer's recommendations for size of door control unit, depending on the size of the door, exposure to weather and anticipated frequency of use. Where parallel arms are indicated for closers, adjust closer unit one size larger than recommended for use with standard arms. Provide arms for closers which allow the closer to be mounted on the room side, unless otherwise noted. Where manual closers are indicated for doors required to be accessible to the physically handicapped, provide adjustable units complying with ANSI A117.1 provisions for door opening force. Fire protection has precedence over handicapped compatibility, check with local jurisdiction.

2.05 WEATHER STRIPPING:

- A. General: Except as otherwise indicated, provide continuous weather stripping at each edge of every exterior door leaf. Provide type, sizes and profiles as indicated as drawn or scheduled. Provide non-corrosive fasteners as recommended by the manufacturer for applications indicated.
- B. Perimeter weather strip: Flexible, hollow neoprene bulb or loop insert, conforming to MIL R 6055, Class II, Grade 40.

2.06 KICK, MOP AND ARMOR PLATES

- A. Plastic, 1/8" thick, laminated double face balance construction, color as selected. Metal: Minimum 0.050 inch thick, finish as specified. Furnish and widths as required to provide 1/4" clearance at sides of doors and stops, heights as specified. Furnish oval head mounting screws and mounting adhesive as recommended by manufacturer.

2.07 STOPS AND HOLDERS

- A. Exterior Doors:
All exterior doors are to be equipped with heavy duty stainless steel latching door stops, which include door stop, holder, hook, ring, and door plate. See exterior door stop detail, sheet C8.1. (Ives Model FS446, Contact Sargent Manufacturing Company, Doug Holderman 425-392-2358)
- B. Interior Doors:
Furnish as required to protect all walls, cabinets, or other hardware operation. Furnish wall stops wherever possible, unless otherwise scheduled. Also furnish plunger door holders for all interior doors. (Ives Model FS1153, Contact Sargent Manufacturing Company, Doug Holderman 425-392-2358).

2.08 OUTDOOR BOOT BRUSH

- A. Provide and install one concrete pad with boot brush adjacent to Blower Building exterior door. Coordinate location with Owner. Boot brush to be securely attached to concrete pad. Item to be: Deluxe Boot and Shoe Brush with Scraper, Gempler's Item # 2900 (phone: 800-382-8473). Style to match image in photo below.



2.09 DOOR SILENCERS

- A. Furnish three for each single door frame, and four for each double door frame.

2.10 FINISHES

- A. All items, unless otherwise specified, US26D Dull Chrome. Exposed closers, factory finished, US26D or spray-painted aluminum to match adjacent hardware.

2.11 THRESHHOLDS

- A. Furnish for each door opening.

2.12 TEMPERED GLASS WINDOWS

- A. Furnish 24" x 24" tempered glass windows with Type C and Type D doors.

PART 3 - EXECUTION

3.01 PREPARATION

- A. Provide solid blocking for all wall stops. Check all conditions and use fastening devices as needed to securely anchor all hardware as per manufacturer's published templates. Self-tapping sheet metal screws are not acceptable. All closers and exit devices on wood doors shall be thru-bolted.

3.02 INSTALLATION

- A. Mounting heights: Mount units at heights recommended in "Recommended Locations for Builders' Hardware" by NBHA, except as otherwise indicated or as required by State Barrier Free regulations.
- B. Install each hardware item in compliance with the manufacturer's instructions. Wherever cutting and fitting are required to install hardware on surfaces which will be painted or finished at a later time, install each item completely and then remove and store in a secure place. After completion of the finishes, re-install each item. Do not install surface-mounted items until finishes have been completed on the substrate.
- C. Adjust and check each operating item of hardware and each door to insure proper operation or function of every unit. Replace units which cannot be adjusted to operate freely and smoothly.
- D. Set units level, plumb and true to line and location. Adjust and reinforce the attachment substrate as necessary for proper installation and operation.
- E. Drill and countersink units which are not factory prepared for fasteners. Space fasteners and anchors in accordance with industry standards.
- F. Set thresholds for exterior doors in a full bed of butyl-rubber or polyisobutylene mastic sealant.

3.03 HARDWARE LOCATIONS

- A. The following shall be used in absence of other specifications from door manufacturers:
 - Center of knob at Rail & Stile: Center on Mid Rail
 - Center of knob to finished floor: 36"
 - Bottom butt: bottom of door to bottom of butt: 10"
 - Top butt - top of door to top of butt: 5"
 - Center butt: equals distance between top and bottom butts.

3.04 ADJUSTMENT

- A. Adjust and check each operating item of hardware and each door, to ensure proper operation or function of every unit. Replace units which cannot be adjusted to operate freely and smoothly as intended for the application made.
- B. Wherever hardware installation is made more than one month prior to acceptance or occupancy, make a final check and adjustment of all hardware items during the week prior to acceptance of occupancy. Clean and lubricate operating items as necessary to restore proper function and finish of hardware and doors. Adjust door control devices to compensate for final operation of heating and ventilating equipment.
- C. Clean adjacent surfaces soiled by hardware installation.
- D. Instruct Owner's personnel in proper adjustment and maintenance of hardware and hardware finishes, during the final adjustment of hardware.
- E. Continued Maintenance Service: Approximately six months after the acceptance of hardware in each area, the installer, accompanied by the representative of the lock and latch manufacturer, shall return to the project and re-adjust every item of hardware to restore proper function of doors and hardware. Consult with an instruct Owner's personnel in recommended additions to the maintenance procedures. Replace hardware items which have deteriorated or failed due to faulty design, materials or installation of hardware units. Prepare a written report of current and predictable problems (of substantial nature) in the performance of the hardware.

3.05 SPECIAL TOOLS

- A. Contractor shall provide 2 sets of any special tools required for installation and maintenance of hardware.

PART 4 - HARDWARE SCHEDULE

4.01 MANUFACTURERS

A. Each hardware symbol is followed by the model number of the first manufacturer listed for the item, unless otherwise identified. Such designation is intended to establish a standard of quality, function, and appearance for the various finish hardware items.

<u>Manufacturer Specified</u>	<u>Approved Substitutions</u>
Stanley	Hager, McKinney, Lawrence
Best	None
Ives	Quality, Rockwood
Tice	Quality, Rockwood
Pemko	National Guard, Reese

4.02 HARDWARE GROUPS (Note: Confirm active door size before ordering doors/hardware)

Hardware Group #1 - (insulated hollow metal doors - exterior single doors)

2 pr Hinges	Stanley FBB179 4-1/2x4-1/2 NRP
1 ea Lockset	Latchbolt by key outside or by grip either side, unless outside grip is locked by toggle-action stop. Auxiliary latch deadlocks latch-bolt. Inside grip always free. Corbin Russwin ML2000 Series Grade Mortise Locksets with Lustra Lever Trim and Escutcheon Plate (satin chrome), include Russwin Mortise Cylinders. Keyway to match District Standard.
1 set Flushbolts	Ives 458-26D manual flush bolts
1 ea Dust Proof Strike	Ives DP2 (US26D)
1 ea Closers	Sargent 421 Series
1 ea Kickplate	PLKP 10x2" LD CAS
1 ea Astragal	Pemko 357SP
1 ea Gasketing	Pemko S88D
1 ea Door Bottom Sweeps	Pemko 315SN
1 ea Threshold	National Guard 713
1 ea Exterior Stops & Holders	Ives Model FS446

Hardware Group #2 - (insulated hollow metal doors - exterior double doors)

4 pr Hinges	Stanley FBB179 4-1/2x4-1/2 NRP
1 ea Lockset	Latchbolt by key outside or by grip either side, unless outside grip is locked by toggle-action stop. Auxiliary latch deadlocks latchbolt. Inside grip always free. Corbin Russwin ML2000 Series Grade Mortise Locksets with Lustra Lever Trim and Escutcheon Plate (satin chrome), include Russwin Mortise Cylinders. Keyway to match District Standard.
1 set Flushbolts	Ives 458-26D manual flush bolts
1 ea Dust Proof Strike	Ives DP2 (US26D)
2 ea Closers	Sargent 421 Series
2 ea Kickplate (Protection Plate)	PLKP 10x2" LD CAS
1 ea Astragal	Pemko 357SP
2 ea Gasketing	Pemko S88D
2 ea Door Bottom Sweeps	Pemko 315SN
2 ea Threshold	National Guard 713
2 ea Exterior Stops & Holders	Ives Model FS446

*****END OF SECTION*****

SECTION 09 25 00 - GYPSUM DRYWALL

PART 1 - GENERAL

1.01 SUMMARY

- A. Work includes furnishing and installing all gypsum drywall shown including taping and spackling.

1.02 QUALITY CONTROL

- A. Gypsum board application and finishing standard: Install and finish gypsum board to comply with ASTM C 840.
- B. Single source responsibility: Obtain each type of gypsum board and related joint treatment materials from a single manufacturer.
- C. Deliver materials in original packages, containers or bundles bearing brand name and identification of manufacturer or supplier.
- D. Handle gypsum boards to prevent damage to edges, ends, and surfaces. Do not bend or otherwise damage metal corner beads and trim.

1.03 SUBMITTALS

- A. Samples: Submit samples of texture finish for approval before application.
- B. Store materials inside under cover and keep them dry and protected against damage from weather, direct sunlight, surface contamination, corrosion, construction traffic and other causes. Neatly stack gypsum boards flat to prevent sagging.

1.04 PROJECT CONDITIONS

- A. Environmental conditions, general: Establish and maintain environmental conditions for application and finishing gypsum board to comply with ASTM C 840 and with gypsum board manufacturer's recommendations.
- B. Minimum room temperatures: For non-adhesive attachment of gypsum board to framing, maintain not less than 40 deg F. For adhesive attachment and finishing of gypsum board maintain not less than 50 deg. F for 48 hours prior to application and continuously thereafter until drying is complete.
- C. Ventilate building spaces to remove water not required for drying joint treatment materials. Avoid drafts during dry, hot weather to prevent materials from drying too rapidly.

PART 2 - PRODUCTS

2.01 MATERIALS

- A. Gypsum wallboard: U.S. Gypsum Co. Aqua-Tough (green board type gypsum drywall). Sheet rock edges tapered. Thickness: 5/8" unless noted otherwise. Conform to ASTM C-36. Provide Type X for fire-rated assemblies. Provide USG Shaftwall accessories for 2-hour rating. Provide fiberrock interior panels suitable for high moisture locations.
- B. Screws: Conform to ASTM C-1002.
- C. Accessories: USG Dur-A-Bead corners, 200-A ("LC") and 200-B ("L") Grounds. Comply with ASTM C-1047.
- D. Tape and compound: U.S. Gypsum Perf-A-Tape. Comply with ASTM C-475 and ASTM C-840.
- E. Spackle texture: USG Imperial QT Spray.

PART 3 - EXECUTION

3.01 PREPARATION

- A. Examine substrates to which drywall construction attaches or abuts, preset hollow metal frames, cast-in-anchors, and structural framing, with installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of drywall construction. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.02 INSTALLATION

- A. Install sound attenuation blankets where indicated, prior to gypsum board unless readily installed after board has been installed.
- B. Install exposed gypsum board with face side out. Do not install imperfect, damaged or damp boards. Butt boards together for a light contact at edges and ends with not more than 1/16 inch open space between boards. Do not force into place.
- C. Locate exposed end-butt joints as far from center of walls and ceilings as possible, and stagger not less than 24 inches in alternate courses of board.
- D. Install ceiling boards across framing in the manner which minimizes the number of end-butt joints, and which avoids end joints in the central area of each ceiling. Stagger end joints at least 24 inches.
- E. Install wall/partition boards in manner which minimizes the number of end-butt joints or avoids them entirely where possible. At stairwells and similar high walls, install boards horizontally with end joints staggered over studs.
- F. Locate either edge or end joints over supports, except in horizontal applications where intermediate supports or gypsum board back-blocking is provided behind end joints. Position boards so that like edges abut, tapered edges against tapered edges and mill-cut or field-cut ends against mill-cut or field-cut ends. Do not place

tapered edges against cut edges or ends. Stagger vertical joints over different studs on opposite sides of partitions.

- G. Attach gypsum board to supplementary framing and blocking provided for additional support at openings and cutouts.
- H. Spot grout hollow metal door frames for solid core wood doors, hollow metal doors and doors over 32 inches wide. Apply spot grout at each jamb anchor clip just before inserting board into frame.
- I. Cover both faces of stud partition framing with gypsum board in concealed spaces (above ceilings, as shown, etc.). Fit gypsum board around ducts, pipes, and conduits.
- J. Space fasteners in gypsum boards in accordance with referenced gypsum board application and finishing standard and manufacturer's recommendations.
- K. Single-layer application: Install gypsum wallboard as follows:
 - 1. On ceilings apply gypsum board prior to wall/partition board application to the greatest extent possible.
 - 2. On partitions/walls apply gypsum board vertically (parallel to framing), unless otherwise indicated, and provide sheet lengths which will minimize end joints.
 - 3. On partitions/walls 8'-1" or less in height apply gypsum board horizontally (perpendicular to framing); use maximum length sheets possible to minimize end joints.
- L. Single-layer fastening methods: Apply gypsum boards to supports as follows:
 - 1. Fasten with self-drilling screws to steel framing at 12" o.c. maximum. Conform to UBC fastener spacing.
 - 2. Double-layer fastening methods: Apply base layer of gypsum board and face layer to base layer as follows: Fasten both base layers and face layers separately to supports with screws.
- M. Installation of drywall trim accessories:
 - 1. General: Where feasible, use the same fasteners to anchor trim accessory flanges as required to fasten gypsum board to the supports. Otherwise, fasten flanges to comply with manufacturer's recommendations.
 - 2. Install corner beads at external corners.
 - 3. Install metal edge trim whenever edge of gypsum board would otherwise be exposed or semi-exposed, and except where plastic trim is indicated. Provide type with face flange to receive joint compound.
 - a. Install "LC" bead where drywall construction is tightly abutted to other construction and back flange can be attached to framing or supporting substrate.
 - b. Install "L" bead where edge trim can only be installed after gypsum board is installed.

3.03 FINISHING

- A. General: Apply joint treatment at gypsum board joints (both directions); flanges of corner bead, edge trim, and control joints; penetrations; fastener heads, surface defects and elsewhere as required to prepare work for decoration.
- B. Prefill open joints and rounded or beveled edges, if any, using setting-type joint compound.
- C. Apply joint tape at joints between gypsum boards, except where trim accessories are indicated.
- D. Finish interior gypsum wallboard by applying the following joint compounds in 3 coats (not including prefill of openings in base), and sand between coats and after last coat:
 - o Embedding and first coat: Ready-mix drying-type all-purpose or taping compound.
 - o Fill (second) coat: Ready-mix drying-type all-purpose or topping compound.
 - o Finish (third) coat: Ready-mix drying-type all-purpose or topping compound.
- E. Partial finishing: Omit third coat and sanding on concealed drywall construction which is indicated for drywall finishing or which requires finishing to achieve fire-resistance rating, sound rating or to act as air or smoke barrier.

3.04 APPLICATION OF TEXTURE FINISH

- A. Surface preparation and primer: Prepare and prime drywall and other surfaces in strict accordance with texture finish manufacturer's instructions. Apply primer to all surfaces to achieve texture finish. See Section 09900 Painting for primer paint coat.
- B. Finish application: Mix and apply finish to drywall and other surfaces indicated to receive finish in strict accordance with manufacturer's instructions to produce a uniform texture matching Architect's sample without starved spots or other evidence of thin application, and free of application patterns.
- C. Remove any texture droppings or overspray from door frames, windows and other adjoining construction.
- D. Provide final protection and maintain conditions, in a manner suitable to Installer, which ensures gypsum drywall construction being without damage or deterioration at time of Substantial Completion.

*****END OF SECTION*****

SECTION 09 90 00 – MISC. PLANT PAINTING

PART 1. GENERAL

1.01 DESCRIPTION

- A. Description of System: The work covered by this section consists of furnishing all plant, labor, equipment and materials necessary for the preparation and application of the paint coatings as specified herein. For new blower building and various masonry/concrete/stucco painting requirements see section 09 90 00 – Paints and Coatings.

1.02 QUALITY ASSURANCE

- A. Requirements of Regulatory Agencies: Comply with the requirements of agencies having jurisdiction over this section of work, including, but not limited to:
 - 1. WISHA, Washington Industrial Safety and Health Act.
- B. Reference Standards: All surface preparation, coating and painting shall conform to the applicable requirements of the:
 - 1. National Association of Corrosion Engineers
 - 2. Steel and Structures Painting Manual, Volume 2, Systems and Specifications (latest revision) published by the Steel Structures Painting Council (SSPC).
 - 3. American Water Works Association (AWWA) Standard D102 "Painting Steel Water Storage Tanks."
- C. Manufacturer: Manufacturer shall be of established good reputation and shall have regularly engaged in the manufacture of such coatings for a minimum of 5 years. This experience shall include a minimum of 20 similar applications in which such coatings have proven satisfactory service for a minimum of 3 years.
- D. Contractor: Contractor shall have 5 years of practical experience and successful history in the application of paint coatings to surfaces of municipal or industrial type equipment.
- E. Paint Film Thickness: All painted surfaces will be inspected by the Contractor with approved wet-film thickness gages. Inspection will include the thickness measurement of each prime and finish coat.
- F. Manufacturer's Representative: The manufacturer shall provide a qualified representative to visit the site from time to time during the paint operations as requested by the Engineer. The manufacturer's representative shall assist the Engineer in monitoring surface preparation and paint application.

1.03 SUBMITTALS

- A. Submittals detailing product data and application procedures shall be submitted in accordance with Section 01 33 00 for each paint service condition.
- B. Color charts for each of the finish coats listed in Part II of this section shall be submitted at least thirty (30) days prior to the starting of painting.
- C. A Schedule of the Painting Work shall be submitted to the Engineer at least fourteen (14) days prior to commencing of any work under this section. A revised schedule shall be submitted as requested by the Engineer to reflect changes or delays in the work.

1.04 JOB CONDITIONS

- A. Environmental:
 - 1. Protective coatings shall not be applied in areas where dust is being generated or in any other areas where disturbances will affect the quality of the work.
 - 2. The Contractor shall comply with the manufacturer's recommendations as to environmental conditions (i.e. temperature, moisture, exposure to sunlight etc.) under which coatings and coating systems must be applied and cured.
- B. Protection: The Contractor shall be responsible for protecting coatings or coating systems from any disturbances during or after application which will affect the quality of the work.

1.05 DELIVERY AND STORAGE

- A. Delivery: All products shall be delivered in sealed containers with labels legible and intact. Labels shall include the following information: Manufacturer's name and stock number, type of paint or protective coating, color, instructions for reducing, label analysis, and federal specification number.
- B. Storage: Products shall be stored in a single location and in a manner complying with all applicable safety, health and fire regulations.

PART 2. PRODUCTS

2.01 GENERAL

- A. Surfaces to receive insulation or other protective materials shall be coated or painted in conformance with the applicable Service Conditions as specified herein.

The intent has not been to identify each and every item to be coated, but only to list the major items. In no case shall any wood, ferrous metal or other surface, requiring protection, be left uncoated or unpainted.
- B. The products specified are those which have been evaluated for the specific service and are given to establish a quality standard for that service. Products of other manufacturers comparable in quality and type to those specified will be acceptable if said paints are offered by the Contractor with satisfactory data on past performance in similar applications. Requests for substitutions shall be in accordance with Section 01 66 00.
- C. The Contractor shall use products of the same manufacturer for all prime and finish coats listed in each separate Service Condition.
- D. Colors to be used shall be as designated by the Owner based upon the color charts provided by the Contractor.
- E. For the paint thickness listed under each Service Condition:
 - 1. WT = wet-film thickness in mils
 - 2. DF = dry-film thickness in mils
 - 3. The number following equals the minimum film thickness required, per coat.
 - 4. Putty: Conform to FS TT-P-791A(3), colored to match paint and stain finishes, as applicable.

- F. Cementitious Filler: Nonshrink formulation, white Portland cement with fine silicate aggregate, zinc-oxide pigment, and reinforcing chemical binder as approved.
- G. Spackling Compound: Standard gypsum board compound.
- H. Unspecified materials such as turpentine, linseed oil, or mineral spirits shall be products of reputable manufacturers and as recommended by paint manufacturers.
- I. Materials for Undercoats and Finish Coats: Ready mixed, and shall not be changed, except thinning of undercoats (when required), reinforcing, or coloring, all of which shall be performed in accordance with manufacturers' recommendations.

1. SERVICE CONDITION A

- a. Generic Type: Epoxy
- b. Applications:

All exposed metal piping (including exposed ductile iron piping), joints, fittings, valves, supports, bollards, yard hydrant accessories, runway beams, runway supports, fasteners, and misc. ferrous metal items for this project shall be painted. Copper, galvanized, aluminum and stainless steel metal surfaces shall not be painted unless specifically specified. Exposed piping shall be defined as all piping that is not backfilled. Exposed piping shall also be defined as all piping that is permanently or intermittently submerged within structures. Exposed piping in vaults is considered to be exposed and shall be painted. All bollards are to be completely protected with shop coatings (primer and finish).

- c. Primer:
One coat Tnemec Series 161- "High Solids Epoxy" or Sherwin Williams Macropoxy 646.
DF = 4.0 to 6.0
- d. Finish:
One coat Tnemec Series 161- "High Solids Epoxy" or Sherwin Williams Macropoxy 646.
DF = 4.0 to 6.0

2. SERVICE CONDITION B

- a. Generic Type: Semi-gloss alkyd enamel.
- b. Application: Interior wood surfaces.
- c. Primer: One coat Sherwin-Williams PrepRite Wall and Wood Interior Oil Primer/Undercoater, alkyd, DF = 1.9.
- d. Finish: Two coats Sherwin-Williams ProMar 200 Interior Alkyd Semi-Gloss, DF = 3.4.

3. SERVICE CONDITION C

- a. Generic Type: Metal, gloss alkyd enamel.
- b. Application: Metal doors and frames.
- c. Primer: Factory baked on enamel.
- d. Finish: Two coats Sherwin-Williams Industrial Enamel, DF = 4.0.

4. SERVICE CONDITION D
 - a. Generic Type: PVC, Vinyl Copolymer
 - b. Application: PVC piping
 - c. Finish: Two coats of Tnemec Series 73 Endurashield or Sherwin Williams High Solids Polyurethane, DF = 3.0
5. SERVICE CONDITION E
 - a. Generic Type: Solid Color Acrylic Stain
 - b. Application: All exterior wood surfaces
 - c. Finish: Two coats Cabot O.V.T. Solid Color Acrylic Stain
6. SERVICE CONDITION H
 - a. Generic Type: Concrete sealer, non-skid/slip-resistant
 - b. Application: Interior concrete slab finish at headworks floor.
 - c. An SSPC SP 1 (Solvent Cleaning) to remove grease oil or any other contaminants, if present.
 - d. Entire area to be coated is to be mechanically etched (diamond grinders, shot blaster, etc.) or chemically etched (Acid etch concrete with a 25% muriatic acid & 75% water solution. Allow to etch for 10 to 15 minutes, and rinse with clean water to neutralize. Allow floor to dry) to supply a minimum 1 mil profile.
 - e. Full Prime coat of American Safety CZ7 primer @ 2-3 mil profile.
 - f. Full Top coat of American Safety 250 Non-Skid Epoxy System @ 20 mils DFT

When applying a nonskid floor, broadcast non-skid into the prime coat allow to dry sweep of excess sand then apply top coat.

PART 3. EXECUTION

3.01 GENERAL

- A. All work shall be performed by skilled craftsmen qualified to perform the required work in a manner comparable with the best standards of practice. The intent of the coating systems is to obtain smooth, clean, dry and well protected surfaces.
- B. All coating and painting shall conform to applicable standards of the National Association of Corrosion Engineers and the Steel Structures Painting Council Manual. Material applied prior to approval of surface by the Engineer shall be removed and reapplied at the expense of the Contractor to the satisfactions of the Engineer.
- C. Dust, dirt, oil, grease or any foreign matter that will affect the adhesion or durability of the finish must be removed by washing with clean rags dipped in a grease solvent and wiped with clean dry rags. Slag and weld metal accumulation and spatters shall be removed by chipping and grinding. All sharp edges shall be peened, ground or otherwise blunted as required by the Engineer.

- D. Painting systems include surface preparations, prime coatings and finish coatings. Unless otherwise specified, prime coat-coatings shall be field applied. Where prime coatings are shop applied, they shall be thoroughly cleaned and touched up in the field as specified. Any off-site work which does not conform to this specification is subject to rejection by the Engineer.
- E. The Contractor's coating and painting equipment shall be designed for the application of the materials specified and shall be maintained in first class working order. The Contractor's equipment shall be subject to approval of the Engineer.
- F. Application of the first coat shall follow immediately after surface preparation and cleaning and within an eight-hour working day. Any cleaned areas not receiving first coat within an eight-hour period shall be recleaned prior to application of first coat.
- G. Prior to assemble, all surfaces that are inaccessible after assembly, shall be prepared as specified herein and shall receive the paint or coating system specified.
- H. Drop cloths shall be used to protect floor and adjoining work from splatter. Any paint surface damaged shall be repaired to the satisfaction of the Engineer before the work will be accepted. The lines formed by changes in color of coatings shall be neat and straight.

3.02 SURFACE PREPARATION

- A. General:
 - 1. All surfaces to be painted shall be prepared in a workmanlike manner with the objective of obtaining a clean and dry surface. Surfaces shall be cleaned of all oil, rust, grease, dust, scale, and other foreign substances that may inhibit bonding.
 - 2. Field blast cleaning for all surfaces shall be dryblasted unless otherwise directed. Maximum particle size of abrasives shall be that which will produce a profile in accordance with recommendations of the coating manufacturer.
- B. All interior and exterior welded and abraded steel shall be sandblast cleaned in conformance with SSPC Section SP10 (Near White Blast Cleaning).
- C. Shop-Primed Ferrous Metal: Contractor shall be responsible for compatibility of the applied shop primer and the proposed field primer. If primers are compatible, all surfaces shall be cleaned in conformance with the manufacturers data sheet. If shop primer is not compatible, or damage to shop applied primer is too extensive for field touch-up, the surfaces shall be sandblasted in conformance to the surface preparation used in the shop.
- D. Shop Finished Ferrous Metal: Factory finished equipment or materials which have suffered damage to the shop applied coatings during shipment or installation shall be roughed-up in the field. All surfaces shall be cleaned in conformance with SSPC Section SP2 and touch-up shall be performed with paint supplied by the manufacturer.
- E. Immersion Service Ferrous Metal: Surfaces shall be sandblasted to a "Near White Clean" SP10 before application of the primer.
- F. Non-Ferrous Metal: Surfaces to be painted shall be cleaned in conformance with SSPC Section SP1 (Solvent Cleaned).
- G. Galvanized Metals: Galvanized surfaces to be painted shall be cleaned in conformance with SSPC Section SP1. Before application of the primer, surfaces shall be treated with one coat Koppers 40 Passivator (WF=4.0, DF=0.4).
- H. PVC Pipe: Surfaces shall be cleaned in conformance with SSPC Section SP1, except hand sanding shall be used to roughen the surface.

- I. Pipe with Factory Applied Bituminous Coatings: All exposed ductile iron pipe shall be painted. Bituminous coating shall be removed by sand blasting to near white before application of primer.
- J. Gypsum Wall Board: Cut out scratches, cracks, and abrasions in surfaces and openings adjoining trim and fill with approved filler. Bring filler flush with adjoining surfaces and when dry, sand smooth.
- K. Masonry: Surfaces shall be reasonably smooth and free of voids, cavities, dirt, dust, oils, grease or other contaminants.
- L. Concrete: Sweep sandblast to provide a surface profile. Allow concrete to cure for 28 days prior to painting.

3.03 COATING SYSTEMS APPLICATION

- A. All coatings shall be applied in strict accordance with the manufacturer's printed instructions and recommendations.
- B. All coatings shall conform to the film thicknesses as specified in Part 2 of this section. Coatings failing to meet the minimum dry film thickness shall be given additional coats until the minimum film thickness is attained.
- C. Undercoats shall be tinted similar to the finish coats. Each coat shall be slightly darker than the preceding coat.
- D. Each coat applied shall be inspected and approved by the Engineer before application of the succeeding coat.
- E. Allow each coat to dry thoroughly before applying the next coat.
- F. Finish coats shall be uniform in color and sheen without streaks, laps, runs, sags or missed areas.

3.04 CONTRACT CLOSE-OUT

- A. The Engineer shall make a detailed inspection of the paint work upon completion. All damage to surfaces resulting from the work of this section shall be cleaned, repaired or refinished as necessary, at no cost to the Owner.
- B. Upon completion of the work, all paint equipment and materials shall be removed from the site. Coating or paint spots, oil or stains upon adjacent surfaces shall be removed and the job site cleaned.

3.05 COLOR REQUIREMENTS

- A. All exposed piping (except stainless steel and galvanized steel) is to be painted and labelled. Piping color charts are to be submitted for Owner review. Piping to be painted and labelled per the following table:

EXPOSED PIPE COLOR & LABEL REQUIREMENTS		
Pipe	Color	Label
Influent Wastewater	Dark Green	-
Activated Sludge (WAS, RAS, SCUM)	Brown	-
Mixed Liquor	Light Brown	ML
Secondary Effluent & Final Effluent	Light Green	-
Air Gap Water (2W)	Light Blue	2W
Domestic Water	Dark Blue	Water
Air	White	AIR

- B. All exposed piping to be labeled with industrial grade water & UV resistant labels. Font color black with background matching piping color and 3" letter size. Labels to be applied by Contractor. In addition, all exposed piping to have direction of flow arrows with similar industrial grade stick-on labels. Coordinate all labels and arrows with the Engineer. Contractor to provide labeling walk-through with Owner and Engineer, and pipe labeling acceptance is required prior to Substantial Completion

3.06 INSPECTION HOLD POINTS INCLUDE:

- A. After surface reparation.
- B. After final coating.

*****END OF SECTION*****

SECTION 09 97 23 – CONCRETE PROTECTIVE COATING

PART 1. GENERAL

1.01 SURFACES TO BE COATED

- A. This specification covers work, materials and equipment required for protecting all interior concrete surfaces of structures specified by spray-application of a monolithic high-build epoxy coating to eliminate infiltration, provide corrosion protection, repair voids and enhance structural integrity. Procedures for surface preparation, cleaning, application and testing are described herein.
- B. Locations to be coated (see drawings for detailed locations):
 - 1. Influent Pump Station

1.02 SUBMITTALS

- A. The following items shall be submitted:
 - 1. Technical data sheet on each product used, including ASTM test results indicating the product conforms to and is suitable for its intended use per these specifications.
 - 2. Material Safety Data Sheets (MSDS) for each product used.
 - 3. Project specific guidelines and recommendations.
 - 4. Applicator Qualifications:
 - a. Manufacturer certification that Applicator has been trained and approved in the handling, mixing and application of the products to be used.
 - b. Certification by the protective coating manufacturer that the equipment to be used for applying the products has been approved and Applicator personnel have been trained and certified for proper use of the equipment.
 - c. Five (5) recent references of Applicator (projects of similar size and scope) indicating successful application of a high-build solvent-free epoxy coating by spray application.
 - d. Proof of any necessary federal, state or local permits or licenses necessary for the project.

1.03 QUALITY ASSURANCE

- A. Applicator shall initiate and enforce quality control procedures consistent with applicable ASTM, NACE and SSPC standards and the protective coating manufacturer's recommendations.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Materials are to be kept dry, protected from weather and stored under cover.
- B. Protective coating materials are to be stored between 50 deg F and 90 deg F. Do not store near flame, heat or strong oxidants.
- C. Protective coating materials are to be handled according to their material safety data sheets.

1.05 SITE CONDITIONS

- A. Applicator shall conform with all local, state and federal regulations including those set forth by OSHA, RCRA and the EPA and any other applicable authorities.

PART 2. PRODUCTS

2.01 EXISTING PRODUCTS

- A. Standard Portland cement or new concrete (not quick setting high strength cement) must be well cured prior to application of the protective coating. A minimum 28 days cure time is required before application.
- B. Cementitious patching and repair materials should not be used unless their manufacturer provides information as to its suitability for topcoating with an epoxy coating. Project specific submittals and procedures should be provided including application, cure time and surface preparation procedures that permit optimum bond strength with the epoxy coating.

2.02 MANUFACTURER

- A. Raven Lining Systems, Inc. or Tnemec Company, Inc.

2.03 REPAIR MATERIALS

- A. Repair materials shall be used to fill voids, structurally reinforce and/or rebuild surfaces, etc. as determined necessary by the engineer and protective coating applicator. Repair materials must be compatible with the specified epoxy coating and shall be applied in accordance with the manufacturer's recommendations.
- B. As an example, the following products may be accepted and approved by the protective coating manufacturer and could be used within the specifications:
 - 1. 100% solids, solvent-free epoxy grout that can be troweled or sprayed and specifically formulated for optimum epoxy topcoating compatibility. The epoxy grout manufacturer shall provide instructions for epoxy topcoating procedures.

2.04 PROTECTIVE COATING MATERIAL

- A. Raven Lining Systems' 405 Series or Tnemec Series G436 PermaShield epoxy coating system - a 100% solids, solvent-free two-component epoxy resin system thixotropic in nature and filled with select fillers to minimize permeability and provide sag resistance.

Product type	Amine cured epoxy
Color	Light Green
Solids Content (vol %)	100

2.05 PROTECTIVE COATING APPLICATION EQUIPMENT

- A. Manufacturer heated plural component spray equipment shall be used in the application of the specified protective coating.

PART 3. EXECUTION

3.01 ACCEPTABLE APPLICATORS

- A. Protective coating must be applied by a Certified Applicator of the protective coating manufacturer and according to manufacturer specifications.

3.02 EXAMINATION & CONDITIONS

- A. Appropriate actions shall be taken to comply with local, state and federal regulatory and other applicable agencies with regard to environment, health and safety.
- B. Installation of the protective coating shall not commence until the concrete substrate has properly cured and been prepared in accordance with these specifications.
- C. Temperature of the surface to be coated shall be maintained between 50° F and 100° F during application and curing. During application and curing the difference between the dew point and the surface temperature shall be no less than 5° F. Prior to and during application, care should be taken to avoid exposure of direct sunlight or other intense heat source to the structure being coated. Where varying surface temperatures do exist, care should be taken to apply the coating when the surface temperature is falling versus rising. The contractor shall be responsible for all heating and covering necessary to maintain the required temperature.

3.03 SURFACE PREPARATION

- A. Applicator shall inspect all surfaces specified to receive a protective coating prior to surface preparation. Applicator shall notify Owner of any noticeable disparity in the surfaces that may interfere with the proper preparation or application of the repair mortar and protective coating.
- B. All contaminants including: oils, grease, unsound or incompatible existing coatings, waxes, form release, curing compounds, efflorescence, sealers, salts, or other contaminants shall be removed.
- C. All concrete that is not sound or has been damaged by chemical exposure shall be removed to a sound concrete surface or replaced.
- D. Surface preparation method(s) should be based upon the conditions of the substrate and the requirements of the epoxy protective coating to be applied.
- E. Surface preparation method(s) should be based upon the conditions of the substrate and the requirements of the epoxy protective coating to be applied. At a minimum, surfaces to receive protective coating shall be cleaned and abraded to produce a sound concrete surface with adequate profile and porosity to provide a strong bond between the protective coating and the substrate. Contractor shall sandblast surfaces to be coated to between a CSP 3 and CSP 5 level. Detergent water cleaning and hot water blasting may be necessary to remove oils, grease, or other hydrocarbon residues from the concrete.
- F. Test prepared surfaces after cleaning but prior to application of the epoxy coating to determine if a specific pH or moisture content of the concrete is required according to manufacturer's recommendations.
- G. All surfaces should be inspected during surface prep and before the repair mortar is applied.

3.04 APPLICATION OF REPAIR MATERIALS

- A. Areas where structural steel has been exposed or removed shall be repaired in accordance with the Project Engineer's recommendations.
- B. Repair materials shall meet the specifications contained herein. The materials shall be trowel or spray applied utilizing proper equipment on to specified surfaces. The material thickness shall be specified by the Project Engineer according to Owner's requirements and manufacturer's recommendations.

- C. Cementitious repair materials shall be troweled to provide a smooth surface with an average profile equivalent to coarse sandpaper to optimally receive the protective coating. No honeycomb surfaces should remain after the final trowel procedure of the repair mortar.
- D. The repair materials shall be permitted to cure according to manufacturer recommendations.
- E. All surfaces should be inspected during and after preparation and before the protective coating is applied.

3.05 APPLICATION OF PROTECTIVE COATING

- A. Application procedures shall conform to the recommendations of the protective coating manufacturer, including material handling, mixing, environmental controls during application, safety, and spray equipment.
- B. The spray equipment shall be specifically designed to accurately ratio and apply the specified protective coating materials and shall be regularly maintained and in proper working order.
- C. The protective coating material must be spray applied by a Certified Applicator of the protective coating manufacturer.
- D. Specified surfaces shall be coated by spray application of a moisture tolerant, solvent-free, 100% solids, epoxy protective coating as further described herein. Spray application shall be to a minimum wet film thickness of 80 mils and an average wet film thickness of 100 mils.
- E. Airless spray application equipment approved by the coating manufacturer shall be used to apply each coat of the protective coating. Air assisted spray application equipment may be acceptable, especially for thinner coats (<10 mils), only if the air source is filtered to completely remove all oil and water.
- F. If necessary, subsequent topcoating or additional coats of the protective coating should occur as soon as the basecoat becomes tack free, ideally within 12 hours but no later than the recoat window for the specified product. Additional surface preparation procedures will be required if this recoat window is exceeded.

3.06 TESTING AND INSPECTION

- A. Coating system thickness shall be inspected to ensure compliance with the specifications herein.
 - 1. During application a wet film thickness gauge, meeting ASTM D4414 - Standard Practice for Measurement of Wet Film Thickness of Organic Coatings by Notched Gages, shall be used. Measurements shall be taken, documented, and attested to by Contractor for submission to Owner.
 - 2. After the coating product(s) have cured in accordance with manufacturer recommendations, coating system thickness may be measured according to SSPC-PA 9 - Measurement of Dry Coating Thickness on Cementitious Substrates Using Ultrasonic Gages.

- B. High voltage holiday detection for coating systems installed in corrosive environments, when it can be safely and effectively employed, shall be performed to ensure monolithic protection of the substrate. After the coating product(s) have cured in accordance with manufacturer recommendations, all surfaces shall be inspected for holidays in accordance with NACE RPO 188-99 Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates or ASTM D4787 Standard Practice for Continuity Verification of Liquid or Sheet Linings Applied to Concrete Substrates. All detected holidays shall be marked and repaired according to the coating product(s) manufacturer's recommendations.
1. Test voltage shall be a minimum of 100 volts per mil of coating system thickness.
 2. Detection of a known or induced holiday in the coating product shall be confirmed to ensure proper operation of the test unit.
 3. All areas repaired shall be retested following cure of the repair material(s).
 4. In instances where high voltage holiday detection is not feasible a close visual inspection shall be conducted and all possible holidays shall be marked and repaired as described above.
 5. Documentation of areas tested, equipment employed, results, and repairs made shall be submitted to the Owner/Engineer by Contractor.
- C. Adhesion of the coating system to the substrate shall be confirmed in a minimum of 10% of the manholes coated, or for large structures once every 1000 square feet of coated area. After the coating product(s) have cured in accordance with manufacturer recommendations, testing shall be conducted in accordance with ASTM D7234 Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers. Owner's representative shall select the areas to be tested.
1. For each test manhole a minimum of three 20 mm dollies shall be affixed to the coated surface; one at the cone area, one at the mid section and one near the bottom of the structure.
 2. For larger structures a minimum of three 20 mm dollies shall be affixed to the coated surface at random locations within each 1000 square foot area or as otherwise agreed upon.
 3. The adhesive used to attach the dollies to the coating shall be rapid setting with tensile strengths in excess of at least twice the anticipated failure point (generally 1000 psi) and permitted to cure in accordance with manufacturer recommendations. The coating and dollies shall be adequately cleaned and prepared to receive the adhesive. Failure of the dolly adhesive shall be deemed a non-test and require retesting.
 4. Prior to performing the pull test, the coating shall be scored to the substrate, or within 10 mils of the substrate surface, by mechanical means without disturbing the dolly or coating system bond within the test area.
 5. Two of the three adhesion pulls in each test area shall exceed 200 psi and shall include substrate adhered to the back of the dolly or no visual signs of the coating product in the test hole. Pulls tests with results between 150 and 200 psi may be acceptable if more than 50 percent of the substrate in the test area is adhered to the dolly.

6. Should a structure, or area, fail to achieve two successful pulls as described above, additional testing shall be performed at the discretion of the Owner or Project Engineer. Any areas detected to have inadequate bond strength shall be evaluated by the Project Engineer. Further bond tests may be performed in that area to determine the extent of potentially deficient bonded area and repairs shall be made by Contractor.
 7. All adhesion testing shall be performed by qualified personnel using calibrated equipment as specified by the applicable ASTM standard(s).
 8. All adhesion testing shall be documented and submitted in a consistent format detailing location, test values, description of the failure point/mode, scoring method employed, adhesive used, cure time of coating and adhesive and other data as deemed necessary by the owner/engineer.
 9. All adhesion test locations shall be repaired by the Contractor at no cost to the Owner.
- D. Visual inspection shall be made by the Project Engineer and/or Inspector. Any deficiencies in the finished coating affecting the performance of the coating system or the operational functionality of the structure shall be marked and repaired according to the recommendations of the coating product(s) manufacturer.
- E. The municipal sewer system may be returned to full operational service as soon as the final inspection has taken place and all coating materials have been adequately cured according to the coating product(s) manufacturer's recommendations.

*****END OF SECTION*****

SECTION 10 52 00 - FIRE EXTINGUISHERS

PART 1 - GENERAL

1.01 SUMMARY

- A. Work includes: Furnishing and installation of all fire extinguishers and cabinets as shown on drawings.
- B. Coordination: Verify that fire extinguisher cabinets are sized to accommodate fire extinguishers of type and capacity indicated.

1.02 SUBMITTALS

- A. Product data: Submit product data for each type of product included in this section. For fire extinguisher cabinets include roughing-in dimensions and details showing mounting methods, relationships of box and trim to surrounding construction, door hardware, cabinet type and materials, trim style and door construction, and panel style and materials.

PART 2 - MATERIAL

2.01 PRODUCTS

- A. Manufacturer: J.L. Industries, Bloomington, Minn. Extinguisher: Model Cosmic 5 E, UL rating 2A-10BC. Standard red finish.
- B. Mounting brackets: Provide manufacturer's standard brackets designed to prevent accidental dislodgment of extinguisher, of sizes required for type and capacity of extinguisher indicated, in manufacturer's standard plated finish.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. Install according to manufacturer's recommendations, providing adequate anchorage into material encountered. Comply with applicable regulations of governing authorities.
- B. Install all fire extinguisher devices where shown in the Drawings and in the approved shop drawings, anchoring all components firmly in place.
- C. Upon completion of the installation, and as a condition of its acceptance, visually inspect all the work under this section, check all components for proper operation and touch up all scratches and abrasions to be completely invisible. Each extinguisher shall bear inspection tag.

*****END OF SECTION*****

SECTION 10 73 05 – FRP LAUNDER COVERS

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install the following fiberglass reinforced plastic launder cover for the new Train 3 Clarifier weir launders.
- B. The Launder Cover Shall consist of a system of molded fiberglass panels that come together to form a continuous cover over the launder trough and weir within the treatment tank. The Cover shall be designed and manufactured to inhibit incident sunlight from striking the surfaces of the launder and weir. Each cover section shall be molded of UV-protected fiberglass and shall be opaque to sunlight. Individual sections shall be a minimum of four feet in length and curved to follow the curvature of the tank. The cover shall extend over the trough and weir as far as possible and may extend to a point immediately outside the scum baffle ring to avoid interference with the sweep arm. The Cover shall be designed such that adjacent panels fit together properly and the completed Cover, when installed, forms a rigid structure, and has a well-engineered and professional appearance.
- C. Launder cover panels shall have a cross-sectional ridge over the width of the launder trough in the direction of the flow> to strengthen the panel and minimize possible deflections against snow loads. Additional reinforcement against snow loads may be incorporated into the design by means of stiffening flanges fastened radially to the cover panel's underside.
- D. The Cover shall be designed to open away from the operator and toward the center of the tank. Each Cover segment shall consist of a single Cover Section fastened to the top or side of the weir wall using two (2) pivoting support brackets. The pivoting support brackets shall provide a rigid mount for the Cover Sections and ensure the proper fixed spacing between them. Cover systems comprised of cover sections with multiple parts connected by piano hinges are unacceptable.
- E. The Launder Cover sections can be designed to open in one of three (3) configurations:
 - i. Consecutively – in sequence
 - ii. Independently; or
 - iii. Alternatively – every other panel opens independently to allow the intermediate panels to open.
- F. Provision shall be made to secure the Cover in the closed position for safety and security. This is accomplished by means of an easily operated, spring-loaded latch mechanism that secures the Launder Cover panels in the closed position. Handles or lift rings may also be required for some panels. A means of limiting the travel of the Launder Cover sections, in the form of a restraint cable or tether, may also be provided to protect against damage. Covers with inspection hatches or cleanout doors are unacceptable.
- G. Where the circumference of the trough is interrupted by a bridge-support or another obstacle, a fixed panel(s) shall be installed over the trough beneath the support such that the surface of the Cover is continuous around the entire tank. Alternatively, vertical panels may be installed on both sides of the bridge supports to block out sunlight.
- H. The Cover system shall be designed to withstand common wind and snow loads but the entire Cover shall not be intended as a “walk-on” Cover designed to support the weight of plant personnel. Adequate stiffeners shall be integral to each panel, but panels reinforced with balsa or foam cores are not acceptable except where a single or double

length reinforced walk-on section is used for safe entry to the launder.

1.02 WARRANTY

- A. In addition to the standard warranty provided by the equipment manufacturer, the Contractor is to provide an extended warranty covering defects in material and workmanship for 1 year following the date of substantial completion of the equipment (per Supplementary Conditions). The warranty shall be in printed form, shall apply to all similar units, and shall include parts and labor.
- B. Supplied equipment to be free from defect in design, manufacture, workmanship and materials

PART 2.

2.01 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials, coatings, dimensions, wiring diagrams, controls, and misc. information.
 - 2. Operations and maintenance information as specified in Section 01 33 00.

PART 3. PRODUCTS

3.01 FIBERGLASS REINFORCED LAUNDER COVERS

- A. The pumps shall be Edson Platinum Peristaltic pumps with 3 HP/460 volt/3 phase motors, or approved equal. Pump capacity is to be 0 to 43 gpm. All materials are to be corrosion resistant and designed for the corrosive environment commonly found at WWTPs. Typical pump set point is anticipated to be between 10 and 30 gpm. Each pump is to include a VFD and to allow local adjustment of pump speed. Pumps are to be configured so that speed can be adjusted to a certain setting and left at that setting. A variable speed dial on local control panel is to be provided. Pump manufacturer is to include an electrical control panel for each pump. Pump manufacturer is to include a 304 stainless steel pump enclosure for each pump. Enclosure is to be approximately 40" x 40" x 40". Contact = Troy Talbot (Edson Pump Out Equipment), phone: 360-739-1056
- B. Each cover panel shall be molded of fiberglass, reinforced plastics. The resins and fiberglass reinforcing materials shall be consistent with environmental conditions and structural requirements of application.
- C. The resin shall be an industrial quality, isophthalic polyester resin with UV suppression additives, or equivalent. The resin shall be pigmented to ensure that the resulting part is opaque. The glass reinforcement shall be continuous fiberglass mats/roving combination, or equivalent. Multiple continuous panels of woven roving may be applicable for extra-large panels. Additional reinforcement in the form of stiffening ribs shall be added when necessary. The glass content of the finished laminate shall be not less than 35% by weight (ASTM D2584-18). The nominal thickness of each panel shall be ¼ inch. The laminate shall consist of resin rich glossy surface finish. The laminations shall be dense and free of voids, dry spots, cracks or crazes. All factory-trimmed edges shall be sanded and sealed. The finished laminate shall have a smooth, even appearance.
- D. FRP Launder Cover panels shall exhibit these minimum properties:

Tensile Strength (ASTM D-638)	26,000 psi
Flexural Strength (ASTM D-790)	30,000 psi
Flexural Modulus (ASTM D-790)	1.9 X 10 ⁶ psi

Barcol Hardness (ASTM D-2853)	50
Notched Izod (ASTM D-256)	20 ft-lbs/in
Water Absorption (ASTM D-570)	0.25% (MAX)

- E. Each cover panel's weight shall not supersede 55 lbs. max.
- F. Fasteners, handles, hinge, and latches shall be stainless steel. The weir wall mounting brackets shall be stainless steel, FRP or a combination of the two. The latch/handle shall be spring-loaded, where the mechanism shall have a positive detent positioned to indicate the closed/locked position of the handle. The spring-loaded latch is activated by pressing down on the handle and turning it.
- G. The tether or restraint cable shall consist of a length of stainless-steel cable secured to the tank wall and the hinged Cover Section by means of stainless-steel eyebolts. The length of the cable is selected to limit the travel of the Cover.

3.02 SPARE PARTS

- A. The FRP Launder Cover manufacturer is to include three complete sets of replacement wear parts for the Launder Covers.

PART 4 - EXECUTION

4.01 INSTALLATION

- A. In accordance with contract drawings, manufacturers' shop drawings and instructions.

4.02 MANUFACTURER'S SERVICES

- A. Installation assistance and certification: As required for proper installation prior to start up.
- B. Start-up and field testing: 2 full days on site, including all travel expenses.
- C. Operator Training: 1 full day on site.
- D. Warranty Service: As required during the warranty period.

***** END OF SECTION *****

SECTION 13 12 20 - METAL BUILDING SYSTEMS

PART 1 - GENERAL

1.01 SECTION INCLUDES

- A. Supplying and installing a rigid steel frame, bolted flange, pre-manufactured building, generally using galvanized cold formed secondary structural members bolted to welded clips with profiled and pre-coated metal roofing and siding screw attached.
- B. Metal building systems included as part of this contract are the:
 - 1. Blower Building
 - 2. Existing Building Modifications as shown on sheet A1.3.

1.02 REFERENCES

- A. AISC Manual of Steel Construction "Allowable Stress Design" (latest edition).
- B. AISI Specifications for the Design of Cold-Formed Steel Structural Members (latest addendum).
- C. MBMA Low Rise Building Systems Manual (latest edition).
- D. International Building Code (2018 edition).

1.03 SYSTEM DESCRIPTION

- A. Jurisdictional Design Requirements
 - 1. Roof collateral load: 5 pounds per square foot. (Note: building dead load shall be added to collateral by design engineer.)
 - 2. Roof snow load: 25 pounds per square foot unless local jurisdictional requirement exceeds value.
 - 3. Roof live load: 20 pounds per square foot minimum
 - 4. Wind speed: 105 miles per hour (ultimate), 81 miles per hour (nominal), exposure "C".
 - 5. Seismic loading: site class D; SS/S1: 1.13/0.40; SDS/SD1: 0.79/0.51; SDC: D; seismic importance factor: 1.25.
 - 6. Frost Depth: 18 inches.
- B. Performance requirements: Design steel frame building in compliance with applicable codes and specified design requirements.
- C. Building engineering and manufacturing is to be consistent with both AISC and MBMA

1.04 SUBMITTALS

- A. Design Calculations and Drawings: Suitable for review by approving building authority. Contractor shall be responsible for submitting to the Engineer, all drawings and calculations necessary to obtain building permits. Structural engineering drawings and calculations submitted by the building manufacturer and installer shall be sealed and signed by a structural engineer registered in the State of Washington. In addition, the anchor bolt embedment design shall be provided by the Contractor, conforming to calculations & details prepared and sealed by a structural engineer registered in the State of Washington (submitted along with metal building submittal).
- B. Erection Drawings: Show method of assembly and erection, bolt setting drawings, base plate connection, and base plate. Show all assembly details and provide detailed material sheets with size, weight and other descriptions.

1.05 QUALITY ASSURANCE

- A. Qualifications: Building fabricator must be certified by American Institute of Steel Construction for Category MB.
- B. Welders: AWS (American Welding Society)
- C. Manufacturer: Member of Metal Building Manufacturers Association (MBMA).

1.06 DELIVERY, STORAGE AND HANDLING

- A. Delivery: Deliver primary frames, secondary framing, panels, trim, flashing, accessories, bolts, nuts, washers and other erection hardware in building manufacturer's unopened packs and containers identified with building manufacturer's name and contents of each carton or pack.
- B. Storage: Store materials off ground and protect from damage. Slope galvanized secondary material and panel packages to avoid moisture accumulation and provide drainage. Cold formed components must be kept dry during storage to avoid corrosion between adjacent parts.
- C. Handling: Handle material properly to protect from damage, particularly during forklift movements.

PART 2 - PRODUCTS

2.01 MANUFACTURER

- A. Garco Building Systems or approved equal.

2.02 FRAMING MATERIALS

- A. Structural framing members (including hoist support columns and beams): manufactured 50 ksi hot rolled AISC shapes or welded, sheared, high strength plate, ASTM A992, ASTM A529, ASTM A570 or ASTM A572, 50,000 pounds per square inch minimum yield stress. All primary steel members, stiffeners, plates and gussets to be hot dipped galvanized after fabrication and mill nominal thickness.
- B. Secondary framing members: manufactured from ASTM A446, grade D, structural quality, G-90 galvanized, minimum 16 gauge.
- C. Structural bolts and nuts:
 - 1. Primary framing: use galvanized or cadmium plated ASTM A325, minimum 5/8-inch diameter, all thread. Length as required with direct tension indicator washers or hardened washers for “turn of the nut” method.
 - 2. Secondary framing: use galvanized or cadmium plated SAE grade 5, minimum 1/2 inch diameter, length as required.
- D. Wind bracing cables and rods:
 - 1. Cables: ASTM A475, galvanized, 7 strand EHS complete with eye bolts, rod shoes, and hardware as required.
 - 2. Rods: ASTM A36 steel, hot dipped galvanized, with manufacturer's standard galvanized hardware, as required.

2.03 ROOF PANELS

- A. Panels: water tight, standing seam with snap together, standing seam on 3-inch-high trapezoidal ribs at each edge.
 - 1. Material: minimum 24-gauge Galvalume/Zincalume conforming to ASTM A792 for coating AZ50. Minimum yield strength is 50 ksi.
 - 2. Panel coverage: 18 or 24 inches.
 - 3. Finish coating: factory applied thermoset modified polyester finish over Zincalume or factory applied Kynar based finish over Zincalume (Owners' Choice).
 - a. Color: Contractor to furnish Owner with color selection chart. Owner will select colors.
 - 4. Attachment: floating clips with 1/4 inch self drilling, self tapping screws.
 - 5. Uplift Rating: UL 90, (ASTM E1592).

2.04 WALL PANELS

- A. Panel: "R" type or "A" type with full bearing leading edge rib, 1-1/8 inch high trapezoidal ribs spaced 12 inches on center and two intermediate stiffener ribs.
 - 1. Material: minimum 26-gauge, Galvalume/Zincalume conforming to ASTM A792 for coating AZ50 or AZ55.
 - 2. Panel coverage: 36 inches.
 - 3. Finish coating: factory thermoset modified polyester finish painted over Galvalume/Zincalume or factory applied Kynar based finish over Galvalume/Zincalume (Owners' Choice).
 - a. Colors: Contractor to furnish Owner with color selection chart. Owner will select colors.
 - 4. Attachment: self-drilling, self tapping, HWH screws with UV resistant sealing washers through flat of panel if "R" type panel or through rib if "A" type panel.

2.05 MANUFACTURED UNITS

- A. Manufacturers building system types
 - 1. Blower Building: RF - Rigid Frame (42' bldg. width, 9.75' bldg. length, and minimum inside frame haunch clearance = 9'-0")
- B. Fabrication.
 - 1. Structural frame.
 - a. Fabricate by continuous submerged arc welding of 50 ksi plate flanges to cut-to shape web sections to steel plate wide flange sections.
 - b. Punch or drill butt-plates and base plates for field bolted assembly and weld into place.
 - c. Hold fabrication tolerances for primary and secondary to those listed by AISC.
 - d. Clips to be welded to frame for purlin and girt attachment with bolts in shear. Bolting of clips to primary flanges or direct bolting of purlins and girts to flanges is not an acceptable attachment method.
 - 2. Secondary
 - a. Fabricate by precise roll-forming or press brake. Purlins and girts with minimum 2-1/2 inch flat flange width with 1" edge stiffening lip.
 - b. All opening framing to be fabricated with minimum 16-gauge galvanized sections, clip bolted to secondary framing.

- C. Finishes
 - 1. All items shall be hot dipped galvanized steel.

PART 3 - EXECUTION

3.01 INSTALLER

- A. The installer shall have a minimum of 5 years experience in installation of similar buildings. The qualifications, including references, of other than the specified installer shall be submitted for review and approval by the Engineer prior to award of the Contract.

3.02 INSPECTION

- A. Site Verification of Conditions: Examine conditions under which work is to be done. Verify anchor bolt placement and base plate elevations are consistent with building manufacturer's "Approved for Construction" erection drawing set. Do not proceed until unsatisfactory conditions have been corrected.

3.03 ERECTION

- A. Use templates for accurate setting of anchor bolts. Level bearing plate area with steel wedges and grout or steel shims.
- B. Erect building frame true and level with vertical members plumb and bracing properly installed. Maintain structural stability of frame during erection.
- C. Ream holes requiring enlargement to admit bolts. Additional holes in structural members are to be drilled or mechanically cut. Burned holes for bolted connections not permitted without written approval by engineer.
- D. Tighten bolts and nuts in accordance with "specification for structural joints using ASTM A325 or A490 bolts" using specified procedure. Turn-of-nut tightening is preferred. (Calibrated wrench tightening; LeJeune type bolts; Direct tension indicator washers may be used).
- E. Set purlins and girts and bolt to appropriate clips. Attach to clips as required to satisfy design loads and as shown on drawings.

- F. Place roof panels at right angle to purlins and girts, assuring that secondary members are held straight and on proper module spacing. Attach and plumb wall panels as shown on Drawings. Maintain consistent 36-inch module coverage for entire length of wall. Pre-drill panels but not the leading profile. Lap panel ends minimum 6 inches on roof and 4 inches on walls. Place end laps over purlins or girts. Apply 3/16 inch butyl roof panel side and end lap seal between panel ends and side laps to provide water tight installation per details furnished.
- G. Place Standing Seam Roof panels at a right angle to purlins. Attach with sliding concealed clip where expansion and contraction must be accounted for. Lap panel ends 3 inches as determined by panel notch. Place end laps above purlin with backup plate and cinch strap so panel endlap fasteners do not penetrate purlin.
- H. Provide trim, flashing and closures as required for weather-tight installation and appearance.

****END OF SECTION****

SECTION 15 08 13 – TEMPORARY CONSTRUCTION SIGN

PART 1. GENERAL

- A. Contractor shall supply and erect a temporary construction sign per Department of Ecology and CDBG Department of Commerce requirements and standards.

1.01 SUBMITTALS

- A. The following items shall be submitted:
 - a. Proof of graphics to be printed.

1.02 PRODUCTS

- A. Dimensions, Materials, and Details
 - 1. Sign shall be 4' x 8' x ¾"
 - 2. Plywood Panel (APA Rated A-B Grade – Exterior)
 - 3. Sign shall consist of black lettering over a white background.
 - 4. Sign shall include the latest logo of the Department of Ecology, CDBG Department of Commerce, Engineer, and Contractor. Logos will be provided.
 - 5. Sign shall include the following information:
 - a. Project Name: **Eastsound Sewer and Water District Wastewater Treatment Plant Upgrade – Phase I (WQC-2021-EaSoWd-00208)**
 - b. Owner: **Eastsound Sewer and Water District**
 - c. County: **San Juan County, WA**
 - d. Engineer: **Wilson Engineering, LLC**
 - e. Contractor: **TBD**

END OF SECTION

SECTION 15 40 00 – MISC. PLUMBING

PART 1. GENERAL

1.01 SYSTEM DESCRIPTION

- A. Providing and installing miscellaneous plumbing for exterior purposes only as described herein and shown on plans,

1.02 SUBMITTALS

- A. Submittals: Shop Drawings and Product Data.

1.03 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with Uniform Building Code (UBC), Uniform Plumbing Code (UPC), and Eastsound Sewer and Water District Standards.

PART 2. PRODUCTS

2.01 MATERIALS

- A. Sanitary Drains:
 - 1. Soil and Waste Piping Above Ground: ABS Drain Waste Vent (DWV).
 - 2. Soil and Waste Piping Below Ground: ABS Drain Waste Vent (DWV).
- B. Unions: Provide at each pipe connection to equipment and fixtures, and where necessary to disconnect piping for repairs. Match pipe in which installed.
- C. Cleanouts: Provide at all locations required to clean piping, including end of main drain, base of all vertical stacks, at crosses, and elsewhere as required by codes and ordinances. Full size of piping served, accessible for use with conventional cleaning equipment. Provide brass lugs, with graphite lubricant on threads.
- D. 1" Yard Hydrants: 1" inlet yard hydrant with stainless steel operating rod, stainless steel pipe, automatic draining, equipped with backflow preventer, and all carbon steel components (if any) protected with fusion epoxy coating. Also, yard hydrant to be frost proof with ¾" male hose threads. Provide with threaded vacuum breaker attachment (Smith Cooper International Model #0190167II, or equal). Yard Hydrant to be non-freezing style with 4' bury depth, Woodford Iowa Model Y2, Merrill CV-1000, or equal.
- E. 2" Post Hydrants: 2" post hydrant with 316 stainless steel hardware, 304 stainless steel lower stem rod, permanently attached hand wheel, brass to brass working parts, and all carbon steel components (if any) protected with fusion epoxy coating. Provide with threaded vacuum breaker attachment. Post Hydrant to be configured for 4' bury depth, American AVK Dry Barrel 67XX, Eclipse Post Hydrant No. 2A, or approved equal. 2" Atmospheric Vacuum Breakers: Danfoss Model AVB, max pressure rating = 125 psi, ASSE Listed 1001, designed to prevent back siphonage.

PART 3. EXECUTION

3.01 INSTALLATION

- A. Slope drainage lines 1/4 inch per foot or as required to meet code.
- B. Provide accessible cleanouts as required by code.
- C. Install in accordance with Manufacturer's instructions and as per local codes and ordinances

*****END OF SECTION*****

SECTION 22 13 29 – SUBMERSIBLE SEWERAGE PUMPS

PART 1. GENERAL

1.01 DESCRIPTION

- A. Furnish complete, tested and operating, submersible pump stations, as shown on the Drawings and as specified herein.
- B. This work shall consist of supplying and installing a new, submersible pump station at the project site (Influent Pump Station), complete with controls, and any other appurtenances as described in these specifications and as shown on the Plans.
- C. Contractor is responsible for taking all necessary field measurements and coordinating with the pump supplier to provide a complete working pump station within the confines of the site configuration. The plans do not purport to show all of the necessary measurements and details for installation of the new submersible pump station equipment. Contractor shall prepare his bid in close coordination with said manufacturers so that no required components are inadvertently excluded from the bid due to miscommunication of intermediate suppliers or manufacturer's representatives. Contractor is ultimately responsible for the complete working operation of the proprietary installations.
- D. This work requires the installation of new pump station equipment, installation of new discharge piping, vaults, flow meter, connection to existing piping, and all of the associated modifications and components necessary for a complete working pump station at the project sites as shown on the Plans.
- E. This is a performance specification which is reliant upon the pump and control system manufacturers to provide, to the Contractor, all necessary components and instructions essential to a complete, working installation of the proprietary package of said manufacturer. The plans and specifications do not provide complete details, yet Contractor is required to provide the Owner with a complete working installation.

1.02 RELATED SECTIONS

- A. Related Requirements Specified Elsewhere:
 - 1. Section 01 33 00 – Submittal Procedures
 - 2. Section 01 50 00 - Temporary Facilities
 - 3. Section 03 30 00 – Cast-in-Place & Precast Concrete
 - 4. Section 31 23 33 – Trenching and Backfilling
 - 5. Section 33 05 00 – Common Works Results for Utilities
 - 6. Section 33 31 00 – Wastewater Piping
 - 7. Division 26 - Electrical

1.03 SUBMITTALS

- A. Submittals shall be made in accordance with SECTION 01 33 00. In addition, the following specific information shall be provided:
 - 1. Manufacturer's brochures and certified pump curves.
 - 2. Shop Drawing of complete pumping assembly, station, and lifting system.
 - 3. Catalog information and cuts.
 - 4. Manufacturer's specifications and equipment drawings.
 - 5. Manufacturer's parts lists, schematic and wiring diagrams.
 - 6. Complete lubrication, maintenance, and operation instructions.
 - 7. Control panel submittals including wiring diagram and panel layout.
 - 8. Interconnection wiring showing field wiring.
 - 9. Copy of manufacturer's warranty for pump.
 - 10. Factory test report.
- B. Affidavits: The Contractor shall furnish affidavits from the manufacturer stating that the pumps have been properly installed and tested, and each is ready for full time operation.
- C. Performance Testing: Certified non-witnessed factory performance tests in accordance with Hydraulics Institute Standards are required for each pump.

1.04 QUALITY ASSURANCE

- A. All equipment furnished under this Section shall: (1) be of a manufacturer who has been regularly engaged in the design and manufacture of the equipment for at least 5 years; and (2) be demonstrated to the satisfaction of the Engineer that the quality is equal to equipment made by those manufacturers specifically named herein.
- B. Field Tests. The pumping units shall be field tested after installation to demonstrate satisfactory operation, without causing excessive noise, vibration, cavitation, or overheating. The field testing shall be performed in the presence of an experience field representative of the manufacturer who shall supervise the startup and checkout of the equipment and who shall certify in writing that the pumps and motors have been properly installed, lubricated, adjusted, and prepared for operation.
- C. The completed pump station shall be given an operational test of all equipment at the factory to check for excessive vibration, leaks in all piping and seals, correct operation of the automatic control system and all auxiliary equipment. All irregularities shall be corrected at the factory. Copies of this factory testing shall be submitted to the Engineer for review and approval PRIOR TO PUMP INSTALLATION.

1.05 WARRANTY

- A. The submersible pumps shall be warranted for a period of five (5) years or 10,000 hours of operation to be of quality construction, free from defects in material and factory workmanship.

- B. Major components that fail to perform as specified by the engineers (or as represented by the manufacturer) or prove defective in service during the warranty period shall be replaced, repaired, or satisfactorily modified by the manufacturer without cost of parts or labor to the owner. After start-up service has been performed, labor to replace accessory items such as blowers, heaters, or other accessible and easily serviced parts, shall be the responsibility of the owner. Such components, parts, or repairs determined by the manufacturer to have failed because of defects in workmanship or materials will be replaced or repaired F.O.B. factory or other designated location.

1.06 WARNING SIGNS

- A. Each piece of driven machinery which can be started manually by any control station not within 15 feet of the machine, or which can be controlled automatically by any means, shall be identified with a warning sign inscribed:

CAUTION
THIS MACHINE IS REMOTELY CONTROLLED
AND MAY START AT ANY TIME

The word "CAUTION" shall be in yellow block letters on black panel at the top of the sign which shall have yellow background and text in black block letters. Signs shall be fabricated from 30-gauge copper bearing steel and finish shall be high baked enamel; the finished signs shall be weather resistant. Signs shall be ten inches wide by seven inches high. Signs shall be located in prominent locations at machines and shall be fastened to surfaces previously specified for nameplates.

PART 2. PRODUCTS

2.01 PUMP STATION - GENERAL

- A. The pump station shall be designed to provide the pumping requirements and functions indicated and implied by the Drawings and these Specifications and to provide trouble-free operation with minimum maintenance. The system shall readily enable manual operation of any and all functions in the event of failure of any one component.
- B. The Pump Station Supplier shall be locally available for onsite response when called within a reasonable time, depending upon the nature of the emergency. The Supplier shall have personnel available 24 hours a day, every day of the year – NO EXCEPTIONS.
- C. All submersible pumps for this project shall be by the same manufacturer – NO EXCEPTIONS.

2.02 SUBMERSIBLE WASTEWATER PUMPS

- A. General: Submersible wastewater pumps shall be heavy-duty, submersible, non-clog, centrifugal, quick disconnect pumps. The pumps shall be capable of operating in the range of capacity specified on a continuous basis with no detrimental effects to the pump or motor.

B. Pump Schedule: The pump operating characteristics shall be as follows.

1. Influent Pump Station – (two small pumps, characteristics below are for each pump)
 - a. Design Condition (gpm) 150 gpm @ 20.5 ft
 - b. Additional operation point 1: 120 gpm @ 22.0 ft
 - c. Additional operation point 2: 220 gpm @ 17.5 ft
 - d. Minimum Shutoff Head 26 ft
 - e. Rated Speed 1800 rpm
 - f. Pump Drive Type single speed
 - g. Nominal Motor Horsepower Size 2.7 HP
 - h. Required minimum hydraulic efficiency at operating point flow rate (percent) 43%
Manufacturer and Pump Type Flygt N3085 MT
Impeller Adaptive N-impeller, non-clog (size = 135 mm)
 - i. Discharge 3 1/8" diameter
2. Influent Pump Station – (two large pumps, characteristics below are for each pump)
 - a. Design Condition (gpm) 385 gpm @ 20.5 ft
 - b. Additional operation point 1: 320 gpm @ 22.5 ft
 - c. Additional operation point 2: 500 gpm @ 15.0 ft
 - d. Minimum Shutoff Head 38 ft
 - e. Rated Speed 1800 rpm
 - f. Pump Drive Type single speed
 - g. Nominal Motor Horsepower Size 5.0 HP
 - h. Required minimum hydraulic efficiency at operating point flow rate (percent) 66%
Manufacturer and Pump Type Flygt N3102 MT
Impeller Adaptive N-impeller, non-clog (size = 152 mm)
 - i. Discharge 3 15/16" diameter

C. Pump Construction:

1. General
 - a. The pumps shall be designed to permit removal of pumping units from the wetwell for inspection or service without disconnecting or disturbing the discharge piping. The pump connection shall be metal to metal or with secondary O ring seal. The design shall permit the pumps when lowered into place to be automatically connected to the discharge piping

by positively locking the volute in position to prevent any axial or lateral movement. There shall be no need for personnel to enter the wetwell when pump inspection or service is required. Pump assembly, including motor, pump and cable accessories must be rated for Class 1, Division 1 hazardous environment, explosion proof, group C & D.

- b. Lifting assemblies and discharge mating flanges shall be spark proofed, Factory Mutual Standards.
2. Piping, Fittings, and Appurtenances: Each pump shall be furnished with quick-disconnect discharge elbow, Type 316 stainless steel Schedule 40 pipe rails, upper guide rail bracket, intermediate guide rail bracket, rail-guided lifting assembly, and stainless steel chain of sufficient strength to raise and lower pump. All guide rail components and fasteners shall be type 316 stainless steel.
3. Pump Castings: Castings shall be of cast iron or semi-steel of uniform quality and free from blowholes, porosity, hard spots, shrinkage defects, cracks and other injurious defects. The casings shall be designed to permit replacement of wearing parts. Joints shall be properly sealed with O rings and shall not leak under a test pressure equal to 50 percent greater than the pump discharge pressure or the total dynamic head, whichever is greater. Passageways shall permit smooth flow and shall be free from sharp turns and projections.
4. Impellers: Impellers shall be high chromium hard iron suitable for the service required. The impellers shall be smooth and free flowing and shall have sufficient clearance to permit objects in the sewage that enter the pump to pass into the discharge pipe. Each impeller shall be accurately fitted and keyed, splined, or threaded on the shaft, and locked in such a manner that lateral movement will be prevented and reverse rotation will not cause loosening.
5. Balance: All rotating parts of the equipment shall be in such balance, mechanically and hydraulically, as to operate throughout the required range without excessive end thrust, vibration or noise.
6. Shafts: Shafts shall be steel, shall be of sufficient size and strength to perform the work required, and shall be adequately provided with alignment bearings.
7. Bearings: Bearings subject to submersion shall be ball bearings manufactured from high-grade bearing alloy. Bearing shall have a minimum B 10 life of 18,000 hours.
8. Mechanical Seals: Each pump shall be equipped with tungsten carbide seals.
9. Electrical Motors: The pump motor shall be induction type with a squirrel cage rotor, housed in an air filled, watertight chamber, NEMA B type. The stator windings and stator leads shall be insulated with moisture resistant Class H insulation rated for 356 °F. The motor shall have a minimum service factor of 1.15. Power cords shall be non-potted and removable. Stator windings shall be trickle impregnated with resin and rated at 180°C (355°F). Stator windings shall be embedded with three (3) thermal switches for overheating protection. The motor shall have integral moisture sensors that shall be monitored by a motor saver relay. The stator shall be heat-shrink fitted into the housing and locked against rotation. Motor speed and power specifications shall be as follows:
 - a. Influent Pump Station – Small Pumps - Submersible, 1800 rpm, 2.7 HP, 460 Volt, 3 phase, 60 Hertz.

- b. Influent Pump Station – Large Pumps - Submersible, 1800 rpm, 5.0 HP, 460 Volt, 3 phase, 60 Hertz.
- 10. Motor Saver Relay: The pumps shall be pre-wired for connection to ITT Flygt MiniCAS – 120 motor saver relay. Contractor shall supply ITT Flygt MiniCAS – 120 to Control Systems Integrator.
- 11. Miscellaneous Metals: Bolts, nuts, anchors, washers, and all other types of supports necessary for the installation of the pumps and drive units shall be furnished and shall be of Type 316 stainless steel.
- 12. Shop Painting: Pump, motor, and accessories shall be factory applied and finish painted in accordance with the manufacturer's standard.
- D. Pumps must be capable of operating within pump manufacturer's recommended operating envelopes through the full range of design flows and speeds. Any restrictions shall be incorporated into the pump control logic to prevent undesirable operating conditions from occurring at no additional cost to the OWNER.
- E. Pump Removal Equipment: Provide and install Type 316 stainless steel guide bars and cable, brackets, and Type 316 Stainless steel lifting chain. There shall be a minimum of two guide bars or cables for each pump. Removal equipment shall be installed in a manner that prevents excessive deflection under normal operating conditions.
- F. PRESSURE GAGE
 - 1. Pressure gages at each pump station shall be Marsh Instruments Series P Pressure Gage, Model No. P0142, or approved equal. Dial face shall be minimum 4-1/2" diameter. Gage mounting shall be selected to conform to available space in the valve vault or valve platform, and provide operator viewing from the surface at the vault opening. Gage range shall be 0-25 psi, ASME Grade 2A rated to +/- 0.5% of full scale.
 - 2. Flow isolator shall be a threaded off-line diaphragm seal with flushing connection, suitable for use on sewage force mains. Unit shall have a replaceable, Type 316 stainless steel diaphragm seated on Buna-N O rings, with stainless steel lower unit and carbon steel upper unit. Flow isolator shall be Branom Instrument Co., Model ICO A1S62Q1SLBC200N or approved equal.
 - 3. Shutoff valve shall be bronze, quarter turn globe valve with threaded ends and stainless ball and stem, Apollo Model No. CON 70-101-01 or approved equal.
 - 4. Gage, isolator and shutoff are available from Branom Instrument Co.

2.03 EQUIPMENT FOR WET WELL INSTALLATION

- A. For each pump the contractor shall supply and install a discharge connection made of cast iron ASTM A-48, Class 35B.
- B. The pump(s) shall be automatically and firmly connected to the discharge connection, guided by no less than two parallel guide bars extending from the top of the station to the wet well mounted discharge connection. The material of the guide bars shall Stainless steel AISI 316.
- C. The length of the guide bars shall be 15 feet (contractor to verify) and they shall be fastened at the top of the station with a guide bar holder made of Stainless steel AISI 316.
- D. The contractor shall supply and install **three** cable holders, each made with 4 hooks of Stainless steel AISI 316.

- E. There shall be no need for personnel to enter the wet-well.
- F. The sealing of the pumping unit to the discharge connection shall be accomplished by a machined metal to metal contact. Sealing of the discharge interface with a diaphragm, O-ring or profile gasket will not be accepted. The entire weight of the pump/motor unit shall be borne by the pump discharge elbow. No portion of the pump/motor unit shall bear on the sump floor directly or on a sump floor mounted stand.

2.04 LIFTING EQUIPMENT FOR PUMPS

- A. Each pump shall be fitted with 20 feet of stainless steel lifting chain with spectra cord for use with the Grip-Eye lifting device. The working load of the lifting system shall be 50% greater than the pump unit weight.

2.05 SPARE PARTS

- A. Each new pump station shall be furnished with the following spare parts, plus any additional spare parts listed as recommended by the manufacturer:
 - 1. Complete replacement mechanical (pump shaft) seal, including all gaskets, seals, sleeves, O-rings, and packings required to be replaced during replacement of the seal.
 - 2. Impeller clearance adjustment shims
 - 3. One plug-in seal unit.
 - 4. One quart seal lubricant
 - 5. One cover plate O-ring
 - 6. One rotating assembly O-ring.
 - 7. One filter element for the seal filter
 - 8. One volute gasket.

PART 3. EXECUTION

3.01 PUMP AND MOTOR

- A. Install pumps and sump floor anchoring in accordance with the manufacturer's recommendations. Pumps must be installed plumb & level.
- B. Each pump shall be installed in accordance with the manufacturer's recommendations by a qualified professional with experience installing similar pumping systems. Mount pumps with motors in concrete wetwells as shown on the drawings. Anchor bolt size and material shall be as per manufacturer's recommendations. Bolts shall be accurately placed with templates and bolt thread projections covered with 2 to 3 wraps of duct-tape.
- C. Control Systems Integrator shall incorporate ITT Flygt MiniCAS – 120 motor saver relay into pump station control panel. Control Systems Integrator shall be responsible for connecting pumps to motor saver relay and confirming operation.

3.02 FLOAT SWITCHES

- A. Float switches shall be suspended at levels as shown on the Plans. Suspension cables shall be secured on the cable holder forming a large loop such that conductors in the cable shall not be squeezed tight. In addition, please note that a submersible level transducer will also be used for pump control.

3.03 START-UP SERVICES

- A. The pump equipment manufacturer shall furnish the services of a qualified factory trained field service engineer for a minimum of one (1) 8-hour day per each pump station (travel time excluded) to inspect the installations and instruct the Owner's personnel on the operation and maintenance of the pumps. The Contractor is responsible for coordinating start-up with the supplier/pump manufacturer. After the pumps have been completely installed and wired, the Contractor shall have the pump manufacturer's field engineer perform the following
1. Inspect megger stator and power cables.
 2. Check seal lubrication.
 3. Check proper rotation.
 4. Check power supply voltage.
 5. Measure motor operating load and no load current.
 6. Check level control operation sequence.
 7. Demonstrate to the Owner each submersible pump being lifted out of the wetwell and then reinstalled using the lifting chain.
 8. Perform flow tests to confirm pump's ability to meet the specified design flow conditions.
- B. The Contractor is required to provide any additional information needed by the pump supplier to validate pump warranties (i.e. voltage and amp readings for each leg of line).

3.04 PAINTING

- A. All painting shall be in accordance with manufacturer's standards appropriate for exterior wastewater applications.

3.05 ELECTRICAL

- A. All electrical work shall be in accordance with State and Local code. Reference Electrical and Controls specifications for additional information regarding the electrical and control system design.

***** END OF SECTION*****

SECTION 22 13 36 – PACKAGED SLUDGE PUMPS

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install the following sludge pumping units: one (1) Waste Activated Sludge (WAS) Pump, one (1) Digested Sludge Pump, and one (1) swing pump which can be used for either application (WAS or Digested Sludge). Three (3) sludge pumps total.
- B. Pumps shall be designed to operate smoothly and efficiently at delivering dilute sludge which has been screened and conveyed through a District wide septic tank effluent system. Solids concentration may range anywhere from 0.5% to 3.0%.

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials, coatings, dimensions, wiring diagrams, controls, and misc. information.
 - 2. Operations and maintenance information as specified in Section 01 33 00.

PART 2. PRODUCTS

2.01 PUMP

- A. The pumps shall be Edson Platinum Peristaltic pumps with 3 HP/460 volt/3 phase motors, or approved equal. Pump capacity is to be 0 to 43 gpm. All materials are to be corrosion resistant and designed for the corrosive environment commonly found at WWTPs. Typical pump set point is anticipated to be between 10 and 30 gpm. Each pump is to include a VFD and to allow local adjustment of pump speed. Pumps are to be configured so that speed can be adjusted to a certain setting and left at that setting. A variable speed dial on local control panel is to be provided. Pump manufacturer is to include an electrical control panel for each pump. Pump manufacturer is to include a 304 stainless steel pump enclosure for each pump. Enclosure is to be approximately 40" x 40" x 40". Contact = Troy Talbot (Edson Pump Out Equipment), phone: 360-739-1056

2.02 CONTROLS

- A. See Section 26 80 00, 3.09 A. WAS Pump Control and H. Rotary Fan Press System.

2.03 SPARE PARTS

- A. The pump manufacturer is to include three complete sets of replacement wear parts for pump equipment.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. In accordance with contract drawings, manufacturers' shop drawings and instructions.

3.02 MANUFACTURER'S SERVICES

- A. Installation assistance and certification: As required for proper installation prior to start up.
- B. Start-up and field testing: 2 full days on site, including all travel expenses.
- C. Operator Training: 1 full day on site.
- D. Warranty Service: As required during the warranty period.

***** END OF SECTION *****

SECTION 23 34 00 - WALL FANS & LOUVERED VENTS

PART 1 - GENERAL

1.01 SUMMARY

- A. Work includes providing and installing the following (as noted herein and shown on the drawings):
- Blower Building: two 3,300 CFM Wall Fans (each wall fan to be controlled by an individual thermostat, two thermostats total), one motorized 36"x84" louvered vent (vent is to open if either of the two fans are powered 'ON'; vent is to remain open until both wall fans are powered 'OFF'), and one manual 24"x24" louvered vent.
 - Existing Dewatering/Train 1&2 Building Area: three 3,300 CFM Wall Fans (each wall fan to be controlled by a manual on/off switch. Three switches total.) and two manual 36"x84" louvered vents.
- B. Each wall fan shall be installed with a gravity combination louver with stainless steel bird screen. In addition, a minimum MFR warranty of 2 years is required.

PART 2 - MATERIALS

2.1 WALL FANS

- A. 3300 CFM Wall Fans
1. Manufacturer: Acme Engineering & Manufacturing Corporation
 2. Distributor: Coast Products, Inc., (800) 735-7026
 3. Type: Model # FQ18G6, 3300 CFM, 1/2 HP Fan, 120V, single phase, with wall collar accessory and guard on back side of fan. 12 sones noise level (max).
 4. Operation: Wall fans to be dual speed and thermostat controlled. Include motor side guard & wall collar (for all fans).
 5. Thermostats: Schaffer model TH109, wall mounted with a 2-speed control that can be used for cooling with 2 speed fan motors. 40F to 100F temperature range dial. All thermostats are to be housed in a NEMA 4X enclosure, to be provided with stainless steel coil sensors, and to be located 4' above floor elevation.

B. Gravity Combination Louvers

1. Manufacturer: Louvers & Dampers, Inc.
2. Distributor: Coast Products, Inc., (800) 735-7026
3. Type: EBE445, with combination backdraft damper and stationary louver (aluminum), size to match each fan individually (approximate dimension = 24" x 24" for 3300 CFM fans). Provide with wall collar accessories and channel frame, unless Contractor prefers otherwise.

2.2 LOUVERED VENTS FOR BLOWER BUILDING

A. Motorized Louvered Vent

1. Manufacturer: Pottoroff
2. Distributor: Coast Products, Inc., (800) 735-7026
3. Type: EXD-645 combination louver/damper, (unit is to be 36" wide x 84" tall; see summary above), all with fixed drainable blades, integral damper with vinyl blade and jamb seals. Include removable/washable stainless steel insect screens – mesh 18 x 16. Include low leakage jamb seals for all units. Provide with channel frame, unless Contractor prefers otherwise. Include all options for automatic operation. Provide with non-protruding actuation system, no obstructions are to protrude into the room.
4. Operation: Automatic louver/damper operation by electric actuator, Pottoroff Model AF120, 120 V, spring return actuator which will power open damper(s) when pre-set thermostat setting is reached. Control to be from auxiliary contact(s) with associated wall fan thermostat(s) in same room.

B. Manually Operated Louvered Vent

1. Manufacturer: Pottoroff
2. Distributor: Coast Products, Inc., (800) 735-7026
3. Type: EXD-645 combination louver/damper, 24"x24", with fixed drainable blades, integral damper with vinyl blade and jamb seals. Size as noted in summary. Include removable/washable stainless steel insect screens – mesh 18 x 16. Include low leakage jamb seals for all units. Provide with channel frame, unless Contractor prefers otherwise. Include all options for manual operation. Provide with non-protruding actuation system, no obstructions are to protrude into the room.

2.3 LOUVERED VENTS FOR EXISTING DEWATERING BUILDING AREA

A. Manually Operated Louvered Vents

1. Manufacturer: Pottoroff
2. Distributor: Coast Products, Inc., (800) 735-7026
3. Type: EXD-645 combination louver/damper, (units are to be 36" wide x 84" tall; see summary above), all with fixed drainable blades, integral damper with vinyl blade and jamb seals. Include removable/washable stainless steel insect screens – mesh 18 x 16. Include low leakage jamb seals for all units. Provide with channel frame, unless Contractor prefers otherwise. Include all options for manual operation. Provide with non-protruding actuation system, no obstructions are to protrude into the room.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Deliver vents in manufacturer's original, unopened cartons. Store in a safe place until ready for installation.
- B. Follow manufacturer's printed instructions for installation.
- C. Remove all identifying marks, labels, etc. from exterior.
- D. Install controls for all fans at 4' above floor.

END OF SECTION

DIVISION 26 – ELECTRICAL SPECIFICATIONS



09/12/2022

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SECTION 26 05 00 – GENERAL ELECTRICAL

PART 1 GENERAL

1.01 DESCRIPTION

- A. The General Conditions, Supplementary Conditions, and related work in other Sections apply for all work in Section 26.

1.02 SCOPE OF WORK

- A. This section specifies general requirements for electrical work. Detailed requirements for specific electrical items are specified in other sections, but are subject to the general requirements of this section. The electrical drawings and schedules included in this project manual are functional in nature and do not specify exact locations of equipment or equipment terminations. It is the intent of this Section of the Specifications and the accompanying drawings to describe and provide for the furnishing, installing, testing and placing in satisfactory and fully operational condition all equipment, materials, devices and necessary appurtenances to provide a complete electrical system, together with such other miscellaneous installations and equipment hereinafter specified and/ or shown on the drawings. The work shall include all materials, appliances and apparatus not specifically mentioned herein or shown on the drawings, but which are necessary to make a complete, fully operational installation of all electrical systems shown on the drawings or described herein. Equipment and devices furnished and installed under other Sections of this specification shall be connected under this Section. The drawings and specifications are complementary and what is called for in either is binding as if called for in both.
- B. This project may include installation of packaged equipment system(s) or sub-system(s) that will require coordination between the Contractor and the manufacturer to determine the detailed installation requirements. The Engineer has shown general installation information for these systems based on the best information available at the time of design. Where indicated on the drawings to ‘provide a complete and operational system’ the Contractor shall provide all materials, installation, and coordination with the manufacturer so the equipment is installed and operates in a satisfactory manner. Minor changes in equipment locations, quantity of terminations or wires, junction boxes, conduit, etc shall be included in the Contract price.
- C. See all other Sections of these specifications for work in other areas and disciplines related to this project.

1.03 GENERAL DESCRIPTION OF ELECTRICAL WORK

- A. The Contractor shall provide all labor, material, tools, equipment and services required to complete the furnishing, installation, wiring, connection, calibration, adjustment, testing and operation of all electrical equipment, devices and components as indicated and implied by the drawings and specifications. General descriptions include:
 - 1. Complete the procurement, installation, wiring, connection, calibration, adjustment, testing and operation of all electrical devices, components, accessories and

equipment that is not shown or specified but which is nonetheless required to make the systems shown and specified function properly.

2. Install all equipment so it shall be readily accessible for maintenance. Installations shall have electrical clearances in accordance with NEC and shall be installed in locations that will provide adequate cooling.
3. Check electrical equipment prior to installation so that defective equipment is not installed.
4. Provide field services of qualified technicians to supervise and check out the installation of the equipment, to supervise and check out interconnecting wiring, to conduct start-up of operation of the equipment, and to correct any problems that occur during start-up.
5. Provide circuit breakers, conduit, wire and installation for all items that require electrical power.

1.04 PROJECT DESCRIPTION

- A. This project shall consist of the construction of process improvements and facility improvements at the Eastsound Waste Water Treatment Plant (WWTP) located on Orcas Island, WA.
- B. The major electrical work shall include, but not limited to:
 1. Selective electrical demolition of existing electrical service area.
 2. Upgrade of existing utility service from 208Y/120V to 480Y/277V.
 3. New standby diesel generator and service rated Automatic Transfer Switch.
 4. New power distribution including panels, raceway, wiring, disconnects, etc.
 5. Lighting and lighting controls.
 6. Motor Control Center with power distribution, motor starters and Variable Frequency Drives (VFD's) as shown on the drawings.
 7. PLC based control panel with operator interface in Blower building,
 8. PLC based control panel for Influent pump station.
 9. Instrumentation including as shown on the drawings including level transducers, float switches, pressure transmitters, DO sensors, gas sensors, door switches, smoke detectors, flow meters and other devices as shown on the drawings.
 10. New Supervisory Control and Data Acquisition computer system, as specified.
 11. Programming and commissioning of all PLC, operator interface and SCADA system hardware and software shall be provided by the System Integrator.

12. Installation of packaged equipment and treatment systems.
13. All other electrical work as shown on the drawings.

1.05 ELECTRICAL CONTRACTOR MINIMUM QUALIFICATIONS

- A. The Electrical Contractor shall have a minimum 5 years of experience with having performed similar construction installations.
- B. Provide a summary of qualifications with the bid form, including Owner, project description, and summary of electrical equipment manufacturer and ratings that shows similar project experience.

1.06 CODES AND REGULATIONS

- A. The electrical systems shall be installed based on the following current Standards:
 1. NFPA 70 - National Electrical Code - current version as adopted by Washington State Department of L&I.
 2. Washington Administrative Code (WAC) chapter 296-46B.
 3. Building Codes - International Conference of Building Officials as adopted and amended by the Local Jurisdiction.
- B. The Contractor is required to familiarize himself with the detailed requirements of these standards and any local codes and ordinances as they affect the installation of specific electrical systems.
- C. Identification of Listed Products
 1. All materials and equipment specified herein shall be within the scope of Nationally Recognized Testing Laboratory (NRTL) examination services, be approved by the NRTL for the purpose for which they are used, and shall bear the appropriate listing label.
 2. Equipment listed/labeled by an NRTL shall be as dictated by the latest printing of the Electrical Testing Laboratories Accreditation Report available from the State of Washington Department of Labor and Industries, Electrical Inspection Division. Any NRTL listing/labeling shall be as accepted by the local authority having jurisdiction.
 3. When a product is not available with a testing laboratory listing for the purpose for which it is to serve, the product may be required by the inspection authority to undergo a special inspection at the manufacturer's place of assembly. All costs and expenses incurred for such inspections shall be included in the original contract price.
 4. All skid mounted and packaged equipment systems shall be labeled in accordance with the requirements of the State of Washington Department of Labor and Industries, Electrical Inspection Division.

5. Where field modifications require field inspection for listing/labeling, the Contractor shall be responsible for all field inspection fees for listing/labeling of all final modified electrical assemblies.

D. Thermal ratings of equipment terminations

1. All materials shall conform to the National Electrical Code Article 110-14C. Wiring and circuit breakers on this project are designed for 75 deg C operation above 100 amperes; 60 deg C for 100 amperes and below.
2. All products furnished on this project shall have electrical terminations rated for 60 deg C for ampacities of 100 amperes and below, and rated for 75 deg C for ampacities above 100 amperes.

1.07 PERMITS AND FEES

- A. The Contractor shall obtain and pay for all licenses, permits and inspections required by laws, ordinances and rules governing work specified herein.
- B. The Contractor shall arrange for inspection of work by the inspectors and shall give the inspectors all necessary assistance in their work of inspection.

1.08 COORDINATION

- A. The Contractor responsible for accomplishing Section 26 work shall coordinate his work with that of the other Contractors and/ or other trades doing work on the project and shall examine all drawings and specifications of other trades for construction details and necessary coordination.
- B. Obtain submittals and shop drawings of all equipment with electrical connections furnished under other Sections of the specifications. Provide all wiring in accordance with requirements indicated. Advise the Engineer of any changes which may affect the contract price.
- C. Special attention is called to the following items and all conflicts shall be coordinated prior to installation:
 1. Location of pipes and equipment so that all electrical equipment, lighting fixtures and other electrical outlets and equipment are clear from and in proper relation to these items.
 2. Recessing and concealing electrical materials in CMU walls, concrete construction and similar construction methods.
 3. Electrical characteristics (HP, KVA, voltage, phase) of actual equipment furnished under other Sections being different from that shown on the electrical drawings.
- D. The Contractor will not be paid for relocation of work, cuttings, patching and finishing required for work requiring reinstallation due to lack of coordination prior to installation.

1.09 SITE FAMILIARIZATION

- A. The Contractor shall become familiar with all features of the site which may affect the execution of the work prior to submitting a bid.
- B. The Contractor shall take all field measurements necessary for the work and shall assume full responsibility for their accuracy.
- C. The Contractor shall take full responsibility for locating and avoiding all substructures and utilities. Any damage to existing equipment or utilities shall be repaired or replaced by the Contractor at the Contractors expense.

1.010 AREA CLASSIFICATIONS

- A. The following classification of areas shall be used as a reference in determining application of material covered by this Section unless specifically shown otherwise on the drawings.
 - 1. Outdoor, Damp or Corrosive Areas:
 - a. Raceways shall be Rigid Galvanized Steel (RGS). Conduit entrances shall be threaded and fittings shall have gasketed covers. Threaded fastening hardware and rods shall be galvanized or stainless steel. Raceway supports such as channel, clamps, and brackets shall be galvanized steel, stainless steel or aluminum. Panels and boxes shall be NEMA 4X, 3R or as shown on the drawings. Enclosures shall be mounted 1 inch from walls to provide an air space unless specifically shown otherwise. Device boxes shall be cast, copper free aluminum.
 - 2. Below Grade Areas:
 - a. Conduits shall be Schedule 40 PVC, or as indicated on the drawings. Sweeps shall be RGS. Transitions from below to above-grade areas in damp areas shall be PVC coated RGS.
 - 3. Hazardous areas: All areas indicated as Hazardous Areas on drawings, or as classified by NFPA 820.
 - a. Raceways, junction boxes and sealing fittings shall be installed in accordance with NFPA 70, article 500.
 - 4. General Purpose Areas: All other areas not described above
 - a. Raceways shall be RGS. Raceways concealed in walls or ceilings for general purpose lighting and receptacle circuits may be EMT. Exposed boxes shall be NEMA 12. Concealed boxes may be NEMA 1.

1.011 CONTRACT DRAWINGS

- A. Raceways, boxes, and ground connections are shown diagrammatically only and indicate the general character and approximate location. The drawings do not necessarily show the total number of raceways or boxes for the circuits required, nor are the locations of indicated runs intended to show the actual routing of the raceways. The Contractor shall furnish, install and place in satisfactory condition all raceways, boxes, conductors, and connections and all

other materials required for the electrical systems shown or noted in the contract documents to be complete, fully operational and fully tested upon completion of the project.

- B. The drawings do not show all requirements of the specifications. The drawings and specifications are complimentary and what is called for (or shown) in either is required to be provided as if called for in both.
- C. The horsepower of motors and apparatus wattages shown on the drawings are estimated requirements of equipment furnished under other Sections of this contract and bid shall be based on these sizes. Overload elements shall be provided to suit actual equipment nameplate current. Advise Engineer of any equipment changes or substitutions affecting electrical systems.
- D. Any minor changes in the location of the raceways, outlets, boxes, devices, wiring, Utility equipment, etc., from those shown on the drawings shall be made without extra charge.
- E. When inserts or sleeves for outlet boxes, conductor, pump cables and/ or raceways are required, Contractor shall provide and shall fully coordinate the installation with other trades.
- F. Electrical drawings shall be used for construction of electrical systems only. The electrical drawings do not show construction features of other trades.

1.012 ELECTRICAL SUBMITTALS

- A. Electrical submittals shall be submitted in electronic .pdf format, labeled with the project and Contractor's name.
- B. The files shall include an index showing each product being submitted. Submittals shall be clearly identified per the electrical specifications by section and paragraph or equipment.
- C. Each equipment submittal sheet shall clearly indicate the individual equipment name and part number. Submittals shall include:
 - 1. Manufacturer's name, address, and telephone number
 - 2. Trade name, catalog model or number, nameplate data and size clearly indicated
 - 3. Layout dimensions, capacity, project specification and paragraph reference
 - 4. Local manufacturers representative
- D. Submittals shall be largely complete prior to the first submittal. Long lead items may be submitted separately. Each item shall be clearly marked and provided with adequate sales and technical information to clearly show conformance with all aspects of the specification. Packages not provided as described above or largely incomplete shall be returned to the Contractor, without review or comment.
- E. The Contractor shall ensure that the material being proposed conforms to the Contract requirements. In the event of any variance, the Contractor shall state specifically which portions vary and shall request a variance in writing.

- F. The Contractor shall certify that all furnished equipment can be installed in the spaces allocated. Any costs for modifications to door openings etc required to move equipment to the final locations shall be the responsibility of the Contractor.
- G. The Contractor shall provide shop drawings in .pdf format scaled for 11" x 17" paper, and shall be scaled using standard engineering or architectural scales. Wiring diagrams shall identify circuit terminals, and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment.
- H. Failure to submit a specified item does not relieve the Contractor from meeting the requirements of the Specification.
- I. The Engineer will review the original submittal and one re-submittal on each item. Subsequent submittal reviews shall be conducted at the Contractor's expense. The Contractor shall be billed at the Engineer's current hourly rates for these subsequent submittal reviews.

1.013 EXISTING SYSTEMS

- A. Prior to bidding, the existing site, exiting site electrical systems and systems adjacent to the work shall be investigated thoroughly.
- B. Any damage resulting from performance of work under this contract shall be repaired to assure continuing operation and integrity during and at completion of the project a no increase in contract cost.
- C. Any existing wiring serving devices to remain in service and which is interrupted by work performed under this contract shall be rerouted to maintain circuit continuity.
- D. Contractor shall assume responsibility for unscheduled interruptions and expedient repair.
- E. The Contractor shall inspect the existing systems prior to bidding and shall make his own judgment as to the work required to provide a complete installation within the intent of the contract documents.

1.014 DEMOLITION WORK

- A. All demolition work required under this contract is not shown on the drawings.
- B. The Contractor shall inspect the existing sites and installations prior to bidding and shall make his own judgment as to the work required to provide complete demolition as shown or within the intent of the contract documents.
- C. Existing equipment, systems, and materials removed during demolition shall be made available for his inspection and decision as to whether the Owner will retain possession. Items selected for retention shall be turned over to the Owner. These items shall be delivered to a location on the premises selected by the Owner.
- D. All material not selected for retention by the Owner and debris shall be legally disposed of by the Contractor.

1.015 CONSTRUCTION POWER

- A. The electrical contractor shall provide a dedicated electrical service for construction power. This includes power for construction equipment, job trailer, and any other construction power needs.
- B. The coordination with the serving electrical utility and all associated costs for the temporary construction power equipment and installation shall be provided and paid for by the Contractor.
- C. The Contractor shall pay for the energy costs as billed by the utility on the construction power meter.

1.016 FACILITY DOWNTIME LIMITATIONS

- A. The work for this project is at an existing operational wastewater treatment facility. Contractor shall maintain power to the facility at all times during construction.
- B. Power interruptions will be required during the upgrade of the electrical service. A minimum of one week notification to the Eastsound Sewer and Water District is required prior to any utility service disruption.
- C. Contractor must provide all required temporary equipment (power generator, pumps, piping, etc.) and submit a schedule and written plan for all wastewater treatment plant disruptions that will last longer than 20 minutes. The plan shall require written approval by the Owner and Engineer prior to the service disruption.

1.017 RECORD DRAWINGS

- A. The Contractor shall record the actual electrical system installation on a set of prints kept readily available at the project during construction. These prints shall be used for this purpose alone. Accurately locate with exact dimensions all underground and under slab raceways and stub-outs. At the completion of the work, Contractor shall furnish the Engineer a set of record drawings and the set of markups. Final payment to the Contractor will not be authorized until these prints have been submitted to and accepted by the Engineer. The contractor shall maintain one set of record drawings at the job showing any deviations in the electrical systems from the original design.
- B. Markings shown on the drawings shall conform to the following color coding conventions:
 - 1. Red - Additions or changes showing placement different than shown on the original drawings
 - 2. Green - Deletions or modifications depicting placements different than shown on the original drawing
 - 3. Blue - Notes and Dimensional data showing exact placement of concealed or buried equipment, raceways, etc.

1.018 WARRANTY

- A. The Contractor shall guarantee all work installed under this specification. He shall repair or replace, at his own expense, defective work, materials or parts which are identified within one year after final acceptance.

PART 2 PRODUCTS

2.01 GENERAL

- A. All materials must be new, free from defects and not less than the quality herein specified.
- B. Each type of materials furnished shall be of the same make and shall be of the standard products or manufacturers regularly engaged in the production of such materials and shall be the manufacturer's latest standard design.
- C. All materials and equipment installed shall have been tested and listed by Underwriters Laboratories or other approved testing organization and shall be so labeled unless otherwise permitted by the Code Inspector (AHJ).

2.02 NAMEPLATES

- A. Nameplates shall be provided on all electrical devices. This includes motor control equipment, MCC buckets, control stations, junction boxes, panels, motors, instruments, switches, indicating lights, meters, and all electrical equipment enclosures.
- B. Nameplates shall be made of 1/16" thick machine engraved laminated phenolic having engraved black filled letters not less than 3/16" high on white background.
- C. Warning nameplates shall be provided on all panels and equipment which contain multiple power sources or which may have energized circuits with the main disconnecting means in the off position. Lettering shall be white on red background.
- D. All nameplates shall be secured to equipment with stainless steel screws or fasteners. Epoxy glue may be used where fasteners are not practical as determined by the Engineer.

2.03 OPERATION AND MAINTENANCE (O&M) MANUALS

- A. The Contractor shall prepare O&M manuals for all equipment furnished under Section 26 of the specifications.
- B. The information included must be the exact equipment installed. Where sheets show the equipment installed and other equipment, the installed equipment shall be neatly and clearly identified on such sheets.
- C. The O&M manuals shall contain all the information needed to operate and maintain all systems and equipment provided in the project. It shall be presented and arranged in the logical manner for efficient use by the Owner's operation personnel. The information provided shall include but not be limited to the following:
 - 1. Equipment manufacturer, make, model number, size, etc.
 - 2. Equipment nameplate data.

3. Description of system configuration and operation including component identification.
 4. Dimensional and performance data for specific unit provided.
 5. Manufacturer's recommended operation instructions.
 6. Manufacturer's recommended lubrication and servicing data.
 7. Complete parts list including reordering information and recommended spares. Parts lists shall give full ordering information assigned by the original parts manufacturer.
 8. Shop drawings and wiring diagrams.
- D. Wiring diagrams for each system shall be complete drawings for the specific system installed under the contract.
- E. The information contained in the manuals shall be grouped in an orderly arrangement by specification index. The manuals shall be bound in a hard cover binder and tabbed with an index. O&M manuals shall not exceed 5" thick. Provide two or more volumes if required. The covers shall be imprinted with the name of the job, Owner, Engineer, Electrical Engineer, Contractor and year of completion. Hard covers and literature contained may be held together with screw post bindings.
- F. A preliminary copy, complete, except for the bound cover, shall be submitted 30 days prior to completion of the project for checking and review. The quantity of manuals shall be as indicated in the General Conditions, but shall not be less than (3) complete sets.

2.04 HOUSEKEEPING PADS

- A. Provide a concrete housekeeping pad under each floor mounted motor control center and other floor mounted electrical equipment. Pad shall be 3" wider on both ends than base of equipment mounted on it. Minimum height is 3".

PART 3 EXECUTION

3.01 STORAGE AND HOUSEKEEPING

- A. The Contractor shall store all electrical equipment in a dry environment free from dust, moisture, sprays or vapors which may be detrimental to their new condition. After installation of equipment, the Contractor shall take care to protect all equipment from all dust, moisture, paint and other sprays, and harmful vapors.
- B. The premises shall be kept free of accumulated materials, rubbish and debris at all times. Surplus material, tools and equipment must not be stored at the job site. Upon completion of the project, all equipment and fixtures shall be cleaned and in proper condition for their intended use.

3.02 SCHEDULING WORK WITH UTILITIES

- A. The Contractor shall be fully and completely responsible for all scheduling and coordination with the Utility companies, including the electrical utility provider, Orcas Power and Light Cooperative (OPALCO).
- B. The Contractor shall coordinate and schedule power outages, power service for operation and construction, telephone and power service as required by the facility prior to Certificate of Occupancy.

3.03 CONTRACTOR / UTILITY INTERFACE RESPONSIBILITIES

- A. The electrical utility providing service to these facilities is Orcas Power and Light Cooperative (OPALCO). The Contractor shall comply with all Utility company standards and Utility requirements.
- B. All Utility company charges for the new service will be paid by the Owner.
- C. The division of responsibilities stated below has been determined by coordination with the Utility, and shall be confirmed by the Contractor prior to bid. The division of responsibility for installation tasks shall be coordinated with the Utility at the time of construction.
- D. Coordinate demolition of existing equipment and services with the Utility. Refer to Contract Documents for additional demolition requirements.
- E. The Contractor shall provide the following service and material for the new 480Y/277V three phase service.
 - 1. Provide new complete secondary underground service, including conduit and conductors, from the existing transformer pad location to the CT enclosure, new Service Rated ATS panel, and new 400 amp distribution panel.
 - 2. Install warning tape for underground electrical utilities.
 - 3. Provide new 480V CT enclosure, landing pads, mounting rack, and meter bases per Utility requirements.
 - 4. Demolition of existing 208V electrical service including secondary raceway and conductors, CT enclosure, meter base, Generator and ATS.
 - 5. Temporary power provisions to limit facility downtime to 20 minutes or less.
 - 6. All other materials and construction shown on the Drawings.
- F. The Utility will provide the following service and materials:
 - 1. Primary Power
 - a. Remove existing 208V Utility transformer from existing concrete pad.
 - b. Install new 480V Utility transformer on existing concrete pad.
 - c. Primary conductor and installation (re-use existing).

d. All primary terminations.

2. Secondary Power

- a. Provide meter in Contractor-provided meter base.
- b. Provide CT's in Contractor-provided CT enclosure.
- c. Terminate Contractors' secondary conductors at transformer.

G. Service equipment shall be listed and labeled by UL as "Suitable for Use as Service Equipment".

3.04 TESTS

- A. The Contractor shall conduct testing for installed feeder cables, motors, and other electrical equipment.
- B. Functional testing of all electrical systems shall be performed. Prior to functional testing, all protective devices shall be adjusted and made operative. Prior to energizing the equipment, the Contractor shall perform a functional checkout of each individual control circuit. Checkout shall consist of energizing each control circuit and operating each control device and verifying that the specified action occurs. The Contractor shall submit a description of the proposed functional test procedures prior to the performance of the functional checkout.

3.05 TRAINING

- A. After substantial completion of the work, O&M manuals have been delivered to the owner, all testing is complete and final inspection of the work by the Authority(s) Having Jurisdiction, the Contractor shall demonstrate the electrical systems and instruct the Owner's designated operation and maintenance personnel in the operations and maintenance of the various electrical systems. The Contractor shall arrange scheduled instruction periods with the Owner. The Contractor's representatives shall be knowledgeable in each system and suppliers representatives, when so specified.
- B. Scheduled minimum instruction period at each location shall be:
 - 1. Electrical Systems 8 Hours
 - 2. Control Systems 24 Hours

3.06 FINAL ACCEPTANCE

- A. Prior to final acceptance, the Engineer will perform one or more site observation trips to develop a punch list of items deemed incomplete. The Electrical Contractor shall be present while these inspections are taking place and shall be available for opening cabinets and operating and adjusting the system as is necessary for the Engineer to verify all equipment is installed and operates to the requirements of the contract documents.
- B. The Contractor shall complete all items of work, including wire markers, nameplates, final tests and final test reports prior to requesting final acceptance inspections. All equipment shall be checked for proper operation and all signals verified for correct calibration and

wiring. Fixtures shall have been cleaned, and burned out or defective lamps shall have been replaced.

3.07 STANDARD ABBREVIATIONS

Abbreviation	Definition
Accepted	Reviewed with no exceptions taken to submittal material.
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
Approved	Inspected and accepted by the Authority Having Jurisdiction
ASTM	American Society for Testing Materials
Boxes	Outlet, Junction or Pull Boxes
Code	All codes currently enforced at project location
Compression	Compressed using a leverage powered crimping tool
Connection	All materials and labor required for equipment to be fully operational
CSI	Construction Specifications Institute
EMT	Electrical Metallic Tubing
Exterior	Outside of outer surfaces of the location building
Fully Operational	Tested and approved and operating to the satisfaction of the AHJ, manufacturer and contract documents
Furnish	Purchase and deliver material
Install	Install and make fully operational
kcmil	Thousand circular mils (also MCM)
Mfr	Manufacturer
NEC	National Electric Code NFPA #70 current revision as adopted by AHJ
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
Noted	Shown or specified in the contract documents
PVC	Polyvinyl Chloride
Provide	Furnish and install
RGS, GRS	Rigid Galvanized Steel
Required	As required by code, AHJ, or contract documents for the installation to be fully operational
Shop Drawing	Hand drafted document which fully details the equipment and intended installation relative to this specified project
Shown	As indicated on the drawings or details
Submittal	Material for Engineer review which may include catalog cuts, shop drawings, wiring diagrams, etc., of the actual material being furnished.
UL	Underwriters Laboratories, Inc.
Wiring	Raceway, conductors and connections

*** END OF SECTION ***

SECTION 26 05 19 – WIRE AND CABLE

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section specifies conductors and cables rated to 600 volts used for power, lighting, receptacle, signal, and control circuits.

PART 2 PRODUCTS

2.01 GENERAL

- A. With the exception of lighting, communication, paging, security and receptacle circuits, the type, size and number of conductors shall be as specified on the drawings or schedules. Lighting and receptacle circuit conductors are unscheduled and shall be sized by the Contractor in accordance with the NEC to limit voltage drop to 3 percent. Number and types of communication, paging, and security cables shall be as required for the particular equipment provided.

2.02 LIGHTING AND RECEPTACLE BRANCH CIRCUIT CONDUCTORS

- A. Lighting and receptacle conductors shall be stranded except for 12 AWG which shall be solid. Minimum conductor size shall be 12 AWG.
- B. Conductors shall be provided with the following characteristics:
 - 1. Voltage: 600 volts
 - 2. Conductor: Bare soft annealed copper, Class B stranded per ASTM-8; solid per ASTM B-3
 - 3. Insulation: THWN/THHN, 90 degree C dry, 75 degree C wet polyvinylchloride (PVC)
 - 4. Jacket: Nylon
 - 5. Flame resistance: UL 83
 - 6. Manufacturer: Okonite; Southwire; or equal

2.03 POWER AND CONTROL CONDUCTORS AND CABLE, 600 VOLT

- A. Single Conductor:
 - 1. Single conductor cable shall be stranded and shall be used in conduits for power and control circuits.
 - 2. Conductor shall be provided with the following characteristics:

- a. Voltage 600 volts
- b. Conductor: Uncoated, soft annealed copper, Class B stranded per ASTM B-8
- c. Insulation: Power #6 AWG and larger: XHHW-2, 90 degrees C cont. rating, wet or dry.
- d. Power up to #6 AWG: THHN/THWN, 90 degrees C cont. rating, wet or dry
- e. Control: THHN/THWN, 90 degrees C cont. rating, wet or dry
- f. Flame resistance: UL 83
- g. Manufacturer: Okonite, Southwire; Anaconda; or equal

2.04 CONNECTORS

- A. Pre-insulated Connectors for splices and taps in conductors 10 AWG and smaller shall be Ideal Industries "Wing Nut" or 3M Company "SCOTCHLOCK", or equal. For 8 AWG and larger conductors shall be T&B compression connectors, or equal. Compress using manufacturers recommended die and tools.
- B. Waterproof silicone filled "wing nut" type connectors or spade/lug type terminations and terminals and coat with liquid insulation shall be used for all connections of wire to cord to removable equipment provided with integral cords (such as floats, transmitters, limit switches, etc.) in junction boxes in underground hand holes or outdoor junction boxes. Insulators shall be Thomas and Betts multi splice insulator MSLT112-4, or equal.

2.05 SPLICE INSULATION

- A. Splice insulation shall be equal or greater than the insulation level of the conductor used.
- B. All permanent splices that are underground or in damp or corrosive environments shall be insulated with cast epoxy type insulation which covers the jacket of all cords and the insulation on all wire. Epoxy splice shall be Scotch #3570 or equal.

2.06 WIRING SCHEDULE

- A. Refer to cable schedule for description of conductors required.

2.07 MOTOR TERMINAL SPLICE INSULATION

- A. Motor terminal splice insulation in the motor connection box shall be provided which will withstand constant vibration and abrasion without degrading the insulation of the splice. A product shall be used that is specifically designed for the purpose of motor terminations in accordance with the following:
 - 1. Motor splices in general purpose areas: bolted splice with a TY-RAP boot type insulator, Thomas and Betts Splice insulator Series MSC, or equal. Splices using

wire larger than 8 AWG may be heat shrinkable motor connection stub splices, Raychem, MCK-V series, or equal.

2. Motors in outdoor, damp, or corrosive environments: waterproof motor stub insulator, Thomas and Betts multi splice insulator MSLT112-4, or equal. Splices using wire larger than 8 AWG may be heat shrinkable motor connection stub splices, Raychem, MCK-V series, or equal.

2.08 WIRE MARKERS

- A. Field installed wire markers shall be T&B SMS pre-printed clip-on markers, or equal.

PART 3 EXECUTION

3.01 GENERAL

- A. Each power and control conductor shall be identified at each terminal to which it is connected.
- B. Pulling wire and cable into conduit or trays shall be completed without damaging or putting undue stress on the cable insulation. Soapstone, talc or UL listed pulling compounds are acceptable lubricants for pulling wire and cable. Grease is not acceptable. Raceway construction shall be complete, cleaned, and protected from the weather before cable is placed in the raceway.

3.02 600 VOLT CONDUCTOR AND CABLE

- A. Conductors in panels and electrical equipment, 6 AWG and smaller, shall be bundled and laced at intervals not greater than 6 inches, spread into trees and connected to their respective terminals. Lacing shall be made up with plastic cable ties. Lacing is not necessary in plastic panel wiring duct. Conductors crossing hinges shall be bundled into groups not exceeding 12 and shall be so arranged that they will be protected from chafing when the hinged member is moved.
- B. Slack shall be provided in junction and pull boxes and hand holes. Slack shall be sufficient to allow cables or conductors to be routed along the walls of the box. Amount of slack shall be equal to largest dimension of the box. Where plastic panel wiring duct is provided for wire runs, lacing is not required.
- C. Solid wire shall not be lugged, nor shall electrical spring connectors be used on any except for solid wires in lighting and receptacle circuits. Lugs and connectors shall be installed with a compression tool.
- D. All splices and terminations are subject to inspection by the Engineer prior to and after insulating. Terminations at 460-volt motors shall be made by bolt-connecting the lugged connectors. Connections shall be insulated and sealed with factory-engineered kits. Bolt connection area shall be kept free of mastics and fillers to facilitate rapid stripping and re-entry. Motor connection kits shall accommodate a range of cable sizes for both in-line and stub-type configurations. Connection kits shall be independent of cable manufacturer's tolerances.

- E. In-line splices and tees, where approved, shall be made with tubular compression connectors and insulated as specified for motor terminations, except that conductors 10 AWG and smaller may be spliced using self-insulating connectors. Splices and tees in underground handholes or pull boxes shall be insulated using Scotch-cast epoxy resin splicing kits. Terminations at devices with 120 volt pigtail leads shall be made using self-insulating tubular compression connectors.
- F. Terminations at solenoid valves, 120 volt motors, and other devices furnished with pigtail leads shall be made using self insulating tubular compression connectors.
- G. In the case where multiple field located instrumentation and control devices require parallel or series wiring configuration, it shall be done at one location in one junction box with terminals. Interconnection of instrumentation and control devices shall not be done within conduit bodies (i.e. LBs, condulets, etc.)
- H. Provide shielded power cable for leads extending from VFDs to motors, where indicated.

3.03 SIGNAL CABLING

- A. Circuit runs shall be of individually shielded twisted pairs or triads. In no case shall a circuit be made up using conductors from different pairs or triads. Triads shall be used wherever 3-wire circuits are required. Terminal blocks shall be provided at instrument cable junctions unless otherwise specified. Signal circuits shall be run without splices between instruments, terminal boxes, or panels.
- B. Shields shall not be used as a signal path, except for coaxial cable circuits operating at radio frequencies.
- C. Unless otherwise specified, shields shall be bonded to the signal ground bus at the control panel and isolated from ground and other shields at other locations. Terminals shall be provided for running signal leads and shield drain wires through junction boxes.
- D. Spare circuits shall be terminated on terminal blocks at both ends of the cable run and be electrically continuous through terminal boxes. Shield drain wires for spare circuits shall not be grounded at either end of the cable run. Terminal boxes shall be provided at instrument cable splices. If cable is buried or in raceway below grade at splice, an instrument stand shall be provided as specified with terminal box mounted approximately 3 feet above grade.
- E. Cable for paging, telephone, and security systems shall be installed and terminated in compliance with the manufacturer's recommendations.

3.04 COLOR CODING

- A. Wiring shall conform to the following color code, unless otherwise specified.
- B. Insulation on phase conductor sizes 8 AWG and smaller shall be colored, 6 AWG and larger may have black insulation with plastic tape of the appropriate color from the table below.

- C. Insulation on the grounded conductor (neutral) sizes 6 AWG and smaller shall be colored; 4 AWG and larger may have black insulation with plastic tape of white or gray in accordance with the table below.

Description	208Y/120V	480Y/277V	Control
Phase A (Left, Top, Front)	Black	Brown	--
Phase B (Center, Center, Center)	Red	Orange	--
Phase C (Right, bottom, Back)	Blue	Yellow	--
Neutral	White	Gray	White
Ground	Green	Green	Green
120 VAC Control	--	--	Red
120 VAC Control	Neutral	--	White
DC Control (+)	--	--	Blue
DC Control (-)	--	--	Blue/Wht Stripe
External Source	--	--	Yellow

- D. All control wiring in control panels or other enclosures that is powered from an external source and is not disconnected by the control panel disconnect shall be terminated at a disconnecting terminal block upon entering the enclosure. The color of the wire shall then be changed to yellow to identify it as being powered from an external source. Provide identification nameplate on exterior of enclosure to indicate sources of external power.

3.05 TERMINAL MARKING

- A. All terminals in instrument and relay compartments, motor control centers, in control panels, instrument panels, field panels and control stations, as well as connections to mechanical equipment shall have reference number and letter.

3.06 WIRE BENDING RADIUS

- A. The radius of bends in all non-shielded wire (conductors and cables) shall not be less than eight (8) times the outside diameter of the wire. Shielded or lead covered wire shall not be bent to a radius less than twelve (12) times the diameter of the wire. Any wire installed with bends less than the allowed diameter and which the Engineer deems has caused that insulation to be damaged, shall be removed and new wire shall be installed.

3.07 GENERAL TESTS

- A. The Contractor shall perform voltage, current and resistance tests as required in this section. Test reports shall be submitted to the Engineer prior to final acceptance by the Owner. The Contractor shall inform the Engineer of scheduled testing a minimum of 5 days prior to the testing.

- B. The Contractor shall undertake all such corrective measures if the test results indicate corrective measures are required. No additional compensation will be paid for corrective measures
- C. Test Scope
 - 1. The Contractor shall provide all material, equipment, labor and technical supervision to perform tests and inspections as specified herein.
 - 2. It is the intent of these tests to assure that all electrical equipment as supplied and installed by the Contractor is operational within the industry and manufacturer's tolerances and is installed in accordance with the design documents.
 - 3. The tests and inspection shall determine the suitability for energization.
- D. Conductor Tests
 - 1. Following the completion of installation, the following conductors shall be tested:
 - a. All power feeders scheduled in Conduit and Cable Schedule.
 - b. Service conductors and feeder conductors
 - c. All new grounding; measure ground resistance at each ground rod.
- E. Visual and Mechanical Inspections
 - 1. Inspect exposed section for physical damage.
 - 2. Verify cable is supplied and connected in accordance with specifications and one line diagram, and that phases are labeled correctly.
- F. Electrical Tests
 - 1. Perform insulation resistance test on each cable in reference to ground and adjacent conductors in the same raceway.
 - 2. Perform continuity test to ensure proper cable connection.
 - 3. Perform thermal imaging inspection on all power terminations.
- G. Test Values
 - 1. Insulation resistance tests shall be performed at 1000 volts DC for one-half minute.
 - 2. Minimum megger readings at 20 degrees C shall be one megohm.

3. The maximum acceptable reading for an individual ground rod shall be 25 ohms as required by the NEC and measured by the three rod method. The composite ground electrode shall have a maximum acceptable reading of 15 ohms.

*** END OF SECTION ***

SECTION 26 05 26 – GROUNDING AND BONDING

PART 1 GENERAL

1.01 SCOPE OF WORK

- A. This specification describes the requirements for the grounding of electrical systems and equipment.
- B. Installation shall be in accordance with the National Electrical Code (NEC).

PART 2 PRODUCTS

2.01 MATERIALS

- A. Ground wire: Soft drawn bare stranded copper wire, sized as noted on the drawings.
- B. Terminals and connectors: Burndy Hyground compression system.
- C. Exothermic type weld: Erico Cadweld process, or Fuseweld/T&B corp. Exothermic welding system.
- D. Rod Electrodes: Copper clad (minimum 0.010 jacket) ground rods minimum ¾" diameter x 8' long.
- E. Grounding Electrode conductors and bonding conductors: Copper conductors, bare or insulated, as shown on drawings.

PART 3 EXECUTION

3.01 PREPARATION

- A. All contacting surfaces of ground connections shall be cleaned to bright metal before connection is made.

3.02 INSTALLATION

- A. Grounding conductors: Install in PVC conduit where subject to damage. All grounding conductors smaller than #6 AWG must be protected.
- B. Connections and splices: Provide as required and as shown on drawings.
- C. Connections, taps, and splices shall be made by compression connectors, Burndy Hyground compression system.
- D. Provide equipment grounding conductor in all PVC conduits.

*** END OF SECTION ***

SECTION 26 05 33 – RACEWAYS AND BOXES

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section specifies raceways for electrical conductors including fittings and supports. Raceways shall be provided for power, control, instrumentation, grounding, lighting, receptacles, and signaling systems. Raceways consist of conduits, tubing, and tray systems. For the purpose of this specification, conduit and tubing is described collectively as conduit.

PART 2 PRODUCTS

2.01 GENERAL

- A. Material
 - 1. All materials shall be new, free from defects, of current manufacture, of quality specified or shown. Each type of material shall be of the same manufacturer throughout the work.
- B. Unscheduled Raceway
 - 1. With the exception of lighting, communication, paging, security and receptacle circuits, the type and size of raceway shall be as specified on the drawings or schedules. Lighting and receptacle raceway are unscheduled and shall be sized by the contractor in accordance with the NEC. Minimum size shall be 1/2 inch for exposed and 1 inch for embedded raceway.
- C. Scheduled Raceway
 - 1. The size and type of raceway shall be as specified on the drawings or schedules. In case of conflicts between the drawings and paragraph 3.1, the drawings shall prevail.

2.02 RACEWAY

- A. Application – also see Area Classifications Section 26 05 00.1.10.
 - 1. All conduits shall be Galvanized Rigid Steel (GRS), unless otherwise noted.
 - 2. All connections to vibrating equipment or motors shall be liquid-tight flexible metallic conduit.
 - 3. PVC installed above grade shall be UV resistant Schedule 80.
 - 4. Underground power, control and telephone conduits shall be schedule 40 PVC. (All sweeps and risers for transition from below grade to above grade shall be PVC coated, GRS), unless indicated otherwise.

5. Underground conduits serving non-critical loads such as site lighting, signs, etc. shall be PVC schedule 40.
 6. Indoor lighting and receptacle circuits may be EMT. Conduit shall be concealed where possible.
- B. Rigid Steel Conduit (RGS)
1. Rigid conduit shall be steel, hot dipped galvanized. Final conduit terminations shall be by means of threaded hubs or double locknuts and insulating grounding type bushings.
- C. Liquid Tight Flexible Metallic Conduit
1. Flexible conduit shall be interlocking single strip, hot dipped galvanized and shall have a polyvinyl chloride jacket extruded over the outside to form a flexible watertight raceway.
- D. Nonmetallic Conduit
1. Nonmetallic conduit shall be rigid PVC, Schedule 40 or 80. Fittings shall be of the same material as the raceway and installed with solvent cement per the manufacturer's instructions. Conduit, fittings and solvent cement shall all be manufactured by the same manufacturer.
- E. PVC Coated Rigid Steel Conduit
1. Conduit shall be hot dip galvanized, then coated with urethane inside and outside, then covered with 40 mil PVC coating.
- F. Aluminum Conduit
1. Aluminum conduit shall be rigid ANSI C80.5, threaded.
- G. Electrical Metallic Tubing (EMT)
1. EMT shall be UL 797 and ANSI C80.3, steel tubing, hot-dip galvanized. EMT fittings shall be ANSI/NEMA FB 1, steel, rain-tight, insulated throat, compression type.

2.03 FITTINGS AND BOXES

- A. Material
1. Materials for fittings and boxes shall be chosen to satisfy the requirements of Paragraph 26 05 00.1.10 - Area Classification. All screws, nuts, bolts, and other hardware used with fittings and boxes shall be stainless steel unless installed in general purpose areas.
- B. Unions

1. All unions of the type designated as UNF and UNY and shall be suitable for use in moist atmospheres. Unions shall be of cast ferrous alloy, electroplated with zinc.

C. Locknuts

1. All locknuts used in general purpose areas shall be extra heavy steel electroplated with zinc for sizes $\frac{3}{4}$ inch to 2 inches. Locknuts larger than 2 inches shall be of malleable iron, electroplated with zinc. Locknuts used in damp and outdoor areas shall be stainless steel. Locknuts in corrosive areas shall be FRP.

D. Bushings

1. All bushings shall be steel or malleable iron threaded type electroplated with zinc or hot-dip galvanized. Bushings shall have a molded-phenolic or nylon insulating collar.
 - a. Grounding Bushings: Grounding-type bushings shall have a projecting portion drilled for the size grounding cable used and shall be provided with a clamp or set screw for securing the cable. In addition, a set screw shall be provided to securely lock the bushing to the conduit. Grounding bushings shall be GEDNEY Type IBC-L-BC, or T&B No. 3870 through 3880, or T&B BG Series, or equal.
 - b. Bushed Openings: Bushings for protection of cables passing through metal boxes or troughs shall all be phenolic type and shall be OZ Type ABB, or equal.
 - c. Hubs for connection of conduit to boxes shall be of zinc. Hubs for use in damp or corrosive areas shall be non metallic or aluminum to match the raceway. The hubs shall provide a liquidtight connection to the box and an insulating bushing for the wiring. Hubs shall be Thomas and Betts bullet type, or equal.

E. Liquidtight Flexible Metallic Conduit Connectors:

1. Connectors for liquidtight flexible metallic conduit shall be electroplated zinc malleable iron. An O-ring gasket and an approved grounding insert shall be part of the unit. Where applicable, 45 degree and 90 degree fittings may be used. Liquid-tight connectors shall be by O.Z. GEDNEY, or equal.

F. Expansion Fittings

1. Expansion fittings in exposed runs shall be weatherproof type and shall be provided with an external bonding jumper. The expansion fittings shall allow for 4 inch longitudinal movement and shall be designed so that when completely assembled the end of each conduit entering the fitting is bushed. Fittings shall be O.Z. GEDNEY Type EX, or equal.
2. Deflection fittings in embedded runs shall be of the watertight type and shall be provided with an internal bonding jumper. The expansion material shall be neoprene and shall allow for $\frac{3}{4}$ inch movement in any direction. Fittings shall be O.Z. GEDNEY Type DX, or equal.

G. Junction Boxes

1. Junction boxes, device boxes, fixture support boxes, oblong, round and rectangular conduit fittings (condulets) shall be of the same material as required by the area classification for the raceway. Junction boxes for use in general purpose areas shall be zinc electroplated cast ferrous alloy. Integrally cast threaded hubs or bosses shall be provided for all conduit entrances and shall provide for full 5 thread contact on tightening. Drilling and threading shall be complete before finishing. Boxes shall be Crouse-Hinds type FS, FD, or approved equal.
2. Cover plates shall be of similar cast ferrous alloy material and finish. Full body neoprene gaskets shall be provided with all covers and shall fastened with stainless steel screws.
3. NEMA 12 boxes shall be of heavy gauge sheet steel, or cast metal. All NEMA 12 boxes shall be provided with a 5 mil thick light gray thermo-epoxy finish, and designed so that moisture will drain away from the gasketed cover joint. Covers for sheet steel boxes shall have turned edges, ground smooth to form a tight seal against the gasket when the cover is closed.

2.04 CONDUIT & CABLE SUPPORTS

A. Conduit Supports

1. Hot-dip galvanized framing channel shall be used to support groups of conduit. Individual conduit supports shall be one-hole galvanized malleable iron pipe straps used with galvanized clamp backs and nesting backs where required. Conduit supports for PVC or epoxy coated rigid steel and PVC conduit systems shall be one hole PVC or epoxy coated clamps or PVC conduit wall hangers.

B. Ceiling Hangers

1. Ceiling hangers shall be adjustable galvanized carbon steel rod hangers as specified. Straps or hangers of plumber's perforated tape are not acceptable. Unless otherwise specified, hanger rods shall be 1/2-inch all-thread rod and shall meet ASTM A193. Hanger rods in corrosive areas and those exposed to weather or moisture shall be stainless steel.

C. Racks

1. Racks shall be constructed from framing channel. Galvanized channels and hanger rods shall be steel, hot dip galvanized, 1.5 oz. / sq. ft. after fabrication. Field cuts shall be re-galvanized by the Galv-A-Weld process or equal. Channels attached directly to building surfaces shall be 14 gauge minimum thickness, 1-5/8 inch deep.
2. Framing channels on all exterior areas and in corrosive areas shall be aluminum stainless steel, or fiberglass. All hardware shall be stainless steel. Framing channel shall be as manufactured by Unistrut or equal.

2.05 CONDUIT SCHEDULE

- A. Refer to conduit schedule on the drawings for raceway sizing and routing description.
- B. Conduit sizing and conductor quantities shall be adjusted as necessary for minor changes based on equipment submittals, at no additional cost to the Owner.

2.06 CONDUIT TAGS

- A. Conduit tags shall be corrosion resistant and remain legible after exposure to abrasion or aggressive fluids. Tags shall be crosslinked polyolefin construction. Manufacturer shall be Impact Industries, or equal.

2.07 HAND HOLES

- A. Hand holes shall be precast concrete with checker plate, galvanized, traffic covers designed for H-20 loading. Dimensions shall be as specified on the drawings. Hand holes shall be provided with precast solid concrete slab bottoms with sumps, with drains, or as shown on the drawings. Hand holes shall be construction of 3000 psi reinforced concrete.

2.08 UNDERGROUND MARKING TAPE (DETECTABLE TYPE)

- A. Underground marking tape shall be for location and early warning protection of buried power and communication lines. Tape shall be detectable by a pipe / cable locator or metal detector from above the undisturbed ground. Tape shall be nominally 2 inches wide with a type B721 aluminum foil core laminated between two layers of 5 mil thickness polyester plastic. The plastic color shall be red for electrical lines and orange for telephone lines. A warning shall be imprinted continuously along the length, with message reading similar to: "CAUTION - STOP DIGGING - BURIED ELECTRIC (TELEPHONE) LINE BELOW." Tape shall be Brady "Detectable Identoline"; Services and Materials "Buried Underground Tape, Detectable", or equal.

PART 3 EXECUTION

3.01 CONDUIT

A. General

1. The Contractor shall limit the number of directional changes of the conduit to a total not more than 270 degrees in any run between pull boxes. Conduit runs shall be limited to 400 feet, less 100 feet or fraction thereof, for every 90 degrees of change in direction. Bends and offsets shall be avoided where possible but, where necessary, shall be made without flattening or kinking, or shall be factory preformed bends. Turns shall be made with case metal fittings or conduit bends. Welding, brazing or otherwise heating of conduit is not acceptable.
2. Where required for pulling cable and as necessary to meet the requirements of the previous Paragraph, the Contractor shall provide cast junction or pull boxes.

3. Conduit entering NEMA 1 type sheet steel boxes or cabinets shall be secured by locknuts on both the interior and exterior of the box or cabinet and shall have an insulating grounding or bonding bushing constructed over the conduit end. Conduit entering all other boxes shall be terminated with a threaded hub. Cast boxes and nonmetallic enclosures shall have threaded hubs. Joints shall be made with standard couplings or threaded unions. Metal parts of nonmetallic boxes and plastic coated boxes shall be bonded to the conduit system. Running threads shall not be used in lieu of conduit nipples, nor shall excessive thread be used on any conduit. The ends of conduit shall be cut square, reamed and threaded with straight threads.
4. Unless otherwise specified, conduit entering field equipment enclosures shall enter the bottom or side of the box. Where conduit comes from above, it shall be run down beside the enclosure and a tee conduit and drip leg shall be installed.
5. When new conduit is added to areas which are already painted, the conduit and its supports shall be painted to match the existing facilities. Where new conduit is used to replace existing conduit, the existing conduit and supports shall be removed, resulting blemishes shall be patched and repainted to match original conditions.

B. Conduit Support

1. Exposed conduit shall be run on supports spaced not more than 10 feet apart and shall be constructed with runs parallel or perpendicular to walls, structural members, or intersections of vertical planes and ceiling.
2. Where three or more conduits are located in a parallel run, they shall be spaced from the wall using framing channel. Support systems shall be galvanized steel unless otherwise specified.
3. Conduit rack and tray supports shall be secured to concrete walls and ceilings by means of cast-in-place anchors. Individual conduit supports shall use cast-in-place anchors, die-cast, rust-proof alloy or expansion shields. Wooden plugs or plastic inserts are not acceptable as a base to secure conduit supports.

C. Conduit Penetrations

1. Unless otherwise specified, conduit routed perpendicular through floors, walls or other concrete structures shall pass through cast-in-place openings wherever possible. In cases where cast-in-place openings are not possible, appropriate size holes shall be bored through the concrete to accommodate the conduit passage. The size and location of the holes shall not impair the structure's integrity. After completion, grout or caulk around conduit and finish to match existing surroundings. Unless otherwise protected, conduits that rise vertically through the floor shall be protected by a 3 1/2-inch high concrete pad with a sloping top.
2. Conduits entering manholes and handholes shall be horizontal. Conduits shall not enter through the concrete bottom of handholes and manholes.

3. Wherever conduits penetrate outdoor concrete walls or ceilings below grade, the Contractor shall provide a watertight seal as manufactured by O.Z. Gedney Co., Type CSM Series; Thunderline Corp., Link Seal; or equal.
4. Wherever conduits enter buildings or structures below grade, seal the conduit opening (after installation of conductors and cables), with conduit sealing material, to prevent water from entering the structure, enclosure, etc. Sealing compound to be a pliable, removable putty-type compound listed for the application.

D. Conduit Separation

1. Signal conduits shall be separated from AC power or control conduits. The separation shall be a minimum of 12 inches.

E. Conduit Seals for Hazardous or Corrosive Areas

1. Each conduit passing from a hazardous or corrosive area into a non-hazardous or non-corrosive area shall be provided with a sealing fitting which may be located on either side of the boundary. The seal shall be located at the boundary in accordance with NEC article 500.
2. Seal fittings for conduit systems in hazardous atmosphere locations shall be hot-dip galvanized case ferrous alloy. Sealing compound shall be hard type, Chico A, or equal, UL listed for explosion-proof sealing fittings. Sealing compound shall be non-hardening type for corrosive areas. Provide reducing bushings and larger seals as required to meet NEC 25% fill.

3.02 HAND HOLES

1. Hand holes shall be set plumb and the tops shall be at finished grade level, or as shown on the drawings.
2. Drainage systems shall be installed to prevent the buildup of standing water inside the hand hole.

*** END OF SECTION ***

SECTION 26 22 13 – DRY TYPE TRANSFORMERS

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section specifies dry-type transformers rated 600 volts and less used for power distribution, lighting and control purposes.

1.02 MATERIALS

- A. All materials shall be new, free from defects, of current manufacture, of quality specified or shown. Each type of material shall be of the same Manufacturer throughout the work.

1.03 STANDARDS AND CODES

- A. All materials and equipment specified herein shall be within the scope of Nationally Recognized Testing Laboratory (NRTL) examination services, be approved by the NRTL for the purpose for which they are used, and shall bear the appropriate listing label.

1.04 EQUIPMENT SIZE

- A. Electrical equipment shall fit in the space provided on the plan drawings or as specified. Equipment heights shall not exceed those shown or specified. Larger equipment shall not be considered equivalent or acceptable.

1.05 SUBMITTALS

- A. Submit all catalog data in accordance with the Submittals requirements in Section 26 05 00. Show material information and confirm compliance with these specifications.

PART 2 PRODUCTS

2.01 ACCEPTABLE PRODUCTS

- A. Reference drawings for transformer size and windings. Transformers shall conform to ANSI/IEEE C57.12.01 and ANSI/ UL 506. The voltage, frequency, number of phases and KVA rating shall be as specified. Transformers shall be General Electric, Square D, Westinghouse, or equal.

2.02 INSULATION

- A. Transformers 15 KVA and above shall have a Class 220 insulation system in accordance with NEMA ST20. Transformers 2 KVA and less shall be designed not to exceed 80 degrees C temperature rise. Transformers 3 KVA and greater shall be designed not to exceed 115 degree C temperature rise.

2.03 COILS

- A. Transformer coils shall be copper. Transformer coils 15 KVA and above shall be

impregnated with varnish. Transformer coils 10 KVA and below shall be encapsulated.

2.04 WINDING CONFIGURATION

- A. Transformers shall have electrically isolated primary and secondary windings. Primary and secondary winding configurations shall be as specified. Provisions shall be made to permit separate grounding of the neutral conductor and enclosure. Single-phase transformers shall be the four winding type.

2.05 TRANSFORMER TAPS

- A. Transformers 15 KVA and above shall be provided with two 2-1/2 percent full capacity taps above normal voltage and four 2-1/2 percent full capacity taps below rated voltage on the primary winding.

2.06 TERMINAL COMPARTMENTS

- A. Terminal compartments shall be sized to permit termination of cables specified. Terminal connections shall be made in the bottom third of the enclosure. The terminals shall be copper and sized for cable specified.

2.07 ENCLOSURES

- A. Transformers rated 15 KVA and smaller shall be provided with weatherproof, non-ventilated enclosures.
- B. Indoor transformers rated greater than 15 KVA shall be provided with drip proof, ventilated enclosures. Outdoor transformers shall have weatherproof enclosures.

2.08 MOUNTING

- A. Transformers 15 KVA and below shall be suitable for wall mounting.
- B. Transformers 20 KVA and larger shall be floor mounting type.

2.09 SOUND LEVELS

- A. The sound levels shall not exceed the following values:

<u>KVA</u>	<u>dB</u>
0-9	40
10-45	42
50-150	45
225-300	50
500	54

2.010 TRANSFORMER EFFICIENCY

- A. Transformers shall be energy-efficient and shall meet the following minimum ratings:

Single Phase		Three Phase	
kVA	Efficiency (%)	kVA	Efficiency (%)
15	97.7	15	97.0
25	98.0	30	97.5
37.5	98.2	45	97.7
50	98.3	75	98.0
75	98.5		

PART 3 EXECUTION

3.01 INSTALLATION

- A. Transformers shall include internal "rubber-in-shear" isolation mounts selected per Manufacturer's recommendations or shall be installed with "KORFUND" or equal external vibration isolators. Wall mounting shall be allowed on masonry. Mounting hardware shall be per Manufacturer's instructions. Transformers with enclosures designed for floor mounting where suspended from above shall be suspended on a trapeze constructed of a minimum of two horizontal structural channels hung from threaded rod attached to structural slab with inserts. Channel rod and inserts shall be sized for not less than 400% load safety factor. Transformer raceway connections shall be flexible metal conduit as specified hereinbefore for equipment subject to vibration.

3.02 CONNECTION

- A. Transformers shall be considered "grounded neutral separately derived systems" and neutral shall be grounded accordingly to the building ground grid utilizing a direct connection.

3.03 IDENTIFICATION

- A. The transformer shall be identified with engraved, phenolic nameplates.

*** END OF SECTION ***

SECTION 26 24 16 – PANELBOARDS

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section specifies panelboards for lighting and power distribution.

1.02 SUBMITTALS

- A. Submit all catalog data in accordance with the Submittals requirements in Section 26 05 00. Show material information and confirm compliance with these specifications.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Panelboards shall be manufactured by Square D, Cutler Hammer, or equal.

2.02 MATERIALS - LIGHTING AND APPLIANCE PANELBOARD

- A. Interior
 - 1. Continuous main current ratings, as indicated on associated panel schedules or drawings, not to exceed 600 amperes maximum for main breaker panelboards and not to exceed 800 amperes for main lug panelboards.
 - 2. Provide one (1) continuous bus bar per phase. Each bus bar shall have sequentially phased branch circuit connectors limited to bolt-on branch circuit breakers. The bussing shall be fully rated. Panelboard bus current ratings shall be determined by heat-rise tests conducted in accordance with UL 67. Bussing rated 100-400 amperes shall be plated copper. Bussing rated for 600 and 800 amperes shall be plated copper as standard construction. Bus bar plating shall run the entire length of the bus bar. Panelboards shall be suitable for use as Service Equipment when application requirements comply with UL 67 and NEC Articles 230-F and -G.
 - 3. All current-carrying parts shall be insulated from ground and phase-to-phase by high dielectric strength thermoplastic.
 - 4. A solidly bonded copper equipment ground bar shall be provided. An additional copper isolated/insulated ground bar shall also be provided.
 - 5. Split solid neutral shall be plated and located in the mains compartment up to 250 amperes so all incoming neutral cable may be of the same length. UL Listed panelboards with 200% rated solid neutral shall be plated copper for non-linear load applications. Panelboards shall be marked for non-linear load applications.
 - 6. Interior trim shall be of dead-front construction to shield user from energized parts. Dead-front trim shall have pre-formed twistouts covering unused mounting space.
 - 7. Nameplates shall contain system information and catalog number or factory order number. Interior wiring diagram, neutral wiring diagram, UL Listed label and

short circuit current rating shall be displayed on the interior or in a booklet format.

8. Interiors shall be field convertible for top or bottom incoming feed. Main circuit breakers in 125A interiors shall be vertically mounted. Main circuit breakers over 125A shall be vertically mounted. Sub-feed circuit breakers shall be vertically mounted. Main lug interiors up to 400 amperes shall be field convertible to main breaker. Interior leveling provisions shall be provided for flush mounted applications.
9. Interior phase bus shall be pre-drilled to accommodate field installable options. (i.e., Sub-Feed Lugs, Sub-Feed Breakers, Thru-Feed Lugs)
10. Interiors shall accept 125 ampere breakers in group mounted branch construction.

B. Main Circuit Breaker

1. Main circuit breakers shall have an overcenter, trip-free, toggle mechanism which will provide quick-make, quick-break contact action. Circuit breakers shall have a permanent trip unit with thermal and magnetic trip elements in each pole. Each thermal element shall be true rms sensing and be factory calibrated to operate in a 40° C ambient environment. Thermal elements shall be ambient compensating above 40° C.
2. Two- and three-pole circuit breakers shall have common tripping of all poles. Circuit breakers frame sizes above 100 amperes shall have a single magnetic trip adjustment located on the front of the breaker that allows the user to simultaneously select the desired trip level of all poles. Circuit breakers shall have a push-to-trip button for maintenance and testing purposes.
3. Circuit breaker handle and faceplate shall indicate rated ampacity. Standard construction circuit breakers shall be UL Listed for reverse connection without restrictive line or load markings.
4. Circuit breaker escutcheon shall have international I/O markings, in addition to standard ON/OFF markings. Circuit breaker handle accessories shall provide provisions for locking handle in the ON or OFF position.
5. Lugs shall be UL Listed to accept solid or stranded copper conductors only. Lugs shall be suitable for 90° C rated wire, sized according to the temperature rating per NEC Table 310-16. Lug body shall be bolted in place; snap-in designs are not acceptable.
6. The circuit breakers shall be UL Listed for use with the following accessories: Shunt Trip, Under Voltage Trip, Ground Fault Shunt Trip, Auxiliary Switch, Alarm Switch, Mechanical Lug Kits, and Compression Lug Kits.

C. Branch Circuit Breakers

1. Circuit breakers shall be UL Listed with ampere ratings, interrupting ratings, and number of poles as indicated on the panelboard schedule.
2. Molded case branch circuit breakers shall have bolt-on type bus connectors.
3. Circuit breakers shall have an overcenter toggle mechanism which will provide quick-make, quick-break contact action. Circuit breakers shall have thermal and magnetic trip elements in each pole. Two- and three-pole circuit breakers shall

have common tripping of all poles.

4. There shall be two forms of visible trip indication. The circuit breaker handle shall reside in a position between ON and OFF. In addition, there shall be a red indicator appearing in the clear window of the circuit breaker housing.
5. The exposed faceplates of all branch circuit breakers shall be flush with one another.
6. Lugs shall be UL Listed to accept solid or stranded copper conductors only.
7. Breakers shall be UL Listed for use with the following factory installed accessories: Shunt Trip, Auxiliary Switch, and Alarm Switch.

D. Enclosures

1. Type 1 Boxes

- a. Boxes shall be galvanized steel constructed in accordance with UL 50 requirements. Galvannealed steel will not be acceptable.
- b. Boxes shall have removable endwalls with knockouts located on one end. Boxes shall have welded interior mounting studs. Interior mounting brackets are not required.
- c. Box width shall not exceed 26" wide.

E. Type 1 Fronts

1. Front shall meet strength and rigidity requirements per UL 50 standards. Shall have ANSI 49 gray enamel electrodeposited over cleaned phosphatized steel.
2. Fronts shall be 1-piece with door, or hinged 1-piece with door. Mounting shall be as indicated on associated schedules and drawings.
3. Panelboards rated 250 amperes and below shall have MONO-FLAT fronts with concealed door hinges and trim screws. Front shall not be removable with the door locked. Panelboards rated above 250 amperes shall have vented fronts with concealed door hinges. Doors on front shall have rounded corners; edges shall be free of burrs.
4. Front shall have flat latch type lock with catch and spring loaded stainless steel door pull. All lock assemblies shall be keyed alike. One (1) key shall be provided with each lock. A clear plastic directory card holder shall be mounted on the inside of door.

F. Type 3R, 5, and 12

1. Enclosures shall be constructed in accordance with UL 50 requirements. Enclosures shall be painted with ANSI 49 gray enamel electrodeposited over cleaned phosphatized steel.
2. All doors shall be gasketed and equipped with a tumbler type vault lock and two (2) additional quarter turn fasteners on enclosures 59 inches or more in height. All lock assemblies shall be keyed alike. One (1) key shall be provided with each lock. A clear plastic directory card holder shall be mounted on the inside of door.
3. Maximum enclosure dimensions shall not exceed 21" wide and 9.5" deep.

G. Circuit Breaker ratings

1. The following interrupting capacity shall be considered minimum. Other ratings shall be as specified on the drawings.
 - a. 240V and 208Y/120V Panelboards 10,000 AIC symmetrical
 - b. 480V and 480Y/ 277V Panelboards 25,000 AIC symmetrical
2. Breakers shall be mounted in panelboards so that breaker handles operate in a horizontal plane. Common trip shall be provided on all multiple pole breakers. Circuit breaker shall be the bolt-on type.
3. Spare breakers shall be provided where indicated, complete for future connection of wiring circuits. Where "Space" is indicated for breakers, bussing and breaker mounting hardware shall be provided in the panelboards; with steel knockouts in dead front metal closure of unused part of panel. If any steel knockouts are removed, breakers shall be provided in such spaces or approved cover plates. Open spaces are not permitted.

H. Surge Arresters

1. Surge arresters, with indicators, shall be provided on all panels to protect against overvoltage transients.

I. Panel schedules

1. Panel Schedules are shown on the drawings.

PART 3 EXECUTION

3.01 MOUNTING

- A. Secure in place with top of cabinet at 6' – 6", unless otherwise noted. Top of cabinet and trim shall be level.

3.02 CIRCUIT INDEX

- A. Each branch circuit panelboard shall be provided with as built information for each panelboard by circuit with its proper load designation. Panelboard index card shall be typed (handwritten not acceptable) and mounted inside the door of each panelboard in a clear plastic sleeve. One spare blank card shall be provided for each card used.

3.03 FIELD QUALITY CONTROL

- A. Inspect complete installation for physical damage, proper alignment, anchorage, and grounding.
- B. Measure steady state load currents at each panelboard feeder; rearrange circuits in the panelboard to balance the phase loads within 20% of each other. Maintain proper phasing for multi-wire branch circuits.

- C. Check tightness of bolted connections and circuit breaker connections using calibrated torque wrench or torque screwdriver per manufacturer's written specifications.

*** END OF SECTION ***

SECTION 26 24 19 – MOTOR CONTROL CENTERS

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section specifies free standing, factory assembled Low Voltage (600 volt) Motor Control Centers (MCC).

1.02 COORDINATION

- A. The Contractor shall be responsible for furnishing the motor control center including all related equipment as shown on the drawings and as specified herein. The Electrical Contractor shall be responsible for installing the motor control center, interconnecting wiring and all related equipment.
- B. The Contractor shall coordinate equipment information with the MCC manufacturer so that the correct type of motor starters and protection equipment are provided and sized properly for the devices being served and to supply such equipment with the proper protection. The Contractor shall verify that all MCC equipment will fit physically within the space allotted per the contract drawings.
- C. The motor control center manufacturer shall equip the assembly with all appurtenances and accessories (including but not limited to control relays, control contacts, control wiring, and terminal strips) as required for interface with the main control system to provide a totally integrated and operable system.

1.03 RELATED SECTIONS

- A. Section 26 29 23 – Variable Frequency Drives
- B. Section 26 80 00 – Control System

1.04 SUBMITTALS

- A. In accordance with the Submittal requirements of Section 26 05 00, the MCC manufacturer shall develop and submit to the following additional information:
- B. System wiring diagrams for each unit in the entire motor control center including but not limited to: all instruments, relays, starters, switches, lights, breakers terminals, etc. Diagrams shall indicate the terminals for remote devices as shown on the wiring diagrams in the contract drawings. Wire and terminal numbers shall be included on the schematic diagrams. Relay contacts shall be indicated for type and number available for each relay used.
- C. Information on ratings and sizes of all equipment such as control transformers, fuses, breakers, etc. on the wiring diagrams for each bucket including time current curves.
- D. Connection diagrams showing physical wiring layout for each unit.

- E. Technical data sheets for all components with the complete part number of the component clearly designated with all required options as specified in part 2.
- F. Scaled arrangement drawings of all panel front- and internal-mounted instruments, switches, devices, and equipment indicated. Show all mounting details required. Deviations from approved arrangements require resubmittal and approval prior to installation.
- G. Descriptive text on wire markers to be used.
- H. Bill of materials showing quantity, manufacturer, catalog number, and the supplier name and phone number for all components of the MCC.
- I. Shop Drawings shall be provided on 11" x 17" sheets (maximum) and shall be scaled using standard engineering or architectural scales.

1.05 COORDINATION OF EQUIPMENT

- A. The Contractor shall coordinate equipment information with the MCC manufacturer so that the correct type of motor starters and protection equipment are provided and sized properly for the devices being served and to supply such equipment with the proper protection.
- B. The Contractor shall verify that all MCC equipment will fit physically within the space allotted per the contract drawings, prior to ordering of equipment.
- C. Coordination with Control System
 - 1. The motor control center manufacturer shall equip the assembly with all appurtenances and accessories (including but not limited to control relays, control contacts, control wiring and terminal strips) as required by the Contractor for interface with the main control system to provide a totally integrated and operable system.
 - 2. The Contractor shall be solely and completely responsible for coordination and integration of control system with the motor control center.
- D. Operation and Maintenance Data
 - 1. Provide operation and maintenance data for all motor control center and related equipment in accordance to the general requirements in Section 26 05 00 and 26 80 00.

PART 2 PRODUCTS

2.01 ACCEPTABLE PRODUCTS

- A. The size of the MCC shown on the drawings is based on MCCs manufactured by Allen-Bradley. Motor control centers and VFDs shall be Allen-Bradley, no exceptions.
- B. The equipment of the manufacturer selected must fit within the space restrictions as shown on the plans.

2.02 CONSTRUCTION

A. General

1. Motor control centers shall be designed, assembled, tested and placed into operation by the MCC supplier. The motor control center shall fit into the space requirements as shown on the drawings. The contract drawings show general MCC layout and space requirements and may be modified by the MCC manufacturer if first approved by the Engineer.

B. Structure

1. The MCC shall consist of vertical sections that can be joined together to form a rigid free-standing, completely enclosed assembly. Vertical sections shall be fabricated of bolted No. 14 gage steel minimum. Each section shall have flange formed doors and/or covers both front and rear. Doors shall include provisions for padlocking all breakers in the open position.

C. MCC construction shall be suitable for the environment installed.

1. Damp or dusty areas: MCCs shall be NEMA 12 construction.
2. General purpose areas (all other areas not described above): MCC construction shall be NEMA 1 gasketed, unless specified otherwise on the plans.

D. Arrangement

1. Motor control center sections shall have a minimum 72-inch working height to accommodate a minimum of six 12-inch compartments. Sections shall be 20 inches deep with a 90-inch height. Minimum width shall be 20 inches. Compartments shall have pan-type doors with quarter turn hold-down latches and neoprene gaskets. Doors for compartments with starter and feeder tap units shall be mechanically interlocked with the unit's disconnect device to prevent unintentional opening of the door while energized and unintentional application of power while the door is open.
2. Starters and feeder tap units shall be draw-out plug-in construction with hardened, plated copper free-floating stabs, steel spring backups, and interference tabs which prevent door closure if unit is improperly installed. Units shall be latched to assure proper bus contact.

E. Finish

1. All steel parts shall be provided with an acrylic baked enamel paint finish, except plated parts used for ground connections. Painted parts shall be primed with a zinc-phosphate primer or undergo a phosphatizing pre-paint treatment for rust resistance and paint bond. Paint shall be applied by electrostatic process and baked to a durable hard finish. Exterior and structure color shall be ASA #49 gray. Starter bucket interiors shall be white.

F. Wiring / Terminals

1. The motor control center shall be suitable for operation on 240 volts, three-wire, 60 hertz. Wiring shall be NEMA Class II, Type B. Each unit shall be completely prewired with all control wiring numbered and terminated on terminal strips.

Terminal's numbering shall be coordinated between units such that like devices shall have the same terminal numbers. Wiring within one bucket shall be labeled with a basic wire numbering scheme.

2. A minimum of 5 spare terminals shall be provided in each unit. Terminal strips are not required for the load wiring. Auxiliary components, such as HOA selector switches, indicating lights and other indicating and/or recording devices, shall be mounted on the compartment door or cover. All control power leads into and out of each unit shall pass through auxiliary contacts of the circuit breaker or be equipped with their own disconnecting device or disconnecting terminal strips, appropriately labeled.

G. Wireways

1. Full height vertical wireway shall be provided in each MCC section that accepts modular plug-in units. The vertical wireway shall connect with both the top and bottom horizontal wireway. The vertical wireway shall be 4" for a 20" wide (9" for a 25") wide section with a separately hinged door. Structures that house a single, full section control unit are not required to have vertical wireways. Those control units must open directly into the motor control center horizontal wireways.

H. Bus

1. The motor control center bus shall be tin-plated copper with connections between vertical and horizontal power bus bars made with 3/8-inch bolts and conical dished steel washers. Access for tightening connections shall be from the front, without the need for tools on the rear of the connection.
2. Unless otherwise specified all sections shall contain horizontal and vertical busses. Each end of the line up shall be provided for connection of future sections.
3. Unless otherwise specified or required by the components installed, main horizontal bus shall be rated a minimum 600 amperes continuous, vertical bus shall be rated a minimum of 300 amperes continuous.
4. A 1/4" X 2" minimum un-plated copper ground bus shall be provided the full length of the motor control center. Ground bus shall be located at the bottom of the motor control center and shall contain lugs to terminate, as a minimum, two (one at each end of ground bus) 4/0 AWG bare copper ground conductors.

I. Motor Starter Units / Contactors

1. Motor starter units shall be of the combination type with components as indicated on the drawings. Magnetic contactors shall be heavy duty NEMA rated and shall be size 1 minimum. All contactors shall be provided with two field convertible auxiliary contacts. An auxiliary switch shall be provided to indicate the circuit breaker is in the "ON" position. Switch shall be open when the circuit breaker is open. Motor starters and associated equipment shall be provided to match the load being served.
2. Overload Relays
 - a. Thermal overload relays on starters shall be ambient compensated bi-

metallic type or solid state type with selector for either auto or manual reset. Overload Relays shall monitor all energized conductors. Overload relay shall be provided with a circuit test button which shall simulate an overload trip, trip indication, and reset pushbutton.

3. Control Transformers

- a. Each control transformer shall be rated 240-120 volt, single phase, 2 wire, 60 hertz. The transformer shall be sized for the load it feeds but shall not be less than the minimum ratings as follows:

<u>NEMA starter size</u>	<u>Minimum transformer volt-ampere rating</u>
1	100
2	150
3	200
4	300
5	500

- b. Each control transformer shall be provided with time-delay, slow blow secondary fuse rated to interrupt 10,000 amperes short circuit at 250 volts AC. Two primary fuses rated to interrupt 200,000 amperes at 600 volts shall be provided on all starters.

4. Transient Suppressors

- a. All contactor coils shall be provided with transient suppressors to limit the high voltage transients produced when power is removed from the coil.

J. Circuit Breakers

1. General

- a. Circuit breakers other than those mounted in the panelboard shall be capable of being padlocked in the open position.

2. Motor Circuit Protectors

- a. Provide motor circuit protectors for all full voltage starters.
- b. The molded case motor circuit protector shall operate on the magnetic principle with a current sensing coil in each of the three poles to provide an instantaneous trip for short circuit protection. The trip setting shall be adjustable from 700 to 1300 percent of the motor full load amperes from the front of the breaker. The motor circuit protector shall be set at its lowest position at the factory. All breakers for motor starters shall include auxiliary contacts which open when the breaker in the OFF position.

3. Thermal Magnetic

- a. Provide thermal magnetic circuit breakers for all reduced voltage starters and variable frequency drives.
- b. Circuit breakers shall be molded case thermal-magnetic type. Circuit breakers shall be quick-make and quick-break type. They shall have wiping type contacts. Each shall be provided with arc chutes, individual

trip mechanisms on each pole. Two and three pole breakers shall be common trip. All breakers shall be calibrated for operation in an ambient temperature of 40°C. Molded case circuit breakers shall be trip-free. Each breaker shall have separate trip indication independent of the ON or OFF positions.

K. Operating and Indicating Devices

1. See section 26 80 00 – Control System
2. Ammeter & Voltmeter
 - a. Voltmeter and voltmeter switch, ammeter and ammeter switch shall be provided where shown on the one-line diagram. Meters shall be analog type 4" nominal size, scaled for the application. Switches shall be instrument grade cam switches, rated for 25 ampere. Volt meter switch shall read phase to phase and phase to ground. Ammeter switch shall read phases A, B, and C. Provide three CTs for each ammeter.
3. Power Monitor and Display (PMD)
 - a. Power monitor and display unit shall be provided for each MCC. The unit shall monitor all three phases and shall display volts (phase to phase and phase to neutral), amperes, power factor, and hertz. The PMD shall be supplied with an Ethernet or equivalent for direct connection and communication to the PLC.
 - b. Power Monitors shall be Allen Bradley Power Monitor 1000.

L. Panelboard

1. Panelboards shall be provided in accordance to applicable requirements of Section 26 24 16.

M. Phase Loss Monitor (PLM)

1. The MCC shall be provided with phase failure relays wired for shutdown of 3-phase motors. It shall monitor phase loss, unbalance, low voltage and reverse phasing, with automatic reset. Each phase fail relay shall have 2 Form C contacts (DPDT).
2. PLM shall have undervoltage adjustment from 75-100%, response time delay set or adjustable between 3 and 5 seconds, phase unbalance set or adjustable between 5% and 10% and an LED indication for relay energized.
3. Sufficient control relays and wiring shall be supplied to provide one shutdown contact for each Full Voltage motor control circuit and 2 additional spare contacts.

N. Operating Mechanisms

1. All circuit breakers in motor control centers shall be provided with external "thru-the-door" operating handles.

PART 3 EXECUTION

3.01 GENERAL

- A. The Contractors shall verify that the motor starters, protection equipment, and other components, etc. provided are suitable (correct phase, voltage, starter type, correct breakers, and overload relays) for the motors and equipment loads being served.
- B. The motor control center shall be assembled per the Contractor's requirements, and shipped to the Contractor's shop. The Contractor shall complete the construction of the motor control center wiring and components per the contract documents. The motor control center shall be completely tested in the Contractor's shop and delivered to the site ready for external connections to field equipment.
- C. All assembly and wiring not completed by the manufacturer or Contractor due to shipping sections, multiple suppliers, etc. shall be the responsibility of the Contractor.
- D. Nameplates shall be mounted in a manner or location such that other equipment or devices do not block them so they are easily viewed.

3.02 TESTING

- A. The motor control center shall be tested in the Contractor's shop along with the control system. The testing shall include, but not be limited to, operation of all input and output (I/O) points, control devices and motor controllers.
- B. Testing and inspection of the motor control center shall include all components. All motor controllers shall be interconnected with the control system and powered with rated incoming voltage.

3.03 INSTALLATION

- A. The motor control system shall be installed in accordance with the installation drawings and manufacturer's instructions. Installation shall be performed by workers who are skilled and experienced in the installation of motor control equipment.

3.04 WIRING

- A. Refer to Section 26 05 19 for acceptable wiring types and methods.
- B. All signal and low voltage wiring shall be separated from 120 volt and 240 volt wiring and shall maintain a minimum of 1 inch separation of conductors.

*** END OF SECTION **

SECTION 26 27 26 – WIRING DEVICES

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section covers furnishing and installing all receptacles, switches and other wiring devices indicated on the drawings.

1.02 PRODUCTS

A. General

- 1. Wiring devices shall be UL approved for the current and voltage specified and shall comply with NEMA WD-1. Devices shall contain provisions for back wiring and side wiring with captively held binding screws. Devices shall be brown except those located in finished areas, which shall be ivory.

B. Receptacles and Plugs

- 1. General:
 - a. Receptacles shall be grounding type.
- 2. 120 Volt Receptacles:
 - a. Indoor Clean Areas: Receptacles shall be duplex 20 ampere, NEMA 5-20R, and shall accept NEMA 5-15P and 5-20P plug caps. Receptacles shall be Hubbell 5362 or equal.
 - b. Ground Fault Circuit Interrupter (GFCI) receptacles: GFCI receptacles shall be provided in all areas as required by local codes and NEC article 210.8. Hubbell GF-5362, or equal.
 - c. Outdoor, Process or Corrosive Areas: Receptacle shall be duplex, 20 ampere, NEMA 5-20R, and shall accept NEMA 5-15P and 5-20P plug caps. Receptacle and plug caps shall be corrosion resistant. Covers shall be rated for “while-in-use” and installed per manufacturer’s directions. Manufacturer shall be Hubbell Series WP, or approved equal.

C. Switches

- 1. General Purpose (Indoor, Clean Areas):
 - a. General purpose switches shall be quiet AC type, specification grade, and shall be provided in accordance with rated capacities as required. Switches shall match receptacles in color. Switches shall be manufactured by General Electric, Hubbell, or equal.
- 2. Switches for Outdoor and Corrosive Areas

- a. Switches shall be 20 amp with weatherproof/corrosion resistant neoprene plate as manufactured by Hubbell, Arrow-Hart, or equal.
- b. Switches shall be mounted in "FS" type copper-free aluminum or PVC mounting boxes.

D. Device Plates

1. Device plates shall be provided with switches. In noncorrosive indoor areas, receptacle device plates shall be made of sheet steel, zinc electroplated with chrome finish. Device plates in corrosive or outdoor areas shall be corrosion-resistant type. Device plates for explosion-proof equipment shall be factory provided with the equipment.
2. Device plates shall be provided with engraved laminated phenolic nameplates with 1/8 inch white characters on black background. Nameplates for switches shall identify panel and circuit number and area served. Nameplates for receptacles shall identify circuit and voltage if other than 120 volts, single phase.

PART 2 EXECUTION

2.01 GENERAL

- A. Boxes shall be independently supported by galvanized brackets, expansion bolts, toggle bolts, or machine or wood screws as appropriate. Wooden plugs inserted in masonry or concrete shall not be used as a base to secure boxes, nor shall welding or brazing be used to attachment. Wiring devices shall be tested for correct connections.
- B. Position of Outlets
 1. All outlets shall be centered with regard to building lines, furring and trim, symmetrically arranged in the room. Set outlets shall be set plumb and extend flush outlets to the finished surface of the wall, ceiling or floor without projecting beyond same. All receptacles, switches and outlets shown on the drawings shall be installed symmetrically along trim and where necessary, set the long dimension of the plate horizontal or gang in tandem.
- C. Mounting Heights
 1. Unless otherwise noted, wall mounted outlet devices shall generally be 24 inches above the floor, 18 inches in architecturally treated areas. Switches shall be 48 inches above the floor. All measurements are to centerline of device.

*** END OF SECTION ***

SECTION 26 28 16 – DISCONNECTS AND SWITCHES

PART 1 GENERAL

1.01 DESCRIPTION

- A. This section specifies all disconnects, fused and unfused, required by code for equipment furnished under this and other Divisions of these specifications.

1.02 STANDARDS AND CODES

- A. All equipment, materials, and the design, construction, installation, and application thereof shall comply with all applicable provisions of the national electrical code (NEC), the occupational safety and health act (OSHA), and any applicable federal, state, and local ordinances, rules and regulations. All materials and equipment specified herein shall be within the scope of nationally recognized testing laboratory (NRTL) examination services, be approved by the NRTL for the purpose for which they are used, and shall bear the appropriate listing label.

1.03 SUBMITTALS

- A. Submit all catalog data In accordance with the Submittals requirements in Section 26 05 00.1.12. Show material information and confirm compliance with these Specifications.

PART 2 PRODUCTS

2.01 DISCONNECTS

- A. Disconnect switches shall be heavy duty type, shall be horsepower rated, quick-make, quick-break construction. Switch blades shall open all ungrounded conductors and shall be single throw, unless otherwise noted.
- B. Disconnect switch enclosures shall be NEMA rated and mounted in accordance with Section 26 05 00.1.10.
- C. Fusible disconnects shall be as specified above with fuse space and clips to accept Class R fuses. Fusible disconnects shall only be utilized where required by equipment manufacturer to meet UL installation requirements.

2.02 MANUFACTURER

- A. Disconnect switches shall be manufactured by Cutler Hammer, Square D, Westinghouse, or equal.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Switches shall be mounted at locations shown on plans. Installation shall be in

accordance with the following methods:

1. Mounting
 - a. Disconnects shall be fastened securely to supporting structure at walls and stands:
 - b. Wood screws or lag screws to wood boards or timbers
 - c. Machine bolt to metal framing or plates
 - d. Expansion anchors to concrete walls
 - e. Expansion toggle wing bolts or sleeve anchors to hollow block
 - f. Provide 1 inch spacers to set enclosure out from concrete or block wall
2. Stands and Supports
 - a. Disconnect stands and support shall be constructed of and secured by:
 - b. Corrosion-resistant materials and finishes
 - c. Unistrut-type materials for fabrication
 - d. Expansion anchors for bolts in concrete floor
 - e. Machine bolt to metal framing or plates
 - f. Wood screws or lag screws to wood boards or timber
 - g. Backing plate for mounting units.
 - h. Fasten stand securely to floor
 - i. Dimensions as required by equipment to be mounted
3. Arrangement
 - a. Disconnects shall be arranged for driven equipment use or function:
 - b. Similar units adjacent
 - c. Adequate space for operation and servicing
4. Mounting Height
 - a. Center of handle shall be 4 feet 6 inch above the finished floor or work platform.

3.02 IDENTIFICATION

- A. Nameplates shall be provided for all disconnects in accordance with Section 26 05 00.2.2.
- B. Nameplate to state load designation and power source equipment.

*** END OF SECTION ***

SECTION 26 29 23 – VARIABLE FREQUENCY DRIVES

PART 1 GENERAL

1.01 DESCRIPTION OF WORK

- A. This section specifies AC pulse width modulated (PWM) Variable Frequency Drives (VFD) for operation on 480 Volt, three phase, 60 cycle power.
- B. Provider of VFDs
 - 1. The VFDs shall be provided by the System Integrator.
 - 2. The VFD's shall be provided as a complete and operational system, integral to the MCC or other control panels, as shown on the drawings.

1.02 SUBMITTALS

- A. Submit all catalog data in accordance with the Submittals requirements in Section 26 05 00. Show material information and confirm compliance with these specifications.
- B. In addition to the requirements of Section 26 05 00, submittals shall include the following information:
 - 1. Outline Dimensions with exterior and interior equipment elevation drawings.
 - 2. Wiring diagrams with all interface points and terminal numbers clearly identified.
 - 3. Specific information on the VFDs components provided for this project and all optional equipment provided.
 - 4. Operations and programming manual.
- C. Warranty
 - 1. The Contractor shall guarantee the VFDs to be free of defects in design, materials and workmanship for a period of one (1) year following the date of acceptance, of all work under the contract. The guarantee shall include all parts and labor and shall be secured by a written guarantee from the Manufacturer to the Owner. The written guarantee shall be delivered to the Owner prior to date of acceptance of all work under the Contract.

PART 2 PRODUCTS

2.01 GENERAL

- A. The Variable Frequency Drive (VFD) motor controller shall convert 480 Volt, three-phase, 60 Hertz power to adjustable voltage (480 volt) and frequency (0 - 60 hertz) three-phase, AC power for motor speed control with a capability of 60:1 speed range. All general options and modifications shall mount within the standard adjustable frequency controller enclosure.
- B. The controller(s) shall be suitable for use with any standard configuration squirrel-cage

induction motor(s) having a 1.15 Service Factor, or with existing standard squirrel-cage induction motor(s) with nameplate data as shown on the plans. At any time in the future, it shall be possible to substitute any standard motor (equivalent horsepower, voltage, and RPM) in the field.

2.02 SERVICE CONDITIONS

- A. The VFD shall be designed and constructed to operate within the following service conditions:
1. Ambient Temperature Range: -10°C to 40°C
 2. Atmosphere: Non-Condensing relative humidity to 95%
 3. AC Line Voltage Variation: +/- 10%
 4. AC Line Frequency Variation: ±3 Hertz

2.03 BASIC DRIVE

- A. Description
1. The VFD shall produce an adjustable AC voltage/frequency output. It shall have an output voltage regulator to maintain correct output V/Hz despite incoming voltage variations.
 2. The VFD shall have a continuous output current rating of 100% of motor nameplate current.
 3. The VFD shall be of the Pulse Width Modulated type and employ a dual full-wave diode bridge converter to convert incoming fixed voltage/frequency to a fixed DC voltage. The Pulse Width Modulation strategy shall be of the space vector type implemented in a microprocessor which generates a sine-coded output voltage.
 4. The inverter output shall be generated by Insulated Gate Bipolar Transistors (IGBT) which shall be controlled by six identical base driver circuits. The worst case RMS motor line current measured at rated speed, torque and voltage shall not exceed 1.05 times the rated RMS motor current for pure sine wave operation.

2.04 KEYPAD OPERATOR INTERFACE

- A. The VFD shall be provided with a door mounted LCD keypad which will indicate (minimum):
1. Frequency output
 2. Voltage output
 3. Current output
 4. Motor RPM
 5. Motor kW
 6. Elapsed Time
 7. Time stamped fault indication

8. DC bus Volts
9. Faults
10. Parameter settings.

2.05 ENCLOSURE

- A. All VFD components shall be factory mounted and wired in an enclosure, meeting the requirements of 26 05 00, Area Classifications.
- B. A ventilation system shall be provided to maintain the internal enclosure temperature within the operating conditions for the VFD.
- C. If a free-standing enclosure is provided, it shall be suitable for mounting on a concrete housekeeping pad.

2.06 PROTECTIVE CIRCUITS AND FEATURES

- A. The VFD shall include the following protective circuits and features:
 1. Overload rating of 110% for 60 seconds, and 150% for 3 seconds.
 2. Output phase to phase short circuit condition.
 3. Total ground fault under any operating condition.
 4. High input line voltage
 5. Low input line voltage
 6. Loss of input or output phase
 7. Metal Oxide Varistors for surge suppression at the VFD input terminals.
- B. Diagnostic Features
 1. The VFD shall include a microprocessor based digital diagnostic system which shall monitor its own control functions and displays faults and operating conditions.
 2. A "FAULT LOG" shall record, store, and display the fifty (50) most recent fault events.
 3. Ethernet communications shall be provided on each VFD for connection to PLC and SCADA systems for monitoring and control.
- C. Acceptable Manufacturers
 1. Acceptable Manufacturers shall provide equipment which meets all of the requirements of these specifications, and fit within the space requirements.
 2. Where VFDs are mounted integral to an MCC assembly, the MCC and VFD manufacturers shall be the same.
 3. VFD's shall be Allen Bradley Powerflex 753 series, or Engineer approved equal.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Installation shall be the responsibility of the Contractor. The Contractor shall install the drive in accordance with the contract drawings and as recommended by the VFD Manufacturer as outlined in the installation manual.
- B. Power and control wiring shall be completed by the Electrical Contractor. The Contractor shall complete all wiring in accordance with the recommendations of the VFD Manufacturer as outlined in the installation manual.

3.02 QUALITY ASSURANCE

- A. The controller shall be subject to, but not limited to, the following quality assurance controls, procedures and tests:
 - 1. Each VFD shall be functionally tested under motor load. During this load test the VFD shall be monitored for correct phase, current, voltages and motor speed. Correct current limiting operation shall be verified by simulating a motor overload. Manufacturing test data shall be recorded and stored by the manufacturer at the time of production.
 - 2. Verification of proper factory presets shall be performed on 100% of all parameters to ensure proper microprocessor settings. Verification that the proper factory settings are loaded correctly in the drive shall be done via the drive serial interface port.
 - 3. The drive assembly shall be tested for shock (15G peak for 11 ms duration) and vibration (-0152 mm displacement, 1G peak).

3.03 START-UP

- A. Certified factory start-up shall be provided for each drive by a factory authorized service center. A certified start-up form shall be filled out for each drive with a copy provided to the Engineer, Owner, and a copy kept on file at the Manufacturer. Cost for this startup support shall be included in the VFD bid price. The Engineer shall be notified a minimum one (1) week in advance of the scheduled start-up.
- B. Where the application utilizes standby power generation, the VFDs shall be tested with both Utility and standby power sources. The VFD Representative shall make any and all adjustments and modifications to the VFDs to operate within specified limits, without additional cost to the owner. The Manufacturer's representative will be present during these tests.
- C. A list of all drive parameters and settings for each drive shall be provided to the Owner upon project completion.

3.04 OPERATION AND MAINTENANCE TRAINING

- A. The Supplier shall conduct specifically organized training sessions in operation and maintenance of the VFD equipment for personnel employed by the Owner. The training

sessions shall be conducted to educate and train the personnel in maintenance and operation of all components of the system. Training shall include, but not be limited to, the following:

1. Preventative maintenance procedures
2. Trouble-shooting
3. Calibration
4. Testing
5. Replacement of components.

- B. At least one (1) training session, of at least four (4) hours in duration, shall be conducted after start-up of the system. The Supplier shall provide specific instruction materials for each training session and shall supply such materials to the Owner at least one (1) week prior to the time of the training.

*** END OF SECTION ***

SECTION 26 32 13 – DIESEL EMERGENCY ENGINE GENERATORS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

- A. Section includes packaged diesel engine generators for emergency use with the following features:
 - 1. Diesel engine.
 - 2. Diesel fuel-oil system.
 - 3. Control and monitoring.
 - 4. Generator overcurrent and fault protection.
 - 5. Generator, exciter, and voltage regulator.
 - 6. Load banks.
 - 7. Outdoor engine generator enclosure.
 - 8. Vibration isolation devices.
 - 9. Finishes.
- B. Related Requirements:
 - 1. Section 263600 "Transfer Switches" for transfer switches, including sensors and relays to initiate automatic-starting and -stopping signals for engine generators.

1.03 DEFINITIONS

- A. Emergency Standby Power (ESP): Per ISO 8528: The maximum power available during a variable electrical power sequence, under the stated operating conditions, for which a generating set can deliver in the event of a utility power outage or under test conditions for up to 200 hours of operation per year with the maintenance intervals and procedures being carried out as prescribed by the manufacturers. The permissible average power output over 24 hours of operation shall not exceed 70 percent of the ESP unless otherwise agreed by the RIC engine manufacturer.
- B. Operational Bandwidth: The total variation, from the lowest to highest value of a parameter over the range of conditions indicated, expressed as a percentage of the nominal value of the parameter.

1.04 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
2. Include thermal damage curve for generator.
3. Include time-current characteristic curves for generator protective device.
4. Include fuel consumption in gallons per hour (liters per hour) at 0.8 power factor at 0.5, 0.75, and 1.0 times generator capacity.
5. Include generator efficiency at 0.8 power factor at 0.5, 0.75, and 1.0 times generator capacity.
6. Include airflow requirements for cooling and combustion air in cubic feet per minute (cubic meters per minute) at 0.8 power factor, with air-supply temperature of 95, 80, 70, and 50 deg F (35, 27, 21, and 10 deg C). Provide Drawings indicating requirements and limitations for location of air intake and exhausts.
7. Include generator characteristics, including, but not limited to, kilowatt rating, efficiency, reactances, and short-circuit current capability.

B. Shop Drawings:

1. Include plans and elevations for engine generator and other components specified. Indicate access requirements affected by height of subbase fuel tank.
2. Include details of equipment assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Identify fluid drain ports and clearance requirements for proper fluid drain.
4. Design calculations for selecting vibration isolators and seismic restraints and for designing vibration isolation bases.
5. Vibration Isolation Base Details: Detail fabrication, including anchorages and attachments to structure and supported equipment. Include base weights.
6. Include diagrams for power, signal, and control wiring. Complete schematic, wiring, and interconnection diagrams showing terminal markings for EPS equipment and functional relationship between all electrical components.

C. Certifications:

1. Submit statement of compliance which states the proposed products are certified to the emissions standards required by the location for Eastsound, WA.

1.05 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For Manufacturer and testing agency.
- B. Source Quality-Control Reports: Including, but not limited to, the following:
 - 1. Certified Test Report: Provide certified test report documenting factory test per the requirements of this specification, as well as certified factory test of generator set sensors per NFPA110 level 1.
 - 2. Certified Test Reports: For components and accessories that are equivalent, but not identical, to those tested on prototype unit.
 - 3. Report of factory test on units to be shipped for this Project, showing evidence of compliance with specified requirements.
 - 4. Report of sound generation.
 - 5. Report of exhaust emissions showing compliance with applicable regulations.
 - 6. Certified Torsional Vibration Compatibility: Comply with NFPA 110.
- C. Field quality-control reports.
 - 1. Warranty: Submit manufacturer's warranty statement to be provided for this Project.

1.06 QUALITY ASSURANCE

- A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.
- B. Manufacturer Qualifications: A qualified manufacturer. Maintain, within 200 miles of Project site, a service center capable of providing training, parts, and emergency maintenance repairs.
- C. Source Limitations: Obtain packaged generator sets and auxiliary components through one source from a single manufacturer.
- D. Comply with NFPA 37 (Standard For the Installation and Use of Stationary Combustion Engines and Gas Turbines).
- E. Comply with NFPA 70 (National Electrical Code. Equipment shall be suitable for use in systems in compliance to Article 700, 701, and 702).
- F. Comply with NFPA 110 (Emergency and Standby Power Systems) requirements for Level 1 emergency power supply system.

1.07 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For engine generators to include in emergency, operation, and maintenance manuals.

1. In addition to items specified in Section 01 33 00 "Submittal Procedures", include the following:
 - a. List of tools and replacement items recommended to be stored at Project for ready access. Include part and drawing numbers, current unit prices, and source of supply.
 - b. Operating instructions laminated and mounted adjacent to generator location.
 - c. Training plan.

1.08 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 1. Fuses: One for every 10 of each type and rating, but no fewer than one of each.
 2. Indicator Lamps: Two for every six of each type used, but no fewer than two of each.
 3. Filters: One set each of lubricating oil, fuel, and combustion-air filters.
 4. Tools: Each tool listed by part number in operations and maintenance manual.

1.09 QUALITY ASSURANCE

- A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer.
- B. Testing Agency Qualifications: Accredited by NETA.
 1. Testing Agency's Field Supervisor: Certified by NETA to supervise on-site testing.

1.010 PROJECT CONDITIONS

- A. Environmental Conditions: Engine-generator system shall withstand the following environmental conditions without mechanical or electrical damage or degradation of performance capability:
 1. Ambient Temperature: 0.0 deg C (32.0 deg F) to 25.0 deg C (77.0 deg F).
 2. Relative Humidity: 0 to 95 percent.
 3. Altitude: Sea level to 100.0 feet.
 4. Close proximity to salt water and related environment.

1.011 WARRANTY

- A. Manufacturer's Warranty: Manufacturer agrees to repair or replace components of packaged engine generators and associated auxiliary components that fail in materials or workmanship within specified warranty period.

- 1. Warranty Period: Two years from date of Substantial Completion.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Manufacturers: The basis for this specification is Cummins Power Generation equipment, generator model C125D6D.
- B. Pre-approved generator equipment manufacturers are Cummins, Kohler, Caterpillar, and Generac.
- C. Equals will be considered if equipment performance is shown to meet the specifications and requirements herein.
- D. Source Limitations: Obtain packaged engine generators and auxiliary components from single source from single manufacturer.

2.02 PERFORMANCE REQUIREMENTS

- A. B11 Compliance: Comply with B11.19.
- B. NFPA Compliance:
 - 1. Comply with NFPA 37.
 - 2. Comply with NFPA 70.
 - 3. Comply with NFPA 99.
 - 4. Comply with NFPA 110 requirements for Level 2 EPSS.
- C. UL Compliance: Comply with UL 2200.
- D. Engine Exhaust Emissions: Comply with EPA Tier 3 requirements and applicable state and local government requirements.
- E. Noise Emission: Comply with applicable state and local government requirements for maximum noise level at adjacent property boundaries due to sound emitted by engine generator, including engine, engine exhaust, engine cooling-air intake and discharge, and other components of installation.

2.03 ENGINE-GENERATOR SET

- A. Factory-assembled and -tested, engine-generator set.
- B. Mounting Frame: Maintain alignment of mounted components without depending on concrete foundation; and have lifting attachments.

1. Rigging Information: Indicate location of each lifting attachment, generator-set center of gravity, and total package weight in submittal drawings.
- C. Capacities and Characteristics:
1. Power Output Ratings: Electrical output power rating for Standby operation of not less than 150.0kW, at 80 percent lagging power factor, 277/480, Series Wye, Three phase, 4 -wire, 60 hertz.
 2. Alternator shall be capable of accepting maximum 920.0 kVA in a single step and be capable of recovering to a minimum of 90% of rated no load voltage. Following the application of the specified kVA load at near zero power factor applied to the generator set.
 3. Nameplates: For each major system component to identify manufacturer's name and address, and model and serial number of component. The engine-generator nameplate shall include information of the power output rating of the equipment.
- D. Generator-Set Performance:
1. Steady-State Voltage Operational Bandwidth: 0.5 percent of rated output voltage from no load to full load.
 2. Transient Voltage Performance: Not more than 14 percent variation for 50 percent step-load increase or decrease. Voltage shall recover and remain within the steady-state operating band within 2 seconds. On application of a 100% load step the generator set shall recover to stable voltage within 10 seconds.
 3. Steady-State Frequency Operational Bandwidth: 0.25 percent of rated frequency from no load to full load.
 4. Steady-State Frequency Stability: When system is operating at any constant load within the rated load, there shall be no random speed variations outside the steady-state operational band and no hunting or surging of speed.
 5. Transient Frequency Performance: Not more than 5 percent variation for 50 percent step-load increase or decrease. Frequency shall recover and remain within the steady-state operating band within 2 seconds. On application of a 100% load step the generator set shall recover to stable frequency within 10 seconds.
 6. Output Waveform: At full load, harmonic content measured line to line or line to neutral shall not exceed 5 percent total and 3 percent for any single harmonic. Telephone influence factor, determined according to NEMA MG 1, shall not exceed 50.
 7. Sustained Short-Circuit Current: (For engine-generator sets using a PMG-excited alternator) For a 3-phase, bolted short circuit at system output terminals, system shall supply a minimum of 300 percent of rated full-load current for not less than 8 seconds without damage to generator system components. For a 1-phase, bolted short circuit at system output terminals, system shall regulate both voltage and current to prevent over-voltage conditions on the non-faulted phases.

8. Start Time: Comply with NFPA 110, Level 1, Type 10, system requirements.
9. Ambient Condition Performance: Engine generator shall be designed to allow operation at full rated load in an ambient temperature under site conditions, based on highest ambient condition. Ambient temperature shall be as measured at the air inlet to the engine generator for enclosed units, and at the control of the engine generator for machines installed in equipment rooms.

2.04 ENGINE

- A. Fuel: ASTM D975 #2 Diesel Fuel
- B. Rated Engine Speed: 1800RPM.
- C. Lubrication System: The following items are mounted on engine or skid:
 1. Lube oil pump: shall be positive displacement, mechanical, full pressure pump.
 2. Filter and Strainer: Provided by the engine manufacturer of record to provide adequate filtration for the prime mover to be used.
 3. Crankcase Drain: Arranged for complete gravity drainage to an easily removable container with no disassembly and without use of pumps, siphons, special tools, or appliances.
- D. Engine Fuel System: The engine fuel system shall be installed in strict compliance to the engine manufacturer's instructions
- E. Main Fuel Pump: Mounted on engine. Pump ensures adequate primary fuel flow under starting and load conditions.
- F. Coolant Jacket Heater: Electric-immersion type, factory installed in coolant jacket system. Comply with NFPA 110 requirements for Level 1 equipment for heater capacity and performance.
 1. Designed for operation on a single 120 VAC, Single phase, 60Hz power connection. Heater voltage shall be shown on the project drawings.
 2. Installed with isolation valves to isolate the heater for replacement of the element without draining the engine cooling system or significant coolant loss.
 3. Provided with a 12VDC thermostat, installed at the engine thermostat housing
- G. Governor: Adjustable isochronous, with speed sensing. The governing system dynamic capabilities shall be controlled as a function of engine coolant temperature to provide fast, stable operation at varying engine operating temperature conditions. The control system shall actively control the fuel rate as appropriate to the state of the engine generator. Fuel rate shall be regulated as a function of starting, accelerating to start disconnect speed, accelerating to rated speed, and operating in various isochronous states.
- H. Cooling System: Closed loop, liquid cooled

1. The generator set manufacturer shall provide prototype test data for the specific hardware proposed demonstrating that the machine will operate at rated standby load in an outdoor ambient condition of 40 deg C.
 2. Coolant: Solution of 50 percent ethylene-glycol-based antifreeze and 50 percent water, with anticorrosion additives as recommended by engine manufacturer.
 3. Size of Radiator overflow tank: Adequate to contain expansion of total system coolant from cold start to 110 percent load condition.
 4. Expansion Tank: Constructed of welded steel plate and rated to withstand maximum closed-loop coolant system pressure for engine used. Equip with gage glass and petcock.
 5. Temperature Control: Self-contained, thermostatic-control valve modulates coolant flow automatically to maintain optimum constant coolant temperature as recommended by engine manufacturer.
 6. Duct Flange: Generator sets installed indoors shall be provided with a flexible radiator duct adapter flange.
- I. Muffler/Silencer: Selected with performance as required to meet sound requirements of the application, sized as recommended by engine manufacturer and selected with exhaust piping system to not exceed engine manufacturer's engine backpressure requirements. For generator sets with outdoor enclosures the silencer shall be inside the enclosure.
- J. Air-Intake Filter: Engine-mounted air cleaner with replaceable dry-filter element and restriction indicator.
- K. Starting System: 12 or 24V, as recommended by the engine manufacturer; electric, with negative ground.
1. Components: Sized so they will not be damaged during a full engine-cranking cycle with ambient temperature at maximum specified in Part 1 "Project Conditions" Article.
 2. Cranking Cycle: As required by NFPA 110 for level 1 systems.
 3. Battery Cable: Size as recommended by engine manufacturer for cable length as required. Include required interconnecting conductors and connection accessories.
 4. Battery Compartment: Factory fabricated of metal with acid-resistant finish.
 5. Battery-Charging Alternator: Factory mounted on engine with solid-state voltage regulation. The battery charging alternator shall have sufficient capacity to recharge the batteries with all parasitic loads connected within 4 hours after a normal engine starting sequence.

6. Battery Chargers: Unit shall comply with UL 1236, provide fully regulated, constant voltage, current limited, battery charger for each battery bank. It will include the following features:
 - a. Operation: Equalizing-charging rate based on generator set manufacturer's recommendations shall be initiated automatically after battery has lost charge until an adjustable equalizing voltage is achieved at battery terminals. Unit shall then be automatically switched to a lower float-charging mode and shall continue to operate in that mode until battery is discharged again.
 - b. Automatic Temperature Compensation: Adjust float and equalize voltages for variations in ambient temperature from minus 20 deg C to plus 40 deg C to prevent overcharging at high temperatures and undercharging at low temperatures.
 - c. Automatic Voltage Regulation: Maintain constant output voltage regardless of input voltage variations up to plus or minus 10 percent.
 - d. Safety Functions: Sense abnormally low battery voltage and close contacts providing low battery voltage indication on control and monitoring panel. Sense high battery voltage and loss of ac input or dc output of battery charger. Either condition shall close contacts that provide a battery-charger malfunction indication at system control and monitoring panel.
 - e. Provide LED indication of general charger condition, including charging, faults, and modes. Provide a LCD display to indicate charge rate and battery voltage. Charger shall provide relay contacts for fault conditions as required by NFPA110.
 - f. Enclosure and Mounting: NEMA, Type 4, mounted in enclosure.

2.05 FUEL OIL STORAGE

- A. Comply with NFPA 30.
- B. Sub Base-Mounted Fuel Oil Tank: Provide a double wall secondary containment type sub base fuel storage tank. The tank shall be constructed of corrosion resistant steel and shall be listed and labeled. The fuel tank shall include the following features:
 1. Capacity: Fuel for 35 hours continuous operation at 100 percent rated power output; 48 hours operation at normal operational output.
 2. Tank rails and lifting eyes shall be rated for the full dry weight of the tank, genset, and enclosure.
 3. Electrical stub up(s)
 4. Normal & emergency vents
 5. Lockable fuel fill

6. Mechanical fuel level gauge
7. High and low level switches to indicate fuel level
8. Leak detector switch
9. Sub base tank shall include a welded steel containment basin, sized at a minimum of 110% of the tank capacity to prevent escape of fuel into the environment in the event of a tank rupture.
10. Fill port with overfill prevention valve (OFPV)
11. 5 gallon fill/spill dam or bucket
12. Tank design shall meet the regional requirements for the Project location
13. Contractor shall provide diesel fuel for all required testing and provide the Owner with a full tank of diesel fuel upon final completion of the project.

2.06 CONTROL AND MONITORING

- A. Engine generator control shall be microprocessor based and provide automatic starting, monitoring, protection and control functions for the unit.
- B. Automatic Starting System Sequence of Operation: When mode-selector switch on the control and monitoring panel is in the automatic position, remote-control contacts in one or more separate automatic transfer switches initiate starting and stopping of generator set. When mode-selector switch is switched to the on position, generator set starts. The off position of same switch initiates generator-set shutdown. (Switches with different configurations but equal functions are acceptable.) When generator set is running, specified system or equipment failures or derangements automatically shut down generator set and initiate alarms. Operation of the local (generator set-mounted) and/or remote emergency-stop switch also shuts down generator set.
- C. Manual Starting System Sequence of Operation: Switching on-off switch on the generator control panel to the on position starts generator set. The off position of same switch initiates generator-set shutdown. When generator set is running, specified system or equipment failures or derangements automatically shut down generator set and initiate alarms. Operation of the local (generator set-mounted) and/or remote emergency-stop switch also shuts down generator set.
- D. Configuration: Operating and safety indications, protective devices, system controls, engine gages and associated equipment shall be grouped in a common control and monitoring panel. Mounting method shall isolate the control panel from generator-set vibration. AC output power circuit breakers and other output power equipment shall not be mounted in the control enclosure.
- E. Indicating and Protective Devices and Controls: As required by NFPA 110 for Level 1 system, and the following:
 1. AC voltmeter (3-phase, line to line and line to neutral values).

2. AC ammeter (3-phases).
3. AC frequency meter.
4. Ammeter-voltmeter displays shall simultaneously display conditions for all three phases.
5. Emergency Stop Switch: Switch shall be a red “mushroom head” pushbutton device complete with lock-out/tag-out provisions. Depressing switch shall cause the generator set to immediately stop the generator set and prevent it from operating.
6. Fault Reset Switch: Supply a dedicated control switch to reset/clear fault conditions.
7. DC voltmeter (alternator battery charging).
8. Engine-coolant temperature gauge.
9. Engine lubricating-oil pressure gauge.
10. Running-time meter.
11. Generator-voltage and frequency digital raise/lower switches. Rheostats for these functions are not acceptable. The control shall adjustment of these parameters in a range of plus or minus 5% of the voltage and frequency operating set point (not nominal voltage and frequency values.) The voltage and frequency adjustment functions shall be disabled when the paralleling breaker is closed.
12. Fuel tank derangement alarm.
13. Fuel tank high-level shutdown of fuel supply alarm.
14. AC Protective Equipment: The control system shall include over/under voltage, reverse kVAR over current, loss of voltage reference, and over excitation shutdown protection. There shall be a overload warning, and overcurrent warning alarm.
15. Status LED indicating lamps to indicate remote start signal present at the control, existing shutdown condition, existing alarm condition, not in auto, and generator set running.
16. A graphical display panel with appropriate navigation devices shall be provided to view all information noted above, as well as all engine status and alarm/shutdown conditions (including those from an integrated engine emission control system). The display shall also include integrated provisions for adjustment of the gain and stability settings for the governing and voltage regulation systems.
17. Panel lighting system to allow viewing and operation of the control when the generator room or enclosure is not lighted.

18. Data Logging: The control system shall log the latest 20 different alarm and shut down conditions, the total number of times each alarm or shutdown has occurred, and the date and time the latest of these shutdown and fault conditions occurred.
19. DC control Power Monitoring: The control system shall continuously monitor DC power supply to the control, and annunciate low or high voltage conditions. It shall also provide an alarm indicating imminent failure of the battery bank based on degraded voltage recover on loading (engine cranking).
20. Paralleling Breaker control switches: The control shall include manual open and close provisions for the paralleling breaker, and LED status lamps indicating whether the breaker is open or closed.

2.07 GENERATOR, EXCITER, AND VOLTAGE REGULATOR

- A. Comply with NEMA MG 1.
- B. Drive: Generator shaft shall be directly connected to engine shaft. Exciter shall be rotated integrally with generator rotor.
- C. Electrical Insulation: Class H
- D. Temperature Rise: 125 / Class H environment.
- E. Construction shall prevent mechanical, electrical, and thermal damage due to vibration, over speed up to 125 percent of rating, and heat during operation at 110 percent of rated capacity.
- F. Permanent Magnet Generator (PMG) shall provide excitation power for optimum motor starting and short circuit performance.
- G. Enclosure: Drip-proof.
- H. Voltage Regulator: SCR type, Separate from exciter, providing performance as specified. The voltage regulation system shall be microprocessor-controlled, full wave rectified, and provide a pulse-width modulated signal to the exciter. No exceptions or deviations to these requirements will be permitted.
- I. Windings: Two-thirds pitch stator winding and fully linked amortisseur winding.
- J. Subtransient Reactance: 15 percent maximum, based on the rating of the engine generator set.

2.08 OUTDOOR GENERATOR-SET ENCLOSURE

- A. Description: Sound Attenuated Aluminum housing. Multiple panels shall be lockable and provide adequate access to components requiring maintenance. Instruments, control, and battery system shall be mounted within enclosure.
- B. Construction:

1. Hinged Doors: With padlocking provisions. Restraint/Hold back hardware to prevent door to keep door open at 180 degrees during maintenance. Rain lips over all doors.
 2. Exhaust System:
 - a. Muffler Location: Within enclosure.
 3. Hardware: All hardware and hinges shall be stainless steel.
 4. Mounting Base: Suitable for mounting on sub-base fuel tank or housekeeping pad.
 5. A weather protective enclosure shall be provided which allows the generator set to operate at full rated load with a static pressure drop equal to or less than 0.5 inches of water.
- C. Engine Cooling Airflow through Enclosure: Housing shall provide ample airflow for engine generator operation at rated load in an ambient temperature of 40 deg C.
1. Louvers: Fixed-engine, cooling-air inlet and discharge.
- D. Sound Performance: Reduce the sound level of the engine generator while operating at full rated load to a maximum of 72 dBA measured at any location 7 m from the engine generator in a free field environment.
- E. Site Provisions:
1. Lifting: Complete assembly of engine generator, enclosure, and sub base fuel tank (when used) shall be designed to be lifted into place as a single unit, using spreader bars.

2.09 VIBRATION ISOLATION DEVICES

- A. Vibration Isolation: Generators installed on grade shall be provided with elastomeric isolator pads integral to the generator, unless the engine manufacturer requires use of spring isolation.

2.010 FINISHES

- A. Indoor and Outdoor Enclosures and Components: Powder-coated and baked over corrosion-resistant pretreatment and compatible primer. Manufacturer's standard color or as directed on the drawings.

2.011 SOURCE QUALITY CONTROL

- A. Prototype Testing: Factory test engine-generator set using same engine model, constructed of identical or equivalent components and equipped with identical or equivalent accessories.
1. Tests: Comply with NFPA 110, Level 1 Energy Converters. In addition, the equipment engine, skid, cooling system, and alternator shall have been subjected

to actual prototype tests to validate the capability of the design under the abnormal conditions noted in NFPA110. Calculations and testing on similar equipment which are allowed under NFPA110 are not sufficient to meet this requirement.

B. Project-Specific Equipment Tests: Before shipment, factory test engine-generator set manufactured specifically for this Project. Perform tests at rated load and power factor. Include the following tests:

1. Test engine generator set manufactured for this Project to demonstrate compatibility and functionality.
2. Full load run.
3. Maximum power.
4. Voltage regulation.
5. Steady-state governing.
6. Single-step load pickup.
7. Simulated safety shutdowns.
8. Provide 14 days' advance notice of tests and opportunity for observation of tests by Owner's representative.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Examine areas, equipment bases, and conditions, with Installer present, for compliance with requirements for installation and other conditions affecting packaged engine generator performance.
- B. Examine roughing-in for piping systems and electrical connections. Verify actual locations of connections before packaged engine generator installation.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PREPARATION

- A. Interruption of Existing Electrical Service: Do not interrupt electrical service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electrical service according to requirements indicated:
 1. Notify Construction Manager and Owner no fewer than two working days in advance of proposed interruption of electrical service.

2. Do not proceed with interruption of electrical service without Owner's written permission.

3.03 INSTALLATION

- A. Comply with packaged engine-generator manufacturers' written installation, application, and alignment instructions and with NFPA 110.
- B. Equipment shall be installed by the contractor in accordance with final submittals and contract documents. Installation shall comply with applicable state and local codes as required by the authority having jurisdiction. Install equipment in accordance with manufacturer's instructions and instructions included in the listing or labeling of UL listed products.
- C. Installation of equipment shall include furnishing and installing all interconnecting wiring between all major equipment provided for the on-site power system. The contractor shall also perform interconnecting wiring between equipment sections (when required), under the supervision of the equipment supplier.
- D. Equipment shall be installed on concrete housekeeping pads. Equipment shall be permanently fastened to the pad in accordance with manufacturer's instructions and seismic requirements of the site.
- E. Equipment shall be initially started and operated by representatives of the manufacturer. All protective settings shall be adjusted as instructed by the consulting engineer.
- F. All equipment shall be physically inspected for damage. Scratches and other installation damage shall be repaired prior to final system testing. Equipment shall be thoroughly cleaned to remove all dirt and construction debris prior to initial operation and final testing of the system.
- G. On completion of the installation by the electrical contractor, the generator set supplier shall conduct a site evaluation to verify that the equipment is installed per manufacturer's recommended practice.

3.04 CONNECTIONS

- A. Ground equipment according to Section 260526 "Grounding and Bonding for Electrical Systems."
- B. Connect wiring according to Section 260519 "Wire and Cable." Provide a minimum of one 90-degree bend in flexible conduit routed to the engine generator from a stationary element.
- C. Balance single-phase loads to obtain a maximum of 10 percent unbalance between any two phases.

3.05 IDENTIFICATION

- A. Identify system components according to Section 260553 "Identification for Electrical Systems."
- B. Install a sign indicating the generator neutral is bonded to the main service neutral at the main service location.

3.06 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect components, assemblies, and equipment installations, including connections.
- B. Perform tests and inspections with the assistance of a factory-authorized service representative.

3.07 ON-SITE ACCEPTANCE TEST

- A. The complete installation shall be tested to verify compliance with the performance requirements of this specification following completion of all site work. Testing shall be conducted by representatives of the manufacturer, with required fuel supplied by Contractor. The Engineer shall be notified in advance and shall have the option to witness the tests. The generator set manufacturer shall provide a site test specification covering the entire system. Tests shall include:
 - 1. Prior to start of active testing, all field connections for wiring, power conductors, and bus bar connections shall be checked for proper tightening torque.
 - 2. Installation acceptance tests to be conducted on site shall include a "cold start" test, a two hour full load (resistive) test, and a one-step rated load pickup test in accordance with NFPA 110. Provide a resistive load bank and make temporary connections for full load test, if necessary.
 - 3. Perform a power failure test on the entire installed system. This test shall be conducted by opening the power supply from the utility service, and observing proper operation of the system for at least 2 hours. Coordinate timing and obtain approval for start of test with site personnel.

3.08 SERVICE AND SUPPORT

- A. The generator set supplier shall maintain service parts inventory for the entire power system at a central location which is accessible to the service location 24 hours per day, 365 days per year. The manufacturer of the generator set shall maintain a central parts inventory to support the supplier, covering all the major components of the power system, including engines, alternators, control systems, paralleling electronics, and power transfer equipment.
- B. The generator set shall be serviced by a local service organization that is trained and factory certified in generator set service. The supplier shall maintain an inventory of critical power system replacement parts in the local service location. Service vehicles shall be stocked with critical replacement parts. The service organization shall be on call 24 hours per day, 365

days per year. The service organization shall be physically located within 200 miles of the site.

- C. The manufacturer shall maintain model and serial number records of each generator set provided for at least 20 years.

3.09 GENERATOR FUEL

- A. All diesel fuel for all testing of the generator shall be provided by the Contractor.
- B. After all testing has been completed to the satisfaction of the Owner and Engineer, the Contractor shall provide a full tank of diesel fuel to the Owner upon final commissioning.

3.010 TRAINING

- A. The equipment supplier shall provide training for the facility operating personnel covering operation and maintenance of the equipment provided.
- B. The training program shall be not less than 4 hours in duration and the class size shall be limited to 5 persons.
- C. Training date shall be coordinated with the facility Owner.

*** END OF SECTION ***

SECTION 26 36 00 – AUTOMATIC TRANSFER SWITCHES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

- A. Section includes Service Rated Automatic Transfer Switches (ATS) rated 600 V and less.

1.03 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for transfer switches.
 - 2. Include rated capacities, operating characteristics, electrical characteristics, and accessories.
- B. Shop Drawings:
 - 1. Include plans, elevations, sections, details showing minimum clearances, conductor entry provisions, gutter space, and installed features and devices.
 - 2. Include material lists for each switch specified.
 - 3. Single-Line Diagram: Show connections between transfer switch, power sources, and load; and show interlocking provisions for each combined transfer switch and bypass/isolation switch.
 - 4. Retain and revise, if necessary, "Riser Diagram" Subparagraph below if retaining remote annunciators, control panels, or bypass/isolation switches.
 - 5. Riser Diagram: Show interconnection wiring between transfer switches, bypass/isolation switches, annunciators, and control panels.

1.04 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For testing agency.
- B. Field quality-control reports.

1.05 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For each type of product to include in emergency, operation, and maintenance manuals.

1.06 QUALITY ASSURANCE

A. Testing Agency Qualifications:

1. Member company of NETA.
 - a. Testing Agency's Field Supervisor: Certified by NETA to supervise on-site testing.

1.07 FIELD CONDITIONS

A. Interruption of Existing Electrical Service: Do not interrupt electrical service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electrical service:

1. Notify Construction Manager and Owner no fewer than two days in advance of proposed interruption of electrical service.
2. Do not proceed with interruption of electrical service without Owner's written permission.

1.08 WARRANTY

A. Manufacturer's Warranty: Manufacturer agrees to repair or replace components of transfer switch or transfer switch components that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Manufacturers: The basis for this specification is Cummins Power Generation equipment, transfer switch model OTECSE with NEMA 12 enclosure and Power Command 40-01 controller.
- B. Switches manufactured by other manufacturers that meet the requirement of this specification are acceptable, if approved not less than two weeks before scheduled bid date. Proposals must include a line-by-line compliance statement based on this specification.
- C. Transfer switches utilizing molded case circuit breakers do not meet the requirements of this specification and will not be accepted.

2.02 GENERAL TRANSFER SWITCH PRODUCT REQUIREMENTS

- A. Provide transfer switches in the number and ratings that are shown on the drawings.
- B. Indicated Current Ratings: Apply as defined in UL 1008 for continuous loading and total system transfer.

- C. Fault-Current Closing and Withstand Ratings: UL 1008 WCR ratings must be specifically listed as meeting the requirements for use with protective devices at installation locations, under specified fault conditions. Withstand and closing ratings shall be based on use of the same set of contacts for the withstand test and the closing test.
- D. Solid-State Controls: All settings should be accurate to +/- 2% or better over an operating temperature range of - 40 to + 60 degrees C (- 40 to + 140 degrees F).
- E. Resistance to Damage by Voltage Transients: Components shall meet or exceed voltage-surge withstand capability requirements when tested according to IEEE C62.41. Components shall meet or exceed voltage-impulse withstand test of NEMA ICS 1.
- F. Electrical Operation: Accomplished by a non-fused, momentarily energized solenoid or electric motor operator mechanism, mechanically and electrically interlocked in both directions.
- G. Switch Characteristics: Designed for continuous-duty repetitive transfer of full-rated current between active power sources.
 - 1. Switches using molded-case switches or circuit breakers, or insulated case circuit breaker components are not acceptable.
 - 2. Transfer switches shall be double-throw, electrically and mechanically interlocked, and mechanically held in the Source 1 and Source 2 positions.
 - 3. Main switch contacts shall be high pressure silver alloy. Contact assemblies shall have arc chutes for positive arc extinguishing. Arc chutes shall have insulating covers to prevent inter-phase flashover.
 - 4. Contacts shall be operated by a high-speed electrical mechanism that causes contacts to open or close within three electrical cycles from signal.
 - 5. Transfer switch shall be provided with flame retardant transparent covers to allow viewing of switch contact operation but prevent direct contact with components that could be operating at line voltage levels.
 - 6. The transfer switch shall include the mechanical and control provisions necessary to allow the device to be field-configured for operating speed. Transfer switch operation with motor loads shall be as is recommended in NEMA MG1.
 - a. Phase angle monitoring/timing equipment is not an acceptable substitute for this functionality
 - 7. Transfer switches designated on the drawings as "3-pole" shall have a full current-rated neutral bar with lugs.
 - 8. Transfer switches designated on the drawings as "service entrance" switches shall meet the requirements of section "SERVICE ENTRANCE TRANSFER SWITCHES" of this specification.

- H. Factory wiring: Transfer switch internal wiring shall be composed of pre-manufactured harnesses that are permanently marked for source and destination. Harnesses shall be connected to the control system by means of locking disconnect plug(s), to allow the control system to be easily disconnected and serviced without disconnecting power from the transfer switch mechanism
- I. Terminals: Terminals shall be pressure type and appropriate for all field wiring. Control wiring shall be equipped with suitable lugs, for connection to terminal strips.
- J. Enclosures: All enclosures shall be third-party certified for compliance to NEMA ICS 6 and UL 508, unless otherwise indicated:
 - 1. The enclosure shall provide wire bend space in compliance to the latest version of NFPA70, regardless of the direction from which the conduit enters the enclosure.
 - 2. Exterior cabinet doors shall provide complete protection for the system's internal components. Doors must have permanently mounted key-type latches. Bolted covers or doors are not acceptable.
 - 3. Transfer switches shall be provided in enclosures that are third party certified for their intended environment per NEMA requirements.

2.03 AUTOMATIC TRANSFER SWITCHES

- A. Comply with requirements for Level 1 equipment according to NFPA 110.
- B. Indicated current ratings:
 - 1. Refer to the Project drawings for specifications on the sizes and types of transfer switch equipment, withstand and closing ratings, number of poles, voltage and ampere ratings, enclosure type, and accessories.
 - 2. Main contacts shall be rated for 600 VAC minimum.
 - 3. Transfer switches shall be rated to carry 100% of rated current continuously in the enclosure supplied, in ambient temperatures of -40 to +60 degrees C (-40 to +140 degrees F), relative humidity up to 95% (non-condensing), and altitudes up to 10,000 feet (3000 meters).
- C. Relay Signal: Control shall include provisions for addition of a pre-transfer relay signal, adjustable from 0 to 60 seconds, to be provided if necessary for elevator operation, based on equipment provided for the project.
- D. Transfer switches that are designated on the drawings as 3-pole shall be provided with a neutral bus and lugs. The neutral bus shall be sized to carry 100% of the current designated on the switch rating.
- E. Automatic Transfer Switch Control Features

1. The transfer switch control system shall be configurable in the field for any operating voltage level up to 600 VAC. Voltage sensing shall be monitored based on the normal voltage at the site. Systems that utilize voltage monitoring based on standard voltage conditions that are not field configurable are not acceptable.
 2. All transfer switch sensing shall be configurable from an operator panel or from a Windows PC-based service tool. Designs utilizing DIP switches or other electromechanical devices are not acceptable.
 3. The transfer switch shall provide a relay contact signal prior to transfer or re-transfer. The time period before and after transfer shall be adjustable in a range of 0 to 60 seconds.
 4. The control system shall be designed and prototype tested for operation in ambient temperatures from - 40 degrees C to + 60 degrees C (- 40 to +140 degrees F). It shall be designed and tested to comply with the requirements of the noted voltage and RFI/EMI standards.
 5. The control shall have optically isolated logic inputs, high isolation transformers for AC inputs and relays on all outputs, to provide optimum protection from line voltage surges, RFI and EMI.
 6. The transfer switch network monitoring equipment, when supplied, shall be provided with a battery-based auxiliary power supply to allow monitoring of the transfer switch when both AC power sources are non-operational.
- F. Transfer Switch Control Panel: The transfer switch shall have a microprocessor-based control with a sealed membrane panel incorporating pushbuttons for operator-controlled functions, and LED lamps for system status indicators. The panel shall also include an alphanumeric display for detailed system information. Panel display and indicating lamps shall include permanent labels. The control shall be PowerCommand 40-01 or approved equal.
1. The indicator panel LEDs shall display:
 - a. Which source the load is connected to (Source 1 or Source 2)
 - b. Which source or sources are available
 - c. When switch is not set for automatic operation, the control is disabled
 - d. When the switch is in test/exercise mode
 2. The indicator shall have pushbuttons that allow the operator to activate the following functions:
 - a. Activate pre-programmed test sequence
 - b. Override programmed delays, and immediately go to the next operation
 - c. Reset the control by clearing any faults

- d. Test all of the LEDs by lighting them simultaneously
- 3. The alphanumeric digital display shall be LCD, clearly visible in both bright sunlight and no-light conditions over an angle of 120 degrees, and shall display the following:
 - a. AC voltage for all phases, normal and emergency
 - b. Source status: connected or not connected.
- 4. The display panel shall be password-protected, and allow the operator to view and make adjustments:
 - a. Set nominal voltage and frequency for the transfer switch
 - b. Adjust voltage and frequency sensor operation set points
 - c. Set up time clock functions
 - d. Set up load sequence functions
 - e. Enable or disable control functions including program transition
 - f. View real-time clock data, operation log (hours connected, times transferred, failures) and service history
- G. Control Functions: Functions managed by the control shall include:
 - 1. Software adjustable time delays:
 - a. Engine start (prevents nuisance genset starts in the event of momentary power fluctuation): 0 to 120 seconds (default 3 sec)
 - b. Transfer normal to emergency (allows genset to stabilize before load is transferred): 0 to 120 seconds (default 3 sec)
 - c. Re-transfer emergency to normal (allows utility to stabilize before load is transferred from genset): 0 to 30 minutes (default 3 sec)
 - d. Engine cooldown: 0 to 30 minutes (default 10 min)
 - e. Programmed transition: 0 to 60 seconds (default 3 sec)
 - 2. Voltage imbalance sensing:
 - a. Dropout: 2 to 10% (default 4%)
 - b. Pickup: 90% of dropout
 - c. Time delay: 2.0 to 20 seconds (default 5 sec)

- d. Bar graph meter panel, to display 3-phase AC Amps, 3-phase AC Volts, Hz, KW load level, and load power factor. The display shall be color-coded, with green scale indicating normal or acceptable operating level, yellow indicating conditions nearing a fault, and red indicating operation in excess of rated conditions for the transfer switch.
 - 3. Phase rotation sensing:
 - a. Time delay: 100 msec
 - 4. Loss of single-phase detection:
 - a. Time delay: 100 msec
- H. Control features shall include:
 - 1. Programmable genset exerciser: A field-programmable control shall periodically start and run the generator with or without transferring the load for a preset time period, then re-transfer and shut down the generator after a preset cool-down period.
 - 2. In event of a loss of power to the control, all control settings, real-time clock setting and the engine start-time delay setting will be retained.
 - 3. The system continuously logs information including the number of hours each source has been connected to the load, the number of times transferred, and the total number of times each source has failed. An event recorder stores information, including time and date-stamp, for up to 50 events.
 - 4. Re-Transfer Inhibit Switch: Inhibits automatic re-transfer control so automatic transfer switch will remain connected to emergency power source as long as it is available regardless of condition of normal source.
 - 5. Transfer Inhibit Switch: Inhibits automatic transfer control so automatic transfer switch will remain connected to normal power source regardless of condition of emergency source.
- I. Control Interface
 - 1. Provide one set Form C auxiliary contacts on both sides, operated by transfer switch position, rated 10 amps 250 VAC.
 - 2. Unassigned Auxiliary Contacts: Two normally open, 1-pole, double-throw contacts for each switch position, rated 10A at 240 VAC.
- J. Engine Starting Contacts
 - 1. One isolated and normally closed pair of contacts rated 10A at 32 VDC minimum.

2.04 SERVICE ENTRANCE TRANSFER SWITCHES

- A. Transfer switches must be specifically intended for service entrance applications, and labeled "Suitable for service entrance use only"
- B. Transfer switch shall meet NEC requirements for emergency, legally required and standby applications as specified in UL 1008.
- C. Entire transfer switch including enclosure must be listed and labeled to UL 1008; switches with only the mechanism listed are not acceptable.
- D. Molded case circuit breaker must be UL 489 listed.

2.05 SOURCE QUALITY CONTROL

- A. Prepare test and inspection reports.
 - 1. For each of the tests required by UL 1008, performed on representative devices, for emergency systems. Include results of test for the following conditions:
 - a. Overvoltage.
 - b. Undervoltage.
 - c. Loss of supply voltage.
 - d. Reduction of supply voltage.
 - e. Alternative supply voltage or frequency is at minimum acceptable values.
 - f. Temperature rise.
 - g. Dielectric voltage-withstand; before and after short-circuit test.
 - h. Overload.
 - i. Contact opening.
 - j. Endurance.
 - k. Short circuit.
 - l. Short-time current capability.
 - m. Receptacle withstand capability.
 - n. Insulating base and supports damage.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Design each fastener and support to carry load indicated by seismic requirements and according to seismic-restraint requirements.
- B. Floor-Mounting Switch: Anchor to floor by bolting.
 - 1. Floor-mounted transfer switches shall be mounted on concrete bases complying with the following requirements:
 - a. Concrete Bases: 4 inches (100 mm) high, reinforced, with chamfered edges. Extend base no more than 4 inches (100 mm) in all directions beyond the maximum dimensions of switch, unless otherwise indicated or unless required for seismic support.
- C. Identify components according to Division 26 Section "Identification for Electrical Systems."
- D. Set field-adjustable intervals and delays, relays, and engine exerciser clock.

3.02 CONNECTIONS

- A. Wiring to Remote Components: Match type and number of cables and conductors to generator sets, control, and communication requirements of transfer switches as recommended by manufacturer. Increase raceway sizes at no additional cost to Owner if necessary to accommodate required wiring.
- B. Wiring Method: Install cables in raceways and cable trays except within electrical enclosures. Conceal raceway and cables except in unfinished spaces.
 - 1. Comply with requirements for raceways and boxes specified in Section 260533 "Raceways and Boxes for Electrical Systems."
- C. Wiring within Enclosures: Bundle, lace, and train conductors to terminal points with no excess and without exceeding manufacturer's limitations on bending radii.
- D. Ground equipment according to Section 260526 "Grounding and Bonding for Electrical Systems."
- E. Connect wiring according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables."
- F. Route and brace conductors according to manufacturer's written instructions. Do not obscure manufacturer's markings and labels.

3.03 SOURCE QUALITY CONTROL

- A. Prior to shipping, factory shall test and inspect components, assembled switches, and associated equipment to ensure proper operation.
- B. Factory shall check transfer time and voltage, frequency, and time-delay settings for compliance with specified requirements.

- C. Factory shall perform dielectric strength test complying with NEMA ICS 1.

3.04 FIELD QUALITY CONTROL

- A. Manufacturer's Field Service: The supplier of the transfer switch(es) and associated equipment shall inspect, test, and adjust components, assemblies, and equipment installations, including connections, and report results in writing.
- B. Manufacturer's representative shall perform tests and inspections and prepare test reports.
- C. After installing equipment and after electrical circuitry has been energized, installer shall test for compliance with requirements.
 - 1. Perform recommended installation tests as recommended in manufacturer's installation and service manuals.
 - 2. After energizing circuits, demonstrate interlocking sequence and operational function for each switch.
 - a. Simulate power failures of normal source to automatic transfer switches and of emergency source with normal source available.
 - b. Verify time-delay settings.
 - c. Verify that the transfer switch is accurately metering AC voltage.
 - d. Verify proper sequence and correct timing of automatic engine starting, transfer time delay, retransfer time delay on restoration of normal power, and engine cool-down and shutdown.
- D. Infrared Scanning: After Substantial Completion, but not more than 60 days after Final Acceptance, perform an infrared scan of each switch. Remove all access panels so joints and connections are accessible to portable scanner.
 - 1. Follow-up Infrared Scanning: Perform an additional follow-up infrared scan of each switch 11 months after date of Substantial Completion.
 - 2. Instrument: Use an infrared scanning device designed to measure temperature or to detect significant deviations from normal values. Provide calibration record for device.
 - 3. Record of Infrared Scanning: Prepare a certified report that identifies switches checked and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.05 DEMONSTRATION

- A. After generator set installation, the generator and transfer switch supplier shall conduct a complete operation, basic maintenance, and emergency service seminar covering generator set and transfer switch equipment, for up to 10 people employed by the Owner.

1. The seminar shall include instruction on operation of the transfer equipment, normal testing and exercise, adjustments to the control system, use of the PC based service and maintenance tools provided under this contract, and emergency operation procedures.
2. The class duration shall be at least 8 hours in length, and include practical operation with the installed equipment.

3.06 SERVICE AND SUPPORT

- A. The manufacturer shall supply the Service Provider with a complete set of the service and maintenance software required to support the product. The software shall be provided at a training class attended by the user, to qualify the user in proper use of the software. The software shall have the following features and capabilities:
 1. The software shall allow adjustment of all functions described herein, adjustment of operating levels of all protective functions, and programming of all optional functions in the controller. Adjustments shall be possible over modem from a facility that is remote from the generator set.
 2. The software shall be capable of storing and displaying data for any function monitored by the generator set control. This data shall be available in common file formats, and on graphical "strip chart" displays.
 3. The software shall automatically record all control operations and adjustments performed by any operator or software user, for tracking of changes to the control.
 4. The software shall display all warning, shutdown, and status changes programmed into transfer switch controller. For each event, the control shall provide information on the nature of the event, when it last occurred, and how many times it has occurred.

END OF SECTION

SECTION 26 51 19 – LED LIGHTING

PART 1 GENERAL

1.01 SCOPE

- A. Provide lighting equipment complete and operational as indicated on the plans. All fixtures are to be new, complete with lamps and fully operational in conformance with code and U.L. listing requirements.

1.02 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including Division 01, General Requirements, apply to this Section.

1.03 CATALOG NUMBERS

- A. Model and series numbers indicate design intent in terms of fixture quality and general requirements. The actual fixtures submitted shall correspond with the number of lamps, wattage, ceiling type and any other specific performance and installation requirements specified or shown on drawings. Verify actual requirements prior to ordering fixtures.
- B. Light fixture schedule series numbers are a design series reference and do not necessarily represent the exact catalog number, size, voltage, wattage, type of lamp, ballast, finish trim, ceiling type, mounting hardware, ceiling trim or special requirements as specified hereinafter or as required by the installations. Provide complete fixtures to correspond with the number of lamps, wattage and/or size specified.
- C. If there are discrepancies between fixture illustrations and the written description in the fixture schedule, the written description in the fixture schedule shall take precedence.

1.04 SUBSTITUTIONS

- A. Substitutions will be allowed for fixtures listed on the lighting fixtures schedule as 'or equal', with Engineer approval.
- B. Proposed substitutions shall include complete photometric calculations in plan view showing lighting levels based on proposed fixtures.
- C. Final determination of equivalence shall be by the Engineer.

1.05 REFERENCES

- A. National Electrical Manufacturer's Association (NEMA) LE5-1993: Procedure for determining Luminaire efficiency ratings.
- B. Underwriters Laboratories, Inc. (UL).
- C. Washington State Energy Code, WSEC latest edition, and WAC 51-11C.

1.06 SUBMITTALS

- A. For standard catalog items, provide original product sheets, to indicate that light fixtures and accessories fully comply with contract documents. Include photometric report by an independent certified testing laboratory when required in fixture schedule. Manufacturer's test report is not acceptable.
- B. Submittals shall have fixture types and project name clearly indicated and shall be prepared by the authorized manufacturer's representative serving the project area. A list of manufacturer's representatives (including address, telephone and fax numbers) identifying which light fixture types they represent shall be included with submittals. Submittals or requests for substitutions not meeting these requirements will be rejected.
- C. Product samples complete with housing, trim, plug, and specified lamp shall be submitted upon request.
- D. Submit lighting control devices as required by Washington State Energy Code

1.07 QUALITY ASSURANCE

- A. Fixtures and components shall be new and listed by Underwriters Laboratories (UL) or other testing lab acceptable to local jurisdiction.

1.08 WARRANTY

- A. LED Lamp Technology: Provide manufacturer's warranty for a period of not less than five years including parts and labor for full replacement of defective product.

PART 2 PRODUCTS

2.01 GENERAL MATERIAL REQUIREMENTS

- A. Fixtures shall be free of light leaks and designed to provide sufficient ventilation of lamps to provide the photometric performance required.
- B. All sheet metal work shall be free from tool marks and dents and shall have accurate angles bent as sharp as compatible with the gauges of the required metal. Intersections and joints shall be formed true and of adequate strength and structural rigidity to prevent any distortion after assembly. Finish exposed edges so no sharp or ragged edges are exposed. All miters shall be in accurate alignment with abutting intersecting members.
- C. Lamp holders shall hold lamps securely against normal vibrations and maintenance handling.
- D. Exterior Fixtures
 - 1. Painted surfaces shall have an outdoor life expectancy of not less than 20 years without any visible rust or corrosion.
 - 2. Diffusing materials shall be UV stabilized.
 - 3. Finish colors shall be as specified.

2.02 LIGHTING CONTROLS

- A. Provide lighting controls including switches, contactors, occupancy sensors etc where indicated on the drawings and schedules.
- B. Lighting controls shall be provided to conform with the latest version of WSEC Energy Code requirements in Washington State. Electrical Contractor shall submit Code-compliant materials, fixtures, and control devices.

PART 3 EXECUTION

3.01 LIGHTING FIXTURES, GENERAL

- A. Provide mounting accessories and trims as required for wall and ceiling construction types shown in Finish Schedule and on Drawings. The lighting fixture schedule does not differentiate between grid and flange mounting. The contractor shall review the architectural reflected ceiling plan to confirm locations where flange or other special mounting conditions are applicable.
- B. Verify weight and mounting method of fixtures and provide suitable supports. Fixture mounting assemblies shall comply with local seismic codes and regulations.
- C. Refer to architectural reflected ceiling plans for coordination of lighting fixtures with mechanical and fire safety equipment. Where conflicts occur, coordinate with Architect prior to installing lighting fixtures.
- D. Install lighting fixtures such that vent holes are free of air blocking obstacles.
- E. Recessed lighting fixtures located in ceilings with fire resistive rating of one hour or more shall be enclosed in an approved fire resistive enclosure with a rating equal to that of the ceiling.
- F. Adjust finish trim rings on all recessed fixtures to be flush and level with the finished ceiling.
- G. Adjust variable position lamp holders for proper lamp position.
- H. Replace all burned out or inoperative lamps at the end of the construction prior to Owner occupancy.
- I. Test and demonstrate lighting control system, as applicable.

*** END OF SECTION ***

SECTION 26 80 00 – CONTROL SYSTEM

PART 1 GENERAL

1.01 SUMMARY

- A. This Section covers includes the general requirements for furnishing, installing, programming, adjusting, testing, documenting and commissioning of the complete and operational Instrumentation, SCADA and PLC based control system.
- B. The control system includes the complete instrumentation, PLC's and control panels, Motor Control Center, operator interface, and SCADA system and related equipment for complete monitoring and control of the entire facility treatment system.
- C. Major components of this system include, but are not limited to, all materials, equipment, and work required to implement a complete and operating system as described herein. The system shall include primary elements for process variable measurements, control elements, analog displays, communication systems, and all hardware and software required to program, calibrate and monitor the instrumentation, communication and control devices.
- D. The Contractor shall provide, program, calibrate, test and commission the complete and operational Instrumentation, SCADA and PLC based control system.
- E. The Contractor shall place the completed system in operation, including programming, tuning loops, testing and adjusting communications and making final adjustments to instruments and equipment as required during system start-up. The Contractor shall provide the services of trained and qualified instrument technicians for these services.

1.02 RELATED SECTIONS

- A. 26 05 00 – Electrical General
- B. 26 29 23– Variable Frequency Drives
- C. 26 24 19 – Motor Control Centers

1.03 DEFINITION OF TERMS

- A. General – Refer to Standard abbreviations, Section 26 05 00.3.7.
- B. PLC – Programmable Logic Controller. Industrial computer system that interfaces with instrumentation and equipment and allows for automated control of process equipment.
- C. OIU – Operator Interface Unit. Local display to display graphical status of instrumentation and equipment status, alarm monitoring, set point control and visualization.
- D. SCADA – Supervisory Control And Data Acquisition computer system. Computer based system with custom software to allow for supervisory monitoring and control, data logging, alarm monitoring and reporting for the overall facility control and treatment system.

- E. Contractor: The party who furnishes and installs all tools, materials, and equipment to complete the work shown and implied in the drawings and these specifications. This includes the Prime Contractor, the Electrical Contractor, System Integrator, Telecom Contractor, and all other Contractors and Subcontractors.
- F. System Integrator: An organization engaged in the business of detail design, component selection and procurement, fabrication, wiring, assembly, programming and testing process control and telemetry systems. The System Integrator will provide the programming, licensing and software development for the PLC and SCADA process control system for this project.
- G. Telecom Contractor: The telecommunications company responsible for all data cabling equipment and testing. This includes all communications cabling shown on the plans. Telecom contractor shall be selected by the Contractor.
- H. Equipment Supplier: The equipment manufacturing company of packaged treatment or other specialty systems and components, furnished by the Equipment Supplier and installed by the Contractor, responsible for providing all materials, equipment, and testing for a fully operational system.

1.04 SPECIAL REQUIREMENTS

- A. The Contractor shall install components including those assembled by the System Integrator at the locations shown in the plans, and in accordance with the requirements of Division 26.
- B. The Contractor shall be responsible for the selection of the System Integrator, and shall be subject to approval by the Engineer. It is recommended that the System Integrator be a control system manufacturing company that conforms to the following requirements:
 - 1. The System Integrator shall be specialized in the design, assembly, programming, testing, installation and service of municipal control and communication systems in the Pacific Northwest for at least the last five years.
 - 2. The System Integrator shall employ technicians and engineers with documented experience in the design, assembly, programming, testing, installation, operation, calibration, troubleshooting, service and repair of control and communication systems for municipal systems and facilities.
 - 3. The System Integrator shall have completed the design, assembly, testing and installation of control systems which include the instruments and devices cited on the Plans by specific manufacturer's name.
 - 4. The System Integrator shall be experienced with the programming and commissioning of Allen Bradley PLC systems, operator interfaces, communications systems and VFD's.
 - 5. The System Integrator shall be an AVEVA registered System Integrator with experience in municipal SCADA system development and shall maintain the required development licenses to develop the project and provide future support

for the project. The System Integrator shall also be experienced with WIN-911 alarm software development.

6. The System Integrators manufacturing and testing facility shall be located within a 150 mile drive from the project location.
 7. The System Integrator shall be a UL508A listed and certified control panel manufacturing facility.
- C. The Contractor shall be responsible for the selection of the System Integrator. Integrators not listed below shall be subject to written approval by the Engineer prior to bid. Pre-approved System Integrators are:
1. Systems Interface Inc, Bothell WA.
 2. Quality Controls Corporation, Lynnwood WA.
 3. Technical Systems Inc, Lynnwood WA.
 4. Process Solutions Inc, Arlington, WA
 5. Taurus Controls, Kent, WA

1.05 DIVISION OF RESPONSIBILITY

- A. All materials and modifications to the existing control system shall be provided under the supervision of a single Contractor, which is regularly engaged in the design and installation of such systems of similar scope and complexity.
- B. The Contractor shall be fully and completely responsible for all work performed and all materials installed under the contract. The contract between the Contractor and subcontractor(s) shall conform to and meet all requirements specified in the contract documents.
- C. Electrical Contractor's Responsibilities:

The Electrical Contractor shall be responsible for the following:

1. Installation of all control system equipment in accordance with these documents, and drawings provided by the Control System Integrator.
2. Provide all electrical permits and inspections as required by the State of Washington.
3. Provide all connections to new and existing electrical equipment including MCC, VFD's, PLC's, instrumentation, control devices, packaged equipment, power generation equipment and other equipment as shown on the drawings.
4. Provide electrical panel modifications, demolition of existing equipment, raceway and other equipment and installation where required.

5. Coordination with the Owner for equipment installation, special operational needs for the facility, and facility scheduling requirements per 260500.1.016.
6. Provide temporary construction power and temporary equipment as listed in 260500.1.015.

D. System Integrator Responsibilities:

The System Integrator shall be responsible for the following:

1. Provide Motor Control Centers, VFD's, Control Panels, PLC and operator interface systems, instrumentation, communications, and control system equipment as shown on the drawings.
2. Provide complete SCADA computer system, per specifications.
3. Provide all software licenses and software development including all system programming including PLC's, instrumentation, VFD's, Operator Interfaces and SCADA system.
4. Provide complete system startup and commissioning, configure, program, test and commission the complete Control System and other process control equipment in coordination with Electrical Contractor, Owner and Engineer.
5. Provide additional system accessories as required for a complete and operational control system.
6. Provide system test forms to identify each test performed and document the completed testing procedures and sign off on operation with Owner and Engineer.
7. Provide final documentation, training, as-built drawings and Operation and Maintenance manuals.

E. Telecomm Contractor's Responsibility:

The Telecomm Contractor shall be responsible for the following equipment and services:

1. Selection of the networking system cabling, patch panels, patch cables, and other equipment as shown on the drawings, and as required for a complete and operational system.
2. Submittal of communications equipment hardware and topology diagrams for review by Engineer. Submittal shall be approved prior to ordering or installation of communications equipment.
3. Installation of communications system cabling, patch panels, patch cables, and other equipment.
4. Testing, certification and commissioning of the complete communications system.
5. Provide the copper cat 6 system cabling, color coordinated wall jacks and cabling and terminations from wall data jacks to data racks and panels.

6. Testing, certification and commissioning of the complete data communications system.

F. Equipment Supplier Responsibilities

The Equipment Supplier shall be responsible for the following:

1. Furnish all packaged equipment and installation requirements for Electrical Contractor installation.
2. Fully program, commission, adjust, test and put the new equipment into operation.
3. The Equipment Supplier shall address PLCs with IP addresses as specified by the Programmer.
4. The Equipment Supplier shall organize and allocate specific memory locations and PLC tags, as required, for complete monitoring of the equipment by the treatment facility PLC and SCADA system. These listings shall be coordinated with and provided to the Programmer prior to startup and commissioning.
5. The Equipment Supplier shall provide UL or ETL certifications for all equipment and equipment packages, as required by the State of Washington, L&I Electrical Inspection division. Field inspections for unlabeled equipment, including all costs and resulting project delays, shall be the complete responsibility of the Contractor.
6. All final programs shall be provided to the Owner with all passwords given to the owner to access and maintain the programs.

1.06 SUBMITTALS

A. Hardware Submittals

1. In addition to the requirements stated elsewhere in these documents, the following information shall be provided:
 - a. Before any components are fabricated, and/or integrated into assemblies, or shipped to the site, the System Integrator shall prepare a complete hardware submittal, including fully detailed shop drawings, catalog cuts, wiring connections, and such other documentation as may be required to fully describe the equipment and to demonstrate its conformity to these plans and specifications. Catalog information shall be submitted for all components and equipment required for the project.
 - b. All submittals shall be complete, organized, and indexed. Partial submittals will not be accepted.

B. System Drawing Submittals

1. Following approval of the Hardware Submittal, the System Integrator shall prepare complete system interconnect wiring diagrams and panel layout drawings for approval.

1.07 OPERATION AND MAINTENANCE MANUALS

- A. Provide Operation and Maintenance (O&M) data for the complete control system and related equipment, in accordance with the general requirements in Section 26 05 00.

PART 2 PRODUCTS

2.01 GENERAL

- A. Material shall be new, free from defects, and of the quality specified. All equipment and materials utilized in the system shall be the products of manufacturers with at least five (5) years experience in the manufacture of similar equipment. Similar items in the system shall be the products of the same manufacturer. All equipment shall be of industrial grade and shall be specifically intended for control, monitoring and operation of motor-driven pumps and equipment. All equipment shall be of modular design to facilitate interchangeability of parts and to assure ease of servicing.

2.02 MAJOR EQUIPMENT LIST

- A. The Contractor shall provide all of the equipment shown on the drawings, including but not limited to, the following major Control System equipment:
 - 1. Motor Control Center (MCC) with power distribution, motors controllers, VFDs, and other features as shown on the drawings.
 - 2. PLC control panels as shown on the drawings. Contractor shall provide detailed design drawings, bill of materials and data sheets for complete control system.
 - 3. Local and remote control panels, as shown on the drawings.
 - a. Main PLC Control Panel with Operator Interface
 - b. Influent Control Panel
 - c. Effluent Flow Meter Panel
 - d. Ventilation Control Panel
 - 4. Instrumentation as shown on the drawings, and specified herein:
 - a. Float Switches:
 - 1) Four floats at 2W pump station
 - 2) Four floats at Influent pump station
 - b. Level Transducers
 - 1) One transducer at Digester Basin

- 2) One transducer at Influent Pump Station
- c. Pressure Transducers
 - 1) One transducer for 2W pressure sensing
- d. Flow Meters
 - 1) INFLUENT FLOW METER. Size: 6". Install in vault with submersible kit. Transmitter located in influent control panel.
 - 2) TRAIN #1 FLOW METER. Size: 8". (future project - not in this project). Provide space for future transmitter in influent control panel.
 - 3) TRAIN #2 FLOW METER. Size: 8". Install in vault with submersible kit. Transmitter located in influent control panel.
 - 4) TRAIN #3 FLOW METER. Size: 8". Install in vault with submersible kit. Transmitter located in influent control panel.
 - 5) PLANT DRAIN FLOW METER. Size: 4". Install in vault with submersible kit. Transmitter located in influent control panel.
 - 6) EFFLUENT FLOW METER. Size: 8". Install in vault with submersible kit. Transmitter located in panel near UV basin. Provide NEMA 4X enclosure for transmitter.
 - 7) WAS FLOW METER. Size: 2". Install on piping near WAS pumps, mount transmitter to flow tube.
 - 8) 2W FLOW METER. Size: 1". Install on piping in hot box #1, mount transmitter to flow tube.
 - 9) Enclosures: Install flow meter transmitters as shown on the drawings. Provide enclosures as shown.
- e. Dissolved Oxygen (DO) Meters
 - 1) Two complete DO sensing systems, one in each Aeration zone of Treatment cell #3.
- f. H2S Sensor
 - 1) One sensor installed near new Dewatering equipment.
- g. Methane Sensor
 - 1) One sensor installed near new Dewatering equipment.
- h. Door Switch

- 1) One switch for Blower building main entrance.
- i. Smoke detector
 - 1) One heat/smoke detector in Blower building.
- j. Solenoid Valves
 - 1) Coordinate to provide solenoids for 2W and WAS control systems, per the drawings and specifications
5. Network equipment, cabling, accessories, and testing.
 - a. Network switches in Main Control Panel, Motor Control Center, Influent Control Panel and at SCADA operator station in Lab.
6. SCADA system hardware and software, as specified.
7. Installation of all packaged equipment system panels, equipment, and instrumentation per manufacturer drawings.
8. Programming and commissioning of all PLC, operator interface and SCADA systems.

2.03 CONTROL PANELS

- A. Control panels shall be designed, assembled, programmed, tested and placed into operation by the System Integrator with support from the Electrical Contractor.
 - B. The control panel shall fit into the space requirements as shown on the drawings. The contract drawings show general control panel layout and space requirements. Final dimensions shall be selected by the System Integrator to adequately install and wire the required control equipment.
 - C. Detailed panel layout and interconnecting drawings shall be submitted prior to ordering of materials, and shall be subject to review and approval by the Engineer.
 - D. Material shall be new, free from defects, and of the quality specified. Similar items in the system shall be the products of the same Manufacturer. All equipment shall be of industrial grade and of standard construction, shall be capable of long, reliable, trouble-free service, and shall be specifically intended for control and monitoring industrial equipment.
1. Indoor Control Panels
 - a. Cabinet shall be a NEMA 12 enclosure, with back panel. Cabinet shall be fabricated from 16 ga. minimum thickness sheet steel, and shall be ANSI 61 gray standard phosphate finish. Panel interiors and back panels shall be white. Cabinet shall be provided with an interior frame or otherwise formed so as to provide a rigid structure. Three-point latch hardware shall be provided for doors exceeding 30 inches high. Hoffman Concept series, or equal.

- b. The panel shall include padlocking quick release L-handles to allow the panel to be opened without the use of tools. Hoffman, Hammond, or equal.

2. Outdoor Control Panels

- a. Cabinet shall comply with NEMA 4X requirements as shown on the drawings and to meet the control equipment manufacturers' environmental requirements. The panel shall be fabricated from Type 316 Stainless Steel, and shall include a hidden hinge and back panel for equipment mounting. Hoffman, Hammond, or equal.
- b. Panels that are mounted outdoors shall be provided with a sun shield, or with a ventilation/cooling system to maintain the internal temperature of the panel interior within the equipment ratings at typical ambient temperatures for the installation conditions. Drip shields shall be installed to prevent water buildup on top of panel doors. Panel fans and/or ventilation systems shall be thermostatically controlled.
- c. The panel shall include thru the door main disconnect and padlocking quick release L-handles to allow the panel to be opened without the use of tools. Hoffman #ELHP, or equal.

E. Operating and Indicating Devices

- 1. Operating and indicating devices minimum rating shall be NEMA 13. Operator devices mounted in outdoor panels, corrosive areas or where exposed to moisture shall be NEMA 4X.
- 2. Selector Switches
 - a. Selector switches shall be for use on 120 volt control circuits. Contacts shall have a continuous current rating of 10 amperes both inductive and resistive. Selector switches shall be of the heavy duty oil tight type. Allen Bradley, Bulletin 800T, 800H, or equal.
- 3. Push buttons
 - a. Push buttons and illuminated push buttons shall be for use on 120 volt and 24 VDC control circuits and shall have continuous current rating of 10 amperes both inductive and resistive. Pushbuttons for "E-STOP" or emergency applications shall have maintained contacts and red mushroom head operators. Allen Bradley, Bulletin 800T, 800H, or equal.
- 4. Indicating Lights
 - a. Indicating lights shall be push-to-test oil tight type. Units shall have LED lamps and shall be of the illuminated pushbutton type with the pushbutton wired for the push-to-test function required. Appropriate LED and lens color shall be provided as shown. Allen Bradley, Bulletin 800T, 800H, or equal.

5. Analog Indicator

- a. Indicator shall have LED process display with 68 x 33 mm minimum dimensions and 0.48" backlit display to indicate wet well level. 24 VDC powered. 4-20 mA process input.
- b. Resolution shall be displayed to one tenth of one foot XX.X ft.
- c. Display shall be Red Lion CUB 4 series, or Engineer approved equal.

6. Control Relays

- a. Relays for general purpose use shall be DPDT, 5 ampere minimum contacts with the appropriate coil voltage for the application. They shall have an 8-pin base, matching socket, and contact status indicator. All relays shall include MOV snubbers (for AC) or diodes (for DC) applied across the relay coils to reduce the surge caused by coil breakdown transients. Relays shall be Idec RH2B-ULD, or equal.

7. Terminal Blocks

- a. Terminal blocks shall be 600 volt modular terminal blocks with tubular screw and pressure plate. Provide a minimum of 20% or four whichever is greater, spare terminals in each panel. Allen-Bradley #1492-J3 series, or equal.

8. Seal Fail and Thermal Relays

- a. Pump seal fail and thermal relays shall be installed for each submersible pump in the system, where required by the pump manufacturer. The relay shall be capable of connecting to a seal failure probe and thermal contacts in the pump housing, and shall have output contacts to energize external indicator lights.
- b. It is the responsibility of the pump control panel manufacturer to select and install the relays with the pump equipment provider for compatibility.
- c. Relays shall be Mini-CAS, or engineer and manufacturer approved equal.

2.04 INSTRUMENTATION

A. Float Switch

- 1. Switch shall be free floating, direct acting float switch designed for operation in raw sewage.
- 2. Mounting hardware shall include fixed installation on a 1" pipe, or suspended with a Kellems cord grip/strain relief and a weighted stainless steel support cable.
- 3. The float cable shall be a PVC coated multicore connecting cable which also contains the conductors, and shall be UL listed.

4. Float shall contain a non-mercury switch with a minimum rating of 4 amps at 120 Volts.
5. Float shall be foam-filled, hermetically sealed and polypropylene coated.
6. Floats shall be supplied with cable of sufficient length to reach the junction box without splices.
7. Intermediate relays and intrinsic safety barriers shall be provided for all wet well instrumentation in accordance with NFPA 820 and NEC article 500 for Class I, Div 1&2 areas.
8. Level switches shall be Orenco, MF series or Anchor Scientific Inc. 'Eco-Float' Model G. Include cable mounting kit and/or float stem and all required accessories, or equal.

B. Submersible Level Transducer for Waste Water applications

1. The level sensor shall be a submersible pressure transducer with flush diaphragm designed for raw waste water and slurry applications.
2. It shall utilize a piezoresistive silicon strain gauge transduction principle with an accuracy of $\pm 0.25\%$. The sensor housing and internal components shall be titanium, and the diaphragm shall be Teflon coated rubber.
3. The sensor shall be selected to provide sensing over the full range of levels in the wet well for the application.
4. The sensor shall include a PVC support structure, stainless steel mounting hardware, and all manufacturer recommended hardware for mounting in the wet well.
5. The sensor shall be FM approved for installation in Class I, Div 1 Hazardous areas, and shall include an Intrinsic Safety Barrier in the Pump Control Panel, and include a 2-wire 4-20ma output compatible with 9-32Vdc excitation.
6. Provide manufacturer dessicant system for vent tube to prevent moisture buildup in the transducer vent line.
7. Level Sensor shall be KPSI Esterline model 750 for raw wastewater, or Engineer approved equal.

C. Pressure Transducer

1. Pressure Transducers shall be of the piezo-resistive bridge design, intended for the purpose of measuring water pressure. The bonded resistance shall provide a linear response proportional to the pressure. The accuracy of the transducer shall be within 0.25% of full scale. Transducer housing and sensor shall be constructed of 316 stainless steel. The transducer shall secure its power from the current loop supply. The system shall be operate on a supply rated from 10-36 VDC, and have a power consumption of no more than 20 watts each.

2. Each transducer shall be wet-calibrated at the manufacturer's facility against the master system. A calibration certificate shall be furnished for each transducer.
3. Pipe mounting hardware, isolation valves, and other manufacturer recommended options shall be included.
4. Pressure ranges shall be coordinated with the pump supplier and submitted for Engineer review.
5. Pressure transducers shall be ASCO Model G2, G27M0242E0200##+CavG, or Engineer approved equal.

D. Electromagnetic Flow Meter

1. Provide and install the magnetic flow meters as shown on the Contract Drawings and specified herein. The flow meters shall be complete with all necessary accessories and hardware for a complete and workable installation.
2. Flow meters shall be for measurement of raw liquid wastewater and sludge.
3. General: The magnetic flow meters shall be of the low frequency and short form coil design. The field principle of electromagnetic induction shall produce a positive DC pulsed signal directly and linearly proportional to the liquid flow rate. The metering tube shall be constructed of carbon steel with ANSI flanged end connections. Electrodes can either be protruding (bullet nose), but shall be of 316 stainless steel construction. The material of construction of the liner shall be hard rubber. The meter shall secure its power from the signal converter. No electronics shall be mounted in the metering tube of the magnetic flow meter.
4. The signal converters shall be integral to the flow head for indoor installations. Signal converters shall be remote mount for meters installed in vaults or where subject to submergence. The flow meter shall be capable of accidental submergence to 30 feet for a period of 24 hours.
5. The signal converters shall convert the output signal from the flow meters into a 4-20 mA signal directly proportional to flow rate. The signal converters shall have automatic zero correction. The accuracy shall be ± 0.5 percent of the actual flow rate. The signal converters shall transmit an isolated analog 4-20 mA D-C signal, directly proportional to flow and a totalizer contact for remote flow totalization. The signal converters shall be designed to operate from a 120 ac, 60 Hz, single phase, power source, or 24VDC as shown on the drawings. The signal converters shall generate power for the magmeter.
6. The magnetic flow meter enclosure shall be NEMA 4X classified, and rated for Class I, Div.2 installation. The units shall be labeled and listed by a recognized electrical testing laboratory for the application, or Approved by the Washington State Department of Labor and Industries for installation on the Project.
7. Each meter system shall be wet-calibrated at the manufacturer's facility against the master system. A calibration certificate shall be furnished for each meter.

8. Type 316 Stainless Steel grounding rings shall be provided, on both sides of the flow meter, per manufacturer's recommendations. Grounding rings shall be provided on both non-metallic and cement lined piping. The Contractor is fully responsible for providing the appropriate grounding rings for the fittings attached to the flow meter. Confirm required materials with manufacturer(s) and submit detailed grounding ring information showing compatibility with flow meter flanges and pipe fittings used for each flow meter location.
9. Flow meter quantities and sizes shall be per the Civil Engineering sheets and corresponding pipe sizes.
10. The magnetic flow meters and signal converters shall be Endress-Hauser, Proline Promag L400 series, with Ethernet communications.

E. Dissolved Oxygen Monitoring system

1. Provide complete Dissolved Oxygen (DO) monitoring systems to be installed in the treatment cell aeration zones. Qty: 2 systems per cell.
2. Probe: Hach LDO series 0-20 PPM range, 10 meter cable with quick disconnect plug
3. Controller: Hach SC200 controller, 0-20 PPM range, 4-20 mA output, 24VDC power.
4. Pole mount kit with service bracket and all required accessories for installation on treatment cell handrail.

F. Hydrogen Sulfide Sensor

1. Hydrogen Sulfide Sensors shall be of the Electrochemical type with a range of 0-100 PPM, accuracy of +/- 1 PPM and a resolution of 0.1 PPM. 120VAC power.
2. Sensor housing shall be 316 stainless steel and rated for the environment it is to be installed in.
3. Hydrogen Sulfide Sensors shall be Sierra Monitor Corporation Model #5100-05-IT, or Engineer approved equal.

G. Combustible Gas Detector (for Methane detection)

1. Combustible Gas Sensors shall provide a range of 0-100 % lower explosive limit (LEL). Alarm Limit = 10% LEL. 120VAC power.
2. Sensor housing shall be 316 stainless steel and rated for the environment it is to be installed in.
3. Combustible Gas Sensors shall be Detcon Model FP-524D, or Engineer approved equal.

H. Smoke Detector

1. Combination smoke and heat detector.
 2. System sensor model #4WTAR-B with form C relay, sounder, 4 wire, or approved equal.
- I. Door Switch
1. Provide door switch for intrusion sensing.
 2. Magnetic reed style, SPDT.
 3. Interlogix Sentrol Model 2807T, or approved equal.

2.05 PROGRAMMABLE LOGIC CONTROLLER AND SUPPORT COMPONENTS

A. PLC hardware

1. A complete PLC and communication system shall be provided with the logic and communications capabilities as shown on the drawings and in these specifications.
2. All PLC hardware shall be manufactured by Allen Bradley, no exceptions.
3. General selection as specified herein. Specific quantities and additional requirements as determined by System Integrator.
4. PLC Platforms:
 - a. CompactLogix:
 - 1) Processor: Model 1769-L33ER
 - 2) Communications: Integral Dual port Ethernet/IP
 - b. Power supply: 1769 series, sized to accommodate all hardware.
 - c. Hardware Chassis: Integral
 - d. Input and Output cards: 1769 series for analog and discrete inputs and outputs. Point count to include a minimum of 20% spare for each card type used.
5. Power Monitors
 - a. Allen Bradley Power Monitor 5000.

B. Programming of PLC and Operator Interface

1. The System Integrator shall provide all programming and functional testing for the entire control system according to the descriptions in this specification and features shown on the drawings and place the complete system into operation.

2. The specifications indicate minimum system requirements. The programmer shall provide all required programming for a complete and functional control system.
3. Field adjustments or additional features added during startup and commissioning shall be included in the bid price.
4. The Contractor shall provide field technicians to assist with the startup and functional checkout of the complete PLC and control system.
5. All final programs shall become the property of the owner and shall be provided on USB drive with full access rights.

C. Operator Interface Unit (OIU):

1. One operator interface unit shall be provided for the Main PLC control panel in the Blower building.
2. The operator interface unit(s) shall be a 9 inch wide aspect color touch panel with Ethernet communications and 24VDC powered.
3. Operator interface shall run Windows CE Pro operating system and shall provide for real time monitoring of the terminal displays from a web browser.
4. Provide all required cables and pre-loaded development software and licenses for a complete and operational system.
5. System Integrator shall program, test and commission the OIU in accordance with these specifications.
6. Manufacturer: Allen-Bradley Panelview Plus 7 performance terminal, #2711P-T9W22D9P, or latest approved model at the time of submittal.

D. Network switches

1. Provide network switches as required for the connection of all Ethernet enabled devices.
2. Provide switches in control panels, MCCs, and other areas as required for a complete Ethernet communication system.
3. Switches shall be Allen Bradley Stratix 5700, 8 or 16 x 10/100 Mbps copper ports, 2 or 4x 10/100/1000 Mbps, DLR, 24-48VDC. Port quantities as required.
4. Network shall be wired in a Ring configuration to provide network redundancy.

E. DC Uninterruptible Power Supply (UPS)

1. DC UPS systems shall be provided for all PLC control panels.
2. A complete DC UPS system shall be supplied for each control panel as shown on the drawings, and shall be manufactured by Allen Bradley.

3. DC systems shall include DC power supply, Allen Bradley 1606-XLE240EN with 1606-XLS240-UPS and 1606-XLSBATASSY2 26 Ah battery assembly. Include all required mounting brackets and accessories.

F. AC Power Line Filter

1. Provide Active Tracking Line Filter in each control panel with 120 VAC input.
2. Manufacturer: Islatrol model IE-120, or Engineer approved equal.

G. PLC Programming Software

1. PLC and OIU programming software shall be provided by the System Integrator for use in developing the software applications for the project, and shall be a current and registered Rockwell Automation programmer with software development licenses and support.
2. Electronic copies of all final PLC and OIU programs will be provided to the Owner at the completion of the project with full ownership rights and all password protection removed or registered to the Owner.
3. The System Integrator may offer the Owner ongoing future technical support following final commissioning of the project, and shall be subject to Owners approval. The Owner shall, at any time after final commissioning, have the ability to hire another registered Rockwell Automation registered integrator to support or modify the existing programs.

2.01 SCADA COMPUTER SYSTEM AND EQUIPMENT

- A. A complete SCADA computer system shall be provided by the Contractor. Computers shall be Dell or equal, with the following minimum requirements:

1. Intel Core i9, 16M Cache, 8 core, 5.2 GHz or latest premium processor
2. NVIDIA GE Force, 12 GB GDDR6 or equal
3. (2) 8 GB DDR3 SDRAM 1600 MHz, 2 DIMMs
4. 2 TB SSHD
5. 16x CD/DVD burner drive
6. Energy efficient power supply
7. Ethernet network adapter, gigabit
8. 27" UHD flat panel display DVI with sound bar
9. Wireless optical mouse and Wireless keyboard
10. 750VA UPS w/surge protection and power strip

11. Local 8 port desktop network switch
12. Short throw projector (to project SCADA screen on wall): Epson Ultra LS300 or equal.
13. Windows 10 Professional operating system, 64 bit
14. Windows Office Professional, latest edition
15. 3 year hardware warranty

B. SCADA Software and Licensing

1. The SCADA software shall be AVEVA (previously Wonderware) and provided by the System Integrator through Industrial Software Solutions, Bothell WA. The local representative is Rick Loya, Sr. Account Development Manager, 425.829.9420.
2. The System Integrator shall be an AVEVA registered Integrator, Certified Integrator preferred.
3. The System Integrator shall maintain all licensing and certification for AVEVA development capacity, including Development Studio.
4. All AVEVA licensing for the SCADA system shall be based on the Annual Subscription model. Software shall be registered in the name of the Owner. The System Integrator shall provide a 3-year subscription for the Owner for all AVEVA software, and the term shall begin when the system is fully tested and commissioned and handed over to the Owner. The Owner shall be responsible for future Annual support beginning at the end of the 3 year term.
5. SCADA software shall include:
 - a. AVEVA InTouch Runtime 500 Tag with I/O
 - 1) Provided for local visualization, monitoring and control
 - 2) Includes AVEVA communications drivers
 - 3) Includes AVEVA Customer First technical software support
 - 4) Includes 3 year annual licensing subscription
 - b. AVEVA Insight Standard with 1 Year History
 - 1) Provided for Cloud-based historical data monitoring and reporting tools.
 - 2) 10,000 data points at 5 second data rate
 - 3) 2 Standard Users included

- 4) 1 Additional Standard User Provided, 3 total users
- 5) Includes AVEVA Customer First technical software support
- 6) Includes 3 year annual licensing subscription
- c. WIN 911 Interactive Alarm Monitoring Software
 - 1) 2 way communication with all notifiers: SMS, email, Voice, and 10 smartphone apps
 - 2) Includes WIN-911 Customer Care technical software support for first year

C. Alarm Monitoring Software and Licensing

- 1. The alarm monitoring software shall be WIN-911 Mobile.
- 2. The alarm software shall reside on the SCADA computer and use the Internet connection to send alarms. The web notifications are encrypted with HTTPS and all connections are outbound initiated, so WIN 911 Mobile is deployed without opening a port on the firewall or introducing threats to the local network. Each operator can run the app on their smartphone and it is compatible with both IOS and Android devices.
- 3. The alarm software licensing shall be a 1 year perpetual license provided by the System Integrator. Software shall be registered in the name of the Owner. The term shall begin when the system is fully tested and commissioned and handed over to the Owner. The Owner shall be responsible for annual WIN-911 support if required, beginning at the end of the first year after commissioning.
- 4. Key features of the WIN 911 Mobile software shall include:
 - a. Configurable Push Notifications:
 - 1) Ringtone selection
 - 2) Alert operator to alarm, report, and chat activity
 - 3) Acknowledge alarm with single click
 - 4) Supports smartwatch notification
 - b. Instant Insight into Alarms:
 - 1) See alarm changes in real time
 - 2) Current alarm states on single screen
 - 3) Multi-selectable alarm acknowledgements
 - c. Alarm Details:

- 1) Gives up to date alarm details and history
 - 2) Configurable properties from within the app
 - 3) Acknowledge alarms with comments
- d. Collaborative chat:
- 1) Gives operators an easy way to collaborate, comment, and resolve problems faster
 - 2) Each alarm instance has its own chat room
- e. Reports:
- 1) Provides the details needed to make informed decisions
 - 2) Display real time process values and alarm states
 - 3) Reports automatically sync with WIN 911
 - 4) Push reports on specific alarm conditions
- f. Team Visibility:
- 1) Displays who has and has not seen the alarm
 - 2) Shows who has been sent the alarm
 - 3) Shows who is in 'do not disturb' mode

2.02 SPARE PARTS

- A. In addition to spare parts mentioned elsewhere in this section, the Contractor shall supply the following spare parts:
1. 4 Spare LED indicators and one cap of each color used for indicating lights.
 2. One spare control, time delay, phase fail, seal fail, etc. relay of each type used, or 20% whichever is the greater number.
 3. Two spare fuses for each fuse provided under 10 amperes and one spare fuse for each fuse provided over 10 amperes.

PART 3 EXECUTION

3.01 GENERAL

- A. Install materials and equipment in a workman-like manner utilizing craftsmen skilled in the particular trade. Provide work which has a neat and finished appearance.

- B. Coordinate Instrumentation and Control work with the Contractor, Owner, and work of other trades to avoid conflicts, errors, delays and unnecessary interference with system operations during construction.

3.02 COORDINATION WITH CONTRACTOR

- A. The System Integrator shall coordinate directly with the Contractor to ensure all requirements within the scope of this Section are satisfied.

3.03 SYSTEM SIMULATION

- A. To the degree possible, the entire control system shall be simulated at the Contractors facility prior to commissioning.
- B. The Engineer, Contractor and Owner personnel shall be invited to witness simulation and approve test results prior to shipment to Contractor.

3.04 FIELD ACCEPTANCE TEST

- A. The System Integrator shall conduct acceptance tests and provide operator training for the complete and functional control system.

3.05 PROTECTION DURING CONSTRUCTION

- A. The Contractor shall provide protection for materials and equipment against loss or damage and the effects of weather.
- B. Prior to installation, store items in an indoor, dry location. Provide heating in storage areas for items subject to corrosion under damp conditions.
- C. Specific storage requirements shall be in accordance with the manufacturer's recommendations.

3.06 MATERIAL AND EQUIPMENT INSTALLATION

- A. Follow manufacturer's installation requirements, unless otherwise indicated. Wherever any conflict arises between manufacturer's instructions and these Contract Documents, follow Engineer's decision, at no additional cost to the Owner. Maintain a copy of manufacturer's installation instructions on the jobsite at all times.

3.07 INSTRUMENTATION INSTALLATION

- A. Process Connections
 - 1. Unless otherwise specified, process taps shall comply with API RP550. Root valves shall be provided at taps, except temperature taps and pump discharge pressure taps. Process connections shall be arranged, where possible, such that instruments may be readily removed for maintenance without disruption of process units or draining of large tanks or vessels. Unions or flange connections shall be provided as necessary to permit removal without rotating equipment. Where process taps are not readily accessible from instrument locations, a block valve

shall be provided at the instrument. Block valves shall also be provided for each instrument where multiple instruments are connected to one process tap.

B. Electrical Connections

1. Final connections between rigid raceway systems and instruments shall be made with jacketed flexible conduit with a maximum length of two feet.

C. Identification

1. Tagging

- a. All field instruments shall be labeled with function and instrument number, i.e. (FIT-301/EFFLUENT FLOW METER). Tag shall be 10ga 316 stainless steel with stamped letters and numbers attached to device with 12ga 316 stainless steel wire.

3.08 CONTROL SYSTEM PROGRAMMING, STARTUP AND TESTING

- A. The Control Panel Manufacturer shall provide all programming and functional testing for the PLC and control system and place the complete system into operation.
- B. The Control System shall be programmed and operate according to the descriptions in this specification and features shown on the drawings. This description indicates minimum system requirements. The Control Panel Manufacturer shall provide all required coordination for a complete and functional control system. Field adjustments or additional features added during startup and commissioning shall be included in the bid price.
- C. The control system shall be put into operation by the System Integrator with support of the Electrical Contractor.
- D. A witnessed Functional Acceptance Test shall be performed on the completed control system.
 1. The System Integrators standard test forms shall be submitted for review by the Engineer, including complete test procedure and forms.
 2. Each feature and function of the control system shall be documented on the test forms and shall be tested and demonstrated to the satisfaction of the Engineer. The System Integrator shall make all revisions to the control system operation noted by the Engineer prior to final acceptance.
 3. The testing program shall be conducted in accordance with the prior approved procedures and shall be witnessed and signed off by both the System Integrator and the Engineer upon satisfactory completion.
- E. All special testing materials and equipment required to demonstrate compliance with the specification shall be provided by under the scope of this Section. Where it is not practical to test with real process variables, provide suitable means of simulation. These simulation techniques shall be subject to the approval of the Engineer.

- F. Coordinate all testing with other associated suppliers and subcontractors.
- G. All final software programs shall become the property of the Owner and shall be provided on USB flash drive with full access rights.

3.09 DESCRIPTION OF OPERATION

The following descriptions describe the general requirements for the Control System functional operation. The PLC, Operator Interface and SCADA system shall be programmed by the System Integrator to provide the monitoring and control for the functions described herein.

Additional detail shall be provided to the Programmer by the Engineer during software development to provide a complete and operational Control System which operates to the intent of the drawings and specifications.

A. WAS Valves and WAS Pump Control:

1. There are two WAS pumps controlled from the MCC with VFD's. Select Primary WAS pump and Backup WAS pump on the OIU or SCADA screen. Primary pump shall operate unless there is a pump failure, where the Backup pump will move into the Primary position.
2. There are three 2" Normally Closed WAS solenoid valves, one for each Train (#1, #2, and #3). The valves operate as full open/full close. Configure controls to open valve X times per day, and run WAS pumps for Y minutes during each open cycle, as set on the OIU or SCADA screen.
3. Confirm WAS flow meter measures flow while pump is running. If there is no flow, generate an alarm and shut down pump. After programmed duration has passed, turn off WAS pump and confirm flow meter measures "no flow". Then close WAS valve.
4. Display total volume wasted each day (gpd) on the OIU or SCADA screen. Normal operation will waste somewhere between 2,000 and 8,000 gallons total (each day), based on operator's preference.
5. Each WAS pump can pump between 10 and 30 gpm. Manually adjust to desired pump speed. Initial set point = 20 gpm. VFD speed in HAND will be set on potentiometer on MCC panel; VFD speed in AUTO will be as set on the OIU or SCADA screen.
6. Offset wasting per Train by open duration so wasting doesn't overlap. i.e. waste from Train #1 for 6 minutes, then Train #2 for 6 minutes, then Train #3 for 6 minutes.
7. WAS valves shall be Normally Closed (Fail Closed).

B. Train #3 Biological Treatment System:

1. A complete biological treatment system, including aeration, mixing, clarifier, blowers, and instrumentation.

2. The complete treatment control panel shall be provided by the Biological Treatment System vendor, and shall be hard-wired controls. Provide monitoring via dry contacts of treatment system panel and display on the OIU and SCADA screen. Program alarms for treatment panel alarm points.
3. There are four 20 HP blowers that will operate to aerate the treatment basins. One aerator for each Train. Fourth blower is redundant backup in case primary blower fails.
4. Program aeration routine to monitor DO levels in each basin and adjust Blower speed based on PID algorithm for automatic aeration control.
5. For each blower, VFD speed in HAND will be set on potentiometer on MCC panel; VFD speed in AUTO will be as determined by the PLC based on the setpoint on the OIU or SCADA screen.
6. A similar system will be added for Train #1 and #2 in Phase 2. Allocate spares in hardware and software systems for future systems.
7. See section 46 53 00 – Biological Treatment System.
8. See section 43 11 00 – Blower Equipment.

C. Carbon Feed to Individual Trains:

1. One peristaltic metering pump.
2. Provide 4-20 mA output from PLC to each metering pump for flow pacing. Output based on set point at the OIU or SCADA system.
3. Set point to be manually adjusted, based on Operator preference. See section 46 53 00 – Biological Treatment System.

D. Influent and Effluent Samplers:

1. Owner-provided samplers require flow pace input.
2. Provide 4-20 mA output from PLC to each sampler.
3. Influent sampler flow pace to match WWTP Influent flow meter.
4. Effluent sampler flow pace to match WWTP Effluent flow meter.

E. UV Disinfection system:

1. There are two UV disinfection units required for the project.
2. Each UV system shall be furnished with a system control panel by the manufacturer. All UV control shall be from the system controllers.
3. A dry contact shall be wired from the UV panel to the PLC control panel for “system status”.

4. An analog output from the UV panel shall be wired to the PLC control panel for “Intensity level”.
 5. See section 46 66 00 – UV Disinfection System.
- F. Aerobic Digester:
1. Level Monitoring: One submersible transducer shall monitoring digester water level and display on the OIU and SCADA screens.
 2. Blower: One 5 HP blower, controlled from VFD in MCC.
 3. Manually adjust to desired pump speed. VFD speed in HAND will be set on potentiometer on MCC panel; VFD speed in AUTO will be as set on the OIU or SCADA screen.
 4. See section 43 11 00 – Blower Equipment.
- G. Rotary Fan Press System:
1. A complete dewatering system per specifications.
 2. System includes the following:
 - a. Rotary Fan Press (1.5 HP), Air Compressor (3 HP), Flocculator (1 HP), & Controls. See Section 46 76 26 – Rotary Fan Press
 - b. Screw Conveyor System. See section 41 12 13 – Screw Conveyor System.
 - c. Emulsion Polymer Make Up Water System. See section 46 33 33 – Emulsion Polymer System
 3. Dewatering system shall include Allen Bradley 1769 series Compact Logix PLC for control of the dewatering system. Vendor shall provide network connection to WWTP SCADA system for remote monitoring and setpoint control.
 4. Control parameters over Ethernet shall be coordinated between the dewatering system programmer and System Integrator for monitoring and control requirements.
 5. Output from the Dewatering PLC System will provide control speed of the Digested Sludge Pump VFD in the MCC to maintain sludge plug pressure.
 6. Dewatering control system shall monitor the water level in the Aerobic Digester, and turn “off” at low set point. Coordinate set points for the OIU and SCADA screens.
- H. 2W Plant Water System:
1. There are two 5 HP 2W pumps controlled from VFDs in the MCC.

2. Design flow range = 0 to 80 gpm. Maintain a constant pressure based on the pressure feedback signal to the PLC, based on pressure setpoint on the OIU and SCADA screens.
3. Initial set point between 60 and 70 psi. Control logic is to turn pump(s) on, adjust speed via PID algorithm, turn pump(s) off, as needed to maintain pressure set point.
4. See section 43 22 10 – 2W Plant Water System.

I. Water Supply to 2W Air Gap:

1. Domestic water feed to two hot boxes, just west of the 2W Plant Water Wet Well.
2. System includes a solenoid valve, 1" flow meter, and an air gap assembly.
3. Program the PLC to operate the solenoid valve to open and close based on 2W wet well water level (float control).
4. See section 43 22 10 – 2W Plant Water System.

J. Ventilation Control Panel

1. Ventilation fans shall be provided in the treatment building to de-classify the dewatering area during dewatering system operation.
2. A ventilation control panel shall be provided for HOA operation of each of (3) wall fans and visual and audible alarm notification.
3. In HAND, the fan operation will be manually controlled. In AUTO, the fans will all turn on together (with 2 second time delay between fans for starting). A signal from the dewatering PLC will be sent to the WWTP PLC to indicate dewatering system operation and turn on the building ventilation fans. The fans will operate continuously while the dewatering system is in operation.
4. Hydrogen Sulfide Sensor: The 4-20 mA signal from the sensor will monitor the gas levels in the dewatering room. If the level reaches the setpoint alarm level as set on the OIU and SCADA screens, activate the alarm light and horn and an alarm on the SCADA system. If the fan selector is in AUTO, activate the fans.
5. Combustible gas detector: The 4-20 mA signal from the sensor will monitor the gas levels in the dewatering room. If the level reaches the setpoint alarm level as set on the OIU and SCADA screens, activate the alarm light and horn and an alarm on the SCADA system. If the fan selector is in AUTO, activate the fans.

K. Generator, ATS and Power Monitor:

1. Status signals shall be wired from the new diesel Generator and ATS to the PLC.
2. Provide indication on the OIU and SCADA screens for Generator Running and alarm status.

3. Provide indication on the OIU and SCADA screens for ATS Normal Power and Emergency position status.
 4. Data from MCC power monitor shall be polled by the PLC over Ethernet and displayed on the OIU and SCADA screens. Display critical power data including current, voltage, KW, power factor, etc.
- L. Intrusion Monitoring:
1. Intrusion logic shall be programmed in the PLC to generate individual alarms for the blower room door. The alarms shall be enabled/disabled at the local operator interface (OIU).
- M. Smoke and Heat Alarm Monitoring:
1. Provide status and alarm monitoring for smoke detector in the Blower building. Provide indication on the OIU and SCADA screens
- N. Influent Pump Station:
1. The Influent pump station shall include Two Small Pumps (3 HP each) and Two Large Pumps (5 HP each). The starters shall be FVNR and located in the Influent pump station control panel.
 2. A PLC system shall be provided as a remote rack to allow control and monitoring from the Main PLC controller.
 3. Primary control of the four pump station pumps shall be by the PLC via setpoints as measured by the wet well level transducer. Set points are shown on sheet E4.2, and shall be adjustable from the OIU and SCADA system.
 4. Each motor control circuit shall have a local Hand-Off-Auto control switch, mounted to the control panel. When in AUTO, the pump shall be enabled for Automatic control by the PLC control system. When in HAND mode, the pump shall run until switched to OFF.
 5. The level transducer shall be monitored for a valid signal. If the signal fails, an alarm will be activated, and the float back up system will operate the pumps.
 6. A high level float switch for each pump and hard wired circuit shall be provided for redundant pump control, as shown on the drawings. If the high level float switch activates, the associated pump shall be called to run for the amount of time as set on the time delay relay.
 7. Seal Fail and overtemperature sensors shall be wired to the control panel. Local alarm and status indicators shall be provided on the control panel and monitored on the OIU and SCADA system.
 8. A step down transformer and load center shall be installed for local 120V control panel as shown in the panel schedules.

9. Provide Intrinsically Safe area and barriers as required to meet Code for Classified locations, as indicated on the drawings.
10. Provisions and space shall be provided for the flow meter transmitters as shown on Sheet E8.1.

O. Main PLC control panel:

1. The Main PLC controller shall be located in the Blower building in a new NEMA 12 control enclosure. Provide all required PLC and I/O as indicated on the drawings. Provide all power distribution, terminals, overcurrent protection etc as required for a complete and operations system. The System Integrator shall provide final documentation, design and selection of components based on the functional requirements shown on the design drawings and specifications.
2. The Main PLC controller shall be programmed by the System Integrator to monitor the complete treatment system, pump station performance, MCC equipment, aeration system, wet well levels, treatment system status, instrumentation, equipment run times, equipment faults, and all related monitoring and control.
3. Main PLC Panel Operator Interface Unit (OIU):

The operator interface shall be programmed with the following minimum features:

- a. System overview screen showing system overview, pump status, wet well level, switch status, and all other available process information.
- b. Alarm screen with each alarm configured with detailed description and time and date stamp.
- c. Trending screen to display graphs of process data including wet well level, pump cycles and other relative information.
- d. Maintenance screen showing equipment run times and start counts.
- e. Intrusion monitoring and control.
- f. Set point screen to allow password-protected system set point changes for all control and alarm points.

3.010 Supervisor Control and Data Acquisition (SCADA) System:

A. SCADA System Software

1. The System Integrator shall develop the custom graphical for the SCADA computer application based on this specification and input from the Owner and Engineer.
2. The functions and screens shall utilize the full software capabilities of the software provided.

3. The SCADA screens shall be developed in coordination with the operator interface screens at the Blower room, for a similar look and operation between the operator interface and new SCADA computer.
4. The proposed SCADA screens will include, but not limited to:
 - a. System Overview: Plan view of the Wastewater Treatment Facility showing key process data and operational status of each motor (running/stopped/failure), instantaneous flow rate for each flow meter, total influent and effluent flow, levels, DO and other critical process data. This screen provides a large-scale overview of the entire treatment system.
 - b. Blower System and Aeration Screen: Provide a detailed graphical screen with all equipment and process information including Blower system, VFD status and speeds, DO measurements, DO control setpoints, alarm setpoints, level, temperature, flow, pump starts, pump runtime, and alarm setpoints.
 - c. Train #3: Provide a detailed graphical screen with all equipment and process information including Treatment system, mixers and scraper arm motor status, DO measurements, DO control setpoints, alarm setpoints, skimmer system, torque limit status, pump starts, pump runtime, and alarm setpoints. Provide provisions for future Trains #2 and #1.
 - d. UV Disinfection and Aerobic Digester: Provide a detailed graphical screen with all equipment and process information including UV system status, flash mixer motor status and control, Digester Level measurement, level alarm setpoints, pump starts, pump runtime, and alarm setpoints.
 - e. WAS Screen: Provide a detailed graphical screen with all equipment and process information including WAS system, VFD status and speeds, WAS flow rate and totals, WAS valve control setpoints, alarm setpoints, pump starts, pump runtime, and alarm setpoints.
 - f. 2W Water System Screen: Provide a detailed graphical screen with all equipment and process information including 2W pumping system, float status, VFD status and speeds, 2W flow rate and totals, 2W system pressure, 2W valve control setpoints, alarm setpoints, pump starts, pump runtime, and alarm setpoints.
 - g. Influent Pump Station Elevation: Shows the water elevation of the wet well as measured by the submersible level transducer, float switch status, and pump status (running/stopped/failure). Indicate control set point levels and set point control.
 - h. Dewatering Screen: Provide a detailed graphical screen with all equipment and process information from the Dewatering PLC panel. Include VFD status and speed for the sludge pump, control setpoints, alarm setpoints, Dewatering flow rate and totals, pump starts, pump runtime, and alarm setpoints.

- i. Ventilation Control Screen: Provide a detailed graphical screen with all equipment and process information for the ventilations system including gas monitors level measurement and alarm setpoints, fan status, fan starts, fan runtime.
- j. Flow meter Screen: Summary of all flow meter instantaneous flow readings, flow totals for each meter, and daily/weekly/month influent and effluent flow totalizers.
- k. Sampler Screen: Display Sampler status and provide set point control for flow pace control and alarms.
- l. Chemical Metering Equipment Screen: Chemical metering equipment status and set point control for flow pace control to metering equipment and alarms.
- m. Maintenance Screen: Summary of pump starts and runtimes for each pump. Allows operators to monitor pump usage and determine maintenance schedules.
- n. Power Screen: Display of power data from the electrical power meter in the MCC. Includes current, voltage and frequency measurements for the 480V three phase system.
- o. Generator Screen: Automatic Transfer Switch (ATS) position, Utility power source status, Generator status (run/fail), low fuel alarm.
- p. Alarm Screen: Shows summary of all active alarms and allows each individual alarm to be enabled or disabled for maintenance.
- q. Trending: Configure AVEVA Insight to log all process data into the online Historian. Configure Historical trend screen allows process data logged into the Historian to be viewed in graphical format.
- r. Reporting: Configure AVEVA Insight to provide interactive reporting tools and coordinate with Owner and operator for the custom reports to meet Agency requirements.
- s. Data trending, logging and reports using the Insight software tools and cloud based system
- t. Common Navigation screen, login tools, time and date display, user and password.
- u. Set point screen to allow password-protected system set point changes for all control and alarm points.
- v. Pop up screens as needed for setpoint control, notifications or other common features.

- w. Additional screens and features not listed herein but required for a complete and operational SCADA application.
 - 5. Configure communication status monitoring and comms loss notification.
 - 6. Configure security functions per AVEVA software capabilities.
 - 7. Configure remote access via VPN to allow for secure remote access by Operator.
- B. Alarm Monitoring Software – WIN 911 Mobile
- 1. Alarm signals shall be programmed for each system alarm. An alarm event logger shall be included to indicate time and date for each alarm, when triggered and when acknowledged.
 - 2. All alarms will be programmed in the PLC for monitoring by the SCADA computer and WIN 911 mobile interactive alarm system.
 - 3. All alarms shall include adjustable de-bounce timers (5 sec) and enable/disable functionality for each alarm.
 - 4. Configure WIN-911 software for treatment system alarm monitoring and remote and interactive notifications. Configure up to 5 remote users, cell phone setup for each, testing and training for system operators to use the system and also adjust and configure schedules and users.

*** END OF SECTION **

SECTION 31 10 00 – SITE CLEARING

PART 1. GENERAL

1.01 DESCRIPTION

- A. Work under this Section includes providing all labor, materials, tools, and equipment necessary for clearing, grubbing, removing and disposing of all vegetation and debris. Prior to the start of clearing and grubbing, the Contractor shall clearly mark the clearing limits in the field.

1.02 RELATED SECTIONS

- A. Demolition: Section 02 41 00.

PART 2. PRODUCTS (NOT USED)

PART 3. EXECUTION

3.01 CLEARING AND GRUBBING

- A. The Contractor shall clear the entire area within the project limits by clearing and grubbing all vegetation to a minimum of 1-foot below the graded surface.
- B. Contractor shall exercise care as to not encroach or disturb vegetation outside of the marked clearing limits.
- C. Vegetation and debris must be hauled to a legal waste site obtained by the Contractor. All costs associated with disposing of grubbed vegetation and debris shall be incidental to the contract.

*****END OF SECTION*****

SECTION 31 20 00 - EARTHWORK

PART 1. GENERAL

1.01 DESCRIPTION

- A. Description of Work: The work covered by this section consists of excavating, hauling and disposal of excavated material, backfilling, placing, compacting and final site grading as specified herein and as shown on the drawings.
- B. Related Documents:
 - Part 4 - Appendix C: *September 7, 2021 Geotechnical Engineering Report*, GeoEngineers. This report summarizes the geotechnical site explorations and design recommendations for the proposed water/wastewater facility improvements.

1.02 SUBMITTALS

- A. Submit, in accordance with requirements of Section 01 33 00 – Submittal Procedures, the following:
 - 1. Gradation and moisture density curves for all imported materials.
 - 2. Separation Geotextile product information
 - 3. Proposed compacting equipment for compacting earth embankments.

1.03 QUALITY ASSURANCE

- A. All field inspections and tests will be conducted by an independent testing lab/agency hired by the Engineer. Contractor is to coordinate and schedule inspections with the Engineer. The Engineer will pay the testing lab/agency. In addition, additional QA/QC testing may be performed by the Owner/Engineer's independent testing lab/agency as desired by the Owner/Engineer. The Contractor shall allow the Owner/Engineer's representative to perform additional QA/QC testing and shall make right to the satisfaction of the Engineer all work found to be deficient in meeting the specifications. If the subgrade or fills which have been placed are below the specified density, additional compaction and testing will be required until satisfactory results are obtained. QA/QC testing and inspection by the Owner/Engineer does not relieve the Contractor from the responsibility to provide all adequate quality control measures and all testing to ensure the quality of his own work.
- B. The Engineer will use an independent soils testing lab to measure ASTM D-1557 Methods A through D dry density. Additional tests will be performed as directed by the Engineer in the field. All material samples will be taken at the time and location as agreed by the Engineer and Contractor. The Contractor shall give the Engineer ample time to perform additional QA/QC testing (if desired) of Contractor stockpiled onsite soil tested – at least 5 working days prior to installation of any embankment or backfill using onsite soils.
- C. The Engineer will use an independent soils testing lab and will perform one in place density measurement per every 100 LF of roadway and per every 1,000 square feet of native subgrade prepared and compacted prior to placement of structural backfill materials. The Engineer will perform one in place density measurement per every 500 cubic yards of structural backfill materials placed. The Engineer may increase the frequency if the properties of the soils being placed change or the equipment or procedures used by the Contractor for compacting the soil change. In place density will be measured using ASTM D-1556 or ASTM D2922 and D3017 (nuclear density) test methods. The Contractor shall give the Owner/Engineer ample time to perform

additional QA/QC testing (if desired) of the subgrade – at least 24 hours advance notice and a minimum of 4 hours for completion of testing after compaction is completed on 100 LF of roadway or 1,000 square feet of native subgrade. The Contractor shall give the Owner/Engineer ample time to perform additional QA/QC structural fill testing (if desired) – at least 24 hours advance notice and a minimum of 4 hours for completion of testing after compaction is completed on a lift.

- D. The Contractor shall pay for any additional testing deemed necessary to provide quality assurance for the work or to corroborate or protest the Owner's test results.

1.04 JOB CONDITIONS

- A. Existing Conditions:

The Contractor shall examine the site before commencing work and shall make his own deductions and conclusions as to the nature of materials to be encountered and difficulties anticipated.

1. Data results of a subsurface investigation of the site and soil conditions are included in the Geotechnical Engineering Report (Appendix C). In addition, the Contractor is encouraged to visually inspect soil conditions at the site.
2. The Contractor is encouraged to visually inspect soil conditions at the site. Contractors shall make whatever investigations as are necessary to determine what measures are necessary to successfully complete the work in accordance with the Contract. The Contractor shall include in the Contract price all work necessary to perform the tasks required to complete the Work as indicated on the Plans and specified herein: including, but not limited to, sheeting, shoring, dewatering, stabilizing slopes, and any other work of temporary nature not a part of the permanent finished structure, lines, and grade.

- B. Subsurface Conditions

The Contractor shall notify the Engineer when excavation for compacted fill or structures is complete. **No forms, reinforcing steel, fill, or concrete shall be placed until the excavation has been inspected by the Engineer.**

Groundwater may be encountered, and dewatering measures may be required for construction activities. Contractor is responsible for any dewatering considered incidental to the project. Contractor to plan accordingly.

Boulders, buried or otherwise, may be found in the project area, and boulder excavation is considered incidental, if needed.

1.05 UNSUITABLE FILL MATERIAL

- A. Unsuitable materials shall be those defined as containing volcanic ash, topsoil, vegetation matter, sludge, peat, organic clays and silts, sod, mulch, rubbish, and materials which are excessively fine or moist not allowing adequate compaction

PART 2. PRODUCTS

2.01 STRUCTURAL FILL

- A. See Processed Structural Fill Material Section 2.03. All import fill is to be imported material per WSDOT specifications or as specified. Contractor to submit proposed aggregate materials information from local supplier(s) prior to construction.
- B. Excavated soils shall not be used unless the contractor can demonstrate that the soils meet the WSDOT specification and gradation requirements indicated for the material specified. All excavated materials not used shall be hauled to a waste site.

PROCESSED STRUCTURAL FILL MATERIAL

- C. CRUSHED SURFACING TOP/BASE COURSE. Per WSDOT 9-03.9(3).
 - 1. Shall be used as indicated in the plans and specs and compacted to 95% MDD.
- D. GRAVEL BASE. Per WSDOT 9-03.10.
 - 1. Shall be used as indicated in the plans and specs and compacted to 95% MDD. When used as pavement section gravel base shall have at least 30 percent retained on the U.S. No. 4 sieve.
- E. QUARRY SPALLS. Per WSDOT 9-13.1(5). Unless otherwise noted on the plans.
- F. GRAVEL BACKFILL FOR PIPE ZONE BEDDING. Per WSDOT 9-03.12(3).
- G. GRAVEL BACKFILL FOR DRAINS. Per WSDOT 9-03.12(4).
- H. SAND. Per WSDOT 9-03.13
- I. CAPILLARY BREAK. Per WSDOT 9-03.1(4)C – AASHTO Grading No. 67
 - 1. Shall be used as indicated in the plans and specs and compacted to 95% MDD.
- J. GRAVEL BORROW. Per WSDOT 9-03.14(1)
 - 1. Shall consist of clean, non-plastic, free-draining sand and gravel free from organic matter, with the exception that the percent passing the U.S. No. 200 Sieve shall not exceed 5% and all materials shall be smaller than 4". The percentage passing the No. 200 sieve should be based on that fraction passing the ¾" sieve. The gradation is summarized below.

Material shall meet the following gradation specifications:

Sieve Size	Percent Passing by Weight
4" Square	100%
2" Square	75-100%
U.S. No. 4	50-80%
U.S. No. 40	30 max.
U.S. No. 200	5% max.
Sand Equivalent	50 minimum

- 2. Alternatively, crushed surfacing base course, in conformance with Section 9-03.9(3) of the WSDOT Standard Specifications may be used.
- K. GRAVEL BORROW FOR STRUCTURAL EARTH WALL. Per 9-03.14(4) – Geosynthetic Reinforcement. – Shall be used as Slope Stabilization Fill where indicated in the plans and specifications.

- L. GRAVEL BACKFILL FOR WALLS. Per 9-03.12(2).
- M. BANK RUN GRAVEL FOR TRENCH BACKFILL. Per 9-03.19.

PART 3. EXECUTION

3.01 GENERAL

- A. Weather Limitations: Construction shall progress only when weather conditions will not adversely affect the quality of the finished work. Soils that are not compactable due to saturation shall be aerated or removed and replaced with a compactable material. Contractor shall bear all costs for rework caused by weather conditions.
- B. Control of Water: See Section 31 23 19 – Dewatering.
- C. Water for Compaction: Contractor shall provide all water as necessary to moisture-condition Structural Fill material to achieve required compaction densities.
- D. Excavation for Structures: Conform to elevations and dimensions shown with a tolerance of plus or minus 0.10 feet and extending a sufficient distance from footings and foundations to permit placing and removal of concrete formwork, installation of services and other construction.

In excavating for footings and foundations, take care not to disturb bottom of excavation. Excavate by hand to final grade prior to placement of concrete reinforcement. Trim bottoms to required lines and grade to provide solid base for structure.

Crushed Surfacing Base Course shall be placed under all concrete slabs, footings, and foundations, including pile foundations, to the minimum depth indicated below, unless otherwise indicated, on exposed, undisturbed, compacted subgrade immediately upon completion of excavation, or greater thickness if needed to protect subgrade and support contractor selected construction equipment.

For Mat Foundations and Vaults – 12 inches minimum depth

For Shallow Foundations – 18 inches minimum depth

For Pile Structures – 24 inches minimum depth

- E. Disposal of Excavated Materials: The Contractor is responsible for ultimate disposal of all excavated material and such disposal shall be incidental to other work. See grading and erosion control plans.
- F. Over Excavation: Excavation of materials beyond the indicated subgrade elevations shall be backfilled with Structural Fill and compacted to provide a firm and stable base at the desired elevation. Work required to remedy over excavation not authorized by the Owner or the Engineer shall be at the Contractor's expense.
- G. Stability of Excavations: Slope sides of excavations to comply with local codes and ordinances having jurisdiction. Shore and brace, as required, to prevent cave-ins. Remove prior to backfilling unless otherwise specified.
- H. Stockpile excavated materials classified as suitable material where directed, until tested and approved for fill. Place, grade, and shape stockpiles for proper drainage and erosion control as approved by the Owner. The temporary stockpile site, if required, shall be coordinated with the Owner.
- I. The Contractor's bid price shall include all costs associated with providing materials and methods for the geogrid-reinforcement where specified, if any.

3.02 SITE PREPARATION

- A. Prior to placement of Structural Fill under buildings, structures, and roadways embankments, scarify, moisture-condition, and compact subgrade soils to at least 95% of the maximum dry density (MDD) based on ASTM D-1557 if not already at 95% of the MDD. Subgrade preparation per WSDOT Standard Specification section 2-06.
- B. Benching subgrade. Slopes to be filled, which are 5:1 or steeper, shall be benched before receiving structural fill. Each bench shall be level in all horizontal directions and shall be at least 8 feet wide (perpendicular to slope contours).

3.03 STRUCTURAL FILL

- A. Structural Fill shall be placed in lifts not exceeding 10-inches in loose thickness before compaction, unless used as Backfill of Structures, Section 3.04. Structural Fill shall be compacted to 95% maximum dry density (MDD) based on ASTM D-1557.
- B. Where material must be moisture-conditioned before compaction, uniformly apply water to surface of subgrade or to layer of material, to prevent free water appearing on surface during or subsequent to compaction operations. Remove and replace, or scarify and air dry soil that is too wet to permit compaction to required density. Material that has been removed due to excessive moisture may be stockpiled or spread and allowed to dry. Assist drying by disking, harrowing or pulverizing until moisture content is reduced to satisfactory value.
- C. Maintain Structural Fill areas as a continuous working surface throughout the project. Fill surfaces are to be graded smooth and sealed or covered as appropriate at the end of each work day to prevent unacceptable wetting. After periods of rain, remove any soft material prior to placement of additional fill.
- D. Provide cut and fill slopes reasonably true to line and grade with a tolerance of plus or minus 3 inches.

3.04 GRADING

- A. General. Uniformly grade areas within limits of project site including adjacent transition areas. Smooth finished surfaces within specified tolerances, compact with uniform levels or slopes between points where finish elevations are shown or between points where finish elevations are shown or between such points and existing grades.
- B. Drainage Ditches. If any existing drainage ditches or swales exist, they should be maintained and/or fully restored to pre-construction conditions if altered in any way during construction, unless otherwise noted on the plans.

3.05 UNSUITABLE MATERIAL (OVER-EXCAVATION)

- A. In the event that during excavation unsuitable material is encountered at the subgrade, the Owner/Engineer shall be notified of such areas prior placing structural fill or pouring concrete. The specific areas of unsuitable material shall be addressed as described herein. Work under this item shall be allowed ONLY upon written authorization of the Owner or Engineer.
- B. Unsuitable material shall be over-excavated 18-inches below the trench neat line and filled with Crushed Surfacing Base Course conforming to WSDOT 9-03.9(3), or locally available approved equal submitted to and reviewed by the Engineer, and compacted to 95 percent of maximum dry density described in ASTM D1557. Fill up to the trench neat line to allow room for the bedding material.

3.06 DISPOSAL OF UNSUITABLE MATERIAL (OVER-EXCAVATION)

- A. Excavated unsuitable material shall be disposed of off site.
- B. All material which is hauled off of job site shall be documented with receipts, documenting weight (or volume agreed upon with Engineer for truck counts) and certification that it was transferred to a legal fill site. Receipts to be provided to Owner.

*****END OF SECTION*****

SECTION 31 23 19 - DEWATERING

PART 1 GENERAL

1.1 SCOPE

The work specified in this Section shall consist of dewatering excavations of any kind and location, including but not limited to groundwater, surface water, and precipitation, until backfilling has been completed to finished grade.

1.2 SUBMITTALS

Prior to the start of construction, the Contractor shall submit a dewatering plan in accordance with Section 01 30 00 containing both a graphical and narrative presentation identifying proposed methods, equipment sizes and contingency plans should dewatering cause settlement of any adjacent facilities. The dewatering plan shall show specific locations, in plan and section, where dewatering is expected as well as a general discussion of methods to be employed should water be encountered in other locations. The plan shall detail the depth, diameter and anticipated flow for dewatering wells, well points or sumps.

In addition, the dewatering plan shall identify all stormwater diversions around the site, discharge locations from the site, treatment, storage, and conveyance facilities needed to manage construction-related stormwater. All stormwater runoff that is encountered on the site and dewatering water from trenches or excavations, is collectively termed "Construction-Related Stormwater." The plan shall be a living document that is kept up to date as construction progresses similar to the temporary erosion and sediment control plans. As adaptive measures are employed, and the plan changes, these changes shall be recorded on the approved plan such that it is always up to date and available for inspection by the Engineer or regulatory authorities.

Acceptance by the Owner of the method, installation, and operation and maintenance details submitted by the Contractor shall not in any way be considered to relieve the Contractor from full responsibility for errors therein or from the entire responsibility for complete and adequate design and performance of the system in controlling the water level in the excavated areas, and for control of the hydrostatic pressures to the depths specified herein. The Contractor shall be solely responsible for the proper design, installation, proper operation, maintenance, and any failure of any component of the dewatering system.

1.3 REFERENCES

"Rossum J.R., 1954, *Control of Sand in Water Systems*, Journal American Water Works Association, Volume 46, pp. 123-132"

Construction Stormwater Permit

Appendix C: *September 7, 2021 Geotechnical Report*, GeoEngineers. This report summarizes the geotechnical site explorations and design recommendations for the proposed wastewater facility improvements.

Appendix B: *Groundwater Level Analysis*

1.4 QUALITY CONTROL

It shall be the sole responsibility of the Contractor to control the rate and effect of the dewatering efforts to avoid all objectionable settlement and subsidence. The Contractor shall comply with local codes and ordinances of governing authorities with regard to disposal of water pumped from dewatering operations.

Proposed discharge points shall be approved by the Owner prior to implementation of dewatering. The Contractor shall be responsible for taking all reasonable precautions necessary to ensure continuous, successful operation of the system.

Groundwater levels were monitored for ten months and the results are included in Appendix B – Groundwater Level Analysis. Significantly higher groundwater levels and dewatering is anticipated during the winter months. Contractor shall review Appendix B and the Geotechnical report and prepare a dewatering plan appropriate for the conditions.

PART 2 PRODUCTS

The Contractor shall have sufficient pumping equipment and/or other machinery available onsite before operations begin to assure that the operation of the dewatering system can be maintained. This shall include providing backup pumps of similar capacity and a standby generator of the capacity required to continuously operate the Contractor's dewatering system.

PART 3 EXECUTION

3.1 GENERAL

To the maximum extent feasible, all clean runoff that would otherwise flow into the project site and become turbid shall be routed around the project site such to not become part of the construction-related stormwater. All Construction-Related Stormwater shall be managed and disposed of in accordance with all local, state, and federal laws. Construction-Related Stormwater that is discharged to the downstream stormwater conveyance system must meet discharge limits for turbidity, pH, and all other discharge limits as specified by the Construction Stormwater General Permit (CSGP) and other applicable permits.

The contractor shall employ all means and methods necessary to clean up Construction-Related Stormwater prior to discharge from the site to meet permit requirements. This

may require the stormwater to be stored, filtered, chemically treated, hauled away, or other means as needed to meet permit requirements. In no case will the contractor be allowed to discharge Construction-Related Stormwater to the sanitary sewer, the sanitary sewer outfall, or into the wastewater treatment plan.

All costs related to management, rerouting, and disposal of stormwater and Construction-Related Stormwater shall be included as part of the unit cost

Contractor is responsible for investigating and becoming familiar with all site conditions that may affect the work including surface water, potential flooding conditions, level of groundwater and the time of year the work is to be done.

3.2 INSTALLATION AND APPLICATION

During excavation, the installation of piping, conduits and structures and during the placing of backfill, excavations shall be kept free of water, subsurface or otherwise. The Contractor shall furnish all equipment necessary to dewater the excavations and shall dispose of the water so as not to cause a nuisance or menace to the public. The dewatering system shall be installed and operated by the Contractor so that the groundwater level outside the excavation is not reduced to the extent that would damage or endanger adjacent structures or property. The release of groundwater to its static levels shall be performed so as to maintain the undisturbed state of the foundation soils, prevent disturbance of backfill and prevent movement of all structures and pipelines.

Design, implementation, and maintenance of any dewatering system shall be the responsibility of the Contractor.

All dewatering shall occur a minimum of 2-feet below the base of any structures.

The Contractor shall construct all dewatering wells in accordance with WAC 173-160. The dewatering system shall be sufficient to maintain the groundwater level at an elevation to protect the surface of the trench bottoms, the base of the bedding course or other foundation, and shall be accomplished prior to pipe laying and jointing or placement of reinforcing steel for concrete.

If well points or wells are used, they shall be adequately spaced to provide the necessary dewatering. The dewatering operation, however accomplished, shall be carried out so that it does not destroy or weaken the strength of the soil under or alongside the excavations.

The Contractor shall design filters and screen slot sizes for all sumps, wells and well points which prevents the movement of fines during pumping. The Contractor shall develop the wells such that they produce no more than 10 ppm silica as measured with a Rossum Sand Tester (Rossum, 1954) or equivalent.

3.3 FIELD QUALITY CONTROL

A continual check by the Contractor shall be maintained to ensure that the subsurface soil is not being removed by the dewatering operation. The Contractor shall test all dewatering discharge using a Rossum Sand Tester or equivalent to determine the silica content of the discharge. The Contractor shall notify the Engineer at least 24 hours prior to testing. Where critical structures or facilities exist immediately adjacent to areas of proposed dewatering, reference points shall be established and observed at frequent intervals to detect any settlement that could develop.

Should settlement be observed, the Contractor shall cease dewatering operations and implement contingency plans as outlined in the Contractor's approved dewatering plan. The responsibility for conducting the dewatering operation in a manner that protects adjacent structures and facilities rests solely on the Contractor. The cost of repairing any damage to adjacent structures and restoration of facilities shall be the responsibility of the Contractor. Permanent piping systems, existing or new, shall not be incorporated into the Contractor's dewatering system.

***** END OF SECTION *****

SECTION 31 23 33 – TRENCHING AND BACKFILL

PART 1. GENERAL

1.01 SECTION INCLUDES

- A. This section specifies trench excavation, bedding, backfilling, and compacting for utilities and related facilities.

1.02 RELATED SECTIONS

- A. Section 00 33 00 – Submittal Procedures
- B. Section 01 41 00 – Regulatory Requirements
- C. Section 01 45 00 – Quality Control
- D. Section 31 20 00 - Earthwork
- E. Section 31 32 11 - Soil Surface Erosion Control
- F. Section 33 31 00 - Sanitary Utility Sewerage Piping
- G. Section 33 41 00 – Storm Utility Drainage Piping

1.03 SPECIFIC STANDARDS

- A. The specific reference standard for this work will be Washington State Department of Transportation (WSDOT)/American Public Works Association (APWA) Standard Specifications/for Road, Bridge, and Municipal Construction, latest edition.
- B. Additional standards may also apply.

1.04 QUALITY ASSURANCE

- A. The Contractor shall comply with the requirements of all applicable regulatory agencies having jurisdiction over this work including 29 CFR PART 1926 -- SAFETY AND HEALTH REGULATIONS FOR CONSTRUCTION.
- B. Material sample, sieve analysis and proctor test results shall be provided in advance for any proposed material not compliant with WSDOT Standard Specs.
- C. Use equipment adequate in size, capacity, and numbers to accomplish the work of this section in a timely manner. The Engineer will reject equipment that repeatedly breaks down or fails to produce results within normal tolerances. The Contractor shall have no claim for additional payment or for extension of time due to rejection and replacement of any equipment.

1.05 JOB CONDITIONS

- A. The Contractor shall provide protection of existing utilities affected by the work and make every effort to minimize disruptions to all utility services.
- B. If, during the course of construction, it is anticipated that excavation will interrupt traffic or parking areas for longer than 10 to 15 minutes the Contractor must provide advance notice to the Owner. For longer intervals or complete shut-downs, the Owner requires 48 hour advance notice. This advance notice allows time to deliver community notices in advance of the route delays or re-routes. In the event of such road closures, the Contractor shall be solely responsible for all traffic control measures including but not limited to flagging, barricades and cones.

- C. The Contractor shall provide a traffic control plan per WSDOT 1-10. Traffic control plan to be submitted for review and acceptance prior to commencing work. See Section 01300 Submittal Procedures.
- D. Trenches shall be closed or covered with steel plates at the end of each work day.

1.06 SUBMITTALS

- A. The Contractor shall furnish the following submittals as part of completing the work associated with this section:
 - 1. Location of disposal sites for excess excavated material.
 - 2. Gradation test results for imported foundation, bedding and backfill material.
 - 3. Proctor tests for proposed imported materials which are not identified as approved for use by WSDOT.
 - 4. Geotextile fabrics cut sheets or WSDOT QPL.

PART 2. PRODUCTS

2.01 TRENCH BACKFILL MATERIAL

- A. Trench Backfill material to be per WSDOT 9-03.19, "Bank Run Gravel for Trench Backfill", and be free from debris and organic matter and other extraneous or objectionable materials. No native backfill to be used.
- B. Excavated soils shall not be used unless the contractor can demonstrate that the soils meet the WSDOT specification and gradation requirements indicated for the material specified.

2.02 BEDDING MATERIAL

- A. Bedding material for utilities shall conform to WSDOT 9-03.12(3), "Gravel Backfill for Pipe Zone Bedding," and be free from debris and organic matter and other extraneous or objectionable materials. No native bedding material to be used.
- B. Pipe bedding material and / or backfill around the pipe shall be placed in layers and tamped around the pipe to obtain complete contact per the project plans.

2.03 UTILITY WARNING TAPE

- A. Shall be APWA color-coded detectable underground marking tape. Tape shall be 2-inch wide plastic-encased aluminum foil tape capable of being located by a metal detector. Message and coding shall be per APWA Standards and shall be as follows:

Message	Color Coding
CAUTION: ELECTRIC LINE BURIED BELOW	Red
CAUTION: WATER LINE BURIED BELOW	Blue
CAUTION: SEWER LINE BURIED BELOW	Green
CAUTION: TELEPHONE LINE BURIED BELOW	Orange
CAUTION: CATV LINE BURIED BELOW	Orange
CAUTION: AIR LINE BURIED BELOW	Light Blue

- B. Provide new continuous warning tape for each type of utility installed. Also provide new replacement warning tape for utilities encountered and replace any/all damaged sections of existing warning tape for those utilities. Should no warning tape exist on encountered utilities, provide a section of new tape at the crossing.

2.04 DRAIN ROCK & GRAVEL BACKFILL FOR DRAINS

- A. Drain Rock for trench drains and retaining wall drains shall conform to WSDOT 9-03.12(4) Gravel Backfill for Drains.

2.05 COBBLES (3-IN TO 4-IN)

- A. Cobbles for trench drains shall conform to WSDOT 9-03.11(2) Streambed Cobbles (4" Cobbles from table gradation).

2.06 GEOTEXTILE MATERIALS

- A. Drainage Geotextile for use in trench drains and retaining wall drains – Non-woven, moderate survivability, per Table 1 of WSDOT Standard Specification 9-33.2(1).

PART 3. EXECUTION

3.01 EXISTING UTILITIES AND RELATED FACILITIES

- A. Keep active utilities intact and in continuous operation. Regarding existing underground utilities, the Contractor shall:
 - 1. Call the utilities underground location center for field location of utilities.
 - 2. Pot hole as necessary to locate existing utilities.
 - 3. Not begin excavation until all known underground facilities in the vicinity of the proposed excavation have been either referenced on Plans, located, and/or marked on the ground.
- B. Location and dimensions shown on the Plans for existing utilities are in accordance with available information without uncovering, measuring, or other verification. Utilities/facilities whose underground location can be reasonably determined from existing above-ground features shall be considered the same as having been individually marked. In the event the Contractor discovers unknown utilities, he shall:
 - 1. Take reasonable and appropriate steps to avoid damage to the utility and/or Commission property.
 - 2. Promptly notify the Engineer for individual directions.
 - 3. All costs to repair damage to above-ground or known subsurface structures due to operator error will be borne by the Contractor.

3.02 TRENCHING

- A. Trench excavation shall conform with the most recent version of the WSDOT Standard Specifications. Special attention shall be paid to the requirements for trench safety noting that all work shall be performed in strict compliance with 29 CFR 1926.
- B. The Contractor shall be solely responsible for any shoring, cofferdams or trench safety systems employed on the project. In no way shall the Owner or Engineer assume any responsibility for trench safety or the protection of life or property implied by the use of trench safety systems.

- C. The width of excavation for utility trenches shall be in accordance with WSDOT Standard Specification. No additional payment will be made for extra excavation required due to poor soil conditions.
- D. See Section 31 20 00 Earthwork for excess material disposal requirements.
- E. The Contractor shall provide and operate all material, equipment and labor necessary to keep excavations and earth embankments free from water during construction. Dewatering shall prevent weakening foundations, undercutting trench walls, or otherwise affecting the stability of sub-grades and foundations. The Contractor shall establish and maintain positive drainage away from excavations to prevent surface water from entering excavations. Water shall be disposed of in a manner which prevents injury to public or damage to property.
- F. The Contractor shall backfill or otherwise cover all trenches at the end of each working day to protect public safety. The length of open trench excavation in advance of pipe laying operations shall not exceed 200 feet unless approved by the Owner. In no case shall the length of an open trench or size of an excavation exceed the Contractor's ability to safeguard the public welfare.

3.03 BEDDING

- A. Pipe bedding and pipe zone backfill installation shall comply with the WSDOT Standard Specifications, Section 7-08 — General Pipe Installation Requirements and per the Plans.
- B. Pipe bedding and pipe zone backfill shall be compacted to 90% of the maximum dry density described in ASTM D1557. Pipe bedding and pipe zone backfill shall be compacted in 6-inch maximum lifts.

3.04 BACKFILL

- A. Trench and structure backfilling shall comply with the most recent version of WSDOT Standard Specifications, Section 7-08 — General Pipe Installation Requirements and per the Plans.
- B. Structure backfilling shall comply with the most recent version of WSDOT Standard Specifications, Section 2-09.3(1)E Backfilling and per the Plans.
- C. In areas beneath driveways, sidewalks, or within 5-feet of the roadway template (including shoulder or structures), backfill shall be compacted to 95% of the maximum dry density described in ASTM D1557. Backfill within the roadway template shall be compacted in 6-inch maximum lifts.
- D. In landscaped or native areas outside roadway templates and not beneath pavement, gravel paving, drives or sidewalks, backfill shall be compacted to 90% of the maximum density described above.
- E. Construction shall progress only when weather conditions will not adversely affect the quality of the finished work. At the same time, the Contractor must be prepared to take such measures as are necessary to complete the construction within the specified contract period. Where soils cannot be compacted due to moisture content, material shall be aerated or removed and replaced with a suitable granular backfill material. Contractor shall bear all costs for necessary extra measures and/or rework if excavated material is made unsuitable by adverse weather conditions and not protected by contractor in accordance with WSDOT Standard Specifications covering contractor requirements for protection of excavated materials.

3.05 UNSUITABLE TRENCH OVEREXCAVATION

- A. In the event that during trenching unsuitable material is encountered at the trench bottom, the Owner shall be notified of such areas prior to placing pipe. The specific areas of unsuitable material shall be addressed as described herein. Work under this item shall be allowed ONLY upon written authorization of the Owner.
- B. Unsuitable material shall be over-excavated 18-inches below the trench neat line and filled with Crushed Surfacing Base Course conforming to WSDOT 9-03.9(3), or locally available approved equal submitted to and reviewed by the Engineer, and compacted to 95 percent of maximum dry density described in ASTM D1557. Fill up to the trench neat line to allow room for the bedding material.

3.06 COMPACTING

- A. Compaction shall be performed in accordance with Section 31 20 00 Earthwork or as detailed on the Contract Plans.

*****END OF SECTION*****

SECTION 31 32 11 - SOIL SURFACE EROSION CONTROL

PART 1. GENERAL

1.01 SECTION INCLUDES

- A. Work includes but is not limited to following:
 - 1. Temporary measures to prevent soil erosion and sedimentation of storm sewers, streams or other bodies of water.
 - 2. A National Pollution Discharge Elimination System Construction Stormwater General Permit (Construction NPDES Permit) will be acquired from the Department of Ecology for this project. The Stormwater Pollution Prevention Plan (SWPPP) provides all the requirements and details to comply with the General Permit.
 - 3. Upon execution of the contract, the Construction NPDES Permit shall be transferred to the Contractor's name, and the Contractor shall be responsible for paying yearly fees as well as maintaining compliance with the permit from that point forward.

1.02 RELATED SECTIONS

- A. Coordinate related work specified in other parts of the Project Manual, including but not limited to following:
 - 1. Section 31 23 33 Trenching and Backfill
 - 2. Section 02 92 00 Landscaping

1.03 REFERENCES

- A. WSDOT Standard Specifications for Road, Bridge and Municipal Construction, latest edition.
- B. Washington State Department of Ecology's 2019 Edition of the Stormwater Management Manual for Western Washington

1.04 SUBMITTALS

- A. Submit in accordance with Sections 01 33 00 and the following
 - 1. Geotextile for temporary silt fence.
 - 2. Compost for compost berm.
 - 3. Inlet protection products.
 - 4. Barrier fence.
 - 5. Track clean plates.
 - 6. Any other erosion control products proposed for use on the site.

1.05 QUALITY ASSURANCE

- A. Installer shall be a Specialist.
- B. Regulatory Requirements: Section 01 41 00.

PART 2. PRODUCTS

2.01 MATERIALS

- A. Silt Fence. Silt Fence shall comply with WSDOT Standard Specification 9-33 – Construction Geotextile and meet the properties described in Table 6 of said specification.
- B. Quarry Spalls. Quarry Spalls shall be per Earthwork Spec.
- C. Straw Mulch. Straw Mulch shall be per WSDOT 9-14.4(1) – Straw
- D. Compost. Compost shall be in accordance with WSDOT Standard Specification 9-14.4(8).
- E. Inlet Protection Insert. Inlet protection insert shall be in accordance with Plans, Specs, and San Juan County requirements for catch basin inserts.
- F. Orange Barrier Fence. Orange barrier fence shall comply with WSDOT Standard Specification 9-14.5(8) – High Visibility Fencing.
- G. Track Clean Plates. Track clean plates shall be Track Clean™ Construction Entrance Plates or approved equal. Plates must adhere to the guidelines of BMP C105: Stabilized Construction Entrance, found in Volume II of the WA State Ecology Stormwater Management Manual of Western Washington.

PART 3. EXECUTION

3.01 PREPARATION

- A. Planning Of Construction: Plan and coordinate to reduce sediment pollution. Install all site BMPs prior to the commencement of land disturbing activities. Minimize the area of disturbance. Keep the area of clearing and grubbing to the minimum necessary for construction.

3.02 INSTALLATION

- A. Install in accordance with “Quality Assurance” provisions, “References,” and Specifications. Where these may be in conflict, the more stringent requirements govern.
- B. Pump Water. Practice sound pump water management to reduce sediment production. Discharge pump water into stabilized surfaces and allow to filter through existing vegetation. Repair discharge areas, upon completion of construction, to pre-existing conditions or better. Do not pump water into adjacent wetlands, creeks, or rivers.
- C. Stabilization. Stabilize all slopes, channels, ditches or any disturbed area as soon as possible after the final grade or final earthmoving has been completed. Upon completion of the project, stabilize all areas which were disturbed by the project to prevent accelerated erosion. Maintain any erosion and sedimentation control facility required or necessary to protect areas from erosion during the stabilization period.
- D. Earthwork.
 - 1. Control excavation for site work operations. Stockpile the material removed from the excavation in area where a minimum of sediment will be generated and where other damage will not result from the piled earth.
 - 2. Stockpile topsoil separately and redistribute where shown on plans uniformly after grading.

3. Protect all stockpiled soil materials from erosion through the use of plastic sheeting or similar temporary measures, secured against wind disturbance.
4. Any area stripped of vegetation, where no further work is anticipated for a period of 14 calendar days, shall be immediately stabilized with an approved erosion control method such as seeding, mulching, netting, erosion control blankets, etc.
5. All disturbed areas shall be promptly and thoroughly stabilized against erosion during periods of wet weather, particularly when work is not being performed at the site.

3.03 MAINTENANCE AND CLEANING

- A. Maintenance. Maintain the erosion control measures and facilities in proper condition so that they will individually and collectively perform the functions for which they were designed. In order to insure the effectiveness and proper maintenance of the measures and facilities, the Contractor and Owner shall make periodic inspections at sufficiently frequent intervals to detect any impairments of the structural stability, adequate capacity, or other requisites of the herein approved measures and facilities which might impair their effectiveness. Take immediate steps to correct any such impairment found to exist.
- B. Cleaning: Leave installations clean; premises free from residue of work of this section.
- C. Street Sweeping: If onsite measures fail to prevent soil migration to street, Contractor shall provide regular sweeping.
- D. Inspection, or lack thereof, shall not relieve the contractor of the responsibility of maintaining erosion control at all times. The contractor should, therefore, check all erosion control periodically on their own to ensure adequacy.

*****END OF SECTION*****

SECTION 31 40 00 – TEMPORARY SHORING AND BRACING

PART 1 - GENERAL

1.01 DESCRIPTION

- A. The work specified in this Section includes the temporary shoring and bracing for excavations including the trench excavation safety systems as shown on the Plans and as specified herein.

1.02 WORK

The extent of temporary shoring and bracing work includes, but is not limited to:

- A. Temporary shoring and bracing necessary to protect the following against loss of ground or caving embankments: existing structures, buildings, roads, walkways, utilities, electrical transmission towers and support wiring, other facilities and improvements where required to comply with codes and authorities having jurisdiction.
- B. Trench excavation safety systems, pursuant to RCW Chapter 49.17 and WAC 296-155-655.
- C. Maintenance of shoring and bracing.

1.03 QUALITY ASSURANCE

A. SHORING CONSULTANT

The Contractor shall engage the services of a qualified geotechnical engineer and qualified structural engineer registered in the State of Washington to design temporary shoring and bracing when required by applicable regulations.

B. SHORING DESIGN

The Contractor shall provide layout and design drawings and specifications for shoring and bracing when a trench box is inadequate for the purpose or will not be used and trench depth exceeds 4 feet and back sloping will not be used. Temporary shoring and bracing system design and calculations shall be prepared, stamped, and signed by a Professional Engineer registered in the State of Washington. Embedded portions of temporary shoring systems shall not be driven into the ground. Embedded portions shall be installed using drilling.

C. REGULATIONS

The Contractor shall design sheeting, shoring, and bracing in accordance with the Washington State Safety Code and any local codes and ordinances of governing authorities having jurisdiction.

1.04 SUBMITTALS

- A. The Contractor shall submit shoring and bracing layout and design drawings, calculations, and other backup data to the Owner for review in accordance with Section 01 33 00 prior to the start of construction. Submit settlement monitoring and contingency plan for temporary shoring at excavations greater than 10-feet in depth.

1.05 PROJECT CONDITIONS

A. SITE SURVEY

1. The background survey information provided on the Plans is shown for clarity only. The Contractor shall determine, before commencing work, the exact location of all existing features that may be disrupted by new construction, including existing underground utilities. The Contractor shall be fully responsible for any and all damages, which might be caused by the Contractor's failure to exactly locate and/or preserve existing site features. Prior to commencing work, the Contractor shall check and verify governing dimensions and elevations.
2. The Contractor shall survey adjacent structures and facilities, establishing exact elevations at fixed points to act as temporary bench marks to monitor potential settlement from the Contractor's ongoing operations. Clearly identify temporary bench marks and record existing elevations from the control points shown on the Plans.
3. During excavation, the Contractor shall resurvey bench marks weekly. The Contractor shall maintain and make available at the job site an accurate log of surveyed elevations for comparison with original elevations, and promptly notify the Owner if changes in elevations occur or if cracks, sags, or other damage is evident.

PART 2 - PRODUCTS

2.01 MATERIALS

- A. The Contractor shall provide suitable shoring and bracing materials, which shall support loads imposed. Materials for shoring systems need not be new, but shall be in serviceable conditions. Embedded portions of temporary shoring systems shall not be driven into the ground. Embedded portions shall be installed using drilling.

PART 3 - EXECUTION

3.01 VERIFICATION OF CONDITIONS

- A. The Contractor shall notify the Owner immediately if, during construction, subsurface conditions are different from those encountered in the exploratory holes.

3.02 INSTALLATION AND APPLICATION

- A. The Contractor shall provide shoring systems adequately anchored and braced to resist earth and hydrostatic pressures at locations as needed to support excavations during construction. The Contractor shall locate required bracing to clear all permanent work. Bracing which must be relocated shall be installed prior to the removal of original bracing. The Contractor shall not place bracing where it will be cast into or included in permanent concrete work, except as otherwise acceptable to the Owner. The Contractor shall maintain bracing until structural elements are rebraced by other bracing or until permanent construction is able to withstand lateral earth and hydrostatic pressures.

3.03 SETTLEMENT MONITORING

- A. Monitoring shall be performed by the Contractor as required to verify that existing items located near temporary shoring systems are not experiencing settlement. Reference points shall be established and surveyed at sufficient frequency to detect any changes prior to adverse impacts occurring. Upon detection of settlement the contractor shall immediately stop work and implement contingency plans per the Contractor's approved temporary shoring monitoring plan. The responsibility for monitoring as required to protect existing items rests solely on the Contractor. See Section 31 23 19 Dewatering for additional monitoring requirements.

3.04 REMOVAL

- A. The Contractor shall remove shoring and bracing in stages to avoid disturbances to adjacent and underlying soils and damage to structures, pavements, facilities and utilities. The Contractor shall repair or replace, as acceptable to the Owner, adjacent work damaged or displaced through the installation or removal of shoring and bracing work.

3.05 EXCAVATION SAFETY SYSTEMS

- A. All work shall be carried out with due regard for public safety. Open trenches shall have proper barricades and at night they shall be distinctly indicated by adequately placed lights, as provided for elsewhere in the Specifications.
- B. The Contractor is reminded that the Owner has not so delegated, and the Owner's Representative does not purport to be a trench or excavation system safety expert, is not so engaged in that capacity under this Contract, and has neither the authority nor the responsibility to enforce construction, safety laws, rules, regulations, or procedures or to order the stoppage of work for claimed violations of trench or excavation safety.
- C. The furnishing by the Owner of resident representation and inspection personnel shall not make the Owner responsible for the enforcement of such laws, rules, regulations, or procedures, nor shall such make the Owner responsible for construction means, methods, techniques, sequences, procedures, or for the Contractor's failure to properly perform the work necessary for proper trench and excavation safety.

*****END OF SECTION*****

SECTION 32 12 16 – HOT MIX ASPHALT PAVING

PART 1. GENERAL

1.01 DESCRIPTION

- A. The work specified in this section includes the construction of hot mix asphalt (HMA) paving as indicated on the Plans in within this Specification. This section also includes temporary asphalt cold-mix for temporary repairs.
- B. Work shall also include the restoration of all asphalt surfaces excavated or disturbed during construction to a condition acceptable to the Owner.
- C. The work in this section shall be accomplished in accordance with the Standard Specifications for Road, Bridge and Municipal Construction, 2022 edition, as issued by the Washington State Department of Transportation (WSDOT) and the American Public Works Association (APWA), Washington State Chapter (hereafter “Standard Specifications”). The Standard Specifications, as modified or supplemented by the Amendments to the Standard Specifications and Technical Specifications shall govern the Work.

1.02 SUBMITTALS

- A. The Contractor shall provide certificates of laboratory tests indicating current sieve analysis data and mix design for asphalt-treated base and hot mix asphalt pavement mix designs in accordance with Section 01 33 00 and as further specified herein. The certificates shall be provided to the Owner at least 5 consecutive calendar days prior to placement of any materials.

PART 2. PRODUCTS

2.01 HOT MIX ASPHALT PAVEMENT

- A. Hot mix asphalt pavement, HMA, CL. 1/2" PG 58H-22, shall conform to Section 5-04.2 of the WSDOT Standard Specifications, 91% minimum density, commercial classification. Prior to the production of HMA, the Contractor shall determine a design aggregate structure and asphalt binder content in accordance with WSDOT Standard Operating Procedure 732. Once the design aggregate structure and asphalt binder content have been determined, the Contractor shall submit the HMA mix design on DOT form 350-042 demonstrating the design meets the requirements of Sections 9-03.8(2) and 9-03.8(6) of the WSDOT Standard Specifications. Mix designs shall be accepted by commercial evaluation. The contractor shall only complete the first page of form 350-042. The contractor shall provide verification of mix design in one of the following processes:
 - 1. Submit samples to WSDOT State Materials Lab for WSDOT verification testing in accordance with WSDOT Standard Specifications.
 - 2. Reference a mix design that has been previously verified by the WSDOT Field Verification Testing Process or verified by WSDOT State Materials Lab on a previous project.

2.02 COLD MIX TEMPORARY REPAIR

- A. Cold-mix material shall be MC-2 asphaltic concrete commonly referred to as “cold-mix”.

PART 3. EXECUTION

3.01 GENERAL

- A. The Contractor shall maintain access to the facility at all times. The Contractor shall coordinate all work with the Owner to ensure the paving plan does not interfere with the Owners on-going operations.
- B. The Contractor shall provide, place and maintain all temporary markings and signage as required to warn and direct facility traffic as necessary during his paving operations.

3.02 ASPHALT CONCRETE PAVEMENT PLACEMENT

- A. Asphalt concrete pavement materials shall be placed on compacted subgrade materials, as shown on the Plans, as indicated elsewhere in these Specifications, and in conformance with the WSDOT Standard Specifications. All existing asphalt pavement excavated or disturbed during construction shall be removed and replaced per this Specification.
- B. No material transfer devices or vehicles shall be used on this project.
- C. The asphalt shall be placed and compacted in 2-inch lifts.
- D. Asphalt pavement replacing existing asphalt shall match existing asphalt thickness or the minimum pavement repair section indicated on the plans whichever is thicker.

3.03 COLD MIX TEMPORARY REPAIR

- A. During the course of construction, it may be necessary to provide temporary vehicle and/or pedestrian access within the project limits. Such temporary access shall be provided by temporarily patching trench crossings or other areas with cold mix, until such time as the permanent surface restoration is installed. Locations shall include those areas specifically indicated on the Plans, directed by the Owner or as further specified herein. This material will be furnished, placed, compacted, and removed and wastehauded at various locations throughout the project. The trenches and/or subgrade shall be thoroughly compacted and brought to a smooth grade prior to placing the material. It shall be placed, maintained (daily), and removed and wastehauded by the Contractor. Provide a minimum compacted thickness of 2 inches.

3.04 ASPHALT JOINTS

- A. All joints of hot mix asphalt pavement shall be sealed with hot poured sealant meeting the requirements of WSDOT Standard Specification 9-04.2

3.05 QUALITY CONTROL

- A. The Contractor shall be responsible for testing the ATB and HMA paving in accordance with the WSDOT Standard Specifications as specified herein, and with Section 01400. Testing shall include asphalt content and grading testing of hot mix asphalt mix samples, aggregate void content, fracture, and equivalence testing, and in-place density testing.

3.06 SAWCUTTING

- A. Where shown on the Plans or where directed in the field by the Owner, the Contractor shall make a neat vertical sawcut at the boundaries of the area to be removed. Care shall be taken during saw-cutting so as to prevent damage to the existing asphalt concrete, or

concrete, to remain in place. Any pavement or concrete damaged by the Contractor outside the area scheduled for removal due to the Contractor's operations or negligence shall be repaired or replaced to the Owner's satisfaction by the Contractor at no additional cost to the Owner.

All cuts shall be continuous, full depth, and shall be made with saws specifically equipped for this purpose. No skip cutting, wheel cutting or jack hammering will be allowed unless specifically approved otherwise in writing by the Owner.

However, even if preapproved as a method of cutting, no payment will be made for this type of work, and it shall be considered incidental and included in the various unit contract and lump sum prices listed in the Proposal.

The location of all pavement cuts shall be preapproved by the Owner in the field before cutting commences.

All water and slurry material resulting from sawcutting operations shall not be allowed to enter the storm drainage or sanitary sewer system and shall be removed from the site and disposed of in accordance with the Washington State Department of Ecology regulations.

All existing asphalt concrete pavement edges shall be saw cut back to sound material, in uniform lines immediately prior to paving operations. Any edges broken between the time of cutting and placement of new paving shall be recut to the satisfaction of the Owner at no additional cost to the Owner. All excess excavated materials shall be hauled to waste.

3.07 ASPHALT TRENCH PATCH

- A. This work shall consist of the preparation, placing and compaction of asphalt trench sections, in accordance with the details included on the plans and the requirements outlined herein. The work shall be in conformance with this specification unless specifically directed otherwise by the Owner.

The Contractor shall restore all asphalt surfaces excavated or disturbed to a condition acceptable to the Owner.

Before any HMA material is placed, all pavement cuts shall be trued so that marginal lines of the patch will form a rectangle with straight edges and vertical faces a minimum of 1 foot back from the maximum trench width.

The asphalt shall be placed and compacted in 2-inch lifts. The asphalt trench patch thickness shall match existing asphalt thickness or the minimum pavement repair section indicated on the plans whichever is thicker.

Seal all joints scheduled to not receive an asphalt overlay.

3.08 ASPHALT OVERLAY

- A. The Contractor shall overlay all pavement areas as shown on the Plans or as required by the Owner. The existing asphalt surface shall be prepared in conformance with WSDOT Standard Specifications Section 5-04.3(5)A.

The asphalt work shall be in conformance with this specification unless specifically directed otherwise by the Owner.

3.09 ADJUSTING STRUCTURES TO GRADE

- A. All utility castings and monuments within the existing and/or new pavement area shall be referenced by the Contractor prior to any pavement removal or planning. The Contractor shall keep a record of such references, and submit a copy to the Owner.

Existing structures and new structures shall be adjusted to the finished grade as shown on the Plans and as further specified herein. Existing boxes, rings, grates, covers, and lids shall be reset in a careful and workmanlike manner to conform to the required grades.

The new and existing utility castings and monuments shall be adjusted to grade in the following manner:

As soon as the street or driveway has been paved past each structure or casting, the asphalt concrete mat shall be scored around the location of the structure or casting. After rolling has been completed and the mat has cooled, it shall be cut along the scored lines. The structure or casting shall then be raised to finished pavement grade and the annular spaces filled as indicated on the Plans. The Contractor shall install the pavement to give a smooth finished appearance. All covers, lids, frames, and grates shall be thoroughly cleaned.

After pavement is in place, all new pavement joints shall be sealed with a 6-inch wide strip of hot asphalt sealer. A sand blanket shall be applied to the surface of the hot asphalt sealer immediately after the placement of the sealer to help alleviate the tracking of the asphalt. The sealer shall meet the requirements of Section 9-04.2 of the WSDOT Standard Specifications.

*****END OF SECTION*****

SECTION 32 32 23 – GRAVITY BLOCK RETAINING WALL

Part 1: GENERAL

1.01 SECTION INCLUDES

- A. Work includes designing, furnishing and construction of a Gravity Block Retaining Wall.

1.02 REFERENCES

- A. American Association of State Highway and Transportation Officials (AASHTO)
 - 1. AASHTO M 252 Corrugated Polyethylene Drainage Pipe
 - 2. AASHTO M 288 Geotextile Specification for Highway Applications
- B. American Society for Testing and Materials (ASTM)
 - 1. ASTM C140 Sampling and Testing Concrete Masonry Units
 - 2. ASTM C1372 Specification for Dry-Cast Segmental Retaining Wall Units
 - 3. ASTM D442 Particle Size Analysis of Soils
 - 4. ASTM D698 Laboratory Compaction Characteristics of Soil – Standard Effort
 - 5. ASTM D1556 Standard Test Method for Density and Unit Weight of Soil In Place by the Sand Cone Method
 - 6. ASTM D1557 Laboratory Compaction Characteristics of Soil – Modified Effort
 - 7. ASTM D2487 Standard Classification of Soils for Engineering Purposes (Unified Soil Classification System)
 - 8. ASTM D2922 Standard Test Methods for Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
 - 9. ASTM D3034 Standard Specification for Type PSM Poly (Vinyl Chloride) (PVC) Sewer pipe and Fittings
 - 10. ASTM D4318 Liquid Limit, Plastic Limit and Plasticity Index of Soils
 - 11. ASTM D4475 Horizontal Shear Strength of Pultruded Reinforced Plastic Rods
 - 12. ASTM D4476 Flexural Properties of Fiber Reinforced Pultruded Plastic Rods
 - 13. ASTM D4595 Standard Test Method for Tensile Properties of Geotextiles by Wide-Width Strip Method
 - 14. ASTM D4873 Standard Guide for Identification, Storage and Handling of Geosynthetics
 - 15. ASTM D5262 Standard Test Method for Evaluating the Unconfined Tension Creep Behavior of Geosynthetics
 - 16. ASTM D5321 Standard Test Method for Determining the Coefficient of Soil and Geosynthetic or Geosynthetic and Geosynthetic Friction by the Direct Shear Method
 - 17. ASTM D5818 Standard Practice for Obtaining Samples of Geosynthetics from a Test Section for Assessment of Installation Damage
 - 18. ASTM D6637 Standard Test Method for Determining Tensile Properties of Geogrids by the Single or Multi-Rib Method
 - 19. ASTM D6638 Standard Test Method for Determining Connection Strength Between Geosynthetic Reinforcement and Segmental Concrete Units
 - 20. ASTM D6706 Standard Test Method for Measuring Geosynthetic Pullout Resistance in Soil
 - 21. ASTM D6916 Standard Test Method for Determining the Shear Strength Between Segmental Concrete Units

C. National Concrete Masonry Association (NCMA)

1. NCMA SRWU-1 Test Method for Determining Connection Strength of SRW
2. NCMA SRWU-2 Test Method for Determining Shear Strength of SRW

1.03 DEFINITIONS

- A. Concrete Unit – a dry-stacked concrete retaining wall unit machine made from Portland cement, water, and aggregates.
- B. Unit Fill – aggregate that is placed within the concrete units.
- C. Retained Soil – the soil mass behind the reinforced backfill.
- D. Foundation Soil – the soil mass below the leveling pad and reinforced backfill.
- E. Leveling Pad – crushed stone, sand and gravel or unreinforced concrete material placed to provide a level surface for placement of the concrete units.

1.04 SUBMITTALS AND CERTIFICATION

- A. Contractor shall submit a Manufacturer’s certification, prior to the start of work, that the retaining wall system components meet the requirements of this specification and the structure design.
- B. Contractor shall submit construction drawings and design calculations for the retaining wall system prepared and stamped by a Professional Engineer registered in the state of Washington.

1.05 QUALITY ASSURANCE

- A. Manufacturer/Installer shall be a Specialist.
- B. Regulatory Requirements - See Section 01 41 00.

1.06 DELIVERY HANDLING AND STORAGE

- A. Contractor shall check all materials upon delivery to assure that the proper type, grade, color, and certification have been received.
- B. Contractor shall protect all materials from damage due to jobsite conditions and in accordance with manufacturer's recommendations. Damaged materials shall not be incorporated into the work.

PART 2: PRODUCTS

2.01 GENERAL

- A. Comply with “Quality Assurance” provisions, “References,” and Specifications. Where these may be in conflict, the more stringent requirements govern.

2.02 CONCRETE UNITS

- A. Concrete units shall be Keystone Compaq II, or approved equal, and conform to the following architectural requirements
 - 1. Face color - concrete gray, unless otherwise specified. The Owner may specify standard manufacturers' color.
 - 2. Finish shall be straight split face. Other face finishes will not be allowed without written approval of Owner.
 - 3. Bond configuration - running with bonds nominally located at midpoint in vertically adjacent units.
 - 4. Exposed surfaces of units shall be free of chips, cracks or other imperfections when viewed from a distance of 20 feet (6 m) under diffused lighting.
- B. Concrete units shall conform to the requirements of ASTM C1372 - Standard Specifications for Segmental Retaining Wall Units.
- C. Concrete units shall conform to the following structural and geometric requirements measured in accordance with ASTM C140 Sampling and Testing Concrete Masonry Units:
 - 1. Compressive strength: ≥ 3000 psi (21 MPa).
 - 2. Absorption: ≤ 8 % for standard weight aggregates.
 - 3. Dimensional tolerances: $\pm 1/8$ " (3 mm) from nominal unit dimensions not including rough split face.
 - 4. Unit Size: 8" (203 mm) (H) x 18" (457 mm) (W) x 12" (304 mm)(D) minimum.
- D. Concrete units shall conform to the following constructability requirements:
 - 1. Vertical setback: $1/8$ inch (3 mm) $\pm$ per course (near vertical) or $1 1/8$ inch (28 mm) + per course, per the design.
 - 2. Alignment and grid attachment mechanism - fiberglass pins, two per unit.
 - 3. Maximum horizontal gap between erected units shall be $\leq 1/2$ inch (13 mm).

2.03 SHEAR AND REINFORCEMENT PIN CONNECTORS

- A. Shear and reinforcement pin connectors shall be 1/2-inch (12 mm) diameter thermoset isophthalic polyester resin pultruded fiberglass reinforcement rods to provide connection between vertically and horizontally adjacent units and geosynthetic reinforcement, with the following requirements:
 - 1. Flexural Strength in accordance with ASTM D4476: 128,000 psi (882 MPa) minimum.
 - 2. Short Beam Shear in accordance with ASTM D4475: 6,400 psi (44 MPa) minimum.

2.04 LEVELING PAD MATERIAL

- A. Material shall be Crushed Surfacing Top Course per WSDOT 9-03.9(3) or approved equal.

2.05 UNIT FILL

- A. Material shall be Crushed Surfacing Top Course per WSDOT 9-03.9(3) or approved equal.
- B. Unit fill shall be placed within the cores of, and between, the units.

PART 3: EXECUTION

3.01 EXCAVATION

- A. Contractor shall excavate to the lines and grades shown on the construction drawings. The Owner or Contractors QA/QC representative shall inspect the excavation and test the foundation soils and approve prior to placement of the leveling pad material or fill soils.

3.02 LEVELING PAD

- A. Leveling pad material shall be placed to the lines and grades shown on the construction drawings to a minimum thickness of 6 inches (150 mm) and extend laterally a minimum of 6 inches in front and behind the Keystone wall unit.
- B. Soil leveling pad materials shall be compacted to a minimum of 95% of Standard Proctor density per ASTM D697 or 92% Modified Proctor density per ASTM D1557.
- C. Leveling pad shall be prepared to insure full contact with the base surface of the concrete units.

3.03 CONCRETE UNIT INSTALLATION

- A. First course of units shall be placed on the leveling pad at the appropriate line and grade. Alignment and level shall be checked in all directions and insure that all units are in full contact with the base and properly seated.
- B. Place the front of units side-by-side. Do not leave gaps between adjacent units. Layout of corners and curves shall be in accordance with manufacturer's recommendations.

- C. Install shear/connecting pins per manufacturer's recommendations.
- D. Place and compact unit fill within and between wall units. Place footing drain and wall backfill.
- E. Maximum stacked vertical height of wall units, prior to unit fill and backfill placement and compaction, shall not exceed two courses.

3.04 CAP INSTALLATION

- A. Prior to placement of the cap units, the upper surface of the top course of wall units shall be cleaned of soil and any other material.
- B. Cap units shall be adequately glued to the underlying wall units with an all-weather exterior construction adhesive.

3.05 AS-BUILT CONSTRUCTION TOLERANCES

- A. Vertical alignment: ± 1.5 inches (40 mm) over any 10 foot (3 m) distance.
- B. Wall batter: within 2 degrees of design batter. Overall wall batter shall be ≥ 0 degrees.
- C. Horizontal alignment: ± 1.5 inches (40 mm) over any 10 foot (3 m) distance.
- D. Corners and curves: ± 1 foot (300 mm) to theoretical location.
- E. Maximum horizontal gap between erected units shall be $\leq 1/2$ inch (13 mm).

*****END OF SECTION*****

SECTION 33 05 00 - COMMON WORK RESULTS FOR UTILITIES

PART 1. GENERAL

1.01 SECTION INCLUDES

- A. This section covers information supplementary to the Drawings and the WSDOT Standard Specifications. The Contractor shall furnish and install pipe and fittings as shown on the Drawings, as specified in these Specifications, and as required for a complete and functional installation. All pipe and fittings shall be new.
- B. Piping systems, including pipe, fittings, anchors, and all other elements, shall be detailed, fabricated, and installed to resist all internal and external loads which will be imposed upon them. Pressure ratings and materials stated in these Piping Specification sections are minimum acceptable standards. Systems shall be suitable for the service intended.
- C. The pipe diameters shown on the Drawings and used in these Specifications are inside diameters unless specific reference is made to outside diameter of the pipe or the pipe is a standardized product normally designated by a nominal size, e.g., ductile iron pipe.
- D. The Contractor shall furnish and install pipe and fittings as shown on the Drawings, as specified in these Specifications, and as required for a complete and functional installation. The pipe shall be new, manufactured in accordance with these Specifications and Drawings.
- E. Work includes but is not limited to following:
 - 1. Trenching, unsuitable trench bottom overexcavation, bedding, backfilling, compacting and disposal of excess materials as required for installation of all underground utilities, conduit and other miscellaneous structures.
 - 2. Providing all material, equipment and labor necessary to complete the excavation and backfill operations necessary to install the underground utilities depicted on the plans.
 - 3. Trench dewatering and pumping.

1.02 RELATED SECTIONS

- A. Coordinate related work specified in other parts of the Project Manual, including but not limited to following:
 - 1. Section 31 23 33 - Trenching and Backfilling
 - 2. Section 33 31 00 – Wastewater Pipe, Fittings and Hoses

1.03 REFERENCES

- A. WSDOT Standard Specifications for Road, Bridge and Municipal Construction, Latest Edition; Sections 2-09 - Structure Excavation, and 7-08 - General Pipe Installation Requirements.
- B. WSDOT Standard Plans for Road Bridge and Municipal Construction, most recent edition.
- C. Washington State Department of Ecology's Latest Edition of the Stormwater Management Manual for Western Washington.

1.04 SUBMITTALS PRIOR TO CONSTRUCTION

- A. Submittals during construction shall be made in accordance with Part 3, Section 01 33 00. In addition, the following specific information shall be provided.
 - 1. Field Fabrication - For pipe lines which are assembled in the field from standard fittings, submit complete data on pipe and fittings, including any manufacturer's installation instructions.
 - 2. Material Certification - Certification of all materials, and manufacturing properly executed by the manufacturer, shall be available to show compliance with the Specification of materials being furnished. Test data on tests performed shall be provided as requested by the Engineer.
 - 3. Existing Pipe Verification - Expose all existing pipes which are to be connected to new pipelines. Verify the size, material, elevation, horizontal location, and pipe service of these existing pipes with sufficient lead time to make accommodations as necessary.

1.05 SYSTEM DESCRIPTION

- A. Provide protection of existing utilities affected by the work and make every effort to minimize disruptions to all utility services.
- B. If, during the course of construction, excavation interrupts traffic for longer than 15 minutes, inform the Owner and make all necessary arrangements for ingress and egress. In the event of such road closures, the Contractor shall be solely responsible for all traffic control measures including but not limited to flagging, barricades and cones.
- C. Trenches shall be closed or covered with steel plates at the end of each work day.

1.06 QUALITY ASSURANCE

- A. Regulatory Requirements:
 - 1. See referenced Codes, ordinances and the like/Section 01 30 00.
 - 2. Comply with the requirements of all applicable regulatory agencies having jurisdiction over this work including the Washington Industrial Safety and Health Act.

PART 2. PRODUCTS

2.01 BOLTS AND NUTS

- A. Bolts and nuts shall be Type 316 stainless steel.

2.02 RUBBER GASKETS

- A. Store all rubber gaskets in a cool, well-ventilated place, and do not expose to the direct rays of the sun. Do not allow contact with oils, fuels, or petroleum solvents.

2.03 JOINT LUBRICANT

- A. Furnish joint lubricant with the pipe. Furnish the amount and type recommended by the pipe manufacturer. The lubricant shall be a water-soluble, nontoxic, vegetable soap compound conforming to United States Pharmacopoeia No. P39.

2.04 TESTING

- A. Provide all hoses, plugs, and other necessary equipment to complete the tests.

2.05 CONCRETE FOR THRUST BLOCKING, THRUST TIES AND ENCASEMENT

- A. The concrete for all thrust blocking and thrust ties shall develop a minimum compressive strength of 3,000 psi at 28 days.

PART 3. EXECUTION

3.01 EXAMINATION

- A. Verify installation conditions as satisfactory to receive work of this Section. Do not install until unsatisfactory conditions are corrected.

3.02 PREPARATION

- A. Utility Location. Make every effort to identify the location of all existing underground utilities. Contact Utility Locate Service 48-hours in advance of any excavation.

3.03 ELECTRICAL AND WATER SERVICE

- A. Contractor to protect existing electrical and water services as outlined on the Contract Plans. Coordinate with OPALCO, Eastsound Water Users Association, Eastsound Sewer and Water District, and the Engineer at least three business days in advance of any underground work in the vicinity of existing electrical services and/or existing water services.

3.04 TRENCH EXCAVATION AND BACKFILL

- A. Trench Excavation and Backfill.
 - 1. All trench excavation and backfill shall be performed in accordance with Section 31 23 33 – Trenching and Backfill.

3.05 SHIPPING AND HANDLING MATERIALS

- A. During transportation, unloading, and storage, pipe and materials shall be protected, supported, and handled in a manner to prevent damage to the materials, especially linings and coatings. Only implements and equipment suitable for proper and safe handling of the materials shall be used. Fabric slings shall be used to lift pipe and fittings, not chains or cables.

3.06 PIPE PREPARATION AND HANDLING

- A. Each pipe and fitting shall be carefully inspected before the exposed pipe or fitting is installed or the buried pipe or fitting is lowered into the trench. The interior and exterior protective coating shall be inspected, and all damaged areas patched in the field with material similar to the original, except damaged glass-lined pipe. Any damaged glass-lined pipe shall not be used and shall be promptly removed from the site. Any pipe which, in the opinion of the Engineer, is damaged beyond repair shall be removed from the site and replaced with another unit. Clean ends of pipe thoroughly. Remove foreign matter and dirt from inside of pipe and keep clean during and after installation.
- B. Use proper implements, tools, and facilities for the safe and proper protection of the pipe. Carefully handle pipe in such a manner as to avoid any physical damage to the pipe. Do not drop or dump pipe from trucks or into trenches under any circumstances.

- C. All pipe fittings and appurtenances shall be installed in accordance with the manufacturer's instructions and these Specifications.

3.07 CUTTING AND FABRICATING

- A. Cut pipe with approved cutters, do not flame cut except for mild steel pipe. Cut perpendicular to axis of pipe. Dress ends to suit type of joint being made, removing burrs, mill scale, and debris before making up. Repair damaged linings and coatings.

3.08 BELL HOLES

- A. Excavate bell holes at each joint to permit proper assembly and visual and feeler gauge inspection of the entire joint.

3.09 EXPANSION PROVISIONS

- A. Provisions shall be made for the expansion and contraction which may occur in pipe due to temperature change. Pipe expansion provisions may not be detailed on the Drawings. The absence of these details on any Drawing shall not relieve the Contractor of the responsibility for providing them where required, and at his sole expense.

3.010 PIPE IN CONCRETE ENCASEMENTS OR CONCRETE BEDDING

- A. Except for welded joints, pipe joints shall not be encased in concrete unless specifically required on the Drawings. Pipe coatings shall be continuous through concrete encasements, thrust blocks, anchors, collars, etc., unless otherwise shown on the Drawings.

3.011 FLEXIBLE JOINTS AT CONCRETE BACKFILL OR ENCASEMENT

- A. Except for welded joint pipe, a flexible joint shall be provided within 18 inches or one-half the pipe diameter, whichever is less, from the terminations of any concrete backfill, or concrete encasement.

3.012 FLEXIBLE JOINTS AT CONCRETE STRUCTURES

- A. A flexible joint shall be provided near the exterior face of all structures. The joint may be flush with the face, may be up to one half pipe diameter away from the face, but shall not be more than 18 inches away from the face.

3.013 LINE AND GRADE

- A. Grade the bottom of the trench by hand, if necessary, to the line and grade to which the pipe is to be laid, with proper allowance for pipe thickness and for base. Remove hard spots that would prevent a uniform thickness of bedding or cause non-uniform pressure on the pipe barrel.
- B. Lay pipe to a uniform grade between indicated elevations. Do not deviate more than 1 inch from line or 1/4 inch from established grade. Measure for grade at the pipe invert.
- C. Before laying each section of pipe, check the grade with a straightedge and correct any irregularities found. The trench bottom with bedding shall form a continuous and uniform bearing and support for the pipe at every point between bell holes, except that the grade may be disturbed for the removal of lifting tackle.

3.014 PERMISSIBLE DEFLECTION AT JOINTS

- A. Wherever it is necessary to deflect pipe from a straight line, either in the vertical or horizontal plane, do not exceed the amount of deflection recommended by the pipe or coupling manufacturer.

3.015 L LAYING AND JOINTING PIPE AND FITTINGS

- A. After a section of pipe has been lowered into the prepared trench with bedding, clean the end of the pipe to be joined, the inside of the joint, and the rubber ring immediately before joining the pipe. Make assembly of the joint in accordance with the recommendations of the manufacturer. Provide all special tools and appliances required for the jointing assembly.
- B. The gasket position shall be checked with a feeler gauge, furnished by the pipe manufacturer, to assure proper seating. After the joint has been made, check pipe for alignment and grade. Provide sufficient pressure in making the joint to assure that the joint is "home", as defined in the standard installation instructions provided by the pipe manufacturer. To assure proper pipe alignment and joint makeup, place sufficient pipe zone material to secure the pipe from movement before the next joint is installed.
- C. Take the necessary precautions required to prevent excavated or other foreign material from entering the pipe during the laying operation. At all times, when laying operations are not in progress, at the close of the day's work, or whenever the workmen are absent from the job, close and block the open end of the last laid section of pipe to prevent entry of foreign material or creep of the gasketed joints.
- D. Take all precautions necessary to prevent the "uplift" or floating of the line prior to the completion of the backfilling operation.
- E. When cutting and/or machining the pipe is necessary, use only tools and methods recommended by the pipe manufacturer.

3.016 UNSUITABLE CONDITIONS FOR LAYING PIPE

- A. Do not lay pipe in water, or when in the opinion of the Engineer, trench conditions are unsuitable.

3.017 PREVENTING TRENCH WATER FROM ENTERING PIPE

- A. When the pipe laying is not in progress, close the open ends of pipe by approved means, and do not permit trench water or other foreign material to enter the pipe. Keep water out of the trench.

3.018 LOCATION OF THRUST RESTRAINT

- A. All pipeline tees, plugs, caps, bends, and other locations where unbalanced forces exist shall be thrust restrained. Thrust blocks and ties are not shown on the Drawings. The Contractor is responsible for providing thrust restraint as incidental items in the Contract Price.

3.019 THRUST TIES

- A. The Contractor shall install thrust ties at all locations necessary to restrain thrust forces. Such thrust ties are subject to the review and approval of the Engineer. The concrete for all thrust ties shall develop a minimum compressive strength of 3,000 psi at 28 days.

3.020 THRUST BLOCKING

- A. Thrust blocking shall be adequate to withstand hydrostatic test pressures as well as normal operating pressures. Place thrust blocking between undisturbed ground and the fitting to be anchored. Place the blocking so that the pipe and fitting joints will be accessible for repairs. The concrete for all thrust blocks shall develop a minimum compressive strength of 3,000 psi at 28 days.

- B. For gravity or low pressure pipeline, when the bearing surface of the fitting against the soil provides an area equal to or greater than the area required for thrust restraint, concrete thrust blocks will no longer be required. The bearing area for fittings without thrust blocks will be determined by the projected area of 70 percent of the internal diameter multiplied by the chord length for the curve along the centerline of the fitting.

3.021 INSTALLATION OF EXPOSED PIPING

- A. Unless shown otherwise, piping shall be parallel to building lines. Hangers on adjacent piping shall be aligned where possible on common size ranges.
- B. All pipe flanges shall be set level, plumb, and aligned. All flanged fittings shall be true and perpendicular to the axis of the pipe. All bolt holes in flanges shall straddle vertical centerline of pipes.
- C. Unions shall be installed where required for piping or equipment installation, even though they are not shown on the Drawings.
- D. Piping shall be installed without springing or forcing the pipe in a manner which would set up stresses in the pipe, valves, or connected equipment.

3.022 ANCHORAGE AND EXPANSION PROVISIONS

- A. All piping shall be anchored against thrust developed by internal pressures. In addition, provisions shall be made for the expansion and contraction which may occur in pipe due to temperature change. Pipe anchorage and expansion provisions are not completely detailed on the Drawings. The absence of these details on any Drawings shall not relieve the Contractor of the responsibility for providing them where required.

3.023 INSTALLATION OF FLEXIBLE COUPLINGS, FLANGED COUPLING ADAPTERS, AND SERVICE SADDLES

- A. Prior to installation, thoroughly clean oil, scale, rust, and dirt from the pipe to provide a clean seat for the gasket. Care shall be taken that the gaskets are wiped clean before they are installed. If necessary, flexible couplings and flanged coupling adapter gaskets may be lubricated with soapy water or manufacturer's standard lubricant before installation on the pipe ends. Install in accordance with the manufacturer's recommendations. Bolts shall be tightened progressively, drawing up bolts on opposite sides a little at a time until all bolts have a uniform tightness. Workmen tightening bolts shall use torque-limiting wrenches.
- B. Flexible couplings with tie rods may be used to tie pipes against thrust. They shall not be used, with or without tie rods, as expansion joints on pipelines with cyclic temperature changes. Readjust tie rod tension after initial filling before pressure testing.

3.024 CORROSION PROTECTION OF PIPE AND ACCESSORIES

- A. Not all corrosion protection details are included, either on the Drawings or in the Specifications. The absence of specific details on corrosion and environmental protection measures shall not relieve the Contractor of the responsibility of providing them, all as part of the Contract price.

3.025 CORROSION PROTECTION FOR BURIED PIPE ACCESSORIES

- A. All buried pipe appurtenances made of steel shall have corrosion protection. Tie rods and similar items shall be heat shrink tube wrapped. Flange bolts, nuts, and similar items shall be coated with a bituminous paint or equal. Flexible couplings, grooved couplings, and similar items shall be heat shrink wrapped or cement coated.

- B. Buried valves and similar elements on wrapped pipelines shall be bituminous paint-coated. On ductile iron or nonmetallic pipelines they shall have exposed nuts and bolts bituminous paint-coated and the entire valve wrapped in 8-mil polyethylene as specified for ductile iron pipe. On cement-coated pipelines they shall be cement-coated similar to detail shown for couplings.

*****END OF SECTION*****

SECTION 33 30 00 – PIPING SYSTEMS

PART 1. GENERAL

1.01 SCOPE OF WORK

- A. This section covers the work necessary to furnish and install the piping systems. The pipe system to be used for each process system is shown on the plans.
- B. For trench excavation, backfill, pipe bedding, and material placed in the pipe zone, see Section 31 23 33 TRENCHING AND BACKFILL.
- C. See Division 1 GENERAL, which contains information and requirements that apply to the work specified herein and are mandatory for this project.
- D. The Contractor shall furnish and install pipe and fittings as shown on the Drawings and as specified in these Specifications. The pipe shall be new, manufactured in accordance with these Specifications and Drawings.
- E. Piping systems, including pipe, fittings, anchors, and all other elements, shall be detailed, fabricated, and installed to resist all internal and external loads which will be imposed upon them. Pressure ratings and materials stated in these Piping Specification sections are minimum acceptable standards. Systems shall be suitable for the service intended.
- F. The pipe diameters shown on the Drawings and used in these Specifications are inside diameters unless specific reference is made to outside diameter of the pipe or the pipe is a standardized product normally designated by a nominal size, e.g., ductile iron pipe.

1.02 SUBMITTALS DURING CONSTRUCTION

Submittals during construction shall be made in accordance with Section 01 30 00. In addition, the following specific information shall be provided.

A. FIELD FABRICATION

- 1. For pipelines which are assembled in the field from standard fittings, submit complete data on pipe, fittings, linings, coatings, any manufacturer's installation instructions, and any required installer certifications.

B. FABRICATION AND LAYING DRAWINGS

- 1. For shop fabricated piping, the Contractor shall furnish the Engineer with pipe design calculations, the required test data, and shop drawings which shall include a laying plan and details of pipe sections, special fittings, and bends. Dimensions, coatings, and other pertinent information shall be shown. The laying plan shall show the location of each pipe section and each special length, with each piece numbered or otherwise designated in sequence. All outlets and bends shall be made up into special lengths so that, when installed, they will be located as indicated. Each pipe and fitting shall be marked on the outside to indicate the class of pipe, location number on the laying plan, size or diameter, manufacturer's identification, and date of manufacture. Pipe shall be furnished and installed in accordance with the reviewed laying plan.

C. PIPE SUPPORT DRAWINGS

1. Drawings of each major piping system locating each support and hanger. Drawings shall identify the support type by catalog number or shop drawing detail number and show anchor locations, identifying them by shop drawing detail number.

D. MATERIAL CERTIFICATION

1. Certification of all materials, and manufacturing properly executed by the manufacturer, shall be available to show compliance with the Specification of materials being furnished. Test data on tests performed shall be provided as requested by the Engineer.

E. FIELD WELDING PROCEDURE

1. Details of welding procedures for each type of field weld, including base metal, welding method, electrodes, preheating requirements, and other data.

F. PLUMBING CODE

1. All sanitary building drainage and vent systems shall conform to the plumbing laws, rules, and regulations of the state and of the District, whichever represents the higher standard.
2. The Owner will obtain any variances imposed by site constraints.

1.03 QUALITY ASSURANCE

- A. Installer shall be a Specialist.

1.04 PROJECT SITE CONDITIONS

- A. Locate and provide protection of existing utilities affected by the work. Minimize disruptions to access existing facilities.

PART 2. PRODUCTS

2.01 MATERIALS

A. GALVANIZING

1. Where galvanizing is specified it shall be hot-dip applied only. Electroplated zinc or cadmium plating is unacceptable.

B. PAINTING

1. All exposed piping, including piping within vaults, basins, and manholes, and except copper, stainless steel, and galvanized piping, shall be painted as specified in Section 09 90 00 PAINTING.

C. PIPE PENETRATIONS

1. Penetrations shall be constructed per specifications and as detailed on drawings.
2. Penetration reinforcing shall be construction as detailed on structural drawings.

3. ABOVE GRADE SLAB, FLOOR, WALL, AND ROOF PENETRATIONS
 - a. All ductile iron penetrations of slabs, floors, walls, and roofs shall be poured in place ductile iron wall pipe. It shall be the Contractor's responsibility to verify the size and location of all building and structure penetrations prior to pouring concrete. All sleeves shall be supported by form work to prevent contact with the reinforcing steel.
4. STRUCTURAL CONCRETE WALL PENETRATIONS IN NEW BASINS
 - a. Wall and floor penetrations shall use Link-Seal modular seals with 316 Stainless Hardware. Install Link-Seal per manufacturers recommendations and as detailed on the plans. PVC pipe wall and floor penetrations shall use GPK Products sand collars. See structural drawings for reinforcing details.
5. STRUCTURAL CONCRETE WALL PENETRATIONS IN EXISTING BASIN WALLS
 - a. Pipe penetrations through existing walls shall be core drilled. Block out all openings. For pipe diameters 6" or less void between pipe and coring shall be sealed completely with Kryton Kim or approved non-shrink grout. For pipe diameters 8" or more penetration shall be constructed by coring and sealing with Link-Seal modular seals with 316 Stainless Hardware. Install Link-Seal per manufacturers recommendations.
6. PRECAST MANHOLE/VAULTS WALL PENETRATION SEALS
 - a. Manhole pipe wall penetrations shall be constructed by coring and sealing with Link-seal modular seals or Kor-N-Seal flexible boots.
- D. RUBBER GASKET STORAGE
 1. Store all rubber gaskets in a cool, well-ventilated place, and do not expose to the direct rays of the sun. Do not allow contact with oils, fuels, or petroleum solvents.
- E. JOINT LUBRICANT
 1. Furnish joint lubricant with the pipe. Furnish the amount and type recommended by the pipe manufacturer. The lubricant shall be a water-soluble, nontoxic, vegetable soap compound conforming to United States Pharmacopoeia No. P39.
- F. HYDROSTATIC TESTING
 1. Provide all hoses, plugs, and other necessary equipment to complete the tests.

PART 3. EXECUTION

3.01 SHIPPING AND HANDLING MATERIALS

- A. During transportation, unloading, and storage, pipe and materials shall be protected, supported, and handled in a manner to prevent damage to the materials, especially linings and coatings. Only implements and equipment suitable for proper and safe handling of the materials shall be used. Fabric slings shall be used to lift pipe and fittings, not chains or cables.

3.02 INSTALLATION

A. PIPE PREPARATION AND HANDLING

1. Each pipe and fitting shall be carefully inspected before the exposed pipe or fitting is installed or the buried pipe or fitting is lowered into the trench. The interior and exterior protective coating shall be inspected, and all damaged areas patched in the field with material similar to the original, except damaged glass-lined pipe. Any damaged glass-lined pipe shall not be used and shall be promptly removed from the plant site. Any pipe which, in the opinion of the Engineer, is damaged beyond repair shall be removed from the site and replaced with another unit. Clean ends of pipe thoroughly. Remove foreign matter and dirt from inside of pipe and keep clean during and after installation.
2. Use proper implements, tools, and facilities for the safe and proper protection of the pipe. Carefully handle pipe in such a manner as to avoid any physical damage to the pipe. Do not drop or dump pipe from trucks or into trenches under any circumstances. All pipe fittings and appurtenances shall be installed in accordance with the manufacturer's instructions and these Specifications.

B. CUTTING AND FABRICATING

1. Cut pipe with approved cutters, do not flame cut except for mild steel pipe. Cut perpendicular to axis of pipe. Dress ends to suit type of joint being made, removing burrs, mill scale, and debris before making up. Repair damaged linings and coatings.

C. BELL HOLES

1. Excavate bell holes at each joint to permit proper assembly and visual and feeler gauge inspection of the entire joint.

D. EXPANSION PROVISIONS

1. Provisions shall be made for the expansion and contraction which may occur in pipe due to temperature change. Pipe expansion provisions are not completely detailed on the Drawings. The absence of these details on any Drawing shall not relieve the Contractor of the responsibility for providing them where required, and at his sole expense.

E. PIPE IN CONCRETE ENCASEMENTS OR CONCRETE BEDDING

1. Except for welded joints, pipe joints shall not be encased in concrete unless specifically required on the Drawings. Pipe coatings shall be continuous through concrete encasements, thrust blocks, anchors, collars, etc., unless otherwise shown on the Drawings.

F. FLEXIBLE JOINTS AT CONCRETE BACKFILL OR ENCASEMENT

1. Except for welded joint pipe, a flexible joint shall be provided within 18 inches or one-half the pipe diameter, whichever is less, from the terminations of any concrete backfill or concrete encasement.

G. FLEXIBLE JOINTS AT CONCRETE STRUCTURES FOR PRESSURE PIPE

1. Unless shown otherwise on the plans, a flexible joint shall be provided at the face of all manholes or other structures. The joint may be flush with the face, may be up to one half pipe diameter away from the face, but shall not be more than 18 inches away from the face.

2. A second flexible joint shall be provided within 18 inches of the first joint for pipelines smaller than 18 inches in diameter or within one pipe diameter of the first joint for pipelines larger than 18 inches in diameter.
3. Flexible joints may be rubber ring joints, mechanical joints, flexible couplings, grooved couplings, or Brico Depend-o-Lok couplings.

H. LINE AND GRADE

1. Grade the bottom of the trench by hand, if necessary, to the line and grade to which the pipe is to be laid, with proper allowance for pipe thickness and for base. Remove hard spots that would prevent a uniform thickness of base or uniform pressure on the pipe barrel.
2. Lay pipe to a uniform grade between indicated elevations. Do not deviate more than 1 inch from line or 1/4 inch from established grade. Measure for grade at the pipe invert.
3. Before laying each section of pipe, check the grade with as straightedge and correct any irregularities found. The trench bottom shall form a continuous and uniform bearing and support for the pipe at every point between bell holes, except that the grade may be disturbed for the removal of lifting tackle.

I. PERMISSIBLE DEFLECTION AT JOINTS

1. Wherever it is necessary to deflect pipe from a straight line, either in the vertical or horizontal plane, do not exceed 75 percent of the amount of deflection recommended by the pipe or coupling manufacturer.

J. LAYING AND JOINTING PIPE AND FITTINGS

1. After a section of pipe has been lowered into the prepared trench, clean the end of the pipe to be joined, the inside of the joint, and the rubber ring immediately before joining the pipe. Make assembly of the joint in accordance with the recommendations of the manufacturer. Provide all special tools and appliances required for the jointing assembly.
2. The gasket position shall be checked with a feeler gauge, furnished by the pipe manufacturer, to assure proper seating. After the joint has been made, check pipe for alignment and grade. After sufficient pressure in making the joint to assure that the joint is "home", as defined in the standard installation instructions provided by the pipe manufacturer. To assure proper pipe alignment and joint makeup, place sufficient pipe zone material to secure the pipe from movement before the next joint is installed.
3. Take the necessary precautions required to prevent excavated or other foreign material from entering the pipe during the laying operation. At all times, when laying operations are not in progress, at the close of the day's work, or whenever the workmen are absent from the job, close and block the open end of the last laid section of pipe to prevent entry of foreign material or creep of the gasketed joints.
4. Take all precautions necessary to prevent the "uplift" or floating of the line prior to the completion of the backfilling operation.
5. When cutting and/or machining the pipe is necessary, use only tools and methods recommended by the pipe manufacturer.

K. UNSUITABLE CONDITIONS FOR LAYING PIPE

1. Do not lay pipe in water, or when in the opinion of the Engineer, trench conditions are unsuitable.

L. PREVENTING TRENCH WATER FROM ENTERING PIPE

1. When the pipe laying is not in progress, close the open ends of pipe by approved means, and do not permit trench water or other foreign material to enter the pipe. Keep water out of the trench.

M. INSTALLATION OF EXPOSED PIPING

1. Unless shown otherwise, piping shall be parallel to building lines. Hangers on adjacent piping shall be aligned where possible on common size ranges.
2. All pipe flanges shall be set level, plumb, and aligned. All flanged fittings shall be true and perpendicular to the axis of the pipe. All bolt holes in flanges shall straddle vertical centerline of pipes.
3. Unions shall be installed where required for piping or equipment installation, even though they are not shown on the Drawings.
4. Piping shall be installed without springing or forcing the pipe in a manner which would set up stresses in the pipe, valves, or connected equipment.
5. Required straight runs of piping upstream and downstream of flow measuring devices shall be smooth.

N. ANCHORAGE AND EXPANSION PROVISIONS

1. All piping shall be anchored against thrust developed by internal pressures. In addition, provisions shall be made for the expansion and contraction which may occur in pipe due to temperature change. Pipe anchorage and expansion provisions are not completely detailed on the Drawings. The absence of these details on any Drawings shall not relieve the Contractor of the responsibility for providing them where required.

O. INSTALLATION OF FLEXIBLE COUPLINGS, FLANGED COUPLING ADAPTERS, GROOVED JOINT COUPLINGS, DEPEND-O-LOK COUPLINGS, AND SERVICE SADDLES

1. Prior to installation, thoroughly clean oil, scale, rust, and dirt from the pipe to provide a clean seat for the gasket. Care shall be taken that the gaskets are wiped clean before they are installed. If necessary, flexible couplings and flanged coupling adapter gaskets may be lubricated with soapy water or manufacturer's standard lubricant before installation on the pipe ends. Install in accordance with the manufacturer's recommendations. Bolts shall be tightened progressively, drawing up bolts on opposite sides a little at a time until all bolts have a uniform tightness. Workmen tightening bolts shall use torque-limiting wrenches.
2. Flexible couplings with tie rods may be used to tie pipes against thrust. They shall not be used, with or without tie rods, as expansion joints on pipelines with cyclic temperature changes. Readjust tie rod tension after initial filling before pressure testing. Depend-o-Lok type FxF couplings as manufactured by Brico Industries may be used in lieu of flexible couplings with tie rods.

P. CORROSION PROTECTION OF PIPE AND ACCESSORIES

1. Not all corrosion protection details are included, either on the Drawings or in the Specifications. The absence of specific details on corrosion and environmental protection measures shall not relieve the Contractor of the responsibility of providing them, all as part of the Contract price.

Q. CORROSION PROTECTION FOR BURIED PIPE ACCESSORIES

1. All buried pipe appurtenances made of steel shall have corrosion protection. Tie rods and similar items shall be heat shrink tube wrapped. Flange bolts, nuts, and similar items shall be coated with a bituminous paint or equal. Flexible couplings, grooved couplings, and similar items shall be heat shrink wrapped or cement coated.
2. Buried valves and similar elements on wrapped pipelines shall be bituminous paint-coated. On ductile iron or nonmetallic pipelines they shall have exposed nuts and bolts bituminous paint-coated and the entire valve wrapped in 8-mil polyethylene as specified for ductile iron pipe. On cement-coated pipelines they shall be cement-coated similar to detail shown for couplings.

R. UV PROTECTION FOR EXPOSED PVC PIPE AND PVC ACCESSORIES

1. All PVC & accessories exposed to sunlight must have stabilizers and UV inhibitors to shield against ultraviolet radiation.

3.03 TRACER WIRE FOR PIPE LOCATION

- A. All new piping is to include tracer wire for future pipe location work. Tracer wire to be attached to top of piping, and tied down at 10-ft intervals. Prior to construction, the Contractor is required to provide submittal with all proposed tracer wire box locations.
- B. Tracer Wire - #12 AWG Solid (0.0808" diameter), steel core soft drawn high strength tracer wire, 380# average tensile break load, 30 mil high molecular weight-high density blue polyethylene jacket complying with ASTM-D-1248, 30 volt rating, (Copperhead Industries Part # 1230-HS Blue, or approved equal).
- C. Tracer Wire Color Requirements -
1. Sewer Pipe: Green
 2. Non-Potable Water Pipe: Purple
 3. Potable Water Pipe: Blue
- D. Tracer Box – (Copperhead Industries)
1. Materials used to construct products in above specifies scope shall be non-corrosive or corrosion resistant.
 2. Tube material shall be of high grade ABS, or equivalent rigid plastic that meets or exceeds ASTM D-1788, Type 1 requirements.
 3. Lid material shall be of cast iron or ductile iron. Tensile strength or ductility of such material shall be equal or superior to hi-tensile cast iron ASTM A-126-B requirements.
 4. Lid-locking bolt material shall be made of aluminum material equal or superior to ASTM B-253.

5. Lid-locking mechanism material shall be made of plastic to meet or exceed ASTM A-126-B requirements.
 6. Box shall be designed to be easily detected by magnetic and electronic locators even when box is covered by a minimum of four (4) inches of soil, sod and / or paving material.
 7. A magnet shall be securely attached at the top of the upper tube of the box for locating purposes. Material used to retain magnet in place shall remain effective at minus 15 degrees Fahrenheit. A magnetized lid or magnet attached to the lid is not acceptable.
 8. Locking Mechanism:
 - a. Lid of valve box shall be designed to employ a locking mechanism that will clamp it to the box collar in a closed position.
 - b. Locking mechanism shall incorporate a standard pentagon-shaped head bolt which when measured from flat to vertex shall not be less than 0.830 inches or greater than 0.875 inches.
 - c. Locking mechanism shall be such that the lid cannot be removed without using the proper wrench.
 9. Copperhead Industries Lite Duty Box #LD14 (landscape areas), Copperhead Industries Concrete/Driveway Box #CD14 (concrete pavement), Copperhead Industries Heavy Duty #HS14 (roadway pavement).
- E. Tracer Wire Connectors – Connectors suitable for direct-bury with moisture displacement silicone filled caps for resistance protection, (Copperhead Industries Snakepit connectors, or equal). Connectors are to be equal to 3M DBR wire connectors.

3.04 TESTING

1. All tests shall be conducted in the presence of the Owner or their appointed designee. The Contractor shall provide the Owner 48 hours notice prior to conducting any tests.
2. Isolate new sections of line to be tested by providing temporary valves, caps, plugs and/or by closing permanently installed valves. Test sections shall normally be limited to 1,500 feet in length or as defined below. Plugs or temporary valves used to close the pipe for the hydrostatic test must be securely braced to prevent the unintentional release of a plug or valve which can become a high velocity projectile. Gauges, air piping manifolds, and valves shall be at the top of the ground. No one is permitted to enter an excavation or structure where a plugged pipe is under test pressure.
3. All leaks shall be repaired and remedied, with tests and repairs repeated until the system meets testing requirements to the satisfaction of the Engineer.
4. All water necessary for testing shall be provided by the Contractor.
5. Upon completion of each portion of the system and prior to connecting to any facility impacted by solids, the system shall be thoroughly flushed to remove and catch all foreign matter such as rocks, misplaced tools, and debris.

6. Valves:
 - a. Valves shall be tested at the same time that the adjacent pipeline is pressure tested. Joints and valve mechanism shall show no visible leakage under test or normal operating conditions. The Contractor shall repair joints that show signs of leakage prior to final acceptance. If there are any special parts of control systems or operators that might be damaged by the pipeline test, they shall be properly protected. The Contractor will be responsible for repairing any damage caused by the testing.
 - b. If requested by the Engineer, the valve manufacturer shall furnish an affidavit stating the material options furnished and/or that he has complied with these and other referenced specifications.
 - c. Valves shall be flushed and disinfected at the same time as adjacent water main, in accordance with WSDOT Section 7-09.3(24).
7. GRAVITY SEWER PIPE TESTING
 - a. All new storm sewer pipes are to be pressure tested per WSDOT 4-04.3(1)
 - b. All new gravity sewer pipes are to be pressure tested per WSDOT 7-17.3(2).
8. PRESSURE SEWER TESTING
 - a. All sewer force main piping shall pass a hydrostatic pressure test of 1.5 times operation pressure or 150 psi, whichever is greater. Test duration to be 15 minutes (per the 2008 Department of Ecology Criteria for Sewage Works).
 - b. Required Pressure Test Pressures:
 - (i) Mixed Liquor Forcemains: 150psi
 - (ii) West Plant Drain Pump Station Forcemain: 150 psi
 - (iii) 3W & 2W Piping: 200 psi
9. WATER LINE TESTING AND DISINFECTION
 - a. All water mains and appurtenances shall pass a hydrostatic pressure test per WSDOT 7-09.3(23).
 - b. Required Pressure Test Pressures:
 - (i) Potable Water Distribution Piping: See WSDOT 7-09.3(23).
 - c. The Contractor shall disinfect, flush and provide a satisfactory bacteriological report to the Owner or their appointed representative in accordance with WSDOT/APWA Section 7-09.3(23). The Contractor shall also provide two chlorine concentration test reports to show the initial chlorine concentration is at least 50 mg/L, and to show the 24-hour residual chlorine concentration is at least 25 mg/L. All tests must be performed by a DOH-certified testing laboratory and sample-taking shall be witnessed by the Owner or their appointed representative. Chlorinated flush water must be de-chlorinated and discharged to nearby vegetated areas in a manner that does not cause flooding or erosion.
 - d. The Contractor shall arrange for water sampling and pay for all chlorine concentration and bacteriological tests associated with water main disinfection and chlorine neutralization.

10. HDPE PIPE TESTING

- a. Pressure testing of the HDPE water mains shall comply with ASTM F2164, “Standard Practice for Field Leak Testing of Polyethylene (PE) Pressure Piping Systems Using Hydrostatic Pressure”, and Performance Pipe Technical Note 802 – Leak Testing of Polyethylene Pipe for Municipal and Industrial Applications.

*****END OF SECTION*****

SECTION 33 31 00 – WASTEWATER PIPING

PART 1. GENERAL

1.01 SCOPE OF WORK

- A. This section covers the work necessary to furnish and install wastewater lines as shown on the plans and described herein.
- B. See Section 33 30 00 PIPING SYSTEMS for additional requirements.

PART 2. PRODUCTS

2.01 MATERIALS

- A. BUILDING / STRUCTURE FOOTING DRAIN PIPE
 - 1. Perforated Corrugated Polyethylene Drainage Tubing Underdrain Pipe:
 - 2. Pipe per WSDOT 9-05.2(7), 4-inch diameter unless otherwise shown on the plans.
- B. PRESSURE SEWER PIPE
 - 1. Exposed:
 - a. Stainless Steel Sch. 80 Pipe or Ductile Iron: AWWA C115, cement lined, thickness class 53 for flanged or grooved piping systems.
 - 2. Buried:
 - a. AWWA C115, Ductile Iron cement lined 8 mil poly wrapped thickness class 50 or 51; unless flanged or grooved piping systems which are required to be thickness class 53.
 - b. PVC, AWWA C900/C905, Class 235 DR 18 (minimum), or
 - c. HDPE (pressure class as specified on drawings) in accordance with WSDOT Section 9-30.1 and Spec Section 33 33 00.
 - d. Or as noted on the plans.
- C. GRAVITY SEWER PIPE & DRAINS
 - 1. PVC for wastewater applications:
 - a. ASTM D3034, SDR 35 for 4" through 15" pipe. ASTM F679, SDR 35 (PS46) for 18" & larger pipe. Or as noted on the plans.
 - 2. Ductile Iron for wastewater applications:
 - a. WSDOT 9-05.13 Ductile Iron Sewer Pipe. Ductile Iron cement lined 8 mil poly wrapped thickness class 50 or 51; unless flanged or grooved piping systems which are required to be thickness class 53
 - 3. PVC drain piping (See Plumbing Spec 22 10 00 for Floor Drain Piping) shall be:
 - a. Schedule 40 PVC for pipe 4" and smaller. Pipe shall meet ASTM D2241 and D1784 and Commercial Standard CS 256. Or as noted on the plans.
 - b. AWWA C900/C905, Class 100 DR 41 (minimum) for 6" and larger.

D. AERATION PIPING

1. Exposed:

- a. Unless otherwise specified, stainless steel pipe 4 inches and smaller shall be Type 316L, seamless, threaded joints conforming to ASTM A312. The minimum wall thickness shall be Schedule 40S.
- b. Unless otherwise specified, stainless steel piping larger than 4 inches shall be manufactured from ASTM A240 annealed and pickled sheets and plates, Type 316L, in accordance with ASTM A778 or ASTM A409 HT-0. Only extra-low carbon (ELC) materials with 0.030 percent maximum carbon shall be used. Pipe shall be manufactured to nominal pipe sizes as listed in ANSI B36.19 and shall have the following nominal wall thickness:
- c. Nominal Pipe Size: 5 – 18 inches Schedule Gage: Schedule 10S

2. Buried:

- a. Ductile Iron wrapped in 8 mil polyethylene wrap; unlined, thickness class 50 unless flanged or grooved piping systems which are required to be thickness class 53 (as described above).
- b. Smaller than 4" Air Pipe: Use Exposed Pipe Spec above.

E. 2W/3W WATER PIPING

- 1. Buried: All buried 2W and 3W water piping shall be PE4710, IPS HDPE DR11. See spec section 33 33 00. Or as noted on the plans.
- 2. Exposed: All above ground exposed 2W and 3W water piping 4" or larger shall be AWWA C115, Ductile Iron cement lined thickness class 50 or 51; unless flanged or grooved piping systems which are required to be thickness class 53.

F. WAS DISCHARGE PIPING

- 1. All exposed WAS discharge piping shall be PE4710, IPS, HDPE DR17. See spec section 33 33 00.
- 2. All buried WAS discharge piping shall be Ductile Iron per 2.01, C, 2.

G. JOINTS

1. BURIED PIPE

- a. Mechanical or push-on joints to be in accordance with AWWA C111 for ductile iron pipe. Anchoring of mechanical joints with external set screws will not be permitted. PVC gravity pipe to be in accordance with ASTM D3212 - Specification for Joints, for Drain and Sewer Plastic Pipe using Flexible Elastomeric Seals.
- b. Joints for foul air PVC piping shall be glued.
- c. PVC drain piping 4" and smaller shall be solvent welded with press fit.

2. EXPOSED PIPE

- a. Flanged joints shall be in accordance with AWWA C115. Grooved and shouldered joints shall be in accordance with AWWA C-606.

H. FITTINGS

1. Cast or ductile iron in accordance with AWWA C110 and/or AWWA C153, short body type, 250 psi working pressure. Where taps are shown on fittings, tapping bosses shall be provided.
2. PVC gravity sewer pipe fittings to be in accordance with ASTM D3034 or ASTM F679.
3. PVC C900 fittings used for drain piping shall be fabricated and supplied in accordance with AWWA C900. Molded fittings shall be manufactured and supplied in accordance with AWWA C907.
4. Fittings for Schedule 40 & 80 PVC shall conform to ASTM D2466 and D2467 for socket type.

I. FLEXIBLE COUPLINGS AND FLANGED COUPLING ADAPTERS

1. Flexible couplings for use with PVC piping shall be Romac Style 501, or approved equal. When restraining is required use Romac 611 Restraining system. Bolts and nuts shall be stainless steel type 316. Center ring shall have fusion bonded epoxy coating and have a length of 2x the pipe diameter. Ductile iron sleeves with mechanical joints at each end may be substituted for flexible couplings on ductile iron pipe. Mechanical joint shall be Romagrip, Megalug 1100, or approved equal.
2. Protection for Buried Couplings and Adaptors:
 - a. Double wrap with polyethylene encasement, AWWA C105 and tape the edges of the encasement with PVC tape.

J. HOSES FOR 2" YARD HYDRANTS

1. Hoses to be 1.5" fire hose, 50' lengths, NPSH couplings and adaptors as needed, rated for 500# test pressure, USA Bluebook item # 26950. Provide two hoses for each 2" yard hydrant.

K. HOSES FOR 1" YARD HYDRANTS

1. Hoses to be 3/4" fire hose, 50' lengths, NPSH couplings and adaptors as needed, rated for 200# test pressure. Provide two hoses for each 1" yard hydrant.

L. CEMENT LINING

1. All ductile iron pipe and fittings shall be cement mortar lined and seal coated in accordance with ANSI A21.4/AWWA C104.

M. COATING

1. All ductile iron pipe to be buried shall receive a coat of bituminous material. All exposed ductile iron pipe (including all piping in valve vaults) is to be cleaned and shop coated with two coats of epoxy per Section 09900 Painting. All bituminous material on exposed piping is to be sand blasted off, prior to application of epoxy.

N. GROOVED

1. Grooved couplings shall be Victaulic style 31 for ductile iron pipe, or equal. Couplings for steel or galvanized pipe shall be Victaulic style 07 for rigid systems or Victaulic style 77 for flexible systems. Grooved dimensions shall be per manufacturer's standards.

O. FLANGES

1. ANSI A21.15/AWWA C115, threaded, 250 psi working pressure, 125-pound ANSI drilling.

P. BOLTS

1. To be ASTM A 307, Grade A hex head bolts and nuts for Class 125 FF Flanges. Required Coatings:
 - a. Flanged Fittings Inside Building 316 Stainless Steel Nuts & Bolts
 - b. Flanged Fittings Outdoor Exposed 316 Stainless Steel Nuts & Bolts
 - c. Flanged Fittings Buried Underground Manufacturer's Standard
 - d. Flanged Fittings in Contact W/ Sewage 316 Stainless Steel Nuts & Bolts
 - e. Buried T-Head Cor-Ten Material
 - f. For Mechanical and Grooved JointsManufacturer's Standard, or provide bolt & nut material as described above, as a minimum requirement.
 - g. Or provide as noted on plans.

Q. GASKETS

1. Gaskets for mechanical or push-on joints shall be rubber conforming to ANSI A21.11, AWWA C111.
2. All gaskets for ductile iron air piping shall be EPDM (Ethylene Propylene-Diene Monomer) material.
3. Gaskets for flanged joints in sewage or water service shall be 1/8 inch thick, SBR rubber conforming to applicable parts of ANSI B16.21 and AWWA C207. Gasket material shall be free from corrosive alkali or acid ingredients. Gaskets shall be one-piece, full-face, with holes to pass bolts. Gaskets for grooved joints shall be Flushseal type, halogenated butyle or nitrile depending on service.
4. Gaskets for PVC joints shall be elastomeric seals conforming to ASTM F477.

R. LUBRICANT

1. Lubricant for mechanical joint end piping shall be manufacturer's standard.

S. RESTRAINED JOINT PIPE

1. Joints for buried pressure pipe may be "restrained type". However, the use of restrained joints in lieu of thrust blocks will be acceptable only if the pipe configuration, soil conditions, and restrained length are suitable in the opinion of the Engineer.

T. PIPE SUPPORTS

1. Stainless steel adjustable saddle supports with 1/4" neoprene cushion under piping. Standon Model S92 Saddle Supports as supplied by Material Resources (503) 693-0727, or approved equal.

U. SERVICE SADDLES

1. Ford Iron Service Saddles, Style FC202 with stainless steel bands and epoxy coating and ¼" neoprene cushion, or approved equal.

V. INSULATION, SHIELDS, AND JACKETING

1. PIPE INSULATION

All above grade pipe which transports wastewater shall be insulated with closed cell polyisocyanurate cellular plastic manufactured by ITW Insulation Systems. Insulation shall be Trymer 2000 XP, 1.25" thick (minimum), designed for cold pipe fitting systems, installed per manufacturers recommendation. Minimum temperature range: -100 degree F to +220 degrees F. K factor at 75 degrees Fahrenheit shall not be more than 0.19 BTU-inch/hour-square feet degrees Fahrenheit. Fire rating flame spread of 25 or less. Joints shall be sealed with manufacturer's recommended contact adhesive to form continuous barrier. Density min 2-lb/ft³.

2. PIPE SHIELDS

All above grade pipe shall have 6" long pipe shields, 1.25 inches thick, at all supports and clamps. Pipe shields shall be Trymer 4000 with a min. 4-lb/ft³ density. Pipe shields shall be installed per manufacturers recommendations. Joints shall be sealed with manufacturer's recommended contact adhesive to form a continuous barrier.

3. PIPE INSULATION JACKETING

Insulation jacketing shall be installed over all exposed insulation and pipe shields to provide complete jacketing for all insulated piping and to provide durability. Jacketing shall be smooth finished aluminum; 0.016-inch (26 gauge) minimum thickness. Overlap circumferential joints 4 inches minimum; overlap longitudinal joints 1-inch minimum; longitudinal joints shall be oriented to minimize water entry. Bands shall be 0.5 inch wide, 0.0508 inch (16 gauge) thick aluminum or 0.0179 inch thick Type 304 stainless steel and shall be installed on 18 inch centers, uniformly spaced and at all fitting joints. Apply waterproof adhesive at joints and overlaps. All fittings shall be of the same jacketing material. Insulation jacketing shall be manufactured by ITW Insulation Systems or approved equal.

PART 3. EXECUTION

3.01 INSTALLATION

A. HANDLING PIPE

1. Handle per manufacturer's recommendations. Take care not to damage lining when handling pipe.

B. CUTTING PIPE

1. Cut pipe with milling type cutter, rolling pipe cutter, or abrasive saw cutter. Do not flame cut. Do not damage linings. Cuts shall leave a smooth end at right angles to the pipe axis.

C. DRESSING CUT ENDS

1. Dress cut ends of pipe in accordance with the type of joint to be made. Dress cut ends of mechanical joint pipe to remove sharp edges or projections which may damage the rubber gasket.
2. Dress cut end of push-on joint pipe by beveling, as recommended by the pipe manufacturer.
3. Dress cut ends of pipe for flexible couplings and flanged coupling adapters as recommended by the coupling or adapter manufacturer.

D. FABRICATION OF FLANGED PIPE AND FITTINGS

1. Flanged pipe and fittings shall be fabricated in the shop, not in the field, and delivered to the job site with flanges in place and properly faced. Threaded flanges shall be individually fitted and machine tightened on the threaded pipe by the manufacturer. Flanges shall be faced after fabrication in accordance with AWWA C115.

E. JOINTING PIPE

1. FLANGED

- a. Prior to connecting flanged pipe, the faces of the flanges shall be thoroughly cleaned of all oil, grease, and foreign material. The rubber gaskets shall be checked for proper fit and thoroughly cleaned. Care shall be taken to assure proper seating of the flange gasket. Bolts shall be tightened so that the pressure on the gasket is uniform. Torque-limiting wrenches shall be used to ensure uniform bearing insofar as possible. If joints leak when the hydrostatic test is applied, the gaskets shall be removed and reset and bolts retightened.

2. MECHANICAL, GROOVED, AND PUSH-ON JOINT

- a. Join pipe with mechanical or push-on type joints in accordance with the manufacturer's recommendations. Tools and devices, such as special jacks, chokers, and similar items required for proper installation. Grooved systems may employ Victaulic field grooving tools, including cut and/or roll groovers as needed. Lubricant for the pipe gaskets shall be furnished by the pipe manufacturer, and no substitutes will be permitted under any circumstances.

F. SPECIAL REQUIREMENTS FOR WATER LINES NEAR SEWER LINES

1. Construction requirements for water and sewer lines near sewer line either running adjacent to or crossing shall be in accordance with all requirements as specified in Washington State Department of Ecology, Criteria for Sewage Works Design, C1-9 Special Requirements.

3.02 TESTING

- A. See 33 30 00 Piping Systems.

*****END OF SECTION*****

SECTION 33 32 00 – WATER DISTRIBUTION PIPING

PART 1. GENERAL

1.01 SUMMARY

- A. Work includes but is not limited to following:
 - 1. Installation of HDPE, PVC, and DI water main and appurtenances;
 - 2. Chlorination, de-chlorination, pressure testing and start-up of water mains and water services;
 - 3. Recording as-built information in accordance with Section 01 70 00;
 - 4. Preparation of record (as-built) drawings as required in Section 01 70 00.
- B. Contractor shall provide a complete working installation with all equipment called for in proper operating condition. Documents do not undertake to show or list every item to be provided. When an item not shown or specified is clearly necessary for proper operation of equipment shown or specified, provide an item which will allow the system to function properly at no increase in Contract Sum.
- C. Piping systems, including pipe, fittings, anchors, and all other elements, shall be detailed, fabricated, and installed to resist all internal and external loads which will be imposed upon them. Pressure ratings and materials stated in these Piping Specification sections are minimum acceptable standards. Systems shall be suitable for the service intended.
- D. The pipe diameters shown on the Drawings and used in these Specifications are inside diameters unless specific reference is made to outside diameter of the pipe or the pipe is a standardized product normally designated by a nominal size, e.g., HDPE pipe.
- E. The Contractor shall furnish and install pipe and fittings as shown on the Drawings, as specified in these Specifications, and as required for a complete and functional installation. The pipe shall be new, manufactured in accordance with these Specifications and Drawings.

1.02 REFERENCES

- A. Comply with the requirements of Section 01 41 00, Regulatory Requirements, and as listed herein.
- B. WSDOT Latest Edition Standard Specifications for Road, Bridge and Municipal Construction.
- C. WSDOT Standard Plans for Road, Bridge and Municipal Construction
- D. American Water Works Association (AWWA).

1.03 SUBMITTALS

- A. Submittals shall be in accordance with requirements of Division 1 and as specified. In addition, the following specific information shall be provided:
 - 1. Field Fabrication
For pipe lines which are assembled in the field from standard fittings, submit complete data on pipe and fittings, including any manufacturer's installation instructions.

2. Material Certification

Certification of all materials and manufacturing, properly executed by the manufacturer, shall be available to show compliance with the Specification of materials being furnished. Test data on tests performed shall be provided as requested by the Engineer.

- B. Submittals for all pipe and appurtenances (fittings, valves, valve boxes, services and appurtenances, et al)
- C. The Contractor shall submit a water main shut-down / temporary bypass plan prior to performing Work. The plan shall include an anticipated schedule for shut-downs, schedule for notification of customers (construction notice, shut-down notice), sample notification (letter, door hanger). Customers shall be notified about construction activities one week in advance and water shut-downs at least 48-hours in advance. This plan shall also include the contingency provisions the Contractor will have in place to provide water to residential customers in the event that a shut-down lasts longer than 6-hours.

1.04 QUALITY ASSURANCE

- A. Quality Assurance shall be in accordance with the requirements of Section 01 45 00 and as specified.
- B. Installer shall be a Specialist and experienced in the installation of potable water systems.
- C. Regulatory Requirements: See referenced Codes, ordinances, and standards in Section 01 41 00 Regulatory Requirements.

1.05 PROJECT SITE CONDITIONS

- A. Locate and provide protection of existing utilities affected by the work.
 - 1. Existing Pipe Verification

Expose all existing pipes which are to be connected to new pipelines. Verify the size, material, elevation, horizontal location, and pipe service of these existing pipes with sufficient lead time to make accommodations as necessary. Connections to the existing pipe system shall be at a restrained joint or fitting.
- B. Minimize disruptions to access, existing water service connections and existing facilities.

PART 2. PRODUCTS

2.01 GENERAL

- A. All materials and equipment shall comply with "Quality Assurance" provisions and Manufacturer's data. Where these may be in conflict, the more stringent requirements govern.
- B. All items of materials in each category of equipment shall be of one manufacturer.
- C. Groups of items having same or similar function shall be by single manufacturer to facilitate maintenance and service.
- D. All materials and equipment shall be compatible with space allocated. Modifications necessary to adjust items to space limitations shall be at Contractor's expense.
- E. All materials and equipment shall conform with conditions shown and specified. Coordinate with other trades for best possible assembly of completed Work.
- F. All materials and equipment installed shall be fully operational and without objectionable noise or vibration.

2.02 PIPE AND JOINTS

A. Buried Pipe

1. PVC potable water piping shall be AWWA C900 Class 235 DR 18 (minimum) with rubber gasketed joints. Fittings shall be ductile iron (AWWA C-110 or AWWA C-153). Buried ductile iron fittings shall be polyethylene encased in accordance with pipe manufacturer's recommendations and WSDOT 9-30.1(2).
2. Ductile Iron (DI) potable water piping shall be standard thickness Class 50 (AWWA C-151) push-on joints or M.J. joints. Pipe shall be cement mortar lined (AWWA C-104). The pipe shall be of 150 psi working pressure, plus 100 psi surge pressure. Fittings shall be ductile iron (AWWA C-110 or AWWA C-153). Buried ductile iron pipe and fittings shall be polyethylene encased in accordance with pipe manufacturer's recommendations and WSDOT 9-30.1(2).

2.03 SERVICE PIPES

- A. Potable water service pipe shall be high-density polyethylene (HDPE) in accordance with the drawings and the following:
1. Polyethylene (PE) piping shall be manufactured from high molecular weight, PE 4710 conforming to the requirements of ASTM D-3350. Piping shall conform to the requirements of AWWA C901 and ASTM F714. Piping shall be installed from continuous coil with fusion weld joints. Fittings shall be compression type fittings suitable for pressure piping applications. Pipe and fittings shall meet requirements as shown on the Plans.

2.04 PIPE FITTINGS

- A. Ductile iron pipe fittings (reducers, bends, elbows, etc.) shall be cast or ductile iron in accordance with WSDOT 9-30.2(1) and ANSI B16.1/AWWA C110, short body type, 250 psi working pressure.
- B. HDPE fittings shall be standard commercial products manufactured by injection molding or by extrusion and machining. The fittings shall be fully pressure rated by the manufacturer to provide a working pressure equal to the pipe. Fitting shall be per WSDOT Standard Specification 9-30.2(10).

Ductile iron fittings may be used provided that the connection is restrained and made using a PE flange adapter or a mechanical joint adapter, and a structural support pad is installed underneath the fittings.

2.05 FLEXIBLE COUPLINGS / MECHANICAL JOINTS

- A. Flexible couplings for use with ductile iron pipe shall be Romac Style 501; or approved equal. Bolts and nuts shall be stainless steel type 316. Center ring shall have fusion bonded epoxy coating and have a length of 2x the pipe diameter. Ductile iron sleeves with mechanical joints at each end may be substituted for flexible couplings on ductile iron pipe. Mechanical joint shall be Romagrip, Megalug 1100, or approved equal.
- B. Restrained flexible couplings shall use Romac 611 Restraining System or approved equal.
- C. Joint Restraint: Provide joint harnesses (tie rod lug or attachment plate assemblies) designed for the test pressure across flexible couplings and flange coupling adaptors. For steel pipe the joint harness shall conform to the requirements of Chapter 13 of AWWA M-11, paragraph 13.10, Table 13-6 - Tie Bolt Schedule for Harnessed Joints. Anchor studs may be used on flange coupling adapters for pipe up to 12 inches in diameter.
- D. Protection for Buried Couplings and Adaptors:
 - 1. Double wrap with polyethylene encasement, AWWA C105 and tape the edges of the encasement with PVC tape.

2.06 BOLTS AND NUTS

- A. Bolts and nuts for buried or exposed conditions shall be Type 304 stainless steel. Bolts and nuts for submerged conditions shall be Type 316 stainless steel.

2.07 RUBBER GASKETS

- A. Store all rubber gaskets in a cool, well-ventilated place, and do not expose to the direct rays of the sun. Do not allow contact with oils, fuels, or petroleum solvents.
- B. Gaskets shall be full faced 1/8-inch thick neoprene having a durometer of 60 plus or minus 5.

2.08 JOINT LUBRICANT

- A. Furnish joint lubricant with the pipe. Furnish the amount and type recommended by the pipe manufacturer. The lubricant shall be a water-soluble, nontoxic, vegetable soap compound conforming to United States Pharmacopoeia No. P39.

2.09 HYDROSTATIC TESTING

- A. Provide all hoses, plugs, and other necessary equipment to complete the tests.

2.010 CONCRETE FOR THRUST BLOCKING, THRUST TIES AND ENCASEMENT

- A. The concrete for all thrust blocking and thrust ties shall develop a minimum compressive strength of 3,000 psi at 28 days.

2.011 FIRE HYDRANTS:

- A. Fire hydrants shall have individually valved two 2-1/2 inch ports and one 5-1/4 inch main valve opening. A 4-1/2 inch NSP pumper nozzle and a 5 inch Storz port with cap and cable shall be supplied. Hydrants shall be either Iowa or M.H. 929T hydrant.
- B. Fire hydrants shall have the Storz port facing the required access and the base flange of the hydrant must not vary more than 1 foot in elevation from the grade level of the required access. The lowest stem shall be a minimum of 14 inches above the ground.
- C. If, in the opinion of the Owner, fire hydrants are vulnerable to vehicular damage, appropriate crash posts shall be provided. No obstructions shall exist within a 3-foot working area of each required access. Crash posts shall be 4-inch cement-filled pipe minimum 3 feet in height with 2 feet of piping below grade.

- D. Hydrant shutoff valves shall be located within between 5 and 20 feet from the hydrant.
- E. Underground supplies to fire hydrants must be inspected. Such inspection shall include visual inspection of piping and hydrostatic pressure test of a minimum of 200 psi or 50 psi in excess of street main pressure, whichever is greater. A flow test will be required when installation is complete.
- F. Fire hydrant installation must comply with Eastsound Sewer and Water District Standard Detail W-1.
- G. Fire hydrants must be maintained in an operable condition at all times and must be repaired or replaced when defective. Hydrants shall be fully operable before construction commences above grade level.

PART 3. EXECUTION

3.01 SHIPPING AND HANDLING OF MATERIALS:

- A. During transportation, unloading, and storage, pipe and materials shall be protected, supported, and handled in a manner to prevent damage to the materials, especially linings and coatings. Only implements and equipment suitable for proper and safe handling of the materials shall be used. Fabric slings shall be used to lift pipe and fittings, not chains or cables.

3.02 EXAMINATION

- A. Verify installation conditions as satisfactory to receive work of this Section. Do not install until unsatisfactory conditions are corrected. Beginning work constitutes acceptance of conditions as satisfactory.

3.03 GENERAL INSTALLATION REQUIREMENTS

- A. Install in accordance with "Quality Assurance" provisions, "References," Specifications, and Manufacturer's recommendations. Where these may be in conflict, the more stringent requirements govern.
- B. All work and materials shall be subject to the approval of the Engineer and the Owner.

3.04 INSTALLATION OF PIPING

- A. Line and Grade.
 - 1. Survey line and grade shall be established prior to laying pipe per Standard Specification 7-09.3(5). Construction layout staking will be provided by the Owner.
 - 2. In areas between survey stakes, transfer line and grade from the Plans to the water line trench and maintain said line and grade using construction practices in conformance with established industry standards.
- B. Handling, Storage and Cutting of Pipe.
 - 1. During transportation, unloading, and storage, pipe and materials shall be protected, supported, and handled in a manner to prevent damage to the materials, especially linings and coatings. Only implements and equipment suitable for proper and safe handling of the materials shall be used. Fabric slings shall be used to lift pipe and fittings, not chains or cables.

2. Each pipe and fitting shall be carefully inspected before the exposed pipe or fitting is installed or the buried pipe or fitting is lowered into the trench. The interior and exterior protective coating shall be inspected, and all damaged areas patched in the field with material similar to the original. Any pipe which, in the opinion of the Engineer, is damaged beyond repair shall be removed from the site and replaced with another unit. Clean ends of pipe thoroughly. Remove foreign matter and dirt from inside of pipe and keep clean during and after installation.
3. Use proper implements, tools, and facilities for the safe and proper protection of the pipe. Carefully handle pipe in such a manner as to avoid any physical damage to the pipe. Do not drop or dump pipe from trucks or into trenches under any circumstances.
4. Cut pipe with approved cutters, do not flame cut except for mild steel pipe. Cut perpendicular to axis of pipe. Dress ends to suit type of joint being made, removing burrs, mill scale, and debris before making up. Repair damaged linings and coatings.

C. Pipe Installation.

1. All pipe shall be installed in conformance with the manufacturer's recommended procedures for the pipe being installed, and in accordance with WSDOT Standard Specifications 7-08 and 7-09. A minimum of 3 feet of cover above the top of pipe shall be provided. Take precautions to prevent water from entering the trench or excavation. Do not lay pipe in water, or when in the opinion of the Owner or Engineer, trench conditions are unsuitable. Prevent excavated or other foreign material from entering the pipe during the laying operation. When laying operations are not in progress, at the close of the day's work, or when labor is absent from the job, close and block all open pipe ends to prevent entry of foreign material or creep of the gasketed joints.
2. Excavating and Backfilling: In accordance with requirements of Division 31. Provide all necessary shoring, sheeting, and pumping as part of Work of this Division.
3. Pipe Laying: After a section of pipe has been lowered into the prepared trench, clean the end of the pipe to be joined, the inside of the joint, and the rubber ring immediately before joining the pipe. Make assembly of the joint in accordance with the recommendations of the manufacturer. Provide all special tools and appliances required for the jointing assembly. All water pipe shall be installed with the bell end upstream. When pipe laying is not in progress, the forward end of the pipe shall be kept tightly closed with an approved temporary plug. All pipe laid in the trench to the specified line shall be kept under longitudinal compression until the backfill has been compacted to the crown of the pipe. Take all precautions necessary to prevent the "uplift" or floating of the line prior to the completion of the backfilling operation. Where field conditions prevent installation per the Plan line, notify the engineer immediately. The joint deflection between segments of the water main shall not exceed 75% of the manufacturer's recommended maximum deflection. Where joint deflection exceeds this limit, install the needed fittings to make the needed horizontal or vertical bend.
4. Service Line Installation: Service lines shall be installed in accordance with WSDOT Standard Specifications Sections 7-08 and 7-15, and the Contract plans.
5. Contractor shall mark all underground water mains and service lines with detector tape in accordance with the Plans and with requirements of Division 31.

- D. Joining Flanged Pipe:
1. Prior to connecting flanged pipe, the faces of the flanges shall be thoroughly cleaned of all oil, grease, and foreign material. The gaskets shall be checked for proper fit and thoroughly cleaned. Care shall be taken to assure proper seating of the flange gasket. Bolts shall be tightened so that the pressure on the gasket is uniform. Torque-limiting wrenches shall be used to ensure uniform bearing insofar as possible. If joints leak when the hydrostatic test is applied, the gaskets shall be removed and reset and bolts retightened.
- E. Except for welded joints, pipe joints shall not be encased in concrete unless specifically required on the Drawings. Pipe coatings shall be continuous through concrete encasements, thrust blocks, anchors, collars, etc., unless otherwise shown on the Drawings.
- F. Except for welded joint pipe, a flexible joint shall be provided within 18 inches or one-half the pipe diameter, whichever is less, from the terminations of any concrete backfill, or concrete encasement.
- G. Corrosion Protection of Pipe and Accessories:
1. Not all corrosion protection details are included, either on the Drawings or in the Specifications. The absence of specific details on corrosion and environmental protection measures shall not relieve the Contractor of the responsibility of providing them, all as part of the Contract price.
- H. Corrosion Protection for Buried Pipe Accessories:
1. All buried pipe appurtenances made of steel shall have corrosion protection. Tie rods and similar items shall be heat shrink tube wrapped. Flange bolts, nuts, and similar items shall be coated with a bituminous paint or equal. Flexible couplings, grooved couplings, and similar items shall be heat shrink wrapped or cement coated.
 2. Buried valves and similar elements on wrapped pipelines shall be bituminous paint-coated. On ductile iron or nonmetallic pipelines they shall have exposed nuts and bolts bituminous paint-coated and the entire valve wrapped in 8-mil polyethylene as specified for ductile iron pipe. On cement-coated pipelines they shall be cement-coated similar to detail shown for couplings.
- I. Thrust Restraint:
1. All pipeline tees, plugs, caps, bends, and other locations where unbalanced forces exist shall be thrust restrained. Thrust blocks and ties may not be shown on the Drawings. The Contractor is responsible for providing thrust restraint as incidental items in the Contract Price.
 2. The Contractor shall install thrust ties at all locations necessary to restrain thrust forces. Such thrust ties are subject to the review and approval of the Engineer. The concrete for all thrust ties shall develop a minimum compressive strength of 3,000 psi at 28 days.
 3. Thrust blocking shall be adequate to withstand hydrostatic test pressures as well as normal operating pressures. Place thrust blocking between undisturbed ground and the fitting to be anchored. Place the blocking so that the pipe and fitting joints will be accessible for repairs. The concrete for all thrust blocks shall develop a minimum compressive strength of 3,000 psi at 28 days.

J. Interface with other products:

1. Exact routing of piping and other items shall be governed by structural conditions or obstruction. Contractor shall make use of data in Contract Documents. In addition, Engineer reserves right, at no increase in Contract Sum, to make any reasonable change in location of pipe and appurtenances.
2. Field verify all dimensions shown on the drawings prior to starting any work. The Contractor shall not be entitled to additional compensation for minor variations in the field dimensions of the existing facilities.

3.05 TESTING AND DISINFECTION

- A. See Piping Systems 33 30 00.

3.06 CONNECTION TO EXISTING SYSTEM

- A. Only Association staff shall operate the isolation valves on the existing system.
- B. Contractor shall provide 48-hour written notice to customers and the Owner for water shut-off. Notification shall be by means of Contractor providing a door hanger at each residence or place of business to which service will be disrupted. The Owner will provide Contractor with the required pre-printed door hangers. It will be the responsibility of the Contractor to plan for, distribute, and fill-in the specific temporary service shut-down information which shall include, at a minimum, the proposed date and time of service disconnection.
- C. Any support provided by the Owner's staff shall be subject to preemption by emergency operations including, but not limited to, power outages and water main breaks.

*****END OF SECTION*****

SECTION 33 33 00 – HDPE PIPING

PART 1 - GENERAL

1.01 DESCRIPTION

Work covered in this section includes all labor, material and equipment required to install High Density Polyethylene (HDPE) Pipe.

1.02 SUBMITTALS PRIOR TO CONSTRUCTION

Submittals shall be made in accordance with Part 3, Section 01 33 00.

The Contractor shall submit the qualifications of the manufacturer-approved fusion operator/installers to the Engineer for approval prior to ordering pipe.

1.03 QUALIFICATIONS

HDPE Fusion Welding Operators – Required Qualifications

- A. Fusion welding operators on the project shall have been trained by the pipe manufacturer and shall have welded a minimum of 5,000 lineal feet of HDPE pipe prior to this job. Fusion operators must also have previous experience in at least one pipe project within the last three years that is equivalent pipe diameter or larger.
- B. Current certification on Title #49 section 192.283 and 192.285 (Fusion Operator requirements)
- C. Current certification on Title #49 section 192.287 (Inspector requirements)
- D. Fusion operator shall have certification on the type of fusion machine being used for the project.

PART 2 – PRODUCTS

2.01 HDPE PIPE

HDPE pressure sewer pipe shall be High Density Polyethylene, manufactured of high molecular weight, high density PE4710 polyethylene pipe grade resin and shall conform to ASTM D3350 in dimensions and workmanship. Pipe shall have DR values as listed in Section 33 31 00 or as noted on the plans. Pipe is to be DriscoPlex 4000/4100 IPS Pipe, or approved equal. It shall have a cell classification of 445574C as specified by ASTM D3350. The Contractor shall furnish the necessary labor, supervision, butt fusion welding equipment and material to install HDPE pipe where indicated on plans. Contractor shall assure there is no miscommunication with pipe suppliers regarding nominal inside and outside pipe diameters.

2.02 HDPE PIPE COLOR REQUIREMENTS

HDPE piping shall be supplied with colored stripping. Color stripe schedule:

- Sewage HDPE Pipe – Green Stripe
- 2W Water Pipe – Light Blue Stripe
- Potable Water Pipe – Dark Blue Stripe

2.03 FITTINGS

HDPE fittings shall be standard commercial products manufactured by injection molding or by extrusion and machining, or shall be fabricated from PE pipe conforming to this specification. The fittings shall be fully pressure rated by the manufacturer to provide a working pressure equal to the pipe.

2.04 REFERENCED DOCUMENTS

- A. ASTM Standards:
D 3350 Specification for Polyethylene Plastics Pipe and Fittings Materials
- B. ANSI Standard:
B 36.10 Standard Dimensions of Steel Pipe (IPS)
- C. ISO Standards:
161 Thermoplastic Pipe for the Transport of Fluids - Nominal Outside Diameters and Nominal Pressures
3607 Polyethylene Pipe: Tolerances on Outside Diameters and Wall Thicknesses
4427 Polyethylene Pipes and Fittings for Water Supply Specification

PART 3 – EXECUTION

3.01 GENERAL

- A. Contractor shall bear complete responsibility for the satisfactory performance of the HDPE pipe system, including materials, joining, installation, inspection and quality assurance. All workmanship shall be in accordance with HDPE manufacturer's requirements.

3.02 INSPECTION OF HDPE PIPE AT JOB SITE

- A. The Contractor shall be responsible for inspection of pipe at the job site. Pipe showing damage from transit shall be marked and rejected by the Contractor for return to the manufacturer.

3.03 JOINING

Sections of polyethylene pipe shall be joined into continuous lengths on the job site above ground. The joining method shall be the butt fusion method and shall be performed in strict accordance with the pipe manufacturer's recommendations. The butt fusion equipment used in the joining procedures shall meet the requirements of 400°F, alignment, and 75 psi interfacial fusion pressure. The joint weld strength shall be equal to or greater than the tensile strength of the pipe. The fused joints shall be watertight, and all fused joints shall be subject to acceptance by the Engineer prior to installation. All defective joints shall be cut out and replaced at no cost to Owner. Any section of the pipe with a gash, blister, abrasion, nick, scar, or other deleterious fault greater in depth than ten percent (10%) of the wall thickness, shall not be used and must be removed from the site. However, a defective area of the pipe may be cut out and the joint fused in accordance with the procedures stated above. In addition, any section of pipe having other defects such as concentrated ridges, discoloration, excessive spot roughness, pitting, variable wall thickness or any other defect of manufacturing or handling, as determined by the Engineer and/or his representative, shall be discarded and prohibited from use.

- A. All fusion joints shall be visually inspected and meet visual requirements set forth by the pipe manufacturer.
- B. 3 each or 3% of fusion joints, which ever is greater, shall be subjected to destructive testing from the actual pipe project.
- C. Data logging of fusion joints shall be used on all welds and be submitted for review and documentation.
- D. No non-standard and/or non-proven equipment or techniques shall be used to install and fuse the polyethylene pipe.

3.04 TESTING

Pressure testing of the HDPE water mains shall comply with ASTM F2164, "Standard Practice for Field Leak Testing of Polyethylene (PE) Pressure Piping Systems Using Hydrostatic Pressure", and Performance Pipe Technical Note 802 – Leak Testing of Polyethylene Pipe for Municipal and Industrial Applications.

END OF SECTION

SECTION 33 41 00 - STORM UTILITY DRAINAGE PIPING

PART 1. GENERAL

1.01 SECTION INCLUDES

- A. Work includes but is not limited to following:
 - 1. Installation of new storm sewer pipes, storm sewer catch basins, and roof drains.
 - 2. Drainage for retaining walls.

1.02 REFERENCES

- A. WSDOT Standard Specifications for Road, Bridge and Municipal Construction, latest edition.
- B. WSDOT Standard Plans for Road, Bridge and Municipal Construction, latest edition.
- C. Washington State Department of Ecology's Stormwater Management Manual for Western Washington, latest edition.

1.03 SUBMITTALS

- A. Submit in accordance with Sections 01 33 00.
 - 1. Catch Basins with frames, grates, lids
 - 2. Storm sewer and roof drain piping & accessories with product installation literature.

1.04 QUALITY ASSURANCE

- A. Manufacturer/Installer shall be a Specialist.
- B. Regulatory Requirements - See Section 01 41 00.

PART 2. PRODUCTS

2.01 GENERAL

- A. Comply with "Quality Assurance" provisions, "References," and Specifications. Where these may be in conflict, the more stringent requirements govern.

2.02 MATERIALS

- A. Storm Sewer Pipe
 - 1. Pipe used for storm sewer shall be double walled (corrugated exterior and smooth interior) polyethylene meeting the requirements of WSDOT Standard Specification 9-05.20 - Corrugated Polyethylene Storm Sewer Pipe. Pipe shall be manufactured by Advanced Drainage Systems, Inc. (model N12), Hancor, Inc. (model HiQ), or engineer approved equal.
 - 2. For storm sewer pipes with less than 18 inches (1.5 feet) of cover, the material shall be ductile iron, which shall comply with WSDOT Standard Specifications 9-05.13 Ductile Iron Sewer Pipe.
 - 3. Couplings for storm sewer pipe shall be classified as water tight.
 - 4. Roof Drain Pipe shall be PVC complying with WSDOT Standard Specification 9-05.1(5).

5. Retaining wall footing drain pipe shall comply with WSDOT Standard Specification 9-05.2(7) – Perforated Corrugated Polyethylene Drainage Tubing Underdrain Pipe. Tight line pipe from wall footing to drainage structure shall be PVC Pipe per WSDOT Standard Specification 9-05.1(5).
- B. Drain Cleanout Box. Cleanout box shall be Highline Series 1015-12” SQ, Highline Series 910L-10” Round or approved equal.
- C. Catch Basins.
1. Type 1 Catch Basins shall comply with WSDOT Standard Specification 9-05.50(3) - Precast Concrete Catch Basins, and WSDOT Standard Plan B-5.20-03.
 2. Type 1L Catch Basins shall comply with WSDOT Standard Specification 9-05.50(3) - Precast Concrete Catch Basins, and WSDOT Standard Plan B-5.40-02.
 3. Adjusting sections shall be precast concrete intended for the application.
 4. Metal frame and solid metal covers shall comply with WSDOT Standard Specification 9-05.15 - Metal Castings, and WSDOT Standard Plan B-30.10-03, and B-30.20-04. Herringbone metal grates shall comply with WSDOT Standard specification 9-05.15-metal castings and WSDOT standard plans B-30.10-03 and B-30.50-03.
- D. Bedding and backfill materials for storm sewer pipe, roof drain pipe, and tight line pipe shall be per Section 31 23 33 – Trenching and Backfilling. Controlled Density Fill (CDF) shall be per WSDOT 2-09.3(1) E.
- E. Bedding and backfill materials for retaining wall footing drain pipe, including geotextile fabric, shall be per the recommendations in Sections 3.5 and 3.7 of the Geotechnical Report dated September 7, 2021.

PART 3. EXECUTION

3.01 EXAMINATION

- A. Verify installation conditions as satisfactory to receive work of this Section. Do not install until unsatisfactory conditions are corrected. Beginning work constitutes acceptance of conditions as satisfactory.

3.02 INSTALLATION

- A. Install in accordance with “Quality Assurance” provisions, “References,” Specifications, and Manufacturer’s directions. Where these may be in conflict, the more stringent requirements govern.
- B. Line and Grade.
1. Survey line and grade shall be established prior to laying pipe per Standard Specification 7-08.3(2)A. Contractor to provide construction layout staking.
 2. In areas between survey stakes, transfer line and grade from the Plans to the storm line trench and maintain said line and grade using construction practices in conformance with established industry standards.
- C. Handling, Storage and Cutting of Pipe.
1. Material shall be stored in such a manner to prevent the pipe from bowing or otherwise being damaged.

2. Saw cut pipe to leave a smooth end at right angles to the pipe axis. Prior to joining pipe, cut ends shall be de-burred, beveled and in accordance with the type of joint to be made.

D. Pipe Installation

1. Storm sewer pipe shall be constructed in accordance with WSDOT Standard Specification 7-08 - General Pipe Installation Requirements.
2. Pipe shall be installed on a well bedded, prepared foundation throughout its length per Section 31 23 33 - Trenching and Backfilling. Grade the bottom of the trench by hand, if necessary, to the line and grade to which the pipe is to be laid, with proper allowance for pipe thickness and for bedding material.
3. The pipe shall be uniformly bedded under and around the entire length of the barrel. The bedding shall be dug out to accommodate pipe bells and prevent the joints from supporting excess weight.
4. Take precautions to prevent water from entering the trench during excavation. Do not lay pipe when, in the opinion of the Owner, trench conditions are unsuitable.
5. Prevent excavated or other foreign material from entering the pipe during the laying operation. When laying operations are not in progress, at the close of the day's work, or when labor is absent from the job, close and block all open pipe ends to prevent entry of foreign material or creep of the joints.
6. Take all precautions necessary to prevent the "uplift" or floating of the line prior to the completion of the backfilling operation.
7. In areas where pipe cover is less than one foot, controlled density fill (CDF) shall be used to encase pipe. The pipe shall have at least one foot of CDF below bottom of pipe and one foot of CDF on either side of pipe, measured from the outer wall of the pipe. CDF shall be used to cover the pipe, bringing it to grade.

E. Catch Basins.

1. All catch basins shall be installed per WSDOT Standard Specification 7-05.3 including excavation, foundation preparation, installation, plumbing and finishing.
2. Pipes entering catch basins shall be trimmed flush with the grouted fillet between the pipe and catch basin structure. Any damage to grouting during trimming operations shall be corrected to the satisfaction of the Owner.
3. The iron frame and grate shall be set to the plan elevation by means of precast concrete grade rings, adjusting rings, or concrete bricks. Scrap lumber, rocks or other material shall not be used in setting the rim elevations. The bearing surface of the all grade rings, bricks, and casting frame shall rest on a full mortar bed to provide a water tight seal between the precast structure and the casting.
4. Prior to final acceptance, remove all debris from catch basins.

F. Footing Drains.

1. Construct footing drain per the recommendations in Section 3.7 of the Geotechnical Report dated September 7, 2021. Establish elevations and grades of footing drains so as to permit gravity drainage to the onsite tight line storm sewer system.

- G. Building Roof Drains and Downspouts.
 - 1. Tight-line roof drains and downspouts to catch basins. Maintain positive slopes to catch basin and 12" minimum cover.
- H. Removal and Abandonment of Storm Sewer.
 - 1. Existing storm sewer identified for removal on the plans shall be fully removed. Any portion left behind shall be cleanly cut. Excavate and remove existing sections of storm drain and catch basins which conflict with proposed grades. Properly dispose of removal debris offsite. Coordinate drain removal work with new drainage construction to provide satisfactory drainage for onsite and upstream areas at all times. Backfill abandoned trench with Structural Fill in accordance with Section 31 23 33 Trenching and Backfilling.
 - 2. Any remaining abandoned piping not designated for removal shall be plugged per WSDOT Standard Specifications 7-08.3(4).
- I. Surface Drainage.
 - 1. During construction, grade, drain, and maintain work site so as to prevent unnecessary construction delays due to ponding. Protect existing drainage structures and adjacent properties from sedimentation, sediment runoff, and damaging drainage velocities.
 - 2. Provide proper coordination of site grading, landscaping, sidewalk construction, surface trench drain and catch basin rim adjustments to assure positive drainage of entire site with no ponding, and all surface stormwater is directed away from buildings.

3.03 ADJUSTING, CLEANING, AND REPLACEMENT

- A. Make any adjustments as required.
- B. Clean: Leave installations clean; premises free from residue of work of this section.

3.04 BACKFILL AND DRAINAGE FOR RETAINING WALLS.

- A. Backfill retaining wall and landscape walls per Construction Plans and Section 32 32 23. Install footing drains and tight line to storm drainage structures as shown on plans.

*****END OF SECTION*****

SECTION 40 05 71 – TELESCOPING VALVE

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install one (1) fully operational rising stem telescoping valve for use on the Aerated Digester outlet as indicated on the Plans.
- B. Valve shall be designed to operate smoothly and efficiently in the control of the fluids at the location shown on the contract drawings. This design is for a 4" telescoping valve, with 48" (minimum) of travel.

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00 Submittal Procedures:
 - 1. Manufacturer's catalog data showing construction materials, required seals and guides, and required coatings.
 - 2. Fabrication drawings with full dimensions and installation drawings showing coordination with outlet piping, liner penetration and booting, the outlet structure and pipe supports.
 - 3. Operations and maintenance information as specified in Section 01 33 00.

PART 2. PRODUCTS

2.01 TELESCOPING VALVE

- A. Materials
 - 1. Tube: Tubes shall be stainless steel and manufactured from seamless pipe or tube. Tube lengths shall be as shown or noted on the drawings and must be of sufficient length to facilitate valve travel and maintain an appropriate insert depth. Valve tubes are to be a minimum 1/8" thick and are attached to connecting stems by use of a lifting bail.
 - 2. Seal Flange: A stainless steel companion flange and neoprene slip seal gasket shall be provided by the valve manufacturer. The gasket must be a minimum 1/4" thick. The inside diameter of the gasket is to be 1/8" smaller than the outside diameter of the valve tube to provide a friction seal. The gasket is to be sandwiched between the riser pipe flange and the companion flange. The gasket and companion flange shall include a 125# standard drilling pattern to match the riser pipe.
 - 3. Lifting Bail: The lifting bail shall be stainless steel construction and be rigidly welded to the tube.
 - 4. Lift and Stems: Lifts shall be handwheel type, with UHMW polyethylene thrust bearings along with a stub acme threaded type 304 stainless steel stem to provide automatic self-locking, infinite valve positioning. The standard rising stem lift shall use a galvanized steel square tube with torque nut design to prevent telescoping valve tube rotation. Alternately, where conditions require, a vee keyed shaft, with torque plate, shall be used to prevent valve tube rotation. Handwheels shall be a minimum of 12" in diameter and shall include a clear plastic Butyrate stem cover with a mylar strip type position indicator, calibrated in 1/4 inch increments to illustrate valve position. The mylar strip, provided by the

manufacturer, will be affixed by the contractor after installation to provide a true and accurate indication of the tube elevation by comparing it to the top of the rising stem. Stainless steel anchor bolts shall be provided for all pedestals. Cleaning and shop prime coat of lift housing and handwheel will be manufacturer's standard.

5. Pedestal: The pedestal shall be of 316 Stainless Steel and designed for mounting to the concrete side wall as shown on the contract drawings.
 - a. Anchor bolts: 5/8"x5", Stainless steel, wedge type anchor bolts (six anchor bolts total).

2.02 MANUFACTURER

- A. Telescoping valve shall be Model TS-2 as manufactured by Waterman Industries, Inc. or equal.

PART 3. EXECUTION

3.01 INSTALLATION

- A. Deliver valve(s) in manufacturer's original, unopened cartons. Store in a safe place until ready for installation.
- B. Follow manufacturer's printed instructions for installation.
- C. Remove all identifying marks, labels, etc. from exterior.

***** END OF SECTION *****

SECTION 41 12 13 – SCREW CONVEYOR SYSTEM

PART 1 -- GENERAL

1.1. DESCRIPTION

- A. The Contractor shall furnish and install one (1) fully operational screw conveyor system as shown on the plans and describe in specifications.
- B. The screw conveyor system shall transport mechanically dewatered wastewater treatment plant sludge from the rotary presses to the final discharge point area as shown on the Drawings.
- C. The screw conveyors system shall be provided complete with all necessary components, drives, motors, instrumentation, controls, ancillary items and supports, except as otherwise shown or specified herein.
- D. The design of the sludge conveying and system shall be based upon the Contract Drawings, these Specifications and the manufacturer's experience with existing systems of similar sizes, capacities and handling material with similar characteristics.
- E. The Contract Drawings, and this Section, provide equipment descriptions, minimum requirements and mandatory features of the equipment to be furnished. It is the Manufacturer's responsibility to design and furnish the equipment complete in all details, performance, and reliability meeting the requirements and intent of the Contract Drawings and these Specifications.
- F. The equipment shall be furnished complete with all accessories, special tools, and other appurtenances as specified herein or as may be required for a satisfactory installation.

1.2. REFERENCE SPECIFICATIONS, CODES AND STANDARDS

- A. The screw conveyors shall be designed and manufactured in accordance with the following listed Standards and Specifications, including applicable addenda in effect as of the date of bid submission. They shall be considered an integral part of this specification and shall govern the design, fabrication, testing and inspection of equipment, except as otherwise shown or specified herein.
 - 1. American Gear Manufacturers Association (AGMA)
 - 2. American Welding Society
 - 3. Conveyor Equipment Manufacturers Association (CEMA)
 - 4. National Electrical Manufacturers Association (NEMA)
 - 5. Anti-Friction Bearing Manufacturers Association (AFBMA)
 - 6. American National Standards Institute (ANSI)

7. National Fire Protection Association (NFPA)
8. American Society for Testing Materials (ASTM)
9. Joint Industrial Council (JIC)
10. American Society of Mechanical Engineers (ASME)
11. Underwriters Laboratories Inc. (UL)
12. Institute of Electrical and Electronic Engineers (IEEE)
13. American Institute of Steel Construction (AISC)

1.3. SUBMITTALS

- A. The following shall be provided in accordance with Section 01340:
 1. Manufacturer's catalog data showing construction materials, required seals and guides, and required coatings.
 2. Fabrication drawings with full dimensions and installation drawings showing coordination with outlet piping, liner penetration and booting, the outlet structure and pipe supports.
 3. Operations and maintenance information as specified in Section 01 30 00.
- B. The manufacturer shall submit the following:
 1. Data sheets with description of the proposed equipment, size, length, type, capacity, arrangement, materials of construction, motor size, motor type and equipment weight, etc.
 2. Provide a detailed list of components and accessories to be furnished within the scope of this Section. Provide catalog information, drawings and specifications for components of the equipment, showing principal dimensions, parts, materials of construction, material thickness, etc.
 3. Significant dimensional differences between the equipment specified herein, indicated on the Drawings and the proposed equipment.

Design loadings to be transmitted to foundations or supports shall be furnished upon request.

1.4. QUALITY ASSURANCE

- A. All parts furnished shall be amply designed and constructed for the maximum stresses occurring during fabrication, erection and continuous operation.
- B. All materials shall be new and shall be of the very best quality, entirely suitable for the service to which the units are to be subjected.

- C. Exposed welds shall be ground smooth to form a neat uniform fillet without weakening the base metal.

1.5. HARDWARE

- A. Design of all support structures, platforms and bracing shall be in accordance with AISC Standards. Structural Welding shall conform to American Welding Society Structural Code D1.1-82.
- B. Hardware requiring special tools or wrenches shall not be used.

PART 2 -- PRODUCTS

1.6. GENERAL REQUIREMENTS

- A. These specifications shall be considered as minimum requirements. The Contractor or Equipment Supplier shall add such additional features as are necessary for satisfactory operation and functioning of the conveying equipment.
- B. Structural design shall be in accordance with American Institute of Steel Construction AISC Standards.
- C. Except as otherwise indicated, welding shall comply with ANSI/AWWA D100 and AWWA C206, and the following:
 - 1. Welding shall be by the metal-arc method or gas-shielded arc method described in the American Welding Society's "Welding Handbook" as supplemented by other AWS standards. Welding shall conform to American Welding Society Structural Code D.1.1-82. Qualification of welders shall comply with AWS Standard AWS D1.198.
 - 2. During assembly and welding the component parts shall be clamped, supported and restrained to minimize distortion and for control of dimensions. Weld reinforcement shall comply with the AWS code. Upon completion of welding, weld splatter, flux, slag and burrs left by attachments shall be removed. Welds shall be repaired to produce a workmanlike appearance with uniform weld contours and dimensions. Sharp corners on material that is to be painted or coated shall be ground to a minimum of 1/32" on the flat.
- D. Material type, thicknesses and dimensions shall conform to the screw conveyor schedule requirements unless otherwise specified, or required for structural considerations.

1.7. SERVICE CONDITIONS FOR SCREW CONVEYORS

- A. Material Characteristics (See Schedule for Details)
- B. Operating Conditions and Performance Requirements (See Schedule for Details)

1.8. TROUGHS, COVERS, END PLATES, CHUTES AND SLIDE GATES

A. Conveyor Troughs

1. Horizontal and inclined conveyor trough bodies shall be “U” type for horizontal conveyors and “O” type for inclined conveyors with trough dimensions conforming to the requirements of CEMA Standard 300, with exceptions as noted herein. The conveyor trough bodies shall be rolled to shape with both side flanges formed with the trough body. Trough end flanges, welded to each trough section, shall be provided for joining adjacent sections or connecting trough end plates to the trough sections. Each conveyors trough sections must not exceed 8 feet (96 inches) from flange to flange.
2. Conveyor troughs shall be equipped with a wear liner across the entire inside length. The wear liner shall be fabricated of high density UHMW polyethylene. The wear liner shall be provided in 4-foot long sections for ease of replacement. The wear liner shall be held in place using clips welded to the inside of the trough. Clips shall be placed away from the conveyor spiral path.
3. Each conveyor shall be equipped with a drainage valve at its lower end. The drainage valves must be stainless steel ball valve of 2 inches in diameter.
4. Conveyor troughs for shaftless screw conveyors shall be furnished with anti-floatation devices. Anti-floatation devices shall be installed at a minimum distance of every sixteen (16) feet of conveyor length.

B. Conveyor Covers (U Shape only)

1. A sectional plate shall cover the entire trough, with exception of the inlet chute connections located. Cover segments at inlet chute connections shall extend, on each side, beyond the inlet chute connection and shall be bolted in place. Each cover section shall be not greater than 6-feet in length. The cover segments shall be arranged so that the trough flange and a trough stiffener provide support to each edge.
2. The covers open for maintenance and cleaning purposes. The cover shall be bolted.

C. Conveyor End Plates

1. End plates shall be bolted to the trough end flange. End plates shall be designed to support the drives, bearings and gear reduction assembly.
2. Where shafts penetrate the end plates, the end plate shall be provided to accommodate the stuffing box provided.

D. Conveyor Chutes

1. Trough inlet and discharge chutes shall be bolted to the adjoining equipment as shown on the Contract Drawings. Inlet and discharge chutes shall be supplied with reinforced rectangular or circular flanges.

1.9. CONVEYOR DRIVE TRAIN EQUIPMENT

- A. The conveyor drive train equipment shall transmit power to the conveyor drive shaft using a shaft mounted speed reducer connected to an electric motor. Drive units shall be rigidly supported to avoid all visible “wobble” movement under any operating conditions. Any leakage coming from the trough passing through the shaft seal shall leak through the adapter end plate to atmosphere rather than into the gear motor.
- B. Drive Shaft Assembly
 1. The drive end assembly shall consist of the following components.
 - a. Drive Shaft
 - b. Seal plate
 - c. Hollow Shaft Gear Reducer
 2. The tail end shall consist of the following components
 - a. Zero Speed Sensor
 3. Drive shafts for the shaftless screw conveyors are to be complete with mating connections appropriate for mating to the shaftless screw assembly. The drive shall be of adequate diameter to handle all radial, thrust and torsion loads. The drive shafts shall be mounted to the shaftless spiral by a flanged connection.
- C. Drive Motor
 1. Drive motors shall be of the high starting torque (NEMA B or C design), TEFC, squirrel cage, AC induction type suitable for continuous severe duty service. The motor shall have Class F, MG 1 Part 31.4.4.2 Inverter Duty insulation, and a Class B temperature rise at 40°C ambient temperature. The drive motor shall be high efficiency with a 1.15 service factor.
 2. The drive motor shall operate at 1800 rpm nominal speed and be sized to operate without overloading when the screw conveyor is operating at, or below, the design capacity. The minimum motor horsepower shall be as specified on the conveyor schedule elsewhere herein.
- D. Speed Reducer
 1. The speed reducer shall be a gearmotor enclosed shaft mount, type unit with a double or triple reduction ratio, as may be required. The speed reducer shall mount directly on the driven shaft and utilize a standard CEMA mounting to support structure. We highly recommend Screw Conveyor Package design flange from Nord Gear Corp.
 2. The speed reducer sealing system shall be a Quadrilip system that has four components for sealing lubricant inside and contaminants outside of the speed

reducer. The system must include a double lip seal, a single lip seal and grease pack barrier (grease lip). The sealing system shall also include diverting ports to avoid any possibilities of contamination to the gear reducer oil, even if the first seals broke down.

3. The speed reducer housing shall be constructed of 1-piece corrosion resistant gray cast iron torsional stiff. All housings bore and mounting faces shall be machined in one step to produce extremely precise tolerances. Thus ensuring accurate positioning of gear teeth, bearing & seals and a longer life of all components
4. All speed reducer gearing shall be of helical design. High speed gears shall be ground and low speed gear skive hobbled. Gears hardness shall be 58 Rc minimum. With an overload capacity of 275%
5. The speed reducer bearings shall be ball or tapered roller type and provide a 50,000-hour B-10 life at the expected design loading rate. All seals shall be double lip, spring-loaded type and made of nitrile rubber. Shaft seals shall be quadrilips two nitrile seals, which consists of 2 sealing lips, a trash guard lip and a grease chamber between the seals.
6. The speed reducer shall be manufactured to Quality Class 8 per AGMA Standard 6001-C88, minimum. The gear reducer shall be selected for AGMA Class 11 service with a 1.4 service factor based on motor nameplate horsepower.
7. The speed reducer shall be as manufactured by Nord Gear Corp., or equal.

1.10. CONVEYOR SUPPORTS

- A. The conveyor troughs shall be supported using saddle type supports shaped to the profile of the troughs and extending to a common fixed distance below the centerline of the screw. Saddle type supports shall be located not greater than 16-feet center-to-center. Separate support points shall be provided (when possible) under the drive end and tail end assemblies. These supports are to be bolted to the trough flange.
- B. Troughs thickness shall be design to maintain through shape under loading. Transverse stiffeners shall not be used in order to allow easy maintenance of the UHMW liner and for the flight maintenance.
- C. Support loadings are to be based on a completely filled trough, weight of the conveyor and the dynamic loading when operating.
- D. The Contractor shall coordinate with the Conveyor Equipment Supplier support locations with the facility structural constraints. Access to other process systems and equipment shall not be restricted by the conveyor supports.
- E. Each conveyor shall be field shimmed as required to conform to the manufacturer's installation tolerances.

1.11. SHAFTLESS SCREW ASSEMBLIES

- A. The shaftless screw flights shall be cold rolled spirals fabricated from high strength carbon spring steel bars with a minimum tensile strength of 100,000 psi and a minimum Brinnell hardness of 200.
- B. Shaftless screw conveyor spiral flights shall be formed in a forming machine to the diameter and pitch required. Each formed section shall be factory welded into full lengths. Where the length of the conveyor precludes shipping, sub-sections of flighting shall be divided into maximum shipping lengths and welded together in the field.
- C. Field splicing of flighting sections shall be by full penetration welds done in strict accordance with the manufacturer's instructions using AWS certified welders.
- D. Spiral flighting shall be connected to the drive shaft using a flanged connection plate that shall be welded to the spiral on one end. Transition between the flighting and connection flange shall be smooth and fabricated to the tolerances listed below. The drive shaft shall have a matching mating flange and shall be bolted to the flight connection plate.

1.12. SAFETY DEVICES AND LIMIT SWITCHES

- A. Zero Speed Switches
 - 1. Motion failure alarm unit shall equip each conveyor. The motion sensor shall be a non-contacting type probe. This probe shall be able to read screw rotation outside of the process, by reading disruption of the magnetic field by the ferrous target (flight).
 - 2. The electronic unit shall operate on 120 volts AC power supply and shall be rated NEMA 4X or IP67. The probe signal shall be relay interconnected with a PLC digital input.
 - 3. The zero speed sensors shall be located at the opposite end of drive. The PLC is always counting pulses, without any pulse count a PLC digital output shall be activated for an alarm or shall stop the whole system. Relay switch contacts shall be rated 120 VAC and interlocked with MCC motor starter.
 - 4. Zero speed detection switches shall be Siemens model Sitrans WM100, or equal.

1.13. CONVEYORS CONTROL

- B. Conveyors shall be controlled by the rotary press PLC and operator interface.

PART 3. EXECUTION

3.01 INSTALLATION

- A. Deliver valve(s) in manufacturer's original, unopened cartons. Store in a safe place until ready for installation.
- B. Follow manufacturer's printed instructions for installation.
- C. Remove all identifying marks, labels, etc. from exterior.

- D. Conveyors shall be set upon level, fully grout foundations so that connecting flanges, screwed connection, or flexible connection shall meet without strain or distortion.

3.02 TESTING

- A. All screw conveyor equipment shall be fully assembled and operated prior to shipment to the job site. Operation shall be smooth, free of vibration or unwarranted noise, or unwarranted high starting or no load running amperage draw.

SCREW CONVEYOR SCHEDULE

Identifier	Receiving Conveyor
Conveyor Type	Horizontal, Mono-Directional, Shaftless
Material handle	Cake from discharge of rotary press
Mass Flow Capacity	0.5 Wet ton/hr
Loading at nominal capacity	45%
Material density	45 lbs / cubic feet
Temperature	40°F to 100°F
Flight nominal diameter	9-inches diameter
Flight Pitch	9-inches pitch
Flight section	2 ½" X ¾"
Flight material	High strength chrome alloy steel, 200 brinnell min
Flight diameter tolerance	+ 0, - 1/4"
Flight pitch tolerance	± 1/4"
Conveyor nominal Length	6'9"
Conveyor slope	0 degrees
Trough body	U-Trough 304 stainless 11 gages thick
Trough cover	304 stainless 12 gage thick
Trough end flanges	304 stainless 3/8" thick
Trough end plates	304 stainless 1/4" thick
Discharges chutes	304 stainless 1/8" thick
Support	304 stainless 5/16" thick
Liner	UHMW 3/8" thick
Inlet	Two (2)
Inlet type	Inlet cake chutes from rotary presses
Outlet	One (1)
Outlet type	To Inclined Conveyor
Gear reducer manufacturer	Nord Gear
Gear reducer model	SK2282 SCP
Power	1 HP 480V/60Hz/3ph
Output speed	24 RPM
Special Requirements	2" drain Screw Conveyor Package gear motor flange

Identifier	Inclined Conveyor
Conveyor Type	Inclined, Mono-Directional, Shaft
Material handle	Cake from discharge of rotary press
Mass Flow Capacity	0.5 Wet ton/hr
Loading at nominal capacity	45%
Material density	45 lbs / cubic feet
Temperature	40°F to 100°F
Flight diameter	9-inches diameter
Flight Pitch	5-inches pitch
Flight section	External section of 2 ½" X ¾" and internal section of 1" X ½"
Flight material	High strength chrome alloy steel, 200 brinnell min
Flight diameter tolerance	+ 0, - 1/4"
Flight pitch tolerance	± 1/4"
Conveyor nominal Length	17'4"
Conveyor slope	35 degrees
Trough body	Tubular O-Trough 304 stainless 10 gage thick
Trough cover	304 stainless 12 gage thick
Trough end flanges	304 stainless 3/8" thick
Trough end plates	304 stainless 1/4" thick
Discharges chutes	304 stainless 1/8" thick
Support	304 stainless 5/16" thick
Liner	UHMW 3/8" thick
Inlet	One (1)
Inlet type	Chute from Receiving Conveyor
Outlet	One (1)
Outlet type	To Bin
Gear reducer manufacturer	Nord Gear
Gear reducer model	SK2282 SCP
Power	5 HP 480V/60Hz/3ph
Output speed	53 RPM
Special Requirements	2" drain Screw Conveyor Package gear motor flange

***** END OF SECTION *****

SECTION 43 11 00 – BLOWER EQUIPMENT

PART 1 - GENERAL

1.01 Description

- A. The Contractor shall furnish and install four (4) fully operational blowers for the Blower Room of the control building. Blowers are to be used to supply air to the Biological Treatment System. In addition, Contractor is to furnish and install one (1) fully functional blower inside the existing metal building (just north of the proposed blower building) as shown on plans. The single blower is to be used to supply air to the Aerobic Digester.
- B. Location for blowers:
 - Blower Room of the Control Building (four identical 20 HP blower assemblies for the Biological Treatment System)
 - Existing Metal Building just west of Train #2 (one 5 HP blower assembly for the Aerobic Digester)

1.02 Submittal Data

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials.
 - 2. Fabrication drawings with full dimensions and installation drawings showing coordination with piping connections.
 - 3. Operations and maintenance information as specified in Section 01 33 00.

PART 2 - PRODUCTS

2.01 Blower

- A. Biological Treatment System Blowers:
Four (4) Blowers. Kaeser Positive Displacement Blower, Com-paK Package, DB 166 C, 20hp/460V/3ph/60Hz
Ventilation Fan: 115V/1ph/60Hz
Design Point: 315 SCFM, 8.9 psig, with 72 dB(A) sound pressure level rated enclosure.

Aerobic Digester Blower:

One (1) Blower. Kaeser Positive Displacement Blower, Com-paK Package, BB 52 C, 5hp/460V/3ph/60Hz

Ventilation Fan: 115V/1ph/60Hz,

Design Point: 120 SCFM, 4.0 psig, with 72 dB(A) sound pressure level rated enclosure.

All blowers are to be variable speed units with VFD drives. Manufacture to be Kaeser Compressors, Aerzen, or an approved equal, that meets all the requirements of this specification.

B. Blower Type

The blowers shall be oil-free, positive displacement, rotary lobe type with three lobes per rotor, designed for air service, and belt driven via electric motor.

C. Casing

The casing shall be made of high strength, close grained, cast iron, and shall be adequately ribbed to resist deflection and facilitate cooling. The casing shall be precision machined to allow for minimum clearances. The design of the housing shall allow for vertical or horizontal blower installation with minimal modification. The casing design shall incorporate channels at the discharge port to minimize pulsation.

D. Rotor Assemblies

The rotor assemblies shall be one piece and precision machined from high strength, close grain, spheroidal graphite ductile iron allowing smooth, efficient operation at all rated speeds and pressures. Rotors shall be closed-end using a threaded plug. The integral shafts shall be designed to carry higher loads than calculated for the maximum design load and shall incorporate replaceable seal ring wear sleeves. The rotor assemblies shall be statically and dynamically balanced. Removal of material from the face of the rotor for balancing purposes is not allowed. The rotors shall be a tri-lobe design in order to minimize pulsation and noise. The rotors shall be balanced according to ISO 1940 class Q 2.5. With respect to acceptable vibration levels, all blowers must conform to VDI 3836 standards and guidelines. Vibration range of 0.11 inches/second to 0.276 inches/second shall be a requirement.

E. End Plates

1. The gear-end plate shall be cast iron. The drive-end plate shall be integral to the blower casing ensuring heavy-duty construction, long life, and maximum machine noise attenuation in all applications. Bearing fits shall be precision machined to ensure accurate positioning of the rotors in the

casing. Replaceable seal wear inserts shall be provided on the drive-end and gear-end plates.

2. The end plate covers shall be heavy-duty cast iron with a precision-machined sealing face. Aluminum oil sump covers are not allowed.

F. Timing Gears

1. The rotor timing gears shall be precision machined from case hardened and ground alloy steel. Each timing gear shall be straight cut and beveled to eliminate axial bearing loads and ensure long life as well as quiet operation.
2. Gears shall be finish ground on a precision grinder to ensure concentricity. The timing gear set shall be taper-mounted on the rotors for improved rebuild ability and accurate timing. Keyed, hub mounted, or taper-pinned timing gear mounting designs are not acceptable.

G. Bearings

All four shaft support locations shall incorporate large, heavy-duty cylindrical roller bearings designed to handle extreme radial loads without sacrificing product integrity and reliability. The minimum acceptable L_{10} design life at the blower's maximum rated speed and maximum rated differential pressure shall be no less than 100,000 hours.

H. Lubrication

Both the gear-end and the drive-end of the blowers shall be oil splash lubricated for minimal maintenance and long service life. Grease lubricated bearings are not acceptable. The lubrication design shall ensure adequate lubrication of the timing gears and bearings. Enough synthetic blower lubricant for up to two oil changes per year. Oil type shall be as recommended by the manufacturer.

I. Seals

1. There shall be four (4) piston ring type labyrinth seals at each end of each rotor to minimize leakage and maintenance costs.
2. The cavity between the air-side and oil-side seals shall be vented through threaded ports to allow external purging or containment of any lubricating oil or process gas that may have migrated past the seals. Enough ports shall be incorporated in the design to ensure that the cavity is vented from the bottom no matter what blower drive-to-driven shaft orientation is utilized.

3. A vent cavity shall be provided on all four (4) sets of piston ring type labyrinth seals. The two (2) vent holes located on the bottom side of the blower shall be left open. The vent holes at other locations shall be closed off with threaded metal plugs.
4. The input drive shaft seal shall be a high temperature radial lip type seal with Viton elastomers. The seal design shall incorporate a replaceable wear sleeve on the input drive shaft.

J. Drive Motors

1. Drive motors shall be designed, manufactured, and tested in accordance with the latest revised editions of NEMA MG 31.4, IEC, IEEE, ANSI, and AFBMMA standards as applicable and shall be capable of continuous operation.
2. The motor nameplate horsepower rating shall not be exceeded at the design speed.
3. The temperature rise of the motor windings shall not exceed IEC and NEMA standards when the motor is operated continuously at the rated horsepower, rated voltage, and rated frequency in ambient conditions of 40 °C.
4. The motors shall conform to the following:

Mounting:	Horizontal
Type:	Squirrel Cage Induction
Enclosure:	Totally Enclosed - Fan Cooled/IP-55
Service Factor:	1.15
Duty Cycle:	Continuos
Ambient Temperature Rating:	40 °C/104°F
Starting Current:	<x10 Full Load Current Maximum
Winding Insulation:	Class F
Temperature Rise:	Class B
Voltage/Frequency/Phase	460/60/3
Bearing Lubrication:	Oil, Oil Mist, or Grease (with grease addition fittings and grease relief fittings)
Maximum Speed:	3,600 rpm
Speeds:	VFD (18 Hz to 60 Hz)
Motor Protection:	PTC thermistors

K. Blower Packages

The packaged blowers are to be standard engineered designs of a CE certified manufacturer regularly engaged in the production of packaged blowers to ensure

single source accountability and shall include the following listed standard features:

- (a) The packages shall be driven through V-belts and sheaves. The drive assembly shall be of the high capacity type, oil and heat resistant, with a minimum service factor of 1.5.
- (b) Automatic tensioning of the V-belts by use of a pivoting, swing frame motor base with adjustable spring assistance and visual indication of V-belt tension shall be provided to ensure the V-belts remain properly tensioned with minimal maintenance and to extend V-belt, sheave, and bearing life. Adjustment of the tensioning device shall be accomplished without removal of the guard or loosening of the motor mounting bolts.
- (c) The drive guard shall be the manufacturer's standard sheet metal with provision for ventilation. The installed guard shall be fully enclosed, easily removable, and designed to meet current OSHA recommendations and CE standards.
- (d) The base shall be an elevated, rigid, fabricated steel design with a solid sub-base. The absorptive type discharge silencer shall be integral to the frame in order to minimize space requirements. The blower shall be mounted horizontally for a compact frame.
- (e) To prevent transmission of vibration and noise, as well as secure the package to the foundation, the base shall include vibration isolators made of rubber in a steel footing equipped with mounting holes for anchoring purposes.
- (f) Oil drains from the blower drive-end and gear-end lubricating oil sumps shall be piped to the front of the base for ease of maintenance. The drain valves shall be a ball valve with a fully retained and gasketed threaded cap.
- (g) The inlet filter shall be integral to the inlet silencer and shall include a polyester element for minimal pressure drop.
- (h) The inlet silencer shall be of the absorptive type, directly connected to the inlet port of the blower, and shall be mounted horizontally.
- (i) The discharge silencer shall be designed specifically for the frequency of the blower for maximum attenuation and shall use a combination of absorption, reflection, and diffusion. The silencer shall be directly connected to the outlet port of the blower. The discharge silencer shall be mounted horizontally and shall be integral to the base frame.
- (j) The relief valve shall be spring loaded and factory installed in a location to protect the blower from excessive differential pressures. The relief valve exhaust shall be piped out of the enclosure.
- (k) An unloaded start valve shall be provided and factory installed to allow the motor to accelerate prior to system pressurization and eliminate pneumatic shock. (When specified)

- (l) An elastomeric compensator/flex connector shall be provided for connection of the packaged blower to the system piping to reduce transmission of structure borne noise as well as prevent unacceptable loading of the silencer connection and blower casing. On packages with tube outlets (4" and below) the compensator shall be hose-type. On packages with outlets larger than 4", the compensator shall be arch-type, flanged with both ANSI and DIN bolting patterns.
- (m) A sound enclosure shall be provided. The sound enclosure shall be sheet steel construction with powder coat finish. It shall have acoustic foam insulation and shall provide sound attenuation of up to 20 dB(A). The enclosure shall have a hinged panel on top and a removable panel on the front of the package to allow maintenance access. Panels shall incorporate locking closures. At least one installed, integral ventilation fan, sized to provide adequate cooling of the package, shall be provided (wiring of fan by others). Unless specified, the fan voltage will be the same as the main motor voltage.
- (n) A pressure gauge shall be provided, pre-piped and panel mounted, on the sound enclosure.
- (o) A temperature gauge, with adjustable switching point and contact, shall be provided pre-piped and panel mounted on the sound enclosure (wiring of switch by others).
- (p) A filter differential pressure gauge shall be provided pre-piped and panel mounted on the sound enclosure.
- (q) A Perspex gauge cover shall be included to protect the gauges.
- (r) The blower package shall be designed to allow all preventive maintenance to be performed from the front of the package. All utility connections and process connections shall be at the rear of the package.
- (s) The blower package shall be capable of being installed directly adjacent to another blower packages of similar design and shall be capable of mounting next to the wall without maintenance interference.

L. Factory Testing

- 1. All critical dimensions of the blower components actually provided by the manufacturer shall be verified and documented prior to assembly.
- 2. The rotating parts of each blower actually provided by the manufacturer shall be statically and dynamically balanced before final assembly. The blower alone shall operate without excessive vibration.
- 3. Each blower actually provided by the manufacturer shall be slip tested. The Slip RPM shall be documented. Each bare blower provided by the manufacturer shall be operated at its maximum rated speed and differential

pressure for thirty (30) minutes. A document certifying that the supplied blowers conform to the design specifications shall be provided.

4. On completion of final assembly of the packaged blower and prior to shipment, each packaged blower shall be mechanically run for a minimum of fifteen (15) minutes.

M. Quality Assurance

1. The blowers and equipment covered by this specification are intended to be standard blower equipment, of proven ability, as manufactured by a reputable CE certified manufacturer having at least two (2) years experience in the production of such blowers. The blowers furnished shall be designed, constructed, and installed in accordance with the best practice and methods and shall operate satisfactorily when installed.
2. All equipment furnished under this specification shall be new, unused, and shall be the standard product of a manufacturer having a successful record of manufacturing and servicing the equipment and systems specified herein for a minimum of two (2) years.

N. Warranty

1. The manufacturer and contractor shall warrant the blower being supplied to the owner against all defects in workmanship and materials for a period of sixty (60) months from date of shipment. All other package components shall be warranted for a period of one year from date of substantial completion.
2. Complete inspection of the equipment. Any work necessary to restore the equipment to "as shipped" condition shall be the responsibility of the Contractor. Blower supplier/distributor must be factory trained and capable of supporting blowers in the field as well as major blower block rebuilding at a local location in the Northwest in the event of an emergency situation.

O. Specific Details

Tag Numbers:	ESWD WWTP – Biological Treatment System Blowers (Trains 1 – 3)
Installation Location:	Blower Building
Blower Manufacturer:	Kaeser Compressors or approved equal.
Blower Package Model:	DB 166 C
Quantity:	(4)
Operation Protocol:	Variable Speed

Tag Numbers:	ESWD WWTP – Aerobic Digester Blower
Installation Location:	Existing Metal Building (Adjacent to Train #2)
Blower Manufacturer:	Kaeser Compressors or approved equal.
Blower Package Model:	BB 52 C
Quantity:	(1)
Operation Protocol:	Variable Speed

P. Service Conditions and Performance Requirements

ESWD WWTP – Biological Treatment System Blowers (Trains 1 – 3)

Elevation: (Std)	25 ft.	Gas:	Air
Ambient Pressure: (Std)	14.7 psia	K-Value:	1.395
Ambient Temperature: (Std)	80 °F	Specific Gravity:	1.000
Relative Humidity: (Std)	50%	Molecular Weight:	28.966
Maximum Allowable Blower Package Noise Level:	72 dB(A) @ 1 Meter	Estimated Blower Package Noise Level:	72 dB(A) @ 1 Meter
Inlet Air Flow at Design:	325 cfm	Max Blower Speed (60Hz):	4270 rpm
Air Flow (std) at Design:	315 scfm	Design Blower Speed (60Hz):	3999 rpm
Max Pressure Differential:	10.1 psig	Min. Blower Speed (18 Hz):	1281 rpm
Discharge Pressure at Design:	23.6 psia	Discharge Temp at Design:	207 °F
Differential Press. at Design:	8.9 psig	Blower Shaft Power at Design:	17.4 hp
Relief Valve Set Pressure:	12.0 psig	Motor Horsepower:	20 hp
Supply Voltage:	480 VAC	Motor Speed:	3600 rpm

ESWD WWTP – Aerobic Digester Blower

Elevation: (Std)	25 ft.	Gas:	Air
Ambient Pressure: (Std)	14.7 psia	K-Value:	1.395
Ambient Temperature: (Std)	80 °F	Specific Gravity:	1.000
Relative Humidity: (Std)	50%	Molecular Weight:	28.966
Maximum Allowable Blower Package Noise Level:	72 dB(A) @ 1 Meter	Estimated Blower Package Noise Level:	72 dB(A) @ 1 Meter
Inlet Air Flow at Design:	124 cfm	Max Blower Speed (60Hz):	4860 rpm
Air Flow (std) at Design:	120 scfm	Design Blower Speed (60Hz):	4729 rpm
Max Pressure Differential:	6.6 psig	Min. Blower Speed (18 Hz):	1458 rpm
Discharge Pressure at Design:	18.7 psia	Discharge Temp at Design:	153 °F
Differential Press. at Design:	4.0 psig	Blower Shaft Power at Design:	3.8 hp
Relief Valve Set Pressure:	8.0 psig	Motor Horsepower:	5 hp
Supply Voltage:	480 VAC	Motor Speed:	3600 rpm

Q. Controls

NEMA 12 steel cabinet for controls. Blower VFDs shall be included in the motor control center (MCC) as shown on the drawings. Contractor shall wire over temp sensors, over pressure sensors, and motor temp sensors to the PLC control panel. See electrical drawings.

PART 3 - EXECUTION

3.01 Installation

- A. Deliver blower in manufacturer's original, unopened cartons. Contractor to provide tees, supports, and fittings (as needed) for construction.
- B. Follow manufacturer's printed instructions for installation.
- C. Remove all identifying marks, labels, etc. from exterior.

***** END OF SECTION *****

SECTION 43 22 10 - 2W PLANT WATER SYSTEM EQUIPMENT

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install a fully operational 2W Plant Water System for the Eastsound WWTP as indicated on the Plans and described in the specifications.
- B. In general, the 2W Plant Water System Equipment will consist of the following components:
 - 1. Two 2W Plant Water Pumps
 - 2. 2W Plant Water Pump Station Accessories
 - 3. One Fiberglass Enclosure with Flow Meter System
 - 4. One Fiberglass Enclosure with Air Gap System
 - 5. Two Air Diaphragm Tanks
 - 6. Yard Hydrants
 - 7. Wall Mount Eye Wash Stations
 - 8. Piping, valving, and other items as described in specifications and shown on drawings.

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials.
 - 2. Fabrication drawings with full dimensions and installation drawings showing coordination with concrete support connections, wall penetrations, and accessories.
 - 3. Operations & maintenance information as specified in Section 01 33 00.

1.03 SPARE PARTS AND SPECIAL TOOLS

- A. The plant water system shall include the following spare parts, at a minimum:
 - 1. One set of new and unused special tools required for installation, operation, and maintenance of Plant Water System Equipment.

PART 2. PRODUCTS

2.01 2W PLANT WATER PUMPS

- A. Two 2W Plant Water Pumps. Pumps are to be sized for a standard 4" diameter well, ITT Goulds submersible pump model # 55GS50 with stainless steel components. Motor to be Centripro 3 phase, 460 Volt, 5 horsepower, model # M50434 (or equal). Design flow range = 20 to 80 gpm. Pumps are to include variable frequency drives (VFDs). Pump control strategy
 - 1. Just maintain a constant pressure (user input pressure, anticipate desired set point will be somewhere between 50 and 75 psi). Turn pump(s) on, adjust speed, turn pump(s) off, as needed.
 - 2. Alternate pumps after each ON/OFF cycle.

2.02 2W PLANT WATER PUMP STATION ACCESSORIES

- A. The Plant Water Pump Stations are to be equipped with the following:
1. External Splice Boxes, Orenco Model # SBEX
 2. Conduit Seals, Orenco Model # SBCS
 3. Float Switch Assembly, Orenco Model # MF2, sized per drawings.
 4. Float Collars, Orenco Model # MFCOLLAR1
 5. Pump Discharge Assemblies, Orenco Model # HV, 2" discharge piping with ball valve, check valve, external flex hose, and quick disconnect for each pump.
 6. Pump Basin, Orenco Model # PB30, sized per drawings.
 7. Pressure Relief Valves (each set at 110 psi), Clayton Mark or equal.
 8. Other Components as shown on drawings, and as needed for complete system assembly.

2.03 2W AIR DIAPHRAGM TANKS

- A. Two (2) Air Diaphragm Tanks. Tanks are to be Amtrol Well-X-Trol, model WX-350D with composite base (or approved equal). Tank size = 119 gallons (26" diameter x 62" height). Additional Tank Accessories
1. Pressure Relief Valves:
 - i. One (1) mounted on tank tee, and set at 100 psi
 - ii. One (1) mounted on pump discharge piping, and set at 120 psi.
 2. Pressure Transducer for controlling pump ON/OFF operation
 3. Pressure Gauge (0 to 150 psi) for visual pressure reading
 4. Pressure Relief Valve (set at 120 psi), Clayton Mark or equal.

2.04 EXTERIOR PIPE INSULATION

- A. Provide at exposed plant water system pipe locations, and locations called out on drawings. Insulation thickness for exterior pipe insulation shall be 1.25 inch.
- B. Insulation Material:
1. Insulation Material shall be closed cell polyisocyanurate cellular plastic.
 2. Manufacturers: One of the following or equal:
 - i. ITW Insulation Systems, Trymer 2000 XP
 - ii. Or equal
 3. Minimum temperature range: Minus 10 degrees Fahrenheit to plus 110 degrees Fahrenheit.
 4. K factor at 75 degrees Fahrenheit: Not more than 0.19 BTU-inch/hour –square feet degrees Fahrenheit.
 5. Fire Rating: flame spread: 25 or less
 6. Joints: Seal with manufacturer's recommended contact adhesive to form continuous water barrier.
 7. Density : Min. 2-lb/ft³

C. Insulation Jacket:

1. Material: Smooth Finish, Aluminum; 0.016-inch (26 gauge) minimum thickness.
2. Overlap: Overlap circumferential joints 4 inches minimum; overlap longitudinal joints 1-inch minimum; longitudinal joints oriented to minimize water entry.
3. Bands: 0.5 inch wide, 0.0508 inch (16 gauge) thick aluminum, same alloy as jacket or 0.0179 inch thick Type 304 stainless steel; install on 18 inch centers, uniformly spaced and at all fitting joints.
4. Joint seal: Apply waterproof adhesive at joints and overlaps.
5. Fittings: Custom fit of same materials.
6. Manufacturers: ITW Insulation Systems or equal.

D. Pipe Insulation Installation:

1. Install insulation and jacket materials in accordance with manufacturer's written instructions.
2. Apply insulation in smooth, clean manner with tight and finished smooth joints. Fit insulation tightly against surfaces. Insulate each continuous run of pipe with full-length sections of insulation with a single piece cut to length to complete the run of pipe. Do not use cut pieces or scraps to complete the installation.
3. Butt longitudinal and circumferential insulation joints firmly together.
4. Maintain the integrity of vapor barrier jacketing. Do not use staples to hold vapor barrier overlaps in place.
5. Apply sealant or cement when previous applications of adhesives and cement have thoroughly dried.
6. Apply insulation to permit upgrade or contraction of pipelines without damage to insulation or jacketing.
7. Fittings:
 - i. Insulate fittings by covering with mitered sections of insulation or utilize factory-made prefabricated fitting shapes.
 - ii. Terminate preformed pipe jackets or covering at sufficient distance from flanges to permit removal of bolts.
 - iii. Overlap flange and flanged fitting insulation on adjacent pipe covering by at least 2 inches.
8. Provide continuous insulation through and over pipe supports and provide protection saddles at supports.
9. Extend insulation against insulation end protection shields or covers so that insulation voids do not exist and provide watertight end seals and covers where insulation terminates.

2.05 EXPOSED PIPE COLOR

- A. See Section 09 91 00 Painting for exposed pipe requirements. All exposed (unburied) plant water piping is to be labelled and coated with paint per Division 9 and 33. This work includes painting plant water yard hydrants, exposed plant water pump discharge piping, and all exposed plant water piping inside the buildings. However, stainless steel pipe does not need to be painted.

PART 3. EXECUTION

3.01 INSTALLATION

- A. Deliver equipment in manufacturer's original, unopened cartons. Contractor to provide supports, anchors, piping, and fittings (as needed) for construction.
- B. Follow manufacturer's printed instructions for installation. Provide anchor bolts and accessories, as needed.
- C. Remove all identifying marks, labels, etc. from exterior.
- D. Verify correct pump rotation direction.

3.02 TRAINING

- A. The equipment supplier(s) shall provide training to the OWNER's personnel as required for a complete understanding of how the Plant Water System Equipment operates.

***** END OF SECTION *****

SECTION 45 05 00 – VALVES

PART 1. GENERAL

1.01 SCOPE

- A. This section covers the work necessary for furnishing and installing the various valves in the plant piping systems.

1.02 SUBMITTALS DURING CONSTRUCTION

- A. Submittals shall be made in accordance with Sections 01 33 00.

PART 2. PRODUCTS

2.01 GENERAL

- A. All valves shall be complete with all necessary operating handwheels, chain wheels, extension stems, floor stands, worm and gear operators, operating nuts, chains, and wrenches which are required for the proper completion of the work included under this section.
- B. For potable water applications, all materials in substantial contact with potable water shall meet NSF 61 certification.

2.02 VALVE OPERATORS

- A. All valve operators shall open by turning counterclockwise. Buried valve operators with 2-inch nuts shall have AWWA C504 2-inch square operating nuts and be full enclosed, grease packed. Depending upon valve type, size, and operating torques, gear operators shall be provided as needed so as to permit operation of the valve under full operating head with a maximum pull of 40 pounds on the handwheel. The valve operators shall be of the self-locking type to prevent the disc or plug from creeping. Self-locking worm gears shall be a one-piece design of gear bronze material, accurately machine cut. Butterfly valve operators shall be provided with position indicators to show the position of the valve disc or plug. Handwheels shall be galvanized and painted the same color as the valve and associated pipeline.
- B. All operating nuts for buried valves are to be located 18" to 24" (approx.) below the surface and with valve boxes. Provide stem extensions as needed.
- C. Operators for Above Ground Valves unless otherwise noted on drawings:
 - 1. All exposed above ground valves 4" or smaller shall have direct-acting lever or handwheel operators on valves within 6 feet of the floor and chainwheel operators on overhead valves.
 - 2. All exposed above ground valves 4" or larger shall have handwheel operators on valves within 6 feet of the floor and chainwheel operators on overhead valves.
- D. Operators for Below Ground Valves unless otherwise noted on drawings:
 - 1. 6" and smaller valves shall be provided with direct 2-inch nut operator.
 - 2. 6" and smaller valves used for flow control shall be provided with worm gear type manual actuators and shall be suited for buried service with 2" nut operator.
 - 3. 8" and larger valves shall be provided with gear type manual actuators and shall be suited for buried service with 2" nut operator.

2.03 VALVE BOXES

- A. Valve boxes shall be two-piece screw type, cast iron, with min. 5-1/4-inch shaft and shall be of appropriate length for the installation. Extension pieces, if required, shall be the manufacturer's standard type. Units shall be Mueller H-10364, Clow Corporation F-2452, or equal. All units shall be complete with all necessary bases and accessories. All buried valves are to be provided with valve box assemblies unless located in a vault per plans.

2.04 SMALL GATE VALVES

- A. 2" to 12" gate valves shall be Kennedy Valve or approved equal. Gate valves shall be non-rising stem, resilient-seated, conforming to AWWA C509 or C515, transition gaskets where needed, fusion bonded epoxy coated inside and out meeting AWWA C550.

2.05 EXPOSED SMALL GATE VALVES

- A. Exposed gate valves smaller than 4" shall be all stainless steel with NPT threads compatible with smaller pipe sizes. Gate valve to be non-rising stem with resilient wedge design for 2" and larger valves. Protect non stainless steel components with fusion bonded epoxy coating.

2.06 CHECK VALVES

- A. Check valves are to be stainless steel or ductile iron body, bronze mounted swing check valves, with external swing arm and spring, fusion bonded epoxy coated inside and out, meeting design per AWWA C-508. Valve shall be MH 259-02 Lever & Spring with Rubber Face and Limit Switch, Model 259-02ALS. External spring and lever shall be relocated to opposite side of valve, if directed by the Owner. Spring and lever relocation, if any, shall be considered incidental to Contractor's bid.
- B. Duckbill check valves shall be Tideflex Model TF-2M for vertical installation only. Duckbill shall be EPDM with 316 Stainless Steel Band compression clamps.

2.07 SMALL PLUG VALVES

- A. Plug valves 2" to 12" shall be Pratt Ballcentric Plug Valves or approved equal with corrosion resistant materials of construction designed for contact with raw sewage, including fusion bonded epoxy coating for all exposed metal. Valves shall be of the non-lubricating, eccentric type and shall be designed for a working pressure of 175 psi. Valves shall provide tight shut off at rated pressure. Plug valve body shall be cast iron ASTM A126 class B with welded-in overlay of 90% nickel alloy content on all surfaces contacting the face of the plug. The valve plug shall be cast iron ASTM A126 class B, with Buna N resilient seating surface. Plug valve bearings shall be furnished with permanently lubricated sleeve type bearings conforming to AWWA C504 with sintered, oil impregnated type 316 stainless steel ASTM A-743 Grade CF-8M.

2.08 LARGE PLUG VALVES

- A. Plug valves 14" and larger shall be of the non-lubricating, eccentric type and shall be designed for a working pressure of 150 psi and have a rectangular port design. Valves shall provide tight shut-off at rated pressure. Valve shall be the Pratt Ballcentric Valve as manufactured by the Henry Pratt Company or approved equal.
- B. Valves shall be constructed of corrosion resistant materials designed for contact with raw sewage, including fusion bonded epoxy coating for all exposed metal.
- C. Valve body shall be cast iron ASTM A126 Class B with welded-in overlay of at least 95% nickel alloy content on all surfaces contacting the face of the plug.

- D. The valve plug shall be ductile iron ASTM A-536, Grade 65-45-12, in valve sizes up to 20" and ASTM A126 Class B cast iron in sizes 24" and larger, with Buna N resilient seating surface to mate with body seat.
- E. Valve flanges shall be in strict accordance with ANSI B16.1, Class 125.
- F. Plug valve shall be furnished with permanently lubricated sleeve type bearings conforming to AWWA C517. Bearings shall be of sintered, oil impregnated type 316 stainless steel ASTM A-743 Grade CF-8M.

2.09 SMALL BALL VALVES

- A. FOR GENERAL USE, NON-POTABLE, EXPOSED: Ball valves smaller than 4" shall be Watts model S-FBV-1, 2 piece full port stainless steel ball valves for air and water applications. Stem shaft shall be 316 Stainless Steel.
- B. INTERIOR POTABLE WATER USAGE: Ball valves 2" and smaller shall be Asahi True Union Type 21, or approved equal. Valves shall be manual operation, meet NSF 60/61 certification, and rated for 200 psi minimum. Construction shall be of CPVC, seat of PTFE, and O rings of FKM.

2.010 FLAP GATES

- A. Flap gates as shown on the drawings and described herein. Flap gates to be M&H Style 47 Flap Valves. Contractor to demonstrate operation of each flap gate during dry test and wet test.

2.011 BUTTERFLY VALVES

- A. FOR INTERIOR PVC PIPE SYSTEMS:
Butterfly valves are to be Asahi Type 57 or approved equal with stainless steel components for all exposed valves. Body and disc shall be PVC. Stem Shall be 316 SS, and seats shall be EPDM. Butterfly Valves shall be supplied with corrosion resistant materials designed for the specific application (such as air pipe applications, water pipe applications, etc.). For potable water applications valves shall meet NSF 60/61 certification.
- B. FOR EXPOSED, DUCTILE IRON, OR STAINLESS AIR PIPE SYSTEMS:
Butterfly valves are to be Bray Series 31 (lug style), Dezurik, or approved equal with stainless steel components for all exposed valves. Body shall material shall match pipe material. Stem shall be 416 stainless steel, disc shall be 316 SS, and seats shall be EPDM – food grade. Butterfly Valves shall be supplied with corrosion resistant materials designed for the specific application (such as air pipe applications, water pipe applications, etc.). For potable water applications valves shall meet NSF 60/61 certification.
- C. FOR BURIED WATER APPLICATION
 - 1. Butterfly valves for mains 10 inches and larger shall be resilient seated and shall meet or exceed AWWA C504, Class 150B. Butterfly valves shall be suitable for direct burial. Shaft seals shall be standard O-ring seals. The size of the butterfly valves shall be the same as the main on which they are located.
 - 2. Valve shall be iron-body, full-bronze mounted, resilient-seat valves and suitable for service with the type and class of pipe used.

2.012 COMBINATION AIR RELEASE VALVES

- A. Combination Air Release Valves shall be A.R.I. D-025 or approved equal. Include stainless steel nipples, adaptors, bottom isolation valve, and accessories.

2.013 GLOBE VALVES

- A. Globe valves 2" and smaller shall be Alloyco stainless steel, Class 150, OS&Y, Plug Type Disc or approved equal. Valves shall be manual operation, meet NSF 60/61 certification, and rated for 200 psi minimum. Construction shall be of CPVC, seat of PTFE, and O rings of FKM.

2.014 HYDRANT ISOLATION VALVES

- A. All plant water distribution system isolation valves (in buried locations and less than 4" diameter) shall be gate valves, 2" to 12" gate valves shall be Kennedy Valve or approved equal. Gate valves shall be non-rising stem, resilient-seated, conforming to AWWA C509 or C515, transition gaskets where needed, fusion bonded epoxy coated inside and out meeting AWWA C550.

2.015 VALVE IDENTIFICATION TAGS

- A. Each shut-off or control valve shall be provided with a 1 ½ inch minimum diameter heavy stainless steel tag. Tags shall bear the valve ID as listed on the valve schedule.
- B. Numbers and letters shall be block type with ½ inch high font stamped on the tags and filled with black enamel.
- C. Attach tags to the valves by split-key rings soldered so that the ring and tag cannot be removed.
- D. Submit to the Engineer for approval, a drawing and a neatly typed valve directory listing each valve number, type of valve and its location.

2.016 VALVE LABEL MARKERS

- A. Each shut-off or control valve shall be provided with a label marker in concrete pad as detailed on the plans and valve schedule.

PART 3. EXECUTION

3.01 INSTALLATION

- A. Bolt holes of flanged valves shall straddle the vertical centerline of the pipe run. Prior to installing flanged valves, the flange faces shall be thoroughly cleaned. After cleaning, insert gasket and bolts, and tighten the nuts progressively and uniformly. If flanges leak under pressure, loosen or remove the nuts and bolts, reseal or replace the gasket, retighten and/or reinstall the nuts and bolts, and retest the joints. Joints shall be watertight at test pressures before acceptance.
- B. Thoroughly clean threads of screwed joints by wire brushing, swabbing, or other approved methods. Apply approved joint compound to threads prior to making joints. Joints shall be watertight at test pressures before acceptance.
- C. Generally, unless otherwise indicated on the Drawings, all valves installed in horizontal runs of pipe having centerline elevations 4 feet 6 inches or less above the finish floor shall be installed with their operating stems vertical. Valves installed in horizontal runs of pipe having centerline elevations 4 feet 6 inches or greater above the finish floor shall be installed with their operating stems horizontal. If adjacent piping prohibits this, the stems and operating handwheel shall be installed above the valve horizontal centerline as close to horizontal as possible.

3.02 TESTING

- A. Valves shall be tested at the same time that the adjacent pipeline is tested. Joints shall show no visible leakage under test. Repair joints that show signs of leakage prior to final acceptance. If there are any special parts of control systems or operators that might be damaged by the pipeline test, they shall be properly protected. The Contractor will be held responsible for any damage caused by the testing.
- B. If requested by the Engineer, the valve manufacturer shall furnish an affidavit stating the materials options furnished and/or that he has complied with these and other referenced specifications.

***** END OF SECTION *****

SECTION 46 33 33 – EMULSION POLYMER MAKE-UP SYSTEM

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install one (1) fully operational emulsion polymer make-up system (liquid polymer system). The polymer system shall be capable of effectively activating and fully blending a homogenous final polymer solution from neat polymers with a 0.1% to 0.5% concentration with an active content from 40% to 55%.

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01340:
 - 1. Manufacturer's catalog data showing construction materials, required seals and guides, and required coatings.
 - 2. Fabrication drawings with full dimensions and installation drawings showing coordination with outlet piping, liner penetration and booting, the outlet structure and pipe supports.
 - 3. Operations and maintenance information as specified in Section 01 33 00.

PART 2. PRODUCTS

1.0 General

- 1.1 Provide all products needed for a complete and functional system.

2.0 Mechanical Equipment

2.1 Water pressure regulator (relief valve)

Clean or filtered water is required at a minimum pressure of 45 PSI. The water pressure must not exceed 100 PSI. Supplied pressure is controlled by a water pressure reducing valve and shall be adjusted to 40 PSI. A pressure gauge, fixed on the water line, will display real time pressure while the system is under operation.

2.2 Proportional Valve (Dilution water flow)

During the process, the water flow is controlled by a proportional valve - an assembly of a ball valve and a positioner. The positioner shall be supplied by a minimum air pressure of 90 PSI. The operational flow range for this valve is selected from 0.1 to 20 GPM (0.4 to 75 LPH) and will be measured by a flow meter installed on the flocculator (already provided with Rotary Press). The regulation of this valve will be based on the ratio Polymer Ratio vs. Sludge, calculated from Rotary Press PLC.

2.3 Chemical Metering Pump (Polymer dosing)

The pump is a peristaltic type. The flow rate range shall be 0.1–500ml/min (0.001–7.93USGPH) and up to 7 bar (100 psi) RMS pressure. A polymer strainer is included and mounted on the inlet suction side of the pump. The control of this pump will be based on Water Flow vs. Final Concentration, calculated from the Rotary Press PLC.

2.4 Static Mixer (Mixing Chamber)

The polymer and the water are injected and mixed inside the static mixer. The energy needed for mixing comes from a loss in pressure as water flow through the static mixer. At the outlet of this motionless mixer, the final solution at the accurate concentration is created.

2.5 Construction

The support frame, brackets, ball valves are made of 304 SS.

The water piping shall be made of HDPE plastic.

The inlet & outlet of neat polymer is blue polyethylene flexible hose

3.0 Electrical Equipment

3.1 Proportional Valve (Dilution water flow)

The proportional valve requires 24 VDC power and 4-20 mA signal for control.

3.2 Chemical Metering Pump (Polymer dosing)

The pump is available 120VAC, 50/60 Hz or 24VDC power, and the control is 4-20 mA.

3.3 Junction Box (NEMA 4X)

The simple construction of these enclosures makes them durable in corrosive environments and have a secure seal. The pump & proportional valves are landed on terminal block. The electrical supply can come from a standard electrical outlet 120VAC or from the Fournier control panel. A power supply 120VAC / 24VDC is included. The control is assumed by a remote or an existing PLC & HMI. It is recommended to have this system controlled by Rotary Press PLC and HMI for full integration.

4.0 Control

The control of the make-up system is provided by the Rotary Press control panel or existing PLC and HMI. The HMI allows the user to manually change the concentration and percent active parameters of the neat polymer as well as the desired final concentration of solution. The HMI allows for the components to be operated in Hand mode as well as to function in Auto (remote) mode, which uses the control signal from the Rotary Press PLC.

5.0 Maintenance

All components are modular, no special tool required and easily accessed.

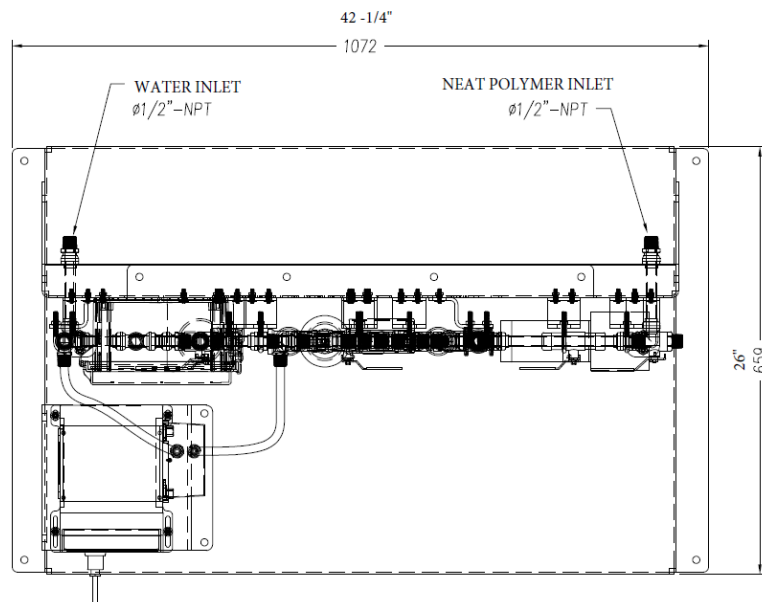
6.0 Spare part

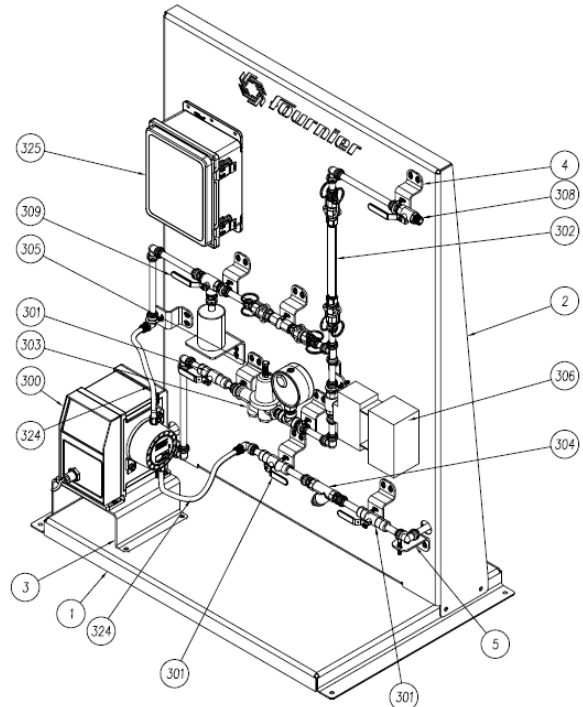
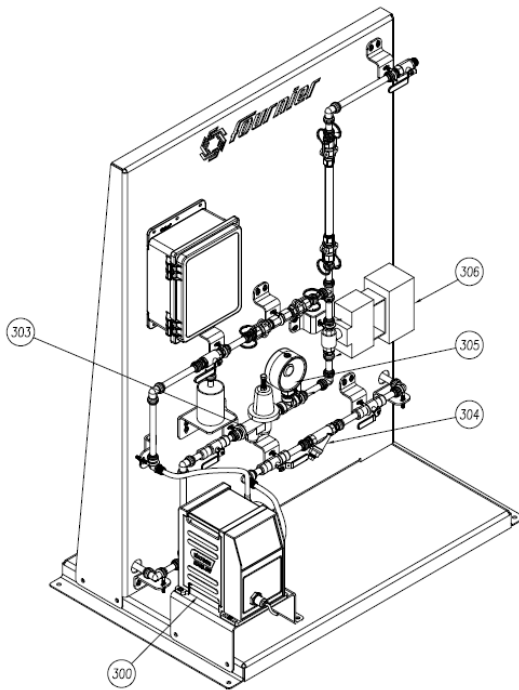
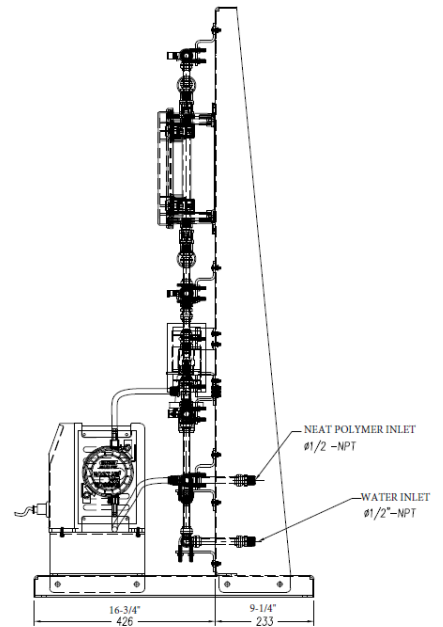
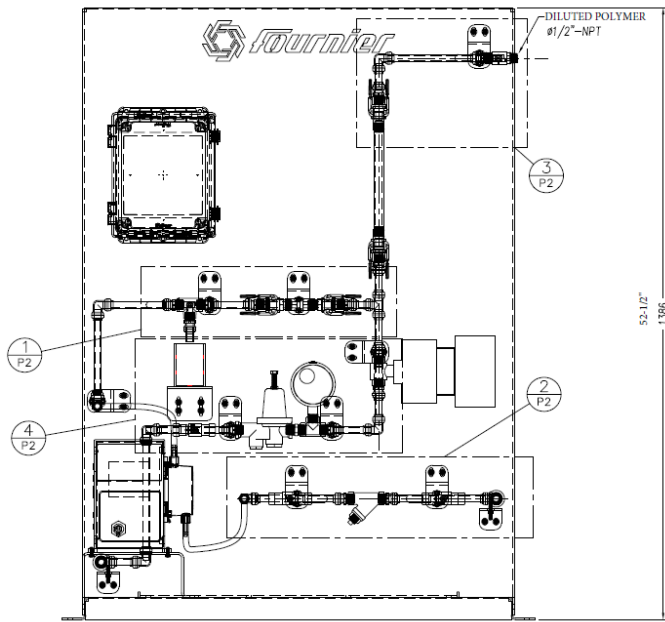
One (1) set of Metering Pump Head replacement kit.

7.0 Materials list

ITEM	DESCRIPTION	MANUFACTURER
1	BASE FRAME	Fournier Industries
3 to 5	BRACKET SUPPORT	Fournier Industries
300	DOSING PUMP	WATSON -MARLOW
301	BALL VALVE ½"	McMASTER
302	STATIC MIXER	KOFLO
303	PRESSURE REGULATOR	WATTS
304	STRAINER ½"	McMASTER
305	PRESSURE GAUGES 0-100 PSI	McMASTER
306	VALVE & ACTUATOR (PROPORTIONAL VALVE)	FESTO
308	BALL VALVE ½"	McMASTER
309	3-WAYS BALL VALVE ½"	McMASTER
324	HOSE ½"	McMASTER
325	JUNCTION BOX, 10 x 10 x 8	HOFFMAN

Example Equipment Drawings:





PART 3. EXECUTION

3.01 INSTALLATION

- A. Deliver valve(s) in manufacturer's original, unopened cartons. Store in a safe place until ready for installation.
- B. Follow manufacturer's printed instructions for installation.
- C. Remove all identifying marks, labels, etc. from exterior.

***** END OF SECTION *****

SECTION 46 41 23 – SUBMERSIBLE MIXERS

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install the following mixing units: two (2) Mixer assemblies, two (2) Masts, and two (2) Davit Cranes. Two (2) mixer assemblies, one in each of the anoxic zones of the new train #3.
- B. Mixers shall be designed to operate smoothly and efficiently in each anoxic zone.

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials, coatings, dimensions, wiring diagrams, controls, and misc. information.
 - 2. Operations and maintenance information as specified in Section 01 33 00.

1.03 WARRANTY

- A. In addition to the standard warranty provided by the equipment manufacturer, the Contractor is to provide an extended warranty covering defects in material and workmanship for 1 year following the date of substantial completion of the equipment (per Supplementary Conditions). The warranty shall be in printed form, shall apply to all similar units, and shall include parts and labor.
- B. Supplied equipment to be free from defect in design, manufacture, workmanship and materials

PART 2. PRODUCTS

2.01

- A. The Submersible mixers will adhere to the following criteria:

Maximum Motor Nameplate Power: 5 HP

Electrical Service: 460 Volt, 60 Hz, 3 phase

- 1. Materials:

Propeller	Stainless Steel, ASTM A276 Type 316 Ti
Motor/Propeller Shaft	Stainless Steel, ASTM A276 Type 316 Ti
Motor Housing	Stainless Steel, ASTM A276 Type 316 Ti
Motor Casing Cover	Duplex Stainless Steel, ASTM A890 CD 4MCU
Casing Cover	Stainless Steel, ASTM A276 Type 316
O-Rings / seals	Viton (FPM)
Fasteners	Stainless Steel
Outer Seal Faces	Silicon Carbide/Silicon Carbide
Inner Seal Faces	Silicon Carbide/Silicon Carbide
Power Cable Jacket	Chloroprene with non-wicking fillers
Guide Rail (Mast)	Stainless Steel, ASTM A276 Type 304 or 316

Support Frame	Stainless Steel, ASTM A276 Type 316 Ti
Mast Mounting Brackets	Stainless Steel, ASTM A276 Type 304 or 316
Lifting Cable	Stainless Steel, ASTM A276 Type 304 or 316
Lifting Davit	Stainless Steel, ASTM A276 Type 304 or 316 with marine-grade winch
Oil (seal lubrication)	Environmentally safe, mineral or paraffin base

B. Accessories

1. **POWER CABLE:** A cable plug and receptacle at motor housing entry shall be an integral design of the submersible mixer allowing the operator to easily replace or service the mixer without needing an electrician. The plug and receptacle eliminates re-wiring errors and ensures operator safety. Provide a suitable length of power/control cable with each mixer, suitable for submersible application, sized in accordance with NEC requirements. Cable entry shall be sealed, with bare strands embedded in resin to insure that no entry of moisture is possible into the motor/ terminal area even if the cable is damaged or severed below water level. Cable entries providing only rubber grommet (external cable jacket) seals will not be accepted. The mixer manufacturer shall provide carabineer clip and a support clamp to secure the power cable at the top of the basin to prevent excessive movement of the cable.
2. **TEMPERATURE PROTECTION:** The stator shall be equipped with three (3) thermal switches (PTC resistor), one embedded in each stator phase for use in conjunction with and supplemental to external motor overload protection. Arrange controls to shut down mixer should any of the monitors detect high temperature and automatically reset (non-XP motors only) once motor temperature returns to normal.
3. **SEAL LEAK DETECTION:** Provide moisture detector in the motor's stator cavity, which allows a control panel mounted relay to indicate leakage into the motor.
4. **MOTOR SENSOR MONITORING RELAY:** The mixer supplier shall furnish all relays required for monitoring all motor sensors. The relays shall be installed by others in the motor control panel and properly wired in accordance with mixer manufacturer's instructions. Relays shall mount in standard 12-pin socket bases (provided with relays) and shall operate on available control voltage of 24-240 VAC. If relays require an input voltage that is not available in the motor control panel an adequate transformer (with fused input) shall be provided. Relays shall have a power consumption of no more than 2.8 watt, and shall be UL approved. Relays shall be modular in design, with each relay monitoring no more than two motor sensor functions. Each relay module shall include a dual color (red/green) LED to indicate the status of each monitored sensor. Green will indicate "status OK"; red will indicate a failure or alarm condition. A self-corrected fault will allow the relay output contacts to reset, and cause the LED to change from a steady alarm indication to a flashing signal. The LED shall continue to flash until locally cleared, providing the operator an indication of a potential intermittent fault. Each relay shall also include a power-on LED and both "test" and "reset" pushbuttons. An independent fail-safe (switch on power loss) form-C output contact shall be included for each monitored sensor to provide a normally-open / normally-closed dry contact to initiate a remote alarm device or shut down the motor. Contacts shall be rated for 5 amps at 120 volt.

- C. **GUIDE MAST ASSEMBLY:** Provide a permanently installed guide mast assembly which allows for mixer installation, operation and retrieval without the need to enter the basin. The mast assembly shall allow for adjustment of mixer vertical location. The mast shall be minimum 4" square tube supplied with factory welded sockets at the top, bottom and intermediate levels as needed to provide support at a maximum of 10ft intervals. Sockets shall be lined with an insert to insure easy mast rotation, and shall work in conjunction with supplied mounting brackets to secure the unit to the tank wall. The top mast bracket shall include a rotary positioning plate which allows the mast to be secured in any of 14 locations with a total of 158 degrees of rotation (for 2" masts there shall be 10 locations with a total of 168 degrees of rotation); adjustments shall be possible without the need to drain the basin. Wall extension brackets, if required, shall be provided to insure a straight and true installation. The mixer shall be guided along the mast through an integral stainless steel mounting bracket fastened to the mixer motor end cap. Support frame to be provided with polymer liner allowing contact only between guide rail and the sliding surface of the polymere (galvanic separation). An adjustable mixer stop bracket shall be furnished to support the mixer at manufacturers recommended height.
- D. **MOUNTING PEDESTAL:** Provide a mounting pedestal designed for permanent mounting of the mixer on the basin floor. The pedestal shall firmly support the mixer at the proper distance from the floor, and shall include a retaining strap for retention of the mixer unit.
- E. **PORTABLE LIFTING DAVIT AND LIFTING CABLE:** Provide two (2) portable lifting davit suitable for use with mixers in this section. The davits shall be designed to be mounted on the top mast mounting bracket (optional floor sockets available for covered tanks). It shall include a manual brake winch, and shall be adequately rated to lift the mixer and support arm off the mast and directly onto the adjacent deck surface. If handrail is fitted to the deck, removable sections or safety chain sections shall be provided such that the mixer does not need to be lifted over the handrail. A lifting cable, permanently attached to each mixer, shall be provided in sufficient length for attachment of the upper end to the winch cable reel when the davit assembly is installed. For installations in which the lifting davit is shared between multiple mixers, equipment shall be provided to securely hold the mixer lifting cable taught.
- F. **Fabrication**
1. **GENERAL:** The mixer supplied with direct drive, close coupled motor must be sealed to prevent ingress of water in wet-well installations with a depth of up to 100 feet. All exposed surfaces shall be corrosion resistant stainless steel approved for both wastewater and water applications.
 2. **MAJOR COMPONENTS:** Furnish major components (motor housing, motor end caps) of solid thick cross section material as specified with smooth surfaces devoid of blow holes and other irregularities. Designs employing sheet-metal sheathing over cast or fabricated components will be considered inferior and will not be acceptable.
 3. **PROPELLER (WITH EVERCLEAN BLADES):** To optimize mixing, and minimize fouling the propeller shall be completely open to the mixed media. Diffuser devices such as jet rings and shrouds should typically be avoided. Propeller shall be made of 316 stainless steels to resist abrasion. Each blade shall be contoured such that the tangent line at any point on the leading edge is equal or less steep relative to the circumferential direction than the resulting force, causing fibrous materials to always slide toward the outer radius of the propeller.

4. PROPELLER SHAFT SEALS: The propeller end of the shaft shall be sealed by two totally independent mechanical shaft seals, each with its own independent single spring system. An environmentally friendly oil-filled chamber shall separate the seals and provide lubrication. Single mechanical seals or rotary lip seals shall not be considered adequate for this critical sealing area.
5. BEARINGS: The motor shaft shall be supported by two bearing sets sized to provide a B10 of greater than 100,000 hours at all anticipated axial and radial loadings. Shaft bearings shall be maintenance free sealed/shielded (permanently lubricated).
6. MAINTENANCE INTERVAL: The complete mixer unit shall be designed for operation with scheduled maintenance intervals not less than 16,000 running hours (or 2 years whichever occurs first). This scheduled maintenance interval shall be clearly stated in the mixer manufacturer's standard operation and maintenance manual supplied with the mixer.
7. MOTOR: Motor shall be squirrel cage, induction in design, housed in a completely watertight and air filled chamber, with a min 1.10 service factor. Insulate the motor stator with, at minimum, Class F insulation rated for 311 Degrees F. Provide temperature protection and seal leak detection as specified above. Motors shall be designed, rated and warranted for continuous operation and shall be capable of sustaining at least 10 evenly spaced starts per hour.

2.02 CONTROLS

- A. See Section 26 80 00, 3.09 B. Train #3 Biological Treatment System.

2.03 SPARE PARTS

- A. The manufacturer is to include three complete sets of replacement wear parts for mixer equipment.

2.04 FACTORY TESTS

- A. Prior to shipment the mixer manufacturer shall perform visual and dimensional inspections, electrical testing, and functional testing which include but are not limited to:
 1. Rotor balance and alignment
 2. Leakage testing of motor chamber and lubricant reservoir
 3. Checking electrical connections for ground conductor, temperature switches, and moisture sensor
 4. Measuring winding and power cable resistances
 5. Testing insulation of windings and power cable
 6. No-load test to ensure proper direction of rotation and electrical characteristics.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. In accordance with contract drawings, manufacturers' shop drawings and instructions.
- B. VERIFICATION OF PERFORMANCE: All mixers shall be field tested by contractor after installation to demonstrate satisfactory operation without excessive noise, or overheating. Any mixer which fails to meet any of the contract specifications will be modified, repaired or replaced by the contractor at no additional cost to the owner.

3.02 MANUFACTURER'S SERVICES

- A. Installation assistance and certification: As required for proper installation prior to start up.
- B. Start-up and field testing: 2 full days on site, including all travel expenses.
- C. Operator Training: 1 full day on site.
- D. Warranty Service: As required during the warranty period.

***** END OF SECTION *****

SECTION 46 51 21 – COARSE BUBBLE DIFFUSERS

PART 1 - GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install fully operation coarse bubble diffusers for the wastewater treatment facility as indicated on the Plans.
- B. Locations for coarse bubble diffusers include:
 - 1. Aerobic Digester

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials.
 - 2. Fabrication drawings with full dimensions and installation drawings showing coordination with piping connections.
 - 3. Operations and maintenance information as specified in Section 01 33 00.

PART 2 - PRODUCTS

2.01 COARSE BUBBLE DIFFUSERS

- A. 24" long, 316 stainless steel, clog free diffusers with open bottom design and rubber check valve in the air inlet to the diffuser to prevent backflow of liquid or sludge into the piping system. Manufactured by Aquarius Technologies, SSI Aeration, or approved equal.
- B. Diffusers shall be removable from above the water line without the need to enter the basin.

PART 3 - EXECUTION

3.01 INSTALLATION

- A. Deliver diffusers in manufacturer's original, unopened cartons. Contractor to provide tees, supports, and fittings (as needed) for construction.
- B. Follow manufacturer's printed instructions for installation.
- C. Remove all identifying marks, labels, etc. from exterior.

***** END OF SECTION *****

SECTION 46 53 00 – BIOLOGICAL TREATMENT SYSTEM

PART 1. GENERAL

1.01 DESCRIPTION OF WORK

- A. The treatment process selected for this project is a multistage (anoxic and aerobic) activated sludge process. The system shall consist of a single bullseye style train (3rd Train), as shown on the project plans. Equipment furnished under this section shall include the in-basin aeration system, mixers, clarifier equipment, process instrumentation and controls as specified in this section to provide a completely functional system. To ensure the integrity of the system, all the components listed above must be supplied and guaranteed by the same system manufacturer.

1.02 DESIGN TREATMENT CAPABILITY

A. Influent Parameters

1. The influent to the treatment system (3rd train) will be as follows:

FLOW	=	63,333 gpd	Daily Avg.
		73,333 gpd	Max. Monthly Flow
		153,333 gpd	Peak Day flow
		178 gpm	Peak Hour Flow

BOD₅ = 151.3 lbs./day

TSS = 35.7 lbs./day

TKN = 49.7 lbs./day

B. Effluent Required

1. When operated according to the manufacturer's instructions by a certified operator, in the absence of toxicity and when sufficient nutrients and alkalinity are present, the treatment system will produce an effluent of the following quality (max monthly average values):

BOD₅ = 30 mg/L

TSS = 30 mg/L

TIN = 8 mg/L

C. Process Design

1. The process shall be an activated sludge process using two anoxic zones and two aeration zones. In addition, blowers will run on VFD drives and adjust, as needed, to maintain a user determined D.O. concentration (such as 2 mg/L).

D. Process Performance Guarantee

1. The treatment system supplier is to provide a written process guarantee, guaranteeing the above effluent quality when equipment is operated and maintained per MFR's recommendations. The guarantee shall cover all equipment components, which affect quality including aeration system and clarifiers.

1.03 BASIS OF DESIGN

- A. Model R Oxigest Multistage Anoxic and Aerobic System as manufactured by Smith and Loveless, Inc.
- B. Approved equals.
- C. Equipment of all manufacturers must be in accordance with these specifications and plans. Being named as a manufacturer does not eliminate their responsibility of providing equipment in compliance with the following specification section. Any deviations without sufficient evidence proving equal or superior quality shall be rejected without further review or comment.
- D. The Oxigest System is the base bid system. The decision of the Engineer with respect to approval of any proposed alternate systems is final.
- E. Contractors may submit an approval package for equipment of alternate manufacturers for review with their bid and included as an Equipment Alternate on the Bid Proposal. A pre-approval review prior to bid will not be done. Contractors proposing alternate manufacturers will be responsible for all costs associated with system evaluation and redesign including all electrical, mechanical, and civil aspects of the installation. Alternate manufacturer shall submit the information in Section 1.04.
- F. The valves, equipment, materials of construction, and controls specified under this section supersede any other specifications located elsewhere in the contract documents.

1.04 ALTERNATE SYSTEMS

- A. SUBMITTAL REQUIREMENTS FOR ALTERNATE TO OXIGEST SYSTEM
 - 1. To be considered acceptable as alternate process equipment, the manufacturer shall send the Engineer sufficient information, which is to include such drawings, specifications, literature, performance data, and other information necessary to describe the proposed equipment as meeting the minimum requirements of the project.
 - 2. The submittal data must include, but not be limited to, a list of at least five Domestic U.S. installations of a similar size and equipment of design similar to the specified aeration/clarification system that have been in operation for five years in similar applications. The information on each plant shall include:
 - Name and location of facility.
 - Brief description of plant size including design flow, peak flow, organic loading, and effluent requirements (must have 10 mg/l nitrate limit or lower).
 - Minimum of twelve months operating data including daily flows, influent and effluent quality for BOD, suspended solids, and ammonia nitrogen, TKN, nitrate nitrogen.
 - Name, address and telephone numbers of owner, plant operator, design engineer and general contractor.

1.05 PERFORMANCE

A. Mechanical Guarantee

1. All equipment manufactured by the supplier shall be guaranteed to be free from defective material and workmanship, under normal use and service, for a period of 12 months from the date of Substantial Completion of the construction of the ESWD WWTP Upgrade (phase I).
2. Any equipment supplied by the supplier, but not of their manufacture, shall be provided with all guarantees and warranties provided by the original equipment manufacturer passed on to the end user.

B. Experience

1. The minimum acceptable experience level is 5 installations in the US provided by the supplier, using a long sludge age, single basin total nitrogen removal process with moving aeration chain equipment, similar to that specified for this project, that have been operating for a minimum of five years.

C. Process Performance Guarantee and Bond

1. The treatment system supplier shall provide a written process performance guarantee (PPG), guaranteeing the final effluent quality in accordance with this specification section. The PPG shall be supplied as part of the equipment submittal.
2. The PPG shall guarantee effluent requirements as listed in this specification and shall remain in effect for a period of two years from start up.

PART 2. PRODUCTS

2.01 TREATMENT SYSTEM

A. GENERAL

1. The biological treatment system shall be designed for BOD, total suspended solids, and total inorganic nitrogen removal. It consists of submerged diffusers, mixers, dissolved oxygen sensors, and control system. To ensure the integrity of the system, all the components listed above must be supplied and guaranteed by the same manufacturer.
2. The aeration system shall include all the aeration system components specified below.
3. All concrete required for support or installation of any equipment supplied as part of the system shall be provided as shown on the plans and as required by the manufacturer.

B. CONSTRUCTION

1. General

Each unit shall form a complete wastewater treatment system. The outer wall shall be reinforced concrete. The inner vertical walls shall be structural grade steel plate not less than 1/4" (6 mm) thick. All welded steel structural members shall be joined by electric arc welding with fillets of adequate section for the joint involved. Where required for additional sectional strength or watertight integrity such welds shall be continuous. Inlet and outlet connections shall be as shown on the drawings.

2. Access Walkway

A 3'-0" (0.9 m) wide access walkway shall span the entire length of the clarifier and extend to the outer tank wall as shown on the drawings. The walkway bridge shall be capable of supporting a live load of 150 lbs. per linear foot (68 kg per m²), in addition to the dead load. Deflection shall not exceed 1/360 of the unsupported span. A 42" (1,068 mm) high handrail and a midrail shall be provided for both sides of the walkway. Openings shall be provided for transition to the clarifier service walkway. A 2'-0" (0.6 m) wide clarifier service walkway shall be provided with outboard and inboard protection as described above. The clarifier service walkway shall cover the full circumference of the clarifier for convenient access to the diffuser assemblies and air control valves. It shall further provide for maintaining the effluent weir and effluent trough. Bar grating shall be aluminum and shall extend over the entire length of walkway. Handrail shall be aluminum.

3. Ladder Access

An aluminum ladder attached to the end of the walkway shall be provided, as shown on the drawings. Treads shall be of nonskid design and handrails on both sides shall be provided.

C. SYSTEM

1. Anoxic and Aeration System.

The aeration and anoxic compartments shall have an outer wall inside diameter of 32 ft, a normal static water depth of 15.5 ft and an overall wall height of 16.9 ft. The total capacity of the annular aeration compartments shall be 43,000 gallons. The annular compartment shall be divided by steel plate partition walls. All necessary internal valves, piping, mixers, supports, etc. shall be provided to permit operation of the unit using the Multistage Anoxic Aerobic Process.

System includes two anoxic compartments and two aerobic compartments as shown on drawings.

The anoxic tanks shall have a combined capacity of 26,508 gallons, and be divided into two (2) compartments. Dimensions shall be as shown on the drawings. Each anoxic compartment shall include a submersible mixer with a 5 HP, 3 phase, 460 volt, 60 Hz motor. A 6 inch recycle airlift shall be installed in the second anoxic compartment, capable of pumping 101 GPM. All airlift accessories such as solenoids, limit switches and controls shall be included by the manufacturer.

It is the intent of this specification that the manufacturer provides a complete system including all equipment and hardware required for the in-basin components.

The aeration system must be easily accessible for service and maintenance without basin dewatering or a complete aeration system shutdown.

System includes two anoxic compartments and two aerobic compartments as shown on drawings.

The design AOR (actual oxygen requirement) of the aeration system is 269 lbs O₂/day for Aerobic Zone 1 and 57 lbs O₂/day for Aerobic Zone 2 (Train #3). This is to be accomplished with an air transfer rate of 240 SCFM using the S&L medium bubble diffusers. In addition, airlift pumping requirements are 75 SCFM, and total air requirement = 315 SCFM.

2. Air Diffusion Equipment

An air distribution header and individual air diffusion assemblies shall be provided. The air header shall be rectangular tubing and the air supply duct shall extend from the header to a point just outside the outer tank. A 125 lb. flanged connection shall be supplied at the outer end of the supply duct. The header shall support drop pipes to which are attached air diffusion devices. Each drop pipe assembly shall include a union to permit convenient removal of the diffuser assembly and an air control valve to permit regulation of the air supply. The air diffusion devices installed on the air drop-pipe assemblies shall be medium bubble style diffusers as manufactured by Smith & Loveless, Inc., designed to minimize clogging as well as to insure proper air diffusion and distribution over the entire aeration tank. The diffusers and piping shall be of corrosion resistant construction and the entire assembly shall be removable for inspection of servicing. The air diffusion efficiency of the devices shall be such that an adequate supply of dissolved oxygen can be maintained in the aeration zones to meet the treatment requirements at the organic load for which the plant is designed

3. Clarifier

The circular clarifier shall have a diameter of 18 ft. The air header shall rest on the top and be welded to the clarifier tank forming part of the freeboard for the tank and also providing reinforcing. The freeboard above the static water level of the aeration compartment at the inner tank shall be 1.4 ft. The clarifier shall be of all-welded construction providing a leak-proof barrier between the clarifier and the aeration tankage. The clarifier bottom shall be formed by the purchaser by placing concrete at a 1 in 12 slope inside the steel tank walls. The finished floor shall conform to the drawings and shall provide a clearance between the floor surface and scraper blades of the sludge collector mechanism.

A complete sludge collector mechanism, support bridge, continuous scum baffle, mechanical skimming system with anti-rotation bar, peripheral effluent trough with adjustable aluminum weir plates, scum baffle and service walkway as specified herein, shall be furnished.

A 10 inches in diameter steel inlet pipe extending from the aeration compartment to the center inlet well shall be furnished as shown on the plans. The collector mechanism shall be complete with center inlet well, drive assembly, drive shaft and adjustable rake arms.

Other connections shall be provided as shown on the drawings.

The inlet well shall be fabricated from 1/4" (6 mm) steel plate, and shall be supported from the drive shaft by a steel frame.

The drive mechanism shall be designed for a rated torque of 800 foot-pounds. It shall consist of a primary gear motor and a secondary speed reducer. Power shall be transmitted from the primary to the secondary unit by means of a roller chain drive, which shall be enclosed for protection from the weather and for operator safety.

The primary and secondary units shall be of the heavy-duty type running in oil. Anti-friction bearings shall be used throughout. The primary unit shall have an integral NEMA C face, ½ horsepower motor, suitable for operation on 460 volt, 3 phase, 60 Hz current. An overload protection system shall be provided to protect the motor should the drive become overloaded. The protection system shall consist of a sprocket on the input shaft of the secondary reducer mounted between two friction facings, which are spring loaded. When an overload occurs, the driven sprocket shall “slip” between the two facings limiting the torque placed on the drive.

The rake arms shall be attached to the drive shaft with a hinged joint to allow easy adjustment of the end elevation of the rake arm. The rakes shall be individually adjustable for height. The rakes shall be made from 3-ply canvas reinforced rubber belting to provide enough rigidity to keep the sludge blanket agitated and flexible enough to pass over any moderate size object.

The clarifier bridge shall be a steel structure as specified herein and shown on the drawings.

A continuous scum baffle shall be provided as shown on the drawings.

The surface of the clarifier shall be skimmed by a mechanical skimming system. A rotating skimming arm shall be attached tangent to the stilling well. This arm shall have a flexible rubber end section to contact the scum baffle. A stationary anti-rotation skimming arm shall be suspended from the clarifier bridge structure to prevent the surface scum from rotating. It shall also act in conjunction with the rotating skimming arm to wedge the floating scum outward toward the airlift scum ejector, located on the circumference of the clarifier. The airlift shall operate intermittently. It shall be controlled by an adjustable microswitch. The airlift shall discharge into the aerobic digester to prevent grease balls and other scum from recycling through the system. The clarified liquid shall pass over the edges of the effluent weirs into the effluent trough which shall be connected to the effluent pipe.

4. Clarifier Reinforcing

The inner tank shall be reinforced to withstand a 15'-6" liquid level differential. Extra structural reinforcing rings shall be installed on the inner steel shell of the wastewater treatment plant structure to withstand external hydrostatic pressure, which will exist if the clarifier is dewatered without dewatering the entire annular aeration compartment. Under these conditions, the inner tank joints shall be continuously welded inside an out.

5. Clarifier Drive Failure Alarm Contacts

A drive failure alarm contact shall be furnished to actuate a suitable alarm device if the drive should become overloaded or stop turning for any reason, i.e., overtorque condition, chain breakage, motor failure, or power failure. A contact shall also be provided to shut off the drive motor when a fault condition occurs.

6. RAS Airlift System

The settled sludge in the clarifier shall be moved continuously across the floor by the sludge collector mechanism to the central sludge well. From the sludge well the collected sludge shall be continuously pumped through a rectangular duct by an airlift pump which shall discharge RAS to Anoxic Zone #1 (as shown on plans). In addition, the sludge sump will discharge WAS into the piping system

which is directed to the digester. The RAS airlift pump shall have a capacity of 51 GPM, using not more than 15 CFM of air. The airlift shall have a 3 inch diameter steel riser pipe with a flow division box on top.

7. MLR (Nitrate Recycle) Airlift System

Mixed liquor is to be recycled from Anoxic Zone #1 to the Aeration Zone #1 as shown on plans. The MLR shall be continuously pumped through a rectangular duct by an airlift pump which shall discharge MLR to Aeration Zone #1 (as shown on plans). Each MLR airlift pump shall have a capacity of 101 GPM, using not more than 30 CFM of air. Two (2) MLR airlift pumps total. Each airlift shall have a 6 inch diameter steel riser pipe with a flow division box on top.

8. Carbon Feed

Provide and install two (2) peristaltic metering pumps for carbon feed (one duty and one standby pump). Pumps are to be Stenner Model S3V02AA302N. Capacity: 17 gpm. Control: Variable speed. Flow pace per Section 26 80 00 3.09 C.

D. AERATION BLOWERS

1. See spec section 43 11 00 Blower Equipment.

E. AIR PIPING

1. General
See Section 33 31 00, 2.01 D. Aeration Piping.
2. Low Pressure Air Test of Air Header

The low-pressure air test determines the rate at which air under pressure escapes from an isolated section of the pipe. The Contractor shall conduct the test prior to complete backfilling of the trenches as follows:

- i. Make sure that all the pipe fittings, T-pieces, elbows, etc. have been securely installed.
- ii. Make sure that an adequate amount of back fill has been tamped under and around the pipe making sure it is securely in place.
- iii. If required, have all concrete thrust blocks in place to immobilize the air header under pressure.
- iv. Pressurize the air main as shown below and record the time.
- v. i) Test Pressure: 15 psi
- vi. ii) Test Duration 60 minutes
- vii. Test Requirement: Test pressure must be held for a minimum of 15 minutes. Test pressure must not fall more than 5% during a one-hour time span.

F. SYSTEM CONTROL PANEL

General.

1. Treatment system manufacturer shall provide complete treatment system electrical and control system per drawings and specifications.
2. Treatment system control panel shall include all wiring and controls for operation of the complete treatment system. Features shall include but not be limited to:
 - a. Treatment system control panel shall accept 460V, three phase, 60 Hz input and shall include Main Circuit Breaker disconnect with rotary handle. Design specifies 30 amp main breaker, but manufacturer shall size main panel breaker based on Code requirements for the equipment supplied, and submit accordingly.
 - b. Treatment system control panel including all electrical devices and accessories shall be UL listed and labeled.
 - c. NEMA 4X Type 316 stainless steel enclosure, sized for the application. The panel will be installed within sight of the treatment system on an exterior wall. Panel shall include interior swing out panel for local switches and indicators. Include drip shield for outdoor panels.
 - d. NEMA style full voltage motor starters for both clarifier drive motor and two submersible mixers.
 - e. Seal fail and overtemp sensors for each submersible mixer motor. Include local pushbuttons and indicator lights in the control panel for status indication and reset capability.
 - f. 480-120v step down transformer for control circuits.
 - g. Hard wired relay logic control circuits for system control and protection.
 - h. Alarm light and horn with silence push button.
 - i. Indicator lights, push buttons, hand-off-auto switches for motor control.
 - j. Wiring terminals for field mounted devices including torque switches, limit switches, solenoids and any other required devices.
 - k. Dry relay contacts for customer connection including: run and fail for each motor, seal fail and overtemp for each mixer, alarm and shutdown for clarifier, and any other equipment status monitoring available.
 - l. Panel heater with fan. Heater shall be sized by the manufacturer to prevent buildup of condensation in panel.
 - m. 120V circuit for pole light and receptacle mounted on treatment cell.
 - n. Detailed wiring diagrams and schematics showing wire and terminal connections, panel layout, oneline diagram and other information required for installation by the Electrical Contractor.

G. SPARE PARTS

1. At a minimum, the following spare parts are to be supplied:
 - i. 1 – Additional Diffuser Assembly
 - ii. 3 – sets of fuses for all types and sizes
 - iii. 1 - torque wrench

PART 3. EXECUTION

3.01 INSPECTION & STORAGE

- A. The Contractor shall inspect and accept all equipment immediately upon delivery to the site to satisfy himself that complete and undamaged equipment package has been delivered for installation.

3.02 INSTALLATION

- A. The Contractor shall install all equipment according to the drawings and specifications and in accordance with the installation instructions and drawings provided by the manufacturer to the Contractor. The bid document drawings are intended to show the Contractor the components of the proposed construction not the details of assembly and installation of the Treatment System equipment. The Contractor is responsible for all costs associated with installing the equipment. The Contractor is to contact the equipment MFR and coordinate responsibilities for field erection services, as needed, for the equipment.
- B. Contractor to replace defective diffusers and/or any defective equipment or components.
- C. If additional information concerning the Smith and Loveless Oxigest equipment is necessary then the Contractor shall contact Joe Buckman, APSCO (425) 822-3335 to obtain additional details on the assembly and installation of said equipment.

3.03 TREATMENT SYSTEM PROGRAMMING, STARTUP AND TESTING

- A. The Treatment System manufacturer shall provide all configuration and functional testing for the control system, and place the complete system into operation.
- B. The Control System shall be configured and adjusted to operate according to the descriptions in this specification and features shown on the drawings.
- C. The Treatment System manufacturer shall provide all required coordination for a complete and functional control system. Field adjustments or additional features added during startup and commissioning shall be included in the bid price.
- D. The Treatment control system shall be put into operation by the Treatment System manufacturer with support of the Electrical Contractor.
- E. A witnessed Functional Acceptance Test shall be performed on the completed control system.
 - 1. The Treatment System manufacturer's standard test forms shall be submitted for review by the Engineer, including complete test procedure and forms.
 - 2. Each feature and function of the Treatment manufacturer's control system shall be documented on the test forms and shall be tested and demonstrated to the satisfaction of the Engineer. The Treatment System manufacturer shall make all revisions to the control system operation noted by the Engineer prior to final acceptance.
 - 3. The testing program shall be conducted in accordance with the prior approved procedures and shall be witnessed and signed off by both the Treatment System manufacturer and the Engineer upon satisfactory completion.

- F. All special testing materials and equipment required to demonstrate compliance with the specification shall be provided by under the scope of this Section. Where it is not practical to test with real process variables, provide suitable means of simulation. These simulation techniques shall be subject to the approval of the Engineer.
- G. Coordinate all testing with other associated suppliers and subcontractors.

*****END OF SECTION*****

SECTION 46 66 00 – ULTRAVIOLET DISINFECTION EQUIPMENT

PART 1 - GENERAL

1.01 DESCRIPTION OF WORK

- A. Furnish all labor, materials, equipment and appurtenances required to provide two open channel, gravity flow, low pressure high intensity ultraviolet lamp (UV) disinfection systems. The UV systems shall be complete and operational with all control equipment and accessories as shown and specified herein. Each system will be capable of disinfecting effluent to meet the water quality standards listed in this section

1.02 RELATED SECTIONS

- A. Section 03 30 00: Cast-in-Place Concrete
- B. Division 26: Electrical

1.03 QUALITY ASSURANCE

- A. Any alternate UV manufacturer that is not named or listed as approved equal must submit the following information to be considered for approval:
 - 1. The manufacturer will be regularly engaged in the manufacture of UV systems with a proven track record of at least five hundred (500) municipal installations in North America.
 - 2. The manufacturer will provide documentation of previous experience with municipal UV disinfection systems in wastewater applications with electronic ballasts.
 - 3. To be considered, the manufacturer will submit a bioassay evaluation for the proposed reactor. This bioassay will have been validated by an independent third party and have followed protocols described in the US EPA Design Manual – Municipal Wastewater Disinfection (EPA/625/1-86/021), without exception. The manufacturer's bioassay report must demonstrate that the proposed UV system design and number of lamps will deliver the specified dose.
 - 4. Documentation of UV manufacturer's service capabilities including location and experience.
 - 5. Sample disinfection performance guarantee including scope and duration of guarantee- minimum of 5 years required.

Design Criteria:

Each of the two (2) UV systems are to be capable of independently disinfecting effluent with the following characteristics:

Max Day Flow:	0.46 MGD
Max Month Flow:	0.22 MGD
Minimum Flow:	0.03 MGD
Total Suspended Solids:	30 mg/L, Maximum

Effluent Temperature Range: 33 to 85 °F (1 to 30 °C)

Ultraviolet Transmittance @ 253.7 nm: 50%, minimum

Effluent standards to be achieved: Monthly Geometric Mean: 200 FC/100mL, Weekly Geometric Mean: 400 FC/100mL of daily samples for the effluent standard as specified in a) through f). Effluent standards will be guaranteed regardless of influent count to UV system.

1. The UV equipment will be installed in a stainless steel channel furnished as part of this system, and having dimensions as shown on the drawings.
2. The system supplied will be arranged in the following manner.
 - a. Number of Lamps in each UV Lamp Module: 4
 - b. Number of UV Lamp Modules per Bank: 6
 - c. Number of UV Banks: 2
 - d. Total Number of Lamps: 48
3. The lamp array configuration will be evenly spaced in both horizontal and vertical rows with all lamps parallel to each other and to the effluent flow.

Performance Requirements:

4. The UV system will be designed to deliver a minimum UV dose of 30,000 μ Ws/cm² or 30 mJ/cm², in effluent with a UV Transmission of 50% after reductions for quartz sleeve absorption, sleeve fouling, and lamp aging. The basis for evaluating the UV dose delivered by the UV system will be the manufacturer's bioassay as carried out by an independent third party. Bioassay validation methodology to follow protocols described in US EPA Design Manual - Municipal Wastewater Disinfection (EPA/625/1-86/021), without exception.
5. The UV system will produce an effluent conforming to the following discharge permit: 200 FC/100 ml. Grab samples will be taken in accordance with the Microbiology Sampling Techniques found in Standard Methods for the Examination of Water and Wastewater, 23rd Ed.
6. The system will be designed for complete outdoor installation.

1.04 SUBMITTALS

Submit for review, shop drawings showing the following:

1. Complete description in sufficient detail to permit an item comparison with the specification.
2. Dimensions and installation requirements.
3. Descriptive information including catalogue cuts and manufacturers' specifications for major components.
4. Electrical schematics and layouts.

5. Hydraulic calculations demonstrating compliance with the required hydraulic characteristics.
6. Independent bioassay validation and dosage calculations demonstrating compliance with the specified RED requirements.
7. Lifetime disinfection performance guarantee.

1.05 WARRANTIES

Equipment:

1. The equipment furnished under this section will be free of defects in material and workmanship, including damages that may be incurred during shipping for a period of 12 months from date of Substantial Completion.

UV Lamps:

1. The UV lamps to be warranted for a minimum of 12,000 hours (non-prorated). Pro-rated lamp warranties will not be accepted. On / off cycles are limited to an average of four (4) per day without exception.

UV Intensity Sensors:

1. UV Intensity Sensors will be warranted for a period of 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

A. MANUFACTURER

- a. Trojan Technologies (UV3000 PTP Equipment), or approved equal.
- b. The physical layout of the system shown on the contract drawings and the equipment specified herein are based on the TrojanUV3000, as manufactured by Trojan Technologies, London, Ontario, Canada.
- c. The Trojan System is the base bid system. The decision of the Engineer with respect to approval of any proposed alternate systems is final.
- d. Contractors may submit an approval package for equipment of alternate manufacturers for review with their bid and included as an Equipment Alternate on the Bid Proposal. A pre-approval review prior to bid will not be done. Contractors proposing alternate manufacturers will be responsible for all costs associated with system evaluation and redesign including all electrical, mechanical, and civil aspects of the installation.

B. GENERAL REQUIREMENTS

- a. Provide a UV disinfection system complete with UV lamp modules, effluent channel, level control, and UV monitoring system as shown on the Contract Drawings and as herein specified.
- b. UV system will be designed for complete outdoor installation, without shelter or supplemental cooling or heating required.

C. DESIGN, CONSTRUCTION AND MATERIALS

General:

1. All material in contact with effluent will be stainless steel type 316 or quartz.
2. All material exposed to UV light will be stainless steel, anodized aluminum, quartz 214, or Teflon™.

UV Module (UVM):

1. Each UV lamp module will consist of four lamps and their corresponding electronic ballast. Each lamp will be enclosed in its individual quartz sleeve, one end of which will be closed and the other end sealed by a lamp end seal and holder.
2. The electrical wires connecting the lamps to the electronic ballasts will be enclosed in the stainless steel frame. Systems where lamp wiring is submerged in the effluent and exposed to UV light will not be allowed.
3. Each UV module will be provided with a standard 120 Volt plug and weatherproof cable for connection to a receptacle. The cable will be 10 feet (3.0 m) long. A total of twelve (6 for each UV system) UV modules will be supplied. Lamp status will be displayed on top of each UV module using watertight LED indicator lights.
4. Modules will be approximately 68 inches long, 20 inches high and 2.8 inches (7.1 cm) wide, weighing approximately 38 lbs. Materials of construction will be stainless steel type 316, anodized aluminum, quartz 214, and Teflon™, with UL rating of Type 6P

UV Lamps:

1. UV system will use low pressure mercury slimline lamps of the hot cathode, instant start design.
2. 90% of UV output will be within the wavelengths of 233.7 to 273.7 nm.
3. The operating life of the lamp will be guaranteed for 12,000 hours, non pro-rated.
4. Independent validation of lamps aging factor is required.

Lamp End Seal and Lamp Holder:

1. The open end of the lamp sleeve will be sealed by means of a sleeve nut which threads onto a sleeve cup and compresses the sleeve 'O' ring.
2. The sleeve nut will have a knurled surface to allow a positive handgrip for tightening. The sleeve nut will not require any tools for removal.

UV Lamp Sleeves:

1. Quartz sleeves to be Type 214 clear fused quartz circular tubing as manufactured

by General Electric or equal.

2. Quartz to be rated for UV transmission of 89% and not subject to solarization.
3. The nominal wall thickness will be 1.0 to 2.0 mm to maximize UV transmission.

Effluent Channel:

1. Each UV unit will be provided with one (1) effluent channel complete with drain, UV module support rack and downstream level control weir.
2. The effluent channel will have the following dimensions as per the drawings.

Level Control Weir:

1. Weir will be located downstream of the UV modules to maintain an average water depth of 12 inches and ensure lamp submergence at all times.
2. Maximum effluent level variance from zero to peak flow will not exceed 1.5 inches (3.8 cm).
3. Weir will be welded water tight and include a drain.

Electrical:

1. Each of the two UV disinfection systems will be divided into 6 UV modules.
2. Interconnecting Cables to be standard 120 Volt, weatherproof, 10 feet (3.0 m) long and will be suitable for outdoor installation.
3. Power Distribution Receptacles:
 - a. 120 Volt receptacles rated for continuous outdoor use will be provided. Receptacles will be of the duplex type complete with ground fault interrupter circuitry.
 - b. Receptacles to be provided by the UV Manufacturer.
4. Power Consumption:
 - a. Maximum power draw to UV System will be 2100 watts.
 - b. All electrical supplies will be 120 Volt, 60 Hz.
 - c. A separate 120 volt, 5 amp supply to be provided for the Monitoring System.

Cleaning Procedure:

The UV lamp modules will be cleaned by removing from the effluent channel and hand wiping the sleeves with an acid solution, using a non-abrasive cloth.

Monitoring System:

1. Two submersible UV sensors (one per bank) will continuously monitor the UV intensity produced in the bank of UV lamp modules. The sensor will measure the germicidal portion of the light emitted by the UV lamps.
2. UV intensity will be indicated on a 3 character display in mW/cm².

3. Elapsed time in hours (lamp age) will be indicated on a 5 character display.
4. Both displays will utilize LEDs and will be visible through the panel door.
5. A dry contact will be provided for remote indication of Low UV intensity alarm.
6. Monitoring System will be enclosed in a fiberglass Type 4X wall mounted panel and is to be located less than twelve (12) feet (3.66 m) from the LED end of the UV Module.

Power Distribution Receptacle (PDR):

1. Duplex ground fault interrupter receptacle(s) will be provided by the UV Manufacturer.
2. Receptacles will be mounted in an individual, impact resistant thermoplastic junction box complete with a Type 3R rain shield for outdoor installation.

Contractor is to supply appropriate 120 Volt, single phase, 60 Hz circuit to power the PDRs which have a total current draw of approximately 19.2 amps. Contractor is to be responsible for distributing the power from the main 120 Volt feed to the individual PDRs. Responsibility to be all encompassing and in accordance with the local electrical codes.

Transition Connections:

1. Two inlet and two outlet transition box(es) will be supplied.
2. The material of construction for the transition boxes will be stainless steel type 304, 14 gauge.
3. Openings on the transition boxes (for flange connections) conform to ANSI standards and have dimensions as shown on the Contract Drawings.

Maintenance Rack:

One (1) Type 304 stainless steel maintenance rack(s) will be supplied. The rack is designed to hold UV modules during service or maintenance.

Spare Parts:

The following additional parts will be furnished:

- Four UV lamps (per bank)
- Four Quartz sleeves (per bank)
- Four Lamp holder seals (per bank)
- One Operators Kit (including face shield, gloves) (per bank)

PART 3 - EXECUTION

3.01 INSTALLATION

- A. In accordance with contract drawings, manufacturers' shop drawings and instructions.

3.02 MANUFACTURER'S SERVICES

- A. Installation assistance and certification: As required for proper installation prior to start up.
- B. Start-up and field testing: 2 full days on site, including all travel expenses.
- C. Operator Training: 1 full day on site.
- D. Warranty Service: As required during the warranty period.

*****END OF SECTION*****

SECTION 46 66 10 – UV ANALYTICAL EQUIPMENT

PART 1. GENERAL

1.01 DESCRIPTION

- A. The Contractor shall furnish and install the following UV Transmissivity Analyzer unit:
Hach Model DR 1900.

1.02 SUBMITTAL DATA

- A. The following shall be provided in accordance with Section 01 33 00:
 - 1. Manufacturer's catalog data showing construction materials, coatings, dimensions, wiring diagrams, controls, and misc. information.
 - 2. Operations and maintenance information as specified in Section 01 33 00.

1.03 WARRANTY

- A. In addition to the standard warranty provided by the equipment manufacturer, the Contractor is to provide an extended warranty covering defects in material and workmanship for 1 year following the date of substantial completion of the equipment (per Supplementary Conditions). The warranty shall be in printed form, shall apply to all similar units, and shall include parts and labor.
- B. Supplied equipment to be free from defect in design, manufacture, workmanship and materials.

PART 2. PRODUCTS

2.01 UV ANALYZER

- A. The pumps shall be Hach Model DR1900.

2.02 SPARE PARTS

- A. The manufacturer is to include standard replacement parts with the equipment assembly.

PART 3 - EXECUTION

3.01 MANUFACTURER'S SERVICES

- A. Installation assistance and certification: As required for proper installation prior to start up.
- B. Start-up and field testing: 1 full days on site, including all travel expenses.
- C. Operator Training: 1 full day on site.
- D. Warranty Service: As required during the warranty period.

***** END OF SECTION *****

SECTION 46 76 26 – ROTARY FAN PRESS SYSTEM

PART 1. GENERAL

1.0 DESCRIPTION OF WORK

- A. The dewatering process selected for this project is a rotary fan press system. The system shall consist of a complete plug and play skid with all components connected, wired, tested, and ready for use. The system shall include a rotary fan press with one dewatering channel, a flocculator system, sludge manifold and piping, a cake chute, a polymer water make up system with dosing pump for an emulsified polymer solution, an air compressor for pneumatic valving, a screw conveyor system with receiving and inclined conveyors (as needed to deliver solids to District's dumpster, as shown on drawings), and control panel (configured for controlling the entire system). In addition, the rotary fan press controls are to control the digested sludge feed pump.
- B. The equipment shall be rated for a solids throughput capacity of 120 dry lbs./hour (min.). In addition, the equipment shall be capable of accepting liquid concentrations of 0.5% to 2.5% solids, and liquid flowrates of 5 gpm to 25 gpm.
- C. To ensure the integrity of the system, all the components listed above must be supplied and guaranteed by the same system manufacturer.
- D. Influent Design Parameters for Dewatering System:

1. Current Parameters – Average Day Flow without digestion:

a. Solids Concentration	=	5,000 mg/L
b. Sludge Production	=	13,261 gal/wk
c. Solids Throughput Capacity	=	120 dry lbs./hr
d. Sludge Flow to Press	=	assume 20 gpm
e. Press Runtime	=	11.1 hours/wk

2. Current Parameters – Max Month Flow without digestion:

a. Solids Concentration	=	5,000 mg/L
b. Sludge Production	=	41,966 gal/wk
c. Solids Throughput Capacity	=	120 dry lbs./hr
d. Sludge Flow to Press	=	assume 20 gpm
e. Press Runtime	=	35.0 hours/wk

3. Current Parameters – Average Day Flow with digestion:

a. Solids Concentration	=	20,000 mg/L
b. Sludge Production	=	3,315 gal/wk
c. Solids Throughput Capacity	=	120 dry lbs./hr
d. Sludge Flow to Press	=	assume 12 gpm
e. Press Runtime	=	4.6 hours/wk

4. Current Parameters – Max Month Flow with digestion:

a. Solids Concentration	=	20,000 mg/L
b. Sludge Production (Max Mon)	=	5,246 gal/wk
c. Solids Throughput Capacity	=	120 dry lbs./hr
d. Sludge Flow to Press	=	assume 12 gpm
e. Press Runtime	=	7.3 hours/wk

5. Future Parameters (Yr-2040) – Avg Day Flow with carbon addition & digestion:

a. Solids Concentration	=	20,000 mg/L
b. Sludge Production (Avg Day)	=	7,890 gal/wk
c. Solids Throughput Capacity	=	120 dry lbs./hr
d. Sludge Flow to Press	=	assume 12 gpm
e. Press Runtime	=	11.0 hours/wk

6. Future Parameters (Yr-2040) – Max Mon Flow with carbon addition & digestion:

a. Solids Concentration	=	20,000 mg/L
b. Sludge Production (Max Mon)	=	12,506 gal/wk
c. Solids Throughput Capacity	=	120 dry lbs./hr
d. Sludge Flow to Press	=	assume 12 gpm
e. Press Runtime	=	17.4 hours/wk

E. Effluent Design Parameters for Dewatering System:

1. When operated according to the manufacturer's instructions by a certified operator, the press will provide the following:

Solids Concentration	=	12% to 16%
Wet Cake Throughput	=	up to 1000 wet lbs. per hour
Dry Cake Throughput	=	up to 120 dry lbs. per hour

1.1 GENERAL INFORMATION

The Rotary Press is supplied with one (1) dewatering channel. Each channel has a diameter of 36 inches (900 mm). Flocculated sludge is fed at a low, constant pressure not exceeding 7 PSI (48 kPa) into the channel where it is dewatered.

A Rotary Press comprises at least the following components:

- 1.1 Drive System
- 1.2 Dewatering Channels
- 1.3 Rotary Press Base
- 1.4 Cake Outlet Chutes
- 1.5 Screw Conveyor System for Sludge Cake Transport (Section 22 13 16 – Screw Conveyor System)
- 1.6 Emulsion Polymer System (Section 46 33 33 – Emulsion Polymer Make-Up System)
- 1.7 Digested Sludge Pump for Feed to Rotary Press (Section 22 13 16 - Packaged Sludge Pumps)

1.1 Drive System

Speed Reducer:

The speed reducer shall be foot mounted type with spiral bevel and helical gears. The speed reducer shall be fabricated according to ANSI/AGMA 6010 standard with a robust cast iron housing. The bevel gears will have a minimum of AGMA quality of 9 and the helical gears a minimum of AGMA 11. Output shaft shall be made of steel with a minimum yield of 59000 PSI (300 MPa). Output shaft will have a minimum diameter of 6-3/16" (158 mm). The speed reducer shall be mounted on a steel base epoxy painted (see section 3.1 for painting). No parts of the speed reducer, including gearbox seals shall be in direct contact with either the sludge filtrate or cake. The output seal shall be an integral part of the speed reducer housing; there shall be no bolt-on output cover to cause leakage, shaft or gear misalignment.

Motor:

The electric motor shall be 480V, 60Hz, 3 PH, 1800 RPM totally enclosed, TEFC, 1.5 HP (1.1KW), "C" flange, premium efficiency, service factor 1.15, insulation "F" class, design "B". The electrical motor is directly coupled to the speed reducer.

1.2 Dewatering channels

The dewatering channel of the Rotary Press shall be mounted on the gear reducer output shaft. Each channel shall be an independent self-contained modular unit which can be interchanged with other Rotary Presses of the same model. The low-speed shaft seal of the speed reducer will, in no way, act as a filtrate seal for the dewatering channel. All material in direct contact with the flocculate sludge or with the cake shall be in stainless steel or plastic made.

Each channel will consist of:

Filtration elements:

The filtration elements shall be made of stainless steel and be hard chrome plated for wear resistance. The filtration elements shall be a non-clogging design which does not require wash water during operation. Filtering element shall be manufactured from thin Stainless steel plate. Filtering element made from wire or wedge wire are not acceptable.

Scrapers:

To clean the filtration elements, each channel shall be equipped with six (6) stainless steel scrapers.

Filtration wheels:

The wheels shall be made of cast iron ASTM A48 Class 40 or equal. The casting will have wall sections with a minimum thickness of 1/2" (12 mm), so designed to withstand internal pressures of up to 300 PSI (2069 kPa). The surface preparation and painting shall be as defined in section 3.1.2. Contact surfaces for the seals shall be made from polished stainless steel 316L and have a surface finish from 16 to 32 RMS.

Deflector:

Deflector shall be fabricated from a molybdenum disulphide-filled nylon (Nylatron).

Fiberglass gland covers, bushings and seals

The gland covers shall be made from 1/4" (6 mm) thick minimum fiberglass with a minimum of 3 to 6 layers depending on the cover section. This cover shall be used to compress the UHMW gland cover seal between the housing and filtration wheel. The cover will have an outlet in its lower section to drain the filtrate.

An additional opening is made for the outside wheel wash water system connection and air bleeding device. For inspection or maintenance, an opening port is located at the rear portion of the cover to permit easy access.

Two (2) UHMW bushings will guide the gland cover on the filtration wheel. Sealing shall be assured by one (1) outside lip seal, four (4) "O-rings": two (2) between bushings and the filtration wheel, and two (2) compressed between the gland cover and the bushing.

Housing:

Each housing is made from 5/8" (16 mm) thick stainless steel 304. The surface in direct contact with the cake shall be stainless steel lined. Housing is supported by the filtration wheel through a set of bushing. Rotating of the housing is restricted by a torque arm, connecting the channel housing to the gear reducer base support. Each housing is interchangeable with other Rotary Presses of the same series.

Wash system:

The wash system shall be provided for use as a daily cleaning of the inside of the channel(s) and the outside of the filtration wheel(s).

One (1) wash manifold is located inside the channel at the sludge inlet. This manifold is made of a stainless-steel tubing assembly, equipped with twenty-eight (28) flush type stainless steel sprinklers.

One (1) set of 2 manifolds (one set for each filtration wheel) consisting of four (4) sprinklers shall be located in between the filtration wheels and the gland covers. Each manifold is made of stainless-steel fittings and brass flat jet sprinklers.

For maintenance purposes, the manifold assembly is assembled to a mounting plate bolted and sealed to the gland cover. All sets of nozzles detach from the water supply using a quick disconnect system for easier maintenance. The system is easily removable from an outside access.

The system shall be equipped with a check valve to prevent the back flow of sludge into the wash system.

Washing system demand is not to exceed 750 gallons every 2.5 hours. For example, 50 gpm for a 15 minute wash interval is acceptable during start-up and shut-down. However, multiple wash water intervals can not exceed 750 gallons of water demand in less than 2.5 hours of equipment run time. The 2W (air gap) water system has limited capacity.

Filtrate Collector:

Each channel shall be equipped with a removable filtrate collector. Filtrate collector shall be made of low linear density polyethylene (LLDPE). Filtrate outlet connection shall be 7.5" (190,5 mm) dia. Each filtrate collector will include two (2) sampling ports to sample filtrate. Sampling port shall be closed by a removable plug.

Filtrate collector manifold:

Each filtrate collector shall be piped to a manifold. Manifold shall be made with Sch. 80, 6" (153 mm) dia. PVC piping. Manifold outlet shall be equipped with a 90 ° elbow.

1.3 Rotary Press Base

The base shall be fabricated from bend 3/8" (9,5 mm) thick epoxy painted steel plate. Base shall be a close shape to optimize rigidity and spread the load evenly to the supporting floor. Anti rotation device fixations are integrated to the base to facilitate the installation.

1.4 Cake outlet chutes

Every cake outlet will have a chute with an inspection door. The chute shall be made in low linear density polyethylene (LLDPE). A hole for mounting of a cake sensor is provided for this purpose.

Every cake outlet will have a chute with an inspection door. The chute shall be made in low linear density polyethylene (LLDPE). A hole for mounting of a cake sensor is provided for this purpose.

1.5 Screw Conveyor System for Sludge Cake Transport

See Section 41 12 13 – Screw Conveyor System

1.6 Emulsion Polymer System

See Section 46 33 33 – Emulsion Polymer Make-Up System

1.7 Digested Sludge Pump

See Section 22 13 36 – Packaged Sludge Pumps

Please note that the control system for the rotary fan press is to control the digested sludge pump feed rate. The variable speed drive will regulate the pump flow, as needed, to maintain a desired pressure into the rotary fan press unit.

2.0 FLOCCULATOR AND FEED MANIFOLD

2.1 The Flocculator

Mechanical agitation using a pressurized vessel is required before the Rotary Press to assure a proper flocculation. Static mixing device or piping used as static mixer is not authorized.

The flocculator model 05-300-900 shall be designed with a tank capacity of 17 gallons (64,3 liters) and for 100 PSI (690 kPa) pressure and a maximum temperature of 140 °F (60 °C). The body and bolted cover shall be fabricated of epoxy painted carbon steel. The thickness of the body shall be 3/8" (9,5 mm). Top cover plate shall be fabricated from 1-1/4" (31,8 mm) epoxy painted carbon steel. Three lifting lugs shall be supplied to facilitate handling.

The agitator shall be variable speed and driven by a gear motor, helical type, hollow shaft, Nord Gear model SK0282. Hollow shaft shall be 1-3/16" (30,2 mm) dia., 4.03 to 1 ratio, 1 HP (0.75 kw) premium efficiency motor, 1750 RPM, 3 PH, 480V, 60Hz, continuous duty, Insulation class F, CSA, CE & UL approved.

The seal at the agitator shaft and the flocculator cover will consist of two (2) lip seal contacts with grease cavity in between. A grease fitting will allow the cavity to be lubricated. The shaft seal wear sleeve shall be fabricated from stainless steel. The wear sleeve shall be polished to a surface finish of 16 rms and shall be hardened by nitriding.

The flocculator will have the following connections:

1. Sludge inlet
2. Sludge outlet
3. Pressure sensor fitting (Ref.: Press inlet pressure)
4. Wash water connection
5. Polymer connection
6. Air purge
7. Spare inlet 3/4" (19 mm)

The sludge inlet and outlet connections shall be 3 inches (75 mm) diameter with machine groove to adapt type 77 Victaulic coupling or equivalent.

Threaded connections for service and instrumentation shall be located on the inlet manifold of the flocculator.

The flocculator shall be equipped with an air breather. The air breather shall be located at the top of the flocculator and hook up to the potable water line for occasional back wash and to the drain line to collect any sludge leak. Breather shall be made of PVC.

2.2 Feed Manifold

2.2.1 Piping for flocculated sludge

The piping between the flocculator and the Rotary Press shall be made of schedule 10, 316 stainless steel.

The feed manifold and other accessories shall be tested in the shop at 100 PSI (690 kPa) hydrostatic pressure.

2.2.2 Pipe fittings

The pipe fittings shall be grooved end type as described below:

On lines less than 10 feet (3 meters) long	Style 77 Victaulic joints or equivalent
On lines more than 10 feet (3 meters) long	Style 07 "Zero Flex" Victaulic joints or equivalent
Flange-to-flange joints	Style 741 "Vic-flange" or equivalent
For butt or field adjust joint coupling	Victaulic style 99 "Roust-A-Bout" or equivalent

2.2.3 Valves and accessories

All valve(s) shall be compatible with the nominal operating pressures of the different systems and shall be designed to resist the operating temperatures and the contained fluids. The valves shall be of equal diameter as the piping except where indicated otherwise.

a) Automatic Sludge feeding and recirculation valve

Two pneumatically-actuated valves shall be installed on the sludge line between the rotary press and the flocculator. The valve shall meet the following characteristics:

Valve

Type	Two-way ball type Full port
Size diameter	3'' (76.2 mm) dia.
End connection	ANSI 125# Flanged
Body material and facing	Painted cast iron
Ball material	Ductile iron / Teflon fuse ball
Shaft	Stainless steel
Seat type	RPTFE
Manufacturer	Quality Certified Valve, American Valve, Sure Flow or equivalent

Actuator

Type	Pneumatic double acting (2-pistons design on geared shaft)
Nominal torque at 81 PSI (560 kPa)	2390 lbs in (270 Nm)
Manufacturer	FESTO or equivalent

Position indicator

Type	Visual Position switches
Switches	SPDT
Protection	Watertight IP67
Manufacturer	FESTO or equivalent

Sludge feed and recirculation valves shall be piloted by solenoid valve located in the Rotary Press Junction Box (ref. section 4.3.6). The valve body shall be made of anodized die cast aluminium and will have an insulation rated IP65. The working pressure shall be in the range of 36 to 116 PSI (248 to 800 kPa). The solenoid valve with manual override shall be 24 VDC. Solenoid valve shall be Festo model no. VMPA1-M1H-M-PI or equivalent.

b) Channel Isolating valve

At the inlet of each channel, a shut-off valve shall be supplied. The valve shall meet the following characteristics:

Valve

Type	Ball type Full port
Size diameter	3 inches (76.2 mm) dia.
End connection	ANSI 125# Flanged
Body material and facing	Painted cast iron
Ball material	Ductile iron / Teflon fuse ball
Shaft	Stainless steel
Seat type	RPTFE
Manufacturers	Quality Certified Valve, American Valve, Sure Flow or equivalent

Actuator

Type	Lever
------	-------

c) Channel Wash water valve

For automatic control purposes, a pneumatic actuator shall be installed to control the wash water valve to each individual channel. The valve shall meet the following characteristics:

Valve

Type	Ball type Full port
Size diameter	1 inch (25.4 mm) dia.
Body material and facing	Type 316 stainless steel
Maximum pressure	1000 PSI (6.9 MPa)
Manufacturer	C.F.F. model no. SS-3DM or equivalent

Actuator

Type	Pneumatic spring return type Aluminium anodized body
Manufacturer	Festo model no. VMPA1-M1H-M-PI or equivalent.

Wash water valve shall be piloted by solenoid valve located in the Rotary Press Junction Box (Ref. Section 4.3.6). The valve body shall be made of anodized die cast aluminium and will have an insulation rated IP65. The working pressure shall be in the range of 36 to 116 PSI (248 to 800 kPa). The solenoid valve with manual override shall be 24 VDC. Solenoid valve shall be Festo model no. VMPA1-M1H-M-PI or equivalent.

d) Other valves

All other valves other than the one above shall meet the following characteristics:

Valve

Type	Ball type Full port
Size diameter	As required
Body material and facing	Type 316 stainless steel
Seat material	PTFE
Manufacturers	Quality Certified Valve, American Valve, C.F.F., Trueline, H&P or equivalent

Actuator

Type	Lever
------	-------

2.2.4 Polymer check valve

A stainless steel 1" (25.4 mm) NPT check valve shall be installed on the polymer line to prevent sludge from entering the polymer feed line. The check valve will start to open at 0.5 psig (3.4 kPa) and shall be fully opened at 1.0 psig (6.8 kPa). The valve shall be spring assisted for silent closing and minimizing the effect of water hammer. The valve temperature limit shall be -20 to 400 °F (-29 to 204°C). The model shall be IFC model no. SC600TSSSSM.

2.2.5 Flocculated sludge sampler

A flocculated sludge sampler shall be supplied. The device shall consist of a transparent PVC tube to inspect the texture of the flocculated sludge. A 3/4" (19 mm) vent and 2" (50 mm) sludge inlet and sludge draining valves will allow filling and draining the sampler without dismantling or adding any pipes, tubes or plugs. A 3/4" (19 mm) vent shall also be installed for wash purpose. The sampler shall be easily removable for maintenance.

2.2.6 Wash water pressure gauge

A pressure gauge shall be installed at the flocculator on the wash water line. The pressure gauge shall be a 2-1/2" (63.5 mm), liquid filled. The scale shall be 0 to 160 PSI (0 to 1.1 MPa).

3.0 MISCELLANEOUS

3.1 Painting and protection

3.1.1 Painting and protection for equipment and steel piping

Surface preparation

Near white blast cleaning according to SSPC-SP10 for the equipment

Pickling SSPC-SP8 for the piping.

Primer

Type: Epoxy Resin (high built) Carboguard 893

Note: According to the ONGC-1 GP165A Standard
(US equivalent Standard SSPC Paint No. 22)

Dry thickness: 6 mls (152 μ m) min.

Finish coat

Type: Polyurethane

Color : Glass Shield Blue # 2862044 Carbothane 134HG blue

Dry thickness: 1.5 mls (38 μ m) by layer (2 layers)= 3 mls (76 μ m) min.

Total system thickness: 9 mls (229 μ m) min.

3.1.2 Painting and protection for all wetted parts

Surface preparation

Near white blast cleaning according to SSPC-SP10

Primer

Type: Epoxy Resin (high built) Plasite # 7122

Note: According to the ONGC-1 GP165A Standard
(US equivalent Standard SSPC-Paint No. 22)

Dry thickness: 9 mls (229 μ m) min.

The surfaces in contact with feed sludge, filtrates and cake will also be painted except when those surfaces are made out of stainless steel or polymer based product.

3.2 Materials

The steel used in the construction will meet the following standards:

- Tubing: CSA G40.21 - 44W (US equivalent ASTM A-500)
- Structural shape: CSA G40.21 - 44W (US equivalent ASTM A-572)
- Plate: CSA G40.21 - 44W (US equivalent ASTM A-36)
- Sheet: ASTM A569
- Shaft: ANSI-1045
- Stainless steel: ANSI 304, ANSI 316

Special tools

Special maintenance tools shall be supplied. Those will include a shaft torque wrench and lifting bolt for wheel and housing. The special maintenance tools shall be delivered into a tool box.

3.3 Fasteners

All fasteners to be used shall be made according the ANSI/ASME B18 standards.

Except if otherwise noted, ASTM F738M stainless steel bolts and ASTM F836M stainless steel nuts (property class A1-50 for both, or equal) shall be used for all assemblies:

- Channel housings
- Gland covers to channel housing

- Filtrate collector to channel housing
- Channel sludge inlet to the channel housing
- Anti-rotation plate to the channel housing
- Channel housings to outer spacers
- Filtration wheels to the inner spacer
- Scraper blades to outer spacer
- Actuator support to the channel assembly
- Lifting devices to the channels
- Interior bushing to the gland cover hub
- Wash water system support to gland cover
- Speed reducer to reducer base
- Deflector
- Actuator support
- Speed reducer output shaft to its cap plate
- Identification plates
- Motor bell housing to the speed reducer input
- Input and output flocculator piping
- Flocculator

3.4 Welding

All the welding shall be performed under the AWS Standard, CSA Standard, or equivalent.

3.5 Noise level

The noise level of each of the equipments will not exceed 75 dBA measured at 3 feet (1 meter) from the source.

3.6 Interchangeability

All parts from press or from channel shall be interchangeable with another.

4.0 INSTRUMENTATION AND CONTROL

4.1 Basic principle of operation

The dewatering unit will consist of one (1) rotary press with its flocculator, along with control devices necessary to maintain an adequate proportion of flow and pressure, resulting in an uniform feed of sludge and diluted polymer to the press.

The feed sludge is pumped to the flocculator as to maintain a constant pressure at the flocculator inlet. To accomplish this, the rotating speed of the feed sludge pump will be adjusted according to the internal pressure of the flocculator as to maintain the pressure close to the set value.

The diluted polymer is pumped to the flocculator in such a way as to obtain a specified polymer-sludge dosage. The dosage is controlled by adjusting the speed of the diluted-polymer pump and is a function of the speed of the sludge pump, and a set value entered by the operator. From the flocculator, the flocculated sludge is fed into each of the channels of the Rotary Press.

The channel outlet pressure is controlled by modulating a restrictor device at the cake outlet. The applied air pressure of the restrictor's actuator is controlled to maintain a constant outlet pressure.

A local control cabinet is used to operate the system and to change the basic parameters.

The control panel shall be CSA certified. All dewatering equipment and peripheral need to be controlled by the Rotary Press Control Panel for better integration.

4.2 Instrumentation

4.2.1 Flowmeters

Sludge flowmeter:

The sludge flowmeter shall be installed at the flocculator inlet.

The instrument shall meet the following specifications:

Type	Electromagnetic
Linearity error	±0.5%
Diameter/measuring range	3" (75mm) - 24-800 gpm (5.5 – 181.6 m3/h)
Process temperature	-4 °F to 176 °F (-20 °C to 80 °C)
Electrical connection	1/2" NPT
Process connection	Flange type
Display/operation	Two line display with backlit / push button
Output signal	4-20mA
CSA/FM approval class	Class 1 division 2
Liner material	Polyurethan
Electrode material	316L SS
Protection	IP67/Nema 4X, IP68/Nema 6P
Company	Endress & Hauser
Series	10W or equivalent

Polymer flowmeter:

The polymer flowmeter shall be installed at the flocculator inlet.

The instrument shall meet the following specifications:

Type	Electromagnetic
Linearity error	±0.5%
Diameter/measuring range	1" / 5-80 GPM (1.1 – 18.1 m3/h)
Process temperature	-40 °F to 266 °F (-40 °C to 130 °C)
Electrical connection	1/2" NPT
Process connection	Flange type
Display/operation	Two line display with backlit / push button
Output signal	4-20mA
CSA/FM approval class	Class 1 division 2
Liner material	PTFE
Electrode material	316L SS
Protection	IP67/Nema 4X, IP68/Nema 6P
Company	Endress & Hauser
Serie	10P or equivalent

4.2.2 Inlet pressure gauge and transmitter

The inlet pressure gauge and transmitter shall be installed at the flocculator inlet. The instrument will meet the following specifications:

Type	Digital pressure transducer, capacitive, ceramic
Linearity error	± 0.075 %
Process temperature	- 4 to 257 °F (- 20 to 125 °C)
Operating pressure range	0-30 PSI (0-207 kPa)
Proof pressure (Without any damage to the transducer)	270 PSI (1,8 MPa)
Electrical connection	1/2" NPT
Calibration	0-30 PSI (0-207 kPa)
Output	4-20mA, with Hart Communication, 2 lines LCD display
Process connection	1/2" NPT
Protection	I, II, III Division 1
Diaphragm material	Ceramic
Company	Endress & Hauser
Series	PMC71

4.2.3 Outlet pressure control

Cake outlet pressure shall be controlled by air pressure on the bellows actuating the vertical restrictor.

Adjustable airline pressure regulator shall be used to control the outlet pressure. Each channel will have its own in line air regulator. Regulator shall be relieving type, pressure range 0 to 100 PSIG (0 to 690 kPa), 1/4 inch port, with 150 PSIG (1 MPa) pressure gage. Regulator shall be Festo Model no. MS4N-LR-1/4-D6-AS or equivalent.

For automation purposes, Electro-pneumatic transducer shall be supplied with the control system and shall be Festo Model no. VPPE-3-1-1/8-6-420-E1 or equivalent.

4.2.4 Cake sensing

Each channel shall have a cake sensor mounted on the outlet chute cover. For easy maintenance, the sensor shall be equipped with a quick connector. Sensor shall be Schneider Electric Model no. XT218A1PCM12 or equivalent

4.2.5 Wash water pressure switch

A pressure switch shall be installed on the wash water line at the flocculator. The pressure switch shall be 1/4" NPT, SPST/NO. The switch shall cover an adjustable range of 30 to 120 PSI (207 to 827 kPa). The switch shall be shop calibrated at 60 PSI (414 kPa) and sealed. The dry contact of the switch shall be used to produce a low pressure alarm on the operator interface.

4.3 Process control cabinets

4.3.1 Prewired control cabinet

Prewired control with connection to field devices and field wiring through terminal strips, in a NEMA 12 enclosure shall be provided. As per customer requirement, the control shall operate on 480 volts, 60Hz, three phases. The control cabinet shall include the following:

- Back-panel mounted components and devices consisting of (but not limited to) the following:
 1. Main fused disconnect switch
 2. Circuit breakers for PLC, HMI and others control devices
 3. Line reactor
 4. Variable frequency drive (VFD)
 5. Programmable logic controller (PLC)
 6. Filter and surge protective device for control voltage
 7. DC power supply
 8. Relay
 9. Safety relay
 10. LAN modem (Internet communication)
- Door mounted components consisting of (but not limited to) the following:
 1. Power warning alert indicator
 2. Operating handle for main fused disconnect switch
 3. Emergency stop push button
 4. Human machine interface (HMI)
- Control cabinet air conditioning system shall be supply if required by the heat load calculation vs room temperature. If required, the air conditioning system shall be Hoffman Proair Harsh Environment CR23 Series or equivalent.

4.3.2 Programmable logic controller (PLC)

A programmable logic controller shall be used to control the Rotary press and the related equipment. PLC shall have Ethernet/IP communication to communicate with the VFD, HMI and the plant network.

Programmable logic controller (PLC) :

The PLC will have the following specifications:

Processor unit	CompactLogix 5370 series (L1X series), including 16 in sink/16out source Communication ports : 2 Ethernet/IP, 1 USB Memory card type : SD Nominal input power : 24Vdc Input power range : 10 to 28.8Vdc Operating temperature : -4 to 140°F (-20 to 60°C)
Additional I/O modules	Point I/O (1734 serie)

Analog inputs	24V DC Analog current input (0-20, 4-20mA selectable), single ended, non-isolated Resolution : 16 bits Absolute accuracy : 0.1% Full scale @ 77°F (25°C) Operating temperature : -4 to 131°F (-20 to 55°C) Data format : Signed integer Maximum overload : Fault protected to 28.8VDC
Analog outputs	24V DC Analog current input (0-20, 4-20mA selectable), single ended, non-isolated Resolution : 16 bits Absolute accuracy : 0.1% Full scale @ 77°F (25°C) Operating temperature : -4 to 131°F (-20 to 55°C) Data format : Signed integer
Discrete inputs	24V DC discrete sink input Min voltage on-state (Nominal) : 10Vdc (24Vdc) Min voltage off-state : 5Vdc Field power nominal voltage : 24Vdc Field power voltage range: 10 to 28.8Vdc Operating temperature : -4 to 131°F (-20 to 55°C)
Discrete outputs	24V DC discrete source output Min voltage on-state (Nominal) : 10Vdc (24Vdc) Output signal delay : Off to On 0.1 ms On to Off 0.1 ms Field power nominal voltage : 24Vdc Field power voltage range: 10 to 28.8Vdc Output current rating : 1.0A per output, not to exceed 3.0A per module Operating temperature : -4 to 131°F (-20 to 55°C)
Certification	cULus Listed Cl 1 Div 2
Company	Allen-Bradley
Serie	CompactLogix

4.3.3 Human machine interface (HMI)

A human machine interface shall be used to operate the Rotary press and the related equipment.

Touchscreen operator interface :

The touchscreen operator interface will have the following specifications:

Display type and size	TFT Color LCD (5.7" - 14.5cm)
Graphic display resolution	600 x 480 (VGA)
Display colors	18-bit Colors
Touch screen	Analog resistive
Operating temperature	32 to 131°F (0 to 55°C)
Communication ports	Ethernet, USB
Programming memory	512 MB Flash EPROM
Memory card type	SD
Input voltage (power consumption)	24 VDC (35W or less)

Certification	CE, cULus Listed Cl 1 Div 2
Protection	IP54, IP66, NEMA 12/13/4X indoor use only
Company	Allen Brandley
Model	2711P serie (Panelview Plus 7)

4.3.4 Variable frequency drive (VFD)

Variable frequency drive shall be used to control the speed of the Rotary press and the flocculator. VFD will have overload protection and ramp regulation capability. Configuration and programming shall be done via a keypad. Run command, status and speed control shall be handle using communication network (VFD Ethernet/IP or RS485/DSI). As per customer requirement, VFD shall be powered as per power inlet described in section 4.3.1. Output shall be 480 Vac 0-60Hz. VFD shall be the following:

Variable frequency drive (VFD) :

The VFD will have the following specifications:

Voltage input	As per Control cabinet Voltage input
Output voltage	Adjustable 0V to rated motor voltage
Input frequency variation	47 to 63 Hz
Communication	Integral RS485 with modbus RTU/DSI, Ethernet/IP
Operating temperature	-4 to 122°F (-20 to 50°C)
Certification	CE, cULus Listed
Protection	IP20 Nema/Open
Company	Allen Bradley
Model	Powerflex 52X Serie
Rotary press rating	As per motor requirement
Flocculator rating	1 HP (0.75 KW)

4.3.5 LAN modem (Internet communication)

A LAN modem is used for remote access in case of programming change needs and troubleshooting. This modem uses common internet as communication link.

LAN modem (internet communication) :

The LAN modem will have the following specifications:

Power supply (consumption)	12 to 24Vdc +/-20% (10W)
Communication port	1 SubD9 serial port RS232 (RS485 not isolated) 1 RJ45 Ethernet 10/100 base Tx
Operating temperature	-4 to 158°F (-20 to 70°C)
Certification	CE, cCSAus
Company	eWON
Model	Cosy 141

4.3.6 Rotary Press Junction Box

A pre-wired Nema 4X FRP enclosure junction box shall be installed on the rear of the rotary press gearbox. Wash valves and cake detection sensor shall be routed to this junction box and connected to terminal blocks. The Rotary press junction box shall include, but not limited to, the following components:

- I/P converter
- Air filter and regulator
- Solenoid valve for pneumatic control of sludge feeding and recirculation valve
- Solenoid valve for pneumatic control of wash water valve

To insure proper air quality and inlet pressure control to system, an air regulator and an air filter are furnished and installed on the air inlet supply of the Rotary press junction cabinet. The air regulator and air filter shall be rated for air purity class 3.7 per DIN ISO 8573-1, high efficiency general purpose protection for a particle removal down to 5 microns, including water and oil aerosols. 1/4" NPT type connection, fully automated drain, maximum pressure 175 PSI, operating temperature between 35 °F & 140 °F (2°C & 60°C), an integrated air regulator shall include a manometer, range 0-175 PSI with lockable rotary knob. Air regulator shall be Festo Model no. MS4N-LFR-1/4-D7-CRV-AS or equivalent.

4.3.7 Emergency stop push button

An emergency stop push button shall be installed in front of the rotary press and wired to the junction box.

4.4 Programming

4.4.1 Control

Process parameters and control shall be accessible to the pre-program PLC via operator interface. The operator will have access to the following:

- Sludge inlet pressure
- Outlet pressure control
- Polymer/Sludge ratio
- Rotary press rotating speed
- Flocculator agitator speed
- Manual control of each equipment

4.4.2 Protection and alarms

Programming will include the following protection/alarm:

- High inlet pressure, stop the dewatering system
- Zero flow on sludge input flow, stop the dewatering system
- Cake flush, stop the dewatering system
- Preventive maintenance notice, alarm only
- Low wash water, alarm only
- Low air pressure, stop the dewatering system
- Any motor did not start, stop the dewatering system

Programming will include at the minimum the following alarms:

- Rotary press faulted
- Rotary press did not start
- Flocculator faulted
- Flocculator did not start
- Sludge pump faulted
- Sludge pump did not start
- Sludge pump low flow
- Polymer pump faulted
- Polymer pump did not start
- High inlet pressure
- Cake flush
- Low air pressure
- Low wash water pressure
- Preventive maintenance warning

4.4.3 Operator interface screens

The operator interface shall be built to give friendly usage to the operator. The following screen page shall be available to the operator:

- Menu page
- Cycle start-stop page
- Parameter page
- Sludge pump control loop page
- Rotary press manual operation page
- Preventive maintenance advise page

4.4.4 Standard features

System programming to incorporate the following features as standard control:

- Individual or grouped channel wash selection (for optimal wash depending on the water supply)
- Automatic system start at selected time of the day
- Automatic adjustable delayed system stop

5.0 TRANSMITTALS

5.0 Format

Transmittal shall be made accordingly to the following format:

- Pages 8-1/2 x 11 inches, or 11 x 17 inches folded to fit 8-1/2 x 11 inch binder.
- Three holes punched and arranged so holes do not destroy text or graphics.
- Tabs for each section.
- Bound in 3-ring binders.
- Binders shall include the following identification on the cover:
 - The text: "OPERATION AND MAINTENANCE MANUAL"
 - Project location.
 - Description of equipment or process system described in manual.

- Name, phone number, and address of equipment manufacturer or Operations and Maintenance subcontractor.
- Material shall be camera-ready quality (suitable to produce quality reproductions).

5.1 Shop drawings

The Supplier shall furnish to the Contractor for review, (x) copies of each submittal. The term "submittal" as used herein shall be understood to include installation drawings, catalog sheets and data sheets. Unless otherwise required, said submittals shall be submitted to the Customer, at a time sufficiently early to allow review. The supplier shall furnish the Customer the submittals in compliance with the Project Specifications.

5.1.1 Mechanical

Mechanical submittals shall include at least the following:

1. Certified shop and erection drawings showing all important details of construction, sludge feed, wash water, high pressure air, and drainage connections, wiring diagrams, itemized motor horsepower, dimensions and anchor bolts.
2. Descriptive literature, bulletins, and/or catalogs of the equipment.
3. Data on the characteristics, features, and performance of the equipment.
4. The total weight of the equipment including the weight of the single largest item.
5. A complete total bill of materials for all equipment.
6. List of recommended spare parts
7. List of spare parts supply with contract

5.1.2 Electrical

Electrical submittals shall include at least the following:

1. Electrical Schematics, panel layouts, point-to-point I/O (input/output) wiring diagrams, control panel network diagram, instrumentation sheets, and product data sheets for electrical equipments being supplied by the Rotary Press® manufacturer.
2. A complete total bill of materials for all equipment.

5.2 Operation and maintenance manual (O&M manual)

O&M Manuals will thoroughly address all items of equipment, components, options, accessories, and ancillary devices provided with equipment components.

O&M Manuals will include the following minimum information in addition to that normally provided or required by the manufacturer; and in addition to the specific requirements identified in equipment specification sections.

- Title Page
 - Project Name, Equipment model
 - Manufacturer Information
- Table of Contents
- Equipment introduction, description of each item of equipment and system addressed in O&M Manual. Use project specific equipment name plate, tag numbers, manufacturing date.
- Rotary Press installation Procedure
 - Lifting instruction

- Storage and installation recommendations
- Rotary Press individual operating Instructions
 - Principle of operation of the rotary press
 - Process overview
 - Zones description
 - Effect of parameters
- Flocculator individual operating instructions
 - Flocculator Operation Instruction
 - Sludge sample operation
 - Jar testing procedure
- Equipment automatic operating instructions
 - Recirculation Cycle
 - Dewatering Cycle
 - Wash Cycle
 - Shutdown sequence
 - Safety instruction
 - Recommended daily check list
 - Trouble shooting information (Problem, causes and solutions chart)
 - Polymer preparation
 - Filtrate sampling procedure
- Maintenance instructions
 - Introduction and safety instructions for maintenance
 - Rotary Press
 - Preventive maintenance
 - Cleaning Exterior/Interior
 - Lubrication chart and mechanical check list
 - Recommended Preventive maintenance with schedule
 - Adjustment and Maintenance: Seals, deflector and inlet spacer
 - Assembly Procedures
 - Disassembly procedures
 - Flocculator
 - Recommended Preventive maintenance with schedule
 - Cleaning
 - Lubrication chart and mechanical check list
 - Replacement of the seals
 - Assembly Procedures
 - Disassembly procedures
- Main maintenance operation
 - For each item or product provided.
 - Identifying only provided options and accessories.
 - Listing complete model number as needed to order an exact duplicate.
- Product Data as follows:
 - Complete technical data sheets for each item or product provided.
 - Identifying only provided options and accessories.
 - Listing complete model number as needed to order an exact duplicate.

- General drawings as follows:
 - Showing all parts, assemblies, and sub-assemblies.
 - Showing arrangement and inter-relationship of all parts and equipment.
 - Showing controls and direction of flows.
- Electrical drawings as follows:
 - Showing all control devices and prefabricated wiring and conduit.
 - Complete and accurate control schematics.
 - Control panel layouts.
 - Identifying all termination's and terminal strips.
 - Settings for all configuration dip switches.
 - Identify each different source of power. Use nomenclature specific to project.
 - Listing and rating of all fuses or breakers.
- A description of the functions including:
 - A description of the automatic cycles.
 - A list of all the alarms.
 - A description of all the screens in the operator interface.
 - Listing of set points for all adjustable devices.
- List of recommended spare parts.
- List of recommended and alternate consumables and expendable such as lubricants and filters.
- Clear and concise written instructions, with illustrations as required, for the following:
 - Normal operating procedures.
 - Long term storage procedures.
 - Adjusting and troubleshooting procedures.
- Copies of all information provided as part of the approved Product Data or Shop Drawing submittals.
- All aspects of press operation and maintenance, including:
 - a. Press and component disassembly and re-assembly, including channels, speed reducer drive, flocculator mixer, system piping, field instruments, pneumatic controls, control valves, seals, etc.
 - b. Maintenance of all dewatering system components, including flocculation system, pneumatic system, dewatering channels, control valves and control elements, such as meters, pressure regulators, pressure sensors, etc., mixers, drives and variable speed controllers.
 - c. Operational controls including pressure adjustments, press speed, sludge and polymer feed, as they relate to control and response to operator variation.
- See other submittal requirements in Section 01 33 00.

6.0 WARRANTY

6.1 Equipment is to include a minimum 1 year, non-prorated, warranty after the date of

Substantial Completion. See general contract requirements.

7.0 EXECUTION

7.1 INSPECTION & STORAGE

7.1.1 The Contractor shall inspect and accept all equipment immediately upon delivery to the site to satisfy himself that complete and undamaged equipment package has been delivered for installation.

7.2 INSTALLATION

7.2.1 The Contractor shall install all equipment according to the drawings and specifications and in accordance with the installation instructions and drawings provided by the manufacturer to the Contractor. The bid document drawings are intended to show the Contractor the components of the proposed construction not the details of assembly and installation of the dewatering system equipment. The Contractor is responsible for all costs associated with installing the equipment.

7.2.2 Contractor to replace any defective equipment or components.

*****END OF SECTION*****

PART 4

REFERENCE DOCUMENTS

APPENDIX A – ANTICIPATED CONSTRUCTION SEQUENCE AND SCHEDULE

See Section 00 31 13 – Preliminary Project Phases. Contractor's schedule is to incorporate all work deadlines shown in Section 00 31 13.

APPENDIX B – GROUNDWATER LEVEL ANALYSIS

Eastsound Sewer and Water District

Well Monitoring Log

Date	Depth to Water (Feet)	Notes/Weather
8/17/2020	No Water	Sunny: 75°F
8/24/2020	13.75	Partly Cloudy: 61°F
9/28/2020	12.67	Sunny: 57°F-5days of rain prior
10/26/2020	7.20	Partly Cloudy: 35°F
11/23/2020	6.50	Raining: 40°F
12/28/2020	5.60	Very Rainy Period: 45°F
1/26/2021	5.20	Sunny: 45°F
2/19/2021	2.50	Snow Covered Sunny: 43°F
3/22/2021	4.10	Sunny: 50°F
4/7/2021	6.67	Sunny: 46°F

APPENDIX C – GEOTECHNICAL REPORT

Geotechnical Engineering Services

Eastsound Wastewater Treatment Plant
Improvements
Eastsound, Washington

for
Wilson Engineering, LLC

November 8, 2022



GEOENGINEERS 
Earth Science + Technology

Geotechnical Engineering Services

Eastsound Wastewater Treatment Plant
Improvements
Eastsound, Washington

for

Wilson Engineering, LLC

November 8, 2022



554 West Bakerview Road
Bellingham, Washington 98226
360.647.1510

Geotechnical Engineering Services
Eastsound Wastewater Treatment Plant
Improvements
Eastsound, Washington

File No. 0407-002-00

November 8, 2022

Prepared for:

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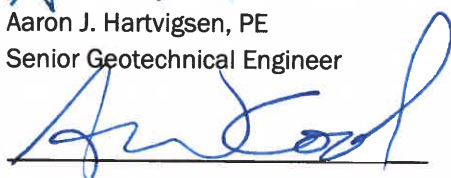
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1.0 INTRODUCTION

This revised report presents the results of GeoEngineers, Inc.'s (GeoEngineers) geotechnical engineering services for the proposed improvements to the Eastsound Sewer & Water District (District) Wastewater Treatment Plant (WWTP) located at the existing treatment plant facility at 143 Cessna Road in Eastsound, Washington located on Orcas Island. The site location is shown in the Vicinity Map, Figure 1. Existing site conditions and the location of the proposed structures are shown in the Site and Exploration Plan, Figure 2.

Our understanding of the project is based on information provided by Jeff Christner with Wilson Engineering, LLC (Wilson), conversations with the project team, and our experience on similar projects. We understand the District plans to construct several new below-grade concrete structures including a new 34-foot-diameter clarifier extending approximately 15 feet below grade; a chlorine contact chamber and concrete digester tank extending approximately 12.5 feet below grade; a new plant drain pump station extending 17.4 feet below grade, a new water pump station 14 feet deep, a new headworks flow splitter structure and screw press constructed at-grade; and miscellaneous piping and site improvements. The new infrastructure will be constructed within and adjacent to the existing active facility.

Our scope of services for this report included reviewing available geotechnical information at and near the site, completing three drilled subsurface explorations and installing one groundwater piezometer, laboratory testing on the samples obtained from the boring, and providing geotechnical conclusions and recommendations for design and construction of the proposed improvements. Our specific scope of services is described in our proposal for the project dated June 4, 2020 and was authorized via Task Order by Wilson on July 14, 2020.

2.0 SITE CONDITIONS

2.1. Surface Conditions

The project site is currently located on the existing WWTP property which supports the surrounding Eastsound community. The site is bounded by Cessna Road to the north and the Orcas Island Airport property to the east, south, and west. The site surface is generally paved around existing enclosed structures and is relatively level.

2.2. Geology

We reviewed a Washington State Department of Natural Resources (DNR) map for the project area, "Geologic Map of the Bellingham 1:100,000 Quadrangle, Washington" by Lapen (2000). This map indicates the site is underlain by undifferentiated glacial deposits. Undifferentiated glacial deposits consist of various material types and include granular outwash and unsorted, unstratified glacial drift. Outwash typically consists of sand with varying silt and gravel content. Glacial drift typically consists of silt and clay with varying amounts of sand, gravel, cobbles and occasional boulders. Glacial outwash soils can have a relatively high permeability and the glacial drift soils typically have a much lower permeability; however, because of the unstratified nature of this deposit, this glacial unit typically has a low effective permeability for the unit as a whole.

Although not mapped at the site, soil interpreted to be glacial till was encountered at depth. Glacial till typically consists of dense, non-sorted mixture of silt, sand and gravel deposited during the last glaciation.

2.3. Subsurface Exploration

Subsurface soil and groundwater conditions were evaluated by completing three borings (B-1 to B-3) on August 10, 2020 using an EC 95 track drill rig subcontracted to GeoEngineers. The borings were completed to depths of 20 to 26½ feet below ground surface (bgs). The locations of the borings are shown in Figure 2. Details of the field exploration program, laboratory testing, and boring logs are presented in Appendix A.

2.4. Subsurface Conditions

2.4.1. Soil Conditions

A general description of each of the soil units encountered at the project site is provided below. Our interpretation is based on soil conditions encountered during our geotechnical borings and our understanding of the local geologic conditions.

- **Asphalt Concrete** – Asphalt concrete surfacing was encountered in all borings. The asphalt thickness ranged from 2 to 3 inches thick.
- **Fill** – A layer of dense to stiff fill consisting of sand with variable silt and gravel content to silt with sand and gravel was encountered underlying the asphalt concrete in the borings. The fill layer generally extended to depths ranging from 2 to 4½ feet bgs in the explorations.
- **Undifferentiated Glacial Drift** – Native undifferentiated glacial drift was encountered underlying the fill layer in all three borings. The undifferentiated glacial drift encountered in the explorations generally consisted of medium stiff to very stiff gray clay with sand and occasional gravel. The undifferentiated glacial drift extended to depths of 19 to 24½ feet bgs in the borings.
- **Glacial Till** – Glacial till was encountered underlying the undifferentiated glacial drift in all three borings. Glacial till at this site consisted of very dense silty sand with occasional gravel. This till unit was not as dense or as hard as Vashon till common in the central Puget Sound region and had a higher clay/silt content. However, the glacial till observed at this site is significantly stiffer than the overlying undifferentiated glacial drift and is interpreted to be glacially consolidated. The borings all terminated in this unit at 20 feet to 26½ feet bgs.

2.4.2. Groundwater Conditions

Wet soils were observed during drilling at depths of 25 feet in B-1 and B-2 and 20 feet in B-3. A groundwater monitoring well was installed in B-3. The groundwater level was measured to be 13.75 feet bgs (approximately Elevation 9.45 feet [NAVD88]) on August 24, 2020 and will continue to be monitored by District staff. The undifferentiated glacial drift and glacial till units commonly have isolated saturated sandier zones or “pods” at variable depths and locations. Although not observed at the time of our explorations, perched groundwater conditions can develop within fill, topsoil, and undifferentiated glacial drift during the wetter portions of the year where underlying low permeability soil limits downward infiltration. The groundwater level is likely representative of the regional groundwater table. The groundwater conditions should be expected to vary as a function of season, precipitation and other factors.

3.0 CONCLUSIONS AND RECOMMENDATIONS

It is our opinion that the site is suitable for the proposed wastewater treatment plant improvements and other associated site improvements. The new structures can be supported on shallow spread footing or

mat foundations. A summary of the primary design and construction considerations for the project is provided below. This summary is presented for introductory purposes and should only be used in conjunction with the complete recommendations presented in this report.

- A seismic Site Class D is appropriate for design. The site soils are not considered liquefiable based on our explorations and local knowledge.
- The undisturbed native; unweathered, medium stiff to very stiff undifferentiated glacial drift; or new structural fill placed directly over these soils will provide adequate support for shallow spread footings designed using an allowable soil bearing pressure of 2,500 pounds per square foot (psf).
- We estimate postconstruction settlement of footings founded on medium stiff to very stiff native soils will be less than $\frac{3}{4}$ -inch. Differential settlement between comparably loaded column footings or along a 50-foot section of continuous wall footing should be less than about $\frac{1}{2}$ -inch.
- Standard slab-on-grade considerations are appropriate based on providing a minimum 6-inch-thick layer of granular capillary break/leveling material.
- The site soils are highly moisture sensitive and will not support rubber-tired or other heavy equipment during wet weather. Earthwork should be planned for the drier summer months to reduce costs.

3.1. Seismic Design Considerations

3.1.1. Seismicity

The site is located within the Puget Sound region, which is seismically active. Seismicity in this region is attributed primarily to the interaction between the Pacific, Juan de Fuca, and North American plates. The Juan de Fuca plate is subducting beneath the North American plate. It is thought that the resulting deformation and breakup of the Juan de Fuca plate might account for the deep focus earthquakes in the region. Hundreds of earthquakes have been recorded in the Puget Sound area. In recent history, four of these earthquakes were large events: (1) in 1946, a Richter magnitude 7.2 earthquake occurred in the Vancouver Island, British Columbia area; (2) in 1949, a Richter magnitude 7.1 earthquake occurred in the Olympia area; (3) in 1965, a Richter magnitude 6.5 earthquake occurred between Seattle and Tacoma; and (4) in 2001, a Richter magnitude 6.8 earthquake occurred near Olympia.

Research has concluded that historical large magnitude subduction-related earthquake activity has occurred along the Washington and Oregon coasts. Evidence suggests several large magnitude earthquakes (Richter magnitude 8 to 9) have occurred in the last 1,500 years, the most recent of which occurred about 300 years ago. No earthquakes of this magnitude have been documented during the recorded history of the Pacific Northwest. Local design practice in Puget Sound and local building codes consider these seismic conditions including known faults for the design of structures. An unnamed concealed thrust fault is mapped at the south end of the airport approximately 1,500 feet from the site. It is our opinion that the risk of surface rupture is low to moderate at the site.

3.1.2. 2018 IBC Seismic Design Information

The 2018 International Building Code (IBC) is the current building code. The 2018 IBC references the 2016 Minimum Design Loads for Buildings and Other Structures (American Society of Civil Engineers [ASCE] 7-16). We recommend that the site be classified as Site Class D – Stiff Soil Profile. Based on the site class and site mapped spectral acceleration, we anticipate the new structures will be designed

in accordance with Exception 2 of Section 11.4.8 in ASCE 7-16. For Site Class D with S_1 greater than or equal to 0.2g, Exception 2 allows the structural designer to use ground motions based on the site class and mapped spectral acceleration rather than completing a site-specific response analysis. Our recommended seismic design parameters for Exception 2 are presented in Table 1 below. If requested, GeoEngineers can complete a site-specific seismic response analysis, which might provide reduced seismic demands from the parameters in Table 1.

TABLE 1. MAPPED 2018 IBC SEISMIC DESIGN PARAMETERS

Seismic Design Parameters	Recommended Value ^{1, 2}
Site Class	D
Mapped Spectral Response Acceleration at Short Period (S_s)	1.128g
Mapped Spectral Response Acceleration at 1 Second Period (S_1)	0.399g
Site Modified Peak Ground Acceleration (PGA_M)	0.539g
Site Amplification Factor at 0.2 second period (F_a)	1.049
Site Amplification Factor at 1.0 second period (F_v)	1.9
Design Spectral Acceleration at 0.2 second period (S_{DS})	0.789g
Design Spectral Acceleration at 1.0 second period (S_{D1})	0.505g

Notes:

¹ Parameters developed based on Latitude 48.707292° and Longitude -122.908679° using the ATC Hazards online tool.

² These values are only valid if the structural engineer utilizes Exception 2 of Section 11.4.8 (ASCE 7-16).

3.1.3. Liquefaction Potential

Liquefaction is a phenomenon where soils experience a rapid loss of internal strength as a consequence of strong ground shaking. Ground settlement, lateral spreading and/or sand boils may result from liquefaction. Structures supported on liquefied soils could suffer foundation settlement or lateral movement that could be severely damaging to the structure. Conditions favorable to liquefaction occur in loose to medium dense, clean to moderately silty sand that is below the groundwater level. Dense soils or soils that exhibit cohesion are less susceptible to liquefaction. The proposed building is underlain by generally cohesive and dense soils which are not considered susceptible to liquefaction.

3.2. Temporary Shoring

Construction of the new clarifier, chlorine contact chamber, and digester structures will require excavations on the order of 10 to 17½ feet below existing site grades. The new flow splitter and screw press structures will be approximately 5 feet or less in depth. The excavation will likely be completed with temporary slopes; however, the size, depth, and location of the structures may require temporary shoring where sufficient space is not available for open cuts adjacent to existing WWTP structures and utilities. For the planned excavation depth, it is likely that cantilevered soldier pile shoring will be sufficient. We anticipate that driven diaphragm type shoring, systems such as driven sheet pile walls, will not adequately penetrate the native soils without pre-drilling. GeoEngineers can provide additional recommendations for temporary shoring prior to final design if necessary. Additional recommendations for temporary cut slopes and excavations shored with trench boxes is provided in Section 3.8.2.

3.3. Shallow Foundation Design

We recommend that the new WWTP structures be supported on shallow spread footings or mat foundations that bear directly on undisturbed, undifferentiated glacial drift deposits, or on compacted structural fill placed directly over undisturbed undifferentiated glacial drift.

3.3.1. Building Footing Subgrade Preparation

As currently planned, we anticipate that all existing non-structural fill will be removed during excavation for project foundation elements. Based on our explorations the medium stiff to very stiff clay encountered below the fill extends to 19½ to 25 feet bgs and is considered a suitable foundation bearing material when undisturbed. We do not anticipate any additional overexcavation will be required at the planned excavation depths; however, the clayey nature of the glacial drift may benefit from placement of a granular layer for protection of the subgrade, especially if construction occurs during periods of wet weather. Overexcavation may be required to reach the native layer for shallower foundation elements. We provide the following specific recommendations:

- We recommend the final excavations for the footings be accomplished with a smooth-bucket to reduce subgrade disturbance. We recommend that a GeoEngineers representative observe the footing subgrade to confirm suitable bearing conditions are present. Any specific overexcavation recommendations can be provided at that time.
- If necessary, any overexcavation of unsuitable or disturbed soils should extend a minimum of 1 foot beyond the edges of the footings or a minimum lateral distance equivalent to one-half the depth below the footings, whichever is greater. Overexcavated areas should be backfilled with structural fill as described later in this report and compacted to 95 percent of the maximum dry density (MDD) in accordance with ASTM International (ASTM) D 1557.
- The native subgrade soils will be susceptible to disturbance when wet. To minimize disturbance of the footing subgrade areas that consist of the native clay during wet weather, a layer of sand and gravel, crushed rock or a lean concrete “mud mat” is recommended. A minimum 12-inch-thick pad of 1½-inch minus clean crushed gravel with negligible sand or silt, as described in Section 3.8.1 will provide uniform support of the foundations and help protect the site subgrade soils from disturbance. Structural fill may be placed directly over the medium stiff to very stiff undisturbed undifferentiated glacial drift deposits in dry conditions. A geotextile fabric for separation place over the subgrade is recommended when the subgrade may become wet.

3.3.2. Foundation Design

For shallow foundation support, we recommend minimum widths of 18 and 24 inches, respectively, for continuous wall and isolated column footings supporting the proposed building. Exterior footings should be founded a minimum of 18 inches below the lowest adjacent grade. Interior footings should be founded a minimum of 12 inches below top of slab.

Provided that footings are supported as recommended above, an allowable bearing pressure of 2,500 psf may be used for the design of footings supported on the native undifferentiated glacial drift deposits. The allowable bearing pressures applies to the total dead and long-term live loads and may be increased up to one-third for short-term live loads such as wind or seismic forces.

For foundations designed as a beam on an elastic foundation (structural mats or grade-beams), a modulus of subgrade reaction of 200 pounds per cubic inch (pci) may be used for foundations bearing on undifferentiated glacial drift soils. This modulus value is for a 1-foot by 1-foot square plate. The actual modulus for a foundation element varies based on the footing size according to the following equation:

$$k_s = k_{s1}[(B+1)/2B]^2$$

Where k_s is the actual footing modulus, k_{s1} is the modulus for 1-foot by 1-foot plate, and B is the width or lateral dimension of the footing.

3.3.3. Foundation Settlement

Based on anticipated loads, we estimate that the postconstruction settlement of footings supported as recommended will be less than ¾-inch. Differential settlement along a 25-foot section of continuous wall footing should be about ½-inch or less. We expect settlement will occur rapidly as loads are applied. Loose or disturbed soils not removed from footing excavations prior to placing concrete will result in additional settlement.

3.3.4. Lateral Resistance

Lateral loads can be resisted by passive resistance on the sides of the footings and by friction on the base of the footings. Passive resistance should be evaluated using an equivalent fluid density of 300 pounds per cubic foot (pcf) where footings are poured neat against native soil or are surrounded by structural fill compacted to at least 95 percent of MDD, as recommended. The value assumes that the soil in front of the foundation element is horizontal for a lateral distance equivalent to 2.5 times the depth of the element. Resistance to passive pressure should be calculated from the bottom of adjacent floor slabs and paving or below a depth of 1 foot where the adjacent area is unpaved, as appropriate. Frictional resistance can be evaluated using 0.35 for the coefficient of base friction against footings. The above values incorporate a factor of safety of about 1.5.

If soils adjacent to footings are disturbed during construction, the disturbed soils must be recompacted or replaced with compacted structural fill. Otherwise the lateral passive resistance value must be reduced.

3.4. Slab-On-Grade Support

If conventional slab-on-grade will be used for the proposed structures, the native undifferentiated glacial drift will provide suitable support for the slab. Slabs-on-grade should be supported on a minimum 6-inch-thick capillary break/leveling layer of gravel or sand and gravel over suitably firm subgrade soils or on structural fill prepared in accordance with the site preparation recommendations. The capillary break/leveling material should consist of 1- or ¾-inch minus clean sand and gravel or crushed rock with less than 3 percent fines and should be uniformly compacted to a minimum of 95 percent of MDD. We do not recommend pea gravel or rounded drain rock for this purpose. A suitable Washington State Department of Transportation (WSDOT) 2020 Standard Specification (hereinafter referred to as WSDOT Standard Specifications) for this material is 9-03.1(4) C Grading No. 67.

3.5. Retaining Wall Considerations

3.5.1. Below-Grade Building Wall Design

Lateral soil pressures acting on the below-grade retaining walls will depend on the nature and density of soil behind the wall, the amount of lateral wall movement which occurs as backfill is placed, the inclination of the backfill surface, surcharge, and seismic loading. For walls free to yield at the top at least one thousandth of the wall height (i.e., wall height times 0.001), soil pressures will be less (termed “active earth pressures”) than if wall movement is restrained (termed “at-rest earth pressures”) by such factors as wall stiffness or bracing. Structures with internal bracing at the time of backfill should be modeled with at-rest earth pressures. We recommend the following lateral earth pressures be considered for design:

- **Active earth pressure for level backfill** – Equivalent fluid pressure (triangular distribution) of 35 pcf for drained conditions and 18 pcf plus hydrostatic pressure for undrained conditions.
- **At-rest earth pressure for level backfill** – Equivalent fluid pressure of 55 pcf for drained conditions and 28 pcf plus hydrostatic pressure for undrained conditions.
- **Passive earth pressure** – Equivalent fluid pressure of 200 pcf for un-drained conditions and 300 pcf for drained conditions.
- **Traffic surcharge pressure** – Typical vehicle traffic surcharge may be modeled assuming an additional 2 feet of soil (approximately 250 psf surcharge), or a uniform (rectangular) lateral pressure of about 70 psf. Other temporary surcharge loading during construction resulting from equipment and soil stockpiles will depend on specific loading type and location and should be evaluated by the contractor.
- **Seismic earth pressure** – Earth pressures resulting from seismic conditions may be modeled as a uniform (rectangular) lateral pressure of $8 \cdot H$ psf (H = height of wall) for the design level earthquake, with a corresponding reduction in the factors of safety to 1.1 or greater. Lateral pressure resulting from seismic surcharge loading is additive to lateral soil pressures computed as recommended above.

All pressures above assume structures are backfilled with granular structural fill and have relatively level grades behind the walls. In order to prevent overstressing the concrete retaining walls and causing bulging or rotation, we recommend that structural fill placed against the back of the wall be compacted with lighter weight hand-operated equipment and within the range of 90 to 92 percent of the MDD. Backfill should only be placed after the concrete has had sufficient time to cure and develop the necessary strength.

3.5.2. Wall Drainage Considerations

Where noted above, the recommended lateral earth pressures assume a free-draining condition behind the wall. For backfilled walls (adjacent to temporary cut slopes) positive drainage should consist of placing a minimum 18-inch-wide zone of free draining gravel backfill immediately adjacent to the walls. Gravel backfill for walls should consist of well-graded sand and gravel with less than 5 percent fines. A suitable WSDOT Standard Specification is Section 9-03.12(2). A 4-inch minimum diameter perforated drainpipe should be embedded in the free-draining sand and gravel zone along the base of the building retaining walls as described in Section 3.7 “Site Drainage Considerations”, of this report. Because of the clayey site soils, granular wall backfill will have a potential “bathtub” effect is not drained. If a drainpipe cannot be provided at the base of below grade walls, hydrostatic buildup should be anticipated to the level of the drainpipe or the seasonal high water level elevation measured in the monitoring well installed, whichever is lower.

3.6. Pipe Design Considerations

Various piping and utility corridors will be required to connect the new and existing infrastructure. The contractor should anticipate, at a minimum, all utility excavations will require a trenchbox for sidewall support and sumps and pumps to control groundwater seepage as necessary, as described in Section 3.8 of this report.

The work areas should be cleared of all surface and subsurface deleterious matter, including debris, vegetation and root wads; and stripped of any organic soil, if present. The cleared/stripped material will not be suitable for use as structural fill, which includes trench backfill.

3.6.1. Pipeline Support

Depending on the pipe depth and location on site, the soils expected to be encountered at the subgrade level for new piping will likely consist of undifferentiated glacial drift. In general, most of the undifferentiated glacial drift will provide adequate support for proposed piping throughout the site. Although not encountered in our explorations, if the pipe subgrade soils are very soft, organic, become disturbed, or are otherwise unsuitable, it may be necessary to over-excavate the unsuitable material and backfill with pipe bedding or crushed surfacing material. General recommendations for pipe bedding support are presented below:

- Pipe bedding material, placement, compaction, and shaping should be in accordance with the project specifications, the pipe manufacturer's recommendations, and Section 7-09 of the WSDOT Standard Specifications. At a minimum, the pipe bedding should meet the gradation requirements for Gravel Backfill for Pipe Zone Bedding Section 9-03.12(3) of the WSDOT Standard Specifications, unless otherwise stated by the project specifications.
- If the trench bottom encounters any soft or organic-rich subgrade soils, it may be necessary to overexcavate the unsuitable material and backfill with additional pipe bedding material. In wet conditions, 1½-inch minus granular fill meeting the gradation requirements for Crushed Surfacing Base Course (WSDOT 9-03.9[3]) or Permeable Ballast (WSDOT 9-03.9[2]) should be used to backfill the over-excavated portion of the trench.
- Pipe bedding material should be used as backfill up to at least the spring line of the pipe or in accordance with the manufacturer's specifications for the type of pipe chosen.
- Prior to the installation of the pipe, the bedding should be shaped to fit the lower part of the pipe exterior with reasonable closeness to provide continuous support along the pipe.
- Pipe bedding material and/or backfill around the pipe should be placed in layers and tamped around the pipe to obtain complete contact per the project plans. In areas where a trench box is used, the bedding material should be placed before the trench box is advanced.

3.6.2. Structural and Trench Backfill

We recommend that trench backfill consist of structural fill as described in Section 3.8.3 of this report. The suitability of the on-site material as structural fill is discussed below. During placement of the initial lifts, the backfill material should not be bulldozed into the trench or dropped directly on the pipe. Furthermore, heavy vibratory equipment should not be permitted to operate directly over the pipe until a minimum of 2 feet of backfill has been placed. Backfill should be placed in lifts to achieve the required compaction

using the mechanical equipment chosen by the contractor. We recommend that the MDD be determined based on ASTM D 1557 or in accordance with WSDOT Standard Specification 2-03.3(14)D.

- In non-settlement sensitive areas, such as landscaping, we recommend that the trench backfill be compacted to at least 85 percent of the MDD.
- In settlement sensitive areas, such as existing or future roadways and driveways, backfill placed at depths greater than 2 feet below the ground surface should be compacted to at least 90 percent of the MDD. The upper 2 feet of backfill should be compacted to at least 95 percent of the MDD in existing/future pavement or yard areas. This is consistent with Method B of WSDOT Standard Specification 2-03.3(14)C.

3.6.3. Pipe Settlement

The likely mechanisms for pipe settlement include poor bearing support immediately below the pipe. Provided pipe subgrade and backfill is completed as recommended above, we estimate that the increased weight of the structural backfill will be negligible and will not lead to significant settlement.

3.7. Site Drainage Considerations

We recommend that final site grading direct water away from the structures to the extent practical. The site may experience seasonally perched groundwater conditions because of the relatively low permeability of undifferentiated glacial drift. Additionally, the excavation for the below-grade structures into the undifferentiated glacial drift and glacial till unit will create a “bathtub” that will retain water when backfilled with granular structural fill. Therefore, we recommend the structures be provided with a perimeter drainage system or be designed for hydrostatic pressures. The footing drains should be installed at the base of footings and retaining walls and sloped to drain. The drains should consist of rigid perforated pipe, a minimum of 4 inches in diameter, enveloped within a minimum thickness of 6 inches of 1-inch washed gravel. A non-woven geotextile fabric such as Mirafi 140N, or similar as approved by GeoEngineers, should be placed between the 1-inch washed gravel and the native soils to prevent movement of the soils into the drainage material. This drainage should be tightlined to the stormwater system. Consideration should be given to the use of clean-outs for drainpipe maintenance. A larger diameter pipe will facilitate maintenance of drainage systems. Additional drainage recommendations are provided below:

- Subsurface slabs and walls should be adequately waterproofed as appropriate for the intended use. As noted previously, some slight seepage was observed during site explorations and the below-grade excavation will create a “bathtub” in the fine-grained glacial drift that retains water if drainage provisions are not included.
- If earthwork will occur during the wet season, perimeter ditching or interceptor trenches may be required to manage surface water and perched groundwater entering the site and excavation.
- We recommend all downspouts be tightlined away from the foundation area to the storm drain system. Downspouts should not be connected to footing drains.

3.8. Earthwork

The site soils within the excavation depths anticipated for the project generally consist of clay with variable sand and gravel content. These soils are very moisture sensitive and susceptible to disturbance by construction equipment during wet weather. These soils are very difficult or impossible to use as structural

fill except during dry weather and on dry subgrades. If practical, the excavation to design subgrades should be performed during extended periods of dry weather. Exposed subgrades should not be left exposed to inclement weather. Earthwork costs will be significantly greater if performed during wet weather.

3.8.1. Subgrade Preparation

The exposed subgrade should be evaluated after the excavation and site grading is complete. GeoEngineers recommends that all subgrade surfaces be evaluated for suitability by a qualified geotechnical engineer or their representative. The structure subgrade should consist of undisturbed medium stiff to very stiff undifferentiated glacial drift. Proof-rolling with heavy, rubber-tired construction equipment should be used for this purpose during dry weather and if access for this equipment is practical. Probing should be used to evaluate the subgrade during periods of wet weather or if access is not feasible for construction equipment. Soft areas noted during proof-rolling or probing should be excavated and replaced with compacted structural fill.

A minimum 12-inch-thick working pad, consisting of 1½-inch minus clean crushed gravel with negligible sand or silt, may be necessary during wet weather construction to protect the site subgrade soils from disturbance. The working pad material should be placed in one lift over the subgrade and be compacted using a smooth-drum roller. The working pad may be left in-place to serve as base course material, provided it can be adequately compacted and does not become contaminated with fines during construction. We recommend keeping construction equipment off the exposed subgrades as much as possible during the wet season or when the subgrade is wet.

A 12-inch working pad will not provide suitable protection of the subgrade for access roads or areas of heavy equipment traffic. For these areas, we recommend a thicker gravel section and use of geotextile fabric placed over the subgrade to provide separation between the fine-grained subgrade and granular fill.

3.8.2. Excavation and Temporary Slopes

Excavation for the proposed site improvements will extend primarily through medium stiff to very stiff undifferentiated glacial drift. Excavation of the site soils can be completed using conventional earthwork equipment. Larger horsepower excavators will be more efficient for excavating the stiffer soils, or if large cobbles and boulders are encountered in the deeper excavations.

Perched shallow groundwater and isolated deeper seepage pockets or conditions could develop and cause water to run into excavations and hamper earthwork. We expect that sumps and pumps should be adequate to handle any groundwater seepage.

3.8.2.1. Temporary Cut Slopes

Regardless of the soil type encountered in the excavation, either shoring, trench boxes or sloped sidewalls will be required for excavations deeper than 4 feet under Washington State Administrative Code (WAC) 296-155, Part N. We expect that the excavations will be made as open cuts in conjunction with the use of a trench box and/or sloped sidewalls for shielding workers. Based on our explorations, the fill soils at the site would be classified as “Type C” and require slopes not exceeding 1.5H:1V (horizontal to vertical). The medium stiff undifferentiated glacial drift encountered at depth in B-1 at the site would be classified as “Type B” and require slopes not exceeding 1H:1V. The stiff to very stiff undifferentiated glacial drift encountered at the site would be classified as “Type A” and require slopes not exceeding 0.75H:1V.

The above regulations assume that surface loads such as construction equipment and storage loads will be kept a sufficient distance away from the top of the cut so that the stability of the excavation is not affected. Flatter slopes and/or shoring will be necessary for those portions of the excavations which are subjected to significant seepage in order to maintain the stability of the cut. It should be expected that unsupported cut slopes will experience some sloughing and raveling if exposed to surface water. Berms, hay bales or other provisions should be installed along the top of the excavation to intercept surface runoff to reduce the potential for sloughing and erosion of cut slopes during wet weather.

In our opinion, the contractor will be in the best position to observe subsurface conditions continuously throughout the construction process and to respond to the soil and groundwater conditions. Construction site safety is generally the sole responsibility of the contractor, who also is solely responsible for the means, methods, and sequencing of the construction operations and choices regarding temporary excavations and shoring. We are providing this information only as a service to our client. Under no circumstances should the information provided below be interpreted to mean that GeoEngineers, Inc. is assuming responsibility for construction site safety or the contractor's activities; such responsibility is not being implied and should not be inferred.

3.8.2.2. Temporary Shoring – Trench Boxes

We recommend that a shoring system be used where excavations are located adjacent to existing utilities, roadways or structures where soil movement or ground loss could result in damage to these facilities. Generally, a trench box is not considered appropriate in areas where soil movement adjacent to the trench is unacceptable. Typically, a trench box is used to protect workers from injury should the sidewall collapse. If a sidewall collapses, a contractor typically will backfill the void space in the trench box to the extent practical. Upon completion of the work the contractor will then pull the trench box out and the sidewalls cave and surface distress or disruption occurs.

A trench box can be used where sidewall support is required provided that it is (1) designed for anticipated earth pressures and hydrostatic pressures if appropriate and (2) the installation, moving and backfilling can be accomplished in such a manner that significant yielding does not occur. Braced or unbraced shoring (several types of adjustable braced trench shoring systems are available) of various types could be used where protection of existing infrastructure is necessary.

The following should be implemented for the shoring systems for the project:

- At locations where settlement could be detrimental to adjacent structures, utilities, or pavements, the shoring system should be designed to prevent significant lateral movement of the existing soils.
- Precautions should be taken during removal of the shoring or sheeting materials to minimize disturbance of the pipe, underlying bedding materials, and natural soils.
- Trench boxes should be of sufficient size, both vertically and horizontally, to support the excavation without excessive deformation of the natural soils.
- The open excavations should be backfilled as soon as practical after the shoring has been removed.
- Heavy construction equipment, construction materials, and excavated soil should not be allowed within a distance, measured from the edge of the excavation, equal to half the depth of the excavation, unless the shoring system has been designed for the additional lateral pressure.

Because of the diversity of available shoring systems and construction techniques, the design of temporary shoring is most appropriately left up to the contractor proposing to complete the installation. We recommend that the shoring be designed by an engineer licensed in Washington, and the PE stamped shoring plans and calculations be submitted to the engineer prior to construction.

3.8.3. Temporary Soldier Pile Walls

We understand that a temporary cantilever soldier pile wall is anticipated between the proposed and existing clarifiers to support existing infrastructure during construction. Soldier pile walls consist of steel beams that are concreted into drilled vertical holes located along the wall alignment, typically spaced about 6 to 8 feet on center. Timber lagging is typically installed behind the flanges of the steel beams to retain the soil located between the soldier piles. Alternatively, this can be achieved using heavy-gauge steel sheets. Geotechnical design recommendations for each of these components of the soldier pile and tieback wall system are presented in the following sections.

We recommend that soldier pile walls be designed using the earth pressure diagram presented in Figure 3. A seismic pressure has not been included in Figure 3 because it is a temporary wall.

We recommend that the embedded portion of the soldier piles be at least 2 feet in diameter and extend a minimum depth of 10 feet below the base of the excavation to resist “kick-out.” The axial capacity of the soldier piles must resist vertical loads, as appropriate. We recommend using an allowable end bearing value of 2.5 ksf. The allowable end bearing value should be applied to the base area of the drilled hole into which the soldier pile is concreted. This value includes a factor of safety of about 2.5. The allowable end bearing value assumes that the shaft bottom is cleaned out immediately prior to concrete placement.

Additional building or construction surcharge loads should be applied to the wall based on applicable scenarios presented in Figure 4.

3.8.3.1. Lagging

We recommend that the timber lagging be sized using the procedures outlined in the Federal Highway Administration’s Geotechnical Engineering Circular No. 4. The site retained soils are best described as competent soils. The following table presents recommend lagging thicknesses (roughcut) as a function of soldier pile clear span and depth.

TABLE 2. LAGGING THICKNESS

Depth (feet)	Recommended Lagging Thickness (roughcut) for Clear Spans					
	5 feet	6 feet	7 feet	8 feet	9 feet	10 feet
0 to 25	2 inches	3 inches	3 inches	3 inches	4 inches	4 inches

Lagging should be installed promptly after excavation, especially in areas where perched groundwater is present or where clean sand and gravel soils are present and caving soil conditions are likely. The workmanship associated with lagging installation is important for maintaining the integrity of the wall excavation.

The space behind the lagging should be filled with soil as soon as practicable. Material used as backfill in voids located behind the lagging should not cause buildup of hydrostatic pressure behind the wall. A suitable material would be select import fill as described below in section 3.8.4.2.

3.8.4. Structural Fill

Structural fill materials should be free of debris, organic contaminants and rocks or rock fragments larger than 6 inches or half the lift thickness, whichever is smaller. All fill placed beneath future structures or pavement should be placed as structural fill. Structural fill should be placed in horizontal lifts and uniformly compacted. The appropriate lift thickness will depend on the material and the compaction equipment being used. Loose lift thicknesses of 8 to 10 inches are typical when using heavy self-propelled vibratory equipment. All excavations should be wide enough to accommodate the appropriate compaction equipment for the thickness of the fill.

3.8.4.1. Suitability of On-site Soil

The onsite soils contain a significant fines content that will make them difficult to compact to specified densities during dry weather and unsuitable during wet weather. For planning purposes we recommend that on-site soils not be considered suitable as structural fill under pavement or structure areas and imported fill, conforming to the recommendations in Section 3.8.3.2 below, should be used as structural fill in these areas.

3.8.4.2. Select Import Fill

We recommend using a select import fill for earthwork on this project site. Unless other specific materials are recommended in this report, the select import should consist of well-graded sand and gravel, with at least 30 percent retained on the No. 4 sieve and less than 5 percent passing the U.S. No. 200 sieve. The percentage passing the No. 200 sieve should be based on that fraction passing the 3/4-inch sieve. Requiring use of this type of material below structures would significantly aid quality control and reduce compaction effort during earthwork procedures.

3.8.4.3. Fill Placement and Compaction Criteria

Structural fill should be mechanically compacted to a firm, non-yielding condition. Structural fill should be placed in lifts not exceeding 10 inches in loose thickness or that necessary to attain the specified compaction. Each lift should be conditioned to the proper moisture content and compacted to the specified density before placing subsequent lifts. Structural fill should be compacted to the following criteria:

- Structural fill placed in structure areas (supporting foundations or slab on grade floors) should be compacted to at least 95 percent of the MDD estimated in accordance with ASTM D-1557.
- Structural fill placed in pavement and hardscape areas should be compacted to at least 90 percent MDD where 2 feet below the pavement subgrade, and 95 percent MDD in the top 2 feet.
- Structural fill placed against retaining walls should be compacted to between 90 and 92 percent MDD. Care should be taken when compacting fill against subsurface walls to avoid over-compaction and overstressing the walls.
- Fill in non-structural areas should be compacted to at least 85 percent MDD to limit excessive postconstruction settlement.

Sufficient earthwork monitoring and a sufficient number of in-place density tests should be performed to evaluate fill placement and compaction operations and to confirm that the required compaction is being achieved.

3.8.5. Weather Considerations

During wet weather, the undifferentiated glacial drift will become muddy and trafficability may become very difficult with rubber tired equipment. We provide the following wet weather considerations:

- Construction activities should be scheduled so that the length of time that soils are left exposed to moisture is reduced to the extent practical and limit the size of areas that are stripped of topsoil or gravel surfacing and left exposed.
- The ground surface in and around the work area should be sloped so that surface water is directed to a sump or discharge location. The ground surface should be graded such that areas of ponded water do not develop.
- Slopes with exposed soils should be covered with plastic sheeting or similar means.
- Providing up-gradient perimeter ditches or low earthen berms and using temporary sumps to collect runoff and prevent water from ponding and damaging exposed subgrades.
- The site soils should not be left uncompacted and exposed to moisture. Sealing the surficial soils by rolling with a smooth-drum roller prior to periods of precipitation will reduce the extent to which these soils become wet or unstable.
- Limiting construction traffic over unprotected soil and by limiting the size and type of construction equipment used.
- Providing gravel “working mats” over areas of prepared subgrade.

3.9. Recommended Additional Geotechnical Services

GeoEngineers should be retained to review the project plans and specifications when complete to confirm that our design recommendations have been implemented as intended. During construction, GeoEngineers should, at minimum, evaluate the suitability of the foundation and slab-on-grade subgrades. The purposes of GeoEngineers construction phase services are to confirm that the subsurface conditions are consistent with those observed in the explorations and other reasons described in Appendix B, Report Limitations and Guidelines for Use.

4.0 LIMITATIONS

We have prepared this report for the exclusive use by Eastsound Sewer & Water District, Wilson Engineering, LLC, and their authorized agents for the proposed Eastsound Wastewater Treatment Plant Improvements project in Eastsound, Washington.

Within the limitations of scope, schedule and budget, our services have been executed in accordance with generally accepted practices in the field of geotechnical engineering in this area at the time this report was prepared. No warranty or other conditions, express or implied, should be understood.

Any electronic form, facsimile or hard copy of the original document (email, text, table, and/or figure), if provided, and any attachments are only a copy of the original document. The original document is stored by GeoEngineers, Inc. and will serve as the official document of record.

Please refer to Appendix B titled Report Limitations and Guidelines for Use for additional information pertaining to use of this report.

5.0 REFERENCES

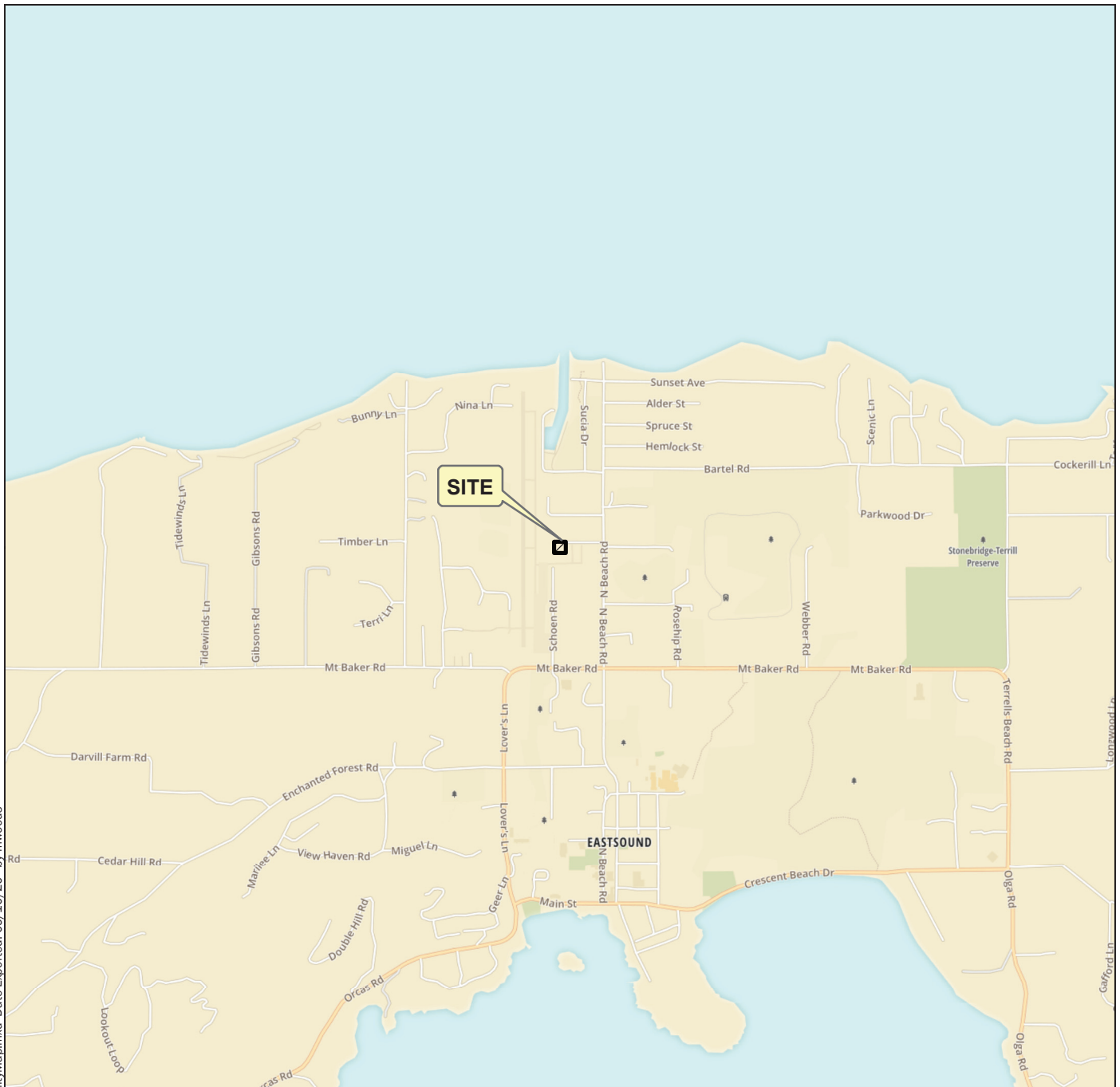
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Vicinity Map

Eastsound Wastewater Treatment Plant Improvements
Eastsound, Washington



Figure 1

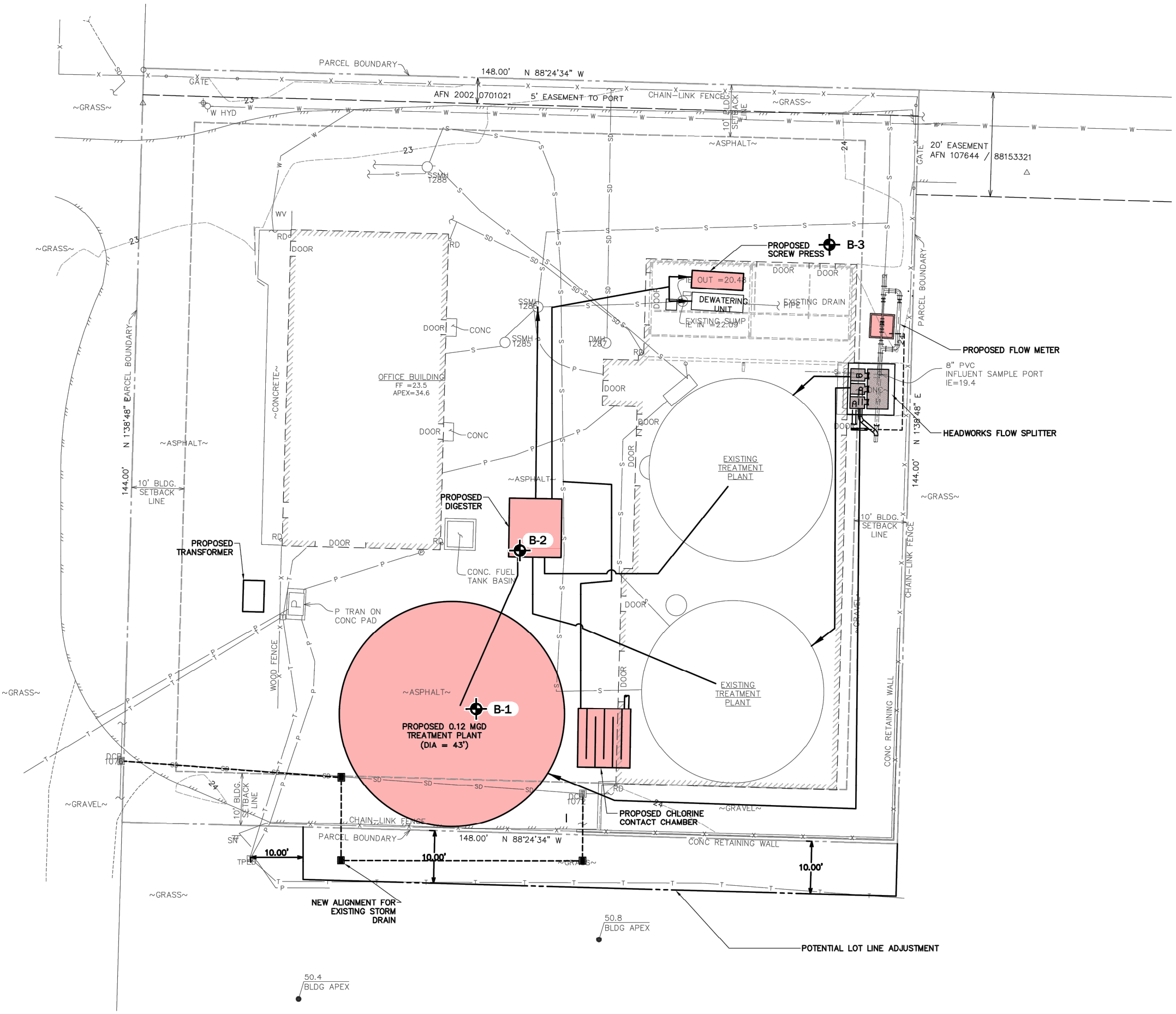
Notes:

1. The locations of all features shown are approximate.
2. This drawing is for information purposes. It is intended to assist in showing features discussed in an attached document. GeoEngineers, Inc. cannot guarantee the accuracy and content of electronic files. The master file is stored by GeoEngineers, Inc. and will serve as the official record of this communication.

Data Source: Mapbox Open Street Map, 2016

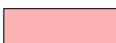
Projection: NAD 1983 UTM Zone 10N

P:\0407002\CAD\00\Geotech\040700200_F02_Site Plan.dwg TAB:F02 Date Exported: 09/22/20 - 10:40 by csticket



Legend

GEI-1  Boring by GeoEngineers, Inc., 2020

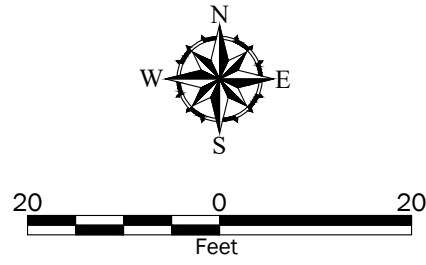
 Proposed Design Features

Notes:

1. The locations of all features shown are approximate.
2. This drawing is for information purposes. It is intended to assist in showing features discussed in an attached document. GeoEngineers, Inc. cannot guarantee the accuracy and content of electronic files. The master file is stored by GeoEngineers, Inc. and will serve as the official record of this communication.

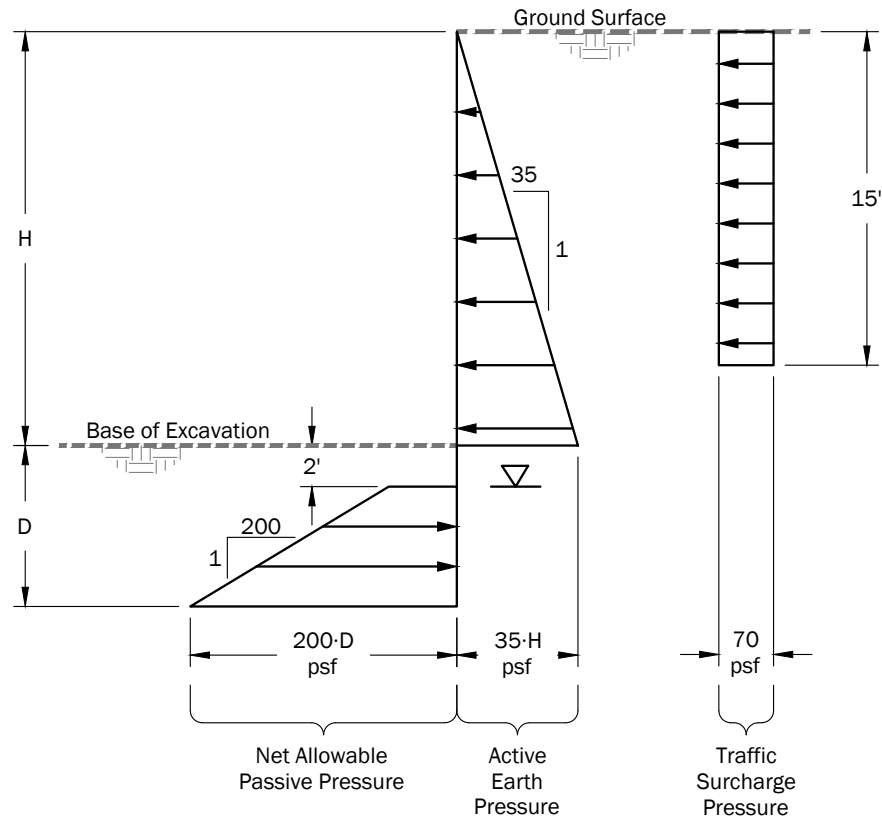
Data Source: Background from Wilson Engineering dated 7/14/2020.

Projection: NAD83 Washington State Planes, North Zone, US Foot



Site Plan	
Eastsound Wastewater Treatment Plant Improvements Eastsound, Washington	
	Figure 2

Cantilever Soldier Pile



Legend

H = Height of Excavation, Feet

D = Vertical Embedment Depth, Feet

▽ Design Groundwater Elevation for Drained Walls/ Passive Resistance Design

Not To Scale

Notes:

1. Apparent earth pressure and traffic surcharge pressure act over the pile spacing above the base of the excavation.
2. Passive earth pressure acts over 2.5 times the concreted diameter of the soldier pile, or the soldier pile spacing, whichever is less.
3. Passive pressure includes a factor of safety of 1.5
4. Additional surcharge from footings of adjacent buildings should be included in accordance with recommendations provided on Figure 4.
5. This pressure diagram is appropriate for temporary cantilever soldier pile walls. If additional surcharge loading (such as from soil stockpiles, excavators, dumptrucks, cranes, or concrete trucks) is anticipated, GeoEngineers should be consulted to provide revised surcharge pressures.

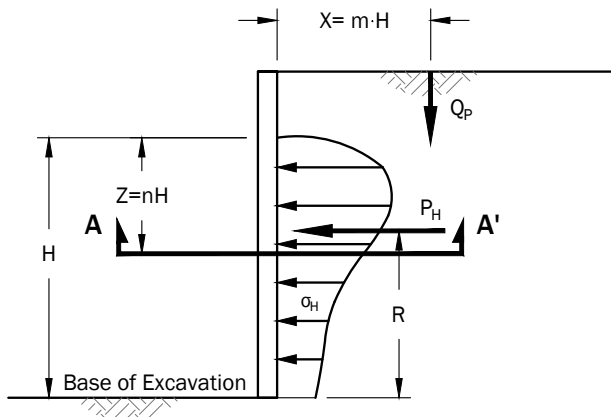
Earth Pressure Diagram Temporary Cantilever Soldier Pile Wall

Eastsound Wastewater Treatment Plant Improvements
Eastsound, Washington

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Figure 3

Lateral Earth Pressure from Point Load, Q_p
(Spread Footing)

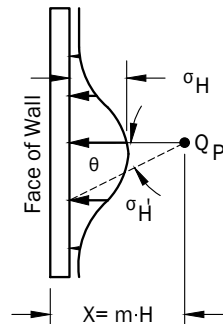


$$\text{For } m \leq 0.4$$
$$\sigma_H = \frac{0.28Q_p n^2}{H^2(0.16+n^2)^3}$$

$$\text{For } m > 0.4$$
$$\sigma_H = \frac{1.77Q_p m^2 n^2}{H^2(m^2+n^2)^3}$$

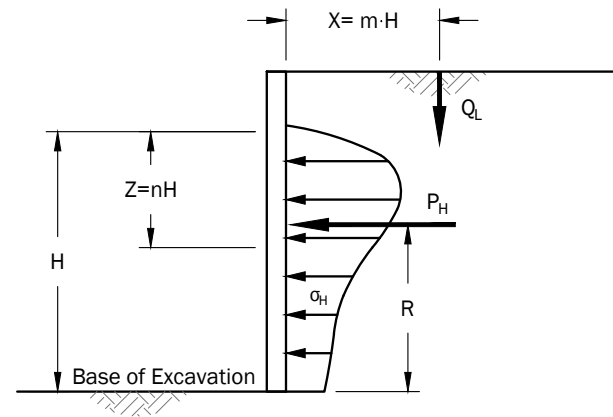
$$\sigma'_H = \sigma \cos^2(1.1\theta)$$

m	$P_H \left(\frac{H}{Q_p} \right)$	R
0.2	0.78	0.59H
0.4	0.78	0.59H
0.6	0.45	0.48H



Section A-A'
Pressures from Point Load Q_p

Lateral Earth Pressure from Line Load, Q_L
(Continuous Wall Footing)



$$\text{For } m \leq 0.4$$

$$\sigma_H = \frac{0.2n \cdot Q_L}{H(0.16+n^2)^2}$$

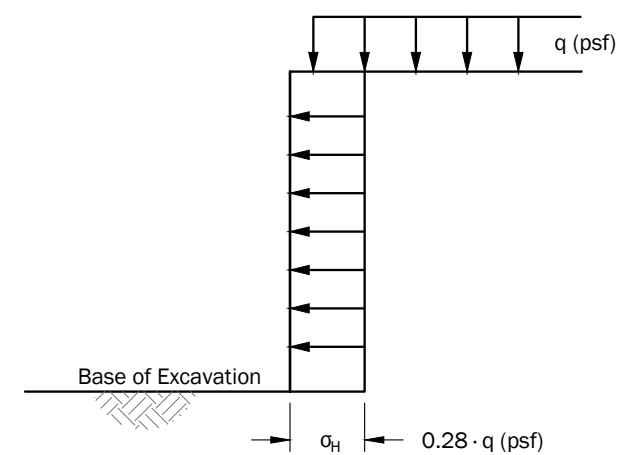
$$\text{For } m > 0.4$$

$$\sigma_H = \frac{1.28m^2 n Q_L}{H(m^2+n^2)^2}$$

$$\text{Resultant } P_H = \frac{0.64Q_L}{(m^2+1)}$$

m	R
0.1	0.60H
0.3	0.60H
0.5	0.56H
0.7	0.48H

Uniform Surcharges,
 q (Floor Loads, Large Foundation
Elements or Traffic Loads)



$$\sigma_H = \text{Lateral Surcharge Pressure from Uniform Surcharge}$$

Notes:

- Procedures for estimating surcharge pressures shown above are based on Manual 7.02 Naval Facilities Engineering Command, September 1986 (NAVFAC DM 7.02).
- Lateral earth pressures from surcharge should be added to earth pressures presented on Figure 3.
- See report text for where surcharge pressures are appropriate.

Definitions:

- Q_p = Point load in pounds
 Q_L = Line load in pounds/foot
 H = Excavation height below footing, feet
 σ_H = Lateral earth pressure from surcharge, psf
 q = Surcharge pressure in psf
 θ = Radians
 σ'_H = Distribution of σ_H in plan view
 P_H = Resultant lateral force acting on wall, pounds
 R = Distance from base of excavation to resultant lateral force, feet
 X = Resultant lateral force acting on wall, pounds
 Z = Depth of σ_H to be evaluated below the bottom of Q_p or Q_L
 m = Ratio of X to H
 n = Ratio of Z to H

Not To Scale

Recommended Surcharge Pressure

Eastsound Wastewater Treatment Plant Improvements
Eastsound, Washington



Figure 4

APPENDIX A

Field Explorations and Laboratory Testing

APPENDIX A

FIELD EXPLORATIONS AND LABORATORY TESTING

Field Explorations

Subsurface conditions at the site were explored by completing two geotechnical borings on August 10, 2020. The borings were completed to depths of 20 to 26½ feet below existing ground surface (bgs) using a track-mounted drill rig subcontracted to GeoEngineers. The approximate locations of the explorations are shown in the Site and Exploration Plan, Figure 2. The locations of the borings were determined using recreational grade global positioning system (GPS) and my measuring from existing site features; therefore, the locations shown in Figure 2 should be considered approximate.

Disturbed soil samples were obtained using Standard Penetration Test (SPT) methodology with the standard split-spoon sampler. The samples were obtained by driving the sampler 18 inches into the soil with a 140-pound hammer free-falling 30 inches using a rope and cathead. The number of blows required for each 6 inches of penetration was recorded. The blow count (“N-value”) of the soil was calculated as the number of blows required for the final 12 inches of penetration. This resistance, or N-value, provides a measure of the relative density of granular soils and the relative consistency of cohesive soils. The samples were placed in plastic bags to maintain the moisture content and transported back to our laboratory for analysis and testing.

The borings were continuously monitored by a geologist from our firm who examined and classified the soils encountered, obtained representative soil samples, observed groundwater conditions and prepared a detailed log of each exploration. Soils encountered were classified visually in general accordance with ASTM International (ASTM) D-2488-90, which is described in Figure A-1. An explanation of our boring log symbols is also shown on Figure A-1.

The logs of the borings are presented in Figures A-2 through A-4. The exploration logs are based on our interpretation of the field and laboratory data and indicate the various types of soils encountered. It also indicates the depths at which these soils or their characteristics change, although the change might actually be gradual. If the change occurred between samples in the boring, it was interpreted.

Observations of groundwater conditions were made during exploration. The groundwater conditions observed are presented in the logs. Groundwater conditions observed during exploration represent a short-term condition and may or may not be representative of the long-term groundwater conditions at the site. A groundwater piezometer well was installed in B-3. A description of the well is shown on the log of B-3.

Laboratory Test Results

Soil samples obtained from the explorations were transported to our laboratory and examined to confirm or modify field classifications, as well as to evaluate index properties of the soil samples. Representative samples were selected for laboratory testing consisting of the determination of the moisture content, and sieve analysis. The tests were performed in general accordance with test methods of ASTM or other applicable procedures.

Moisture Content Testing

The natural moisture contents of selected soil samples obtained from the exploratory borings were determined in general accordance with ASTM D 2216 test procedures. The results from the moisture content determinations are displayed shown on the exploration logs in Appendix A in the column labeled “Moisture Content %” adjacent to the corresponding samples.

Percent Passing U.S. No. 200 Sieve

Selected samples were “washed” through the U.S. No. 200 mesh sieve to determine the relative percentages of coarse- and fine-grained particles in the soil. The percent passing value represents the percentage by weight of the sample finer than the U.S. No. 200 sieve. These tests were conducted to verify field descriptions and to determine the fines content for analysis purposes. The tests were conducted in general accordance with ASTM D 1140, and the results are shown on the exploration logs in Appendix A in the column labeled “Fines Content %” adjacent to the corresponding samples.

Atterberg Limits Testing

Atterberg limits test was performed on a selected fine-grained soil sample. The test was used to classify the soils as well as to evaluate index properties. The liquid limit and the plastic limit were estimated through a procedure performed in general accordance with ASTM D 4318. The results of the Atterberg limits test is summarized in Figure A-5.

SOIL CLASSIFICATION CHART

MAJOR DIVISIONS			SYMBOLS		TYPICAL DESCRIPTIONS	
			GRAPH	LETTER		
COARSE GRAINED SOILS	GRAVEL AND GRAVELLY SOILS	CLEAN GRAVELS		GW	WELL-GRADED GRAVELS, GRAVEL - SAND MIXTURES	
		(LITTLE OR NO FINES)		GP	POORLY-GRADED GRAVELS, GRAVEL - SAND MIXTURES	
		GRAVELS WITH FINES		GM	SILTY GRAVELS, GRAVEL - SAND - SILT MIXTURES	
		(APPRECIABLE AMOUNT OF FINES)		GC	CLAYEY GRAVELS, GRAVEL - SAND - CLAY MIXTURES	
	MORE THAN 50% OF COARSE FRACTION RETAINED ON NO. 4 SIEVE	SAND AND SANDY SOILS	CLEAN SANDS		SW	WELL-GRADED SANDS, GRAVELLY SANDS
			(LITTLE OR NO FINES)		SP	POORLY-GRADED SANDS, GRAVELLY SAND
MORE THAN 50% RETAINED ON NO. 200 SIEVE	SANDS WITH FINES	(APPRECIABLE AMOUNT OF FINES)		SM	SILTY SANDS, SAND - SILT MIXTURES	
		(APPRECIABLE AMOUNT OF FINES)		SC	CLAYEY SANDS, SAND - CLAY MIXTURES	
FINE GRAINED SOILS	SILTS AND CLAYS	LIQUID LIMIT LESS THAN 50		ML	INORGANIC SILTS, ROCK FLOUR, CLAYEY SILTS WITH SLIGHT PLASTICITY	
				CL	INORGANIC CLAYS OF LOW TO MEDIUM PLASTICITY, GRAVELLY CLAYS, SANDY CLAYS, SILTY CLAYS, LEAN CLAYS	
				OL	ORGANIC SILTS AND ORGANIC SILTY CLAYS OF LOW PLASTICITY	
	MORE THAN 50% PASSING NO. 200 SIEVE	SILTS AND CLAYS	LIQUID LIMIT GREATER THAN 50		MH	INORGANIC SILTS, MICACEOUS OR DIATOMACEOUS SILTY SOILS
					CH	INORGANIC CLAYS OF HIGH PLASTICITY
					OH	ORGANIC CLAYS AND SILTS OF MEDIUM TO HIGH PLASTICITY
HIGHLY ORGANIC SOILS				PT	PEAT, HUMUS, SWAMP SOILS WITH HIGH ORGANIC CONTENTS	

NOTE: Multiple symbols are used to indicate borderline or dual soil classifications

Sampler Symbol Descriptions

	2.4-inch I.D. split barrel
	Standard Penetration Test (SPT)
	Shelby tube
	Piston
	Direct-Push
	Bulk or grab
	Continuous Coring

Blowcount is recorded for driven samplers as the number of blows required to advance sampler 12 inches (or distance noted). See exploration log for hammer weight and drop.

"P" indicates sampler pushed using the weight of the drill rig.

"WOH" indicates sampler pushed using the weight of the hammer.

NOTE: The reader must refer to the discussion in the report text and the logs of explorations for a proper understanding of subsurface conditions. Descriptions on the logs apply only at the specific exploration locations and at the time the explorations were made; they are not warranted to be representative of subsurface conditions at other locations or times.

ADDITIONAL MATERIAL SYMBOLS

SYMBOLS		TYPICAL DESCRIPTIONS
GRAPH	LETTER	
	AC	Asphalt Concrete
	CC	Cement Concrete
	CR	Crushed Rock/Quarry Spalls
	SOD	Sod/Forest Duff
	TS	Topsoil

Groundwater Contact



Measured groundwater level in exploration, well, or piezometer



Measured free product in well or piezometer

Graphic Log Contact



Distinct contact between soil strata



Approximate contact between soil strata

Material Description Contact



Contact between geologic units



Contact between soil of the same geologic unit

Laboratory / Field Tests

%F	Percent fines
%G	Percent gravel
AL	Atterberg limits
CA	Chemical analysis
CP	Laboratory compaction test
CS	Consolidation test
DD	Dry density
DS	Direct shear
HA	Hydrometer analysis
MC	Moisture content
MD	Moisture content and dry density
Mohs	Mohs hardness scale
OC	Organic content
PM	Permeability or hydraulic conductivity
PI	Plasticity index
PL	Point load test
PP	Pocket penetrometer
SA	Sieve analysis
TX	Triaxial compression
UC	Unconfined compression
VS	Vane shear

Sheen Classification

NS	No Visible Sheen
SS	Slight Sheen
MS	Moderate Sheen
HS	Heavy Sheen

Key to Exploration Logs



Figure A-1

Date:9/21/20 Path:\\GEOENGINEERS.COM\\WAN\\PROJECTS\\0407002\\GINT\\040700200.GPJ DBLibrary\\Library\\GEOENGINEERS_DF STD_US_JUNE_2017.GLB\\GEI8 GEOTECH_STANDARD_%F_NO_GW

Note: See Figure A-1 for explanation of symbols.
Coordinates Data Source: Horizontal approximated based on Topographic Survey. Vertical approximated based on Topographic Survey.

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Figure A-2
Sheet 1 of 1

Drilled	Start 8/10/2020	End 8/10/2020	Total Depth (ft)	26.5	Logged By Checked By	NBD AJH	Driller	Borettec1, Inc.	Drilling Method	Hollow-stem Auger
Surface Elevation (ft) Vertical Datum	23 NAVD88				Hammer Data	Rope & Cathead 140 (lbs) / 30 (in) Drop			Drilling Equipment	EC Track 95
Easting (X) Northing (Y)	1139360 629537				System Datum	WA State Plane North NAD83			See "Remarks" section for groundwater observed	
Notes:										

Elevation (feet)	FIELD DATA				Graphic Log	Group Classification	MATERIAL DESCRIPTION	Moisture Content (%)	Fines Content (%)	REMARKS
	Depth (feet)	Interval Recovered (in)	Blows/foot	Collected Sample						
0						AC	Approximately 3 inches of asphalt concrete pavement			
						SM	Light brown silty fine to medium sand with gravel (dense, moist) (fill)	2	18	
5		3	16			GP	Gray gravel with silt and sand (dense, moist)			
		18	14			CL	Gray clay with trace sand and occasional gravel (stiff, moist) (undifferentiated glacial drift)	19		
10		18	15					19		
		18	9					21		AL (LL = 37, PI = 17)
15		18	12					26		
20		18	12			CL	Gray clay with fine sand and occasional gravel (stiff, moist to wet)	17		
25		0	73			SM	Gray silty fine to medium sand with occasional gravel (very dense, wet) (glacial till)	12	39	Groundwater observed at approximately 25 feet during drilling

Note: See Figure A-1 for explanation of symbols.

Coordinates Data Source: Horizontal approximated based on Topographic Survey. Vertical approximated based on Topographic Survey.

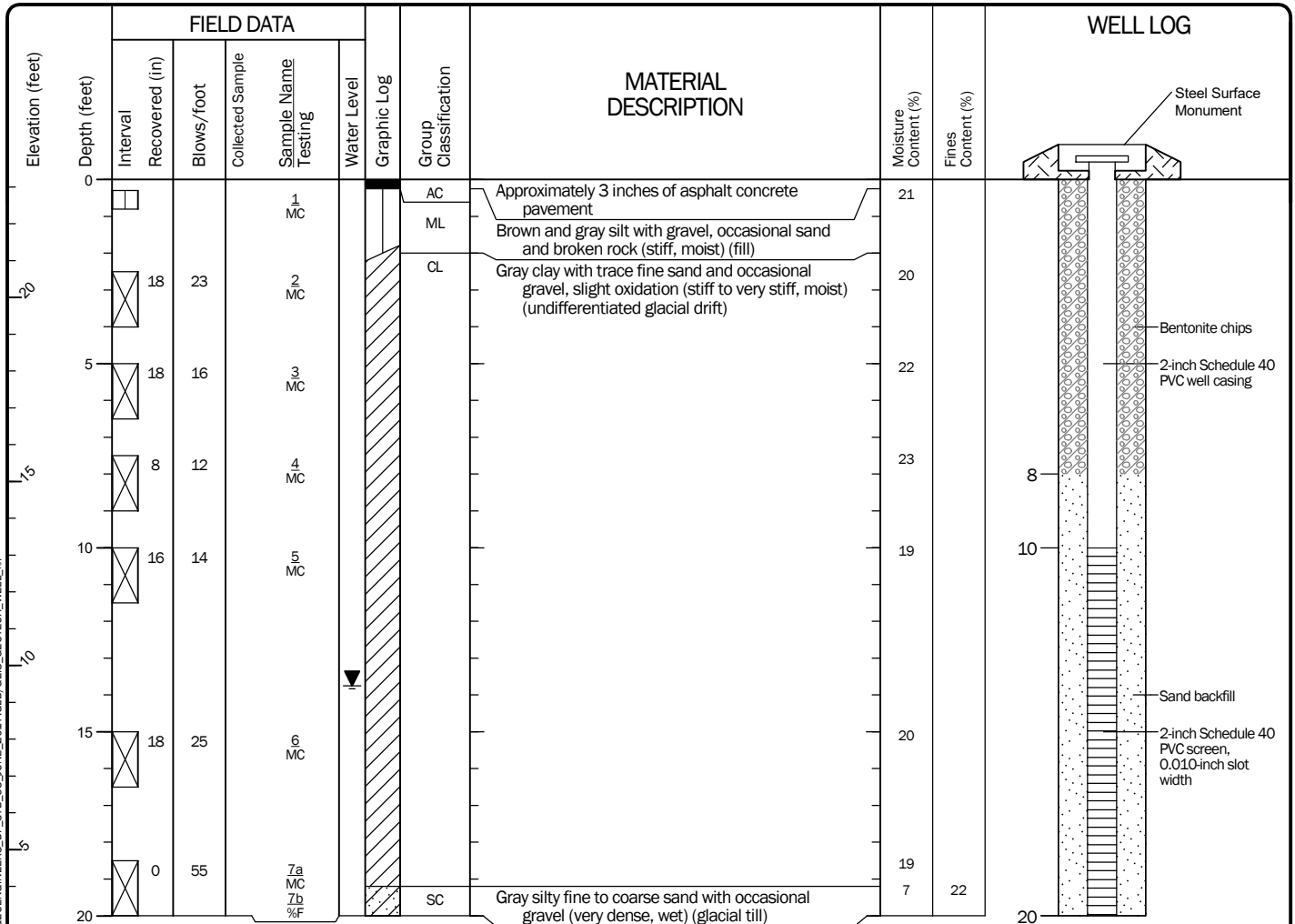
Log of Boring B-2



Project: Eastsound Wastewater Treatment Plant Improvements
Project Location: Eastsound, Washington
Project Number: 0407-002-00

Figure A-3
Sheet 1 of 1

Start Drilled 8/10/2020	End 8/10/2020	Total Depth (ft) 20	Logged By Checked By NBD AJH	Driller Borettec1, Inc.	Drilling Method Hollow-stem Auger
Hammer Data	Rope & Cathead 140 (lbs) / 30 (in) Drop		Drilling Equipment EC Track 95	DOE Well I.D.: BKH 369 A 2-in well was installed on 8/10/2020 to a depth of 20 ft.	
Surface Elevation (ft) Vertical Datum	23.2 NAVD88		Top of Casing Elevation (ft) 22.67	<u>Groundwater</u> <u>Date Measured</u> 8/24/2020	
Easting (X) Northing (Y)	1139419 629595		Horizontal Datum WA State Plane North NAD83	<u>Depth to Water (ft)</u> 13.75	<u>Elevation (ft)</u> 9.45
Notes:					



Note: See Figure A-1 for explanation of symbols.
Coordinates Data Source: Horizontal approximated based on Topographic Survey. Vertical approximated based on Topographic Survey.

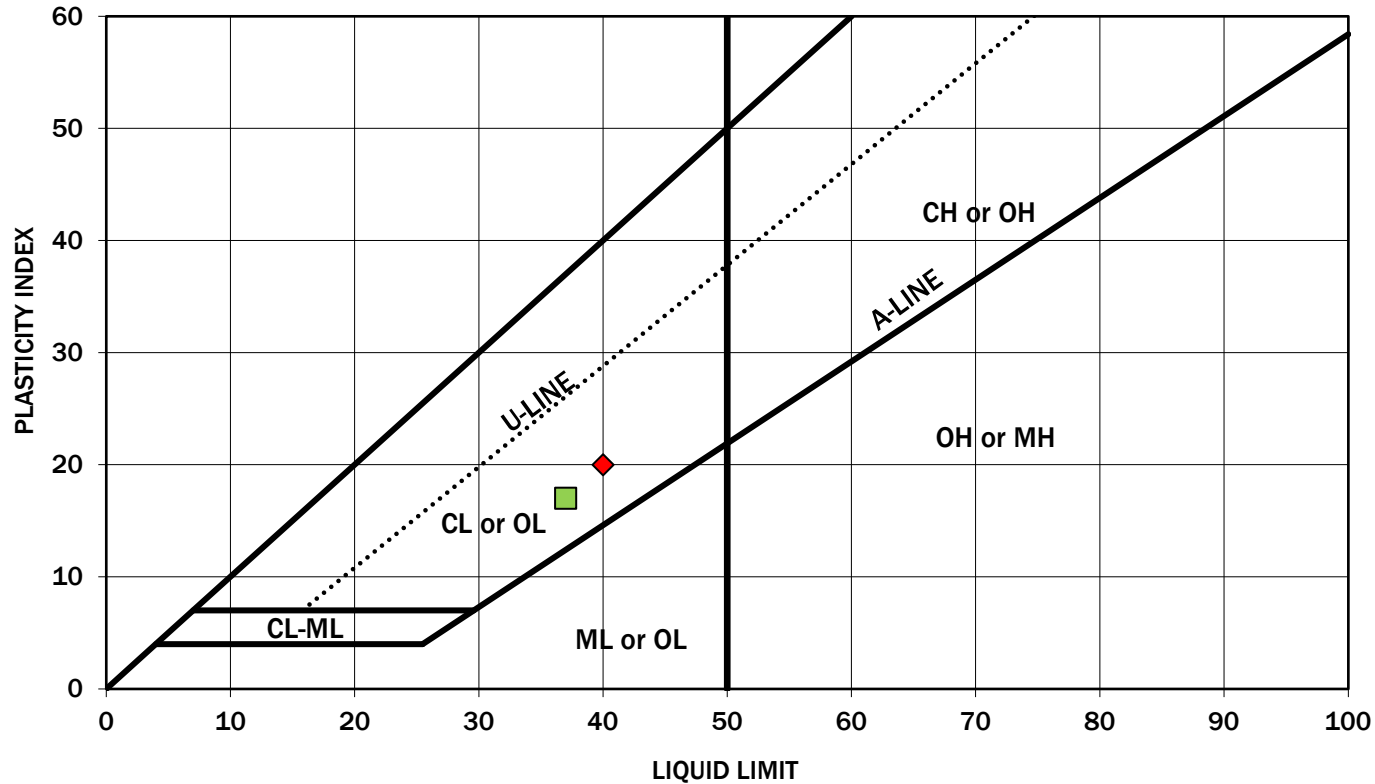
Log of Boring with Monitoring Well B-3



Project: Eastsound Wastewater Treatment Plant Improvements
Project Location: Eastsound, Washington
Project Number: 0407-002-00

Figure A-4
Sheet 1 of 1

PLASTICITY CHART



Symbol	Boring Number	Depth (feet)	Moisture Content (%)	Liquid Limit (%)	Plasticity Index (%)	Soil Description
◆	B-1	15	26	40	20	Gray clay (CL)
■	B-2	10	21	37	17	Gray clay (CL)

Note: This report may not be reproduced, except in full, without written approval of GeoEngineers, Inc. Test results are applicable only to the specific sample on which they were performed, and should not be interpreted as representative of any other samples obtained at other times, depths or locations, or generated by separate operations or processes. The liquid limit and plasticity index were obtained in general accordance with ASTM D 4318. GeoEngineers 17425 NE Union Hill Road Ste 250, Redmond, WA 98052

Atterberg Limits Test Results

Eastsound WWTP Improvements
Eastsound, Washington



Figure A-5

APPENDIX B

Report Limitations and Guidelines for Use

APPENDIX B

REPORT LIMITATIONS AND GUIDELINES FOR USE¹

This appendix provides information to help you manage your risks with respect to the use of this report.

Geotechnical Services are Performed for Specific Purposes, Persons and Projects

This report has been prepared for the exclusive use of the Eastsound Water & Sewer District, Wilson Engineering, LLC., and their authorized agents. This report may be made available to other members of the design team. This report is not intended for use by others, and the information contained herein is not applicable to other sites.

GeoEngineers structures our services to meet the specific needs of our clients. For example, a geotechnical or geologic study conducted for a civil engineer or architect may not fulfill the needs of a construction contractor or even another civil engineer or architect that are involved in the same project. Because each geotechnical or geologic study is unique, each geotechnical engineering or geologic report is unique, prepared solely for the specific client and project site. Our report is prepared for the exclusive use of our Client. No other party may rely on the product of our services unless we agree in advance to such reliance in writing. This is to provide our firm with reasonable protection against open-ended liability claims by third parties with whom there would otherwise be no contractual limits to their actions. Within the limitations of scope, schedule and budget, our services have been executed in accordance with our Agreement with the Client and generally accepted geotechnical practices in this area at the time this report was prepared. This report should not be applied for any purpose or project except the one originally contemplated.

A Geotechnical Engineering or Geologic Report is Based on a Unique Set of Project-specific Factors

This report has been prepared for the proposed Eastsound Wastewater Treatment Plant Improvements project in Eastsound, Washington. GeoEngineers considered a number of unique, project-specific factors when establishing the scope of services for this project and report. Unless GeoEngineers specifically indicates otherwise, do not rely on this report if it was:

- Not prepared for you,
- Not prepared for your project,
- Not prepared for the specific site explored, or
- Completed before important project changes were made.

For example, changes that can affect the applicability of this report include those that affect:

- The function of the proposed structure;
- Elevation, configuration, location, orientation or weight of the proposed structure;
- Composition of the design team; or
- Project ownership.

¹ Developed based on material provided by ASFE, Professional Firms Practicing in the Geosciences; www.asfe.org.

If important changes are made after the date of this report, GeoEngineers should be given the opportunity to review our interpretations and recommendations and provide written modifications or confirmation, as appropriate.

Subsurface Conditions Can Change

This geotechnical or geologic report is based on conditions that existed at the time the study was performed. The findings and conclusions of this report may be affected by the passage of time, by manmade events such as construction on or adjacent to the site, or by natural events such as floods, earthquakes, slope instability or groundwater fluctuations. Always contact GeoEngineers before applying a report to determine if it remains applicable.

Most Geotechnical and Geologic Findings are Professional Opinions

Our interpretations of subsurface conditions are based on field observations from widely spaced sampling locations at the site. Site exploration identifies subsurface conditions only at those points where subsurface tests are conducted or samples are taken. GeoEngineers reviewed field and laboratory data and then applied our professional judgment to render an opinion about subsurface conditions throughout the site. Actual subsurface conditions may differ, sometimes significantly, from those indicated in this report. Our report, conclusions and interpretations should not be construed as a warranty of the subsurface conditions.

Geotechnical Engineering Report Recommendations are Not Final

Do not over-rely on the preliminary construction recommendations included in this report. These recommendations are not final, because they were developed principally from GeoEngineers' professional judgment and opinion. GeoEngineers' recommendations can be finalized only by observing actual subsurface conditions revealed during construction. GeoEngineers cannot assume responsibility or liability for this report's recommendations if we do not perform construction observation.

Sufficient monitoring, testing and consultation by GeoEngineers should be provided during construction to confirm that the conditions encountered are consistent with those indicated by the explorations, to provide recommendations for design changes should the conditions revealed during the work differ from those anticipated, and to evaluate whether or not earthwork activities are completed in accordance with our recommendations. Retaining GeoEngineers for construction observation for this project is the most effective method of managing the risks associated with unanticipated conditions.

A Geotechnical Engineering or Geologic Report Could be Subject to Misinterpretation

Misinterpretation of this report by other design team members can result in costly problems. You could lower that risk by having GeoEngineers confer with appropriate members of the design team after submitting the report. Also retain GeoEngineers to review pertinent elements of the design team's plans and specifications. Contractors can also misinterpret a geotechnical engineering or geologic report. Reduce that risk by having GeoEngineers participate in pre-bid and preconstruction conferences, and by providing construction observation.

Do Not Redraw the Exploration Logs

Geotechnical engineers and geologists prepare final boring and testing logs based upon their interpretation of field logs and laboratory data. To prevent errors or omissions, the logs included in a geotechnical engineering or geologic report should never be redrawn for inclusion in architectural or other design

drawings. Only photographic or electronic reproduction is acceptable, but recognize that separating logs from the report can elevate risk.

Give Contractors a Complete Report and Guidance

Some owners and design professionals believe they can make contractors liable for unanticipated subsurface conditions by limiting what they provide for bid preparation. To help prevent costly problems, give contractors the complete geotechnical engineering or geologic report, but preface it with a clearly written letter of transmittal. In that letter, advise contractors that the report was not prepared for purposes of bid development and that the report's accuracy is limited; encourage them to confer with GeoEngineers and/or to conduct additional study to obtain the specific types of information they need or prefer. A pre-bid conference can also be valuable. Be sure contractors have sufficient time to perform additional study. Only then might an owner be in a position to give contractors the best information available, while requiring them to at least share the financial responsibilities stemming from unanticipated conditions. Further, a contingency for unanticipated conditions should be included in your project budget and schedule.

Contractors are Responsible for Site Safety on Their Own Construction Projects

Our geotechnical recommendations are not intended to direct the contractor's procedures, methods, schedule or management of the work site. The contractor is solely responsible for job site safety and for managing construction operations to minimize risks to on-site personnel and to adjacent properties.

Read These Provisions Closely

Some clients, design professionals and contractors may not recognize that the geoscience practices (geotechnical engineering or geology) are far less exact than other engineering and natural science disciplines. This lack of understanding can create unrealistic expectations that could lead to disappointments, claims and disputes. GeoEngineers includes these explanatory "limitations" provisions in our reports to help reduce such risks. Please confer with GeoEngineers if you are unclear how these "Report Limitations and Guidelines for Use" apply to your project or site.

Geotechnical, Geologic and Environmental Reports Should Not be Interchanged

The equipment, techniques and personnel used to perform an environmental study differ significantly from those used to perform a geotechnical or geologic study and vice versa. For that reason, a geotechnical engineering or geologic report does not usually relate any environmental findings, conclusions or recommendations; e.g., about the likelihood of encountering underground storage tanks or regulated contaminants. Similarly, environmental reports are not used to address geotechnical or geologic concerns regarding a specific project.

Topsoil

For the purposes of this report, we consider topsoil to consist of generally fine-grained soil with an appreciable amount of organic matter based on visual examination, and to be unsuitable for direct support of the proposed improvements. However, the organic content and other mineralogical and gradational characteristics used to evaluate the suitability of soil for use in landscaping and agricultural purposes was not determined, nor considered in our analyses. Therefore, the information and recommendations in this report, and our logs and descriptions should not be used as a basis for estimating the volume of topsoil available for such purposes.

Biological Pollutants

GeoEngineers' Scope of Work specifically excludes the investigation, detection, prevention or assessment of the presence of biological pollutants. Accordingly, this report does not include any interpretations, recommendations, findings, or conclusions regarding the detecting, assessing, preventing or abating of biological pollutants and no conclusions or inferences should be drawn regarding biological pollutants, as they may relate to this project. The term "biological pollutants" includes, but is not limited to, molds, fungi, spores, bacteria, and viruses, and/or any of their byproducts.

If Client desires these specialized services, they should be obtained from a consultant who offers services in this specialized field.

December 16, 2022

Wilson Engineering, LLC
805 Dupont Street, Suite 7
Bellingham, Washington 98225

Attention: Jeff Christner, PE

Subject: Report Addendum
Geotechnical Engineering Services
Eastsound Wastewater Treatment Plant
Eastsound, Washington
File No. 0407-002-00

INTRODUCTION

GeoEngineers, Inc. (GeoEngineers) presents this report addendum to our “Geotechnical Engineering Services, Eastsound Wastewater Treatment Plan Improvements, Eastsound, Washington” report dated November 8, 2022. We understand that some of the pavement areas are being replaced. The purpose of the addendum is to provide pavement thickness and subgrade preparation recommendations.

Pavement Recommendations

We do not have specific design parameters (traffic data) to develop pavement sections. The owner can choose the pavement section based on the available capital budget for the project and desired pavement performance. We recommend that any site grading be accomplished with gravel base. The existing gravel base or exposed subgrade should be proofrolled, and any unsuitable or yielding soils should be removed or remediated prior to placing gravel base. We recommend that the pavement section consist of gravel base (existing sand and gravel fill, minus organics, or clean import meets this purpose), a layer of crushed rock, and hot mix asphalt (HMA) pavement. The pavement materials should be in conformance with the most recent version of the Washington State Department of Transportation (WSDOT) Standard Specifications for Bridge and Municipal Construction (hereinafter referred to as the WSDOT Standard Specifications). We suggest the following typical pavement sections for commercial sites:

Automobile Parking Areas

- 2½ to 3 inches of HMA, Class ½-inch, PG 58H-22 per the WSDOT Standard Specifications 5-04 and 9-03.8
- 3 to 4 inches of crushed surfacing base course (CSBC) WSDOT Standard Specifications 9-03.9(3)
- 8 to 12 inches of gravel base, as needed based on thickness and/or proof roll of existing fill



Access Roads, Truck Routes, and Loading Areas

- 3 to 4 inches of HMA
- 4 to 6 inches of CSBC
- 12 to 18 inches of gravel base, as needed based on thickness and/or proof roll of existing fill

LIMITATIONS

This report addendum is subject to the same limitations provided in our "Geotechnical Engineering Services, Eastsound Wastewater Treatment Plan Improvements, Eastsound, Washington" report dated November 8, 2022.

Please contact Sean Cool or Aaron Hartvigsen at 360.647.1510 if you have any questions or need to discuss the contents of this report addendum.

Sincerely,
GeoEngineers, Inc.



Aaron J. Hartvigsen, PE
Senior Geotechnical Engineer

AJH:SWC:leh

One copy submitted

Disclaimer: Any electronic form, facsimile or hard copy of the original document (email, text, table, and/or figure), if provided, and any attachments are only a copy of the original document. The original document is stored by GeoEngineers, Inc. and will serve as the official document of record.



Sean W. Cool, PE
Associate

APPENDIX D – WASHINGTON STATE PREVAILING WAGE RATES

State of Washington
Department of Labor & Industries
Prevailing Wage Section - Telephone 360-902-5335
PO Box 44540, Olympia, WA 98504-4540

Washington State Prevailing Wage

The PREVAILING WAGES listed here include both the hourly wage rate and the hourly rate of fringe benefits. On public works projects, worker's wage and benefit rates must add to not less than this total. A brief description of overtime calculation requirements are provided on the Benefit Code Key.

Journey Level Prevailing Wage Rates for the Effective Date: 1/4/2023

<u>County</u>	<u>Trade</u>	<u>Job Classification</u>	<u>Wage</u>	<u>Holiday</u>	<u>Overtime</u>	<u>Note</u>	<u>*Risk Class</u>
San Juan	Asbestos Abatement Workers	Journey Level	\$56.80	<u>5D</u>	<u>1H</u>		View
San Juan	Boilermakers	Journey Level	\$72.54	<u>5N</u>	<u>1C</u>		View
San Juan	Brick Mason	Journey Level	\$66.32	<u>7E</u>	<u>1N</u>		View
San Juan	Brick Mason	Pointer-Caulker-Cleaner	\$66.32	<u>7E</u>	<u>1N</u>		View
San Juan	Building Service Employees	Janitor	\$15.74		<u>1</u>		View
San Juan	Building Service Employees	Shampooer	\$15.74		<u>1</u>		View
San Juan	Building Service Employees	Waxer	\$15.74		<u>1</u>		View
San Juan	Building Service Employees	Window Cleaner	\$15.74		<u>1</u>		View
San Juan	Cabinet Makers (In Shop)	Journey Level	\$40.05		<u>1</u>		View
San Juan	Carpenters	Acoustical Worker	\$71.53	<u>15J</u>	<u>4C</u>		View
San Juan	Carpenters	Bridge, Dock And Wharf Carpenters	\$71.53	<u>15J</u>	<u>4C</u>		View
San Juan	Carpenters	Floor Layer & Floor Finisher	\$71.53	<u>15J</u>	<u>4C</u>		View
San Juan	Carpenters	Journey Level	\$71.53	<u>15J</u>	<u>4C</u>		View
San Juan	Carpenters	Scaffold Erector	\$71.53	<u>15J</u>	<u>4C</u>		View
San Juan	Cement Masons	Application of all Composition Mastic	\$70.09	<u>15J</u>	<u>4U</u>		View
San Juan	Cement Masons	Application of all Epoxy Material	\$69.59	<u>15J</u>	<u>4U</u>		View
San Juan	Cement Masons	Application of all Plastic Material	\$70.09	<u>15J</u>	<u>4U</u>		View

San Juan	Cement Masons	Application of Sealing Compound	\$69.59	15J	4U		View
San Juan	Cement Masons	Application of Underlayment	\$70.09	15J	4U		View
San Juan	Cement Masons	Building General	\$69.59	15J	4U		View
San Juan	Cement Masons	Composition or Kalman Floors	\$70.09	15J	4U		View
San Juan	Cement Masons	Concrete Paving	\$69.59	15J	4U		View
San Juan	Cement Masons	Curb & Gutter Machine	\$70.09	15J	4U		View
San Juan	Cement Masons	Curb & Gutter, Sidewalks	\$69.59	15J	4U		View
San Juan	Cement Masons	Curing Concrete	\$69.59	15J	4U		View
San Juan	Cement Masons	Finish Colored Concrete	\$70.09	15J	4U		View
San Juan	Cement Masons	Floor Grinding	\$70.09	15J	4U		View
San Juan	Cement Masons	Floor Grinding/Polisher	\$69.59	15J	4U		View
San Juan	Cement Masons	Green Concrete Saw, self-powered	\$70.09	15J	4U		View
San Juan	Cement Masons	Grouting of all Plates	\$69.59	15J	4U		View
San Juan	Cement Masons	Grouting of all Tilt-up Panels	\$69.59	15J	4U		View
San Juan	Cement Masons	Guniting Nozzleman	\$70.09	15J	4U		View
San Juan	Cement Masons	Hand Powered Grinder	\$70.09	15J	4U		View
San Juan	Cement Masons	Journey Level	\$69.59	15J	4U		View
San Juan	Cement Masons	Patching Concrete	\$69.59	15J	4U		View
San Juan	Cement Masons	Pneumatic Power Tools	\$70.09	15J	4U		View
San Juan	Cement Masons	Power Chipping & Brushing	\$70.09	15J	4U		View
San Juan	Cement Masons	Sand Blasting Architectural Finish	\$70.09	15J	4U		View
San Juan	Cement Masons	Screed & Rodding Machine	\$70.09	15J	4U		View
San Juan	Cement Masons	Spackling or Skim Coat Concrete	\$69.59	15J	4U		View
San Juan	Cement Masons	Troweling Machine Operator	\$70.09	15J	4U		View
San Juan	Cement Masons	Troweling Machine Operator on Colored Slabs	\$70.09	15J	4U		View
San Juan	Cement Masons	Tunnel Workers	\$70.09	15J	4U		View

San Juan	Divers & Tenders	Bell/Vehicle or Submersible Operator (Not Under Pressure)	\$126.05	15J	4C		View
San Juan	Divers & Tenders	Dive Supervisor/Master	\$89.94	15J	4C		View
San Juan	Divers & Tenders	Diver	\$126.05	15J	4C	8V	View
San Juan	Divers & Tenders	Diver On Standby	\$84.94	15J	4C		View
San Juan	Divers & Tenders	Diver Tender	\$77.16	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 0-30.00 PSI	\$89.09	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 30.01 - 44.00 PSI	\$94.09	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 44.01 - 54.00 PSI	\$107.09	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 54.01 - 60.00 PSI	\$103.09	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 60.01 - 64.00 PSI	\$105.59	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 64.01 - 68.00 PSI	\$110.59	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 68.01 - 70.00 PSI	\$112.59	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 70.01 - 72.00 PSI	\$114.59	15J	4C		View
San Juan	Divers & Tenders	Hyperbaric Worker - Compressed Air Worker 72.01 - 74.00 PSI	\$116.59	15J	4C		View
San Juan	Divers & Tenders	Manifold Operator	\$77.16	15J	4C		View
San Juan	Divers & Tenders	Manifold Operator Mixed Gas	\$82.16	15J	4C		View
San Juan	Divers & Tenders	Remote Operated Vehicle Operator/Technician	\$77.16	15J	4C		View
San Juan	Divers & Tenders	Remote Operated Vehicle Tender	\$71.98	15J	4C		View
San Juan	Dredge Workers	Assistant Engineer	\$76.56	5D	3F		View
San Juan	Dredge Workers	Assistant Mate (Deckhand)	\$75.97	5D	3F		View
San Juan	Dredge Workers	Boatmen	\$76.56	5D	3F		View
San Juan	Dredge Workers	Engineer Welder	\$78.03	5D	3F		View

San Juan	Dredge Workers	Leverman, Hydraulic	\$79.59	5D	3F		View
San Juan	Dredge Workers	Mates	\$76.56	5D	3F		View
San Juan	Dredge Workers	Oiler	\$75.97	5D	3F		View
San Juan	Drywall Applicator	Journey Level	\$71.53	15J	4C		View
San Juan	Drywall Tapers	Journey Level	\$70.61	5P	1E		View
San Juan	Electrical Fixture Maintenance Workers	Journey Level	\$15.74		1		View
San Juan	Electricians - Inside	Cable Splicer	\$86.71	7H	1E		View
San Juan	Electricians - Inside	Construction Stock Person	\$41.31	7H	1D		View
San Juan	Electricians - Inside	Journey Level	\$81.23	7H	1E		View
San Juan	Electricians - Motor Shop	Craftsman	\$15.74		1		View
San Juan	Electricians - Motor Shop	Journey Level	\$15.74		1		View
San Juan	Electricians - Powerline Construction	Cable Splicer	\$88.89	5A	4D		View
San Juan	Electricians - Powerline Construction	Certified Line Welder	\$81.65	5A	4D		View
San Juan	Electricians - Powerline Construction	Groundperson	\$52.91	5A	4D		View
San Juan	Electricians - Powerline Construction	Heavy Line Equipment Operator	\$81.65	5A	4D		View
San Juan	Electricians - Powerline Construction	Journey Level Lineperson	\$81.65	5A	4D		View
San Juan	Electricians - Powerline Construction	Line Equipment Operator	\$70.02	5A	4D		View
San Juan	Electricians - Powerline Construction	Meter Installer	\$52.91	5A	4D	8W	View
San Juan	Electricians - Powerline Construction	Pole Sprayer	\$81.65	5A	4D		View
San Juan	Electricians - Powerline Construction	Powderperson	\$60.75	5A	4D		View
San Juan	Electronic Technicians	Electronic Technicians Journey Level	\$51.68	5B	1B		View
San Juan	Elevator Constructors	Mechanic	\$103.81	7D	4A		View
San Juan	Elevator Constructors	Mechanic In Charge	\$112.09	7D	4A		View
San Juan	Fabricated Precast Concrete Products	Journey Level	\$15.74		1		View
San Juan	Fabricated Precast Concrete Products	Journey Level - In-Factory Work Only	\$15.74		1		View
San Juan	Fence Erectors	Fence Erector	\$48.14	15J	4V	8Y	View

San Juan	Fence Erectors	Fence Laborer	\$48.14	15J	4V	8Y	View
San Juan	Flaggers	Journey Level	\$48.14	15J	4V	8Y	View
San Juan	Glaziers	Journey Level	\$75.91	7L	1Y		View
San Juan	Heat & Frost Insulators And Asbestos Workers	Journey Level	\$84.58	15H	11C		View
San Juan	Heating Equipment Mechanics	Mechanic	\$88.45	7F	1E		View
San Juan	Hod Carriers & Mason Tenders	Journey Level	\$59.85	15J	4V	8Y	View
San Juan	Industrial Power Vacuum Cleaner	Journey Level	\$15.74		1		View
San Juan	Inland Boatmen	Boat Operator	\$61.41	5B	1K		View
San Juan	Inland Boatmen	Cook	\$56.48	5B	1K		View
San Juan	Inland Boatmen	Deckhand	\$57.48	5B	1K		View
San Juan	Inland Boatmen	Deckhand Engineer	\$58.81	5B	1K		View
San Juan	Inland Boatmen	Launch Operator	\$58.89	5B	1K		View
San Juan	Inland Boatmen	Mate	\$57.31	5B	1K		View
San Juan	Inspection/Cleaning/Sealing Of Sewer & Water Systems By Remote Control	Cleaner Operator, Foamer Operator	\$15.74		1		View
San Juan	Inspection/Cleaning/Sealing Of Sewer & Water Systems By Remote Control	Grout Truck Operator	\$15.74		1		View
San Juan	Inspection/Cleaning/Sealing Of Sewer & Water Systems By Remote Control	Head Operator	\$15.74		1		View
San Juan	Inspection/Cleaning/Sealing Of Sewer & Water Systems By Remote Control	Technician	\$15.74		1		View
San Juan	Inspection/Cleaning/Sealing Of Sewer & Water Systems By Remote Control	Tv Truck Operator	\$15.74		1		View
San Juan	Insulation Applicators	Journey Level	\$71.53	15J	4C		View
San Juan	Ironworkers	Journeyman	\$82.03	7N	1Q		View
San Juan	Laborers	Air, Gas Or Electric Vibrating Screed	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Airtrac Drill Operator	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Ballast Regular Machine	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Batch Weighman	\$48.14	15J	4V	8Y	View

San Juan	Laborers	Brick Pavers	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Brush Cutter	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Brush Hog Feeder	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Burner	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Caisson Worker	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Carpenter Tender	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Cement Dumper-paving	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Cement Finisher Tender	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Change House Or Dry Shack	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Chipping Gun (30 Lbs. And Over)	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Chipping Gun (Under 30 Lbs.)	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Choker Setter	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Chuck Tender	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Clary Power Spreader	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Clean-up Laborer	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Concrete Dumper/Chute Operator	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Concrete Form Stripper	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Concrete Placement Crew	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Concrete Saw Operator/Core Driller	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Crusher Feeder	\$48.14	15J	4V	8Y	View
San Juan	Laborers	Curing Laborer	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Demolition: Wrecking & Moving (Incl. Charred Material)	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Ditch Digger	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Diver	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Drill Operator (Hydraulic, Diamond)	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Dry Stack Walls	\$56.80	15J	4V	8Y	View

San Juan	Laborers	Dump Person	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Epoxy Technician	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Erosion Control Worker	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Faller & Bucker Chain Saw	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Fine Graders	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Firewatch	\$48.14	15J	4V	8Y	View
San Juan	Laborers	Form Setter	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Gabian Basket Builders	\$56.80	15J	4V	8Y	View
San Juan	Laborers	General Laborer	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Grade Checker & Transit Person	\$59.85	15J	4V	8Y	View
San Juan	Laborers	Grinders	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Grout Machine Tender	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Groutmen (Pressure) Including Post Tension Beams	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Guardrail Erector	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Hazardous Waste Worker (Level A)	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Hazardous Waste Worker (Level B)	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Hazardous Waste Worker (Level C)	\$56.80	15J	4V	8Y	View
San Juan	Laborers	High Scaler	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Jackhammer	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Laserbeam Operator	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Maintenance Person	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Manhole Builder-Mudman	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Material Yard Person	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Mold Abatement Worker	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Motorman-Dinky Locomotive	\$59.95	15J	4V	8Y	View
San Juan	Laborers	nozzleman (concrete pump, green cutter when using combination of high pressure	\$59.85	15J	4V	8Y	View

		air & water on concrete & rock, sandblast, gunite, shotcrete, water blaster, vacuum blaster)					
San Juan	Laborers	Pavement Breaker	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Pilot Car	\$48.14	15J	4V	8Y	View
San Juan	Laborers	Pipe Layer (Lead)	\$59.85	15J	4V	8Y	View
San Juan	Laborers	Pipe Layer/Tailor	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Pipe Pot Tender	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Pipe Reliner	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Pipe Wrapper	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Pot Tender	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Powderman	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Powderman's Helper	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Power Jacks	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Railroad Spike Puller - Power	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Raker - Asphalt	\$59.85	15J	4V	8Y	View
San Juan	Laborers	Re-timberman	\$58.56	15J	4V	8Y	View
San Juan	Laborers	Remote Equipment Operator	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Rigger/Signal Person	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Rip Rap Person	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Rivet Buster	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Rodder	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Scaffold Erector	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Scale Person	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Sloper (Over 20")	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Sloper Sprayer	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Spreader (Concrete)	\$57.84	15J	4V	8Y	View

San Juan	Laborers	Stake Hopper	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Stock Piler	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Swinging Stage/Boatswain Chair	\$48.14	15J	4V	8Y	View
San Juan	Laborers	Tamper & Similar Electric, Air & Gas Operated Tools	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Tamper (Multiple & Self-propelled)	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Timber Person - Sewer (Lagger, Shorer & Cribber)	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Toolroom Person (at Jobsite)	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Topper	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Track Laborer	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Track Liner (Power)	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Traffic Control Laborer	\$51.48	15J	4V	9C	View
San Juan	Laborers	Traffic Control Supervisor	\$54.55	15J	4V	9C	View
San Juan	Laborers	Truck Spotter	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Tugger Operator	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 0-30 psi	\$158.87	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 30.01-44.00 psi	\$163.90	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 44.01-54.00 psi	\$167.58	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 54.01-60.00 psi	\$173.28	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 60.01-64.00 psi	\$175.40	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 64.01-68.00 psi	\$180.50	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 68.01-70.00 psi	\$182.40	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 70.01-72.00 psi	\$184.40	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Compressed Air Worker 72.01-74.00 psi	\$186.40	15J	4V	9B	View
San Juan	Laborers	Tunnel Work-Guage and Lock Tender	\$59.95	15J	4V	8Y	View
San Juan	Laborers	Tunnel Work-Miner	\$59.95	15J	4V	8Y	View
San Juan	Laborers	Vibrator	\$57.84	15J	4V	8Y	View

San Juan	Laborers	Vinyl Seamer	\$56.80	15J	4V	8Y	View
San Juan	Laborers	Watchman	\$43.76	15J	4V	8Y	View
San Juan	Laborers	Welder	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Well Point Laborer	\$57.84	15J	4V	8Y	View
San Juan	Laborers	Window Washer/Cleaner	\$43.76	15J	4V	8Y	View
San Juan	Laborers - Underground Sewer & Water	General Laborer & Topman	\$56.80	15J	4V	8Y	View
San Juan	Laborers - Underground Sewer & Water	Pipe Layer	\$57.84	15J	4V	8Y	View
San Juan	Landscape Construction	Landscape Construction/Landscaping Or Planting Laborers	\$43.76	15J	4V	8Y	View
San Juan	Landscape Construction	Landscape Operator	\$75.51	15J	11G	8X	View
San Juan	Landscape Maintenance	Groundskeeper	\$32.83		1		View
San Juan	Lathers	Journey Level	\$71.53	15J	4C		View
San Juan	Marble Setters	Journey Level	\$66.32	7E	1N		View
San Juan	Metal Fabrication (In Shop)	Fitter	\$15.74		1		View
San Juan	Metal Fabrication (In Shop)	Laborer	\$15.74		1		View
San Juan	Metal Fabrication (In Shop)	Machine Operator	\$15.74		1		View
San Juan	Metal Fabrication (In Shop)	Painter	\$15.74		1		View
San Juan	Metal Fabrication (In Shop)	Welder	\$15.74		1		View
San Juan	Millwright	Journey Level	\$73.08	15J	4C		View
San Juan	Modular Buildings	Journey Level	\$15.74		1		View
San Juan	Painters	Journey Level	\$49.46	6Z	11J		View
San Juan	Pile Driver	Crew Tender	\$77.16	15J	4C		View
San Juan	Pile Driver	Journey Level	\$71.98	15J	4C		View
San Juan	Plasterers	Journey Level	\$67.49	7Q	1R		View
San Juan	Plasterers	Nozzleman	\$71.49	7Q	1R		View
San Juan	Playground & Park Equipment Installers	Journey Level	\$15.74		1		View
San Juan	Plumbers & Pipefitters	Journey Level	\$83.47	5A	1G		View

San Juan	Power Equipment Operators	Asphalt Plant Operators	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Assistant Engineer	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Barrier Machine (zipper)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Batch Plant Operator: concrete	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Boat Operator	\$80.33	7A	11H	8X	View
San Juan	Power Equipment Operators	Bobcat	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Brokk - Remote Demolition Equipment	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Brooms	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Bump Cutter	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Cableways	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Chipper	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Compressor	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Concrete Finish Machine - Laser Screed	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Concrete Pump - Mounted Or Trailer High Pressure Line Pump, Pump High Pressure	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Concrete Pump: Truck Mount With Boom Attachment Over 42 M	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Concrete Pump: Truck Mount With Boom Attachment Up To 42m	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Conveyors	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Cranes Friction: 200 tons and over	\$82.76	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes, A-frame: 10 tons and under	\$75.55	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes: 100 tons through 199 tons, or 150' of boom (including jib with attachments)	\$81.12	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes: 20 tons through 44 tons with attachments	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes: 200 tons- 299 tons, or 250' of boom including jib with attachments	\$81.97	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes: 300 tons and over or 300' of boom including jib with attachments	\$82.76	7A	11H	8X	View

San Juan	Power Equipment Operators	Cranes: 45 tons through 99 tons, under 150' of boom(including jib with attachments)	\$80.33	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes: Friction cranes through 199 tons	\$81.97	7A	11H	8X	View
San Juan	Power Equipment Operators	Cranes: through 19 tons with attachments, a-frame over 10 tons	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators	Crusher	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Deck Engineer/Deck Winches (power)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Derricks, On Building Work	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Dozers D-9 & Under	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Drill Oilers: Auger Type, Truck Or Crane Mount	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Drilling Machine	\$77.54	15J	11G	8X	View
San Juan	Power Equipment Operators	Elevator and man-lift: permanent and shaft type	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Finishing Machine, Bidwell And Gamaco & Similar Equipment	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Forklift: 3000 lbs and over with attachments	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Forklifts: under 3000 lbs. with attachments	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Grade Engineer: Using Blue Prints, Cut Sheets, Etc	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Gradechecker/Stakeman	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Guardrail Punch	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Hard Tail End Dump Articulating Off- Road Equipment 45 Yards. & Over	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Hard Tail End Dump Articulating Off-road Equipment Under 45 Yards	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Horizontal/Directional Drill Locator	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Horizontal/Directional Drill Operator	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Hydralifts/Boom Trucks Over 10 Tons	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators	Hydralifts/boom trucks: 10 tons and under	\$75.55	7A	11H	8X	View
San Juan	Power Equipment Operators	Leverman	\$78.33	15J	11G	8X	View
San Juan	Power Equipment Operators	Loader, Overhead, 6 Yards. But Not Including 8 Yards	\$76.77	15J	11G	8X	View

San Juan	Power Equipment Operators	Loaders, Overhead Under 6 Yards	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Loaders, Plant Feed	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Loaders: Elevating Type Belt	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Locomotives, All	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Material Transfer Device	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Mechanics: All (Leadmen - \$0.50 per hour over mechanic)	\$77.54	15J	11G	8X	View
San Juan	Power Equipment Operators	Motor Patrol Graders	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Mucking Machine, Mole, Tunnel Drill, Boring, Road Header And/or Shield	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Oil Distributors, Blower Distribution & Mulch Seeding Operator	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Outside Hoists (Elevators and Manlifts), Air Tuggers, Strato	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Overhead, bridge type Crane: 20 tons through 44 tons	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators	Overhead, bridge type: 100 tons and over	\$81.12	7A	11H	8X	View
San Juan	Power Equipment Operators	Overhead, bridge type: 45 tons through 99 tons	\$80.33	7A	11H	8X	View
San Juan	Power Equipment Operators	Pavement Breaker	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Pile Driver (other Than Crane Mount)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Plant Oiler - Asphalt, Crusher	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Posthole Digger, Mechanical	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Power Plant	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Pumps - Water	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Quad 9, Hd 41, D10 And Over	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Quick Tower: no cab, under 100 feet in height base to boom	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Remote Control Operator On Rubber Tired Earth Moving Equipment	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Rigger and Bellman	\$75.55	7A	11H	8X	View
San Juan	Power Equipment Operators	Rigger/Signal Person, Bellman(Certified)	\$79.00	7A	11H	8X	View

San Juan	Power Equipment Operators	Rollagon	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Roller, Other Than Plant Mix	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Roller, Plant Mix Or Multi-lift Materials	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Roto-mill, Roto-grinder	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Saws - Concrete	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Scraper, Self Propelled Under 45 Yards	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Scrapers - Concrete & Carry All	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Scrapers, Self-propelled: 45 Yards And Over	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Service Engineers: Equipment	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Shotcrete/Gunite Equipment	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Shovel, Excavator, Backhoe, Tractors Under 15 Metric Tons	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Shovel, Excavator, Backhoe: Over 30 Metric Tons To 50 Metric Tons	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Shovel, Excavator, Backhoes, Tractors: 15 To 30 Metric Tons	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Shovel, Excavator, Backhoes: Over 50 Metric Tons To 90 Metric Tons	\$77.54	15J	11G	8X	View
San Juan	Power Equipment Operators	Shovel, Excavator, Backhoes: Over 90 Metric Tons	\$78.33	15J	11G	8X	View
San Juan	Power Equipment Operators	Slipform Pavers	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Spreader, Topsider & Screedman	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Subgrader Trimmer	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Tower Bucket Elevators	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Tower Crane: over 175' through 250' in height, base to boom	\$81.97	7A	11H	8X	View
San Juan	Power Equipment Operators	Tower crane: up to 175' in height base to boom	\$81.12	7A	11H	8X	View
San Juan	Power Equipment Operators	Tower Cranes: over 250' in height from base to boom	\$82.76	7A	11H	8X	View
San Juan	Power Equipment Operators	Transporters, All Track Or Truck Type	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Trenching Machines	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators	Truck Crane Oiler/Driver: 100 tons and over	\$79.62	7A	11H	8X	View

San Juan	Power Equipment Operators	Truck crane oiler/driver: under 100 tons	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators	Truck Mount Portable Conveyor	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Vac Truck (Vactor Guzzler, Hydro Excavator)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators	Welder	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators	Wheel Tractors, Farmall Type	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators	Yo Yo Pay Dozer	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Asphalt Plant Operators	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Assistant Engineer	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Barrier Machine (zipper)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Batch Plant Operator, Concrete	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Boat Operator	\$80.33	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Bobcat	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Brokk - Remote Demolition Equipment	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Brooms	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Bump Cutter	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cableways	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Chipper	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Compressor	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Concrete Finish Machine - Laser Screed	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Concrete Pump - Mounted Or Trailer High Pressure Line Pump, Pump High Pressure	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Concrete Pump: Truck Mount With Boom Attachment Over 42 M	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Concrete Pump: Truck Mount With Boom Attachment Up To 42m	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Conveyors	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes Friction: 200 tons and over	\$82.76	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes, A-frame: 10 tons and under	\$75.55	7A	11H	8X	View

San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: 100 tons through 199 tons, or 150' of boom (including jib with attachments)	\$81.12	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: 20 tons through 44 tons with attachments	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: 20 tons through 44 tons with attachments	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: 200 tons- 299 tons, or 250' of boom including jib with attachments	\$81.97	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: 300 tons and over or 300' of boom including jib with attachments	\$82.76	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: 45 tons through 99 tons, under 150' of boom(including jib with attachments)	\$80.33	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: Friction cranes through 199 tons	\$81.97	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Cranes: through 19 tons with attachments, a-frame over 10 tons	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Crusher	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Deck Engineer/Deck Winches (power)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Derricks, On Building Work	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Dozers D-9 & Under	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Drill Oilers: Auger Type, Truck Or Crane Mount	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Drilling Machine	\$77.54	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Elevator and man-lift: permanent and shaft type	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Finishing Machine, Bidwell And Gamaco & Similar Equipment	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Forklift: 3000 lbs and over with attachments	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Forklifts: under 3000 lbs. with attachments	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Grade Engineer: Using Blue Prints, Cut Sheets, Etc	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Gradechecker/Stakeman	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Guardrail Punch	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Hard Tail End Dump Articulating Off- Road Equipment 45 Yards. & Over	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Hard Tail End Dump Articulating Off-road	\$76.09	15J	11G	8X	View

		Equipment Under 45 Yards					
San Juan	Power Equipment Operators-Underground Sewer & Water	Horizontal/Directional Drill Locator	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Horizontal/Directional Drill Operator	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Hydralifts/boom trucks: 10 tons and under	\$75.55	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Hydralifts/boom trucks: over 10 tons	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Leverman	\$78.33	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Loader, Overhead, 6 Yards. But Not Including 8 Yards	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Loaders, Overhead Under 6 Yards	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Loaders, Plant Feed	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Loaders: Elevating Type Belt	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Locomotives, All	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Material Transfer Device	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Mechanics: All (Leadmen - \$0.50 per hour over mechanic)	\$77.54	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Motor Patrol Graders	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Mucking Machine, Mole, Tunnel Drill, Boring, Road Header And/or Shield	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Oil Distributors, Blower Distribution & Mulch Seeding Operator	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Outside Hoists (Elevators and Manlifts), Air Tuggers, Strato	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Overhead, bridge type Crane: 20 tons through 44 tons	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Overhead, bridge type: 100 tons and over	\$81.12	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Overhead, bridge type: 45 tons through 99 tons	\$80.33	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Pavement Breaker	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Pile Driver (other Than Crane Mount)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Plant Oiler - Asphalt, Crusher	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Posthole Digger, Mechanical	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Power Plant	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Pumps - Water	\$72.20	15J	11G	8X	View

San Juan	Power Equipment Operators-Underground Sewer & Water	Quad 9, Hd 41, D10 And Over	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Quick Tower: no cab, under 100 feet in height base to boom	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Remote Control Operator On Rubber Tired Earth Moving Equipment	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Rigger and Bellman	\$75.55	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Rigger/Signal Person, Bellman(Certified)	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Rollagon	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Roller, Other Than Plant Mix	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Roller, Plant Mix Or Multi-lift Materials	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Roto-mill, Roto-grinder	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Saws - Concrete	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Scraper, Self Propelled Under 45 Yards	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Scrapers - Concrete & Carry All	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Scrapers, Self-propelled: 45 Yards And Over	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Shotcrete/Gunite Equipment	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Shovel, Excavator, Backhoe, Tractors Under 15 Metric Tons	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Shovel, Excavator, Backhoe: Over 30 Metric Tons To 50 Metric Tons	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Shovel, Excavator, Backhoes, Tractors: 15 To 30 Metric Tons	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Shovel, Excavator, Backhoes: Over 50 Metric Tons To 90 Metric Tons	\$77.54	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Shovel, Excavator, Backhoes: Over 90 Metric Tons	\$78.33	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Slipform Pavers	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Spreader, Topsider & Screedman	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Subgrader Trimmer	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Tower Bucket Elevators	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Tower Crane: over 175' through 250' in height, base to boom	\$81.97	7A	11H	8X	View

San Juan	Power Equipment Operators-Underground Sewer & Water	Tower crane: up to 175' in height base to boom	\$81.12	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Tower Cranes: over 250' in height from base to boom	\$82.76	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Transporters, All Track Or Truck Type	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Trenching Machines	\$75.51	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Truck Crane Oiler/Driver: 100 tons and over	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Truck Crane Oiler/Driver: 100 tons and over	\$79.62	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Truck crane oiler/driver: under 100 tons	\$79.00	7A	11H	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Truck Mount Portable Conveyor	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Vac Truck (Vactor Guzzler, Hydro Excavator)	\$76.09	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Welder	\$76.77	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Wheel Tractors, Farmall Type	\$72.20	15J	11G	8X	View
San Juan	Power Equipment Operators-Underground Sewer & Water	Yo Yo Pay Dozer	\$76.09	15J	11G	8X	View
San Juan	Power Line Clearance Tree Trimmers	Journey Level In Charge	\$57.22	5A	4A		View
San Juan	Power Line Clearance Tree Trimmers	Spray Person	\$54.32	5A	4A		View
San Juan	Power Line Clearance Tree Trimmers	Tree Equipment Operator	\$57.22	5A	4A		View
San Juan	Power Line Clearance Tree Trimmers	Tree Trimmer	\$51.18	5A	4A		View
San Juan	Power Line Clearance Tree Trimmers	Tree Trimmer Groundperson	\$38.99	5A	4A		View
San Juan	Refrigeration & Air Conditioning Mechanics	Journey Level	\$83.96	5A	1G		View
San Juan	Residential Brick Mason	Journey Level	\$46.20		1		View
San Juan	Residential Carpenters	Journey Level	\$46.70		1		View
San Juan	Residential Cement Masons	Journey Level	\$15.74		1		View
San Juan	Residential Drywall Applicators	Journey Level	\$49.92	15J	4C		View
San Juan	Residential Drywall Tapers	Journey Level	\$24.48		1		View
San Juan	Residential Electricians	Journey Level	\$28.97		1		View
San Juan	Residential Glaziers	Journey Level	\$21.60		1		View
San Juan	Residential Insulation Applicators	Journey Level	\$15.74		1		View

San Juan	Residential Laborers	Journey Level	\$28.00		<u>1</u>		View
San Juan	Residential Marble Setters	Journey Level	\$46.20		<u>1</u>		View
San Juan	Residential Painters	Journey Level	\$32.20		<u>1</u>		View
San Juan	Residential Plumbers & Pipefitters	Journey Level	\$24.09		<u>1</u>		View
San Juan	Residential Refrigeration & Air Conditioning Mechanics	Journey Level	\$15.74		<u>1</u>		View
San Juan	Residential Sheet Metal Workers	Journey Level	\$18.54		<u>1</u>		View
San Juan	Residential Soft Floor Layers	Journey Level	\$28.92		<u>1</u>		View
San Juan	Residential Sprinkler Fitters (Fire Protection)	Journey Level	\$15.74		<u>1</u>		View
San Juan	Residential Stone Masons	Journey Level	\$25.69		<u>1</u>		View
San Juan	Residential Terrazzo Workers	Journey Level	\$15.74		<u>1</u>		View
San Juan	Residential Terrazzo/Tile Finishers	Journey Level	\$35.85		<u>1</u>		View
San Juan	Residential Tile Setters	Journey Level	\$15.74		<u>1</u>		View
San Juan	Roofers	Journey Level	\$59.05	<u>5A</u>	<u>3H</u>		View
San Juan	Roofers	Using Irritable Bituminous Materials	\$62.05	<u>5A</u>	<u>3H</u>		View
San Juan	Sheet Metal Workers	Journey Level (Field or Shop)	\$88.45	<u>7F</u>	<u>1E</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Boilermaker	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Carpenter	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Crane Operator	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Electrician	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Heat & Frost Insulator	\$84.58	<u>15H</u>	<u>11C</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Laborer	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Machinist	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Operating Engineer	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Painter	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Pipefitter	\$39.58	<u>7V</u>	<u>1</u>		View
San Juan	Shipbuilding & Ship Repair	New Construction Rigger	\$39.58	<u>7V</u>	<u>1</u>		View

San Juan	Shipbuilding & Ship Repair	New Construction Sheet Metal	\$39.58	7V	1		View
San Juan	Shipbuilding & Ship Repair	New Construction Shipfitter	\$39.58	7V	1		View
San Juan	Shipbuilding & Ship Repair	New Construction Warehouse/Teamster	\$39.58	7V	1		View
San Juan	Shipbuilding & Ship Repair	New Construction Welder / Burner	\$39.58	7V	1		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Boilermaker	\$50.35	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Carpenter	\$50.95	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Crane Operator	\$45.06	7Y	4K		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Electrician	\$50.42	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Heat & Frost Insulator	\$84.58	15H	11C		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Laborer	\$50.95	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Machinist	\$50.95	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Operating Engineer	\$45.06	7Y	4K		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Painter	\$50.95	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Pipefitter	\$50.95	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Rigger	\$50.35	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Sheet Metal	\$50.35	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Shipwright	\$50.95	7X	4J		View
San Juan	Shipbuilding & Ship Repair	Ship Repair Warehouse / Teamster	\$45.06	7Y	4K		View
San Juan	Sign Makers & Installers (Electrical)	Journey Level	\$15.74		1		View
San Juan	Sign Makers & Installers (Non-Electrical)	Journey Level	\$15.74		1		View
San Juan	Soft Floor Layers	Journey Level	\$55.56	5A	3J		View
San Juan	Solar Controls For Windows	Journey Level	\$15.74		1		View
San Juan	Sprinkler Fitters (Fire Protection)	Journey Level	\$69.97	7J	1R		View
San Juan	Stage Rigging Mechanics (Non Structural)	Journey Level	\$15.74		1		View
San Juan	Stone Masons	Journey Level	\$66.32	7E	1N		View
San Juan	Street And Parking Lot Sweeper Workers	Journey Level	\$15.74		1		View

San Juan	Surveyors	Assistant Construction Site Surveyor	\$79.00	7A	11H	8X	View
San Juan	Surveyors	Chainman	\$75.55	7A	11H	8X	View
San Juan	Surveyors	Construction Site Surveyor	\$80.33	7A	11H	8X	View
San Juan	Surveyors	Drone Operator (when used in conjunction with survey work only)	\$75.55	7A	11H	8X	View
San Juan	Surveyors	Ground Penetrating Radar Operator	\$75.55	7A	11H	8X	View
San Juan	Telecommunication Technicians	Telecom Technician Journey Level	\$51.68	5B	1B		View
San Juan	Telephone Line Construction - Outside	Cable Splicer	\$39.15	5A	2B		View
San Juan	Telephone Line Construction - Outside	Hole Digger/Ground Person	\$26.29	5A	2B		View
San Juan	Telephone Line Construction - Outside	Telephone Equipment Operator (Light)	\$32.72	5A	2B		View
San Juan	Telephone Line Construction - Outside	Telephone Lineperson	\$37.00	5A	2B		View
San Juan	Terrazzo Workers	Journey Level	\$60.36	7E	1N		View
San Juan	Tile Setters	Journey Level	\$60.36	7E	1N		View
San Juan	Tile, Marble & Terrazzo Finishers	Finisher	\$51.19	7E	1N		View
San Juan	Traffic Control Stripers	Journey Level	\$51.90	7A	1K		View
San Juan	Truck Drivers	Asphalt Mix Over 16 Yards	\$71.70	15J	11M	8L	View
San Juan	Truck Drivers	Asphalt Mix To 16 Yards	\$70.86	15J	11M	8L	View
San Juan	Truck Drivers	Dump Truck	\$70.86	15J	11M	8L	View
San Juan	Truck Drivers	Dump Truck & Trailer	\$71.70	15J	11M	8L	View
San Juan	Truck Drivers	Other Trucks	\$71.70	15J	11M	8L	View
San Juan	Truck Drivers - Ready Mix	Transit Mix	\$71.70	15J	11M	8L	View
San Juan	Well Drillers & Irrigation Pump Installers	Irrigation Pump Installer	\$15.74		1		View
San Juan	Well Drillers & Irrigation Pump Installers	Oiler	\$15.74		1		View
San Juan	Well Drillers & Irrigation Pump Installers	Well Driller	\$15.74		1		View

APPENDIX E – FEDERAL PREVAILING WAGE RATES

Superseded General Decision Number: WA20210057

State: Washington

Construction Type: Heavy
including water and sewer line construction

County: San Juan County in Washington.

HEAVY CONSTRUCTION PROJECTS (including sewer/water construction).

Note: Contracts subject to the Davis-Bacon Act are generally required to pay at least the applicable minimum wage rate required under Executive Order 14026 or Executive Order 13658. Please note that these Executive Orders apply to covered contracts entered into by the federal government that are subject to the Davis-Bacon Act itself, but do not apply to contracts subject only to the Davis-Bacon Related Acts, including those set forth at 29 CFR 5.1(a)(2)-(60).

If the contract is entered into on or after January 30, 2022, or the contract is renewed or extended (e.g., an option is exercised) on or after January 30, 2022:	. Executive Order 14026 generally applies to the contract. . The contractor must pay all covered workers at least \$15.00 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2022.
If the contract was awarded on or between January 1, 2015 and January 29, 2022, and the contract is not renewed or extended on or after January 30, 2022:	. Executive Order 13658 generally applies to the contract. . The contractor must pay all covered workers at least \$11.25 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on that contract in 2022.

The applicable Executive Order minimum wage rate will be adjusted annually. If this contract is covered by one of the Executive Orders and a classification considered necessary for performance of work on the contract does not appear on this wage determination, the contractor must still submit a conformance request.

Additional information on contractor requirements and worker protections under the Executive Orders is available at <https://www.dol.gov/agencies/whd/government-contracts>.

1	02/25/2022
2	06/24/2022
3	08/05/2022
4	12/23/2022

CARP0030-001 06/01/2021

	Rates	Fringes
CARPENTER.....	\$ 49.18	19.01

(HOURLY ZONE PAY: WESTERN AND CENTRAL WASHINGTON - ALL CLASSIFICATIONS EXCEPT MILLWRIGHTS AND PILEDRIVERS

Hourly Zone Pay shall be paid on jobs located outside of the free zone computed from the city center of the following listed cities:

Seattle	Olympia	Bellingham
Auburn	Bremerton	Anacortes
Renton	Shelton	Yakima
Aberdeen-Hoquiam	Tacoma	Wenatchee
Ellensburg	Everett	Port Angeles
Centralia	Mount Vernon	Sunnyside
Chelan	Pt. Townsend	

Zone Pay:

0 -25 radius miles	Free
26-35 radius miles	\$1.00/hour
36-45 radius miles	\$1.15/hour
46-55 radius miles	\$1.35/hour
Over 55 radius miles	\$1.55/hour

(HOURLY ZONE PAY: WESTERN AND CENTRAL WASHINGTON - MILLWRIGHT AND PILEDRIVER ONLY)

Hourly Zone Pay shall be computed from Seattle Union Hall, Tacoma City center, and Everett City center

Zone Pay:

0 -25 radius miles	Free
26-45 radius miles	\$.70/hour
Over 45 radius miles	\$1.50/hour

ELEC0191-013 06/01/2020

	Rates	Fringes
ELECTRICIAN		
DOUGLAS, CHELAN, and OKANOGAN Counties.....	\$ 46.15	26.10
ISLAND, SAN JUAN, SKAGIT, SNOHOMISH and WHATCOM Counties.....	\$ 47.95	26.16

ENGI0302-001 06/01/2022

	Rates	Fringes
Power equipment operators:		
Group 1A.....	\$ 54.20	24.47
Group 1AA.....	\$ 54.98	24.47
Group 1AAA.....	\$ 55.78	24.47
Group 1.....	\$ 53.40	24.47

Group 2.....	\$ 52.72	24.47
Group 3.....	\$ 52.12	24.47
Group 4.....	\$ 48.78	24.47

Zone Differential (Add to Zone 1 rates):

Zone 2 (26-45 radius miles) - \$1.00

Zone 3 (Over 45 radius miles) - \$1.30

BASEPOINTS: Aberdeen, Bellingham, Bremerton, Everett, Kent,
Mount Vernon, Port Angeles, Port Townsend, Seattle,
Shelton, Wenatchee, Yakima

POWER EQUIPMENT OPERATORS CLASSIFICATIONS

GROUP 1AAA - Cranes-over 300 tons, or 300 ft of boom
(including jib with attachments)

GROUP 1AA - Cranes 200 to 300 tons, or 250 ft of boom
(including jib with attachments); Tower crane over 175 ft
in height, base to boom; Excavator/Trackhoe, Backhoes: Over
90 metric tons

GROUP 1A - Cranes, 100 tons thru 199 tons, or 150 ft of boom
(including jib with attachments); Crane-overhead, bridge
type, 100 tons and over; Tower crane up to 175 ft in height
base to boom; Loaders-overhead, 8 yards and over;
excavator/Trackhoe, backhoes: over 50 metric tons to 90
metric tons

GROUP 1 - Cranes 45 tons thru 99 tons, under 150 ft of boom
(including jib with attachments); Crane-overhead, bridge
type, 45 tons thru 99 tons; Derricks on building work;;
Excavator/Trackhoe, backhoes: over 30 metric tons to 50
metric tons; Loader- overhead 6 yards to, but not including
8 yards; Dozer D-10

GROUP 2 - Cranes, 20 tons thru 44 tons with attachments;
Crane-overhead, bridge type-20 tons through 44 tons;;
Excavator/Trackhoe, backhoe: 15 to 30 metric tons;
Loaders-overhead under 6 yards; Mechanic; Drilling Machine;
Grader (finishing)

GROUP 3 - Cranes-thru 19 tons with attachments; A-frame crane
over 10 tons;; Dozers-D-9 and under; Roller-Plant Mix;
Excavator/Trackhoe, backhoe: under 15 metric tons;
Forklift: 3000 lbs and over with attachments; Oiler; Grader
(non-finishing); Boom Truck over 10 tons

GROUP 4 - Cranes-A frame-10 tons and under; Roller-other than
plant mix; Forklift: under 3000 lbs with attachments; Boom
Truck 10 tons and under

IRON0086-012 07/04/2022

	Rates	Fringes
Ironworker (REINFORCING & STRUCTURAL).....	\$ 49.90	31.82

LAB00292-001 06/01/2022

ZONE 1:

	Rates	Fringes
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Laborers:

GROUP 2.....	\$ 34.20	13.80
GROUP 3.....	\$ 42.86	13.80
GROUP 4.....	\$ 43.90	13.80
GROUP 5.....	\$ 44.62	13.80

ZONE DIFFERENTIAL (ADD TO ZONE 1 RATES):

ZONE 2 - \$1.00

ZONE 3 - \$1.30

BASE POINTS: BELLINGHAM, MT. VERNON, EVERETT, SEATTLE, KENT,
TACOMA, OLYMPIA, CENTRALIA, ABERDEEN, SHELTON, PT.
TOWNSEND, PT. ANGELES, AND BREMERTON

ZONE 1 - Projects within 25 radius miles of the respective
city hall

ZONE 2 - More than 25 but less than 45 radius miles from the
respective city hall

ZONE 3 - More than 45 radius miles from the respective city
hall

LABORERS CLASSIFICATIONS

GROUP 2: Flagger

GROUP 3: General or Common Laborer; Chipping Guns (Under 30
lbs)

GROUP 4: Pipe Layer; Chipping Guns (Over 30 lbs); Groutmen

GROUP 5: Mason Tender-Brick; Mason Tender-Cement/Concrete;
Grade Checker

* PAIN0005-008 07/01/2022

	Rates	Fringes
PAINTER (Brush, Roller and Spray).....	\$ 28.41	14.50

PLAS0528-004 06/01/2022

	Rates	Fringes
CEMENT MASON/CONCRETE FINISHER...	\$ 50.00	19.59

* TEAM0174-002 06/01/2019

	Rates	Fringes
Truck drivers:		
ZONE A:		
GROUP 1:.....	\$ 40.38	20.46
GROUP 2:.....	\$ 39.54	20.46

ZONE B (25-45 miles from center of listed cities*): Add \$.70
per hour to Zone A rates.

ZONE C (over 45 miles from centr of listed cities*): Add
\$1.00 per hour to Zone A rates.

*Zone pay will be calculated from the city center of the
following listed cities:

BELLINGHAM	CENTRALIA	RAYMOND	OLYMPIA
EVERETT	SHELTON	ANACORTES	BELLEVUE
SEATTLE	PORT ANGELES	MT. VERNON	KENT
TACOMA	PORT TOWNSEND	ABERDEEN	BREMERTON

TRUCK DRIVERS CLASSIFICATIONS

GROUP 1 - Dump Trucks, side, end and bottom dump, including semi-trucks and trains or combinations thereof with 16 yards to 30 yards capacity: Over 30 yards \$.15 per hour additional for each 10 yard increment.

GROUP 2 - Dump trucks, side, end and bottom dump, including semi-trucks and trains or combinations thereof with less than 16 yards capacity.

HAZMAT PROJECTS

Anyone working on a HAZMAT job, where HAZMAT certification is required, shall be compensated as a premium, in addition to the classification working in as follows:

LEVEL C: +\$.25 per hour - This level uses an air purifying respirator or additional protective clothing.

LEVEL B: +\$.50 per hour - Uses same respirator protection as Level A. Supplied air line is provided in conjunction with a chemical "splash suit."

LEVEL A: +\$.75 per hour - This level utilizes a fully-encapsulated suit with a self-contained breathing apparatus or a supplied air line.

* SUWA2009-048 08/07/2009

	Rates	Fringes
LABORER: Landscape.....	\$ 14.67 **	0.00
PIPEFITTER.....	\$ 30.00	8.35
TRUCK DRIVER: Water Truck.....	\$ 24.36	8.30
TRUCK DRIVER: 10 Yard Truck.....	\$ 24.61	8.34

WELDERS - Receive rate prescribed for craft performing operation to which welding is incidental.

=====

** Workers in this classification may be entitled to a higher minimum wage under Executive Order 14026 (\$15.00) or 13658 (\$11.25). Please see the Note at the top of the wage determination for more information.

Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other

health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (ii)).

The body of each wage determination lists the classification and wage rates that have been found to be prevailing for the cited type(s) of construction in the area covered by the wage determination. The classifications are listed in alphabetical order of ""identifiers"" that indicate whether the particular rate is a union rate (current union negotiated rate for local), a survey rate (weighted average rate) or a union average rate (weighted union average rate).

Union Rate Identifiers

A four letter classification abbreviation identifier enclosed in dotted lines beginning with characters other than ""SU"" or ""UAVG"" denotes that the union classification and rate were prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2014. PLUM is an abbreviation identifier of the union which prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. 07/01/2014 is the effective date of the most current negotiated rate, which in this example is July 1, 2014.

Union prevailing wage rates are updated to reflect all rate changes in the collective bargaining agreement (CBA) governing this classification and rate.

Survey Rate Identifiers

Classifications listed under the ""SU"" identifier indicate that no one rate prevailed for this classification in the survey and the published rate is derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As this weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SULA2012-007 5/13/2014. SU indicates the rates are survey rates based on a weighted average calculation of rates and are not majority rates. LA indicates the State of Louisiana. 2012 is the year of survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. 5/13/2014 indicates the survey completion date for the classifications and rates under that identifier.

Survey wage rates are not updated and remain in effect until a new survey is conducted.

Union Average Rate Identifiers

Classification(s) listed under the UAVG identifier indicate that no single majority rate prevailed for those classifications; however, 100% of the data reported for the classifications was union data. EXAMPLE: UAVG-OH-0010 08/29/2014. UAVG indicates that the rate is a weighted union average rate. OH indicates the state. The next number, 0010 in the example, is an internal number used in producing the wage determination. 08/29/2014 indicates the survey completion date for the classifications and rates under that identifier.

A UAVG rate will be updated once a year, usually in January of each year, to reflect a weighted average of the current negotiated/CBA rate of the union locals from which the rate is based.

WAGE DETERMINATION APPEALS PROCESS

1.) Has there been an initial decision in the matter? This can be:

- * an existing published wage determination
- * a survey underlying a wage determination
- * a Wage and Hour Division letter setting forth a position on a wage determination matter
- * a conformance (additional classification and rate) ruling

On survey related matters, initial contact, including requests for summaries of surveys, should be with the Wage and Hour National Office because National Office has responsibility for the Davis-Bacon survey program. If the response from this initial contact is not satisfactory, then the process described in 2.) and 3.) should be followed.

With regard to any other matter not yet ripe for the formal process described here, initial contact should be with the Branch of Construction Wage Determinations. Write to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2.) If the answer to the question in 1.) is yes, then an interested party (those affected by the action) can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Write to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and by any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3.) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

4.) All decisions by the Administrative Review Board are final.

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END OF GENERAL DECISIO"

APPENDIX F – PLAN OF INTERIM OPERATION

PLAN OF INTERIM OPERATION

FOR THE

EASTSOUND SEWER AND WATER DISTRICT WASTEWATER TREATMENT PLANT UPGRADES PROJECT

Prepared for:

EASTSOUND SEWER AND WATER DISTRICT

By:

WILSON ENGINEERING, LLC

December 1, 2022

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LIST OF FIGURES

- 5-1 Organization Chart

1.0 INTRODUCTION

The purpose of this plan for interim operation is to identify and describe, in chronological order, the key operational milestones and actions from beginning of construction to the end of start-up necessary to integrate the improvements into the existing system for the Eastsound Sewer and Water District (District) Wastewater Treatment Plant (WWTP) Improvements project. It is to be used for guidance by the District and their representatives responsible for quality assurance during this project.

1.1 Scope

This Plan:

- Identifies Key Construction-Related Dates
- Plan for staffing during construction
- Plan for Maintaining Interim Operation During Construction
- Startup Plan
- Training Plan for Startup

1.2 Reference Documents

This plan is not part of the construction Contract Documents for the project. However, the Contract Documents issued for construction are referenced extensively by this plan. Additionally, several other documents may be referenced for information pertaining to the project and the system/facilities. A list of available reference documents is provided below.

- ***Construction Contract Documents (plans and specifications)***, Eastsound Sewer and Water District WWTP Upgrades, Wilson Engineering, December 2022.

A copy of these references will be kept in the ESWD's onsite office.

2.0 SCOPE OF CONSTRUCTION

This project involves civil, structural, electrical, and mechanical construction aspects for the improvements detailed within the construction Contract Documents (plans and specifications) for the Eastsound Sewer and Water District WWTP Upgrades. The general scope of the project is to make several new improvements at the existing WWTP, including constructing a third bullseye style treatment train with a Bardenpho treatment process with anoxic basins, aeration basins, and a new clarifier; constructing an aerobic digester and ultraviolet disinfection system; adding new dewatering equipment; constructing a new influent flow splitter; constructing a new influent pump station and 2W pump station, and constructing a new blower building, and improving the stormwater system.

Contract Documents issued for construction of this project were prepared by Wilson Engineering. These documents detail work associated with the improvements, the Contractor's responsibilities, the Engineer's authority and the Owner's requirements.

3.0 PRELIMINARY CONSTRUCTION SCHEDULE

A preliminary construction schedule for the Eastsound Sewer and Water District WWTP Improvements project is illustrated in Figure 3-1.

A critical path method (CPM) construction schedule reflecting detailed activities, sequencing and project duration is required by Section 01 31 00 of the Specifications. The Contractor will provide a construction schedule within 10 days of Contract Award. A copy of the CPM schedule will be kept on file in the Contractor's field office.

CPM schedule shall be updated monthly, with each pay request. No monthly progress payments will be made to the Contractor until an updated schedule is received. A copy of each updated CPM schedule will be kept in the Contractor's field office.

Table 3-1 – Construction Schedule

Pre-Bid Meeting	January 18, 2023
Bid Opening	February 8, 2023
Bid Award	February 15, 2023
Contract Executed - Contract Times Commence	March 15, 2023
Notice to Proceed	March 15, 2023
O&M Completion Date	June 2024
Substantial Completion	July 2024
Final Completion	October 2024

4.0 STAFFING

On-site construction observation will be performed by Eastsound Sewer and Water District and Wilson Engineering personnel. Duties of on-site personnel will primarily involve observation and documentation of construction activities, verification of conformance with Contract Documents, performance and coordination of CQA activities, as well as communication with the office engineering staff, Owner, Contractor, and others.

Neither the District nor Wilson Engineering plan to change staffing from the existing staffing plan during construction, during startup, or after construction is completed.

District Contacts:

Jason Bradshaw, General Manager
620-441-4006
JasonB@eswd.org

Michael Hidalgo, Treatment Operator I
360-376-2720
mikeh@eswd.org

Wilson Contacts:

Jeff Christner, P.E.
360-312-3145
jgc@wilsonengineering.com

Sam Rodriguez, E.I.T
360-312-3165
srodriguez@wilsonengineering.com

5.0 PLAN FOR MAINTAINING INTERIM OPERATION DURING CONSTRUCTION

5.1 General

The construction of the WWTP Upgrades will generally take place while maintaining continuous treatment through the plant. The treatment process will transition from a two train process to a three train process with minimal disruptions in treatment. Temporary piping will be provided by the contractor to maintain continuous flow from the existing chlorine disinfection chamber outlet to the existing 8-inch effluent piping just upstream of the existing effluent manhole (SSMH #1288). The duration of each disruption will be a maximum of one (1) hour, during which time flow will either be carefully plugged/stored/monitored or will be piped/pumped around the disruption.

Disruptions for bringing the influent pump station online will be a maximum of (1) hour.

Potential Disruptions:

- Bringing influent pumpstation online
 - o Plug influent pipe at North Beach Rd and Cessna Rd intersection
 - o Bring on influent flow splitter at the same time
- Connecting the new effluent 10" pipe to the existing effluent pipe
- Connection of the new transformer. See Specification 26 05 26 Section 1.013 and 1.016.

The District intends to work with the Contractor to coordinate the work that requires disruption in treatment during periods of low flow. All work that requires disruption shall be coordinated with the plant to occur at low flow periods and as quickly as possible.

Temporary pumps and temporary piping will also be required to maintain operation during various phases of construction, such as lowering Train #2 water level so that the new influent splitter box can be connected to Train #2 influent piping.

A detailed description of the proposed construction sequence plan, including strategies for maintaining plant operation, can be found on sheet C0.7.

Figure 5-1 shows the project construction organization chart.

See Operation Plan on sheet G0.3 for emergency contact information. In addition, the Contractor will be required to prepare and maintain a Safety Plan which will be at the jobsite and available for all Construction Staff to view.

O&M Manual will be complete by the time the plant is brought online. Start-up planning meetings will be scheduled when the plant reaches 60% completion.

5.2 Coordination

Installation of the new facilities and conversion from the existing systems to the newly constructed facilities will require coordination and clear communication. The Contractor must coordinate their

efforts so that construction activities do not adversely affect existing systems and utilities. Clear communication will facilitate the coordination process.

The Contractor is required to coordinate with the Owner for scheduling various components of the WWTP improvements that impact the treatment process. The Contractor is also required to submit a bypass plan which will be reviewed by the Engineer and Owner.

5.3.1 Coordination with Owner's Facilities

It is the responsibility of the Contractor to coordinate his work so that impacts to the Owner's facilities are minimized. The District must continue to provide uninterrupted wastewater treatment throughout the duration of the construction contracts.

The Contractor will be responsible for coordinating with the Owner prior to demolition of the existing generator and generator switch gear. The Owner will have full use of new generator equipment prior to demotion of the old generator equipment.

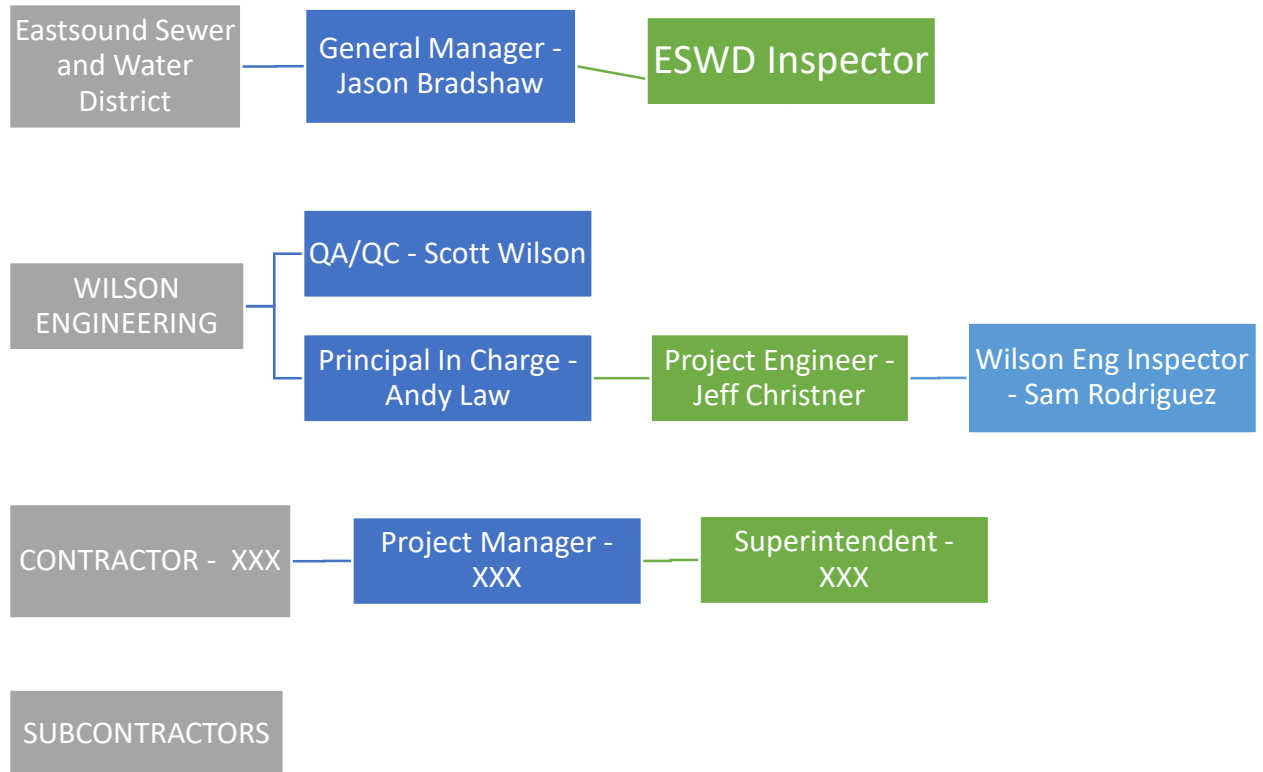
5.3.2 Coordination with Utilities

Several existing utilities, including water service, power, and fiber optic telephone service utilities are located within the project areas. It is the responsibility of the Contractor to coordinate his construction activities so that interruptions in service are minimized. The Contractor may address correspondence directly to the affected utility; however, the Engineer must receive copies to verify that the intent of the design is maintained and that the interests of the Owner are not adversely affected.

6.0 START-UP

New concrete basins will be subject to clean water leakage testing, and equipment installed will be subject to operational testing as part of the start-up procedures. Testing requirements are specified in Section 01 45 00 of the Specifications, as well as in each specified equipment specification section. Start-up and testing shall be performed by the Contractor and equipment manufacturers as required, in the presence of the Engineer, prior to acceptance of the work. Per the project Specifications, the Contractor and equipment manufacturers will provide startup training to the District WWTP staff.

Figure 5-1 – Organization Chart



APPENDIX G – INADVERTENT DISCOVERY PLAN

PLAN AND PROCEDURES FOR THE UNANTICIPATED DISCOVERY OF CULTURAL RESOURCES AND HUMAN SKELETAL REMAINS¹

PROJECT TITLE: Eastsound Sewer and Water District Wastewater Treatment Plant Upgrades Project

COUNTY WASHINGTON: San Juan County

Section, Township, Range: 11, 37N, 2W

1. INTRODUCTION

The following Inadvertent Discovery Plan (IDP) outlines procedures to perform in the event of discovering archaeological materials or human remains, in accordance with state and federal laws.

2. RECOGNIZING CULTURAL RESOURCES

A cultural resource discovery could be prehistoric or historic. Examples include:

- a. An accumulation of shell, burned rocks, or other food related materials.
- b. Bones or small pieces of bone.
- c. An area of charcoal or very dark stained soil with artifacts.
- d. Stone tools or waste flakes (i.e. an arrowhead. or stone chips).
- e. Clusters of tin cans or bottles, logging or agricultural equipment that appears to be older than 50 years.
- f. Buried railroad tracks, decking, or other industrial materials.

When in doubt, assume the material is a cultural resource.

3. ON-SITE RESPONSIBILITIES

STEP 1: Stop Work. If any employee, contractor or subcontractor believes that he or she has uncovered a cultural resource at any point in the project, all work must stop immediately. Notify the appropriate party(s). Leave the surrounding area untouched, and provide a demarcation adequate to provide the total security, protection, and integrity of the discovery. The discovery location must be secured at all times by a temporary fence or other onsite security.

STEP 2: Notify Archaeological Monitor or Licensed Archaeologist. If there is an Archaeological Monitor for the project, notify that person. This would be Garth Baldwin,

¹ If you need this document in a format for the visually impaired, call Water Quality Reception at Ecology, (360) 407-6600. Persons with hearing loss can call 711 for Washington Relay Service. Persons with a speech disability can call 877-833-6341.

Drayton Archaeology at 360-739-3921. If there is a monitoring plan in place, the monitor will follow the outlined procedure.

STEP 3: *Notify the Project Manager* of this project and contact the Ecology Staff Project Manager, or other applicable contacts:

Project Manager: Name: Jason Bradshaw Phone: 360-376-2720 Email: jasonb@eswd.org	Ecology Staff Project Manager Name: Ken Ziebart Phone: 206-594-0163 Email: kenneth.ziebart@ecy.wa.gov
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Assigned Alternates:

Assigned Project Manager Alternate: Name: Susie Chapman Phone: 360-376-2720 Email: susiec@eswd.org	Ecology Cultural Resource Specialist (Alternate): Name: Liz Ellis Phone: 360-407-6429 email: lell461@ECY.WA.GOV
---	--

The Project Manager or applicable staff will make all calls and necessary notifications.

If human remains are encountered, treat them with dignity and respect at all times.

Cover the remains with a tarp or other materials (not soil or rocks) for temporary protection and to shield them from being photographed. **Do not call 911 or speak with the media. Do not take pictures unless directed to do so by DAHP. See Section 5.**

4. FURTHER CONTACTS AND CONSULTATION

A. Project Manager's Responsibilities:

- *Protect Find:* The Project Manager is responsible for taking appropriate steps to protect the discovery site. All work will stop immediately in a surrounding area adequate to provide for the complete security of location, protection, and integrity of the resource. Vehicles, equipment, and unauthorized personnel will not be permitted to traverse the discovery site. Work in the immediate area will not resume until treatment of the discovery has been completed following provisions for treating archaeological/cultural material as set forth in this document.
- *Direct Construction Elsewhere on-Site:* The Project Manager may direct construction away from cultural resources to work in other areas prior to contacting the concerned parties.
- *Contact Senior Staff:* If the Senior Staff person has not yet been contacted, the Project Manager must do so.

B. Senior Staff Responsibilities:

- *Identify Find:* The Senior Staff (or a delegated Cultural Resource Specialist), will ensure that a qualified professional archaeologist examines the area to determine if there is an archaeological find.
 - If it is determined not to be of archaeological, historical, or human remains, work may proceed with no further delay.
 - If it is determined to be an archaeological find, the Senior Staff or Cultural Resource Specialist will continue with all notifications.
 - If the find may be human remains or funerary objects, the Senior Staff or Cultural Resource Specialist will ensure that a qualified physical anthropologist examines the find. **If it is determined to be human remains, the procedure described in Section 5 will be followed.**
- *Notify DAHP:* The Senior Staff (or a delegated Cultural Resource Specialist) will contact the involved federal agencies (if any) and the Washington Department of Archaeology and Historic Preservation (DAHP).
- *Notify Tribes:* If the discovery may be of interest to Native American Tribes, the DAHP and Ecology Supervisor or Coordinator will coordinate with the interested and/or affected tribes.

General Contacts

Federal Agencies:

Agency: Name Title Number Email

State Agencies:

Agency: Name Title Number Email

Department of Archaeology and Historic Preservation:

Dr. Allyson Brooks State Historic Preservation Officer 360-586-3066 Assigned Alternate:	Rob Whitlam, Ph.D. Staff Archaeologist 360-586-3050 Assigned Alternate:
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The DAHP or appropriate Ecology Staff will contact the interested and affected Tribes for a specific project.

Tribes consulted on this project are:

Tribe:	Tribe
Name:	Name

Title:	Title
Phone:	Phone
Email:	Email

Further Activities

- Archaeological discoveries will be documented as described in Section 6.
- Construction in the discovery area may resume as described in Section 7.

5. SPECIAL PROCEDURES FOR THE DISCOVERY OF HUMAN SKELETAL MATERIAL

Any human skeletal remains, regardless of antiquity or ethnic origin, will at all times be treated with dignity and respect. Do not take photographs by any means, unless you are pre-approved to do so.

If the project occurs on federal lands or receives federal funding (e.g., national forest or park, military reservation) the provisions of the Native American Graves Protection and Repatriation Act of 1990 apply, and the responsible federal agency will follow its provisions. Note that state highways that cross federal lands are on an easement and are not owned by the state.

If the project occurs on non-federal lands, the Project Manager will comply with applicable state and federal laws, and the following procedure:

A. In all cases you must notify a law enforcement agency or Medical Examiner/Coroner's Office:

In addition to the actions described in Sections 3 and 4, the Project Manager will immediately notify the local law enforcement agency or medical examiner/coroner's office.

The Medical Examiner/Coroner (with assistance of law enforcement personnel) will determine if the remains are human, whether the discovery site constitutes a crime scene, and will then notify DAHP.

Enter contact information below:

San Juan County Sheriff's Office
360-378-4151 (non-emergency number)

B. Participate in Consultation:

Per RCW 27.44.055, RCW 68.50, and RCW 68.60, DAHP will have jurisdiction over non-forensic human remains. Ecology staff will participate in consultation.

C. Further Activities:

- Documentation of human skeletal remains and funerary objects will be agreed upon through the consultation process described in RCW 27.44.055, RCW 68.50, and RCW 68.60.

- When consultation and documentation activities are complete, construction in the discovery area may resume as described in Section 7.

6. DOCUMENTATION OF ARCHAEOLOGICAL MATERIALS

Archaeological deposits discovered during construction will be assumed eligible for inclusion in the National Register of Historic Places under Criterion D until a formal Determination of Eligibility is made.

Project staff will ensure the proper documentation and field assessment will be made of any discovered cultural resources in cooperation with all parties: the federal agencies (if any), DAHP, Ecology, affected tribes, and a contracted consultant (if any).

All prehistoric and historic cultural material discovered during project construction will be recorded by a professional archaeologist on a cultural resource site or isolate form using standard and approved techniques. Site overviews, features, and artifacts will be photographed; stratigraphic profiles and soil/sediment descriptions will be prepared for minimal subsurface exposures. Discovery locations will be documented on scaled site plans and site location maps.

Cultural features, horizons and artifacts detected in buried sediments may require further evaluation using hand-dug test units. Units may be dug in controlled fashion to expose features, collect samples from undisturbed contexts, or to interpret complex stratigraphy. A test excavation unit or small trench might also be used to determine if an intact occupation surface is present. Test units will be used only when necessary to gather information on the nature, extent, and integrity of subsurface cultural deposits to evaluate the site's significance. Excavations will be conducted using state-of-the-art techniques for controlling provenience, and the chronology of ownership, custody and location recorded with precision.

Spatial information, depth of excavation levels, natural and cultural stratigraphy, presence or absence of cultural material, and depth to sterile soil, regolith, or bedrock will be recorded for each probe on a standard form. Test excavation units will be recorded on unit-level forms, which include plan maps for each excavated level, and material type, number, and vertical provenience (depth below surface and stratum association where applicable) for all artifacts recovered from the level. A stratigraphic profile will be drawn for at least one wall of each test excavation unit.

Sediments excavated for purposes of cultural resources investigation will be screened through 1/8-inch mesh, unless soil conditions warrant 1/4-inch mesh.

All prehistoric and historic artifacts collected from the surface and from probes and excavation units will be analyzed, catalogued, and temporarily curated. Ultimate disposition of cultural materials will be determined in consultation with the federal agencies (if any), DAHP, Ecology and the affected tribes.

Within 90 days of concluding fieldwork, a technical report describing any and all monitoring and resultant archaeological excavations will be provided to the Project Manager, who will forward the report for review and delivery to Ecology, the federal agencies (if any), DAHP, and the affected tribe(s).

If assessment activity exposes human remains (burials, isolated teeth, or bones), the process described in Section 5 will be followed.

7. PROCEEDING WITH WORK

Work outside the discovery location may continue while documentation and assessment of the cultural resources proceed. A professional archaeologist must determine the boundaries of the discovery location. In consultation with Ecology, DAHP and any affected tribes, the Project Manager will determine the appropriate level of documentation and treatment of the resource. If there is a federal nexus, Section 106 consultation and associated federal laws will make the final determinations about treatment and documentation.

Work may continue at the discovery location only after the process outlined in this plan is followed and the Project Manager, DAHP, any affected tribes, Ecology (and the federal agencies, if any) determine that compliance with state and federal law is complete.

8. RECIPIENT/PROJECT PARTNER RESPONSIBILITY

The Project Recipient/Project Partner is responsible for developing an IDP. The IDP must be immediately available onsite, be implemented to address any discovery, and be available by request by any party. The Project Manager and staff will review the IDP during a project kickoff or pre-construction meeting.

We recommend that you print images in color for accuracy.

Implement the IDP / UDP if ...

You see chipped stone artifacts.

- Glass-like material
- Angular
- “Unusual” material for area
- “Unusual” shape
- Regularity of flaking
- Variability of size



Implement the IDP / UDP if ...

You see ground or pecked stone artifacts.



- Striations or scratching
- Unusual or unnatural shapes
- Unusual stone
- Etching
- Perforations
- Pecking
- Regularity in modifications
- Variability of size, function, and complexity

Implement the IDP / UDP if ...

You see bone or shell artifacts.



- Often smooth
- Unusual shape
- Carved
- Often pointed if used as a tool
- Often wedge shaped like a “shoehorn”



Implement the IDP / UDP if ...

You see bone or shell artifacts.



- Often smooth
- Unusual shape
- Perforated
- Variability of size



Implement the IDP / UDP if ...

You see fiber or wood artifacts.



- Wet environments needed for preservation
- Variability of size, function, and complexity
- Rare



Implement the IDP / UDP if ...

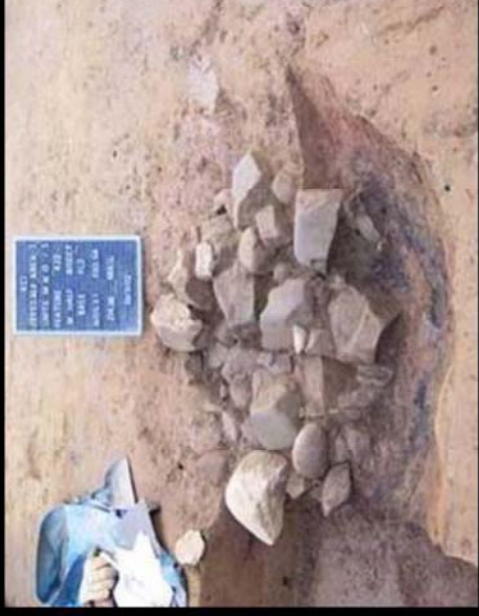
You see historic period artifacts.



Implement the IDP / UDP if ...

You see strange, different or interesting looking dirt, rocks, or

- Human activities leave traces in the ground that may or may not have artifacts associated with them
- “Unusual” accumulations of rock (especially fire-cracked rock)
- “Unusual” shaped accumulations of rock (e.g., similar to a fire ring)
- Charcoal or charcoal-stained soils
- Oxidized or burnt-looking soils
- Accumulations of shell
- Accumulations of bones or artifacts
- Look for the “unusual” or out of place (e.g., rock piles or accumulations in areas with few rock)



Implement the IDP / UDP if ...

You see strange, different or interesting looking dirt, rocks, or



- “Unusual” accumulations of rock (especially fire-cracked rock)
- “Unusual” shaped accumulations of rock (e.g., similar to a fire ring)
- Look for the “unusual” or out of place (e.g., rock piles or accumulations in areas with few rock)

Implement the IDP / UDP if ...

You see strange, different or interesting looking dirt, rocks, or

- Often have a layered or “layer cake” appearance
- Often associated with black or blackish soil
- Often have very crushed and compacted shells



Layers of shell
midden



Historic Debris

Implement the IDP / UDP if ...

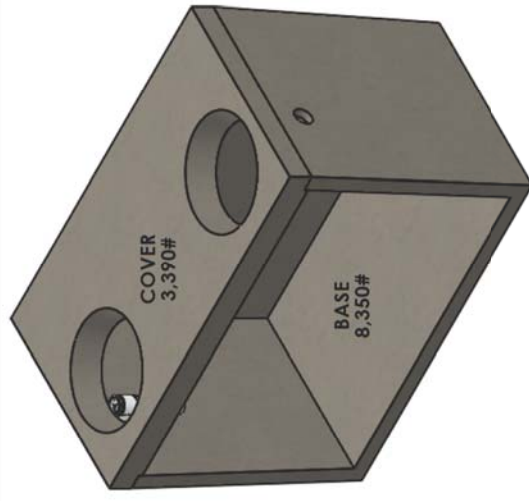
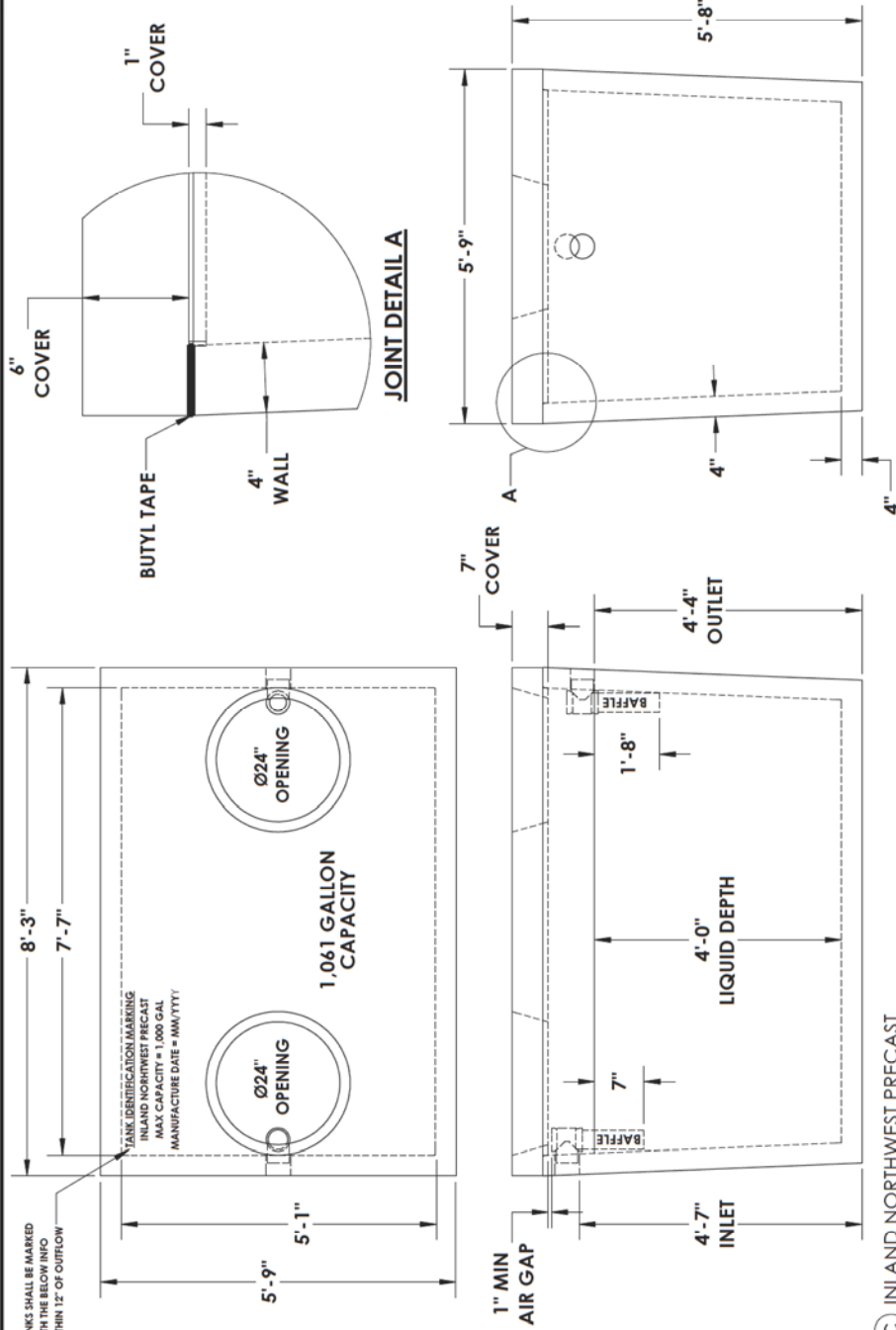
You see historic foundations or buried structures.



APPENDIX H – SEPTIC TANK LAYOUT DRAWING

See attached 1000 gallon single compartment septic tank. Contractor's proposed septic tank is to match this layout. Installed structure is to be complete and functional with traffic rated risers, manhole lids, piping, etc.

1000 GALLON SINGLE COMPARTMENT SEPTIC TANK - TRAFFIC RATED



* CONCRETE:
4,000 PSI MINIMUM

* RISER/FLANGE ADAPTERS:
AVAILABLE UPON REQUEST

NOTES:
~ BEDDING MUST BE COMPACTED AND AREA AROUND TANK MUST BE BACKFILLED PRIOR TO PUTTING INTO SERVICE
~ PRESS SEAL BOOTS PROVIDED AT INLET & OUTLET
~ STRUCTURAL ENGINEERING CALCULATIONS AVAILABLE UPON REQUEST
~ 2-1/2" TAPER FROM TOP TO BOTTOM OF TANK

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REINFORCEMENT

COVER:

~ TRANSVERSE #4 @ 6"
~ LONGITUDINAL #4 @ 1'-0"

BASE WALLS:

~ 4x4W4xW4 WIRE MESH IN WALLS

BASE FLOOR:

~ TRANSVERSE #4 @ 9"
~ LONGITUDINAL #4 @ 1'-0"

* ALL TANKS ARE MADE IN ACCORDANCE WITH ASTM C1227 - 13 (STANDARD SPECIFICATION FOR PRECAST CONCRETE SEPTIC TANKS)

APPENDIX I – CDBG FUNDING: REQUIRED DOCUMENTS

See attached documents:

- Attachment 7-A (See Section 00 54 00 – CDBG Conditions)
- Attachment 7-B San Juan County Section 3 Labor Hours Summary Form
- Attachment 7-C Contract Subcontract Activity
- Attachment 8-B Notification of Contract Award and Start of Construction
- Attachment 8-C Labor Standards Compliance Report
- Attachment 8-D Authorization to Sign Certified Payrolls

San Juan County 20-62210-038 Eastsound Sewer Project

Section 3 Labor Hours Summary Report (24 CFR Part 75)

This form must be submitted by the contractor by the end of the project (generally after all certified payrolls are complete).

The work to be performed on this project is subject to the requirements of Section 3 of the Housing and Urban Development Act of 1968, as amended, 12 U.S.C. 1701u (Section 3). The purpose of Section 3 is to ensure that employment and other economic opportunities generated by the U.S. Department of Housing and Urban Development (HUD) assistance or HUD assisted projects covered by Section 3, shall, to the greatest extent feasible, be directed to low- and very low-income persons.

Section 3 Summary - All Project Labor Hours form: Total labor hours reported for the project using subtotals from individual contractor and subcontractors forms.

Section 3 Labor Hours - Contractor form: Contractor reports hours for all workers and for each Section 3 worker.

Section 3 Labor Hours - Subcontractor form: Each subcontractor reports hours for all workers and for each Section 3 worker.

Definitions

Section 3 Business Concern: A Section 3 business concern is defined in as a business that meets at least one of the following criteria, documented within the last six-month period:

1. At least 51 percent owned and controlled by low-income persons (see table below),
2. More than 75 percent of the labor hours performed for the business over the previous 3-month period are performed by Section 3 workers, or
3. At least 51 percent owned and controlled by current residents of public housing or Section 8-assisted housing.

Section 3 Worker :

1. The worker is a low income person (see table below);
2. Employed by a Section 3 business concern; or
3. A YouthBuild participant.

Targeted Section 3 Worker:

1. Employed by a Section 3 business concern; or
2. Employee lives within the project area . Within [one mile of the project location](#) for which this report is being submitted (or if fewer than 5,000 people live within one mile of the project, within a circle centered on the Section 3 project that is sufficient to encompass a population of 5,000 people according to the most recent U.S. Census.)

Low-income Person: An individual whose income for a previous or annualized calendar year (any year since January 2021) is less than the income limit shown in the table below (established by HUD) based on the person's county of residence.

Adams	\$ 39,550	Grays Harbor	\$ 39,550	Pierce	\$ 50,900
Asotin	\$ 40,050	Island	\$ 44,700	San Juan	\$ 43,700
Benton	\$ 43,750	Jefferson	\$ 39,950	Skagit	\$ 46,100
Chelan	\$ 39,950	King	\$ 63,350	Skamania	\$ 54,150
Clallam	\$ 39,550	Kitsap	\$ 52,750	Snohomish	\$ 63,350
Clark	\$ 54,150	Kittitas	\$ 42,600	Spokane	\$ 43,200
Columbia	\$ 39,550	Klickitat	\$ 39,550	Stevens	\$ 39,550
Cowlitz	\$ 39,950	Lewis	\$ 39,550	Wahkiakum	\$ 39,550
Douglas	\$ 39,950	Lincoln	\$ 39,550	Thurston	\$ 50,550
Ferry	\$ 39,550	Mason	\$ 39,550	Walla Walla	\$ 40,950
Franklin	\$ 43,750	Okanogan	\$ 39,550	Whatcom	\$ 45,450
Garfield	\$ 39,550	Pacific	\$ 39,550	Whitman	\$ 41,400
Grant	\$ 39,950	Pend Oreille	\$ 39,550	Yakima	\$ 39,550

Contract and Subcontract Activity

U.S. Department of Housing and Urban Development

OMB Approval No.: 2577-0088

OMB Approval No.: 2502-0355

Public reporting burden for this collection of information is estimated to average .5 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. The Information is voluntary. HUD may not collect this information, and you are not required to complete this form, unless it displays a currently valid OMB Control Number.

Executive Orders dated July 14, 1983, directs the Minority Business Development Plans shall be developed by each Federal Agency and these annual plans shall establish minority business development objectives. The information is used by HUD to monitor and evaluate MBE activities against the total program activity and the designated minority business enterprise (MBE) goals. The Department requires the information to provide guidance and oversight for programs for the development of minority business enterprise concerning Minority Business Development. If the information is not collected HUD would not be able to establish meaningful MBE goals nor evaluate MBE performance against these goals.

Privacy Act Notice = The United States Department of Housing and Urban Development, Federal Housing Administration, is authorized to solicit the Information requested in this form by virtue of Title 12, United States Code, Section 1701 et seq., and regulation. It will not be disclosed or released outside the United States Department of Housing and Urban Development without your consent, except as required or permitted by Law.

[illegible]

**Washington State
Community Development Block Grant**

NOTIFICATION OF CONTRACT AWARD AND START OF CONSTRUCTION

This form must be completed within 10-days after contract award. The form must be filled out for the general contractor working on the CDBG awarded project.

1. CDBG Contract Information

Grantee:	CDBG Contract #:
Description of work:	

2. Construction Procurement

Bid Opening/Closing Date:	# Of Bids Received:	*SAM.gov Verification Date:
Construction Contract Award Date:	Construction Contract Amount:	Construction Start Date:
Contractor Name:		Contractor UBI #:

*Date verification was made at SAM.gov that contractor is not excluded from receiving federal funds.

3. Labor Standards

Federal Wage Decision #:	Modification #:	Published Date:	*10-Day Wage Verification Date:
**Person Responsible for Labor Standards Compliance:			
Email Address:		Phone #:	

*Date the grantee (or representative) determined if modifications were made to the federal wage decision contained in the bid document (must occur 10 days prior to the scheduled bid opening.)

**Person responsible for enforcing the labor standards requirements for the CDBG project.

FOR CDBG STAFF USE:

☐ **Copy routed to Labor Standards specialist (CDBG report collection file)**

LABOR STANDARDS COMPLIANCE REPORT

This form must be used for restitution (underpayments) of **\$10.00** to **\$999.99**.

For amounts totaling **\$1,000** or more per prime or subcontractor, contact your CDBG Project Manager.

PROJECT INFORMATION

CDBG Contract #:	HUD Program or Source of Funds: WA State CDBG
Grantee Name and Address:	
Description of Work:	
Prime Contractor:	Contract Amount: \$

VIOLATIONS

Number of Violations: _____

Number of Employees Involved in Violations: _____

Number of Subs Involved-Violations: _____

Total Amount of **Straight Time** Underpayments (restitution): \$ _____

Total Amount of **Overtime** Underpayments (restitution): \$ _____

(Federal Contract Work Hours & Safety Standards Act violations only)

Briefly Describe Any Labor Standards Complaints & Resolutions:

Is any contractor retainage held at this time? ☐ Yes ☐ No Amount of retainage held: \$

SUBCONTRACTOR(S) WHO HAD VIOLATIONS:

(Please attach more pages, if needed, for any additional information)

Name	Address	Contract Amount	Type of Work
		\$	
		\$	
		\$	

CERTIFICATION

I CERTIFY TO THE BEST OF MY KNOWLEDGE THAT ALL LABORERS AND MECHANICS EMPLOYED ON THIS CONSTRUCTION CONTRACT WERE PAID IN ACCORDANCE WITH THE LABOR PROVISIONS, EXCEPT AS NOTED. THERE ARE NO OUTSTANDING OR UNRESOLVED LABOR STANDARDS, UNDERPAYMENTS, COMPLAINTS, OR DISPUTES.

Signature:

Name:

Title:

Date:

Authorization to Sign Certified Payroll

Project Name

Project location

I, _____, *(Name of Owner or Officer)*,

authorize _____, *(Name of Payroll Supervisor/Signer)*,

to supervise the payment of employees for the above-referenced project and that they are in a position to have full knowledge of the facts set forth in the payroll documents and in the statement of compliance which they are to execute with my authority and approval.

The willful falsification of any information on the federal certified payrolls may subject the contractor or subcontractor to civil or criminal prosecution under Section 1001 of Title 18 and Section 231 of Title 31 of the United States Code.

(Name of Firm or Corporation)

(Name of Owner or Officer and Title)

(Signature)

(Date)

This certification must be executed prior to and be submitted with the first certified payroll. Should the appointee change, a new certification must accompany the first payroll for which the new appointee executes a statement of compliance.